
Thickness-Dependent Behaviour of Ultrathin W-Doped In_2O_3 Thin-Film Transistors

A Source-Traceable TCAD Study: Literature Basis, Inputs, Derivations, Simulation
Campaigns, Mechanism Analysis and Validation Status

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Silvaco ATLAS 5.28.1.R / DeckBuild 5.0.10.R drift-diffusion simulations (52 recorded launches)
of the 2.0 nm, 6.3 nm and 13.2 nm devices, with the 31.8 nm device as context

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Every simulated number in this report comes from a real ATLAS launch listed in Chapter A; every material
number carries a provenance label and a page-exact source (Section 2.16).

Abstract

Ultrathin tungsten-doped indium oxide (IWO) channels are candidates for back-end-of-line thin-film transistors, but the physical origin of their thickness dependence has not been established. This report presents a physics-constrained Silvaco ATLAS drift-diffusion model of bottom-gate IWO TFTs (TiN/HfO₂/Al₂O₃ gate stack, Pd contacts, $L = 20\ \mu\text{m}$) with channel thicknesses of 2, 6.3 and 13.2 nm, based on one measured device per thickness, and it examines which mechanisms the data can and cannot identify. The model combines an exponential acceptor tail, interface and deep states, a fixed front charge, an effective donor density, a confinement shift of the conduction band taken from first-principles slab calculations on pure In₂O₃, and a single constant band mobility for each film. Sixty ATLAS launches were recorded, of which 54 produced currents; they include one-at-a-time parameter controls, six hypothesis tests for the 6.3 nm film, a converged recalibration (DOS 384/192 levels) and the three runs of model V2. The measured devices follow the pattern $2\ \text{nm} \approx 6.3\ \text{nm} \neq 13.2\ \text{nm}$, with constant-current thresholds of 0.662, 0.644 and 0.083 V and field-effect mobilities of 12.2, 11.0 and 50.2 cm²/V s. This pattern cannot be described by a power law in thickness. On a single transfer curve, the confinement shift, the gate work function, the electron affinity and the fixed charge act together as one rigid offset, and a single shared offset fits the thresholds equally well with and without confinement (rms 0.136 and 0.133 V, respectively). Confinement is therefore an assumed input and not a result of the study. Within the calibrated trap-limited model, the approximately 4.6-fold increase in apparent mobility between 6.3 and 13.2 nm cannot be attributed to contact resistance, roughness, electrostatics or tail filling. Its origin is not determined, and a structural transition is adopted as the working hypothesis. The shared-parameter model misses the 6.3 nm threshold by $-0.294\ \text{V}$, and no tested variant reproduces the on-state shape of this device without degrading the subthreshold slope, which points to a limitation of the constant-mobility model form. The model is calibrated, not validated. Output characteristics and transfer curves at 338 and 358 K were registered with SHA-256 hashes before any such data were available. They were registered in two exact variants, because ATLAS had applied a default temperature exponent for the mobility, and they define explicit falsification rules.

Density-functional calculations of this work (Quantum ESPRESSO, PBE, checked against HSE06) reproduce the published confinement of 1 and 2 nm In₂O₃ films and show that 84–94 % of the gap opening lies in the conduction band. In a second model version (V2), these values replace the literature laws. They move the calibrated curves by at most 75 mV, and a pre-registered held-out test of the 6.3 nm film fails again (threshold $-0.23\ \text{V}$). A gate-independent strip of film outside the gate’s control, computed with the model’s own material values, reproduces the $t^{3.85}$ growth of the off-state floors with one fitted factor; its location is not determined. The report closes with a list of the measurements that are required before claims about mechanisms can be made.

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Nomenclature

This list fixes the meaning of every symbol, abbreviation, provenance label and evidence class used in the report. Values given here are those of the device and of the model as used throughout; the chapter or section in which a quantity is derived is given in the last column where applicable. All currents are per micrometre of channel width ($A/\mu m$); the simulated two-dimensional cross-section has a width of $1 \mu m$ (Section 3.2), and the total current of the $290 \mu m$ wide device is 290 times the quoted value.

Device and bias symbols

Symbol	Meaning	Unit / value
t_{IWO}, t	IWO channel thickness (nominal film thickness of the measured series)	nm; 2.0, 6.3, 13.2, 31.8
L	channel length (gap between the Pd source and drain)	$20 \mu m$
W	physical channel width	$290 \mu m$
V_G, V_D	gate and drain voltage (source grounded)	V; $V_D = 0.7 V$ for all transfer curves
I_D, I_G, I_S	drain, gate and source terminal currents	$A/\mu m$
C_{ox}	gate-stack capacitance per area (HfO ₂ 15 nm + Al ₂ O ₃ 2 nm in series)	$8.955 \times 10^{-7} F cm^{-2}$
EOT	equivalent SiO ₂ thickness of the stack	3.86 nm
q/C_{ox}	gate-voltage equivalent of a sheet charge of $10^{12} cm^{-2}$	0.179 V
T	lattice temperature (not recorded for the measurement; assumed)	300 K
$k_B T$	thermal energy at 300 K	25.85 meV

Extracted metrics

The same extractor (`scripts/extract_metrics.py`) is applied to measured and simulated curves; the definitions are derived in Section 5.10.

Symbol	Meaning	Unit
$V_{th,cc}$	constant-current threshold voltage, V_G at $I_D = 10^{-9} A/\mu m$	V
$V_{th,lin}$	linear-extrapolation threshold voltage (tangent at the transconductance maximum)	V
$g_m, g_{m,max}$	transconductance $\partial I_D / \partial V_G$ and its maximum	$A/(V \mu m)$

Symbol	Meaning	Unit
$SS_{[I_1, I_2]}$	fixed-current subthreshold swing: gate-voltage span per decade between the currents I_1 and I_2 ($A/\mu m$)	mV/dec
SS_{cc}	fixed-current swing between 10^{-10} and 10^{-8} A/ μm	mV/dec
SS_{min}	minimum local swing of a curve; not a reliable metric (grid-phase and floor-window artefacts, Section 6.6); reported for history only	mV/dec
μ_{FE}	linear-regime field-effect mobility, $L g_{m,max}/(WC_{ox} V_D)$	$cm^2/V s$
μ_{sat}	saturation-formula mobility, $(2L/WC_{ox}) (d\sqrt{I_D}/dV_G)_{max}^2$, applied at $V_D = 0.7 V$	$cm^2/V s$
I_{on}	drain current at $V_G = 3 V$	A/ μm
$I_{off, off-floor}$	median measured current in the off-band $-2 \leq V_G \leq -0.5 V$	A/ μm
active region	measured points with $I_D > 5 \times$ the off-floor (all points for $t \geq 20$ nm); used for the fit residual	–
active RMSE	root-mean-square of $\log_{10} I_D^{sim} - \log_{10} I_D^{meas}$ over the active region	dec

Material and model parameters

Symbol	Meaning	Unit / section
E_g	band gap (bulk value plus confinement increment)	eV, Section 4.3
χ	electron affinity	eV, Section 4.4
$\Delta E_c, \Delta E_g^{QC}$	confinement shift of the conduction-band edge; equal to the gap increment (E_v unchanged)	eV, Section 4.5
m^*	electron effective mass (enters only N_c)	m_0 , Section 4.7
N_c, N_v	effective conduction- and valence-band densities of states	cm^{-3} , Sections 4.8 and 4.9
ϵ_{IWO}	relative permittivity of IWO	9.3, Section 4.2
N_t	integrated acceptor-like tail density, $N_{TA} W_{TA}$	cm^{-3} , Section 4.10
N_{TA}, W_{TA}	tail DOS at E_c and its characteristic energy ($g_{TA} = N_{TA} \exp[(E - E_c)/W_{TA}]$)	$cm^{-3} eV^{-1}$, eV
N_{GA}, E_{GA}, W_{GA}	amplitude, depth below E_c and width of a Gaussian acceptor band	$cm^{-3} eV^{-1}$, eV, eV
D_{it}	interface acceptor sheet at the IWO/ Al_2O_3 interface (regularized in a 0.25 nm layer)	$cm^{-2} eV^{-1}$, Section 4.12
Q_f	fixed front-interface charge density (in units of q)	cm^{-2} , Section 4.13
Q_b	fixed charge on the exposed back (IWO/air) surface	cm^{-2} , Section 4.15
$N_{d,eff}$	effective ionized background donor density (not the W concentration)	cm^{-3} , Section 4.14
μ_{band}	constant band mobility of a film (fitted per thickness)	$cm^2/V s$, Section 5.1
MUN	ATLAS constant electron mobility, $\mu_{band}(1 - \Delta_{sr}/t)^2$	$cm^2/V s$
Δ_{sr}	surface-roughness length of the roughness factor	0.287 nm, Section 5.2
R_{sd}	source/drain series resistance (zero in the model: ideal Ohmic contacts)	$\Omega \mu m$, Section 5.6
E_a	activation energy of I_D at fixed V_G	meV, Section 5.9

Symbol	Meaning	Unit / section
tmu, TMUN	ATLAS exponent of the constant-mobility temperature law $\mu(T) = \mu(300)(T/300)^{-\text{TMUN}}$	1.5 (default, Section 5.8)
NUMA/NUMD	number of energy levels used by ATLAS to integrate the continuous acceptor/donor DOS	96/48 or 384/192, Section 5.11

Abbreviations

Abbreviation	Meaning
IWO	tungsten-doped indium oxide (about 2 at. % W in In_2O_3 ; composition definition not determined from the available data)
TFT	thin-film transistor
TCAD	technology computer-aided design (here: Silvaco ATLAS 5.28.1.R drift-diffusion)
ALD	atomic layer deposition
RTA	rapid thermal annealing
DOS	density of states
DFT, PBE	density functional theory; Perdew-Burke-Ernzerhof functional
EMA	effective-mass approximation
SP	Schrödinger-Poisson (1-D surrogate of specialist S3)
MTR	multiple trapping and release
SRH	Shockley-Read-Hall recombination
KCL	Kirchhoff current law check, $\max I_D + I_S + I_G $
CNL	charge-neutrality level
TLM	transfer-length method (not available)
PBS	positive bias stress
SI	supporting information of Januar et al. [1] (not bundled)
V0, V1	earlier per-thickness package (IWO_ATLAS_Model_claude, 34 launches) and the present physics-constrained package
Astra	earlier independent package (astra_IWO_PRIORITY_THREE_20260912), used for cross-checks
S1–S6	specialist analyses of 2026-09-25 (S1 electrostatics, S2 transport, S3 confinement, S4 defects, S5 contacts/validation, S6 numerics/predictions)

Run identifiers

Every simulated number carries its run identifier `run_NNNN` (NNNN = 0001 . . . 0052), the name of the folder `results/runs/run_NNNN_<film>_<label>` in the package; the film key is written 2p0, 6p3, 13p2, 31p8. The complete list is Table A.2 (Chapter A). Run identifiers of the earlier V0 package are always written “V0 run_NNNN” to avoid confusion. Campaign codes of 2026-09-25 map onto run identifiers as follows: A1 = run_0031, A2 = run_0032, A3 = run_0030, A4 = run_0033, A5 = run_0035, A6 = run_0034, A7 = run_0036; B0 = run_0037, B1–B3 = run_0038–run_0040, B4–B7 = run_0048–run_0051 (isolation tests), B8–B10 = run_0041–run_0043 (I_D – V_D predictions), B11–B14 = run_0044–run_0047 (elevated temperature); C1 = run_0052.

Provenance labels

Every parameter carries one label; the labels are those of `config/iwo_material_model.yaml` and the master table Table 2.6.

Label	Meaning
MEASURED_TARGET_DEVICE	measured on, or stated for, the target devices of Januar et al. [1] or the user's schematic and workbook
LITERATURE_IWO	value reported for IWO, but for a different device or process
MEASURED_RELATED_MATERIAL	measured on a related material (e.g. bulk In_2O_3)
DFT_PURE_IN2O3_PROXY	computed by DFT for pure In_2O_3 slabs and used as a proxy for IWO
LITERATURE_IN2O3_PROXY	literature value for an In_2O_3 device used as a proxy (e.g. D_{it} of $\text{HfO}_2/\text{In}_2\text{O}_3$)
FITTED	adjusted to the measured transfer curves of this study (calibration)
ASSUMED	chosen without direct evidence; the choice and its sensitivity are stated
CALCULATED	computed from other labelled quantities by a stated formula

Evidence classes and verdicts

Class	Meaning
DATA	a measured curve or a number extracted from it
NUM	converged ATLAS result; a statement about the model, not about the device
CAL	agreement with the same curve that was used to fit the parameters (calibration, not validation)
OOS	out-of-sample test designed before the outcome was known
POST	test designed or evaluated after the outcome was known
PRED	registered prediction, not yet tested against measurement
SURR	result of a specialist surrogate model (1-D Poisson, SP, charge sheet), not of ATLAS
THEORY	analytic argument
LIT	statement of the literature, with page
CORR	correction of an earlier claim of this project

Verdicts are **demonstrated**, **strongly supported**, **plausible**, **unverified**, **rejected** and **not determined**. “Calibrated” (CAL) and “validated” (OOS passed) are never used interchangeably.

Chapter 1

Introduction

When the channel of an oxide-semiconductor transistor is thinned to a few nanometres, almost every quantity that determines its current changes. Quantum confinement moves the band edges, the growing weight of the surfaces changes the trap density, roughness, microstructure and scattering change the mobility, and the contacts and the electrostatics of any fixed charge are modified as well. A measured thickness series can therefore be explained in many ways at the same time. This report examines which of these explanations a physics-constrained TCAD model, calibrated on one transfer curve per thickness, is able to distinguish. It documents every input, derivation, simulation run and test that is relevant to this question.

1.1 Ultrathin oxide-semiconductor channels and the IWO system

The target devices are the bottom-gate tungsten-doped In_2O_3 (IWO) TFTs of Januar et al. [1, pp. 2–3], which have a TiN gate, a $\text{HfO}_2/\text{Al}_2\text{O}_3$ gate stack and Pd source/drain contacts. The IWO series of that paper spans 2–13 nm [1, p. 2]. The measured curves modelled here have nominal channel thicknesses of 2, 6.3, 13.2 and 31.8 nm and an air-exposed back channel, as recorded in the device schematic and in the experimental workbook (Chapter 3). The ultrathin films do not have long-range crystallinity and show nanoscale grains [1, p. 2], and the AFM roughness is higher at 2 nm than at 10 nm [1, p. 8]. The same work reports positive-bias-stress trap densities that differ strongly between the 2 and 10 nm devices [1, p. 9], as well as supplementary transfer curves of a 2 nm device at 65 and 85 °C [1, SI Fig. S12, caption on p. 11]. The literature basis of every model input, including first-principles slab calculations on pure In_2O_3 and measured properties of related materials, is reviewed page by page in Chapter 2.

1.2 The measured thickness dependence

In the measured single devices at $V_D = 0.7$ V (Section 3.4), the 2 and 6.3 nm TFTs are electrically similar, whereas the 13.2 nm TFT differs strongly. The constant-current thresholds $V_{\text{th,cc}}$ are 0.662, 0.644 and 0.083 V, the linear mobilities μ_{FE} are 12.15, 10.95 and 50.17 $\text{cm}^2/\text{V s}$, the slopes SS_{cc} are 269.9, 295.4 and 160.0 mV/dec, the on-currents I_{on} are 5.09×10^{-7} , 4.03×10^{-7} and 3.07×10^{-6} A/ μm , and the off-state floors are 4.6×10^{-15} , 1.5×10^{-13} and 6.6×10^{-12} A/ μm . Since $\mu_{\text{FE}}(6.3) < \mu_{\text{FE}}(2)$, no monotonic law in t can pass through the three points. The published mobilities (5.1 and 27.4 $\text{cm}^2/\text{V s}$)

are reproduced exactly when the saturation formula is applied to these curves (Section 5.5). The 31.8 nm curve is not described by the model and is reported only as a documented failure.

1.3 Research question

Which physical mechanisms can explain the thickness dependence of the measured IWO TFTs, which of them are required, identified or excluded by the available data when represented in a physics-constrained ATLAS model, and what additional measurement would decide the rest?

Three subsidiary questions structure the work. The first is whether the quantum-confinement shift expected at 2 nm is necessary or identifiable. The second concerns the cause of the failure of the shared-parameter model at 6.3 nm. The third concerns what the calibrated model predicts for quantities that were not used in its calibration.

1.4 Competing explanations

Five families of mechanisms are examined in Chapter 9. The first family covers electrostatics and band alignment, namely confinement of the conduction band, work function, electron affinity, fixed and back-surface charge and donors (Section 9.1). The second covers transport, including band mobility, roughness, trap-limited conduction and microstructure (Section 9.2). The third is quantum confinement itself (Section 9.3), the fourth comprises tail, deep and interface traps together with the off-state floors (Section 9.4), and the fifth comprises contacts and series resistance (Section 9.5). The couplings between these mechanisms and the identifiability of their parameters are treated in Sections 9.6 and 9.7.

1.5 Integrity rules of the study

The study follows a fixed set of integrity rules. Every simulated number names the run from which it comes, and every analysis number names its report or script; no value is invented, and where evidence is missing the result is stated as “not determined”. Every statement taken from the literature carries the exact printed page. Evidence is classified into the classes DATA, NUM, CAL, OOS, POST, PRED, SURR, THEORY, LIT and CORR, and conclusions are graded as demonstrated, strongly supported, plausible, rejected or not determined. Calibration is never described as validation, and tests that were designed after the outcome was known are labelled post hoc. Superseded claims are corrected rather than repeated: run_0027 is retracted, $SS_{\min}$ is replaced by the fixed-current SS, and $t_{\text{mu}} = 1.5$ is stated for the elevated-temperature runs. Finally, predictions are registered with a timestamp and SHA-256 hashes before comparison data are obtained (Chapters D and 10).

1.6 Methodology

The workflow of the study is shown in Figure 1.1. It proceeds from the research question to the evidence, the model, the derivations and the implementation, and then to the calibration, the falsification tests,

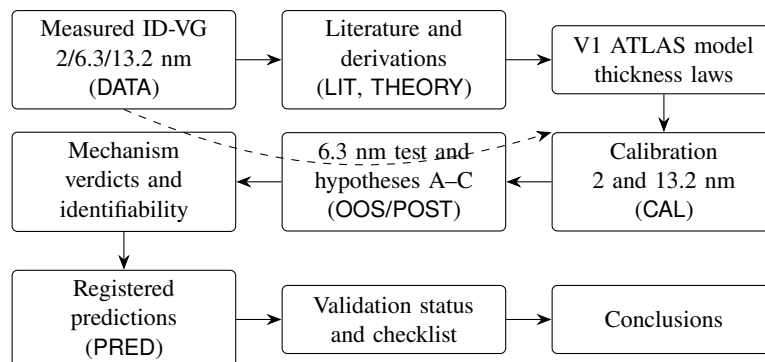


Figure 1.1. Workflow of the study: from the measured curves and the literature to the V1 ATLAS model, its calibration, the out-of-sample and post hoc tests, the mechanism verdicts, the registered predictions and the validation status. Evidence classes as in ?? .

the mechanism verdicts, the predictions, the validation status and finally the conclusions. The model is a classical drift-diffusion description for electrons only, with Fermi statistics, in the physical device structure (Chapter 6). All 52 recorded ATLAS launches (run_0001–run_0052), of which 49 produced scored currents, are catalogued in Chapter A. This catalogue includes the runs that do not fit the data; they are used to show how each parameter controls the curves (Section 7.4).

1.7 What is new relative to the earlier reports

Earlier documents of the package described the model as a thickness-resolved parameter extraction in which confinement accounted for the threshold step between 2 and 13.2 nm and the 6.3 nm run validated the shared laws. This report replaces those statements with the evidence-graded results of the analysis of 2026-09-25. According to these results, the thresholds are neutral with respect to confinement, the 6.3 nm test failed and its miss can be decomposed into a rigid part and a shape part, and the mobility step is located in the fitted band mobility. In addition, the numerics are converged where this was checked for 300 K transfer after a recalibration at DOS 384/192, and the predictions are registered before any comparison. The corrections are listed in Section 14.6.

1.8 Structure of the report

?? defines the symbols and evidence classes, and Chapter 2 reviews the literature basis. Chapter 3 describes the device and the data, Chapters 4 and 5 derive every electrostatic and transport input, and Chapter 6 documents the ATLAS implementation and its convergence. Chapter 7 presents the calibration and the parameter-control runs, Chapter 8 discusses the 6.3 nm problem, and Chapter 9 gives the mechanism verdicts. Chapter 10 contains the registered predictions, Chapter 11 describes the validation status and the work still required, and Chapter 14 states the conclusions. The appendices list all runs (Chapter A), the code (Chapter B), the input decks (Chapter C), the hashes of the prediction register (Chapter D) and the instructions for reproduction (Chapter E).

Summary. The study is based on three single devices with one transfer curve each and on an ATLAS study of 52 recorded launches, of which 49 produced scored currents. The report establishes what

these data identify, what they exclude and what must be measured next; it does not claim a validated model.

The next chapter reviews, page by page, the literature on which every model input is based, beginning with the target devices themselves.

Chapter 2

Literature basis of the TCAD model

The credibility of a drift–diffusion model of an ultrathin oxide-semiconductor transistor depends on the numbers that enter it. The thickness-dependent indium–tungsten oxide (IWO) model of this report contains about forty inputs: a gate stack, a band structure that changes with film thickness, an electron effective mass, a trap density of states, fixed and interface charges, a band mobility, contact assumptions and numerical settings. Some of these inputs were measured on the target device, some were taken from measurements on a related material (pure In_2O_3) and some from density-functional (DFT) slab calculations on pure In_2O_3 ; others were assumed, and the remainder were fitted to the measured transfer curves. For every source that supplied a number or a modelling choice, this chapter addresses three questions: (i) what exactly was taken, together with the printed page (and figure or equation) at which it can be checked; (ii) how the item enters the ATLAS model, with a reference to the parameter section in which it is derived; and (iii) what was considered but deliberately not used, and where the sources disagree. The chapter closes with a master provenance table (Section 2.16), which lists every model input with its status label, its source page or the run in which it was fitted, and the runs that test its sensitivity.

Table 2.1 gives an overview of all sources. Each bundled paper (text available locally with page markers) is treated in a section of its own, in Sections 2.2 to 2.7 and 2.9 to 2.13. The sources that are not bundled appear only through the bundled record that cites them (Section 2.14), and the Silvaco manual pages behind every deck statement are listed in Section 2.15.

2.1 Citation and page conventions

Page-exact citations. Every statement taken from a paper carries the printed page of the article, written as [2, p. 21539]; figures and equations are added where useful, as in [1, p. 8, Fig. 5f]. The local text files contain one block per PDF page. Because the mapping from the PDF page to the printed article page differs between journals, it is fixed in Table 2.2. Where a journal uses article-number pagination (APL, JAP), the page is written as *article number–page*, as in [3, p. 173507-6]. The publisher cover pages of Kim2024 and Stokey2021 carry no article page and are never cited.

Sources that are not bundled. The supporting information of the target paper and eight secondary works (Section 2.14) are not available locally, and they are never cited as if they had been read. Each of them appears either together with the bundled record that cites it and that record’s page (“as cited in Si

Table 2.1. Map of all literature and reference sources used by the model. “Bundled” means that the text is available locally with page markers and every page cited in this chapter was located in it. Status labels follow Section 2.1.

Key	Work	Bundled	Role in the model	Section
Januar2026	Target paper: IWO and In ₂ O ₃ TFTs vs thickness, Small Struct.	yes	gate stack and process; roughness Δ_{sr} ; trap trend $N_t(t)$; mobility extraction convention; analytic framework	Section 2.2
JanuarSI2026	Supporting information of the target paper	no	permittivities of IWO, In ₂ O ₃ , HfO ₂ (Fig. S1b); grains; temperature data (Fig. S12)	Section 2.14
Si2021	In ₂ O ₃ transistors down to 0.7 nm, Nano Lett.	yes	confinement point ΔE_c (1.5 nm); E_c -only partition; CNL above E_c	Section 2.3
Lin2022	nm-thick In ₂ O ₃ with high drain current, ACS Nano	yes	PBE slab gaps and masses; D_{it} of HfO ₂ /In ₂ O ₃	Section 2.4
Si2022	Scaled ALD In ₂ O ₃ transistors, Nat. Electron.	yes	CNL and low contact resistance (Ohmic S/D, qualitative)	Section 2.5
Wang2022	Traps in HfO ₂ /In ₂ O ₃ MOS capacitors, Front. Mater.	yes	D_{it} bracket; subgap DOS and TCAD trap conventions	Section 2.6
Kim2024	Thickness-dependent stability of IWO TFTs, APL	yes	thickness trend of bulk/border traps (support for $N_{d,eff}(t)$ rise)	Section 2.7
Stokey2021	Dielectric constants and electron mass of In ₂ O ₃ , JAP	yes	bulk $m^* = 0.208 m_0$; $\epsilon_{DC} = 10.55$ (sensitivity bound)	Section 2.9
Pandey2025	Analytical short-channel model of oxide TFTs, TED	yes	method reference only	Section 2.10
WangX2025	Physics-based compact model of IGZO FETs, TED	yes	method reference (Gaussian shallow donors)	Section 2.11
Anusha2024	Review of IGZO TCAD/compact modelling, Meas. Sens.	yes	SS–trap relation (theory)	Section 2.12
Giannozzi2009	Quantum ESPRESSO, JPCM	yes	software reference of the unexecuted DFT workflow	Section 2.13
Fan2021, Hamberg1986, Yoo2024, Lee2018, King2008, Robertson2011, Hinuma2019, Ando1982	secondary sources	no	only through a citing bundled record or the Astra audit	Section 2.14
AtlasManual, DeckBuildManual	Silvaco ATLAS 5.28.1.R / DeckBuild manuals	local PDF	syntax and physics of every deck statement	Section 2.15
ExperimentalWorkbooks, De-viceSchematic	measured ID–VG workbook; device schematic	local	ground truth; geometry	Chapter 3

Table 2.2. Page convention: printed page to be cited for the N -th PDF page of each bundled source.

Key	Local text file	Printed page for PDF page N	Printed range
Januar2026	Januar2026_SmallStruct_IWO.txt	N (“ N of 12”)	1–12
Si2021	Si2021_NanoLett_In2O3_0p7nm.txt	$499 + N$	500–506
Lin2022	Lin2022_ACSNano_thick_In2O3.txt	$21535 + N$	21536–21545
Si2022	NatElec2022_s41928-022-00718-w.txt	$163 + N$	164–170
Wang2022	Wang2022_FrontMater_traps_HfO2In2O3V.txt	$13507 - (N - 1)$; PDF p. 1 is a cover	1–7
Kim2024	Kim2024_APL_IWO_thickness_stability13507-1 to -7	$13507 - (N - 1)$; PDF p. 1 is a cover	173507-1 to -7
Stokey2021	Stokey2021_JAP_In2O3_dielectric_masse25102-1 to -15	$25102 - (N - 1)$; PDF p. 1 is a cover	225102-1 to -15
Pandey2025	Pandey2025_TED_SCE_TFT.txt	N (early-access pagination)	1–9
WangX2025	Wang2025_TED_IGZO_compact.txt	$2389 + N$	2390–2398
Anusha2024	Anusha2024_MeasSens_IGZO_review.txt	N	1–16
Giannozzi2009	Giannozzi2009_QE.txt	N	1–19
AtlasManual	atlas_users1.pdf (ATLAS 5.28.1.R)	printed footer page; PDF page stated where they differ	see Table 2.5

et al. [4, p. 501]”), or with an explicit note that the value was taken over from the audit of the earlier Astra package (inputs/astra_IWO_PRIORITY_THREE_20260912/.../audit/docs/paper_revision_20260911/).

Status labels. Every model input carries one label (typeset as small capitals, e.g. FITTED): MEASURED_TARGET_DEVICE (measured on the target process or device); LITERATURE_IWO (reported for IWO, possibly a different process or device); MEASURED_RELATED_MATERIAL (measured on pure In_2O_3); LITERATURE_IN2O3_PROXY (reported for In_2O_3 and transferred as a proxy); DFT_PURE_IN2O3_PROXY (PBE slab-minus-bulk differences for pure In_2O_3); CALCULATED (computed from other inputs by a stated formula); ASSUMED (chosen without a direct measurement; a way to measure it is given in the parameter section); FITTED (adjusted to the measured transfer curves, with the run that fixed it). The evidence classes of Chapter 11 (DATA, NUM, CAL, OOS, POST, PRED, SURR, THEORY, LIT, CORR) are used when a claim, rather than an input, is qualified. A literature value used as a model input belongs to the class LIT; a fitted value belongs to the class CAL and never counts as validation.

Structure of each paper section. Each paper section contains (a) a short summary; (b) a box that lists every item used, with the value, the printed page, the way in which it enters the model, the parameter section of Chapter 4 or Chapter 5 in which it is derived, and its caveat; (c) the items that were considered but not used; and (d) the contradictions with other sources or within the paper.

2.2 Januar et al. 2026: the target device and its analytic framework

Summary. Januar et al. [1] fabricate bottom-gate thin-film transistors with sputtered IWO (“~2% W”) and In_2O_3 channels on a $\text{TiN}/\text{HfO}_2/\text{Al}_2\text{O}_3$ gate stack with Pd source/drain contacts [1, pp. 2–3], and study how the channel thickness controls mobility, threshold, subthreshold swing and bias stability. They report peak field-effect mobilities of about 5.1 to $27.4 \text{ cm}^2/\text{Vs}$ for the IWO thickness series [1, p. 3]. The decline for thinner films is interpreted within a trap-limited transport framework in which the tail-trap density N_t grows roughly fourfold from 13.2 to 2 nm [1, p. 7], and the gate-dependent mobility is described by a surface-roughness (Ando-type) term with an effective roughness $\Delta_{\text{sr}} \approx 2.87 \text{ \AA}$ for IWO [1, pp. 7–8, Fig. 5f]. PBE-GGA calculations are used for qualitative trends in the band structure

and at the IWO/HfO₂ interface [1, p. 3], and positive-bias-stress (PBS) devices are analysed with the same compact model [1, p. 9]. The measured transfer curves of the workbook used in this report belong to the thickness-series devices of the paper; this identification was established in the S5 report, as described below.

Device and process facts used. The gate is a 50 nm TiN electrode deposited by physical vapour deposition, followed by the HfO₂/Al₂O₃ gate dielectric and the IWO or In₂O₃ channel [1, p. 2]. The channel lacks long-range crystallinity and consists of nanoscale grains, which are smaller for IWO than for In₂O₃ at comparable thickness [1, p. 2]. Pd source/drain electrodes of 70 nm complete the device, which is then annealed by rapid thermal annealing [1, p. 3]. The paper states that the trap metrics improve after annealing at 150 °C, and that higher annealing temperatures can raise the mobility further but make the device harder to turn off, through a crystallinity-driven increase of the carrier concentration [1, p. 3]. The HfO₂ and Al₂O₃ thicknesses (15 and 2 nm), $W/L = 290/20 \mu\text{m}$ and $V_D = 0.7 \text{ V}$ are taken from the device schematic [5] and are consistent with the stack description of the paper (details in Sections 3.1 and 3.2).

Items used in the TCAD model.

Items used from Januar et al. [1] (and its SI, not bundled).

Quantity	Value	Page	How it enters the model	Section	Caveat
Gate metal	TiN, 50 nm	p. 2	gate electrode region; work function assumed 4.70 eV	Sections 4.1 and 4.16	TiN work function not reported
Gate dielectric	HfO ₂ /Al ₂ O ₃ bilayer	p. 2	two insulator regions; $C_{\text{ox}} = 8.955 \times 10^{-7} \text{ F cm}^{-2}$	Section 4.1	thicknesses from the schematic
Channel	IWO (~2% W)	p. 2	material label only	Section 4.14	composition basis not defined; never converted to a donor density
S/D metal	Pd, 70 nm	p. 3	ideal Ohmic contacts	Section 5.6	no TLM data
ϵ_{IWO}	9.30	SI Fig. S1b (caption list p. 11)	permittivity of the channel	Section 4.2	SI not bundled; value recorded by the Astra audit
ϵ_{HfO_2}	19.57	SI Fig. S1b (p. 11)	gate-stack capacitance	Section 4.1	same
Δ_{sr}	2.87 Å	p. 8, Fig. 5f	roughness factor $(1 - \Delta_{\text{sr}}/t)^2 = 0.734, 0.911, 0.957, 0.982$	Section 5.2	fitted by the authors on μ_{sat} values; unit ambiguity between prose (Å) and figure axis (nm)
$N_{\text{t}}(13.2)/N_{\text{t}}(2)$	$\approx 1/4$	p. 7, Fig. 5a	trend constraint for $N_{\text{t}}(t) \propto t^{-0.75}$	Section 4.10	a compact-model output, not a measurement
N_{t} convention	$n_{\text{t}} \approx \theta_{\text{t}} N_{\text{t}} \exp[(E_{\text{F}} - E_{\text{c}})/k_{\text{B}} T_{\text{t}}]$	pp. 4, 9	mapping to ATLAS $N_{\text{TA}} W_{\text{TA}}$	Sections 4.10 and 5.3	absolute T_{t} not given (only changes)
μ_{max}	5.1 / 27.4 cm ² /V s (2 / 13.2 nm)	pp. 3, 7	defines the extraction convention; reproduced by the saturation formula (5.09 / 27.40, S5 report)	Section 5.5	not a model input; linear μ_{FE} is 12.15 / 50.17
Roll-off $\mu_0/(1 + \theta_{\text{r}} V_{\text{ov}})$	form only	p. 7	NOT inserted; approximation gap	Sections 5.1 and 5.4	ATLAS ignores MOBILITY updates between SOLVES
Smoother IWO/HfO ₂ interface	qualitative	pp. 2, 8	supports the low assumed D_{it}	Section 4.12	the target interface is IWO/Al ₂ O ₃
Grain/crystallinity trend	qualitative	pp. 2–3	hypothesis for the μ_{band} step 6.3 $\rightarrow$ 13.2 nm	Sections 5.1 and 9.2	not quantified

How the paper’s analytic framework is used. The paper bases its interpretation on a trap-limited conduction model in which the free-carrier fraction of the induced charge sets the apparent mobility [1, p. 7, Eq. 5], and on a tail-trap density written with a factor θ_{t} and a characteristic temperature T_{t} [1, p. 4, Eq. 3]; the same convention is used for the PBS extraction [1, p. 9]. These closed forms are not inserted into the TCAD model. The model solves the Poisson and drift–diffusion equations with an explicit acceptor-like exponential tail $g_{\text{TA}}(E) = N_{\text{TA}} \exp[(E - E_{\text{c}})/W_{\text{TA}}]$, whose integral $N_{\text{t}} \approx N_{\text{TA}} W_{\text{TA}}$ plays the role of N_{t} in the paper. The partition between free and trapped charge then follows self-consistently from the occupancy of the discretised trap levels (Section 5.3). Only the *trend* reported in the paper ($\times 4$ from 13.2 to 2 nm) constrains the model; the absolute magnitudes differ by the convention factor θ_{t} and are therefore compared but not equated (Section 4.10).

Mobility definition (resolved discrepancy). The mobilities of the paper (5.1 and 27.4 cm²/V s; 1, pp. 3, 7) are not the linear-regime field-effect mobilities of the workbook curves (12.15, 10.95 and 50.17 cm²/V s for 2.0, 6.3 and 13.2 nm, docs/MATERIAL_PARAMETER_EXTRACTION.md section 10; the 49.2 cm²/V s extracted at 31.8 nm is not a mobility, because g_m has several maxima there). Applying the saturation formula $\mu = (2L/WC_{ox})(d\sqrt{I_D}/dV_G)_{\max}^2$ to the workbook curves at $V_D = 0.7$ V gives 5.09 and 27.40 cm²/V s (S5 report, analysis_2026-09-25/S5_contacts_experimental_validation.md). This result identifies the workbook curves with the thickness series of the paper, and the $\mu_{\max}$ of the paper with a peak of the saturation formula. All comparisons in this report use one extraction, applied in the same way to measured and simulated curves (Sections 5.5 and 5.10).

Considered but not used. The absolute values $N_t(2\text{ nm}) = 5.3 \times 10^{19} \text{ cm}^{-3}$ and $N_t(10\text{ nm}) = 3.39 \times 10^{18} \text{ cm}^{-3}$ of the unstressed PBS devices [1, p. 9] were not used, because they belong to a different device set and depend on a θ_t convention; the model anchor is instead the real-ATLAS tail integral $N_{TA}W_{TA} = 2.0 \times 10^{19} \text{ cm}^{-3}$ at 2 nm (Section 4.10). The stress-induced changes of N_t and T_t under PBS [1, p. 9] were also not used, since the model describes unstressed DC transfer curves only. The PBE band structures and interface structures [1, pp. 3, 5] were used only qualitatively (W raises m^* ; the IWO/HfO₂ interface is smoother). No number was taken from them, because absolute PBE values are not transferable and the effect of W on m^* cannot be determined numerically from the available data. The oxygen-vacancy donor excess of undoped In₂O₃ ($n > 10^{19} \text{ cm}^{-3}$; 1, p. 2) serves as the motivation for W doping and is not a model input; the fitted $N_{d,\text{eff}}$ of IWO is 2–3 orders of magnitude lower (Section 4.14). The In₂O₃ reference devices of the paper consist of a different material and were not modelled. Finally, the elevated-temperature transfer curves of the 2 nm device at 338 and 358 K (SI Fig. S12, referred to on p. 9 and captioned on p. 11) were not used for calibration; they form the natural first test of the registered temperature predictions run_0044–run_0047 (Section 10.2).

Contradictions and open points.

1. *$N_t(t)$ exponent.* The thickness-series ratio (≈ 4 from 13.2 to 2 nm, p. 7) implies $N_t \propto t^{-0.73}$, whereas the two PBS devices (5.3×10^{19} at 2 nm, $3.39 \times 10^{18} \text{ cm}^{-3}$ at 10 nm, p. 9) imply $t^{-1.71}$ (S2 report). The two come from different device sets; the model follows the thickness series (exponent 0.75).
2. *Mobility extraction.* The $\mu_{\max}$ of the paper and the linear μ_{FE} differ by a factor of about 2 (see above); the difference arises from the extraction formula, not from the data.
3. *Roughness units.* Δ_{st} is printed as 2.87 Å in the text (p. 8), whereas the fitted figure uses a nm axis (a unit ambiguity recorded by the Astra audit); the Å value is used.
4. *Composition.* The specification “~2% W” (p. 2) does not state whether an atomic, cation or WO₃ fraction is meant; it is therefore treated as a label only.
5. *Equation conventions.* The Astra audit records that some main-text equations differ in convention from the SI. Only the N_t trend and the θ_t definition are used here, so the discrepancy does not propagate into the model.
6. *Interface.* The paper discusses the IWO/HfO₂ interface (pp. 2, 5, 8), but the target stack has a 2 nm Al₂O₃ interlayer between HfO₂ and IWO; statements about the interface are therefore

transferred only qualitatively.

2.3 Si et al. (2021): why In_2O_3 can make 0.7 nm transistors

Summary. Si et al. [4] demonstrate atomic-layer-deposited In_2O_3 transistors with channels down to sub-nanometre thickness and explain why an oxide with a very high bulk electron density (of order 10^{20} cm^{-3} ; 4, p. 500) can still be switched off when it is thin enough. Their argument has two parts. First, the charge-neutrality (transition) level of bulk In_2O_3 lies about 0.4 eV above the conduction-band edge [4, pp. 503–504], so surfaces and interfaces tend to donate electrons. Second, PBE slab calculations of In_2O_3 on Al_2O_3 show that thinning the film to 1.5 nm raises E_c by about 0.6 eV while E_v stays almost unchanged [4, p. 504, Fig. 4]; the charge-neutrality level then falls below E_c and the film becomes depletable. The paper also uses a relatively low In_2O_3 dielectric constant of 8.9 (taken from Hamberg and Granqvist) to argue for good gate control [4, p. 501].

Items used in the TCAD model.

Items used from Si et al. [4].

Quantity	Value	Page	How it enters the model	Section	Caveat
ΔE_c at 1.5 nm	$\approx 0.6 \text{ eV}$	p. 504, Fig. 4b	one of three fit points of $\Delta E_g^{\text{QC}}(t) = 0.9205 t^{-1.3815} \text{ eV}$	Sections 4.5 and 4.6	pure In_2O_3 , crystalline slab on Al_2O_3 , PBE; $\pm 50\%$ uncertainty assigned
Band-edge partition	E_v almost unchanged $\Rightarrow$ $\Delta E_c / \Delta E_g^{\text{QC}} = 1$	p. 504	the whole confinement shift is put on E_c : $\chi(t) = \chi_{\text{bulk}} - \Delta E_c(t)$	Sections 4.3 and 4.4	one slab thickness only
CNL/TNL above bulk E_c	$\approx 0.4 \text{ eV}$	pp. 503–504	consistency check: $\text{TNL} - E_c(t) = 0.4 - \Delta E_c(t)$ (+0.05 eV at 2 nm) predicts E_F deeper and V_{th} more positive for thinner films; qualitative support for Ohmic S/D	Sections 4.4, 4.18 and 5.6	not a deck parameter; pure In_2O_3

Considered but not used. The dielectric constant 8.9 (4, p. 501, from 6) is kept only as a lower bound of the permittivity range; the model uses the C–V value 9.30 measured on the target process (Section 4.2), and the one sensitivity run on permittivity used the upper bound 10.55 (run_0026). The bulk electron density of order 10^{20} cm^{-3} [4, p. 500] refers to undoped ALD In_2O_3 and is not transferred to IWO. The PBE absolute band gaps are not used (PBE underestimates them); only the slab-minus-bulk shift enters.

Contradictions. The Si2021 shift at 1.5 nm (0.6 eV) lies between the Lin2022 shifts at 1.98 and 0.95 nm (0.33 and 0.94 eV; Section 2.4) and is consistent with them, although the slabs differ in termination (Al_2O_3 versus hydrogen/vacuum) and in the code used. The quoted 8.9 conflicts with the bulk-crystal static value 10.55 of Stokey et al. [7, p. 225102-12] and with the target-process 9.30; the spread (8.9–10.55) is carried as the permittivity uncertainty.

2.4 Lin et al. (2022): nanometre-thick In_2O_3 with ultra-high drain current

Summary. Lin et al. [2] report ALD In_2O_3 transistors on HfO_2 with very high drain currents and large sheet densities (an n_{2D} of about $7.6 \times 10^{13} \text{ cm}^{-2}$ is estimated for one device; 2, p. 21537). From the subthreshold swing they infer a high-quality $\text{HfO}_2/\text{In}_2\text{O}_3$ interface with $D_{\text{it}} \approx 6 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$ and attribute the ease of accumulation to a charge-neutrality level about 0.4 eV above E_c [2, p. 21539]. DFT (PBE) calculations of bulk and slab In_2O_3 give conduction-band effective masses of 0.17 (bulk), 0.19 (3.52 nm slab), 0.23 (1.98 nm) and 0.30 m_0 (0.95 nm) [2, p. 21540], and the Methods list the PBE band gaps of bulk In_2O_3 , the 1.98 nm slab and the 0.95 nm slab as 0.94, 1.27 and 1.88 eV [2, p. 21542].

Items used in the TCAD model.

Items used from Lin et al. [2].

Quantity	Value	Page	How it enters the model	Section	Caveat
PBE gap, bulk / 1.98 nm / 0.95 nm	0.94 / 1.27 / 1.88 eV	p. 21542	shifts +0.33 and +0.94 eV = two fit points of $\Delta E_g^{\text{QC}}(t)$	Sections 4.5 and 4.6	PBE absolute gaps not used; H-passivated slab in vacuum; the 3.52 nm slab gap is not printed
m^* bulk / 3.52 / 1.98 / 0.95 nm	0.17 / 0.19 / 0.23 / 0.30 m_0	p. 21540	increments 0.02, 0.06, 0.13 m_0 over bulk give $b = 0.1311$, $q = 1.4121$ in $m^*(t) = 0.208 + b t^{-q}$; m^* also sets $N_c(t)$	Sections 4.7 and 4.8	PBE bulk 0.17 is below experiment 0.208; only the increment is used
D_{it} ($\text{HfO}_2/\text{In}_2\text{O}_3$)	$6 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$	p. 21539	upper literature bracket for the assumed Gaussian interface sheet $3 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$	Section 4.12	different interface (the target has $\text{Al}_2\text{O}_3/\text{IWO}$)
CNL above E_c	$\approx 0.4 \text{ eV}$	p. 21539	same consistency use as Si2021; context for Ohmic S/D	Sections 4.4 and 5.6	qualitative

Considered but not used. The very high sheet density and drain current [2, p. 21537] describe thick, highly conducting ALD In_2O_3 and are outside the range of the measured IWO devices. The PBE bulk mass 0.17 is replaced by the measured 0.208 (Section 2.9); only the slab-minus-bulk increment is kept, on the assumption that PBE errors cancel in the difference.

Contradictions. (i) The D_{it} unit is printed as $\text{cm}^{-1} \text{ eV}^{-1}$ at p. 21539; it is read as the usual $\text{cm}^{-2} \text{ eV}^{-1}$, consistent with the value $6.3 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$ of Wang et al. [8, p. 4]. (ii) PBE bulk $m^* = 0.17 m_0$ vs the experimental 0.208 m_0 [7, p. 225102-12] and the zero-density extrapolation 0.18 (Feneberg et al., as cited in 7, p. 225102-2); V1 uses 0.208 plus the DFT increment (docs/INPUT_INVENTORY.md section G, item 1).

2.5 Si et al. (2022): scaled ALD In₂O₃ transistors

Summary. Si et al. [9] scale ALD In₂O₃ transistors with high- k gate stacks (the device structure includes Ni gate metal and HfO₂, and EDX maps of a W/HfO₂/Al₂O₃/In₂O₃/Ni stack are shown; 9, pp. 164–165). They attribute the low contact resistance, below 0.1 Ω mm, to the charge-neutrality level of bulk In₂O₃ lying about 0.4 eV above the conduction-band edge [9, p. 167]. From capacitor $1/C^2$ - V analysis they estimate a donor density $N_D \approx 9.0 \times 10^{19} \text{ cm}^{-3}$, using an In₂O₃ permittivity of about 8.9 [9, p. 168]; the devices are annealed in O₂ at 250 °C [9, p. 169].

Items used in the TCAD model.

Items used from Si et al. [9].

Quantity	Value	Page	How it enters the model	Section	Caveat
Contact resistance	< 0.1 Ω mm	p. 167	qualitative support for ideal Ohmic Pd S/D contacts (no Schottky barrier, $R_{sd} = 0$ for the thin films)	Section 5.6	different metal (Ni), pure In ₂ O ₃ ; no TLM exists for the target device
CNL above E_c	≈ 0.4 eV	p. 167	mechanism behind the low contact barrier	Section 5.6	qualitative

Considered but not used. The donor density $9.0 \times 10^{19} \text{ cm}^{-3}$ [9, p. 168] is that of undoped ALD In₂O₃; the electrically active background donor density needed by the IWO model is $N_{d,\text{eff}} \approx 2.5 \times 10^{17} \text{ cm}^{-3}$ for the thin films (Section 4.14), 2–3 orders lower, consistent with W suppressing the oxygen-vacancy donors. The permittivity 8.9 (p. 168) is the same Hamberg value as in Si2021 and is treated as a lower bound only.

Contradictions. Apart from the donor density, which refers to a different material, no contradiction with the model exists. The low contact resistance supports, but does not prove, the Ohmic assumption for the IWO/Pd contacts; the Y-function analysis of the measured curves provides the check at device level (Section 5.7).

2.6 Wang et al. (2022): interface and bulk traps in HfO₂/In₂O₃ MOS capacitors

Summary. Wang et al. [8] characterise ALD In₂O₃ MOS capacitors with HfO₂ dielectric by C- V and conductance methods. They extract a donor density of about 10^{20} cm^{-3} and a high subgap density of states of $3.3 \times 10^{20} \text{ cm}^{-3} \text{ eV}^{-1}$ by the conductance method [8, pp. 1–2], recall that the In₂O₃ charge-neutrality level lies above E_c [8, p. 2], and report an interface-trap density of $6.3 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$ for the HfO₂/In₂O₃ interface, quoted from transistor data on p. 4 and estimated with the subthreshold method on p. 5 [8, pp. 4–5]. For the extraction they use $\epsilon_S = 8.9$ from optical measurements [8, p. 4]. A TCAD model with tail and Gaussian trap densities (bulk trap DOS up to $6 \times 10^{21} \text{ cm}^{-3} \text{ eV}^{-1}$) reproduces their C- V curves [8, p. 6].

Items used in the TCAD model.

Items used from Wang et al. [8].

Quantity	Value	Page	How it enters the model	Section	Caveat
D_{it} (HfO ₂ /In ₂ O ₃)	$6.3 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$	pp. 4–5	literature bracket for the assumed interface sheet (3×10^{11} at $E_c - 0.3 \text{ eV}$, width 0.12 eV)	Section 4.12	HfO ₂ contact, not Al ₂ O ₃ ; undoped In ₂ O ₃
TCAD trap representation	exponential tails + Gaussians	p. 6	supports the ATLAS DEFECTS representation (tail + deep Gaussian)	Sections 4.10 and 4.11	method only; their magnitudes are for undoped In ₂ O ₃

Considered but not used. The subgap DOS $3.3 \times 10^{20} \text{ cm}^{-3} \text{ eV}^{-1}$ [8, pp. 1–2] and the donor density $\sim 10^{20} \text{ cm}^{-3}$ (p. 1) describe undoped ALD In₂O₃ and are not transferred. For comparison, the model’s tail intercept N_{TA} is 5.0×10^{20} (2 nm) to 6.3×10^{19} (31.8 nm) $\text{cm}^{-3} \text{ eV}^{-1}$ at the band edge, falling exponentially into the gap (Section 4.10); the magnitudes are therefore of the same order near E_c but not comparable in energy resolution.

Contradictions. The D_{it} values of Lin2022 and Wang2022 agree ($\approx 6 \times 10^{11}$). The value 3×10^{11} assumed in the model is a factor of 2 lower, because the target interface has an Al₂O₃ interlayer and the target paper reports a smoother interface for IWO [1, p. 8]. The sensitivity to this value was tested with factors of $\times 3$ (run_0025, 2 nm) and $\times 5$ (run_0033, 6.3 nm); neither variant explains the threshold offset of the 6.3 nm film (Section 8.2).

2.7 Kim et al. (2024): thickness-dependent stability of IWO transistors

Summary. Kim et al. [3] study RF-sputtered IWO transistors (target In₂O₃:WO₃ 99:1 by weight, SiO₂ gate oxide, anneal at 300 °C; 3, p. 173507-2) with channel thicknesses of 10, 20 and 30 nm. AFM roughness increases with thickness (values up to 0.258 nm at 30 nm; 3, p. 173507-3), and low-frequency-noise analysis gives bulk/border trap densities N_{bt} of 2.37, 3.12 and $14.6 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$ for 10, 20 and 30 nm [3, p. 173507-6]. The work is the only bundled IWO thickness series other than the target paper, but it is a different process on a different gate dielectric.

Items used in the TCAD model.

Items used from Kim et al. [3].

Quantity	Value	Page	How it enters the model	Section	Caveat
N_{bt} at 10 / 20 / 30 nm	2.37 / 3.12 / $14.6 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$	p. 173507-6	qualitative support for the steep rise of $N_{d,eff}(t) = 2.5 \times 10^{17} [1 + (t/20 \text{ nm})^4]$ beyond $\approx 20 \text{ nm}$ (value $1.85 \times 10^{18} \text{ cm}^{-3}$ at 31.8 nm)	Section 4.14	a trap density, not a donor density; different process and gate dielectric

Considered but not used. The AFM roughness (p. 173507-3) is not used for Δ_{sr} ; the target paper’s own fit ($\Delta_{sr} = 2.87 \text{ \AA}$) applies to the target process (Section 5.2). The package extraction note also records a performance optimum near 20 nm from this paper as support for the crystallinity hypothesis of the μ_{band} step. The paper does report that its 20 nm devices combine the highest mobility with a

low trap density [3, pp. 173507-1, 173507-6], but it does not attribute this to crystallinity, so it is not used as evidence for that hypothesis here.

Contradictions. The N_{bt} trend (traps *increase* with thickness) runs opposite to the tail-trap trend of the target paper (N_t *decreases* with thickness, 1, p. 7). The two quantities are different (border traps from noise vs band-tail states from transfer curves) and the processes differ, so no conflict is resolved or implied; Kim2024 is used only as a qualitative indication that thick IWO films can carry a much larger density of electrically active defects.

2.8 Cross-source comparison of the In_2O_3 and IWO literature parameters

Table 2.3 collects the quantities reported by more than one source, with the value adopted by the model. Two general observations follow from it. First, every band-structure number is a pure- In_2O_3 proxy (from DFT or from measurement), and none was measured on IWO. Second, the donor and trap densities reported for undoped ALD In_2O_3 ($\sim 10^{20} \text{ cm}^{-3}$) are 2–3 orders of magnitude above the values required by the IWO transfer curves, so that the literature densities constrain the model only through trends and brackets.

Table 2.3. Quantities reported by several sources, and the value adopted by the V1 model.

Quantity	Reported values (source, page)	Adopted
CNL/TNL above bulk E_c	$\approx 0.4 \text{ eV}$ [4, pp. 503–504]; 0.4 eV [2, p. 21539]; 0.4 eV [9, p. 167]; above 0.4 eV [8, p. 2]	consistency check only
Confinement shift	$\Delta E_c \approx 0.6 \text{ eV}$ at 1.5 nm [4, p. 504]; $\Delta E_g^{\text{QC}} = 0.33$ (1.98 nm), 0.94 eV (0.95 nm) [2, p. 21542]	$0.9205 t^{-1.3815} \text{ eV}$ (Section 4.5)
Bulk m^*	0.17 (PBE) [2, p. 21540]; 0.208 [7, p. 225102-12]; 0.18 (zero density, Feneberg, cited in 7, p. 225102-2)	$0.208 m_0 + \text{DFT increment}$
Permittivity	8.9 (Hamberg, cited in 4, p. 501; 8, p. 4; 9, p. 168); 10.55 [7, p. 225102-12]; 9.30 IWO and 9.03 In_2O_3 (SI Fig. S1b of 10, not bundled; values via the Astra audit, caption printed in 1, p. 11)	9.30 (Section 4.2); 10.55 tested in run_0026
D_{it}	6×10^{11} [2, p. 21539]; 6.3×10^{11} [8, pp. 4–5] ($\text{cm}^{-2} \text{ eV}^{-1}$, $\text{HfO}_2/\text{In}_2\text{O}_3$)	3×10^{11} ASSUMED
Donor density	$\sim 10^{20}$ [4, p. 500]; $\sim 10^{20}$ [8, p. 1]; 9.0×10^{19} [9, p. 168] (undoped ALD In_2O_3)	$N_{d,\text{eff}}(t) = 2.5 \times 10^{17} - 1.85 \times 10^{18} \text{ cm}^{-3}$ FITTED
Trap trend with t	$N_t \times 4$ from 13.2 to 2 nm [1, p. 7]; N_{bt} rising $10 \rightarrow 30 \text{ nm}$ [3, p. 173507-6]	$N_t \propto t^{-0.75}$

2.9 Stokey et al. (2021): dielectric constants and effective mass of cubic In_2O_3

Summary. Stokey et al. [7] determine the infrared dielectric function of bulk single-crystal In_2O_3 by ellipsometry combined with density-functional perturbation theory, and measure the conduction-band effective mass by the optical Hall effect. They review earlier work, including the strongly nonparabolic conduction band and the zero-density mass parameter $m^*(0) = (0.18 \pm 0.02) m_e$ of Feneberg et al. [7, p. 225102-2], and the static dielectric constant of Hamberg and Granqvist [7, p. 225102-3]. From the phonon-mode parameters (Table II) and the Lyddane–Sachs–Teller relation

they obtain $\varepsilon_\infty = 4.05 \pm 0.05$ and $\varepsilon_{\text{DC}} = 10.55 \pm 0.07$ [7, p. 225102-12, Table II], and from the optical Hall data an effective mass $m^* = (0.208 \pm 0.006) m_e$ at a free-electron density of $2.81 \times 10^{17} \text{ cm}^{-3}$ [7, pp. 225102-12–13].

Items used in the TCAD model.

Items used from Stokey et al. [7].

Quantity	Value	Page	How it enters the model	Section	Caveat
Bulk m^*	$0.208 m_0$	pp. 225102-12–13	constant term of $m^*(t) = 0.208 + 0.1311 t^{-1.4121} (m_0)$; sets $N_c(t) = 2.509 \times 10^{19} (m^*/m_0)^{3/2} \text{ cm}^{-3} = 3.27, 2.55, 2.44, 2.40 \times 10^{18}$ for 2.0, 6.3, 13.2, 31.8 nm	Sections 4.7 and 4.8	pure crystalline In_2O_3 at low density; W and disorder may raise m^* ; tested by $m^* \times 1.3$ (run_0050, run_0051)
ε_{DC}	10.55	p. 225102-12, Table II	upper sensitivity bound of the channel permittivity (run_0026)	Section 4.2	bulk crystal; the model value is the C–V 9.30

Considered but not used. The zero-density extrapolation $0.18 m_0$ and the nonparabolicity of the conduction band [7, p. 225102-2] were not used, because the model employs a parabolic band (ATLAS continuum DOS). The effect of nonparabolicity is absorbed in the exponent of the confinement law, which is smaller than 2, and is discussed in Section 4.6. The high-frequency constant $\varepsilon_\infty = 4.05$ (p. 225102-12) has no role in a static device simulation. The bulk mobilities of the crystals (p. 225102-13) are not transferable to nanometre-thick amorphous/nanocrystalline IWO.

Contradictions. Three values of the bulk mass are available: 0.208 (optical Hall, $n = 2.8 \times 10^{17} \text{ cm}^{-3}$), 0.18 (zero-density extrapolation, Feneberg) and 0.17 (PBE, 2, p. 21540). Their spread of $\sim 20\%$ is smaller than the $\times 1.3$ sensitivity test, which shifted $V_{\text{th,cc}}$ by -44 mV at 2 nm (run_0050: 0.637 versus 0.680 V in run_0038) and by -29 mV at 13.2 nm (run_0051: 0.027 versus 0.055 V in run_0040).

2.10 Pandey et al. (2025): analytical short-channel model of BEOL oxide TFTs

Summary. Pandey et al. [11] derive a two-dimensional analytical potential model for oxide-semiconductor thin-film transistors with a Green’s-function approach, motivated by the high carrier densities (10^{19} – 10^{20} cm^{-3}) of In_2O_3 -type channels [11, p. 1]. They compare the Green’s function and scale-length approaches and approximate a Gaussian trap distribution by an integrable function [11, p. 2], then build the potential from Green’s identity [11, p. 3]. The model is checked against Sentaurus simulations calibrated to In_2O_3 data, in which the channel trap concentration is 10^{20} cm^{-3} at 0.4 eV above the conduction-band edge, with a Gaussian trap distribution and an intrinsic density of $5 \times 10^{19} \text{ cm}^{-3}$ [11, p. 5]; the trap density in the source/drain regions is set to $4 \times 10^{20} \text{ cm}^{-3}$, and it stays fixed there in the later channel-trap tests [11, pp. 5, 7].

Items used in the TCAD model.

Items used from Pandey et al. [11].

No number from this paper enters the model. It is used as a method reference: (i) a calibrated TCAD treatment of In_2O_3 places a large trap density at the charge-neutrality-like level ≈ 0.4 eV above E_c (p. 5), which is the same physical picture as the CNL argument used as a consistency check in Section 4.4; (ii) Gaussian trap distributions are standard in oxide-TFT modelling (p. 2), as used for the deep acceptor Gaussian and the interface sheet (Sections 4.11 and 4.12).

Considered but not used. The short-channel analysis itself was not used. The target devices have $L = 20\text{ }\mu\text{m}$, so that two-dimensional short-channel effects are negligible and the ATLAS model is in effect a long-channel simulation (Section 3.2). The trap magnitudes $10^{20} - 4 \times 10^{20}\text{ cm}^{-3}$ (pp. 5, 7) describe undoped In_2O_3 and are not transferred.

Contradictions. The HfO_2 relative permittivity used in their example is 15 [11, p. 2], lower than the 19.57 measured on the target process (SI Fig. S1b, via the Astra audit); the model uses 19.57 (Section 4.1).

2.11 Wang X. et al. (2025): physics-based compact model of IGZO FETs

Summary. Wang et al. [12] propose a compact model for IGZO-channel FETs in which the shallow donors are described by a Gaussian density of states that acts as positive charge in depletion [12, pp. 2390–2391]. The charge of the Gaussian distribution requires a Gauss–Fermi integral; they take over a published error-function approximation of it and replace the error function by a tanh function so that the model can be written in Verilog-A [12, pp. 2391–2392]; the electrostatics apply Gauss’s law across the gate oxide [12, p. 2393], and device-to-device variability is analysed with Gaussian-distributed oxide thickness, channel thickness and donor density [12, p. 2396].

Items used in the TCAD model.

Items used from Wang et al. [12].

No number is used. The paper supports the modelling choice recorded in `config/iwo_material_model.yaml`: a donor Gaussian at a fixed absolute energy (ATLAS NGD/EGD) is the natural V2 replacement for the empirical $N_{d,\text{eff}}(t)$ law, because confinement would then ionise or neutralise the donors automatically as E_c moves (Section 4.14). In V1 the donor Gaussian is set to zero (NGD = 0, ASSUMED).

Considered but not used. The closed-form Gauss–Fermi approximation (pp. 2391–2392) was not used, because ATLAS integrates the discretised DOS numerically and no analytical approximation is needed. The IGZO parameter values are not transferable to IWO.

Contradictions. No contradiction relevant to the model was found.

2.12 Anusha and Dwivedi (2024): review of TCAD and compact modelling of IGZO TFTs

Summary. Anusha and Dwivedi [13] review the simulation and compact modelling of amorphous IGZO transistors, including the commercial structure editors of Sentaurus and Silvaco [13, p. 2]. They

summarise work on W-doped InZnO (WIZO) transistors in which the field-effect mobility declines as the W content increases [13, p. 4, Fig. 2a], and quote the standard relation between the subthreshold swing and the bulk and interface trap densities, $SS \propto (k_B T/q) (N_{ss} t_{ch} + D_{it})$ (full form with C_{ox} in the source), used to separate N_{ss} and D_{it} by fixing one of them [13, p. 5]. Typical IGZO swings of about 120 mV/dec are mentioned in the outlook [13, p. 14].

Items used in the TCAD model.

Items used from Anusha and Dwivedi [13].

Quantity	Value	Page	How it enters the model	Section	Caveat
SS–trap relation	$N_{ss} t_{ch} + D_{it}$ form	p. 5	theory used to interpret the fixed-current SS and the trap-DOS inversion; not a deck input	Section 4.19	valid only where the traps are in equilibrium and uniformly distributed; bulk and interface terms are degenerate from one curve
W doping lowers mobility (WIZO)	qualitative	p. 4	context for the low μ_{band} of IWO vs In_2O_3	Section 5.1	different host (InZnO)

Considered but not used. The IGZO parameter sets of the review were not used, because they do not describe IWO.

Contradictions. No contradiction was found.

2.13 Giannozzi et al. (2009): Quantum ESPRESSO and the unexecuted DFT workflow

Summary. Giannozzi et al. [14] describe Quantum ESPRESSO, an open-source suite for electronic-structure calculations [14, p. 1] based on density-functional theory, plane-wave basis sets and norm-conserving, ultrasoft or projector-augmented-wave pseudopotentials, distributed under the GNU General Public License [14, p. 2].

Items used in the TCAD model.

Items used from Giannozzi et al. [14].

The paper serves as a software reference only. The package contains a prepared QE slab workflow (`qe_workflow/`) for computing the vacuum-aligned band edges and slab masses of IWO directly; its results would replace the pure- In_2O_3 proxies of Lin2022 and Si2021 and fix the absolute electron affinity. The workflow was not executed, because `pw.x` is absent on this host (Windows and WSL) and its installation required an interactive password (`docs/INPUT_INVENTORY.md`). No DFT number in this report was computed by the project; all DFT numbers are literature PBE values (`DFT_PURE_IN2O3_PROXY`).

2.14 Secondary sources that are not bundled locally

The works in Table 2.4 were not available as text, and none of them was read for this report. Each appears only through the bundled record that cites it (with the printed page of that record) or through the Astra audit notes (`audit/docs/paper_revision_20260911/defect_papers.md` and `ald_paper.md` in the Astra package), and is labelled accordingly. Values that exist only in the Astra audit are marked “via Astra” and could not be verified locally.

Table 2.4. Secondary (not bundled) sources: what is attributed to them, the citing record, and their use.

Key	Attributed content	Citing record	Use in the model
Januar et al. [10]	MIM/MISM C–V permittivities: IWO 9.30 ± 0.04 , In_2O_3 9.03, HfO_2 19.57 (Fig. S1b); grain sizes (Fig. S4); annealing series (Figs. S7, S8); 2 nm transfer curves at 338/358 K (Fig. S12)	captions printed in the main text, Januar et al. [1, p. 11]; values recorded by the Astra audit	$\epsilon_{\text{IWO}} = 9.30$ and $\epsilon_{\text{HfO}_2} = 19.57$ (MEASURED_TARGET_DEVICE, Sections 4.1 and 4.2); Fig. S12 is the first test of <code>run_0044–run_0047</code>
Fan et al. [15]	bulk $E_g = 3.05$ eV and $\chi = 4.30$ eV of a sputtered a-IWO (Sec. 3; χ a composition-mixing estimate); deep-acceptor range $2.5\text{--}3.7 \times 10^{16}$ (p. 9); N_c used as a fit knob; printed inconsistencies noted by Astra	via Astra (<code>defect_papers.md</code>)	E_g , χ bulk priors (LITERATURE_IWO, Sections 4.3 and 4.4); $N_{\text{GA}} = 5 \times 10^{16}$ (ASSUMED, Section 4.11)
Hamberg and Granqvist [6]	static permittivity ≈ 8.9 of evaporated (Sn-doped) In_2O_3 films	Si et al. [4, p. 501]; Wang et al. [8, p. 4]; Si et al. [9, p. 168]; Stokey et al. [7, p. 225102-3]	lower bound of the permittivity range only
Feneberg et al. (no key)	zero-density mass $m^*(0) = 0.18 \pm 0.02 m_e$, nonparabolicity	Stokey et al. [7, p. 225102-2]	considered; 0.208 used
King et al. [16]	charge-neutrality level above E_c , surface electron accumulation in In_2O_3	cited for TNL ≈ 0.4 eV in Si et al. [4, p. 504] (ref. 26, listed p. 506)	consistency check only (Section 4.4)
Robertson and Clark [17]	limits to doping in oxides; CNL alignment	Si et al. [4, p. 504] (ref. 24); Wang et al. [8, p. 5]	consistency check only
Hinuma et al. [18]	band alignment at surfaces and interfaces	listed as ref. 27 in Si et al. [4, p. 506]	none numeric
Ando et al. [19]	surface-roughness scattering and the mobility roll-off $\mu_0/(1 + \theta_r V_{\text{ov}})$	Januar et al. [1, p. 7] (ref. 39, listed p. 11)	form of the roughness factor; roll-off not inserted (Sections 5.1 and 5.2)
Yoo et al. [20]	ALD IWO channel	ref. 14 of Januar et al. [1, p. 10]; reviewed in the Astra <code>ald_paper.md</code>	none numeric (process context)
Lee et al. [21]	gate-current effects in oxide TFTs (title only; content not read here)	no bundled record cites it; listed via the Astra package bibliography	none numeric; attribution to be confirmed

2.15 The Silvaco ATLAS 5.28.1.R user manual: every statement used by the decks

The decks (Section 6.1, full listings in Chapter C) use a small set of native ATLAS statements. Table 2.5 gives, for each statement group, the printed manual page(s) where its syntax and parameters are defined, and what the decks set with it. Page numbers are the printed page numbers of `atlas_users1.pdf` (ATLAS 5.28.1.R); the PDF page is equal to the printed page at p. 179, 4 higher at p. 513 and 6 higher throughout the statement chapter (pp. 1117–1647) (`digest/atlas_manual_pages.md`).

Two facts about the simulator are relevant to the physics; both were established by runs rather than taken from the manual. First, ATLAS 5.28.1.R ignores MOBILITY updates between SOLVE statements (V0 Claude run 0008, recorded in `config/iwo_material_model.yaml`); for this reason the gate-dependent roll-off of Januar et al. [1, p. 7] could not be inserted. Second, the constant-mobility temperature exponent $tmu = 1.5$ is active in the elevated-temperature runs (SYNTHESIS correction), so that the registered temperature predictions exist in a phonon-like and an MTR-like variant (Section 10.2).

2.16 Master provenance table

Table 2.6 lists every input of the V1 ATLAS model: the material description (`config/iwo_material_model.yaml`), the geometry, the numerical settings, and the per-film fitted values, including the DOS 384/192 recalibration. For each input the table gives the value(s) per film (2.0 / 6.3 / 13.2 / 31.8 nm where they differ), the status label, the source with its printed page or the run that fixed the value, the section in which the input is derived, and the runs that test its sensitivity. Run identifiers refer to `results/RUN_INDEX.csv` (catalogue in Table A.2). Three conventions apply. (i) “Seed” denotes the value written in the YAML before calibration. (ii) The fitted values were set in stages: Q_f in stage 3 (run_0003, run_0004), μ_{band} in stage 6 (run_0008, run_0009), the 6.3 nm device-specific pair in run_0007/run_0015, and the DOS 384/192 recalibration in run_0038–run_0040; because I_D is linear in a uniform constant mobility (Section 5.4), the recalibrated μ_{band} of 6.3 and 13.2 nm were finally rescaled exactly by $\times 1.015074$ and $\times 1.020113$, and these rescaled values (17.6897 / 11.582 / 61.6046) are the ones used in the prediction runs run_0041–run_0047. (iii) The retracted run run_0027 (malformed mesh, contact overlap) is listed but is not evidence.

Table 2.6. Master provenance of every model input. Values per film are given as 2.0 / 6.3 / 13.2 / 31.8 nm where they differ. Status labels as in Section 2.1.

Input	Value(s)	Status	Source (page) or run	Derivation	Sensitivity runs
Geometry and gate stack					
Channel length, width	$L = 20 \mu\text{m}$, $W = 290 \mu\text{m}$; simulated width $1 \mu\text{m}$ ($A/\mu\text{m}$)	MEASURED_TAS	schematic [6]	Section 3.2	–
Film thickness t_{WO}	2.0 / 6.3 / 13.2 / 31.8 nm	MEASURED_TAS	workbook headers [23]	Section 3.3	6.3 nm as 5.3 nm (run_0031), 5.8 nm (run_0035)

continued on next page

Table 2.6 continued

Input	Value(s)	Status	Source (page) or run	Derivation	Sensitivity runs
Gate electrode	TiN, 50 nm	MEASURED_TARGET_DEVICE;	Januar et al. [1, p. 2]	Section 4.1	–
HfO ₂ thickness, permittivity	15 nm; 19.57	MEASURED_TARGET_DEVICE;	schematic; SI Fig. S1b [10] via Astra (caption 1, p. 11)	Section 4.1	–
Al ₂ O ₃ thickness, permittivity	2 nm; 9.0	thickness MEA-SURED_TARGET_DEVICE; ϵ ASSUMED	schematic; –	Section 4.1	–
Gate capacitance C_{ox} , EOT	$8.955 \times 10^{-7} \text{ F cm}^{-2}$; 3.86 nm; $C_{ox}/q = 5.59 \times 10^{12} \text{ cm}^{-2} \text{ V}^{-1}$	CALCULATED	$\epsilon_0/(15/19.57 + 2/9.0) \text{ nm}$	Section 4.1	–
S/D contacts	Pd 70 nm; overlap $2 \mu\text{m}$	metal MEA-SURED_TARGET_DEVICE; overlap ASSUMED	Januar et al. [1, p. 3]; –	Section 5.6	overlap $4 \mu\text{m}$ run_0027 (malformed, retracted)
Top of channel	air, no cap	MEASURED_TARGET_DEVICE	schematic	Section 3.2	–
Composition	“~2% W”, label only	not determined	Januar et al. [1, p. 2]	Section 4.14	–
Band structure and electrostatics					
Bulk band gap E_g	3.05 eV (± 0.2)	LITERATURE_IW	Fan et al. [15] Sec. 3 via Astra	Section 4.3	through confinement off (run_0016, run_0017)
Bulk affinity χ	4.30 eV (± 0.3)	LITERATURE_IW	Fan et al. [15] Sec. 3 via Astra	Section 4.4	± 0.1 eV: run_0020, run_0021
Confinement $\Delta E_g^{QC}(t)$	$0.9205 t^{-1.3815} \text{ eV}$; 0.353 / 0.072 / 0.026 / 0.008 eV	DFT_PURE_IN2D	Lin et al. [2, p. 21542]; Si et al. [4, p. 504]	Sections 4.5 and 4.6	off: run_0016, run_0017; 6.3 nm off with Q_f from 2 nm: run_0034
Partition $\Delta E_c/\Delta E_g^{QC}$	1.0	DFT_PURE_IN2D	Si et al. [4, p. 504]	Section 4.4	–
$E_g(t), \chi(t)$	$3.05 + \Delta E_g^{QC}(t)$; $4.30 - \Delta E_c(t)$	CALCULATED	from the two rows above	Sections 4.3 and 4.4	as above
Effective mass $m^*(t)$	0.208 + $0.1311 t^{-1.4121} m_0$	MEASURED_RELATIVE_EFFECTIVE_MASS + DFT increment	Stokoy et al. [7], pp. 225102-12-13; Lin et al. [2, p. 21540]	Section 4.7	$\times 1.3$: run_0050 (2 nm), run_0051 (13.2 nm)
$N_c(t)$ (300 K)	$3.27 / 2.55 / 2.44 / 2.40 \times 10^{18} \text{ cm}^{-3}$	CALCULATED	$2(2\pi m^* kT/h^2)^{3/2}$	Section 4.8	coupled to m^* in run_0050, run_0051
N_v	10^{19} cm^{-3}	ASSUMED	– (holes not solved)	Section 4.9	–
Channel permittivity ϵ_{TWO}	9.30	MEASURED_TARGET_DEVICE;	SI Fig. S1b [10] via Astra	Section 4.2	10.55: run_0026

continued on next page

Table 2.6 continued

Input	Value(s)	Status	Source (page) or run	Derivation	Sensitivity runs
Gate work function	4.70 eV	ASSUMED (TiN 4.5–4.9 eV)	–	Section 4.16	+0.1 eV: run_0048 (2 nm), run_0049 (13.2 nm)
Charges and traps					
Tail density $N_t(t)$	$2.0 \times 10^{19} (2/t)^{0.75} \text{ cm}^{-3}$; $N_{TA} = 5.0 / 2.1 / 1.2 /$ $0.63 \times 10^{20} \text{ cm}^{-3} \text{ eV}^{-1}$	FITTED law (anchor CAL at 2 nm; trend LIT)	trend Januar et al. [1, p. 7]; anchor from real-ATLAS lineages	Section 4.10	$\times 1.5$: run_0023; 6.3 nm $N_t = 2.66 \times 10^{19}$: run_0037
Tail width W_{TA}	0.040 eV ($T_t = 464 \text{ K}$), shared	FITTED	V0/Astra calibrations	Section 4.10	+5 meV: run_0024
Deep acceptor Gaussian	$N_{GA} =$ $5 \times 10^{16} \text{ cm}^{-3} \text{ eV}^{-1}$, $E_{GA} = E_c - 0.6 \text{ eV}$, $W_{GA} = 0.15 \text{ eV}$	ASSUMED	range of Fan et al. [15] via Astra	Section 4.11	– (insensitive)
Donor tail, donor Gaussian	NTD = 0, NGD = 0	ASSUMED	Fan et al. [15] via Astra; V2 option Wang et al. [12, p. 2391]	Section 4.14	–
Interface acceptor sheet D_{it}	$3 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$ at $E_c - 0.3 \text{ eV}$, width 0.12 eV; volume $1.2 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$ in 0.25 nm	ASSUMED	bracket Lin et al. [2, p. 21539], Wang et al. [8, pp. 4–5]	Section 4.12	$\times 3$: run_0025; $\times 5$ at 6.3 nm: run_0033; near- E_c band at 6.3 nm: run_0052
Fixed charge Q_f (shared)	seed 0; fitted $1.73 \times 10^{12} \text{ cm}^{-2}$ ($\Delta V_{fb} = -0.310 \text{ V}$) for 2.0, 13.2 (and 31.8) nm	FITTED	run_0003, run_0004; kept in run_0012, run_0013, run_0038, run_0040	Section 4.13	0 vs 1.73×10^{12} : run_0001/run_0003, run_0002/run_0004
Fixed charge Q_f (6.3 nm)	$8.7 \times 10^{10} \text{ cm}^{-2}$ ($\Delta V_{fb} = -0.016 \text{ V}$)	FITTED device- specific	run_0007, run_0015, run_0039	Sections 4.13 and 8.4	shared value held out: run_0005, run_0014; 4.76×10^{10} : run_0034; 9.8×10^{11} : run_0037
Required Q_f per film (analysis)	with confinement 1.81 / 0.09 / 1.56×10^{12} ; without 0.047 / $-0.27 /$ $1.43 \times 10^{12} \text{ cm}^{-2}$	analysis (POST)	SYNTHESIS D3	Section 9.1	–
Background donors $N_{d,eff}(t)$	$2.5 \times 10^{17} [1 + (t/20)^4]$: 2.5 / 2.5 / 3.0 / 18.5 $\times 10^{17} \text{ cm}^{-3}$	FITTED law	trend support Kim et al. [3, p. 173507-6]	Section 4.14	$\times 2$: run_0022; 10^{16} at 6.3 nm: run_0032, run_0035

continued on next page

Table 2.6 continued

Input	Value(s)	Status	Source (page) or run	Derivation	Sensitivity runs
Back-surface charge / donors	0 for thin films; 6.3 nm hypotheses -1.64×10^{12} , $-1.0 \times 10^{12} \text{ cm}^{-2}$; 31.8 nm donor sheet hypothesis	HYPOTHESIS	–	Section 4.15	run_0030, run_0035; 31.8 nm run_0010 (aborted)
Capture cross sections	10^{-15} cm^2	ASSUMED	–	Section 4.17	not identifiable from DC
SRH lifetime	10^{-6} s	ASSUMED	–	Section 4.17	–
Transport					
Roughness Δ_{sr}	0.287 nm; factor $(1 - \Delta_{\text{sr}}/t)^2 = 0.734 / 0.911 / 0.957 / 0.982$	LITERATURE	Januar et al. [1, p. 8, Fig. 5f]	Section 5.2	–
Band mobility μ_{band} , seed	16.8 / 12.4 / 57.6 / 84 $\text{cm}^2/\text{V s}$	FITTED (V0)	YAML seed; 31.8 nm run run_0006	Section 5.1	–
μ_{band} , stage 6 (DOS 96/48)	18.13 / 11.69 / 61.9 / – $\text{cm}^2/\text{V s}$	FITTED	run_0008, run_0015, run_0009; final run_0012, run_0013	Sections 5.1 and 7.3	run_0003 vs run_0008; run_0004 vs run_0009
μ_{band} , DOS 384/192	17.69 / 11.41 / 60.39 $\text{cm}^2/\text{V s}$; rescaled 17.6897 / 11.582 / 61.6046	FITTED	run_0038, run_0039 ($\times 1.015074$), run_0040 ($\times 1.020113$); rescaled values used in the predictions run_0041–run_0047; the isolation runs run_0048–run_0051 keep 17.69 / 60.39 as their baselines run_0038 / run_0040	Sections 5.1 and 5.4	–
μ_{band} , 6.3 nm hypotheses	19.6 (S2 re-partition); 15.4 (S4 near- E_c band)	HYPOTHESIS	run_0037; run_0052	Section 8.2	–
Gate-dependent roll-off	not inserted	approximation	form: Januar et al. [1, p. 7]	Section 5.4	–
Hole mobility	0.01 $\text{cm}^2/\text{V s}$	ASSUMED	–	Section 4.9	–
Temperature, tmu	300 K; predictions 338.15 / 358.15 K with tmu = 1.5	ASSUMED	[22, p. 179]	Section 5.8	run_0044–run_0047
Contacts					
S/D model	ideal Ohmic	ASSUMED	context Si et al. [9, p. 167], Lin et al. [2, p. 21538]	Section 5.6	Y-function check (Section 5.7)
Series resistance	0 / 0 / 0 / 92000 $\Omega \mu\text{m}$	FITTED hypothesis (31.8 nm only)	bound: total device 672 Ω at 3 V (Astra measurement constraints)	Section 5.6	–

continued on next page

Table 2.6 continued

Input	Value(s)	Status	Source (page) or run	Derivation	Sensitivity runs
Numerics					
DOS discretisation	96/48 (run_0012–run_0015); choice	numerical	[22, pp. 1168–1174]	Section 5.11	192/96 run_0019; 384/192 run_0029
NUMA/NUMD	384/192 (run_0038–run_0052)				
Mesh	spacing factor 1.0	numerical choice	[22, pp. 1373–1377]	Section 6.2	0.7: run_0028 (13.2 nm), run_0036 (6.3 nm); run_0018 timed out
Interface-layer regularisation	0.25 nm layer	numerical choice	–	Section 4.12	layer/2 changes I_D by 0.009 dec (V0)
Bias sweep	$V_G = -3 \rightarrow +3$ V (measured grid 0.05 V); $V_D = 0.7$ V; stepped VSTEP/VFINAL (0.2 / 0.05 / 0.1 V)	MEASURED_TARGET (bias); numerical (stepping)	workbook [22, pp. 1593–1611]	Section 6.4	–
Solver	Newton, electrons only	numerical choice	[22, pp. 1383–1389]	Section 6.3	–

Reading the table. Among the inputs that control the transfer curves most strongly (Section 7.4), only the gate stack was measured on the target device. The confinement law and the effective mass are pure-In₂O₃ proxies, the tail density follows a literature trend but has a fitted anchor, and Q_f and μ_{band} are fitted per film (with one Q_f shared by 2.0 and 13.2 nm). The model is therefore calibrated, but not validated, on the three thin films. The held-out test at 6.3 nm with shared parameters (run_0014) failed in threshold and could be repaired only by a device-specific Q_f (Section 8.4).

Summary. Literature basis: what the sources do and do not fix.

- Measured on the target process: gate stack (TiN 50 nm and Pd S/D, 1, pp. 2–3; HfO₂ 15 nm and Al₂O₃ 2 nm from the device schematic), the HfO₂ permittivity 19.57 and the channel permittivity 9.30; the two permittivities come from the non-bundled SI (Fig. S1b, caption printed in 1, p. 11) via the Astra audit.
- Pure-In₂O₃ proxies: the confinement shift $\Delta E_g^{\text{QC}}(t)$ from PBE slabs [2, p. 21542], [4, p. 504], and the mass $0.208 m_0$ plus a PBE increment (7, p. 225102-12; 2, p. 21540).
- Literature trends and brackets, not values: $N_t(t)$ ($\times 4$, 1, p. 7), $D_{\text{it}} \approx 6 \times 10^{11}$ (2, p. 21539; 8, pp. 4–5), the rise of defect densities in thick IWO [3, p. 173507-6], the CNL above E_c .
- Fitted to the measured curves: Q_f (1.73×10^{12} shared; $8.7 \times 10^{10} \text{ cm}^{-2}$ at 6.3 nm), μ_{band} (17.69 / 11.58 / 61.60 cm²/V s at DOS 384/192: run_0038, run_0039 $\times 1.015074$, run_0040 $\times 1.020113$), the N_t anchor, W_{TA} and $N_{\text{d,eff}}(t)$.
- Not determined from the available data: the absolute electron affinity of IWO, the W composition basis, the absolute tail temperature, the contact resistance of the thin films, and the effect of W

Table 2.5. ATLAS statements used by the decks, with the printed manual pages and the model content.

Statement (parameters)	Manual pages	What the decks set	Section
MESH, X.MESH, Y.MESH	[22, pp. 1117–1118, 1373–1377]	structured 2-D mesh; fine vertical spacing in the film; spacing factor 1.0 (0.7 in the checks <code>run_0028</code> , <code>run_0036</code>)	Section 6.2
REGION	[22, pp. 1548–1560]	TiN gate, HfO ₂ 15 nm, Al ₂ O ₃ 2 nm, IWO film, 0.25 nm interface layer, back-surface layer, air	Sections 3.2 and 4.1
ELECTRODE	[22, pp. 1207–1211]	gate, source, drain (Pd, 2 μ m overlap)	Section 5.6
DOPING UNIFORM	[22, pp. 1183–1193]	$N_{d,eff}(t)$ as uniform n-type background	Section 4.14
MATERIAL (EG300, AFFINITY, PERMITTIVITY, NC300, NV300, MUN, MUP, TAUN0, EGALPHA, TMUN)	[22, pp. 1307–1355]	$E_g(t)$, $\chi(t)$, ϵ_{IWO} , $N_c(t)$, N_v , $\mu_0(t) = \mu_{band}(t)(1 - \Delta_{sr}/t)^2$, hole mobility 0.01, SRH lifetime	Sections 4.3, 4.4, 4.8 and 5.1
Constant mobility temperature dependence (TMUN)	[22, p. 179]; [22, p. 513]	$\mu \propto (T/300)^{-tmu}$ with $tmu = 1.5$ in the elevated-temperature runs <code>run_0044</code> – <code>run_0047</code>	Section 5.8
CONTACT (WORKFUNCTION, RESISTANCE)	[22, pp. 1145–1156]	gate work function 4.70 eV; Ohmic S/D (no work function); lumped series resistance for 31.8 nm only	Sections 4.16 and 5.6
MODELS (FERMI, SRH, TEMPERATURE, PRINT)	[22, pp. 1426, 1433, 1444–1448, 1453, 1462, 1477–1478]	Fermi–Dirac statistics, SRH, lattice temperature (300 K; 338.15/358.15 K in predictions)	Sections 5.8 and 6.3
DEFECTS (CONTINUOUS, NUMA, NUMD, NTA, WTA, NGA, EGA, WGA, NGD, EGD, SIGTAE)	[22, pp. 1167–1174]	acceptor tail $N_{TA}(t)$, $W_{TA} = 0.040$ eV; deep Gaussian; interface Gaussian in the 0.25 nm layer; back-surface donor Gaussian (31.8 nm hypothesis); NUMA/NUMD = 96/48, 192/96, 384/192	Sections 4.10 to 4.12 and 5.11
INTERFACE (QF)	[22, pp. 1252–1264]	fixed charge Q_f at the Al ₂ O ₃ /IWO interface	Section 4.13
METHOD (NEWTON, TRAP, MAXTRAPS, ITLIMIT, CLIMIT, IR.TOL, CR.TOLER, XANDRNORM, CARRIERS)	[22, pp. 1378–1390, 1395]	Newton solver, electrons only, bias-step reduction on failure	Section 6.3
SOLVE (INIT, VSTEP, VFINAL, NAME)	[22, pp. 1587–1611]	stepped gate sweeps $-3 \rightarrow +3$ V (steps 0.2 / 0.05 / 0.1 V) at $V_D = 0.7$ V; ID–VD sweeps	Section 6.4
LOG, SAVE, PROBE, OUTPUT, EXTRACT	[22, pp. 1216, 1299–1305, 1504–1515, 1524–1540, 1566–1579]	terminal currents, structure files, band diagrams (EXTRACT is only listed in the ATLAS manual, p. 1216, and documented in the DeckBuild manual)	Section 6.1

on m^* .

With the source of every input established, the next chapter considers the measured devices themselves: their geometry, the transfer curves and the metrics extracted from them.

Chapter 3

Device, measured data and inputs

A TCAD model cannot be more specific than the information on which it is built. This chapter therefore states, item by item, what is known about the measured transistors, where each item comes from, and what is not known. It addresses four questions: (i) which device and process are modelled (Section 3.1); (ii) how the physical stack is translated into the two-dimensional simulation domain (Section 3.2); (iii) what the measured data consist of (Section 3.3) and which figures of merit they yield when the extractor used for the simulations is applied to them (Section 3.4); and (iv) which gaps in the data limit every later conclusion (Section 3.5). No quantity in this chapter is fitted. Every number is either measured on the target devices, read from the user's schematic, or explicitly labelled as assumed.

3.1 Device and process

The devices are the bottom-gate IWO thin-film transistors of the thickness series of Januar et al. [1]. The primary source for the stack is Sec. 2.1 of that paper [1, pp. 2–3] together with the user's device schematic [5]; the measured transfer curves are taken from the experimental workbook [23].

From the substrate upwards, the stack consists of an n^+ Si support carrying a 50 nm TiN bottom gate deposited by physical vapour deposition. On the gate lie a HfO_2 gate dielectric (nominal 15 nm) and an Al_2O_3 interlayer (nominal 2 nm), both grown by ALD at 250 °C. The paper reports growth rates of about 0.92 and 1.04 Å per cycle for the two oxides [1, p. 2], while the nominal thicknesses of 15 and 2 nm are taken from the schematic; the Al_2O_3 layer is in contact with the channel. The channel is a sputtered IWO film (about 2 % W; the definition of the composition cannot be determined from the available data) of nominal thickness 2.0, 6.3, 13.2 or 31.8 nm. 70 nm Pd source and drain electrodes are placed on top of the channel and patterned with contact holes and metallization. There is no passivation or capping layer, so the channel between source and drain is exposed to the ambient.

A rapid thermal anneal at 150 °C for 600 s completes the process (process record of the package, docs/DEVICE_STRUCTURE.md); it serves as process context only and does not enter any model parameter directly. The channel length and width, $L = 20 \mu\text{m}$ and $W = 290 \mu\text{m}$, and the drain bias of the transfer measurement, $V_D = 0.7 \text{ V}$, are read from the schematic, because the workbook columns carry no units.

Table 3.1 lists every structural item with its provenance label. The HfO_2 permittivity 19.57 and the IWO permittivity 9.30 are C-V results reported in the SI of Januar et al. [1] (Fig. S1; the SI is not

bundled, and the values were recorded by the Astra audit; the main text gives only “approximately 9” for IWO, 1, p. 2). The Al_2O_3 permittivity 9.0 is assumed. The consequences of each label for the gate capacitance are derived in Section 4.1.

Table 3.1. Structural items of the modelled device and their provenance (config/geometry.yaml, docs/DEVICE_STRUCTURE.md).

Item	Value	Provenance
Gate	TiN 50 nm on n^+ Si (Si screened by the continuous TiN, omitted)	MEASURED_TARGET_DEVICE
Gate dielectric	HfO_2 15.0 nm, $\epsilon_r = 19.57$ (SI Fig. S1)	MEASURED_TARGET_DEVICE
Interlayer	Al_2O_3 2.0 nm, $\epsilon_r = 9.0$	thickness MEASURED_TARGET_DEVICE; ϵ_r ASSUMED
Channel	IWO, $t = 2.0/6.3/13.2/31.8$ nm, $\epsilon_{\text{IWO}} = 9.3$	MEASURED_TARGET_DEVICE
Source/drain	Pd 70 nm, top contacts	MEASURED_TARGET_DEVICE
S/D overlap	2 μm per side	ASSUMED (not reported)
Top of channel	air, no cap	as reported (no fictitious cap added)
W/L	290 μm / 20 μm	MEASURED_TARGET_DEVICE (schematic)
V_D (transfer)	0.7 V	MEASURED_TARGET_DEVICE (schematic)
Temperature	300 K	ASSUMED (not recorded)
Current unit	A/ μm	interpretation of the schematic axis

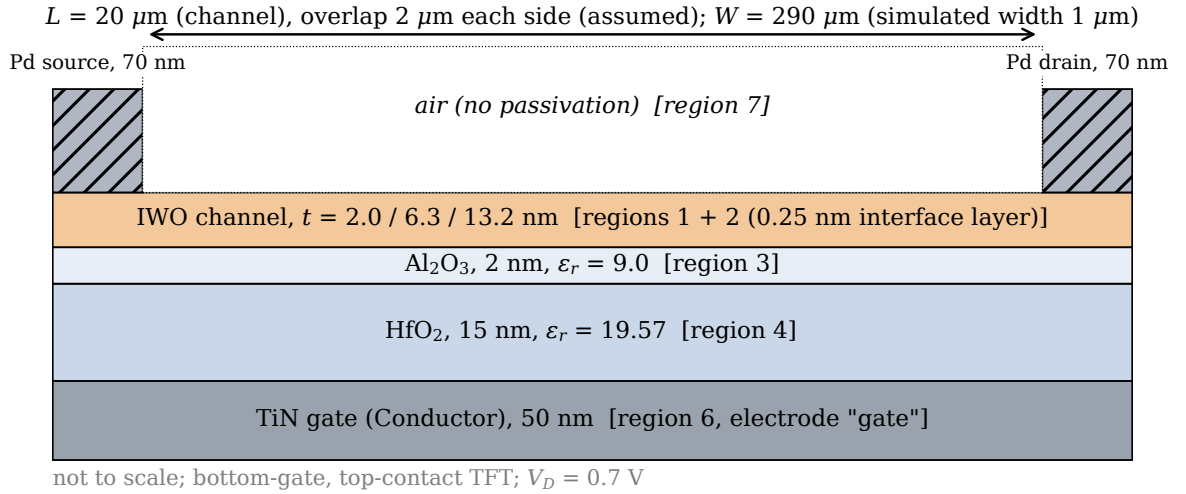


Figure 3.1. Cross-section of the modelled device (not to scale): TiN gate 50 nm, HfO_2 15 nm, Al_2O_3 2 nm, IWO channel of thickness t , Pd source/drain 70 nm overlapping the channel by 2 μm on each side, air above the channel; $L = 20 \mu\text{m}$, $W = 290 \mu\text{m}$. The ATLAS region numbers of the physical-structure deck are indicated (1: IWO bulk; 2: 0.25 nm IWO interface layer; 3: Al_2O_3 ; 4: HfO_2 ; 6: Conductor gate volume; 7: Air). Drawn from config/geometry.yaml and docs/DEVICE_STRUCTURE.md.

3.2 Geometry and the simulated structure

Coordinates. The simulation domain is a two-dimensional cross-section along the channel. In micrometres, x runs from 0 to 24: the source contact occupies $0 \leq x \leq 2$, the channel $2 < x < 22$ and the drain $22 \leq x \leq 24$. The vertical coordinate y increases downward from the IWO/ Al_2O_3 interface at $y = 0$. The IWO film occupies $-t \leq y \leq 0$, the Al_2O_3 $0 \leq y \leq 0.002$, the HfO_2 $0.002 \leq y \leq 0.017$ and the TiN gate volume $0.017 \leq y \leq 0.067$; the Pd electrodes and the air region

occupy $-t - 0.07 \leq y \leq -t$. The $2\text{ }\mu\text{m}$ overlap of the contacts is not reported. With contacts on the top surface of a $20\text{ }\mu\text{m}$ long channel, the drain current is expected to be insensitive to it. However, the one launch intended to test this (run_0027, overlap $4\text{ }\mu\text{m}$) is malformed, and its result is retracted (Section 7.4).

Width convention. The mesh is given the width $1\text{ }\mu\text{m}$ (mesh width=1.0), so that every logged current is numerically in $\text{A}/\mu\text{m}$ and can be compared directly with the workbook. The current of the physical device is $290\times$ the logged value; a division by 290 is never applied. This convention was verified in the V0 package (V0 run_0007 with WIDTH=2 against V0 run_0005: raw currents exactly $\times 2.000000$, currents per micrometre identical; check J of Table 6.2).

Two electrically equivalent representations. The deck generator scripts/build_iwo_decks.py writes the stack in one of two forms (Section 6.1):

1. *Reduced structure* (runs run_0001–run_0009): IWO regions, Al_2O_3 ($\epsilon_r = 9.0$) and HfO_2 ($\epsilon_r = 19.57$). The gate is a Dirichlet boundary at the bottom of the HfO_2 with the TiN work function, and source and drain are Dirichlet boundaries on the IWO top surface over 0–2 and 22–24 μm . In DC a continuous metal is an equipotential, so replacing its volume by a boundary with the same work function changes neither the field in the dielectrics nor the contact boundary condition.
2. *Physical structure* (structure: full, every run from run_0012 on): the same IWO regions; explicit Al_2O_3 and HfO_2 regions whose permittivities are set on material region= statements (the ATLAS defaults for these material names are not the measured values); a 50 nm Conductor volume carrying the gate electrode (TiN is not in the ATLAS metal table; work function 4.70 eV set on contact, ASSUMED); a 70 nm Air region above the channel; and 70 nm $\times$ 2 μm Palladium source/drain electrode volumes without a work function (Ohmic, ASSUMED; no TLM data).

The equivalence of the two forms was verified in the real solver. run_0012 (physical structure, stepped sweep) and run_0008 (reduced structure, pointwise sweep), with identical parameters, differ by +0.08, +0.05, +0.01, -0.003 and -0.002% in I_D at $V_G = 0.5, 0.7, 1, 2$ and 3 V (check M in Table 6.2).

Gate capacitance. With $\epsilon_0 = 8.854 \times 10^{-14}\text{ F cm}^{-1}$, the series capacitance of the two oxides is

$$C_{\text{ox}} = \frac{\epsilon_0}{t_{\text{HfO}_2}/\epsilon_{\text{HfO}_2} + t_{\text{Al}_2\text{O}_3}/\epsilon_{\text{Al}_2\text{O}_3}} = \frac{8.854 \times 10^{-14}}{(1.5 \times 10^{-6}/19.57) + (2 \times 10^{-7}/9.0)}\text{ F cm}^{-2} = 8.955 \times 10^{-7}\text{ F cm}^{-2}, \quad (3.1)$$

which is equivalent to $\text{EOT} = 3.9\epsilon_0/C_{\text{ox}} = 3.86\text{ nm}$. A sheet charge of 10^{12} cm^{-2} corresponds to $q \times 10^{12}/C_{\text{ox}} = 0.179\text{ V}$. These three numbers are used throughout the report; their derivation and error budget are given in Section 4.1.

3.3 The measured data

The measurement consists of a single transfer curve per thickness, stored in the package as data/experimental_clear with the columns thickness_nm, source_row, vg_V and id_A_per_um. Each curve spans $-3 \leq$

$V_G \leq 3$ V in steps of 0.05 V at $V_D = 0.7$ V. No other electrical data exist for these devices (Section 3.5).

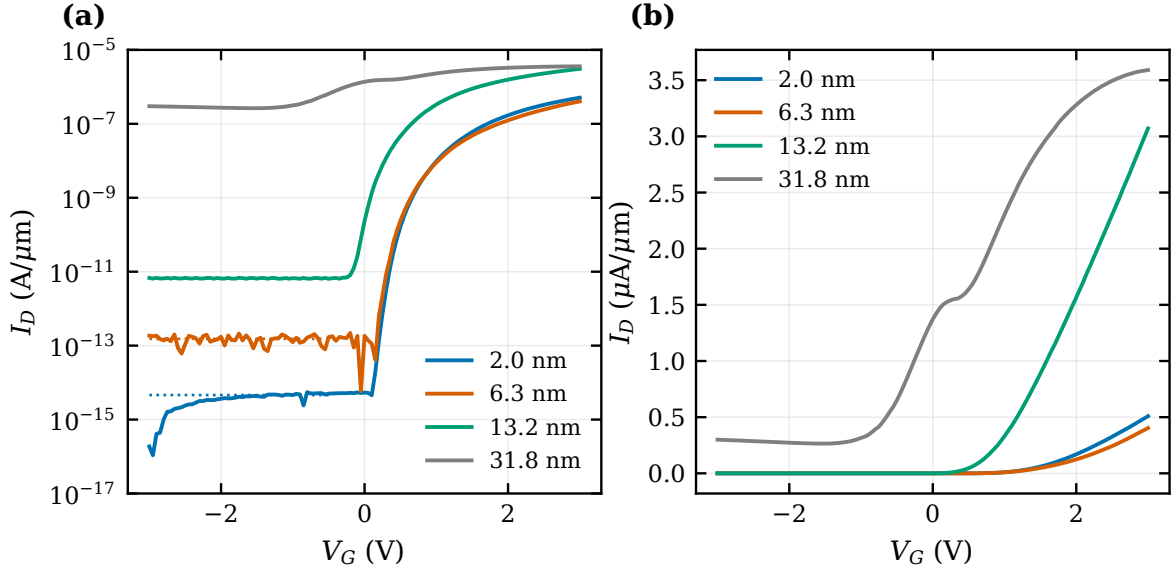


Figure 3.2. Measured transfer curves of the 2.0, 6.3, 13.2 and 31.8 nm devices at $V_D = 0.7$ V (data/experimental_clean.csv). (a) Logarithmic scale; the dotted lines mark the off-band medians ($-2 \leq V_G \leq -0.5$ V): 4.61×10^{-15} (2.0 nm), 1.53×10^{-13} (6.3 nm) and 6.60×10^{-12} $\text{A}/\mu\text{m}$ (13.2 nm). (b) Linear scale. The 2.0 and 6.3 nm curves are close to each other, the 13.2 nm curve is shifted to lower gate voltage and carries about six times more on-current, and the 31.8 nm device does not turn off within the sweep.

Four features of Figure 3.2 determine the course of the whole study.

1. *Non-monotonic thickness dependence.* The 2.0 nm and 6.3 nm curves nearly coincide in threshold and on-current, whereas the 13.2 nm curve differs strongly; the 6.3 nm film even has the lower field-effect mobility of the two thin films (Section 3.4). A smooth thickness law cannot describe this behaviour without a device-specific parameter (Chapter 8).
2. *Off-state floors.* Each curve saturates at a flat plateau in the off-band. The plateaus increase steeply with thickness: the 13.2 nm floor is 1432 times the 2 nm floor, which corresponds to a log-log slope of 3.78 over the three thin films (the S4 report quotes 3.85; Figure 9.4). Drift-diffusion with thermal generation does not produce such floors; the Astra audit bounded the thermal-generation current at 13 to 17 decades below them. Gate leakage is disfavoured as the dominant floor at 6.3 and 13.2 nm, because the floors do not follow the drain–gate bias (Section 9.4). Their origin (a measurement floor or an ungated channel path) cannot be determined from the available data, because I_G and I_S were not recorded.
3. *An isolated low point.* The minimum of the 2 nm curve, about 1.1×10^{-16} $\text{A}/\mu\text{m}$ at -3 V, is a single point about 40 times below the floor of that curve. It is treated as an outlier, and the KCL acceptance limit of the runner was redefined on the median off-band current instead of the minimum (Section 6.5).
4. *The 31.8 nm device is always on* ($I_D \geq 2.6 \times 10^{-7}$ $\text{A}/\mu\text{m}$ at $V_G = -3$ V). It lies outside the 2–13 nm series published by Januar et al. [1, p. 2], was excluded by the user from the final calibration set, and appears only in the early runs `run_0006` and `run_0010`. The V0 package required an additional back-surface donor sheet and a series resistance of about $9 \times 10^4 \Omega \mu\text{m}$ to describe it.

3.4 Measured metrics

Table 3.2 lists the figures of merit of the measured curves. They are extracted by `scripts/extract_metrics.py`, the same code that scores every ATLAS run, so that measured and simulated numbers are directly comparable; the definitions and their derivations are given in Section 5.10. The swing is reported over fixed current windows, because the minimum local swing $SS_{\min}$ depends on the voltage grid and on the floor window and is therefore not a reliable metric (Section 5.10); it is listed only for comparison with earlier documents.

Table 3.2. Measured metrics at $V_D = 0.7$ V (DATA; `data/experimental_clean.csv`, extractor `scripts/extract_metrics.py`; numbers from `figures/MANIFEST.md` and the evidence brief of 2026-09-25). Currents in $\text{A}/\mu\text{m}$, swings in mV/dec , mobilities in $\text{cm}^2/(\text{V s})$.

Metric	2.0 nm	6.3 nm	13.2 nm	31.8 nm
$V_{\text{th,cc}}$ at $10^{-9} \text{ A}/\mu\text{m}$ (V)	0.662	0.644	0.083	always on
$V_{\text{th,lin}}$ (V)	1.663	1.820	1.044	–
$V_{\text{th,lin}} - V_{\text{th,cc}}$ (V)	1.001	1.176	0.962	–
$SS_{[10^{-11}, 10^{-10}]}$	121.2	124.5	136.1	–
$SS_{[10^{-10}, 10^{-9}]}$	194.8	210.0	121.6	–
$SS_{\text{cc}} = SS_{[10^{-10}, 10^{-8}]}$	269.9	295.4	160.0	–
$SS_{\min}$ (not robust)	84.5	114.7	130.8	–
$g_{\text{m,max}}$ ($\text{A}/(\text{V } \mu\text{m})$)	3.807×10^{-7}	3.431×10^{-7}	1.572×10^{-6}	–
μ_{FE} (linear, g_{m} -based)	12.15	10.95	50.17	–
μ_{sat} (saturation formula)	5.09	3.86	27.40	–
$I_{\text{on}} = I_D(V_G = 3 \text{ V})$	5.09×10^{-7}	4.03×10^{-7}	3.07×10^{-6}	3.59×10^{-6}
Off-floor (median, -2 to -0.5 V)	4.61×10^{-15}	1.53×10^{-13}	6.60×10^{-12}	–

Interpretation of the table. Three observations are essential for the later chapters. (i) $V_{\text{th,cc}}$, μ_{FE} and I_{on} are non-monotonic or step-like: the two thinnest films are similar, whereas the 13.2 nm film differs by about 0.58 V in threshold and by a factor of about four in mobility. (ii) The 6.3 nm film has the largest gap $V_{\text{th,lin}} - V_{\text{th,cc}}$ (1.176 V); this on-state shape is the quantity that no parameter set of the model reproduces (Chapter 8). (iii) The swing in the window 10^{-10} – $10^{-8} \text{ A}/\mu\text{m}$ is 270–295 mV/dec for the two thin films and 160 mV/dec for the 13.2 nm film. These values are far above the thermal limit of about 60 mV/dec at 300 K, which indicates a large density of states in the gap.

Mobility definitions and the values of the paper. Januar et al. [1] quote peak mobilities of about 5.1 and 27.4 $\text{cm}^2/(\text{V s})$ for the thinnest and thickest films of the series [1, p. 3] and relate the decrease from about 27.4 to about 5.1 to a roughly fourfold increase of the trap density [1, p. 7]. The linear, transconductance-based μ_{FE} of the workbook curves is 12.15 and 50.17 $\text{cm}^2/(\text{V s})$, i.e. a factor of 2.4 and 1.8 higher. Specialist S5 showed that applying the saturation formula to the same curves at $V_D = 0.7$ V gives 5.09 and 27.40 $\text{cm}^2/(\text{V s})$, which reproduces the numbers of the paper. The workbook curves are therefore the thickness series of the paper, and the mobility of the paper is a saturation-formula peak taken near threshold rather than a linear-regime field-effect mobility (CORR; Section 5.5, Figure 5.1). The model is calibrated against the curves themselves and never against a mobility number, so the choice of definition affects the comparison with the literature but not the calibration.

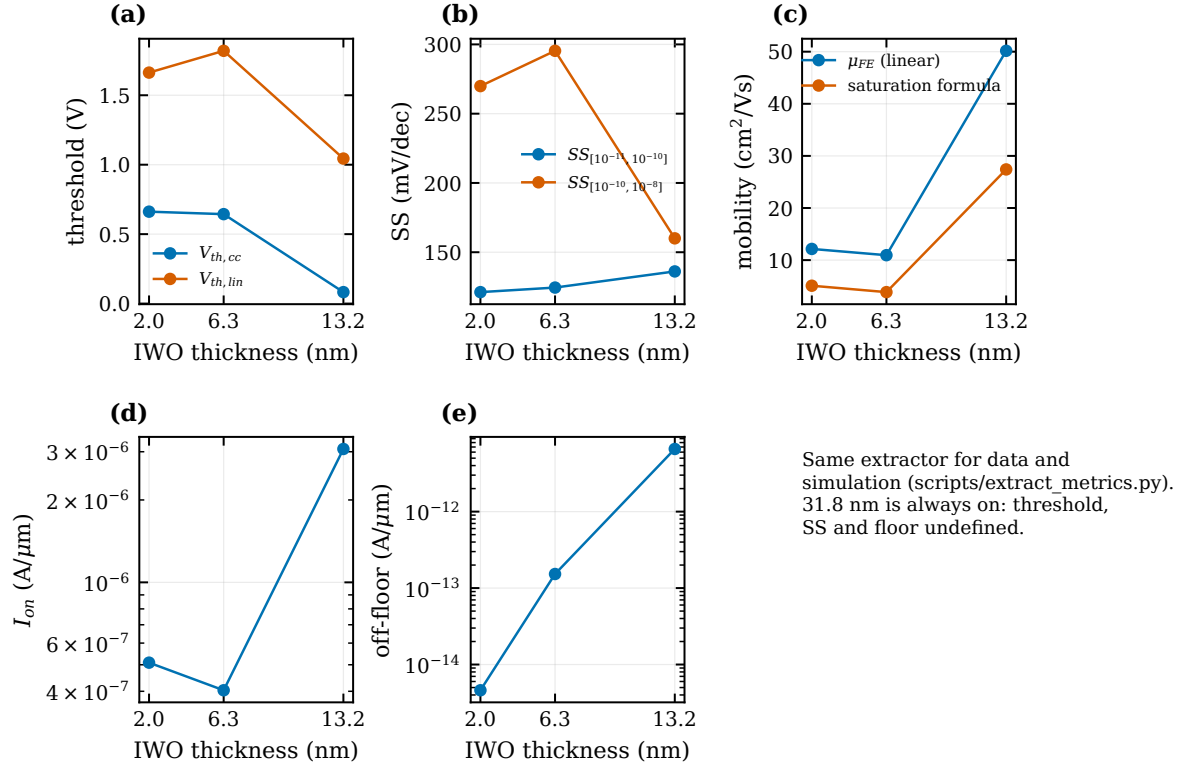


Figure 3.3. Measured metrics versus channel thickness (2.0, 6.3, 13.2 nm; 31.8 nm as a hollow symbol where defined): $V_{th,cc}$ and $V_{th,lin}$, fixed-current swings $SS_{[10^{-11}, 10^{-10}]}$ and $SS_{[10^{-10}, 10^{-8}]}$, linear μ_{FE} and saturation-formula mobility, I_{on} and the off-floor. Same numbers as Table 3.2. The step between 6.3 and 13.2 nm in threshold, mobility and on-current, and the relation $\mu_{FE}(6.3) < \mu_{FE}(2.0)$, are visible.

3.5 Data limitations and their consequences

The following items are not available. Each limitation is stated together with the conclusion that it prevents.

1. One device per thickness. The number of devices per thickness and their spread are unknown. As a consequence, a difference between two films that is smaller than the (unknown) device-to-device spread cannot be attributed to thickness; in particular, the 6.3 nm anomaly may be a property of the device rather than of the thickness.
2. No gate or source current. The off-state floors therefore cannot be assigned directly to an instrument floor or to channel conduction (gate leakage is only disfavoured indirectly, from the bias independence of the floors), and the KCL check cannot be compared with the measurement.
3. No sweep direction, hysteresis or dwell time. Slow traps and bias-stress shifts therefore cannot be separated from the static DOS, and the fitted Q_f may absorb an offset caused by the sweep history.
4. No output (I_D – V_D) curves. Series resistance and contact behaviour are therefore under-constrained, and the I_D – V_D curves of Chapter 10 are predictions, not fits.
5. No C-V on these devices and no TLM. C_{ox} therefore relies on the SI permittivity of HfO₂ and an assumed Al₂O₃ permittivity, and the Ohmic contact assumption cannot be tested directly (Section 5.6).

6. No temperature record. The measurement temperature is assumed to be 300 K, so the thermal voltage in all swing and occupancy arguments carries this assumption. Temperature-dependent data exist in the SI of Januar et al. [1, SI Fig. S12, caption on p. 11] (2 nm at 338 and 358 K) but are not available to the project, which is why the temperature runs of Chapter 10 are registered as predictions.
7. No thickness uncertainty. The method used to measure the film thickness and its uncertainty are not reported. A thickness error of the order of a nanometre at 6.3 nm is therefore an open hypothesis (tested as A1, run_0031, in Chapter 8).
8. No ambient record. The channel is uncapped, and adsorbates on the exposed back surface can carry charge. A back-surface charge Q_b therefore cannot be excluded and is tested as a hypothesis (A3, run_0030).
9. Units inferred from the schematic. The workbook headers carry no units; $A/\mu\text{m}$ and $V_D = 0.7\text{ V}$ are taken from the schematic. The consistency of the saturation-formula mobility with the paper (Section 3.4) supports this reading.
10. Composition. “About 2 % W” is not defined as an atomic or a weight fraction in the available sources. No parameter depends on it directly, and $N_{d,\text{eff}}$ is an electrically active donor density, not the W concentration (Section 4.14).
11. The 31.8 nm column. It lies outside the published series and cannot be turned off within the sweep; it is excluded from the final calibration set by the user’s decision.

Summary. The model rests on one measured transfer curve per thickness (DATA), on a gate stack whose capacitance $C_{\text{ox}} = 8.955 \times 10^{-7} \text{ F cm}^{-2}$ (EOT = 3.86 nm) follows from a measured HfO_2 and an assumed Al_2O_3 permittivity, and on an assumed Ohmic top-contact geometry. The measured metrics are non-monotonic in thickness: the 2.0 and 6.3 nm films are similar ($V_{\text{th,cc}}$ 0.662 and 0.644 V, μ_{FE} 12.15 and 10.95 $\text{cm}^2/(\text{V s})$), whereas the 13.2 nm film is very different ($V_{\text{th,cc}} = 0.083\text{ V}$, $\mu_{\text{FE}} = 50.17 \text{ cm}^2/(\text{V s})$). The mobilities of the paper are saturation-formula values (5.09 and 27.40 are reproduced). The absence of I_G/I_S , hysteresis, I_D-V_D , C-V, TLM, temperature and device statistics limits every later conclusion.

The next two chapters derive, one input at a time, how this device and these data are translated into the parameters of the ATLAS model, beginning with the electrostatic inputs.

Chapter 4

Physical model I: electrostatics, band structure, density of states and charges

This chapter documents, one input at a time, every parameter that determines the electrostatics of the simulated IWO thin-film transistor: the gate stack, the dielectric constant and band structure of the channel, the thickness-dependent conduction-band shift attributed to quantum confinement, the effective masses and effective densities of states, the localized-state (trap) spectrum, and the fixed and interface charges. It closes with three derivations that relate these inputs to the measured quantities: the threshold charge budget, the subthreshold-slope relations (including the trap-density inversion used in the analysis), and the Debye and depletion length scales that decide whether a film is fully depleted. For each electrostatic input the chapter states the number that enters the ATLAS deck, its origin, the way in which it could be derived or measured, the strength of its control over the simulated transfer curve, and the confidence that it deserves.

Every section has the same internal structure: definition and role, values per thickness, source, derivation, ATLAS implementation, sensitivity evidence and status. Status labels follow the provenance vocabulary (MEASURED_TARGET_DEVICE, LITERATURE_IWO, MEASURED_RELATED_MATERIAL, DFT_PURE_IN2O3_PROXY, FITTED, ASSUMED, CALCULATED); evidence classes use DATA, NUM, CAL, OOS, POST, PRED, SURR, THEORY, LIT as defined in the nomenclature. The configuration values are those of `config/iwo_material_model.yaml`. The analysis reports of 2026-09-25 are abbreviated as S1 (analysis_2026-09-25/S1_electrostatics_6p3nm.md), S3 (S3_quantum_confinement.md), S4 (S4_defects_traps.md), S6 (S6_tcad_numerics_predictions.md), the evidence brief (EVIDENCE_BRIEF.md), REVIEW and SYNTHESIS, all in analysis_2026-09-25/. Sensitivity numbers at 2 nm are one-at-a-time perturbations of the calibrated deck run_0012 (DOS 96/48) or, for the isolation tests, of run_0038 (DOS 384/192); they are summarised graphically in Figures 7.4 and 7.5 and discussed as a whole in Section 7.4. Three standing corrections of SYNTHESIS section K apply throughout: run_0027 (contact overlap) is malformed and retracted and is never used here; $SS_{\min}$ is not a reliable metric (Section 5.10), so slopes are quoted at fixed current; and the confinement shift is not identified by the transfer curves, because it is degenerate with a fixed charge (Sections 4.5 and 4.13).

Throughout, q is the elementary charge, $k_B T = 25.85$ meV ($T = 300$ K assumed; the measurement temperature was not recorded), t_{IWO} (or t) the nominal channel thickness, and energies of localized states are referenced to the local conduction-band edge E_c unless stated otherwise.

4.1 Gate stack: thicknesses, permittivities, oxide capacitance and EOT

Definition and role. The bottom gate is a TiN electrode (50 nm) under a bilayer dielectric of HfO₂ (15 nm) and Al₂O₃ (2 nm); the IWO channel sits on the Al₂O₃, Pd source/drain contacts (70 nm) are on top, and the channel is exposed to air (no passivation); $W/L = 290/20 \mu\text{m}$, $V_D = 0.7 \text{ V}$ (Figure 3.1, Section 3.2). The gate-stack capacitance C_{ox} converts every sheet charge into a gate-voltage shift and is therefore the most important electrostatic constant of the model; the ratio q/C_{ox} converts charge into threshold voltage in every section below.

Values used. Thickness-independent (the same stack for all films): $t_{\text{HfO}_2} = 15 \text{ nm}$, $\varepsilon_{\text{HfO}_2} = 19.57$, $t_{\text{Al}_2\text{O}_3} = 2 \text{ nm}$, $\varepsilon_{\text{Al}_2\text{O}_3} = 9.0$; hence $\text{EOT} = 3.86 \text{ nm}$, $C_{\text{ox}} = 8.955 \times 10^{-7} \text{ F cm}^{-2}$ and $q/C_{\text{ox}} = 0.179 \text{ V per } 10^{12} \text{ cm}^{-2}$ (evidence brief).

Source. Layer thicknesses and device geometry: the device schematic [5]. The HfO₂ permittivity 19.57 is the C–V value of the target process in the supporting information of the target paper [10, SI Fig. S1b]; the SI is not bundled and the value was taken over from the Astra audit (digest/used_items.csv). The Al₂O₃ permittivity 9.0 is ASSUMED (not measured on the target process).

Derivation. The two dielectrics are capacitors in series,

$$\frac{1}{C_{\text{ox}}} = \frac{t_{\text{HfO}_2}}{\varepsilon_0 \varepsilon_{\text{HfO}_2}} + \frac{t_{\text{Al}_2\text{O}_3}}{\varepsilon_0 \varepsilon_{\text{Al}_2\text{O}_3}}, \quad \text{EOT} = 3.9 \varepsilon_0 / C_{\text{ox}} = t_{\text{HfO}_2} \frac{3.9}{\varepsilon_{\text{HfO}_2}} + t_{\text{Al}_2\text{O}_3} \frac{3.9}{\varepsilon_{\text{Al}_2\text{O}_3}}. \quad (4.1)$$

With the values above, $\text{EOT} = 15 \times 3.9/19.57 + 2 \times 3.9/9.0 = 2.989 + 0.867 = 3.856 \text{ nm}$ and $C_{\text{ox}} = 3.9 \times 8.854 \times 10^{-14} \text{ F/cm} / 3.856 \times 10^{-7} \text{ cm} = 8.955 \times 10^{-7} \text{ F cm}^{-2}$. A sheet charge ΔQ (in cm^{-2}) placed at the channel/dielectric interface shifts the gate voltage by

$$\Delta V = \frac{q \Delta Q}{C_{\text{ox}}} = 0.1789 \text{ V} \times \frac{\Delta Q}{10^{12} \text{ cm}^{-2}}. \quad (4.2)$$

The dominant uncertainty is $\varepsilon_{\text{Al}_2\text{O}_3}$: evaluating Equation (4.1) with 8.0 gives $\text{EOT} = 3.96 \text{ nm}$ (+2.8 %) and with 7.0 gives 4.10 nm (+6.4 %). REVIEW quotes “about 3 %” for the range 7–9; the calculation above shows that this holds for the upper half of the range, and that the lower end reaches about 6 %. Every charge derived through q/C_{ox} in this report (e.g. Q_f) scales with it. An accumulation C–V measurement on a MIM capacitor of the same stack (TiN/HfO₂/Al₂O₃/metal) would fix C_{ox} directly, independently of both permittivities.

ATLAS implementation. The HfO₂ and Al₂O₃ layers are insulator REGIONs whose permittivity is set with the PERMITTIVITY parameter of the MATERIAL statement [22, p. 1340]; the TiN gate is an electrode with a work function (Section 4.16). The full statement list is reproduced in Chapter C.

Sensitivity evidence. No run varies C_{ox} ; its effect enters analytically through Equation (4.2). The rigid-shift property of C_{ox} -weighted charges was verified in ATLAS: a change of Q_f reproduces $\Delta V_{\text{th}} = -q \Delta Q_f / C_{\text{ox}}$ to 1 mV (runs run_0001→run_0003 and run_0005→run_0007, evidence brief).

Status. Thicknesses MEASURED_TARGET_DEVICE (nominal); ϵ_{HfO_2} MEASURED_TARGET_DEVICE; $\epsilon_{\text{Al}_2\text{O}_3}$ ASSUMED; C_{ox} , EOT CALCULATED. Caveat: the thickness-measurement method is not reported, and 3–6 % of C_{ox} uncertainty propagates into every fitted charge.

4.2 Relative permittivity of the IWO channel

Definition and role. ϵ_{IWO} is the static relative permittivity of the channel. It enters Poisson’s equation inside the film and therefore only the *internal* potential drop across the film: the back-surface lever arm t/ϵ_s (Section 4.15), the donor term $qN_d t^2/2\epsilon_s$ (Section 4.14), the Debye and trap-screening lengths (Section 4.20) and the image-charge correction to confinement (Section 4.6).

Values used. $\epsilon_{\text{IWO}} = 9.3$ for all thicknesses; no thickness dependence is imposed (no evidence for one).

Source. C–V of IWO of the target process, 9.30 ± 0.04 [10, SI Fig. S1b] (SI not bundled; value recorded by the Astra audit). The same SI gives 9.03 for In_2O_3 of the target process. Alternatives: the static (DC) permittivity of single-crystal In_2O_3 , 10.55 [7, p. 225102-12], and 8.9 for In_2O_3 as quoted by Si et al. [4, p. 501] (their source, Hamberg and Granqvist, concerns evaporated Sn-doped In_2O_3 films; reference list p. 506).

Derivation (how it is measured). For a MIM or MIS structure with a film of thickness t , $\epsilon_r = Ct/(\epsilon_0 A)$ in depletion/insulating conditions; for a semiconducting film it is extracted from the series combination with the known dielectric. A THz/Lyddane–Sachs–Teller route gives the crystal value, $\epsilon_{\text{DC}} = \epsilon_\infty \prod (\omega_{\text{LO}}/\omega_{\text{TO}})^2$, used by Stokey et al. [7, p. 225102-12]. The model uses the device-process C–V value because it contains the actual film microstructure.

ATLAS implementation. MATERIAL . . . PERMITTIVITY=9.3 on the IWO region(s) [22, p. 1340].

Sensitivity evidence. run_0026 ($\epsilon_{\text{IWO}} = 10.55$ vs run_0012, 2 nm): $\Delta V_{\text{th,cc}} = -0.001$ V, fixed-current SS (10^{-10} – 10^{-9} A/ μm) -0.4 mV/dec, $\Delta I_{\text{on}} = +0.7$ % (figures/MANIFEST.md, param_control_delta). At 2 nm the permittivity is electrostatically irrelevant (SYNTHESIS: “donors and permittivity change the 2 nm result by < 2 %”, runs run_0022, run_0026). The internal term grows with t (it is $\propto t/\epsilon_s$), so the insensitivity is not established at 13.2 nm, where no permittivity run exists.

Status. MEASURED_TARGET_DEVICE (via an unbundled SI; page-level verification of the SI not possible here); NUM for the insensitivity at 2 nm.

4.3 Band gap

Definition and role. $E_g(t) = E_g^{\text{bulk}} + \Delta E_g^{\text{QC}}(t)$ is the fundamental gap of the channel. In the electrons-only drift–diffusion model holes are not solved, so E_g enters only (i) the intrinsic density $n_i = \sqrt{N_c N_v} \exp(-E_g/2k_B T)$, which is negligible, and (ii) the energy range over which the tail and

Gaussian defect distributions are integrated (Sections 4.10 and 4.11). The thickness dependence is carried entirely by the conduction band (Section 4.5).

Values used. $E_g^{\text{bulk}} = 3.05 \text{ eV } (\pm 0.2 \text{ eV})$. With $\Delta E_g^{\text{QC}} = \Delta E_c$ from Equation (4.3): $E_g = 3.403 / 3.122 / 3.076 \text{ eV}$ at $2.0 / 6.3 / 13.2 \text{ nm}$ (evaluated from the configuration law).

Source. Sputtered amorphous IWO of a different device, Fan et al. [15, Sec. 3] (not bundled; value taken over from the Astra audit, `digest/used_items.csv`). PBE absolute gaps (0.94 eV bulk In_2O_3 ; 1.27 and 1.88 eV for 1.98 and 0.95 nm slabs, 2, p. 21542) are *not* used because semi-local DFT underestimates absolute gaps; only slab-minus-bulk differences enter (Section 4.5). The partition $\Delta E_c / \Delta E_g^{\text{QC}} = 1$ follows the observation that the valence-band edge is almost unchanged under confinement in the $\text{In}_2\text{O}_3/\text{Al}_2\text{O}_3$ slabs [4, p. 504].

Derivation (how it could be measured). Optical absorption (Tauc analysis), spectroscopic ellipsometry or REELS on films of each thickness would give $E_g(t)$; S3 lists the expected signature: $E_g(2) - E_g(13.2) \approx 0.2\text{--}0.45 \text{ eV}$ if the film confinement is real, $< 0.02 \text{ eV}$ if not (S3 falsification table).

ATLAS implementation. MATERIAL EG300 on each IWO region [22, p. 1338], set per thickness by the deck builder.

Sensitivity evidence. No run varies E_g alone. Since holes are not solved and $n_i \lesssim 10^{-7} \text{ cm}^{-3}$ (evaluated from the formula with the values of Sections 4.8 and 4.9), E_g has no first-order effect on the transfer curve; its only role is the integration range of the defect distributions, whose integrands are negligible at mid-gap.

Status. E_g^{bulk} LITERATURE_IWO (different device, not bundled); $\Delta E_g^{\text{QC}}(t)$ DFT_PURE_IN2O3_PROXY; the transfer curves are insensitive to it (THEORY).

4.4 Electron affinity

Definition and role. $\chi(t) = \chi^{\text{bulk}} - \Delta E_c(t)$ places the conduction-band edge relative to the vacuum level and thereby relative to the gate work function ϕ_M . The combination $\phi_M - \chi$ is the flat-band offset of the gate with respect to the channel conduction band; it enters the threshold voltage one-to-one (Section 4.18).

Values used. $\chi^{\text{bulk}} = 4.3 \text{ eV } (\pm 0.3 \text{ eV})$; $\chi = 3.947 / 4.228 / 4.274 \text{ eV}$ at $2.0 / 6.3 / 13.2 \text{ nm}$ (configuration law).

Source. Fan et al. [15, Sec. 3], a composition-mixing (96/4) estimate for IWO (not bundled; recorded by the Astra audit). Consistency check: the charge-neutrality level of In_2O_3 lies about 0.4 eV *above* E_c [4, pp. 503–504], which is why low-barrier contacts and electron accumulation at In_2O_3 surfaces

are expected; the thin film's E_c rises by ΔE_c , so the neutrality level appears deeper relative to E_c in the thinnest film.

Derivation (how it could be measured). UPS (work function and $E_F - E_v$) combined with the optical gap, or IPES directly, gives χ ; an alternative electrical route is the flat-band voltage of a MIS capacitor with a known gate metal, which measures $\phi_M - \chi$ plus fixed charge – exactly the degenerate combination discussed in Section 4.13.

ATLAS implementation. MATERIAL AFFINITY on each IWO region [22, p. 1337]; the confinement shift is imposed by lowering AFFINITY by $\Delta E_c(t)$ and raising EG300 by the same amount (Section 4.5).

Sensitivity evidence. run_0020 ($\chi - 0.1$ eV) and run_0021 ($\chi + 0.1$ eV) vs run_0012: $\Delta V_{th,cc} = +0.100 / -0.100$ V, fixed-current SS unchanged ($+0.0$ mV/dec), $\Delta I_{on} = -6.9 / +7.0$ %. S6 shows that work function, electron affinity and Q_f are each an exact rigid shift at every thickness tested (max $|\Delta \log_{10} I_D| \leq 1.6 \times 10^{-5}$ after a 0.100 V shift), hence they cannot alter the thickness dependence. The I_{on} change is the rigid shift evaluated at fixed $V_G = 3$ V.

Status. LITERATURE_IWO (bulk) + DFT_PURE_IN2O3_PROXY (thickness dependence). Not identifiable from I_D - V_G : degenerate with ϕ_M and Q_f (1 V $\leftrightarrow 5.6 \times 10^{12}$ cm $^{-2}$, S3).

4.5 Confinement shift of the conduction-band edge: the thickness law

Definition and role. $\Delta E_c(t)$ is the upward shift of the conduction-band edge of a film of thickness t relative to the bulk, attributed to geometric quantum confinement. In the classical drift–diffusion model it is inserted as a rigid shift of the band edge (and of the gap, since $\Delta E_c / \Delta E_g^{QC} = 1$). Because all localized-state distributions are referenced to E_c (Sections 4.10 to 4.12), the whole DOS moves with it, and the shift acts on the transfer curve as an almost exact rigid translation in V_G .

Values used.

$$\Delta E_c(t) = \Delta E_g^{QC}(t) = a t_{nm}^{-p}, \quad a = 0.9205 \text{ eV}, \quad p = 1.3815, \quad (4.3)$$

giving $\Delta E_c = 0.353 / 0.072 / 0.026$ eV at 2.0 / 6.3 / 13.2 nm (evidence brief).

Source. Least-squares fit of $\ln \Delta E_c$ vs $\ln t$ through three PBE slab-minus-bulk shifts of pure In $_2$ O $_3$: 0.94 eV at 0.95 nm and 0.33 eV at 1.98 nm (2, p. 21542: slab gaps 1.88 and 1.27 eV minus the bulk 0.94 eV), and $\Delta E_c = 0.6$ eV for a 1.5 nm In $_2$ O $_3$ layer on Al $_2$ O $_3$ with E_v almost unchanged [4, p. 504]. The fit reproduces the points to $+5 / -12 / +8$ % (0.988, 0.526, 0.358 eV at 0.95, 1.5, 1.98 nm). The stated uncertainty is ± 50 %.

Derivation. The full derivation, including the effective-mass, finite-barrier and Schrödinger–Poisson cross-checks, is given in Section 4.6. Two properties of the law matter for its use. (i) It is a *local* power law of three points below 2 nm; the effective-mass limit is t^{-2} , so the fitted exponent 1.38 necessarily

overestimates the shift when extrapolated to thicker films ($\times 1.45$ at 6.3 nm and $\times 2.15$ at 13.2 nm relative to the DFT-anchored effective-width estimate E_1 of S3, Table 4.1; $\times 1.59 / \times 2.51$ relative to the infinite-well upper bound, SYNTHESIS Q3). The two pairs of factors describe the same overshoot against different references and do not conflict. (ii) The law describes a subband energy, whereas a classical model needs the *subthreshold-equivalent* shift, which is smaller by the 2-D density-of-states gain (Equation (4.5)).

ATLAS implementation. MATERIAL AFFINITY = $\chi^{\text{bulk}} - \Delta E_c(t)$ and EG300 = $E_g^{\text{bulk}} + \Delta E_c(t)$ on the IWO region [22, pp. 1337–1338]; no Schrödinger or quantum-correction model is enabled in V1 (the MODELS statement offers a Schrödinger–Poisson solver whose quantization direction and masses are set by SP.DIR and the band masses, 22, p. 1469; it was not used in the V1 decks).

Sensitivity evidence. Removing confinement at 2 nm (run_0016 vs run_0012, same Q_f and mobility) shifts $V_{\text{th,cc}}$ by -0.315 V ($0.676 \rightarrow 0.361$ V), $V_{\text{th,lin}}$ by -0.299 V and raises I_{on} by 21.4 %; at 13.2 nm (run_0017 vs run_0013) the shift is -0.023 V (evidence brief; S3). The shift is rigid: the horizontal offset is constant at 0.345 V from 10^{-14} to 10^{-11} A/ μm , equal to $\Delta E_c - k_B T \ln(N_c \text{ ratio}) = 0.3451$ V, and the local SS vs $\log I_D$ profiles of the on/off pairs agree within 1–2 mV/dec (S3). The apparent change of SS_{min} ($73 \rightarrow 84$ and $151 \rightarrow 139$ mV/dec) is an extraction artefact of the floor-dependent window (Section 5.10). At 6.3 nm the confinement contribution is 0.064 V (run_0034 minus run_0014, corrected for the exact Q_f shift; S1).

Status and caveats. DFT_PURE_IN2O3_PROXY, extrapolated beyond 2 nm. Verdicts (S3, SYNTHESIS): film confinement at 2 nm is physically expected (THEORY; not measured on these films, and not identified from I_D – V_G); the ATLAS value 0.353 eV (realised 0.315–0.345 V) is *plausible, unverified* and high relative to Schrödinger–Poisson (by 0.09 V against the central SP case); the $t^{-1.38}$ law beyond 3 nm is *rejected as physics* (kept as a numerical interpolation, its Vth impact at 6.3 nm is ≤ 0.035 V); “the transfer data require confinement” is *rejected* (held-out run_0034, POST); and ΔE_c is *not identifiable* from one I_D – V_G per thickness: at 2 nm it is equivalent to 1.68 – $1.76 \times 10^{12} \text{ cm}^{-2}$ of fixed negative charge (S3), i.e. it trades one-to-one against Q_f (Section 4.13; Figure 9.3).

4.6 Derivation of the confinement shift

This section derives $\Delta E_c(t)$ from first principles at increasing levels of realism and compares each level with the law of Equation (4.3). All numbers are from S3 (S3_quantum_confinement.md and its scripts in scratch/S3/), which is a specialist 1-D surrogate (SURR) checked against ATLAS.

(1) Infinite square well, parabolic band. An electron with isotropic effective mass m^* confined between impenetrable walls at $z = 0$ and $z = t$ obeys $-(\hbar^2/2m^*) \psi''(z) = E\psi(z)$ with $\psi(0) = \psi(t) = 0$, so $\psi_j \propto \sin(j\pi z/t)$ and

$$E_j = \frac{\hbar^2 \pi^2 j^2}{2m^* t^2} \quad \Rightarrow \quad E_1 = \frac{0.376 \text{ eV}}{(m^*/m_0) t_{\text{nm}}^2}. \quad (4.4)$$

For $t = 2$ nm this gives 0.522 / 0.452 / 0.366 / 0.269 eV for $m^* = 0.18 / 0.208 / 0.257 / 0.35 m_0$ (S3), bracketing the ATLAS value 0.353 eV. The t^{-2} scaling is exact in this limit; any fitted exponent

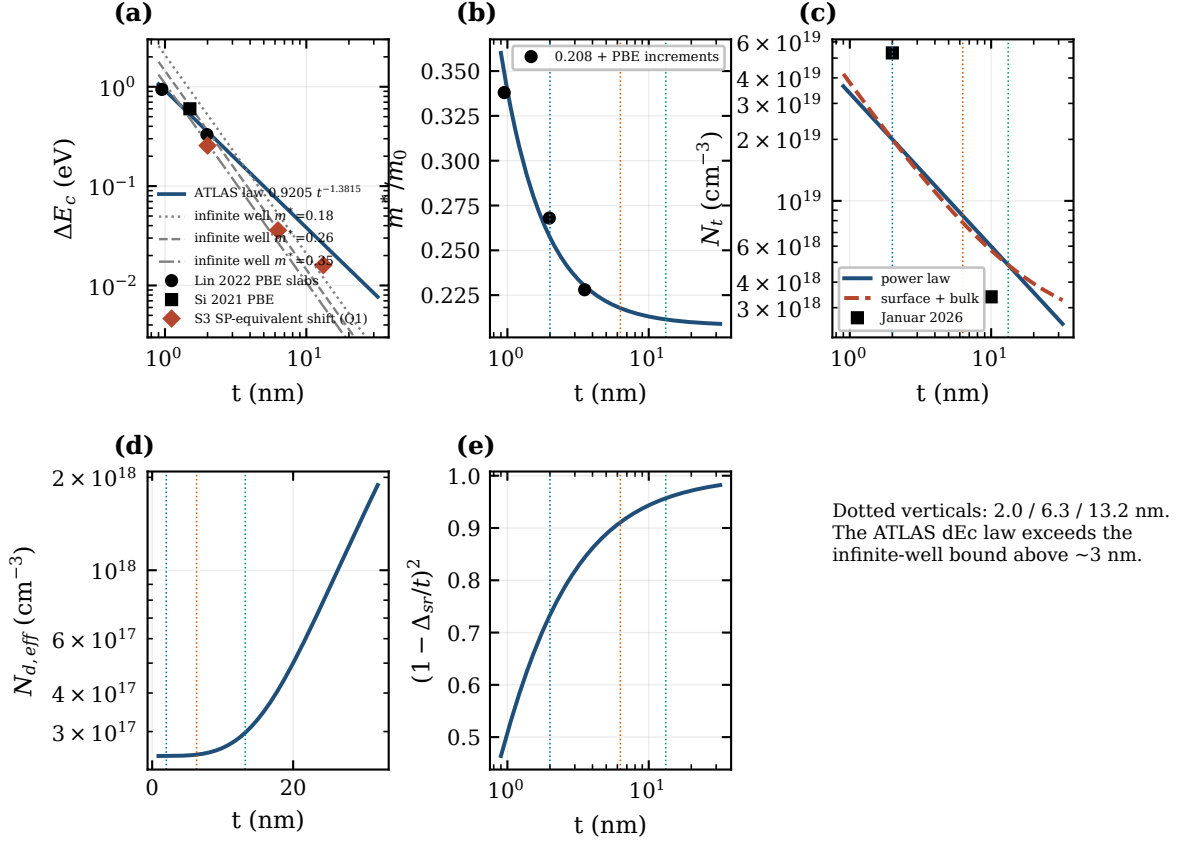


Figure 4.1. Thickness laws of the V1 model. (a) $\Delta E_c(t)$: the ATLAS law $0.9205 t^{-1.3815}$ eV (Equation (4.3)) against the infinite-well effective-mass E_1 for $m^* = 0.18, 0.26$ and $0.35 m_0$, the PBE slab points of Lin et al. [2, p. 21542] and Si et al. [4, p. 504], and the Schrödinger–Poisson subthreshold-equivalent shifts of S3 (case Q1 at $n_s = 10^{10}$ cm⁻²). Above 3 nm the power law diverges from the t^{-2} limit. (b) $m^*(t)$ law with the PBE slab points. (c) $N_t(t)$ power law against the surface + bulk alternative through the same anchors; squares: PBS devices of Januar et al. [1, p. 9]. (d) $N_{d,eff}(t)$. (e) Roughness factor $(1 - \Delta_{sr}/t)^2$ (used in Section 5.2). Laws from `config/iwo_material_model.yaml`; DFT points from `tables/THICKNESS_LAW_EVIDENCE.csv`; SP shifts from `scratch/S3/s3_sp1d_notraps_out.txt`.

Table 4.1. Effective-mass ground-state energy E_1 of the S3 effective-width model anchored on the PBE slab shifts, vs the ATLAS law (S3). The ratios are relative to this DFT-anchored effective-width estimate; relative to the infinite-well upper bound they are $\times 1.59 / \times 2.51$ at 6.3 / 13.2 nm (SYNTHESIS Q3). The law agrees at 2 nm by construction and overestimates increasingly with thickness because its exponent (1.38) is flatter than the effective-mass limit (2).

t (nm)	2	3	4	5	6.3	13.2
E_1 (eV)	0.357	0.187	0.114	0.076	0.050	0.012
ATLAS law, Equation (4.3) (eV)	0.353	0.202	0.136	0.100	0.072	0.026
ratio law/EM	0.99	1.08	1.19	1.30	1.45	2.15

smaller than 2 is a sign of finite barriers and band non-parabolicity at small t .

(2) Non-parabolicity. In a Kane-type band $E(1 + \alpha E) = \hbar^2 k^2 / 2m^*$ the effective mass grows with energy, $m(E) = m^*(1 + 2\alpha E)$, so a large confinement energy is partly self-limiting. Replacing $k = \pi/t$ gives $E_1 = [-1 + \sqrt{1 + 4\alpha E_1^{(0)}}] / (2\alpha)$ with $E_1^{(0)}$ from Equation (4.4); S3 finds 0.430 / 0.380 / 0.316 / 0.240 eV at 2 nm for the same four masses. The same physics explains why the PBE slab masses rise as the slab thins (Section 4.7).

(3) Finite barriers. The film is bounded by Al_2O_3 (below) and air (above). With BenDaniel–Duke boundary conditions (ψ and ψ'/m continuous) the wave function penetrates the barriers and E_1 falls: S3 obtains 0.231–0.252 eV at 2 nm for a heavy barrier mass and 0.305 eV for equal masses ($m^* = 0.18$, with non-parabolicity), 0.039–0.040 eV at 6.3 nm and 0.010–0.011 eV at 13.2 nm.

(4) DFT anchoring. PBE slab calculations include the true band structure, the surface termination and the non-parabolicity, but not the device environment; they underestimate absolute gaps, so only the slab-minus-bulk difference is used. S3 constructs an “effective width” anchored on the slab shifts and obtains $E_1 \approx 0.35$ eV at 2 nm, i.e. agreement with Equation (4.3) at 2 nm by construction (Table 4.1, ratio 0.99). The slab terminations (H-passivated/vacuum in 2, p. 21542; In_2O_3 on one Al_2O_3 layer with an H-terminated top surface in 4, pp. 504–505) differ from the device (Al_2O_3 below, air above), and W is absent from the slabs; hence the $\pm 50\%$ uncertainty.

(5) From a subband energy to a classical band-edge shift. A classical 3-D model with band edge $E_c + \Delta E_c$ has an electron sheet density $n_s^{\text{cl}} = N_c t e^{-(E_c + \Delta E_c - E_F)/k_B T}$ (non-degenerate), whereas the quantized film has $n_s^{\text{Q}} = \sum_i g_{2D,i} k_B T e^{-(E_c + E_i - E_F)/k_B T}$ with $g_{2D} = m^* / \pi \hbar^2$ per subband (spin included). Equating the two gives the shift that a classical model must use to reproduce the subthreshold charge,

$$\Delta E_c^{\text{eq}} = -k_B T \ln \left[\frac{\sum_i g_{2D,i} k_B T e^{-E_i/k_B T}}{N_c t} \right]. \quad (4.5)$$

Because a 2-D subband carries more states near its edge than $N_c t$ does, $\Delta E_c^{\text{eq}} < E_1$: at 2 nm ($m^* = 0.18$, non-parabolic, finite barrier) $\Delta E_c^{\text{eq}} = 0.207$ – 0.215 eV while $E_1 = 0.248$ eV; the 2-D DOS term subtracts about 0.04 V at 2 nm and 0.01 V at 6.3 nm (S3). This is the first reason the ATLAS rigid shift (which uses E_1 -like DFT shifts) is on the high side.

Table 4.2. Schrödinger–Poisson onset shifts (V) relative to the classical no-confinement case at $n_s = 10^{10} \text{ cm}^{-2}$, trap-free (S3, scratch/S3/s3_sp1d_notraps.json). C1: classical with the ATLAS law; Q1: central quantum case ($m^* = 0.18$, finite barriers, non-parabolic); Q2: hard wall; Q3: BenDaniel–Duke with heavy barrier mass; Q4/Q5: $m^* = 0.257 / 0.35$.

t (nm)	C1	Q1	Q2	Q3	Q4	Q5
2.0	0.345	0.256	0.384	0.202	0.205	0.139
3.0	0.197	0.131	0.179	0.105	0.090	0.052
4.0	0.133	0.078	0.102	0.065	0.047	0.021
5.0	0.097	0.052	0.065	0.045	0.028	0.007
6.3	0.070	0.036	0.043	0.031	0.016	−0.002
13.2	0.026	0.016	0.018	0.015	0.001	−0.012

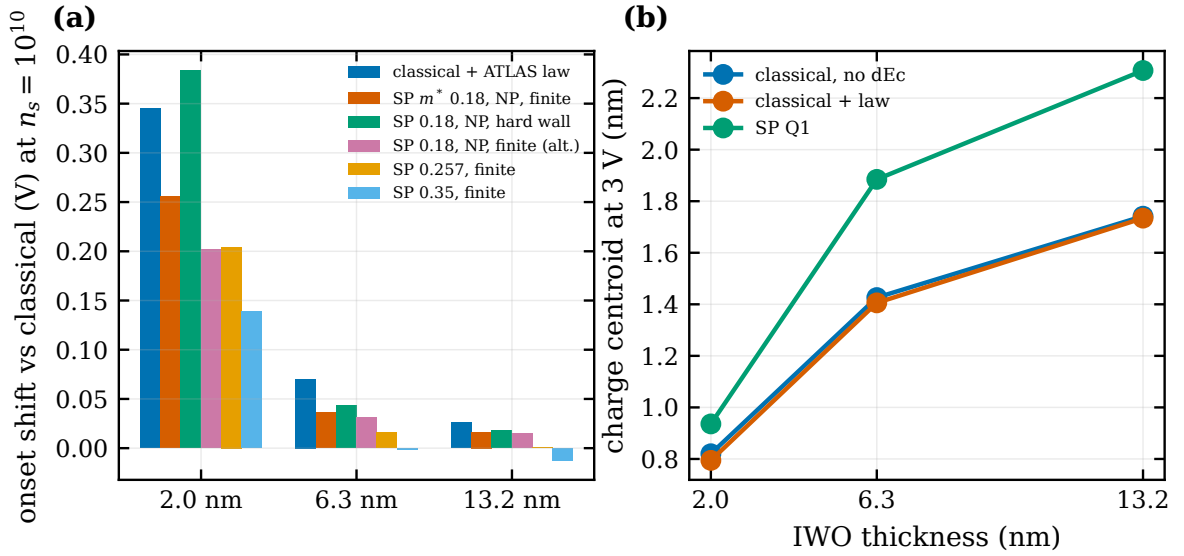


Figure 4.2. S3 1-D Schrödinger–Poisson surrogate (trap-free, scratch/S3/s3_sp1d_notraps.json): electron sheet density $n_s(V_G)$, quantum vs classical, at 2 nm and 6.3 nm, with the onset shift and the electron centroid. Shifts vs the classical case C0 at 10^{10} cm^{-2} : C1 (ATLAS law) 0.345 / 0.070 / 0.026 V and Q1 0.256 / 0.036 / 0.016 V at 2 / 6.3 / 13.2 nm. The shift is rigid at 2 nm, and at 6.3 nm the law exceeds the quantum shift by about a factor 2.

(6) Self-consistent Schrödinger–Poisson. S3 solves the 1-D Schrödinger and Poisson equations across gate stack and film with Fermi–Dirac filling, $Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$ and the V1 $N_{d,\text{eff}}(t)$, trap-free (Gauss-law error $\leq 2 \times 10^{-18} \text{ C/cm}^2$). The onset shifts relative to the classical no-confinement case (C0) at $n_s = 10^{10} \text{ cm}^{-2}$ are listed in Table 4.2; they are shown with the $n_s(V_G)$ curves in Figure 4.2. The surrogate reproduces the ATLAS relative shifts within 11 mV (0.326 V at 10^{10} vs ATLAS $\Delta V_{\text{th,cc}} = 0.315 \text{ V}$ at 2 nm, run_0012–run_0016; 0.023 V at 13.2 nm, run_0013–run_0017). The central SP case Q1 gives 0.256 V at 2 nm, the DFT-anchored width about 0.30 V and the hard wall 0.384 V; if W raises m^* to 0.26–0.35 the shift falls to 0.14–0.20 V. The resulting bracket is 0.14–0.38 V, with 0.26–0.30 V central for $m^* = 0.18$.

(7) Field quantization and capacitance. Above threshold the gate field F forms a triangular well at the front interface, whose ground state in the triangular-well approximation is $E_1^{\text{tri}} \approx (\hbar^2/2m^*)^{1/3} (9\pi qF/8)^{2/3}$. S3 finds $E_1^{\text{tri}} = 0.10 \text{ eV}$ and centroid 3.45 nm at $n_s = 10^{12} \text{ cm}^{-2}$ ($F = 1.95 \times 10^5 \text{ V/cm}$), and 0.56 eV with centroid 1.47 nm at $n_s = 1.3 \times 10^{13} \text{ cm}^{-2}$ (the bound $C_{\text{ox}}(3 - V_{\text{th,cc}})/q$ at $V_G = 3 \text{ V}$). The 2 nm

film is never field-quantized (centroid $0.94\text{--}0.98\text{ nm} \approx t/2$ up to $1.3 \times 10^{13}\text{ cm}^{-2}$); the 6.3 and 13.2 nm films are, above about $2\text{--}3 \times 10^{12}$ and $1\text{--}3 \times 10^{11}\text{ cm}^{-2}$ respectively. The effective capacitance $C_{\text{eff}}/C_{\text{ox}}$ over $2\text{--}3\text{ V}$ is already $0.87\text{--}0.88$ classically (centroid and degenerate DOS), and quantum dark space lowers it by $< 1\%$ at 2 nm and $3\text{--}5\%$ at 6.3 and 13.2 nm (S3). This on-state field quantization is not represented in V1; its implication for the mobility is discussed in Section 5.1.

(8) Two neglected corrections. *Image (dielectric) confinement:* the air side ($\varepsilon = 1$) against IWO ($\varepsilon = 9.3$) adds about $+31\text{ meV}$ at 1 nm from the surface, roughly $+0.02\text{--}0.03\text{ eV}$ at 2 nm (S3 estimate; *plausible, unverified*). *Energy reference of the tail:* if the localized tail states follow the confined edge the 2 nm onset shift is 0.26 V (Q1 with traps); if they stay at the bulk edge (localization length shorter than t) the whole $4 \times 10^{12}\text{ cm}^{-2}$ tail must fill before the subband populates and the onset moves by 0.91 V (S3). This is the dominant model-form uncertainty of the confinement treatment and is *not determined* from the available data. With local tails, field quantization at 6.3 and 13.2 nm raises $V_G(n_s = 10^{12})$ by $0.15\text{--}0.19\text{ V}$ relative to classical while $V_G(10^{10})$ moves by $\leq 0.04\text{ V}$ – a change of curve shape the rigid law cannot represent (S3; SURR, registered as a prediction).

Summary of the derivation. The law of Equation (4.3) is consistent with effective-mass theory at 2 nm (inside the $0.14\text{--}0.38\text{ V}$ bracket) but on its high side, and it overestimates the S3 subthreshold-equivalent shift by about $2\times$ at 6.3 nm (0.070 vs 0.036 V) and $1.6\times$ at 13.2 nm (0.026 vs 0.016 V). The shift could be measured through the thickness dependence of the optical gap (expected $+0.2\text{--}0.45\text{ eV}$ at 2 nm vs 13.2 nm), through IPES or UPS/XPS for the CBM, and through a denser thickness series ($2 \rightarrow 3\text{ nm}$: $\Delta V_{\text{th,cc}} +0.125\text{--}0.205\text{ V}$ predicted; S3 falsification table).

4.7 Conduction-band effective mass

Definition and role. $m^*(t)$ is the density-of-states effective mass of the conduction band. In the V1 decks (no Schrödinger solver, constant mobility) it enters *only* through the effective density of states $N_c \propto m^{*3/2}$ (Section 4.8); its influence on the mobility (Drude $\mu = q\tau/m^*$) is absorbed into the fitted band mobility (Section 5.1), and its influence on the confinement energy is already contained in the DFT-based ΔE_c law.

Values used.

$$m^*(t) = m_{\text{bulk}} + b t_{\text{nm}}^{-q}, \quad m_{\text{bulk}} = 0.208 m_0, \quad b = 0.1311 m_0, \quad q = 1.4121, \quad (4.6)$$

giving $m^* = 0.257 / 0.218 / 0.211 m_0$ at $2.0 / 6.3 / 13.2\text{ nm}$ (evidence brief).

Source. Bulk value: optical-Hall measurement on a single In_2O_3 crystal ($n = 2.8 \times 10^{17}\text{ cm}^{-3}$), $0.208 m_0$ [7, p. 225102-1, abstract]; the zero-density extrapolation quoted there is $0.18 m_0$ [7, p. 225102-2]. Thickness increment: PBE slab masses of pure In_2O_3 , 0.17 (bulk), 0.19 (3.52 nm), 0.23 (1.98 nm) and $0.30 m_0$ (0.95 nm) [2, p. 21540]. The effect of W on m^* is reported only qualitatively in the target paper (flatter CBM, heavier mass; text lines 238–241 as quoted in S3); it is *not determined* numerically and is not included.

Derivation. Only PBE *differences* are used, as for ΔE_c : $\Delta m(t) = m_{\text{slab}}(t) - m_{\text{PBE,bulk}} = 0.13, 0.06$ and $0.02 m_0$ at 0.95, 1.98 and 3.52 nm. A least-squares fit of $\ln \Delta m$ vs $\ln t$ gives b and q of Equation (4.6); the fitted increments are 0.141 / 0.050 / 0.022 m_0 at the three slab thicknesses. The increment is added to the *experimental* bulk mass. Physically the increase is the non-parabolicity of Section 4.6: a confined electron sits at a higher kinetic energy, where $m(E) = m^*(1 + 2\alpha E)$ is heavier. The mass could be measured by optical Hall or infrared ellipsometry (plasma frequency with known n) on films of each thickness, or from the temperature dependence of Shubnikov–de Haas oscillations in thick films.

ATLAS implementation. Not a direct deck parameter in V1: m^* is converted into NC300 of the MATERIAL statement [22, p. 1340] (Section 4.8). The band masses of the MATERIAL statement would matter only if the Schrödinger model (SP.DIR masses, 22, p. 1469) were enabled, which it is not.

Sensitivity evidence. $m^* \times 1.3$ (isolation tests, DOS 384/192, vs the same-mobility baselines): $\Delta V_{\text{th,cc}} = -0.044$ V (2 nm, run_0050 vs run_0038) and -0.029 V (13.2 nm, run_0051 vs run_0040); $SS_{\text{cc}} -17.7 / -13.2$ mV/dec; $\Delta I_{\text{on}} = +4.1 / +5.8$ % (S6; SYNTHESIS). The shift is *not* rigid (0.17 dec change of shape at 2 nm, S6), in contrast to the work-function and affinity shifts. The evidence brief quotes -0.028 V and $+3.7$ % for run_0051; S6 and SYNTHESIS, which are later and state the baseline explicitly, are used here. In the budget, $m^*(t)$ contributes $+14.9$ mV to the 2→13.2 nm $V_{\text{th,cc}}$ step (S6). The reason a 30 % heavier mass (only $k_B T \ln 1.3^{3/2} = 10$ mV of Fermi-level shift at fixed free density) moves $V_{\text{th,cc}}$ by 44 mV is the tail: at fixed free density a lower E_F empties part of the tail charge, and the tail term is ten times larger than the free-carrier term at threshold (Table 4.3).

Status. MEASURED_RELATED_MATERIAL (bulk, pure In_2O_3) + DFT_PURE_IN2O3_PROXY (increment). Non-parabolicity is otherwise ignored; W-induced mass change *not determined*.

4.8 Effective density of states of the conduction band

Definition and role. N_c sets the free-electron density for a given $E_F - E_c$, $n = N_c \mathcal{F}_{1/2}[(E_F - E_c)/k_B T]$ (Fermi–Dirac statistics, $\mathcal{F}_{1/2}$ the normalised Fermi integral). It fixes where the Fermi level sits at threshold and therefore the partition between free and trapped electrons.

Values used.

$$N_c = 2 \left(\frac{2\pi m^* k_B T}{h^2} \right)^{3/2} = 2.51 \times 10^{19} \text{ cm}^{-3} \left(\frac{m^*}{m_0} \right)^{3/2} \left(\frac{T}{300 \text{ K}} \right)^{3/2}, \quad (4.7)$$

which with Equation (4.6) evaluates to $3.28 / 2.55 / 2.44 \times 10^{18} \text{ cm}^{-3}$ at 2.0 / 6.3 / 13.2 nm ($2.38 \times 10^{18} \text{ cm}^{-3}$ for the bulk mass 0.208).

Source and derivation. Equation (4.7) is the standard result for a parabolic 3-D band: the number of states per volume below energy E above the edge is $(1/3\pi^2)(2m^*E/\hbar^2)^{3/2}$ per spin, so $g(E) = (1/2\pi^2)(2m^*/\hbar^2)^{3/2}\sqrt{E}$ including spin; integrating $g(E)e^{-E/k_B T}$ gives N_c . At 2 nm the film is

quasi-2D and N_c is an effective continuum parameter; the correct 2-D occupation is the subject of Equation (4.5). The ratio $N_c(2\text{ nm})/N_c(\text{bulk}) = 1.376$ corresponds to $k_B T \ln 1.376 = 8\text{ mV}$ of Fermi-level shift, the value S1 quotes for the analytic $k_B T \ln N_c$ term at 2 nm. The same ratio explains why removing confinement (which also removes the mass increment) shifts the 2 nm curve by $\Delta E_c - k_B T \ln(N_c \text{ ratio}) = 0.345\text{ V}$ rather than 0.353 V (S3).

ATLAS implementation. MATERIAL NC300 per IWO region [22, p. 1340]; Fermi–Dirac statistics are enabled on the MODELS statement (FERMIDIRAC, 22, p. 1433). NC300 is the 300 K value; its temperature scaling in the elevated-temperature predictions is treated in Section 5.8.

Sensitivity evidence. Through m^* : runs run_0050/run_0051 (Section 4.7). In the 1-D surrogate, “ m^* through N_c only” contributes $-0.039 / -0.008 / -0.002\text{ V}$ to $V_{th,cc}$ at 2 / 6.3 / 13.2 nm (S1). REVIEW (O2) notes that the N_c part of the confinement removal gives 43 mV and 14.6 mV/dec, so the confinement degeneracy with Q_f is near-exact, not exact.

Status. CALCULATED from m^* (itself a proxy). Caveat: continuum 3-D DOS used for a quasi-2-D film.

4.9 Effective density of states of the valence band

Definition and role. N_v enters only the intrinsic density $n_i = \sqrt{N_c N_v} e^{-E_g/2k_B T}$, because the model solves electrons only (holes are not transported).

Values used. $N_v = 1.0 \times 10^{19}\text{ cm}^{-3}$ at 300 K, all thicknesses; hole mobility $0.01\text{ cm}^2/\text{V s}$ (both ASSUMED).

Source and derivation (how it could be derived). No source; the value is a generic placeholder. With the values of Sections 4.3 and 4.8, $n_i \approx 2 \times 10^{-10}$ (2 nm) to $7 \times 10^{-8}\text{ cm}^{-3}$ (13.2 nm) (evaluated from the formula), so no physically plausible N_v (10^{18} – 10^{20} cm^{-3}) can influence an n -type transfer curve. If needed, N_v would follow from Equation (4.7) with the valence-band DOS mass, which for In_2O_3 is large and poorly known; DFT hole masses of the slabs could supply it.

ATLAS implementation. MATERIAL NV300 [22, p. 1340].

Sensitivity evidence. None run; insensitive by construction (THEORY).

Status. ASSUMED; irrelevant to the fitted curves.

4.10 Acceptor-like band-tail states

Definition and role. The exponential tail of localized, acceptor-like states below the conduction-band edge is the dominant charge reservoir of the model near threshold. When filled, the states are

negatively charged; they pin the Fermi level, set the subthreshold slope in the 10^{-10} – 10^{-8} A/ μ m window and the trapped/free partition above threshold. The density of states is

$$g_{\text{TA}}(E) = N_{\text{TA}} \exp\left(\frac{E - E_c}{W_{\text{TA}}}\right), \quad E < E_c, \quad \int_{E_c - E_g}^{E_c} g_{\text{TA}} dE = N_{\text{TA}} W_{\text{TA}} \left(1 - e^{-E_g/W_{\text{TA}}}\right) \simeq N_{\text{TA}} W_{\text{TA}} \equiv N_t. \quad (4.8)$$

The model is parameterised by the integral $N_t(t)$ and the width W_{TA} , and $N_{\text{TA}} = N_t/W_{\text{TA}}$.

Values used. $N_t(t) = N_t^{(2)}(2/t_{\text{nm}})^\alpha$ with $N_t^{(2)} = 2 \times 10^{19} \text{ cm}^{-3}$ and $\alpha = 0.75$; $W_{\text{TA}} = 0.040 \text{ eV}$ shared (tail temperature $T_t = W_{\text{TA}}/k_B = 464 \text{ K}$). Per film:

t (nm)	2.0	6.3	13.2
N_t (cm^{-3})	2.0×10^{19}	8.5×10^{18}	4.9×10^{18}
$N_{\text{TA}} = N_t/W_{\text{TA}}$ ($\text{cm}^{-3} \text{ eV}^{-1}$)	5.0×10^{20}	2.1×10^{20}	1.2×10^{20}
sheet $N_t t$ (cm^{-2})	4.0×10^{12}	5.3×10^{12}	6.4×10^{12}
filled at $V_{\text{th,cc}}$ (cm^{-2} ; S1, 1-D)	1.43×10^{12}	1.27×10^{12}	5.5×10^{11}
qn_t/C_{ox} at $V_{\text{th,cc}}$ (V; S1)	0.256	0.226	0.098

(6.3 nm columns for the validation configuration run_0014.)

Source. $N_t^{(2)}$ is FITTED: the product $N_{\text{TA}} W_{\text{TA}}$ of the 2 nm calibration in ATLAS (Chapter 7). The exponent comes from a two-point ratio of about 4 between the 13.2 and 2 nm trap densities of the target paper's thickness series [1, Sec. 2.8, Fig. 5a, p. 7]: $\alpha = \ln 4 / \ln(13.2/2) = 0.73 \approx 0.75$. The same paper's compact-model values for its bias-stress devices, 5.3×10^{19} (2 nm) and $3.39 \times 10^{18} \text{ cm}^{-3}$ (10 nm) [1, p. 9], use a different (θ_t) convention and different devices; they imply a ratio 15.6 and a sheet density that *decreases* with t , which forces a negative bulk term and is unphysical for this series (S4), so they are not used as anchors. Context: the conductance-method bulk DOS of ALD In_2O_3 is $3.3 \times 10^{20} \text{ cm}^{-3} \text{ eV}^{-1}$ at $N_d \approx 10^{20} \text{ cm}^{-3}$ [8, pp. 4–5]. W_{TA} is fitted on the 2 nm subthreshold slope; the target paper reports only the stress-induced change of the tail temperature, so the absolute T_t is *not determined* from the literature.

Derivation. *Occupation.* The trapped density at quasi-Fermi level $u = E_{Fn} - E_c$ is $n_t(u) = \int g_{\text{TA}}(E) f(E - u) dE$ with the Fermi function f (Section 4.17 shows why the steady-state occupancy is Fermi–Dirac regardless of capture cross sections). For u several $k_B T$ below E_c and $W_{\text{TA}} > k_B T$, extending the integral to $+\infty$ gives

$$n_t(u) \simeq N_{\text{TA}} W_{\text{TA}} e^{u/W_{\text{TA}}} \frac{\pi k_B T / W_{\text{TA}}}{\sin(\pi k_B T / W_{\text{TA}})}, \quad (4.9)$$

where the last factor is the thermal enhancement over a step-function occupation; for $W_{\text{TA}} = 40 \text{ meV}$ it is 2.26 (the value used by S4). Near E_c the truncation of the tail at E_c reduces it. Equation (4.9) shows the two regimes of the subthreshold: when $W_{\text{TA}} > k_B T$ the trapped charge grows more slowly with u than the free charge ($e^{u/W_{\text{TA}}}$ vs $e^{u/k_B T}$), so it dominates at low current and yields to the free charge at high current. Its trap capacitance $C_t = q^2 t \partial n_t / \partial u = q^2 t n_t / W_{\text{TA}}$ enters the slope (Equation (4.18)). *Thickness law.* The power law is an interpolation. S4 shows that the surface + bulk form $N_t t = N_s + N_b t$ through the same anchors ($N_s = 3.57 \times 10^{12} \text{ cm}^{-2}$, $N_b = 2.15 \times 10^{18} \text{ cm}^{-3}$)

predicts $N_t(6.3) = 7.8 \times 10^{18}$, 8 % below the power law, and diverges only beyond the fitted range (31.8 nm: 3.3 vs $2.5 \times 10^{18} \text{ cm}^{-3}$). The trap-DOS inversion (Section 4.19) supports the surface + bulk form: $D(E_c - 0.1 \text{ eV}) = 1.05 / 1.67 / 2.16 \times 10^{13} \text{ cm}^{-2} \text{ eV}^{-1}$, i.e. $\propto t^{0.38 \pm 0.15}$, consistent with $D_s \approx 8.5 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$ plus $g_b \approx 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$ (S4). Suitable measurements are temperature-dependent transfer curves (activation energy vs u), quasi-static C–V (free/trapped partition), photo-excited charge collection, and Hall measurements on unpatterned films.

ATLAS implementation. DEFECTS CONTINUOUS with NTA (the density at the conduction-band edge, 22, p. 1171) and WTA (the characteristic decay energy of the acceptor tail, 22, p. 1173); the DEFECTS statement admits one exponential and one Gaussian distribution per state type [22, p. 1169]. The continuous distribution is discretised on NUMA/NUMD energy levels (default 12, 22, p. 1168); the V1 decks use 96/48 and, from campaign B, 384/192 (Section 5.11). The tail capture cross section is 10^{-15} cm^2 (Section 4.17).

Sensitivity evidence. $N_t \times 1.5$ (run_0023 vs run_0012): $\Delta V_{\text{th,cc}} = +0.082 \text{ V}$, fixed-current SS (10^{-10} – $10^{-9} \text{ A}/\mu\text{m}$) $+49.0 \text{ mV/dec}$, $\Delta I_{\text{on}} = -25.1 \%$. $W_{\text{TA}} + 5 \text{ meV}$ (run_0024): $+0.020 \text{ V}$, $+4.6 \text{ mV/dec}$ in the same window ($SS_{\text{cc}} + 12 \text{ mV/dec}$, evidence brief), $\Delta I_{\text{on}} = +0.4 \%$. The tail is the largest non-confinement term of the threshold budget (0.10–0.26 V, 25 % of the 2→13.2 nm step; S1). The pre-registered S2 re-partition with $N_t(6.3) = 2.66 \times 10^{19} \text{ cm}^{-3}$ (B0, run_0037) met the on-state targets but gave $SS_{\text{cc}} = 410.9$ vs 295.4 mV/dec measured and was *rejected* (OOS, failed; SYNTHESIS). A broader tail ($W_{\text{TA}} 60 \text{ meV}$) was disfavoured in the S4 surrogate ($SS_{\text{cc}} + 126 \text{ mV/dec}$).

Status. FITTED ($N_t^{(2)}$, W_{TA}) with a LITERATURE_IWO trend. Verdicts (S4): an exponential acceptor tail near E_c controls threshold and SS_{cc} in all films – *strongly supported* (inversion ratio measured/ATLAS 0.81–1.17 for $u = -0.125$ to -0.025 eV at 2 and 13.2 nm); $N_t \propto t^{-0.75}$ as physics – *rejected* (kept as interpolation); surface + bulk – *plausible, unverified* (preferred form). Caveat: N_t , W_{TA} and Q_f trade off on a single curve.

4.11 Deep acceptor Gaussian

Definition and role. A Gaussian band of acceptor-like states deep in the gap,

$$g_{\text{GA}}(E) = N_{\text{GA}} \exp \left[- \left(\frac{E - (E_c - E_{\text{GA}})}{W_{\text{GA}}} \right)^2 \right], \quad \int g_{\text{GA}} dE = \sqrt{\pi} N_{\text{GA}} W_{\text{GA}}, \quad (4.10)$$

representing deep (e.g. oxygen-vacancy related) states. Below threshold these states are full, so they act as a fixed negative charge; they matter only through their total number.

Values used. $N_{\text{GA}} = 5 \times 10^{16} \text{ cm}^{-3} \text{ eV}^{-1}$, $E_{\text{GA}} = 0.6 \text{ eV}$ below E_c , $W_{\text{GA}} = 0.15 \text{ eV}$, all thicknesses. Total density $\sqrt{\pi} N_{\text{GA}} W_{\text{GA}} = 1.33 \times 10^{16} \text{ cm}^{-3}$, i.e. sheets of $2.7 \times 10^9 / 8.4 \times 10^9 / 1.75 \times 10^{10} \text{ cm}^{-2}$ at 2.0 / 6.3 / 13.2 nm and threshold contributions $\leq 0.003 \text{ V}$ (Equation (4.2)), matching S1's 0.000 / 0.001 / 0.003 V.

Source. ASSUMED; the magnitude is within the order of the range $2.5\text{--}3.7 \times 10^{16}$ reported by Fan et al. [15] for a different IWO device (not bundled; value recorded in the configuration). Donor-like tail and donor Gaussian are set to zero (valence tail negligible for n -type transfer curves; the donor background is represented by $N_{d,\text{eff}}$, Section 4.14).

Derivation (how it could be measured). Photo-excited C–V or sub-gap photocurrent spectroscopy resolve deep states; the transfer curve is blind to them because they never change occupancy in the swept range. The trap-DOS inversion of Section 4.19 does see the states at $E_c - 0.15$ to -0.2 eV and finds that V1 underestimates them by $\times 2.2\text{--}2.8$ (2 nm) and $\times 2.6$ (6.3 nm) (S4), consistent with the model slope being too soft in the $10^{-10}\text{--}10^{-9}$ A/ μm window (Section 7.3); the lower SS_{min} of run_0012 (73 vs 84.5 mV/dec) is not used as evidence, because SS_{min} is not reliable (Section 5.10).

ATLAS implementation. DEFECTS NGA, EGA, WGA (acceptor Gaussian; NGA is the density of acceptor-like states in the Gaussian, 22, p. 1171; defaults NGA $5 \times 10^{17} \text{cm}^{-3} \text{eV}^{-1}$, WGA 0.1 eV, 22, pp. 1168–1169).

Sensitivity evidence. No run varies the deep Gaussian (insensitive by the estimate above). A *near-edge* acceptor Gaussian was tested as a hypothesis for the 6.3 nm on-state shape (campaign C, run_0052): $2 \times 10^{12} \text{cm}^{-2}$ at $E_c - 0.03$ eV, $W_{\text{GA}} = 0.02$ eV (i.e. $N_{\text{GA}} = 2 \times 10^{12} / (6.3 \times 10^{-7} \cdot 0.02 \sqrt{\pi}) = 9.0 \times 10^{19} \text{cm}^{-3} \text{eV}^{-1}$ by Equation (4.10)), with $\mu_{\text{band}} = 15.4 \text{cm}^2/\text{Vs}$. Against the S4 prediction (gap +0.28 V, g_m +24 %, SS_{cc} +23 mV/dec, $I_{\text{on}} \pm 3$ %) it gave gap +0.119 V, g_m +25.5 %, SS_{cc} +15.6 mV/dec and I_{on} +16.0 %: *partly failed* (OOS; S6, SYNTHESIS), recovering only 33 % of the gap (Chapter 8).

Status. ASSUMED; insensitive (THEORY). Deeper states are underestimated (*strongly supported*, S4).

4.12 Front interface acceptor sheet and its regularisation

Definition and role. Acceptor-like interface states at the IWO/ Al_2O_3 interface, described as a Gaussian in energy per unit area, $D_{\text{it}}(E) = D_{\text{it}}^{\text{pk}} \exp\{-[(E - (E_c - E_0))/W_{\text{it}}]^2\}$. They add a sheet charge that is full below threshold and a capacitance $C_{\text{it}} = q^2 D_{\text{it}}$ where they cross the Fermi level.

Values used. $D_{\text{it}}^{\text{pk}} = 3 \times 10^{11} \text{cm}^{-2} \text{eV}^{-1}$, $E_0 = 0.3$ eV, $W_{\text{it}} = 0.12$ eV, shared across thickness. Total sheet $\sqrt{\pi} D_{\text{it}}^{\text{pk}} W_{\text{it}} = 6.4 \times 10^{10} \text{cm}^{-2}$.

Source. ASSUMED. Literature for the $\text{HfO}_2/\text{In}_2\text{O}_3$ interface: $6 \times 10^{11} \text{cm}^{-2} \text{eV}^{-1}$ from an SS of 63.5 mV/dec [2, p. 21539] and 6.3×10^{11} from the subthreshold method [8, pp. 4–5]. The device interface here is IWO/ Al_2O_3 , for which no value is available; the assumed peak is half the $\text{HfO}_2/\text{In}_2\text{O}_3$ proxy.

Derivation: regularisation of a sheet. ATLAS defect distributions are volumetric (per cm^3). A sheet density is represented by a thin IWO layer of thickness d_{reg} adjacent to the interface carrying the volume density

$$N_{\text{GA}}^{\text{vol}} = \frac{D_{\text{it}}^{\text{pk}}}{d_{\text{reg}}}, \quad d_{\text{reg}} = 0.25 \text{ nm} \Rightarrow N_{\text{GA}}^{\text{vol}} = \frac{3 \times 10^{11}}{2.5 \times 10^{-8}} = 1.2 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}. \quad (4.11)$$

The representation is exact in the limit $d_{\text{reg}} \rightarrow 0$, and its error is of order the potential drop across the layer, $\Delta\psi \approx qN_{\text{it}}d_{\text{reg}}/2\epsilon_s$, which must be small compared with $k_{\text{B}}T/q$: with the full sheet $6.4 \times 10^{10} \text{ cm}^{-2}$, $\Delta\psi \approx 1.6 \times 10^{-19} \cdot 6.4 \times 10^{10} \cdot 2.5 \times 10^{-8} / (2 \cdot 8.2 \times 10^{-13}) \approx 0.2 \text{ mV}$. The condition is equivalently $d_{\text{reg}} \ll L_{\text{D}}$ (Section 4.20). Numerical check: halving the layer changes I_{D} by ≤ 0.009 dec and -0.06% (S6 check E; configuration note). At threshold (E_{F} about 0.04 eV below E_{c} , S1) the whole Gaussian at $E_{\text{c}} - 0.3 \text{ eV}$ is full, giving $q \cdot 6.4 \times 10^{10} / C_{\text{ox}} = 0.011 \text{ V}$, consistent with S1's 0.012 V . The interface-state density could be measured by conductance or charge-pumping methods on the target stack, or by the trap-DOS inversion of Section 4.19 at energies where the tail is small; the SS formula (Equation (4.18)) gives only an upper bound when bulk traps are present.

ATLAS implementation. A 0.25 nm IWO region at the Al_2O_3 interface with a DEFECTS acceptor Gaussian (NGA, EGA, WGA; 22, p. 1171) of amplitude Equation (4.11).

Sensitivity evidence. $D_{\text{it}} \times 3$ (run_0025 vs run_0012): $\Delta V_{\text{th,cc}} = +0.025 \text{ V}$, fixed-current SS $+0.1 \text{ mV/dec}$ (10^{-10} – 10^{-9} ; $SS_{\text{cc}} +9 \text{ mV/dec}$, evidence brief), $\Delta I_{\text{on}} = -1.8\%$. $D_{\text{it}} \times 5$ at 6.3 nm (A4, run_0033 vs run_0014): $+0.051 \text{ V}$, gap $V_{\text{th,lin}} - V_{\text{th,cc}}$ unchanged (0.857 vs 0.862 V ; the SS_{min} change, $+8.3 \text{ mV/dec}$, is quoted for history only). About 0.012 – 0.013 V per multiple of the assumed value; closing the 0.29 V 6.3 nm miss would need a peak of about 7×10^{12} ($\times 23$ – 24) at 6.3 nm only, on a gate stack shared with the other films, so front D_{it} is *rejected* as that cause (S1, S4; SYNTHESIS D; POST).

Status. ASSUMED (value), CALCULATED (regularisation, NUM-verified). Interface-trap charge *demonstrated small* (0.012 V , thickness-independent).

4.13 Fixed interface charge

Definition and role. Q_{f} is a fixed positive sheet charge (per cm^2) at the front IWO/ Al_2O_3 interface. It is the model's alignment parameter: it shifts the whole transfer curve rigidly along V_{G} and absorbs every other rigid offset (gate work function, affinity, C_{ox} uncertainty, fixed oxide charge, ionised donors within about 1 nm of the interface).

Values used. $Q_{\text{f}} = 1.73 \times 10^{12} \text{ cm}^{-2}$, shared by 2.0 and 13.2 nm (and by the 6.3 nm validation configuration run_0014); $8.7 \times 10^{10} \text{ cm}^{-2}$ for the device-specific “tuned” 6.3 nm configuration (run_0015, and the DOS 384/192 recalibration run_0039) (evidence brief; predictions register).

Source. FITTED: stage 3 of the calibration (Section 7.2), one value shared across thickness. No literature value exists for this interface.

Derivation. Gauss's law across the gate dielectric, with the sheet at the semiconductor side of the interface, requires the gate to image $-qQ_f$; for fixed channel potential this is a flat-band shift

$$\Delta V_{FB} = -\frac{q Q_f}{C_{ox}}, \quad (4.12)$$

i.e. -0.310 V for 1.73×10^{12} and -0.016 V for $8.7 \times 10^{10} \text{ cm}^{-2}$ (Equation (4.2)). Because the charge sits at the interface it does not alter the potential profile inside the film, so the shift is exact at every V_G (confirmed to 1 mV in ATLAS, Section 4.1). The charge is modelled as a sheet rather than a volume for the following reason. Spread through the film, $1.73 \times 10^{12} \text{ cm}^{-2}$ corresponds to $8.7 \times 10^{18} / 2.7 \times 10^{18} / 1.3 \times 10^{18} \text{ cm}^{-3}$; a *volumetric* donor of 8.7×10^{18} would add about -2.0 V at 13.2 nm (Equation (4.14)), so the shared fit requires the charge to be sheet-like at the front. It is electrostatically identical to a layer of ionised oxygen vacancies of about $1.7 \times 10^{19} \text{ cm}^{-3}$ within 1 nm of the Al_2O_3 (S4) – the two are indistinguishable on I_D – V_G . The required Q_f is degenerate with, and therefore depends on, the confinement assumption. Per film, with confinement the values that would fit each device alone are $1.81 / 0.09 / 1.56 \times 10^{12} \text{ cm}^{-2}$ and without confinement $0.047 / -0.27 / 1.43 \times 10^{12} \text{ cm}^{-2}$ (S1; SYNTHESIS D3; Figure 9.3); the best single Q_f over the three films is 1.153×10^{12} (rms residual 0.136 V) with and 0.403×10^{12} (0.133 V) without confinement (SYNTHESIS D3). Both readings need one film-specific charge of about 1.4 – $1.6 \times 10^{12} \text{ cm}^{-2}$. The charge could be measured through the flat-band voltage of MIS capacitors with several EOTs of the same stack, $V_{FB} = (\phi_M - \chi - \dots) / q - qQ_f \text{ EOT} / (3.9\epsilon_0)$, separates Q_f (slope) from the work-function difference (intercept).

ATLAS implementation. INTERFACE QF= 1.73×10^{12} at the IWO/ Al_2O_3 interface; QF is given in electronic charges per cm^2 , a positive value introduces positive charge [22, p. 1260]; the statement applies by default to semiconductor–insulator interfaces [22, p. 1264] (example of use, 22, p. 1263).

Sensitivity evidence. $Q_f 0 \rightarrow 1.73 \times 10^{12}$ at 2 nm (run_0001→run_0003): $\Delta V_{th,cc} = -0.309$ V, fixed-current SS $+0.4 \text{ mV/dec}$, $\Delta I_{on} = +27.1 \%$; at 13.2 nm run_0002→run_0004. S6: an exact rigid shift ($\max |\Delta \log_{10} I_D| \leq 1.6 \times 10^{-5}$ for the equivalent 0.1 V work-function shift). At 6.3 nm the tuned Q_f (run_0015) fixes $V_{th,cc}$ ($+0.007$ V) and I_{on} but leaves $V_{th,lin}$ -0.35 V and $g_{m,max}$ -23% (Chapter 8).

Status. FITTED, classified as an effective electrostatic correction. REVIEW: with ϕ_M , χ , D_{it} , the deep Gaussian and $\epsilon_{\text{Al}_2\text{O}_3}$ all inside degenerate sets, Q_f is “an alignment absorber, not a charge”; ± 0.2 eV of TiN work function is equivalent to $\pm 1.12 \times 10^{12} \text{ cm}^{-2}$. The 6.3 nm device-specific value is *plausible, unverified* (calibration; no independent observable).

4.14 Effective donor density

Definition and role. $N_{d,eff}(t)$ is the effective density of electrically active, ionised shallow donors in the channel (the net of oxygen vacancies, W donors and compensating acceptors). It is *not* the W concentration: the target paper gives “ $\sim 2 \%$ W” without specifying atomic, cation or molar basis, so the composition is treated as a label and never converted into a donor density.

Values used.

$$N_{d,\text{eff}}(t) = N_{d0} \left[1 + \left(\frac{t}{t_c} \right)^k \right], \quad N_{d0} = 2.5 \times 10^{17} \text{ cm}^{-3}, \quad t_c = 20 \text{ nm}, \quad k = 4, \quad (4.13)$$

i.e. $2.50 / 2.52 / 2.97 \times 10^{17} \text{ cm}^{-3}$ at $2.0 / 6.3 / 13.2 \text{ nm}$ and $1.85 \times 10^{18} \text{ cm}^{-3}$ at 31.8 nm .

Source. FITTED law. The configuration cites as support the bulk/border trap densities $2.37 / 3.12 / 14.6 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$ at $10 / 20 / 30 \text{ nm}$ of a different IWO process [3, p. 173507-6]. S4 corrects this: those numbers are gate-oxide *border traps* from the carrier-number-fluctuation noise model, not donors or IWO tail states, and the configuration “support” conflates the two. The credible support is S4’s reading of the same paper’s turn-on voltage, which shifts linearly with t by about -2 V per 10 nm and implies $N_d \approx 4.3\text{--}4.5 \times 10^{17} \text{ cm}^{-3}$, within a factor 2 of V1. The target paper attributes electron densities above 10^{19} cm^{-3} to undoped In_2O_3 and the suppression of vacancies to W [1, p. 2]; ALD In_2O_3 reaches about 10^{20} cm^{-3} [8, pp. 4–5].

Derivation. For a fully depleted film (Section 4.20) with uniform donors, Poisson’s equation $d^2\psi/dz^2 = -qN_d/\epsilon_s$ with zero field at the free back surface ($z = t$) gives a parabolic potential whose maximum (the conduction-band minimum) is at the back surface, $qN_d t^2/2\epsilon_s$ above the front value. The gate must image the sheet $N_d t$. Keeping the band minimum at the threshold position therefore requires

$$\Delta V_{\text{th}} = -\frac{qN_d t}{C_{\text{ox}}} - \frac{qN_d t^2}{2\epsilon_s}, \quad (4.14)$$

which is $-0.040 / -0.151 / -0.406 / -1.55 \text{ V}$ per 10^{18} cm^{-3} at $2 / 6.3 / 13.2 / 31.8 \text{ nm}$ (S4; e.g. at 13.2 nm $0.236 + 0.170 \text{ V}$). With the V1 law the first term is only $0.009 / 0.028 / 0.070 \text{ V}$ (S1). Consequently the donor density is *unidentifiable* at 2 nm (0.036 V per 10^{18} cm^{-3} , S4) and the steep $k = 4$ law matters only above about 20 nm . The donor density could be measured by Hall or capacitance–voltage profiling on thick unpatterned films, or from the slope of $V_{\text{on}}(t)$ over a dense thickness series, which is Equation (4.14) in differential form.

ATLAS implementation. A uniform n -type DOPING statement on the IWO region (DOPING UNIFORM, 22, p. 1189); the donor Gaussian of DEFECTS (NGD, 22, p. 1171) is zero in V1 and is the V2 candidate for a donor band at fixed absolute energy.

Sensitivity evidence. $N_d \times 2$ (run_0022 vs run_0012): $\Delta V_{\text{th,cc}} = -0.009 \text{ V}$, fixed-current SS $+0.7 \text{ mV/dec}$, $\Delta I_{\text{on}} = +0.7 \%$. A2 at 6.3 nm ($N_{d0} = 10^{16}$, run_0032 vs run_0014): $+0.030 \text{ V}$ (analytic 0.028 V), SS_{min} $+6.5 \text{ mV/dec}$, $I_{\text{on}} -1.8 \%$ – donors are electrostatically irrelevant at 6.3 nm and reduced donors are *rejected* as the cause of the 6.3 nm miss (S1, S4). A thickness-independent donor density fitted to the $6.3 \rightarrow 13.2 \text{ nm}$ drop needs $2.2 \times 10^{18} \text{ cm}^{-3}$ and then predicts -0.25 V from 2 to 6.3 nm against -0.02 V measured (S4). With the V1 law, 31.8 nm gives $V_{\text{th}} = -2.71 \text{ V}$ and $I_D(-3 \text{ V}) = 1.2 \times 10^{-12} \text{ A}/\mu\text{m}$, i.e. not always-on (S1 1-D); its always-on state needs $N_d \gtrsim 3 \times 10^{18} \text{ cm}^{-3}$ or a back donor sheet (Section 4.15).

Status. FITTED. Low $N_{d,\text{eff}} \approx 2.5 \times 10^{17} \text{ cm}^{-3}$: *plausible, unverified* (correlation with Kim2024); a donor *step* ($\approx 2.5 \times 10^{17}$ for $t \leq 6.3 \text{ nm}$, $\approx 10^{18}$ at 13.2 nm , $\gtrsim 2 \times 10^{18}$ at 31.8 nm) is *plausible, unverified* and degenerate with ΔE_c (S4).

4.15 Back-surface charge and its lever arm

Definition and role. The top surface of the channel is exposed to air. Adsorbates (O_2 , H_2O) can form acceptor-like (negative) or donor-like (positive) surface states. A sheet Q_b at the back surface acts on the threshold through a larger lever arm than a front charge, because its field crosses the film as well as the dielectric.

Values used. $Q_b = 0$ for 2.0, 6.3 and 13.2 nm in the calibrated decks. Hypothesis tests at 6.3 nm: $-1.64 \times 10^{12} \text{ cm}^{-2}$ (A3, run_0030) and $-1.0 \times 10^{12} \text{ cm}^{-2}$ within the combination A5 (run_0035). At 31.8 nm a donor-like back sheet was a V0 device-specific hypothesis (configuration: centre $E_c - 0.02 \text{ eV}$, width 0.05 eV, amplitude zero in V1).

Source. ASSUMED zero; the $-1.64 \times 10^{12} \text{ cm}^{-2}$ magnitude is the charge that closes the 6.3 nm $V_{\text{th,cc}}$ miss and has no independent source (“adsorbate scale” is an assumption, S1).

Derivation: the lever arm. A sheet Q_b is assumed at $z = t$ and a front sheet Q_f at $z = 0$, with a fully depleted film free of other charge. The gate images $-(Q_b + Q_f)$; the field in the film between the two sheets is qQ_b/ϵ_s . Holding the potential at the back surface fixed, a back sheet shifts the gate voltage by

$$\Delta V_b = -qQ_b \left(\frac{1}{C_{\text{ox}}} + \frac{t}{\epsilon_s} \right), \quad (4.15)$$

while a front sheet gives only $-qQ_f/C_{\text{ox}}$. The ratio $1 + C_{\text{ox}}t/\epsilon_s$ is 1.22 / 1.68 / 2.44 at 2 / 6.3 / 13.2 nm (analytic back lever 0.218 / 0.301 / 0.436 V per 10^{12} cm^{-2} vs 0.179 front). Equation (4.15) holds for the potential *at the back surface*. The threshold current, however, flows where the electrons are: at threshold the electron centroid is 0.8 / 2.1 / 5.7 nm from the front (S1), and the filled tail screens the back field over the trap-screening length $L_t = 1.7 / 3.2 / 7 \text{ nm}$ (Section 4.20). The realised lever is therefore close to the front value: in the 1-D model with the full DOS it is 0.185 / 0.172 / 0.215 V per 10^{12} on $V_{\text{th,cc}}$ (0.195 / 0.220 / 0.261 without the tail) and only 0.168 / 0.077 / 0.064 on $V_{\text{th,lin}}$, where the front accumulation layer dominates; the back charge also lowers SS_{cc} by 11 / 38 / 47 mV/dec (S1).

ATLAS implementation. For A3/A5 a fixed sheet charge on the exposed IWO/air surface was added (campaign A configuration config/campaign_A_6p3_hypotheses.json; statement in Chapter C). The V0 donor-like back sheet is a regularised DEFECTS donor Gaussian (NGD, EGD; 22, pp. 1170–1171) in a 0.25 nm IWO layer, analogous to Equation (4.11).

Sensitivity evidence. A3 (run_0030): $-1.64 \times 10^{12} \text{ cm}^{-2}$ at the back gives +0.278 V against +0.293 V for the same charge at the front (ratio 0.95, S1): the thick-film lever of Equation (4.15) is *not* realised at threshold. A3 reproduces $V_{\text{th,cc}}$ (error -0.017 V) but moves SS_{cc} by -64.1 mV/dec and $V_{\text{th,lin}}$ by -0.47 V away from the data; the gap $V_{\text{th,lin}} - V_{\text{th,cc}}$ becomes 0.72 V vs 1.176 V measured. A5

(run_0035) gives $V_{th,cc} -0.079$ V and $SS_{cc} -45.9$ mV/dec. The same charge at 2 or 13.2 nm would shift those films by 0.30 or 0.35 V, so it cannot be specific to 6.3 nm without a sample-specific cause. At 31.8 nm the always-on state would need a back donor sheet $\gtrsim 4 \times 10^{12} \text{ cm}^{-2}$ (0.80 V per 10^{12} at 31.8 nm; S4).

Status. ASSUMED zero. Back-surface negative charge as the sole cause of the 6.3 nm miss: *rejected* (model-conditional, POST); back-referenced lever arm at threshold: *rejected* (0.96–1.2× front, not 1.7–2.4×). The air-side charge of the real devices is *not determined* (measurement ambient not recorded).

4.16 Gate work function

Definition and role. ϕ_M of the TiN gate. Together with χ it sets the flat-band offset $\phi_M - \chi$, which enters $V_{th,cc}$ one-to-one; with the bulk affinity it contributes +0.400 V to every film’s budget (S1).

Values used. $\phi_M = 4.70$ eV, all thicknesses.

Source. ASSUMED, centre of a TiN literature range 4.5–4.9 eV quoted in the configuration (TiN work functions depend on stoichiometry, deposition and the dielectric underneath); no measurement on the target stack.

Derivation (how it could be measured). The terraced-oxide (multi-EOT) flat-band method of Section 4.13 gives $\phi_M - \chi$ as intercept; Kelvin-probe or UPS on the TiN gives ϕ_M directly (with the caveat of interface dipoles). On the transfer curve it is exactly degenerate with χ and Q_f : ± 0.2 eV $\equiv \pm 1.12 \times 10^{12} \text{ cm}^{-2}$ (Equation (4.2); REVIEW).

ATLAS implementation. CONTACT NAME=gate WORKFUN=4.7 (the CONTACT statement specifies the physical attributes of an electrode; without it an electrode is charge-neutral/Ohmic, 22, p. 1145; WORKFUN, default 0, 22, p. 1148). The Pd source/drain are left Ohmic (Section 5.6).

Sensitivity evidence. $\phi_M + 0.1$ eV: $\Delta V_{th,cc} = +0.100$ V at 2 nm (run_0048 vs run_0038) and 13.2 nm (run_0049 vs run_0040), exact rigid shift (max $|\Delta \log_{10} I_D| = 1.6 \times 10^{-5}$ and 4.1×10^{-6}), SS unchanged (0.0 mV/dec), $\Delta I_{on} = -6.8 / -5.0$ % at fixed $V_G = 3$ V (S6, REVIEW).

Status. ASSUMED; held fixed because it is degenerate with Q_f and χ (NUM).

4.17 Capture cross sections and recombination lifetimes

Definition and role. Capture cross sections σ_n, σ_p of the defect distributions and the Shockley–Read–Hall (SRH) lifetimes of the background recombination set the *rates* of trapping and generation. In a DC transfer simulation they would matter only if trap occupancy departed from equilibrium with the electron quasi-Fermi level, or if generation current were significant.

Values used. Tail capture cross section 10^{-15}cm^2 ; background SRH lifetime $\tau = 10^{-6}\text{ s}$; both ASSUMED, all thicknesses (configuration). Other cross sections are those listed in the decks (Chapter C); the ATLAS defaults are, e.g., SIGTAE 10^{-16} and SIGTAH 10^{-14}cm^2 [22, p. 1169].

Derivation: why DC results are independent of σ . For a trap level E with capture coefficients $c_n = \sigma_n v_{\text{th}}$, c_p and emission rates $e_n = c_n n_1$, $e_p = c_p p_1$ (detailed balance, $n_1 = N_c e^{(E-E_c)/k_B T}$), the steady-state occupancy is

$$f = \frac{c_n n + e_p}{c_n n + e_n + c_p p + e_p} \xrightarrow{p, p_1 \rightarrow 0} \frac{n}{n + n_1} = \frac{1}{1 + e^{(E-E_{Fn})/k_B T}}. \quad (4.16)$$

In the electrons-only, wide-gap n -type channel the hole terms vanish, σ_n cancels, and the occupancy is the Fermi function of the electron quasi-Fermi level. The cross sections therefore set only time constants, $\tau_{\text{trap}} = [\sigma_n v_{\text{th}}(n + n_1)]^{-1}$, which matter for hysteresis, frequency dispersion and transients – none of which is simulated. Likewise, thermal SRH generation $q n_i t / \tau$ with $n_i \lesssim 10^{-7}\text{cm}^{-3}$ is negligible: as an origin of the measured off-state floors it falls 13–17 decades short (Astra audit, quoted by S4). The cross sections could be measured by transient (pulsed- V_G) current recovery, deep-level transient spectroscopy, frequency-dependent C–V or the time dependence of bias-stress shifts.

ATLAS implementation. DEFECTS SIGTAE/SIGTAH (tail) and SIGGAE/SIGGAH (Gaussian) [22, pp. 1168–1169]; MATERIAL TAUN0 [22, p. 1349] with SRH on the MODELS statement.

Sensitivity evidence. None run: by Equation (4.16) the DC results are invariant (THEORY). The bias-stress literature for the same material family (a threshold shift of 0.72–1.2 V after 1200 s at 6 V, as summarised by S1/S4 from the target paper) shows that slow trapping exists; extrapolated to one $-3 \dots +3\text{ V}$ sweep it gives 0.003–0.24 V, a *not determined* confounder of the 6.3 nm offset (S4).

Status. ASSUMED; not identifiable from DC data.

4.18 Threshold charge budget

Purpose. This derivation decomposes the simulated constant-current threshold $V_{\text{th,cc}}$ (at $10^{-9}\text{ A}/\mu\text{m}$, $V_D = 0.7\text{ V}$; Section 5.10) into additive electrostatic terms, so that every input of this chapter has a quantified weight and the thickness dependence can be attributed term by term.

Derivation. At threshold the film is fully depleted and nearly flat-band inside (band bending across the film 0.022 / 0.045 / 0.016 eV in the ATLAS profiles of run_0012 / run_0015 / run_0013, S1). Gauss's law across the dielectric, with the gate Fermi level at $-qV_G$ relative to the source electron quasi-Fermi level, gives

$$V_{\text{th,cc}} = \underbrace{\frac{\phi_M - \chi^{\text{bulk}}}{q}}_{\text{alignment}} + \underbrace{\frac{\Delta E_c}{q}}_{\text{confinement}} + \underbrace{\frac{(E_{Fn} - E_c)_{\text{cc}}}{q}}_{\text{sets } n_s} - \frac{qQ_f}{C_{\text{ox}}} - \frac{qN_{\text{dt}}}{C_{\text{ox}}} + \frac{q(n_t + N_{\text{it}}^- + n_{\text{GA}}^- + n_s)}{C_{\text{ox}}}, \quad (4.17)$$

Table 4.3. Threshold charge budget at $V_{th,cc}$ (V), S1 1-D surrogate (s1_poisson1d.out.txt); Figure 9.1 shows the same terms graphically. Sheet densities in parentheses (cm^{-2}). Columns: 2.0 nm run_0012; 6.3 nm validation run_0014; 6.3 nm tuned run_0015; 13.2 nm run_0013.

term	2.0 nm	6.3 nm val.	6.3 nm tuned	13.2 nm	status
$(\phi_M - \chi^{\text{bulk}})/q$	+0.400	+0.400	+0.400	+0.400	assumed
$\Delta E_c/q$	+0.353	+0.072	+0.072	+0.026	DFT proxy
$(E_{Fn} - E_c)_{cc}/q$	-0.038	-0.039	-0.037	-0.111	model
$-qQ_f/C_{ox}$	-0.310	-0.310	-0.016	-0.310	fitted
$-qN_d t/C_{ox}$	-0.009	-0.028	-0.028	-0.070	fitted law
tail qn_t/C_{ox}	+0.256 (1.43×10^{12})	+0.226 (1.27×10^{12})	+0.231 (1.29×10^{12})	+0.098 (5.5×10^{11})	fitted
interface traps	+0.012	+0.012	+0.012	+0.012	assumed
free electrons	+0.014 (7.6×10^{10})	+0.016 (9.1×10^{10})	+0.017	+0.005 (2.6×10^{10})	model
deep Gaussian	0.000	+0.001	+0.001	+0.003	assumed
$V_{th,cc}$ 1-D / ATLAS / measured 0.679 / 0.676 / 0.662 0.351 / 0.350 / 0.644 0.652 / 0.651 / 0.644 0.052 / 0.053 / 0.083					

where n_t , N_{it}^- , n_{GA}^- are the ionised tail, interface and deep-Gaussian sheets (Equations (4.9) to (4.11)), n_s the free sheet at the source, and $(E_{Fn} - E_c)_{cc}$ the Fermi-level position at which the constant-current criterion is met. The last term depends on mobility: a film with higher μ_{band} reaches $10^{-9} \text{ A}/\mu\text{m}$ at a lower n_s and hence a lower E_F , which also empties part of the tail. S1 evaluates Equation (4.17) with a 1-D Poisson surrogate that matches ATLAS $V_{th,cc}$ within 3 mV (runs run_0015, run_0030, run_0013); its closure is exact to 10^{-6} V .

Attribution of the thickness dependence. Differencing Table 4.3 between films gives the model's thickness steps (S1): 2→13.2 nm, -0.627 V in total, of which confinement -0.327 (52 %), tail -0.158 (25 %), $(E_{Fn} - E_c) -0.073$ (12 %) and donors -0.061 (10 %), against -0.579 V measured; 2→6.3 nm, -0.328 V , of which confinement -0.281 (86 %), tail -0.030 , $(E_{Fn} - E_c) -0.001$ and donors -0.019 , against -0.018 V measured. Hence the V1 6.3 nm miss (run_0014: -0.294 V) is essentially the confinement difference $\Delta E_c(2) - \Delta E_c(6.3)$ (*demonstrated within the model*, S1). Only two analytic terms are fixed by laws rather than fitted: $qN_d t/C_{ox}$ and ΔE_c . The constant-current definition itself contributes: if the 13.2 nm film had the 2 nm mobility its $V_{th,cc}$ would be 0.108–0.115 V higher (analytic on data), so 0.11–0.14 V of the measured -0.58 V step is a consequence of the threshold definition, not of charge (*strongly supported*, S1; naive estimate $SS_{cc} \log_{10}(\mu \text{ ratio}) = 0.10 \text{ V}$ at 160 mV/dec). One-at-a-time removal in the 1-D model confirms the weights: confinement (ΔE_c and m^*) 0.315 / 0.064 / 0.023 V, tail 0.221 / 0.215 / 0.112 V, m^* via N_c only -0.039 / -0.008 / -0.002 V , N_d -0.009 / -0.031 / -0.094 V , D_{it} 0.012 / 0.012 / 0.011 V (2 / 6.3 tuned / 13.2 nm), consistent with the ATLAS differences run_0012–run_0016 (0.315 V), run_0013–run_0017 (0.023 V), run_0022 and run_0025.

Degenerate rigid shifts. ϕ_M , χ , Q_f , ΔE_c (to within the N_c term), D_{it} charge, deep-Gaussian charge and C_{ox} errors all enter Equation (4.17) as rigid offsets at first order; on one I_D – V_G curve only their weighted sum is identified (at 6.3 nm about $1.64 \times 10^{12} \text{ cm}^{-2} \equiv 0.29 \text{ V}$; S1). This is the formal statement of the non-identifiability of the confinement shift.

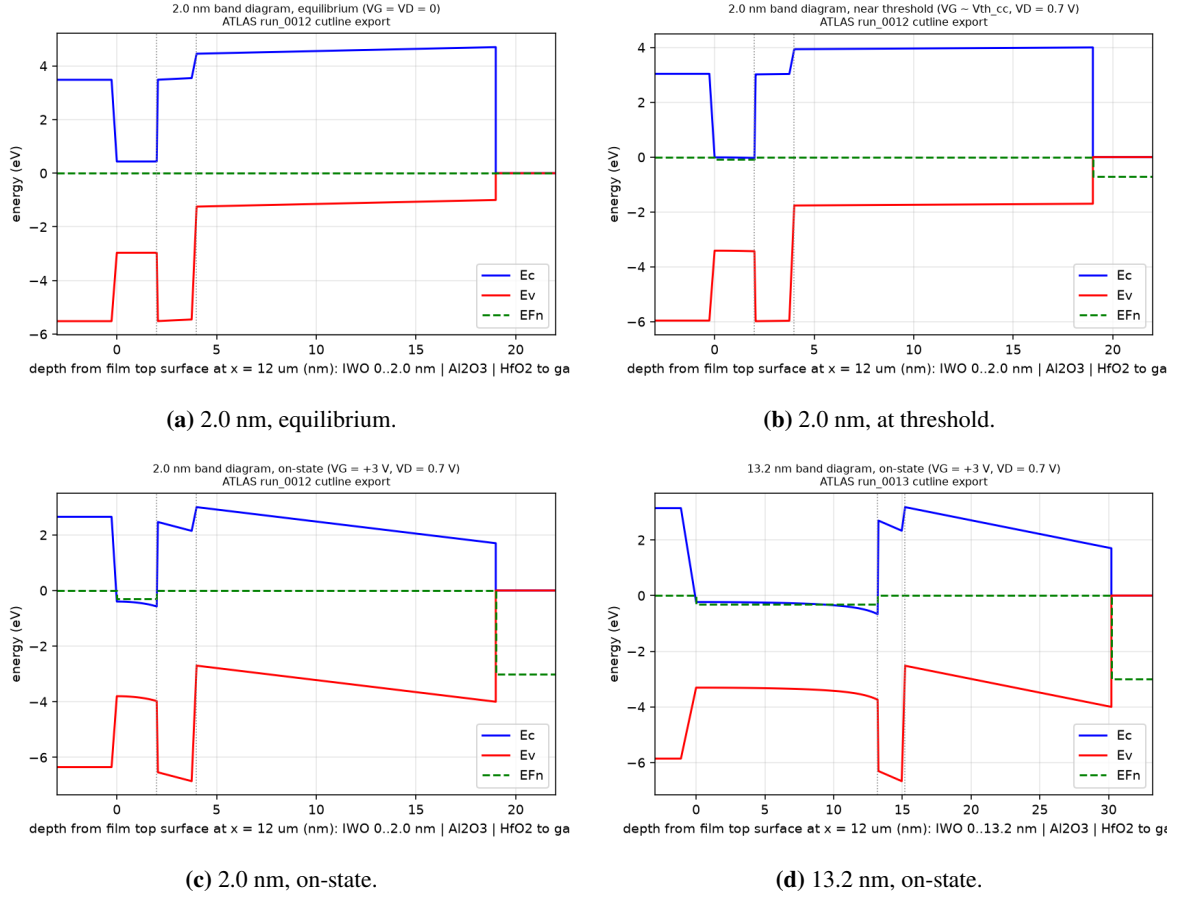


Figure 4.3. Energy-band diagrams of the calibrated decks `run_0012` (2.0 nm, panels a–c) and `run_0013` (13.2 nm, panel d), as exported by the package plotting scripts (`plots/2p0/band_diagram_{eq,vth,on}.png`, `plots/13p2/band_diagram_on.png`). The panels show the conduction-band edge of the 2 nm film raised by confinement; the almost flat bands inside the film at threshold (ATLAS band bending across the film 0.022 eV at 2 nm and 0.016 eV at 13.2 nm, S1), which justifies the additive budget of Table 4.3; and the front-interface band bending in the on-state.

Band diagrams. Figures 4.3a to 4.3d show the ATLAS band diagrams of the calibrated decks, on which the budget is based: the film is nearly flat inside at threshold, so the budget terms are additive, and the gate field bends the bands only in the on-state.

Status. SURR checked against NUM (3 mV). The budget explains the model, not the device: the measured $2 \approx 6.3 \gg 13.2$ nm pattern of $V_{th,cc}$ is not reproduced by any single-material law (Chapter 9).

4.19 Subthreshold slope relations and trap-DOS inversion

Charge-control derivation. Below threshold the current is proportional to the free sheet density at the source, $I_D \propto n_s \propto e^{u/k_B T}$ with $u = E_{Fn} - E_c$ (non-degenerate). Differentiating the Gauss law $C_{ox}(V_G - V_{FB} - u/q) = q(n_s + n_t + N_{it}^- + \dots)$ gives $C_{ox} dV_G = (C_{ox}/q) du + q dn_s + q dn_t + q dN_{it}^-$, so

$$SS \equiv \frac{dV_G}{d \log_{10} I_D} = \ln 10 \frac{k_B T}{q} \left[1 + \frac{C_{it} + C_t + C_{free}}{C_{ox}} \right], \quad C_{it} = q^2 D_{it}, \quad C_t = q^2 t \frac{\partial n_t^{vol}}{\partial u}, \quad C_{free} = q^2 \frac{\partial n_s}{\partial u} = \frac{q^2 n_s}{k_B T}. \quad (4.18)$$

For the exponential tail, $C_t = q^2 t n_t^{\text{vol}} / W_{\text{TA}}$ (Equation (4.9)), so the tail contribution grows exponentially as E_F approaches E_c ; the free-carrier term takes over once $n_s / k_B T$ exceeds C_{ox} / q^2 , the crossover “ $C_{\text{free}} = C_{\text{ox}}$ ”. The textbook inversion of Equation (4.18),

$$D_{\text{tot}} = \frac{C_{\text{ox}}}{q^2} \left(\frac{SS}{\ln 10 k_B T / q} - 1 \right), \quad (4.19)$$

attributes the whole excess slope to traps. It is the method used by the literature values of Section 4.12 (e.g. the $\text{HfO}_2/\text{In}_2\text{O}_3$ D_{it} from an SS of 63.5 mV/dec, 2, p. 21539).

Why the naive inversion fails here. Applied to the measured slopes, Equation (4.19) gives 2.35 / 5.18 / $6.69 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$ from SS_{min} and 1.98 / 2.22 / 0.94×10^{13} from SS_{cc} (2 / 6.3 / 13.2 nm; S4). Two confounders make these numbers unreliable (S4): (i) *mobility*: the crossover $C_{\text{free}} = C_{\text{ox}}$ sits at 3.2×10^{-10} / 4.0×10^{-10} / $1.8 \times 10^{-9} \text{ A}/\mu\text{m}$ for $\mu = 10.7$ / 13.3 / $59 \text{ cm}^2/\text{V s}$, so the 10^{-10} – 10^{-8} window straddles it; a trap-free 2 nm film already gives $SS_{\text{cc}} = 122 \text{ mV/dec}$ at $\mu = 13.3$ against 75 at $\mu = 59$, i.e. about 60 mV/dec of the thin films’ SS_{cc} (about 15 at 13.2 nm) is mobility, not traps. (ii) *energy window*: the low 13.2 nm SS_{cc} reflects a window about $k_B T \ln 4.5 \approx 39 \text{ meV}$ deeper in energy, not fewer traps. The naive formula therefore ranks 13.2 nm lowest (9.4×10^{12} vs 2.0×10^{13}), whereas the proper inversion ranks it highest. The floor-dependent SS_{min} window (Section 5.10) adds a further artefact: it reaches $u \approx -0.21$ / -0.17 / -0.13 eV for 2 / 6.3 / 13.2 nm because the off-floors rise $\times 1432$ (Figure 9.4).

Charge-sheet trap-DOS inversion. S4 removes both confounders with an exact identity. In the gradual-channel approximation $I_D = (W/L) q \mu \int_0^{V_D} n_s(V_G - V) dV$, hence $g_m = (W/L) q \mu [n_s(V_G) - n_s(V_G - V_D)]$ and, by telescoping,

$$n_{s0}(V_G) = \frac{L}{q \mu W} \sum_{k \geq 0} g_m(V_G - k V_D), \quad (4.20)$$

which removes the drain term. With u obtained from n_{s0} through N_c and Fermi–Dirac statistics, differentiating the Gauss law gives the trapped DOS

$$D_{\text{trap}}(u) = \frac{C_{\text{ox}}}{q^2} \left(q \frac{dV_G}{du} - 1 \right) - \frac{dn_f}{du}, \quad (4.21)$$

in which V_{FB} cancels. The method was validated on ATLAS curves before use (measured/ATLAS DOS ratio 0.81–1.17 for $u = -0.125$ to -0.025 eV at 2 and 13.2 nm) and requires the mobility, which shifts the energy axis by $k_B T \ln(\mu \text{ ratio})$, ± 10 – 15 meV between the V1 and S2 mobilities (S4). Results: near E_c the sheet DOS grows sub-linearly with thickness, $D(E_c - 0.1 \text{ eV}) = 1.05$ / 1.67 / $2.16 \times 10^{13} \text{ cm}^{-2} \text{ eV}^{-1}$ ($\propto t^{0.38 \pm 0.15}$), a surface + bulk signature (Section 4.10); V1 underestimates the states at $E_c - 0.15$ to -0.2 eV by $\times 2.2$ – 2.8 (Section 4.11); and the 6.3 nm film carries more states in the last $\sim 50 \text{ meV}$ below E_c (ratios 1.25 / 1.46 / 1.72 at -0.05 / -0.025 / 0 eV relative to run_0015), which fill by $V_{\text{th,lin}}$ and then saturate – a later, steeper turn-on, as measured (S4; Chapter 8).

Sensitivity evidence. Slope-changing inputs at 2 nm (vs run_0012; fixed-current SS in 10^{-10} – $10^{-9} \text{ A}/\mu\text{m}$): $N_t \times 1.5$ +49.0, $W_{\text{TA}} + 5 \text{ meV}$ +4.6, $D_{\text{it}} \times 3$ +0.1, $m^* \times 1.3$ –18.0 mV/dec (run_0050

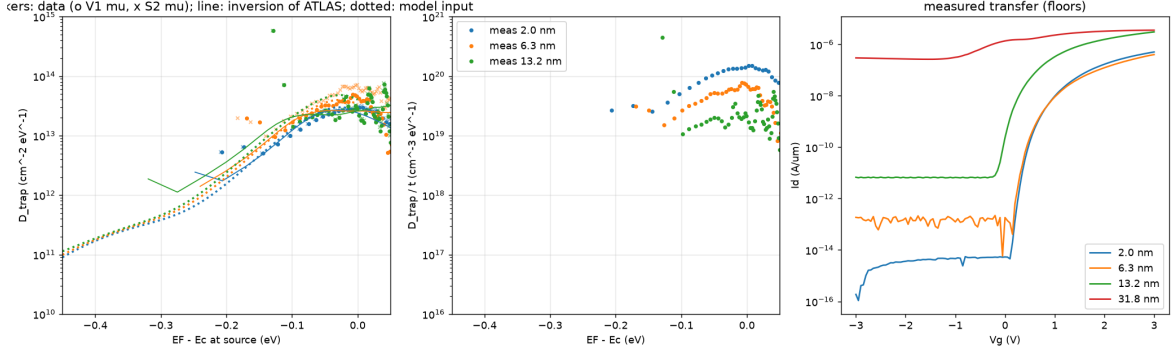


Figure 4.4. Trap-DOS spectroscopy by the charge-sheet inversion (Equations (4.20) and (4.21)) of S4 (copied from scratch/S4/s4_trap_spectroscopy.png): trapped-state density vs $u = E_{Fn} - E_c$ from the measured curves and from the simulated curves of the calibrated decks. The figure shows the agreement of measured and ATLAS DOS within 0.81–1.17 for $u = -0.125$ to -0.025 eV (2 and 13.2 nm); the sub-linear growth of the near-edge sheet DOS with thickness; the underestimate of the deeper states by V1.

vs run_0038); rigid inputs (χ , ϕ_M , Q_f) change it by 0.0. The surrogate reproduces ATLAS SS_{cc} as 294 / 303 / 165 vs 289 / 289 / 193 mV/dec (S4 part C).

Status. THEORY (Equations (4.18), (4.20) and (4.21)); inversion SURR validated against NUM. $SS_{cc}(t)$ read as a trap trend with Equation (4.19): *rejected* (S4). SS_{min} is retired as a metric (Section 5.10).

4.20 Debye length, depletion width and full depletion

Debye and trap-screening lengths. Linearising Poisson's equation about a uniform electron density n gives the screening length

$$L_D = \sqrt{\frac{\epsilon_s k_B T}{q^2 n}}, \quad L_t = \sqrt{\frac{\epsilon_s W_{TA}}{q^2 n_t^{vol}}}, \quad (4.22)$$

the second for screening by the exponential tail ($\partial n_t / \partial u = n_t / W_{TA}$). With $\epsilon_s = 9.3$: $L_D = 7.3 / 6.7 / 3.7 / 2.1$ nm at $n = 2.5 \times 10^{17} / 3 \times 10^{17} / 10^{18} / 3 \times 10^{18} \text{ cm}^{-3}$, and $L_t = 1.7 / 3.2 / 7$ nm at the threshold trapped densities of 2 / 6.3 / 13.2 nm ($7.2 \times 10^{18} / 2.0 \times 10^{18} / 4.2 \times 10^{17} \text{ cm}^{-3}$) (S1). The regularisation layer of Equation (4.11) (0.25 nm) is well below both.

Depletion width and full depletion. A surface sheet Q_s is compensated by depleting donors over $W = Q_s / N_d$; the maximum depletion width for a band bending ψ is $W_{max} = \sqrt{2\epsilon_s \psi / (qN_d)}$. At the V1 N_d ($2.5\text{--}3 \times 10^{17} \text{ cm}^{-3}$) a sheet of 10^{12} cm^{-2} depletes 40 nm, so every film ≤ 31.8 nm is fully depleted (S1). A neutral (ungated) layer between 6.3 and 13.2 nm would require $7.6 \times 10^{17} < N_d < 1.6 \times 10^{18} \text{ cm}^{-3}$ (with $Q_s \approx 10^{12}$). The data exclude it at 13.2 nm: the off-band current $6.6 \times 10^{-12} \text{ A}/\mu\text{m}$ bounds any ungated free sheet to $n_s = IL / (q\mu V_D W) \leq 2 \times 10^7 \text{ cm}^{-2}$ ($\mu = 59 \text{ cm}^2/\text{V s}$), whereas a neutral sheet of only 10^{10} cm^{-2} would carry $3.3 \times 10^{-9} \text{ A}/\mu\text{m}$ (S1). The always-on 31.8 nm film needs at least $5.6\text{--}9.2 \times 10^{11} \text{ cm}^{-2}$ ungated, i.e. uniform $N_d \geq 3 \times 10^{18} \text{ cm}^{-3}$, which would put 13.2 nm at -1.0 V; the 13.2→31.8 nm transition therefore requires a $\geq 10\times$ change of effective donor density or

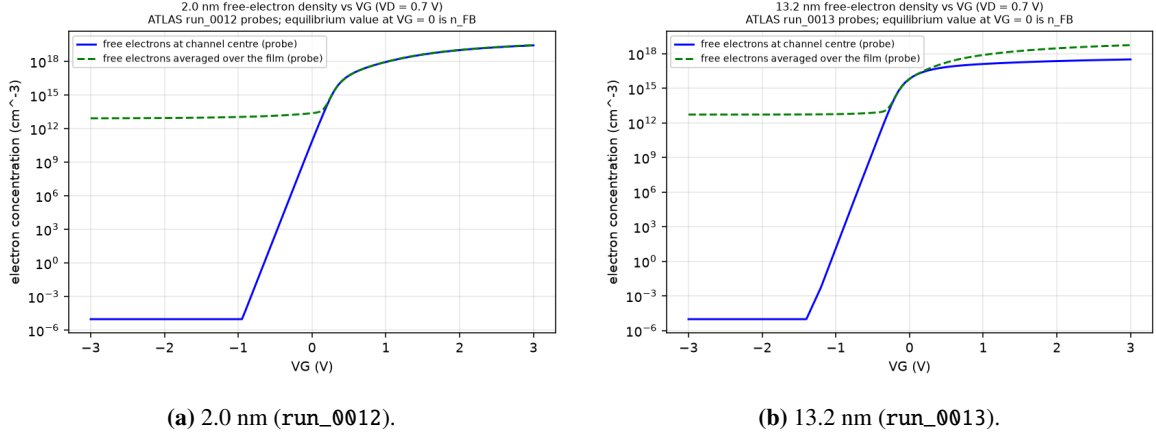


Figure 4.5. Electron density vs gate voltage in the calibrated decks, as exported by the package plotting scripts (plots/2p0/electron_density_vs_vg.png, plots/13p2/electron_density_vs_vg.png). The thin film fills uniformly (volumetric conduction, centroid $\approx t/2$), whereas the 13.2 nm film develops a front accumulation layer of about 1 nm at high V_G (S1).

a surface donor layer – a material step, not dimensional electrostatics (S1).

Where the electrons are. Subthreshold conduction is volumetric: the electron centroid at threshold is 0.8 / 2.1 / 5.7 nm from the front (2 / 6.3 / 13.2 nm). In the on-state at 3 V all films carry a 0.7–1.1 nm front accumulation layer (ATLAS profiles; S1), which is where the field quantization of Section 4.6 acts. Figures 4.5a and 4.5b show the evolution of the electron density with gate voltage in the calibrated decks.

Status. THEORY with SURR/NUM values. Full depletion of all films ≤ 31.8 nm at the V1 N_d : *demonstrated within the model*; a neutral bulk layer as the 6.3→13.2 nm step: *rejected* (S1).

4.21 Summary

Table 4.4 collects the inputs. Only the gate stack (geometry and ϵ_{HfO_2} , hence most of C_{ox}) and ϵ_{IWO} are measured on the target process; the band-structure inputs are literature values of other IWO devices or pure-In₂O₃ DFT proxies; the charge-and-trap inputs are fitted or assumed. The model is nonetheless well conditioned for what it is used for, because the inputs separate into two classes with different signatures: *rigid shifts* (ϕ_M , χ , Q_f , ΔE_c within the N_c term, D_{it} and deep-Gaussian charge, C_{ox} errors), of which only a weighted sum is identified per curve; and *shape parameters* (N_t , W_{TA} , m^* through N_c , near-edge bands), which the slope and the $V_{\text{th,lin}} - V_{\text{th,cc}}$ gap constrain.

Summary. The electrostatics of the V1 IWO model is fixed by a measured gate stack ($C_{\text{ox}} = 8.955 \times 10^{-7}$ F/cm², so 0.179 V per 10^{12} cm⁻²), a DFT-proxy confinement law ($\Delta E_c = 0.353 / 0.072 / 0.026$ eV) and a fitted exponential acceptor tail ($N_t = 2 \times 10^{19} (2/t)^{0.75}$ cm⁻³, $W_{\text{TA}} = 40$ meV) with a shared fixed charge $Q_f = 1.73 \times 10^{12}$ cm⁻². The threshold budget (1-D surrogate within 3 mV of ATLAS) attributes 52 % of the model's 2→13.2 nm step and 86 % of its 2→6.3 nm step to ΔE_c , and 25 % of the former to the tail (SURR/NUM). Confinement at 2 nm is theoretically expected (Schrödinger–Poisson 0.14–0.38 V, central 0.26–0.30 V), but it is exactly the kind of rigid shift that

Table 4.4. Electrostatic inputs of the V1 model: values, status and controlling runs.

input	2.0 / 6.3 / 13.2 nm	sensitivity (runs)	status
C_{ox} , EOT	$8.955 \times 10^{-7} \text{ F/cm}^2$, 3.86 nm	analytic, $q/C_{\text{ox}} = 0.179 \text{ V}$ per 10^{12}	CALCULATED
ϵ_{IWO}	9.3	$< 1 \%$ at 2 nm (run_0026)	MEASURED_TARGET_DEVICE
E_{g}	3.403 / 3.122 / 3.076 eV	none (holes not solved)	LITERATURE_IWO + proxy
χ	3.947 / 4.228 / 4.274 eV	$\pm 0.100 \text{ V}$ rigid (run_0020/run_0021)	LITERATURE_IWO + proxy
ΔE_{c}	0.353 / 0.072 / 0.026 eV	$-0.315 / -0.023 \text{ V}$ if removed (run_0016, run_0017)	DFT_PURE_IN2O3_PROXY
m^*	0.257 / 0.218 / 0.211 m_0	$\times 1.3$: $-0.044 / -0.029 \text{ V}$ (run_0050/run_0051)	proxy
N_{c}	$3.28 / 2.55 / 2.44 \times 10^{18} \text{ cm}^{-3}$	via m^*	CALCULATED
N_{v}	10^{19} cm^{-3}	none	ASSUMED
N_{t} , W_{TA}	$2.0 / 0.85 / 0.49 \times 10^{19} \text{ cm}^{-3}$; 40 meV	$\times 1.5$: $+0.082 \text{ V}$, -25% I_{on} (run_0023); $+5 \text{ meV}$: $+0.020 \text{ V}$ (run_0024)	FITTED
deep Gaussian	5×10^{16} , $E_{\text{c}} - 0.6, 0.15 \text{ eV}$	$\leq 0.003 \text{ V}$ (analytic)	ASSUMED
D_{it}	3×10^{11} at $E_{\text{c}} - 0.3, 0.12 \text{ eV}$	$\times 3$: $+0.025 \text{ V}$ (run_0025); $\times 5$: $+0.051 \text{ V}$ (run_0033)	ASSUMED
Q_{f}	1.73×10^{12} (6.3 tuned 8.7×10^{10})	exact rigid -0.309 V (run_0001 $\rightarrow$ run_0003)	FITTED
$N_{\text{d,eff}}$	$2.50 / 2.52 / 2.97 \times 10^{17} \text{ cm}^{-3}$	$\times 2$: -0.009 V (run_0022); A2 $+0.030 \text{ V}$ (run_0032)	FITTED
Q_{b}	0	A3 -1.64×10^{12} : $+0.278 \text{ V}$, SS -64 (run_0030)	ASSUMED
ϕ_{M}	4.70 eV	$+0.1 \text{ eV}$: $+0.100 \text{ V}$ rigid (run_0048/run_0049)	ASSUMED
σ , τ_{SRH}	10^{-15} cm^2 , 10^{-6} s	none (DC-invariant)	ASSUMED

one transfer curve cannot distinguish from $\approx 1.7 \times 10^{12} \text{ cm}^{-2}$ of fixed charge, and the $t^{-1.38}$ law overestimates the subthreshold-equivalent shift by $\approx 2\times$ at 6.3 nm. ϕ_{M} , χ and Q_{f} are one parameter (exact rigid shifts, runs run_0020, run_0021, run_0048, run_0049); donors, permittivity, interface traps and the deep Gaussian are electrostatically small at $t \leq 6.3 \text{ nm}$ (runs run_0022, run_0026, run_0025, run_0032); the tail controls slope and threshold. Status: *calibrated, not validated*.

The electrostatics fixes where the Fermi level sits for a given gate voltage; how much current flows then depends on the mobility, the contacts and the extraction of the metrics, which are derived in the next chapter.

Chapter 5

Physical model II: transport, extraction, contacts, temperature and numerical discretization

Chapter 4 established where the charge sits in the gate stack and in the IWO film. The present chapter treats the subsequent steps. It describes how V1 converts that charge into a drain current, how the current is reduced to the metrics that are compared with the measurement, and which parts of this chain are assumed, fitted or derived. Six topics are addressed. The first is the set of transport parameters used for each film and their origin (Sections 5.1 to 5.3). The second is the exact relation that reduces the mobility calibration to a single linear rescaling (Section 5.4). The third is the definition of the measured and simulated metrics, the identification of those that are reliable, and the reason why the mobilities of the target paper differ from those of this report (Sections 5.5 and 5.10). The fourth is the information that the data contain about contacts and series resistance (Sections 5.6 and 5.7). The fifth is what was actually simulated at elevated temperature and what an activation energy measures in this model (Sections 5.8 and 5.9). The sixth is the convergence of the discretized trap density of states (Section 5.11) and the question of whether three films can identify a thickness law at all (Section 5.12). Every section follows the structure fixed for parameter sections: definition and role, values, source, derivation, ATLAS implementation, sensitivity evidence, and status with caveats. Every simulated number names its run. Analysis numbers name the 2026-09-25 report (S2, S5, S6, SYNTHESIS, REVIEW, PREDICTIONS_REGISTER) they come from.

5.1 Band mobility per film, $\mu_{\text{band}}(t)$

Definition and role in the model. V1 describes electron transport in the IWO region with the ATLAS constant low-field mobility model. The mobility is uniform in space and does not depend on electric field, carrier density or doping. The value handed to ATLAS is

$$\mu_n^{\text{ATLAS}}(t) = \mu_{\text{band}}(t) R(t), \quad (5.1)$$

where $R(t)$ is the roughness factor of Section 5.2. In the multiple-trapping-and-release (MTR) picture that V1 implements, μ_{band} is the mobility of the *free* (band) electrons. Electrons captured in the

Table 5.1. Transport, extraction and numerical inputs treated in this chapter, with their final values and status. “Recalibrated” refers to the DOS 384/192 state (run_0038, run_0039 rescaled $\times 1.015074$, run_0040 rescaled $\times 1.020113$; SYNTHESIS D8).

Section	Quantity and final value(s)	Status
5.1	$\mu_{\text{band}} = 17.69 / 11.58 / 61.60 \text{ cm}^2/\text{V s}$ (2 / 6.3 / 13.2 nm)	FITTED
5.2	$R(t) = (1 - \Delta_{\text{sr}}/t)^2$, $\Delta_{\text{sr}} = 0.287 \text{ nm}$; $R = 0.734 / 0.911 / 0.957$	LITERATURE_IWO (form assumed)
5.3	Free/trapped partition $P = 0.847 / 0.791 / 0.826$ (V1 runs)	CALCULATED
5.4	$I_{\text{D}} \propto \mu$ exactly; checked to 6×10^{-6}	demonstrated (NUM)
5.5	μ_{FE} (linear) 12.15 / 10.95 / 50.17; saturation formula 5.09 / 3.86 / 27.40 $\text{cm}^2/\text{V s}$	DATA
5.6	Ideal Ohmic Pd S/D, $R_{\text{sd}} = 0$, overlap 2 μm	ASSUMED
5.7	$\theta_{\text{net}} < 0$ in all films; R_{sd} not identifiable	not determined
5.8	$T = 300 \text{ K}$; 338.15 / 358.15 K with ATLAS default $\text{tmu} = 1.5$	ASSUMED; configuration error
5.9	$E_{\text{a}}(V_{\text{G}})$ from MTR; two exact variants differ by 42.3 meV	PRED
5.10	$V_{\text{th,cc}}$ at $10^{-9} \text{ A}/\mu\text{m}$; fixed-current SS; SS_{min} retired	convention
5.11	NUMA/NUMD 384/192; Richardson residual +0.16 % I_{on}	demonstrated (NUM)
5.12	Power laws in t for μ_{FE} , I_{on} , V_{th} : LOO miss $\times 2.63 / \times 3.76$	rejected

acceptor-like tail (Section 4.10) do not conduct. The field-effect mobility extracted from a transfer curve is therefore lower than μ_n^{ATLAS} , by the partition factor P and the capacitance ratio $C_{\text{eff}}/C_{\text{ox}}$ (Section 5.3). μ_{band} is the only transport parameter that V1 fits. It sets the magnitude of the on-state current. Because the current is exactly linear in a uniform constant mobility (Section 5.4), it has no effect on the electrostatics.

Values used (per thickness). Table 5.2 lists the values. Two calibrated states exist. The V1 state at DOS 96/48 (run_0012, run_0015, run_0013) was superseded by the recalibrated state at DOS 384/192 (run_0038, run_0039, run_0040; S6 section 1, SYNTHESIS D8). run_0039 and run_0040 were simulated at 11.41 and 60.39 $\text{cm}^2/\text{V s}$. Their currents were then rescaled by $\times 1.015074$ and $\times 1.020113$, which is exact because of Equation (5.8). That gives the final values 11.5820 and 61.6046 $\text{cm}^2/\text{V s}$. The earlier stages of the calibration used 16.8 (2 nm, run_0003) and 57.6 $\text{cm}^2/\text{V s}$ (13.2 nm, run_0004) before I_{on} was matched (run_0008, run_0009).

Table 5.2. Band mobility per film. Measured μ_{FE} is the linear-regime estimator of Equation (5.11) applied to the $V_{\text{D}} = 0.7 \text{ V}$ data. Simulated μ_{FE} is the same estimator applied to the final runs (MANIFEST, calibrated_overlays). All mobilities in $\text{cm}^2/\text{V s}$.

t (nm)	V1, DOS 96/48 (run)	recalibrated, DOS 384/192 (run)	μ_{FE} measured	μ_{FE} simulated
2.0	18.13 (run_0012)	17.6897 (run_0038 at 17.69, $\times 0.999983$)	12.15	11.10
6.3	11.69 tuned (run_0015); 12.4 (run_0014)	11.5820 (run_0039 at 11.41, $\times 1.015074$)	10.95	8.39
13.2	61.9 (run_0013)	61.6046 (run_0040 at 60.39, $\times 1.020113$)	50.17	48.85

Source. FITTED, separately for each film, to the measured on-current $I_{\text{on}} = I_{\text{D}}(V_{\text{G}} = 3 \text{ V})$ at $V_{\text{D}} = 0.7 \text{ V}$. With the final runs the I_{on} error is 0.00 % by construction (S6 section 1.6). No measured band or Hall mobility of these IWO films is available. The 6.3 nm value 12.4 $\text{cm}^2/\text{V s}$ used in the shared-parameter run run_0014 is not independent: it is the V0 per-film fit of the same 6.3 nm curve (REVIEW; SYNTHESIS D4). Hence, run_0014 tests the threshold only and is not a parameter-free prediction.

Derivation. Since I_D is exactly proportional to μ_n^{ATLAS} (Equation (5.8)), the fit takes a single step:

$$\mu_{\text{band}}^{\text{new}} = \mu_{\text{band}}^{\text{old}} \frac{I_{\text{on}}^{\text{meas}}}{I_{\text{on}}^{\text{sim}}(\mu_{\text{band}}^{\text{old}})}. \quad (5.2)$$

For example, $11.41 \times 1.015074 = 11.582$ and $60.39 \times 1.020113 = 61.60 \text{ cm}^2/\text{V s}$ (S6 section 1: `run_0039` was 1.48 % and `run_0040` 1.97 % low in I_{on} before rescaling). The measured field-effect mobility can be split into the factors V1 carries (S2 report):

$$\mu_{\text{FE}}^{\text{meas}} = \mu_{\text{band}} \cdot R(t) \cdot P(t) \cdot \frac{\mu_{\text{FE}}^{\text{meas}}}{\mu_{\text{FE}}^{\text{sim}}}, \quad P \equiv \frac{\mu_{\text{FE}}^{\text{sim}}}{\mu_n^{\text{ATLAS}}}. \quad (5.3)$$

For the V1 runs `run_0012` / `run_0015` / `run_0013`, S2 finds $R = 0.734 / 0.911 / 0.957$, $P = 0.847 / 0.791 / 0.826$, $\mu_{\text{FE}}^{\text{sim}} = 11.27 / 8.42 / 48.95$ and a measured-to-simulated ratio of $1.078 / 1.300 / 1.025$. On a logarithmic scale the $6.3 \rightarrow 13.2 \text{ nm}$ step ($\times 4.58$ in μ_{FE}) splits into +109 % from the fitted μ_{band} , +3 % from R , +3 % from P and -16 % model residual. The $2 \rightarrow 13.2 \text{ nm}$ step ($\times 4.13$) splits into +87 / +19 / -2 / -4 %. Hence, the thickness dependence of the mobility sits almost entirely in the fitted parameter (S2, “demonstrated, model-internal”). Figure 9.2 shows the partition.

A band mobility could be measured directly instead of being fitted. A gated Hall measurement (van der Pauw) on each film would separate μ_{band} from the trap partition, and quasi-static C–V would give $n_{\text{free}}/n_{\text{total}}$. A temperature series would separate band-limited from barrier-limited transport, and TLM would remove R_{sd} from μ_{band} (S2 report; S5 report).

ATLAS implementation. The product in Equation (5.1) is passed as the electron mobility MUN of the IWO user material, set with the `MATERIAL` statement [22, p. 1343]. The constant low-field model is $\mu_{n0} = \text{MUN} (T_L/300)^{-T_{\text{MUN}}}$ [22, p. 179, Eq. 3-215]. At 300 K the temperature factor is unity. At elevated T it was *not* unity (Section 5.8). As an example, $\text{MUN} = 17.6897 \times 0.73359 = 12.977 \text{ cm}^2/\text{V s}$ for 2 nm is the 300 K value printed in `run_0045/deckbuild.out` (REVIEW V7). The deck listings are in Chapter C.

Sensitivity evidence.

- `run_0003` $\rightarrow$ `run_0008` (2 nm, μ_{band} $16.8 \rightarrow 18.13$, $\times 1.0792$): the current ratio is 1.079171 against 1.079167 expected (S6 section 1.1). $\Delta V_{\text{th,cc}} = -0.009 \text{ V}$, $\Delta SS_{[10^{-10}, 10^{-9}]} = -3.8 \text{ mV/dec}$, $\Delta I_{\text{on}} = +7.9$ % (MANIFEST, `param_control_delta`; Figures 7.4 and 7.5).
- `run_0004` $\rightarrow$ `run_0009` (13.2 nm, $57.6 \rightarrow 61.9$): the same pure rescaling (Figure 7.6).
- Constant-current threshold confounding (SYNTHESIS D6): giving the 13.2 nm curve the 2 nm mobility moves $V_{\text{th,cc}}$ by +0.108 to +0.115 V. That is 19–20 % of the measured 0.579 V step. $V_{\text{th,cc}}$ is therefore partly a mobility metric.
- Degeneracy with the tail density (S2 report): on I_{on} alone, $N_t \times 1.5$ (`run_0023`) is exactly compensated by $\mu \times 1.33$. Only the on-state shape (gap, g_m) separates them.
- Campaign A (`run_0030`–`run_0034`, 6.3 nm, μ_{band} 12.4) changes $g_{m,\text{max}}$ by at most 2 %. Hence, the 6.3 nm g_m miss is not electrostatic (S2 report).

Status and caveats. FITTED per thickness. No thickness law is claimed (Section 5.12). Caveats: (1) The fitted sequence 17.69 / 11.58 / 61.60 is non-monotonic. S2 rates the 6.3 nm dip “strongly supported” as an artefact of the $N_t(t)$ law partition. S2’s shape-consistent alternative (μ_{band} 19.3 / 19.6 / 62.7 with N_t 2.4 / 2.7 / $0.48 \times 10^{19} \text{ cm}^{-3}$) was tested prospectively in **run_0037** (B0). The on-state passed, but SS_{cc} rose to 410.9 against 295.4 mV/dec measured, so the re-partition is **rejected** (SYNTHESIS conflict table; S6 P1). (2) μ_{band} lumps together any series resistance (Section 5.6), the ratio $C_{\text{eff}}/C_{\text{ox}} \approx 0.87$ (S3), the Drude $m^*(t)$ term that V1 does not carry (−18 % at 2 nm at fixed scattering time; S2 report), and the roughness factor. (3) The 6.3 nm on-state shape (gap −0.36 V, g_m −23 %) is not reproduced by any tested V1 variant (Section 8.3). The most plausible cause is the constant, density-independent mobility itself. That inference is **plausible, unverified** (SYNTHESIS section A.2).

5.2 Roughness factor and Δ_{sr}

Definition and role in the model. V1 multiplies the band mobility by a thickness-dependent reduction factor,

$$R(t) = \left(1 - \frac{\Delta_{\text{sr}}}{t}\right)^2, \quad (5.4)$$

where Δ_{sr} is a roughness length. It is the only explicit thickness dependence of transport in V1 that does not come from a fit.

Values used (per thickness). $\Delta_{\text{sr}} = 0.287 \text{ nm}$ (2.87 Å). This gives $R = 0.73359$ (2.0 nm), 0.911 (6.3 nm) and 0.957 (13.2 nm) (EVIDENCE_BRIEF section 3; S2 report).

Source. Both are taken from the target paper: the functional form is the thickness factor of its Eq. 6 [1, p. 7], and $\Delta_{\text{sr}} \approx 2.87 \text{ Å}$ is its extracted effective roughness for IWO [1, p. 8]; the page mapping is listed in Section 2.2. The package classifies the value as literature (EVIDENCE_BRIEF section 3).

Derivation. The package documentation does not give a microscopic derivation of Equation (5.4), and none is claimed here. Two exact rewritings show what the form implies. For $\Delta_{\text{sr}} \ll t$,

$$R(t) \approx 1 - \frac{2\Delta_{\text{sr}}}{t} \iff \frac{1}{\mu_n^{\text{ATLAS}}} \approx \frac{1}{\mu_{\text{band}}} + \frac{1}{\mu_x}, \quad \mu_x = \mu_{\text{band}} \frac{t}{2\Delta_{\text{sr}}}. \quad (5.5)$$

Hence, to first order the factor acts like an extra Matthiessen scattering channel whose mobility grows only linearly with t . The comparison with thickness-fluctuation scattering is instructive. In that mechanism the subband energy fluctuates by $\delta E_1 = 2E_1\Delta/t$ and $\mu_{\text{SR}} \propto t^6$. S2 evaluates $\delta E_1 = 105 / 3.96 / 0.44 \text{ meV}$ for the three films. If such a channel alone produced the step, with $\mu_{\text{SR}}(6.3) = 17 \text{ cm}^2/\text{V s}$ (the extra channel Matthiessen’s rule needs for $\mu(13.2) \approx 62$), then $\mu_{\text{SR}}(2)$ would be $0.017 \text{ cm}^2/\text{V s}$, against about 12–19 $\text{cm}^2/\text{V s}$ extracted. That contradiction is $\times 700$ (S2 report). A roughness factor can therefore only be a weak correction.

Δ_{sr} could be measured by AFM or X-ray reflectivity of the film roughness at each thickness. A Hall mobility at several thicknesses would test whether the reduction exists.

ATLAS implementation. No separate surface-roughness mobility model is used. $R(t)$ is folded into MUN through Equation (5.1) [22, p. 1343].

Sensitivity evidence. No run varies Δ_{sr} alone. None is needed: by Equation (5.8), any change in R is absorbed exactly by the refit of μ_{band} , so the two are perfectly degenerate on one curve. The ratios that R contributes are $\times 1.051$ (13.2/6.3 nm) and $\times 1.305$ (13.2/2 nm), i.e. 3 % of the logarithm needed for the 6.3 $\rightarrow$ 13.2 nm step (S2 report; SYNTHESIS section D).

Status and caveats. Value LITERATURE_IWO; functional form ASSUMED. As the mobility step, the roughness factor is **rejected** ($\times 1.05$ against $\times 4.58$ needed). Whether the 2 nm film is continuous is **not determined** (SYNTHESIS referee responses). Because R is degenerate with μ_{band} , its presence does not change any fitted curve. It only changes how the fitted μ_{band} is read.

5.3 Trap-limited conduction: the free/trapped partition

Definition and role in the model. V1 contains no mobility that depends on density. The apparent density dependence of the field-effect mobility comes from multiple trapping. The acceptor-like exponential tail (Section 4.10, Equation (4.8)) takes part of the gate-induced charge, and only the free part conducts. The partition factor P turns μ_n^{ATLAS} into μ_{FE} .

Values used (per thickness). There is no input of its own. P follows from the tail parameters $N_t(t) = 2 \times 10^{19} (2/t)^{0.75} \text{ cm}^{-3}$ (2.0/0.85/0.49 $\times 10^{19}$) and $W_{TA} = 0.040 \text{ eV}$ (both FITTED; EVIDENCE_BRIEF section 3). The resulting values are $P = 0.847 / 0.791 / 0.826$ for run_0012 / run_0015 / run_0013 (S2 report). They come from a 1-D surrogate that reproduces the ATLAS μ_{FE} within 3 % (11.56 vs 11.27, 8.48 vs 8.42, 49.44 vs 48.95). At $V_G = 3 \text{ V}$ the free fraction is 0.68 / 0.67 / 0.76, with trapped sheet densities $n_{trap} \approx 4.0 / 4.2 / 3.4 \times 10^{12} \text{ cm}^{-2}$.

Source. CALCULATED from the fitted tail and the model electrostatics. The trap-limited partition of the target paper [1, p. 7, Eq. 5] motivated it. The page mapping of that equation is listed in Section 2.2.

Derivation. *Derivation.* Let n_f and n_t be the free and trapped sheet densities, and let C_{eff} be the effective gate capacitance per unit area. Gauss's law above flat band gives $q d(n_f + n_t) = C_{eff} dV_G$. In the linear regime (small V_D) only free electrons drift: $I_D = (W/L) q \mu_n n_f V_D$. Hence

$$g_m = \frac{W}{L} q \mu_n V_D \frac{dn_f}{dV_G} = \frac{W}{L} \mu_n C_{eff} V_D \frac{dn_f}{d(n_f + n_t)}, \quad \mu_{FE} \equiv \frac{L g_m}{W C_{ox} V_D} = \mu_n \frac{C_{eff}}{C_{ox}} P, \quad P = \left(1 + \frac{dn_t}{dn_f} \right)^{-1}. \quad (5.6)$$

For an exponential tail with $W_{TA} > k_B T$ and the Fermi level below E_c , the trapped density is $n_t \simeq t N_{TA} W_{TA} \exp[(E_F - E_c)/W_{TA}]$. The free density is non-degenerate, $n_f \propto \exp[(E_F - E_c)/k_B T]$. Eliminating E_F gives

$$n_t \propto n_f^{T/T_t}, \quad T_t \equiv W_{TA}/k_B = 464 \text{ K} \quad (W_{TA} = 0.040 \text{ eV}), \quad \frac{dn_t}{dn_f} = \frac{T}{T_t} \frac{n_t}{n_f}. \quad (5.7)$$

Where trapping dominates ($n_t \gg n_f$, with $n_t \propto V_G - V_0$), Equation (5.7) gives $n_f \propto (V_G - V_0)^{T_t/T}$. The linear-regime current is then supralinear, $I_D \propto (V_G - V_0)^\alpha$ with $\alpha = T_t/T \approx 1.55$ at 300 K. This is a thin-sheet estimate with no vertical band bending (THEORY). For comparison, the threshold-free H-function exponents are $\alpha_H = 1.35\text{--}1.46$ for run_0012, and $1.59 / 2.14 / 1.29$ for the measured $2 / 6.3 / 13.2$ nm curves (S5 report; Section 5.7). As the tail fills, n_f/n_t grows and $P \rightarrow 1$. This is the origin of the negative mobility-degradation coefficient θ seen in all films.

Linearising the partition shows what the trap model can do. Moving P from 0.791 (6.3 nm) to 0.826 (13.2 nm) contributes only +4 %. Producing the $\times 4.6$ step through P alone would need N_t to change by $\times 6.7 \times 10^3$. This falls short by a factor of 30 in the logarithm (S2 report). The trapped sheet charge is nearly the same in every film ($4.0 / 4.2 / 3.4 \times 10^{12} \text{ cm}^{-2}$), so the partition cannot carry a thickness step.

ATLAS implementation. The tail is a DEFECTS CONTINUOUS acceptor tail with NTA and WTA [22, pp. 1169, 1171, 1173]. Its occupancy is computed in steady state from the local quasi-Fermi level. Trapped electrons contribute to Poisson's equation but not to the current. No parameter for P exists: the partition emerges from Poisson's equation and Fermi–Dirac occupancy. The discretization of the tail integral is treated in Section 5.11.

Sensitivity evidence.

- run_0012 $\rightarrow$ run_0023 ($N_t \times 1.5$, 2 nm): $\Delta V_{th,cc} = +0.082 \text{ V}$, $\Delta SS_{[10^{-10}, 10^{-9}]} = +49.0 \text{ mV/dec}$, $\Delta I_{on} = -25.1 \%$ (MANIFEST). Per unit $\ln N_t$ the Jacobian is $dV_{th,cc} = +0.202 \text{ V}$ and $d(\text{gap}) = +0.495 \text{ V}$ (S2 report).
- run_0037 (B0, S2 re-partition at 6.3 nm): $V_{th,cc}$ 0.654 V, $V_{th,lin}$ 1.856 V and I_{on} -7.6% all fall inside the predicted ranges, but $SS_{cc} = 410.9 \text{ mV/dec}$ against 295.4 measured, so the prediction **failed** (S6 P1).
- run_0052 (C1, near- E_c band, μ 15.4): recovers 28 % of the g_m miss and 33 % of the 0.356 V gap miss at a moderate SS cost ($SS_{cc} + 15.6 \text{ mV/dec}$). The gap opens by +0.119 against +0.28 V predicted, and I_{on} rises +16 % against $\pm 3 \%$ predicted (S6 P3; **partly failed**, SYNTHESIS section D).
- Trade-off: every tested change that opens the 6.3 nm gap costs 3.2–8.5 mV of gap per mV/dec of SS_{cc} (S6 section 3.3). Even at the best efficiency, closing the missing +0.36 V would give $SS_{cc} \approx 332 \text{ mV/dec}$ against 295 measured.

Status and caveats. CALCULATED (model-internal). The underlying N_t anchor and W_{TA} are FITTED. The trap-limited partition as the primary cause of the mobility step is **rejected** (S2). The tails are rigid, in thermal equilibrium and independent of T . Above-threshold trapping and sweep history are not modelled (S6 section 3).

5.4 Drain current is linear in a uniform constant mobility

Definition and role in the model. This section proves, and verifies numerically, the property that makes the V1 mobility calibration and both temperature variants exact. For a uniform, constant mobility, the drain current at every bias is proportional to that mobility:

$$I_D(V_G, V_D; \lambda \mu_n) = \lambda I_D(V_G, V_D; \mu_n) \quad \text{for all } \lambda > 0. \quad (5.8)$$

Values used. The identity has no input of its own. It holds for μ_n^{ATLAS} of Equation (5.1), and therefore also for $R(t)$ and for the temperature factor $(T/300)^{-\text{TMUN}}$.

Source. The analytic argument is given below (THEORY). The numerical verification is from S6 section 1.1 (NUM).

Derivation. *Derivation.* The steady-state electron equations in quasi-Fermi form are

$$\nabla \cdot (\varepsilon \nabla \psi) = -q [p - n - n_t(\psi, \phi_n) + N_D^+ + \dots], \quad (5.9)$$

$$\mathbf{J}_n = -q \mu_n n \nabla \phi_n, \quad \nabla \cdot \mathbf{J}_n = q U, \quad (5.10)$$

with $n = N_c F_{1/2}[(E_{Fn} - E_c)/k_B T]$. Because $D_n = \mu_n k_B T/q$ (Einstein relation), the quasi-Fermi form $\mathbf{J}_n = -q \mu_n n \nabla \phi_n$ contains both drift and diffusion. Three conditions are needed. (a) μ_n is uniform and constant in the only conducting region. (b) The net recombination U is negligible. The IWO film is unipolar ($p \approx 0$), and thermal generation is 13–17 decades below the measured floors (S4). (c) The trap occupancy is a function of the local (ψ, ϕ_n) only, which holds in steady state. Under these conditions, dividing Equation (5.10) by the constant μ_n gives $\nabla \cdot (n \nabla \phi_n) = 0$, which does not contain μ_n . Equation (5.9) and the Ohmic and gate boundary conditions do not contain μ_n either. The solution (ψ, ϕ_n, n, n_t) is therefore independent of μ_n , and $\mathbf{J}_n$, hence I_D , is proportional to it. Subthreshold diffusion current scales in the same way, because $D_n \propto \mu_n$.

Consequences used throughout the report:

1. The fit of μ_{band} takes one rescaling step (Equation (5.2)). The rescaled runs `run_0039` ($\times 1.015074$) and `run_0040` ($\times 1.020113$) are exact representatives of runs at 11.582 and 61.605 cm²/V s.
2. Every ratio metric is independent of μ_n : $V_{\text{th,lin}}$, I_{on}/g_m , the gap $V_{\text{th,lin}} - V_{\text{th,cc}}$ in its $V_{\text{th,lin}}$ part, α_H and θ . Hence, is the local slope $dV_G/d \log I_D$ at fixed V_G . Constant-current quantities are not: $V_{\text{th,cc}}$ moves by $SS_{\text{local}} \log_{10} \lambda$, and fixed-current SS windows slide along the curve. This is why μ_{band} 16.8 $\rightarrow$ 18.13 moves $V_{\text{th,cc}}$ by -9 mV (a factor of 1.079 is 0.033 decade, at a local slope of order 270 mV/dec) and changes $SS_{[10^{-10}, 10^{-9}]}$ by -3.8 mV/dec.
3. μ_{band} , $R(t)$, any constant factor on W/L , and the temperature factor of the constant-mobility model are indistinguishable on one curve.
4. The elevated-temperature runs computed with $\text{TMUN} = 1.5$ can be converted exactly into the intended T -independent-mobility variant (Section 5.8).

ATLAS implementation. The identity holds for the constant low-field model [22, p. 179] as used in V1. That means no concentration-, field- or density-dependent mobility is enabled in the IWO region. The IWO user material carries further silicon defaults in the parameter dump, such as $v_{\text{sat}} = 9.78 \times 10^6$ cm/s and SRH/Auger coefficients. REVIEW judges them inactive or irrelevant here, but they are still to be audited (SYNTHESIS item 0.3).

Sensitivity evidence.

- run_0029 vs run_0038 (2 nm, DOS 384/192, μ_{band} 18.13 vs 17.69): the current ratio is 1.024879 against $18.13/17.69 = 1.024873$. Over 43 points with $I_D > 10^{-14}$ A/ μm it ranges from 1.024817 to 1.025135 (S6 section 1.1).
- run_0008 vs run_0003 (2 nm, 18.13 vs 16.8): the ratio is 1.079171 against 1.079167.
- Path independence: the I_D - V_D runs at $V_D = 0.7$ V reproduce the rescaled transfer curves at $V_G = 1$ –3 V to 0.000 % (S6 section 1.1).

The ratio error is 6×10^{-6} (S6 findings table). The small min–max spread reflects solver tolerance.

Status and caveats. **Demonstrated** (analytic identity plus NUM check). The property holds only because V1’s mobility is uniform and constant. It would fail with a field- or density-dependent mobility, with a series resistance, with a different mobility in another conducting region, or with significant recombination. The constancy of the mobility is itself an assumption of V1, and the most plausible cause of the unexplained 6.3 nm on-state shape (Section 5.1). Equation (5.8) is a property of the model, not of the device.

5.5 Field-effect mobility: linear estimator versus the saturation formula

Definition and role in the model. Two estimators of mobility from a transfer curve are used in this report and in the target paper. They differ by up to a factor of 2.8 on the same data. The linear-regime estimator is

$$\mu_{\text{FE}} = \frac{L g_{\text{m,max}}}{W C_{\text{ox}} V_D}, \quad g_{\text{m}} = \left. \frac{\partial I_D}{\partial V_G} \right|_{V_D}, \quad (5.11)$$

and the saturation formula is

$$\mu_{\text{sat}} = \frac{2L}{WC_{\text{ox}}} \left(\frac{\partial \sqrt{I_D}}{\partial V_G} \right)_{\text{max}}^2. \quad (5.12)$$

The report uses Equation (5.11), identically for measured and simulated curves, as the apparent field-effect mobility. It is neither μ_{band} nor μ_n^{ATLAS} (Equation (5.6)).

Values used (per thickness). Constants in `scripts/extract_metrics.py`: $C_{\text{ox}} = 8.955 \times 10^{-7}$ F cm⁻², $L = 20$ μm , $W = 1$ μm (the currents are normalised in A/ μm), and $V_D = 0.7$ V. The measured values (MANIFEST, `mufe_extraction`; S6 section 1.6) are:

t (nm)	$g_{m,\max}$ (A/V/ μm)	μ_{FE} , Equation (5.11)	μ_{sat} , Equation (5.12)	V_G at μ_{sat} peak (V)	peak $-V_{\text{th,lin}}$ (V)	paper
2.0	3.807×10^{-7}	12.15	5.09	1.75	+0.09	5.1
6.3	3.431×10^{-7}	10.95	3.86	2.55	+0.73	not quoted
13.2	1.572×10^{-6}	50.17	27.40	0.95	-0.09	27.4

Mobilities are in $\text{cm}^2/\text{V s}$. The final simulated values of Equation (5.11) are 11.10 (run_0038), 8.39 (run_0039 rescaled) and 48.85 (run_0040 rescaled). These are -8.6, -23 and -2.6 % from the measured values (S6 section 1.6).

Source. Measured curves data/experimental_clean.csv (Section 3.3) at $V_D = 0.7 \text{ V}$. The V_D value and the $A/\mu\text{m}$ normalisation come from the device schematic (Section 3.1). The saturation-formula reproduction of the paper's numbers is from S2 report and S5 report, key finding 4, with the regime reading corrected by SYNTHESIS (section C, D12).

Derivation. *Derivation.* With a constant mobility μ and threshold V_t , the gradual-channel current in the linear regime ($V_D < V_{\text{ov}} \equiv V_G - V_t$) is $I_D = \mu C_{\text{ox}} \frac{W}{L} \left(V_{\text{ov}} V_D - \frac{1}{2} V_D^2 \right)$. Hence $g_m = \mu C_{\text{ox}} (W/L) V_D$, which gives Equation (5.11). In saturation ($V_D \geq V_{\text{ov}}$), $I_D = \mu C_{\text{ox}} \frac{W}{2L} V_{\text{ov}}^2$, so $\sqrt{I_D}$ is linear in V_G with slope $\sqrt{\mu C_{\text{ox}} W/2L}$, which gives Equation (5.12). If Equation (5.12) is applied to the linear-regime expression, then with $k = \mu C_{\text{ox}} W/L$

$$\left(\frac{\partial \sqrt{I_D}}{\partial V_G} \right)^2 = \frac{k^2 V_D^2}{4 I_D} = \frac{k V_D}{4 V_{\text{ov}} - 2 V_D} \Rightarrow \frac{\mu_{\text{sat}}}{\mu} = \frac{V_D}{2 V_{\text{ov}} - V_D} \quad (V_D < V_{\text{ov}}). \quad (5.13)$$

Hence, in the square-law model the saturation formula underestimates the mobility wherever the device is in its linear regime. At the saturation boundary $V_{\text{ov}} = V_D$ the two estimators coincide.

The measured ratios $\mu_{\text{FE}}/\mu_{\text{sat}}$ are 2.39 (2 nm), 2.84 (6.3 nm) and 1.83 (13.2 nm). Equation (5.13) cannot be applied quantitatively, for three reasons. The curves are supralinear ($\alpha_H > 1$, Section 5.3). The peaks of $\partial \sqrt{I_D}/\partial V_G$ lie at +0.09, +0.73 and -0.09 V relative to $V_{\text{th,lin}}$. The earlier readings of the peak positions as “linear regime” (S2) or “ $V_{\text{ov}} \approx 1.1 \text{ V}$ ” (S5) were withdrawn (SYNTHESIS conflict table and referee responses). The conclusion that can be supported is narrower. Applied to the workbook curves, the saturation formula reproduces the paper's 5.1 and 27.4 $\text{cm}^2/\text{V s}$ [1, p. 7] to within rounding (5.09 and 27.40). The paper's mobilities are therefore saturation-formula peaks of these same curves, not linear-regime μ_{FE} (an inference of this analysis; the paper calls the quantity field-effect mobility in the Fig. 1b caption on p. 3 but does not state its extraction formula). This is provenance evidence. It fixes the product $L/(WC_{\text{ox}})$ together with the $A/\mu\text{m}$ normalisation, but not V_D , because Equation (5.12) does not contain V_D (SYNTHESIS D12). The same method predicts $\mu_{\text{sat}} = 3.86 \text{ cm}^2/\text{V s}$ at 6.3 nm. Checking this against the paper's figure requires figure values that are not available locally (S6 P10, not yet tested).

Three further properties of the measured μ_{FE} matter for its use as a calibration target:

- At 2 nm, g_m is largest at the last point of the sweep ($V_G = 3.00 \text{ V}$), so 12.15 $\text{cm}^2/\text{V s}$ is a lower bound. The roll-off $g_m(3 \text{ V})/g_{m,\max}$ is 1.000 / 0.981 / 0.978, which limits series-resistance or mobility-degradation effects to at most 2.2 % (S2 report).

- The classical centroid and degenerate DOS alone give $C_{\text{eff}}/C_{\text{ox}} \approx 0.87$. Hence, even with no trapping, Equation (5.11) with C_{ox} understates the mobility by about 13 % (S3). ATLAS includes this, so the fitted μ_{band} already absorbs it.
- The apparent 49.2 cm²/V s of the 31.8 nm film is not a mobility. Its g_m has maxima at -0.30 , 0.85 , 1.15 and 1.70 V, and $g_m(3 \text{ V})/g_{m,\text{max}} = 0.067$ (SYNTHESIS conflict table). It is reported only as a documented model failure.

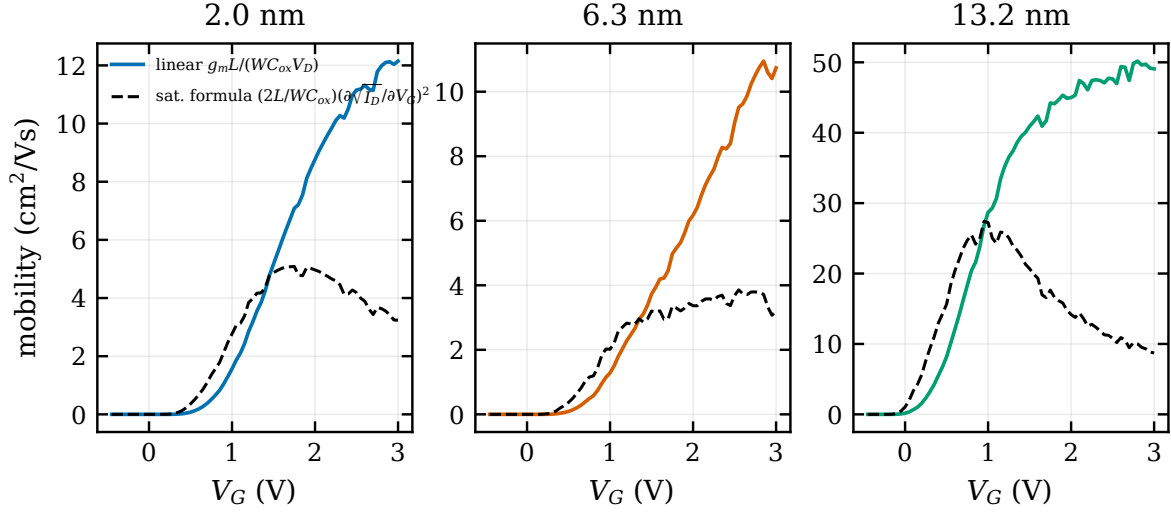


Figure 5.1. Mobility extraction from the measured transfer curves at $V_D = 0.7$ V for 2.0, 6.3 and 13.2 nm. The linear-regime estimator $\mu_{\text{FE}}(V_G) = Lg_m/(WC_{\text{ox}}V_D)$ (Equation (5.11)) is compared with the saturation formula $(2L/WC_{\text{ox}})(\partial\sqrt{I_D}/\partial V_G)^2$ (Equation (5.12)). Derivatives use `numpy.gradient` on the measured 0.05 V grid. The peaks are 12.15 / 10.95 / 50.17 cm²/V s (linear) against 5.09 / 3.86 / 27.40 cm²/V s (saturation formula). The target paper quotes 5.1 and 27.4 cm²/V s for 2 and 13.2 nm. Note that the 2 nm linear estimator is still rising at the end of the sweep.

ATLAS implementation. None. Both estimators are post-processing. Equation (5.11) is implemented in `scripts/extract_metrics.py` (`metrics: gm = np.gradient(idv, vg); mu_FE_cm2Vs = gm_max*L_CM/(W_CM*COX*VD)`). The code is listed in Chapter B. Simulated curves are first interpolated onto the 0.05 V measurement grid (S6 section 1).

Sensitivity evidence. The numerical uncertainty of $g_{m,\text{max}}$ and μ_{FE} is $\pm 0.7\%$ (S6 section 1.5; Table 5.7). The disagreement between the two estimators is a factor of 1.8–2.8. That is a definitional difference, not an uncertainty.

Status and caveats. Measured μ_{FE} : DATA, one device per thickness. The equality of the paper’s mobilities with the saturation formula on the workbook curves is **demonstrated** (independent retrodiction; S6 findings row 21). Caveat: the thickness dependence of any “mobility” depends on the estimator. Between 2 and 13.2 nm, μ_{FE} changes by $\times 4.13$ but μ_{sat} by $\times 5.38$. Hence, an exponent quoted for a mobility law without its estimator is meaningless (S5 report).

5.6 Source/drain contacts, overlap and series resistance

Definition and role in the model. The Pd source and drain (70 nm) sit on top of the IWO film and overlap the gate by 2 μm on each side. $L = 20 \mu\text{m}$ and $W = 290 \mu\text{m}$ (Section 3.2, Figure 3.1). In V1 the contacts only inject carriers. They are ideal Ohmic boundaries with no barrier, no specific contact resistivity ρ_c and no lumped series resistance. Hence, any access or contact resistance present in the device is absorbed into the fitted μ_{band} .

Values used (per thickness). The same for all films: ideal Ohmic S/D, $R_{\text{sd}} = 0$, $\rho_c = 0$, overlap 2 μm . The measured on-resistances $R_{\text{on}}W = V_{\text{D}}/I_{\text{D}}(3 \text{ V})$ set the scale for any contact contribution: 1.38×10^6 (2.0 nm), 1.74×10^6 (6.3 nm), 2.28×10^5 (13.2 nm) and $1.95 \times 10^5 \Omega \mu\text{m}$ (31.8 nm) (S5 report).

Source. Geometry from the device schematic (MEASURED_TARGET_DEVICE as drawn). The Ohmic boundary and $R_{\text{sd}} = 0$ are ASSUMED. The literature bound on metal/ In_2O_3 contact resistance from TLM, $R_c < 0.1 \Omega \text{ mm}$, is taken from Si et al. [9, p. 167] as quoted in S5 (page mapping in Section 2.5).

Derivation. For a top contact of length L_{ov} on a film of sheet resistance R_{sh} , the transmission-line model gives

$$R_c W = \sqrt{\rho_c R_{\text{sh}}} \coth\left(\frac{L_{\text{ov}}}{L_T}\right), \quad L_T = \sqrt{\rho_c / R_{\text{sh}}}, \quad (5.14)$$

which reduces to $R_c W \approx \sqrt{\rho_c R_{\text{sh}}}$ for $L_{\text{ov}} \gg L_T$. S5 evaluates Equation (5.14) with the accumulated sheet resistance at the source at $V_{\text{G}} = 3 \text{ V}$: $R_{\text{sh}} = 8.2 \times 10^4 / 1.0 \times 10^5 / 1.26 \times 10^4 \Omega/\square$ for 2 / 6.3 / 13.2 nm (Table 5.3). Two consequences follow. (a) For $\rho_c \leq 10^{-4} \Omega \text{ cm}^2$, $L_T \leq 0.35 \mu\text{m} \ll L_{\text{ov}}$. Hence,

Table 5.3. Transfer length and contact share of the on-resistance for assumed specific contact resistivities (S5 report, Equation (5.14)).

$\rho_c (\Omega \text{ cm}^2)$	L_T at 2 / 6.3 / 13.2 nm (μm)	$2R_c/R_{\text{on}}$ at 2 / 6.3 / 13.2 nm
10^{-6}	0.035 / 0.031 / 0.089	0.4 / 0.4 / 1.0 %
10^{-5}	0.11 / 0.10 / 0.28	1.3 / 1.2 / 3.1 %
10^{-4}	0.35 / 0.31 / 0.89	4.2 / 3.7 / 10 %
10^{-3}	1.1 / 0.98 / 2.8	14 / 12 / 51 %

R_c does not depend on the overlap, and current crowding matters only above about $10^{-3} \Omega \text{ cm}^2$. (b) A contact penalty scales as $R_{\text{sh}}^{-1/2}$ relative to R_{on} , so it hurts the thick, conductive films more. That is the opposite of what is needed to explain the low currents of the thin films. Explaining the I_{on} gap between 2 and 13.2 nm by contacts alone would need $R_{\text{sd}}W \approx 1.15 \times 10^6 \Omega \mu\text{m}$ at 2 nm. That means $\rho_c \approx 0.26 \Omega \text{ cm}^2$ and $L_T \approx 45 \mu\text{m}$ (longer than L), and 0.58 V of the 0.7 V drain bias dropped across the contacts. The implied R_c is about 5700 times the TLM bound quoted above (S5 report). Vertical access through the film is also negligible: the series resistivities are 3.3×10^{-9} (2 nm, accumulated), 2.2×10^{-8} (13.2 nm, accumulated) and $4.5 \times 10^{-7} \Omega \text{ cm}^2$ (13.2 nm, neutral top layer with $N_{\text{d,eff}} = 2.97 \times 10^{17} \text{ cm}^{-3}$). All are below 1 % of R_{on} and grow with t (S5 report).

The contact parameters could be measured by TLM or an L -series, by gated four-probe measurements, and by low- V_{D} output characteristics. Superlinear output below $V_{\text{D}} \approx 0.1 \text{ V}$ at 2 nm would

indicate a barrier. The registered ideal-contact I_D – V_D predictions are the reference (Section 10.1).

ATLAS implementation. An electrode without a CONTACT statement is treated as charge-neutral (Ohmic) [22, p. 1145]. V1 attaches no CON.RESIST (distributed contact resistance, $\Omega \text{ cm}^2$) and no RESISTANCE (lumped resistor) to the source and drain. Both are available [22, p. 1153], with an example of a lumped and a distributed contact resistor on p. 1155 of the manual. With ideal Ohmic boundaries the model cannot represent ρ_c at all (S5; SYNTHESIS section F).

Sensitivity evidence.

- **run_0027** is malformed and has been retracted. Its mesh collapsed the drain, so its -2.15% I_{on} change is not an overlap effect, and contact geometry and resistance are untested in V1 (full statement in Section 7.4).
- The bounds from the calibrated model are $R_{\text{sd}}W \lesssim 5 \times 10^4 \Omega \mu\text{m}$ at 2 nm ($\leq 4\%$ of R_{on}) and $\lesssim 1\text{--}2 \times 10^4 \Omega \mu\text{m}$ at 13.2 nm (4–9%). They come from the Y- and H-function analysis of Section 5.7 and hold only within the model (S5 report).
- The 31.8 nm series resistance of V0 ($9.2 \times 10^4 \Omega \mu\text{m}$ total) cannot be carried over to 13.2 nm. There it would give $\theta \approx +0.17 \text{ V}^{-1}$, -27% I_{on} and α_H lower by 0.25. None of these is observed: $\theta_{\text{net}} = -0.046 \text{ V}^{-1}$, and **run_0013** reproduces α_H with $R_{\text{sd}} = 0$ (S5 report).

Status and caveats. ASSUMED (ideal Ohmic, $R_{\text{sd}} = 0$). The mechanism verdicts of the S5 report are as follows. A thickness-independent R_{sd} as the cause of the $I_{\text{on}}/\mu_{\text{FE}}$ trend is **rejected**. A confinement-raised Pd/IWO barrier at 2 nm is **plausible, unverified**. It would need $\Delta\phi \geq k_{\text{B}}T \ln 2600 = 0.20 \text{ eV}$, and it is rejected as the cause of the 2/6.3 vs 13.2 nm step because $\Delta E_{\text{c}}(6.3) = 0.07 \text{ eV}$ cannot explain $I_{\text{on}}(6.3) \leq I_{\text{on}}(2)$. Vertical access and current crowding are **rejected** as thickness mechanisms. R_{sd} as a minor contributor is **not determined** and bounded only within the model. Whether the gate extends under the S/D in the real device is **not determined**.

5.7 Y-function and H-function analysis

Definition and role in the model. The Y-function method separates an intrinsic low-field mobility μ_0 from the mobility attenuation θ , which contains the series-resistance term βR_{sd} . The threshold-free H-function measures the power-law exponent of the linear-regime current. In this report both serve as tests. They show whether R_{sd} can be extracted from a single transfer curve, and whether ATLAS with ideal contacts reproduces the measured curvature.

Values used (per thickness). Fit windows $V_{\text{G}} \geq V_{\text{th,lin}} + V_{\text{D}}$ (2.40–3.00, 2.55–3.00 and 1.75–3.00 V for 2 / 6.3 / 13.2 nm; 13, 10 and 26 points). Results in Table 5.4.

Source. S5 report (scripts `scratch/S5/s5_yfunction.py`, `s5_checks.py`), applied to the measured curves and to the V1 runs **run_0012**, **run_0015**, **run_0013**.

Derivation. *Derivation.* Assume a linear-regime current with first-order mobility attenuation, $I_D = \mu_0 C_{\text{ox}} \frac{W}{L} V_D \frac{V_{\text{ov}}}{1 + \theta V_{\text{ov}}}$ with $\theta = \theta_0 + \beta R_{\text{sd}}$ and $\beta = \mu_0 C_{\text{ox}} W/L$. Then $g_m = \mu_0 C_{\text{ox}} \frac{W}{L} V_D / (1 + \theta V_{\text{ov}})^2$ and

$$Y \equiv \frac{I_D}{\sqrt{g_m}} = \sqrt{\mu_0 C_{\text{ox}} \frac{W}{L}} V_D V_{\text{ov}}, \quad (5.15)$$

which is independent of θ . Hence, μ_0 follows from the slope of $Y(V_G)$, and θ from a direct fit of the attenuated expression. R_{sd} follows from $\theta - \theta_0$ if θ_0 is known. For a power-law current $I_D \propto (V_G - V_0)^\alpha$ the H-function is

$$H \equiv \frac{\int_{V_0}^{V_G} I_D dV}{I_D} = \frac{V_G - V_0}{\alpha + 1}, \quad (5.16)$$

whose slope $1/(\alpha + 1)$ gives α without any threshold. An ideal square-law device has $\alpha_H \approx 1$ at small V_D . Series resistance lowers α_H and makes $\theta > 0$. Tail filling (Equation (5.7)) raises α_H and makes $\theta_{\text{net}} < 0$.

Table 5.4. Y- and H-function results (S5 report). The GCA baseline is the α_H expected from the gradual-channel model at $V_D = 0.7$ V without traps. Ranges span windows shifted by ± 0.2 V and two g_m methods (central difference and Savitzky–Golay, which agree within 2 %).

curve	$\mu_{0,Y}$ (cm ² /V s)	θ_{net} (V ⁻¹)	α_H	$\mu_{0,Y}/\mu_{\text{band}}$
measured 2.0 nm	8.8 (8.0–8.9)	-0.100 ± 0.007	1.59 (1.55–1.71) vs GCA 1.1–1.2	–
measured 6.3 nm	7.9 (5.7–8.0)	–0.12 to –0.20	2.14 (2.14–2.23) vs GCA 1.1–1.3	–
measured 13.2 nm	41.2 (39.2–41.9)	-0.046 ± 0.002	1.29 (1.27–1.35) vs GCA 1.05–1.13	–
run_0012 (2 nm)	9.2–9.9	–0.045 to –0.070	1.35–1.46	0.53
run_0015 (6.3 nm)	6.6–7.1	–0.054 to –0.077	1.34–1.42	0.58
run_0013 (13.2 nm)	41.5–43.0	–0.032 to –0.047	1.23–1.28	0.68

All three measured films are net supralinear, with $\theta_{\text{net}} < 0$. The ideal-contact, constant- μ_{band} ATLAS runs also give negative θ . The measured-minus-ATLAS residuals are -0.044 , -0.053 and -0.010 V⁻¹, every one with the sign opposite to what R_{sd} would produce. When the tail exponent and θ are fitted together, they are perfectly correlated ($r = +1.00$). Hence R_{sd} is *not identifiable* from one transfer curve at one channel length. The Y-function mobility is not an intrinsic mobility here: $\mu_{0,Y}/\mu_{\text{band}} = 0.53$ – 0.68 on the model’s own curves. The method’s premise, a density-independent μ_0 , fails in trap-limited films. The free fraction is still rising at +3 V at 2 and 6.3 nm (S5 report).

ATLAS implementation. Post-processing only. The ATLAS runs enter as ideal-contact references (run_0012, run_0015, run_0013). A forward check (gradual-channel model plus R_{sd} , s5_checks.py) shows what a real R_{sd} would do. At $R_{\text{sd}}W = 9.2 \times 10^4 \Omega \mu\text{m}$, a 13.2-nm-like device loses 27 % of I_{on} , its α_H falls by 0.25 and the IR drop is 0.19 V. A 2-nm-like device loses 5 % and its α_H falls by 0.06.

Sensitivity evidence. The measured 13.2 nm α_H is the model value plus 0.04 ± 0.04 , which gives $R_{\text{sd}}W \lesssim 1$ – $2 \times 10^4 \Omega \mu\text{m}$. The 2 nm scatter across windows ($\Delta\theta \approx 0.02$ V⁻¹) gives $\Delta\theta/\beta \approx 5 \times 10^4 \Omega \mu\text{m}$ (S5 report). At 31.8 nm g_m has two maxima and the Y curvature is ≥ 1.6 , so the positive θ there

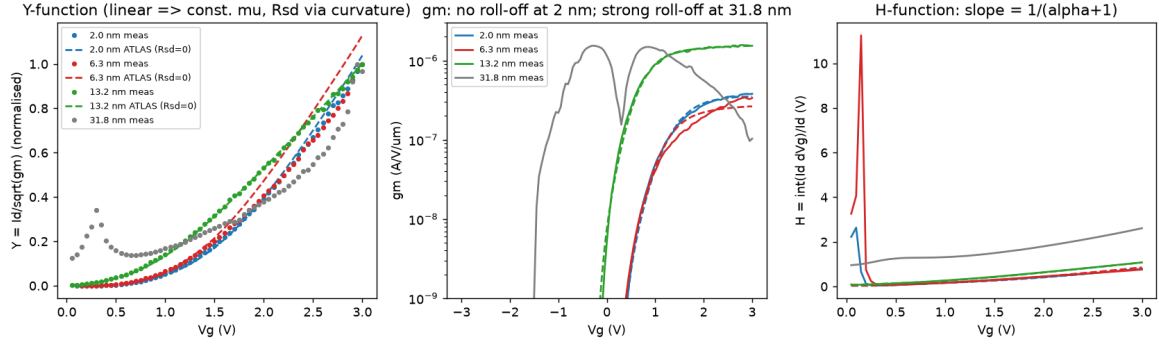


Figure 5.2. Y-function and H-function analysis of the measured transfer curves at $V_D = 0.7$ V (existing analysis figure, analysis_2026-09-25/scratch/S5/s5_yfunction_hfunction.png). $Y = I_D/\sqrt{g_m}$ gives μ_0 from its slope, and the H-function slope $1/(\alpha + 1)$ gives the exponent without a threshold. All films show $\theta_{\text{net}} < 0$ and $\alpha_H > 1$ (1.59 / 2.14 / 1.29), the tail-filling signature rather than the series-resistance signature. The 31.8 nm film has two g_m maxima, so its Y-function is invalid.

(+0.17 to +0.28 V^{-1}) cannot be assigned to contacts.

Status and caveats. Y-function R_{sd} and intrinsic mobility for these films: **not determined** (method invalid; α - θ correlation +1.00). The R_{sd} bounds are calibration-class. They assume the V1 trap model and are not independent measurements. Near threshold the device is in saturation at $V_D = 0.7$ V (below $V_{\text{th,lin}} + V_D = 2.36 / 2.52 / 1.74$ V), which leaves only 9–17 usable points at 2 and 6.3 nm. Transfer curves at $V_D = 0.05$ –0.1 V and a TLM series would make the method applicable.

5.8 Temperature: T , $\text{tmu} = 1.5$, $N_c(T)$ and the band gap

Definition and role in the model. The lattice temperature enters V1 through the Fermi–Dirac statistics and the effective densities of states $N_c(T)$ and $N_v(T)$ (Sections 4.8 and 4.9). It also enters through the occupation of the tails relative to the fixed tail width, and through the temperature factor of the constant mobility. All calibration is done at 300 K. The elevated-temperature runs are predictions (Section 10.2).

Values used.

- $T = 300$ K for all calibration and sensitivity runs.
- $T = 338.15$ K (2 nm) and 358.15 K (2, 6.3 and 13.2 nm) in run_0044–run_0047 (campaign B). run_0045 is the 2 nm run at 358.15 K.
- Mobility temperature exponent: *the ATLAS default $\text{TMUN} = 1.5$ was applied*, although the plan declared a T -independent band mobility (PREDICTIONS_REGISTER; SYNTHESIS D10).
- $N_c \propto T^{3/2}$, a T -independent band gap, a T -independent tail ($W_{\text{TA}} = 40$ meV, $T_t = 464$ K) and a T -independent μ_{band} are the declared assumptions of campaign B (S2 report).

Source. $T = 300$ K is ASSUMED as the measurement temperature. The TMUN value is the silicon default of the manual [22, p. 179, Table 3-43]; the default is also listed on p. 1413 of the manual, and the parameter is defined on p. 1355. It is printed in the run_0045 run log: deckbuild.out

l. 391–393 reads “Temperature = 358 Kelvin, mu = 12.977, tmu = 1.5” (PREDICTIONS_REGISTER; REVIEW V7).

Derivation. The constant low-field model [22, p. 179, Eq. 3-215] is $\mu_{n0}(T) = \text{MUN} (T/300)^{-\text{TMUN}}$. By Equation (5.8) the simulated current at temperature T is exactly

$$I_D^{\text{P-phonon}}(T) = \left(\frac{T}{300} \right)^{-1.5} I_D^{\text{P-MTR}}(T), \quad (5.17)$$

where “P-MTR” denotes the intended variant with T -independent band mobility and “P-phonon” the variant as simulated. The phonon-like variant has $\mu \propto T^{-3/2}$. Both variants are therefore exact, and both are registered. The factors are:

T (K)	$(T/300)^{-1.5}$	P-MTR multiplier $(T/300)^{1.5}$
338.15	0.83564	1.19669
358.15	0.76663	1.30441

The multipliers are the exact values recomputed by REVIEW (V7). The register prose quoted $\times 1.19665$ and $\times 1.30426$, which is 0.012 % low; the code uses the exact expression. As a check, $I_{\text{on}}(\text{run_0045})/I_{\text{on}}(\text{run_0038}) = 0.7742$, and $0.7742 \times 1.30441 = 1.0098$. Hence, the P-MTR on-current rises by +0.98 % at 2 nm (6.3 nm: +2.23 %; 13.2 nm: +1.52 %). The “mu = 12.977” in the log is the 300 K parameter (17.6897×0.73359), not the effective mobility at 358 K. Hence, the log line alone is not proof that the factor was applied. The proof is the exact current ratio.

$N_c(T) = N_c(300) (T/300)^{3/2}$ follows from the parabolic effective-mass density of states (Section 4.8). The band gap is assumed T -independent. Whether the deck sets the ATLAS band-gap temperature coefficients (EGALPHA, described with the MATERIAL parameters on p. 1352 of the manual) to zero explicitly, or inherits defaults, is part of the default audit (SYNTHESIS item 0.3). It is **not determined** from the material read for this chapter. Because the IWO film is unipolar, a small $E_g(T)$ shift would affect only the valence side and the deep states.

ATLAS implementation. The temperature is set by the TEMPERATURE parameter of the MODELS statement [22, p. 1478]. The mobility exponent is TMUN on the MATERIAL (or MOBILITY) statement [22, p. 1355]. V1 did not set TMUN, so the default 1.5 applied. The same parameter dump carries further silicon defaults ($v_{\text{sat}} = 9.78 \times 10^6$ cm/s, SRH and Auger coefficients, lattice constant 5.43 Å). REVIEW judges them inactive or irrelevant here.

Sensitivity evidence. Table 5.5 summarises the registered changes from 300 K to 358.15 K (S6 report; Figure 10.2). At 338.15 K (2 nm), $\Delta V_{\text{th,cc}}$ is -49.3 mV (P-MTR) and -27.9 mV (P-phonon). The S2 1-D surrogate had predicted $\Delta V_{\text{th,cc}}$ of $-64/-66$ mV, $\mu_{\text{FE}} \times 0.993/\times 0.999$ and $I_{\text{on}} \times 1.004/\times 1.011$ at 350 K for 2/13.2 nm. ATLAS P-MTR reproduces these within 1 mV and 0.4 % (S6 P7). That is a numerical cross-check between two implementations of the same assumptions, not a physical test. The factor is common to all films, so the thickness ratios do not depend on the variant. $\mu_{\text{FE}}(13.2)/\mu_{\text{FE}}(2)$ goes from 4.403 to 4.440 and $I_{\text{on}}(13.2)/I_{\text{on}}(2)$ from 6.033 to 6.065 (PREDICTIONS_REGISTER).

Table 5.5. Registered 358.15 K predictions relative to 300 K, both exact variants (S6 report; PREDICTIONS_REGISTER). Runs run_0044–run_0047; 300 K references run_0038, run_0039 ($\times 1.015074$), run_0040 ($\times 1.020113$).

quantity	2 nm	6.3 nm	13.2 nm
$\Delta V_{\text{th,cc}}$, P-MTR (mV)	−73.1	−78.9	−76.3
$\Delta V_{\text{th,cc}}$, P-phonon (mV)	−40.5	−46.7	−56.6
$\Delta V_{\text{th,lin}}$, both (mV)	−26.5	−34.5	−30.1
$\Delta SS_{[10^{-10}, 10^{-9}]}$, P-MTR / P-phonon (mV/dec)	−12.1 / +2.0	−12.5 / +2.4	−10.6 / −3.3
μ_{FE} ratio, P-MTR / P-phonon	0.992 / 0.760	1.000 / 0.766	1.000 / 0.767
ΔI_{on} , P-MTR / P-phonon	+1.0 / −22.6 %	+2.2 / −21.6 %	+1.5 / −22.2 %

Status and caveats. $T = 300$ K: ASSUMED. $\text{TMUN} = 1.5$ in the elevated- T runs: accepted as a **configuration-control failure** (SYNTHESIS referee response 2.7), with no loss of information because of Equation (5.17). The planned fix is to re-run the 2 nm, 358.15 K case with $\text{TMUN} = 0$. The P-MTR variant is expected to agree within ± 0.2 % in current (SYNTHESIS item 2.1). DOS convergence (Section 5.11) was verified only at 300 K. The tails are assumed T -independent. A rise of $SS_{[10^{-10}, 10^{-9}]}$ by more than 20 mV/dec at 358 K would falsify that assumption (PREDICTIONS_REGISTER).

5.9 Activation energy in the multiple-trapping model

Definition and role in the model. The apparent activation energy at fixed bias is

$$E_a(V_G) \equiv -k_B \left. \frac{\partial \ln I_D}{\partial (1/T)} \right|_{V_G, V_D} \approx k_B \frac{\ln[I_D(T_2)/I_D(T_1)]}{1/T_1 - 1/T_2}. \quad (5.18)$$

It is the observable that best separates the transport pictures that three room-temperature curves cannot tell apart: MTR with a T -independent band mobility, phonon-limited band transport, and barrier- or percolation-limited transport. V1 has no activation parameter. E_a emerges from the statistics.

Values used (per thickness). Registered predictions for the two exact variants, from run_0044–run_0047 and the 300 K references (S6 report; MANIFEST activation_energy). Least-squares Arrhenius fits use 300 / 338.15 / 358.15 K at 2 nm and 300 / 358.15 K for the other films. At $V_G = 1$ V, P-MTR gives $E_a = 57.3 / 53.4 / 20.8$ meV and P-phonon 15.1 / 11.1 / −21.5 meV for 2 / 6.3 / 13.2 nm. The full P-MTR table (meV) is:

film	$V_G = 0.5$	1	1.5	2	3 V
2 nm	124	57	23	8	1.5
6.3 nm	128	53	20	9	3.5
13.2 nm	63	21	8	5	2.4

P-phonon is lower by exactly 42.3 meV at every V_G (S6 report; PREDICTIONS_REGISTER).

Source. CALCULATED from the V1 model at elevated T (Section 5.8). The 1-D surrogate expectation is from S2 report.

Derivation. *Derivation.* In the non-degenerate limit the free sheet density is $n_f \propto N_c(T) \exp[-(E_c - E_F)/k_B T]$, and $I_D \propto \mu_n(T) n_f$. With $N_c \propto T^{3/2}$ and $\mu_n \propto T^{-\text{TMUN}}$, Equation (5.18) gives

$$E_a = (E_c - E_F)_{\text{eff}} + \frac{3}{2} k_B T - \text{TMUN} k_B T, \quad (5.19)$$

because a prefactor T^a contributes $a k_B T$ to $-k_B \partial \ln I_D / \partial (1/T)$. The first term is the Arrhenius slope of $\exp[-(E_c - E_F)/k_B T]$ at fixed V_G . It includes the temperature shift of E_F caused by the redistribution between free and trapped electrons (Equation (5.7)). In subthreshold it follows $E_c - E_F$ at the surface. Once E_F enters the band it falls to about zero (S2 report). The two registered variants differ only in the last term. For the finite-difference form of Equation (5.18) the difference is $1.5 k_B T_{\text{eff}}$ with $T_{\text{eff}} = \ln(T_2/T_1)/(1/T_1 - 1/T_2) = 327.4 \text{ K}$ for 300–358.15 K. That equals 42.3 meV, exactly the offset found between the variants.

For comparison, the S2 surrogate gives, at 350 K, $E_a = 504 / 120 / 56 / 23 / 7 / 1 \text{ meV}$ (2 nm) and $156 / 61 / 20 / 8 / 4 / 2 \text{ meV}$ (13.2 nm) at $V_G = 0 / 0.5 / 1 / 1.5 / 2 / 3 \text{ V}$. At equal V_G the 2 nm film is more activated. At equal overdrive the two films are similar, about 20 meV at $V_{\text{th,lin}}$. Two alternative transport pictures give different signatures (S2 report): (a) *Percolation* over a Gaussian barrier distribution, $\mu \propto \exp[-(\phi_m - \sigma^2/2k_B T)/k_B T]$. For $\phi_m = 20\text{--}60 \text{ meV}$ and $\sigma = 0\text{--}20 \text{ meV}$ this gives $\mu(350)/\mu(300) = 1.03\text{--}1.39$ (E_a 6–60 meV). The activation persists above threshold, and the Arrhenius plot is curved. (b) *Grain-boundary barriers* as the source of the $\times 4.6$ step. A barrier difference of $k_B T \ln 4.6 = 39 \text{ meV}$ suffices. The thin films would then gain $\times 1.24$ more than 13.2 nm between 300 and 350 K, and $\mu_{\text{FE}}(13.2)/\mu_{\text{FE}}(2)$ would fall from 4.13 to about 3.3.

ATLAS implementation. None beyond Section 5.8: E_a is evaluated in post-processing from runs at different MODELS TEMPERATURE [22, p. 1478]. The numerical uncertainty is below 0.5 meV (S6 report).

Sensitivity evidence. The ATLAS P-MTR variant reproduces the S2 surrogate within 1–7 meV in E_a and within 1 mV in $\Delta V_{\text{th,cc}}$ (S6 P7). The registered discriminators (S6 report; S5 report; Section 10.3) are the following. A change $\Delta I_{\text{on}} > \sim +5\%$ at 358 K would point to an activated band mobility or to percolation, whereas $\Delta I_{\text{on}} \approx -22\%$ would point to phonon-limited band transport. A thin-film $E_a(3 \text{ V}) > E_a(13.2) + 10 \text{ meV}$, or a μ_{FE} ratio shrinking toward 3.3, would indicate that there is no common transport mechanism. An activation energy of $\mu_{\text{FE}}(3 \text{ V})$ below about 5 meV at 2 nm would falsify trap-limited conduction as the driver of $\mu_{\text{FE}}(t)$.

Status and caveats. PRED, registered and untested. The numerical cross-check (S6 P7) **passed**. The physical MTR signature is **not determined**. Caveats: the TMUN variant choice (Section 5.8); T -independent tails; no measured data at $T \neq 300 \text{ K}$. Figures: Figures 10.2 and 10.3.

5.10 Extraction of the device metrics

Definition and role in the model. Every comparison between the model and the data in this report goes through a fixed set of metrics. They are extracted with identical code from the measured and the simulated curves. The target paper does not define its extraction, so the definitions below are

stated conventions (`scripts/extract_metrics.py`, `docstring`). Section 3.4 and Table 3.2 list the measured values.

Values used (definitions).

Constant-current threshold.

$V_{\text{th,cc}}$ is the gate voltage at which $I_D = I_{\text{cc}} = 10^{-9} \text{ A}/\mu\text{m}$, interpolated log-linearly between adjacent grid points (V_i, I_i) , (V_{i+1}, I_{i+1}) that bracket I_{cc} , with no extrapolation:

$$V_{\text{th,cc}} = V_i + (V_{i+1} - V_i) \frac{\log_{10} I_{\text{cc}} - \log_{10} I_i}{\log_{10} I_{i+1} - \log_{10} I_i}. \quad (5.20)$$

The same routine gives $V(I)$ at any current level.

Linear-extrapolation threshold.

$V_{\text{th,lin}} = V_g^* - I_D(V_g^*)/g_{\text{m,max}}$ at the point V_g^* of maximum g_{m} . g_{m} is `numpy.gradient` on the 0.05 V grid (central differences in the interior).

Fixed-current SS.

$SS_{[I_1, I_2]} = [V(I_2) - V(I_1)]/\log_{10}(I_2/I_1)$, with windows 10^{-11} – 10^{-10} and 10^{-10} – $10^{-9} \text{ A}/\mu\text{m}$ (S6 section 1).

SS_{cc} .

$[V(10^{-8}) - V(10^{-10})]/2$ decades.

SS_{min} (retired).

The minimum of $\Delta V_G/\Delta \log_{10} I_D$ over 5-point (0.2 V) windows. Each window's first point must exceed 5 times the measured off-band median of the same thickness, and its current must be below $0.01 I_{\text{on}}$.

On-state shape.

$\text{gap} = V_{\text{th,lin}} - V_{\text{th,cc}}$; $\text{gap7} = V(10^{-7}) - V_{\text{th,cc}}$; $I_{\text{on}}/g_{\text{m}}$.

μ_{FE} .

Equation (5.11).

Currents.

$I_{\text{on}} = I_D(V_G = 3 \text{ V})$; I_{off} = minimum I_D over the sweep. For simulations it is the minimum positive I_D , with the count of native zeros reported and never replaced by a floor. The off-floor is the median of I_D over $V_G \in [-2, -0.5] \text{ V}$.

Simulated curves are interpolated with `np.interp` from the native solver grid onto the 0.05 V measurement grid before extraction (S6 section 1).

Source. Conventions of this study (ASSUMED); code in Chapter B. The retirement of SS_{min} and the fixed-current replacements are from S6 section 1.4 and SYNTHESIS D7.

Derivation. The failure of SS_{min} has the following cause. The simulated SS rises steeply with current, by about 10–25 mV/dec per 0.5 decade (S3). SS_{min} is therefore just the SS of the first window above the $5\times$ floor gate, and it depends on where that window falls on the 0.05 V grid. At 2 nm the

gate is $2.30 \times 10^{-14} \text{ A}/\mu\text{m}$. `run_0012` has $I_D(0.15 \text{ V}) = 2.34 \times 10^{-14}$, so its first window starts at 0.15 V and gives 73.2 mV/dec. `run_0038` (B1) has 2.28×10^{-14} , so its window jumps to 0.20 V and gives 83.1 mV/dec. The two runs have the same I_{on} within 0.01 % and differ by 2.5–2.6 % at that one point (S6 section 1.4; SYNTHESIS D7). `run_0029` (2.34×10^{-14}) behaves like `run_0012`. The pure mobility rescaling `run_0003` $\rightarrow$ `run_0008` flips SS_{min} the other way ($83.1 \rightarrow 73.2$). At 13.2 nm, `run_0013` and `run_0017` give $SS_{\text{min}} = 151.3$ and 138.6 but fixed-current $SS = 90.8$ and 92.0 mV/dec. The earlier claim that “removing confinement moves SS toward the data” is therefore an artefact. The fixed-current SS agrees within 1–2 mV/dec in these pairs: 122.5 / 123.5 (10^{-11} – 10^{-10}) and 212.9 / 215.0 (10^{-10} – 10^{-9}) for `run_0012` / `run_0038`.

A second extraction subtlety concerns the floor. The measured 13.2 nm window 10^{-11} – $10^{-10} \text{ A}/\mu\text{m}$ sits only 1.5 times above the off-floor ($6.6 \times 10^{-12} \text{ A}/\mu\text{m}$). Subtracting the floor changes its SS from 136.1 to 91.8 mV/dec. The floor-subtracted value is used, and the subtraction is flagged as an assumption (S6 section 1.4; SYNTHESIS, referee response 2.6). The measured SS_{min} values are 84.5 / 114.7 / 130.8 mV/dec. The paper’s “near-thermal-limit” 60–70 mV/dec is reachable on these curves only next to the noise floor. The 5-point SS without the floor gate is 70.6 mV/dec at 0.20 V for 2 nm (REVIEW).

Table 5.6. Measured / simulated metrics of the final calibrated state (`run_0038`; `run_0039` $\times 1.015074$; `run_0040` $\times 1.020113$) with the revised definitions (S6 section 1.6). *13.2 nm measured SS values are floor-subtracted. SS in mV/dec, voltages in V, g_m in $\text{A}/\text{V}/\mu\text{m}$, μ_{FE} in $\text{cm}^2/\text{V s}$.

t	$V_{\text{th,cc}}$	$SS_{[10^{-11}, 10^{-10}]}$	$SS_{[10^{-10}, 10^{-9}]}$	SS_{cc}	$V_{\text{th,lin}}$	gap7	μ_{FE}	I_{on}/g_m
2.0	0.662 /	121.2 /	194.8 /	269.9 /	1.663 /	1.051 /	12.15 /	1.337 /
	0.680	123.5	215.0	290.9	1.537	1.024	11.10	1.463
6.3	0.644 /	124.5 /	210.0 /	295.4 /	1.820 /	1.224 /	10.95 /	1.176 /
	0.653	124.5	215.6	289.7	1.466	1.085	8.39	1.534
13.2	0.083 /	91.8* /	118.8* /	158.4* /	1.044 /	0.571 /	50.17 /	1.953 /
	0.054	90.9	143.4	193.0	0.995	0.630	48.85	2.005

I_{on} matches to 0.00 % by construction. $V_{\text{th,cc}}$ is within +19 / +9 / –29 mV. The deep subthreshold slope (10^{-11} – 10^{-10}) is within 2.3 mV/dec in all films, where 13.2 nm holds only with floor subtraction (REVIEW O7). In 10^{-10} – 10^{-9} the model is too soft by +20 / +6 / +25 mV/dec. The measured off-floors are 4.61×10^{-15} , 1.53×10^{-13} and $6.60 \times 10^{-12} \text{ A}/\mu\text{m}$ (ratio 1432 between 13.2 and 2 nm). Their origin is **not determined** (Figure 9.4). The simulated native zeros occur only at $V_G \leq -0.10 \text{ V}$ with $|I_D| \leq 6.3 \times 10^{-18} \text{ A}/\mu\text{m}$, three decades below every measured floor. No metric uses them.

ATLAS implementation. None (post-processing). The solver grid matters in one place: the 0.1 V solver step above 1.5 V biases the simulated $V_{\text{th,lin}}$ by –5.4 / –0.8 / –11.7 mV relative to the data (S6 section 1.5; Section 6.4).

Sensitivity evidence. Table 5.7 gives the numerical uncertainty budget per metric (S6 section 1.5).

Status and caveats. Definitions: conventions, applied identically to data and model. SS_{min} : **rejected** as a metric (numerical audit; 73.2 vs 83.1 mV/dec at identical I_{on}). It survives as the key $SS_{\text{mV/dec}}$ in `scripts/extract_metrics.py` for backward compatibility, and must not be quoted as a result. $V_{\text{th,cc}}$ is partly a mobility metric (Section 5.1). The floor subtraction at 13.2 nm is an assumption. With one device per thickness, no metric carries a device-to-device uncertainty.

Table 5.7. Numerical uncertainty of the extracted metrics (S6 section 1.5). DOS: residual beyond 384/192. Mesh: $\times 0.7$ refinement. Interpolation: log-linear vs PCHIP and grid effects.

metric	DOS	mesh	interpolation / grid	total
$V_{th,cc}, V(I)$	< 0.5 mV	≤ 0.4 mV	≤ 1.5 mV	± 1.5 mV
fixed-current SS	< 0.3	≤ 0.1	≤ 1.1 (meas. 13.2 nm at 10^{-11} : 4.6)	± 1 mV/dec
SS_{min}	—	—	window phase ± 10	invalid
$V_{th,lin}$	≈ -1.4 mV (2 nm)	0.15 mV	solver-grid bias $-5.4/-0.8/-11.7$ mV	± 2 mV + bias
$g_{m,max}, \mu_{FE}$	≈ 0.1 %	0.02 %	$-0.4/0/-0.7$ %	± 0.7 %
I_{on}, I_D	$+0.16/\sim 0.06/\sim 0.03$ %	0.013 %	exact	± 0.2 %

5.11 Discretization of the trap density of states (NUMA/NUMD)

Definition and role in the model. The continuous tail and Gaussian defect densities (Sections 4.10 and 4.11) enter Poisson’s equation through energy integrals of density times occupancy. ATLAS evaluates these on a finite set of discrete energy levels: NUMA for acceptor-like and NUMD for donor-like states. The discretization error propagates directly into the trapped charge, hence into P (Section 5.3), I_{on} and the fitted μ_{band} .

Values used. NUMA/NUMD = 96/48 in the V1 calibration and campaign-A runs (for example run_0012–run_0015, run_0030–run_0035) and in B0 (run_0037). 192/96 in run_0019. 384/192 in run_0029, the recalibrated runs run_0038–run_0040, C1 (run_0052) and the prediction runs run_0041–run_0047 (S6 section 1.6; MANIFEST). The ATLAS default is 12 [22, p. 1168].

Source. Numerical choice (CALCULATED by convergence study; S6 section 1; SYNTHESIS D9).

Derivation. *Derivation.* The trapped acceptor density is $n_t(\mathbf{r}) = \int g_A(E) f(E; E_{Fn}(\mathbf{r})) dE$, evaluated as a sum over N levels. The integrand varies on the scales $W_{TA} = 40$ meV and $k_B T = 25.9$ meV, so the levels must resolve these scales. For a smooth integrand and a midpoint- or trapezoid-type rule, the error behaves as $e(N) \simeq CN^{-p}$ with $p = 2$. Given a quantity Q at N , $2N$ and $4N$, the observed order and the extrapolated limit are

$$p = \log_2 \frac{Q_{2N} - Q_N}{Q_{4N} - Q_{2N}}, \quad Q_\infty \approx Q_{4N} + \frac{Q_{4N} - Q_{2N}}{2^p - 1}. \quad (5.21)$$

At 2 nm (run_0012 $\rightarrow$ run_0019 $\rightarrow$ run_0029), I_{on} changes by $+1.98$ % (96 $\rightarrow$ 192) and $+0.48$ % (192 $\rightarrow$ 384). That gives $p = 2.03$ and a residual beyond 384/192 of $0.48/(2^{2.03} - 1) \approx +0.16$ % in I_{on} , and about -1.4 mV in $V_{th,lin}$ (S6 section 1). The observed order matches the expected second-order quadrature.

The DOS offset between 96/48 and 384/192 at equal mobility depends on thickness and on bias (Table 5.8). At $V_G = 3$ V it is $+2.48 / +0.95 / +0.50$ %, in the ratio 1 : 0.38 : 0.20. That is close to the tail densities $N_t = 2.0 : 0.85 : 0.49 \times 10^{19} \text{ cm}^{-3}$ (1 : 0.43 : 0.25). The offset is therefore a quadrature error of the discretized tail. Its sign flips between threshold, where trapped charge is over-counted, and the on-state, where it is under-counted. Where ATLAS places the levels is **not determined** (S6 section 1). Because I_D is linear in μ_{band} (Equation (5.8)), the recalibration at 384/192 absorbs these

Table 5.8. Effect of DOS 96/48 $\rightarrow$ 384/192 at equal mobility (S6 section 1). 6.3 and 13.2 nm compare run_0015 and run_0013 with the recalibrated runs scaled back to the V1 mobilities ($\times 11.69/11.41$, $\times 61.9/60.39$).

film (pair)	ΔI_D at V_G 0.7 / 1.5 / 2 / 3 V (%)	$\Delta V_{th,cc}$ (mV)	$\Delta V_{th,lin}$ (mV)	Δg_m	ΔSS fixed (mV)
2 nm (run_0012 $\rightarrow$ run_0029)	-0.95 / +2.56 / +3.64 / +2.48	+1.1	-22.0	+0.9 %	$\leq +0.8$
6.3 nm (run_0015 $\rightarrow$ run_0039)	-0.63 / +0.78 / +1.12 / +0.95	+0.7	-4.8	+0.6 %	$\leq +0.6$
13.2 nm (run_0013 $\rightarrow$ run_0040)	-0.04 / +0.68 / +0.64 / +0.50	+0.2	-4.5	+0.3 %	$\leq +0.3$

offsets into μ_{band} : 18.13 / 11.69 / 61.9 $\rightarrow$ 17.69 / 11.58 / 61.60 cm²/V s, which is -2.4 / -0.9 / -0.5 %, the I_{on} offsets with the sign reversed. For the historical 96/48 runs (run_0014, run_0015, A1–A6, B0) the effect on the metrics of Table 5.6 is at most 1 mV, 0.6 mV/dec and 1 % (S6 section 3.1).

ATLAS implementation. DEFECTS CONTINUOUS . . . NUMA=384 NUMD=192 [22, pp. 1168–1169]. NUMA and NUMD cannot be changed with MODIFY [22, p. 1171], so every DOS level change requires a new deck. Mesh convergence is a separate check (Sections 6.2 and 6.6): $\times 0.7$ refinement changes the 6.3 nm curve by at most 0.0039 decade ($\Delta V_{th,cc} = -0.32$ mV; run_0036 vs run_0015) and the 13.2 nm curve by at most 0.0029 decade (-0.36 mV; run_0028 vs run_0013).

Sensitivity evidence. Table 5.8 and Figure 6.1.

Status and caveats. DOS 384/192 converged: **demonstrated** (NUM; residual +0.16 % I_{on} at 2 nm, $p = 2.0$). The DOS offset is thickness-dependent: **demonstrated**. Convergence was checked at 300 K only, and at 2 nm only for the three-level sequence. The 6.3 and 13.2 nm residuals are expected to be smaller in proportion to N_t , but that is inferred, not computed.

5.12 Identifiability of thickness power laws; leave-one-out tests

Definition and role in the model. Several V1 inputs are written as power laws in t . Examples are $\Delta E_c \propto t^{-1.38}$ (Section 4.5), $N_t \propto t^{-0.75}$ (Section 4.10), $m^* \propto t^{-1.41}$ (Section 4.7) and the $N_{d,eff}$ law (Section 4.14). Earlier package documents also described the measured metrics with power laws. This section asks what three films can identify.

Values used. Measured $\mu_{FE} = 12.15 / 10.95 / 50.17$ cm²/V s and $I_{on} = 5.09 \times 10^{-7} / 4.03 \times 10^{-7} / 3.07 \times 10^{-6}$ A/ μ m at 2 / 6.3 / 13.2 nm. The 31.8 nm film serves as context only ($I_{on} = 3.59 \times 10^{-6}$ A/ μ m).

Source. DATA; analysis in S2 report, REVIEW V12 and MANIFEST powerlaw_loo.

Derivation. A two-parameter law $y = A t^b$ through two anchors has

$$b = \frac{\ln(y_2/y_1)}{\ln(t_2/t_1)}, \quad \hat{y}(t_3) = y_1 (t_3/t_1)^b. \quad (5.22)$$

With three films and two parameters there is exactly one residual degree of freedom. Any two-point law passes exactly through its anchors, so it cannot be falsified there. The only test the data allow is to leave

one film out and predict it. More flexible forms (three parameters) cannot be tested at all. The measured sequence is also non-monotonic: $\mu_{\text{FE}}(6.3) < \mu_{\text{FE}}(2) < \mu_{\text{FE}}(13.2)$ and $I_{\text{on}}(6.3) < I_{\text{on}}(2) < I_{\text{on}}(13.2)$. Every monotonic law in t therefore fails at one film at least.

Fitting through 2 and 13.2 nm and predicting 6.3 nm (measured $\mu_{\text{FE}} = 10.95 \text{ cm}^2/\text{V s}$):

form through (2, 13.2 nm)	parameters	predicted $\mu_{\text{FE}}(6.3)$	miss
power law At^b	$b = 0.7516$	28.78	$\times 2.63$
$A + B/t$	$A = 56.96, B = -89.62$	42.73	$\times 3.90$
exponential $Ae^{t/\tau}$	$1/\tau = 0.1266 \text{ nm}^{-1}$	20.94	$\times 1.91$
step, $t_c \in (6.3, 13.2)$	–	12.15	$\times 1.11$

The predicted values are from MANIFEST (powerlaw_loo). The parameters follow from Equation (5.22) and its analogues through the anchors 12.15 and 50.17. For I_{on} the same power-law test misses 6.3 nm by $\times 3.76$ (SYNTHESIS section D). An I_{on} power law through 6.3 and 13.2 nm (exponent 2.74–2.75) predicts $3.4 \times 10^{-5} \text{ A}/\mu\text{m}$ at 31.8 nm. The measured value is 3.59×10^{-6} , $\times 9.6$ lower. The prediction also exceeds the series-resistance ceiling $0.7 \text{ V}/9 \times 10^4 \Omega \mu\text{m} = 7.8 \times 10^{-6} \text{ A}/\mu\text{m}$ (S2 report).

The step form wins the leave-one-out test ($\times 1.11$), but that result is circular. The step location was chosen after the data were seen, and with $t_c \in (2, 6.3)$ the same form misses by $\times 4.58$ (REVIEW V12). SYNTHESIS concedes the point: all laws are two-point laws or extrapolations, the step is equally unsupported, and no functional form is claimed.

The same limitation applies to the input laws. The N_t exponent 0.75 comes from a two-point ratio of about 4 between 13.2 and 2 nm stated by Januar et al. [1, p. 7]. The same paper's PBS devices (5.3×10^{19} at 2 nm and $3.39 \times 10^{18} \text{ cm}^{-3}$ at 10 nm) imply 1.71 [1, p. 9]. A surface-plus-bulk form through the same anchors ($N_t = 2.15 \times 10^{18} \text{ cm}^{-3} + 3.57 \times 10^{12} \text{ cm}^{-2}/t$) predicts 7.8×10^{18} at 6.3 nm, against $8.5 \times 10^{18} \text{ cm}^{-3}$ from the power law. The data cannot tell them apart (S2 report; S4). The ΔE_c exponent is a local three-point slope of DFT slab data. Beyond about 3 nm it exceeds the infinite-well upper bound ($\times 1.59$ at 6.3 nm, $\times 2.51$ at 13.2 nm); relative to the DFT-anchored effective-width estimate of S3 the same overshoot is $\times 1.45 / \times 2.15$ (Table 4.1), a different reference, not a different result. Its effect on V_{th} is at most 35 mV (SYNTHESIS; Section 4.6).

ATLAS implementation. The thickness laws enter the decks only as per-film numbers written by the deck generator (Chapter B). ATLAS itself knows no law. The fitted per-film μ_{band} is deliberately *not* a law (Section 5.1).

Sensitivity evidence. Leave-one-out misses are $\times 2.63$ for μ_{FE} and $\times 3.76$ for I_{on} . Applied to the V1-fitted μ_{band} , the power-law test misses by $\times 3.27$ (S2 report). The 31.8 nm extrapolation overshoots by $\times 9.6$.

Status and caveats. Power laws in t for μ_{FE} , I_{on} and V_{th} : **rejected** as descriptions of the data. Thickness laws in general: not identifiable from three films with $N = 1$ device each. Of the input laws, $N_t \propto t^{-0.75}$ is **not determined** (empirical interpolation). $m^* \propto t^{-1.41}$ is **plausible, unverified**. $\Delta E_c \propto t^{-1.38}$ beyond 3 nm is **rejected** as physics (S2 findings). The measured pattern, $2 \approx 6.3 \text{ nm} \neq 13.2 \text{ nm}$ in every metric, is more consistent with a change of material between 6.3 and 13.2 nm

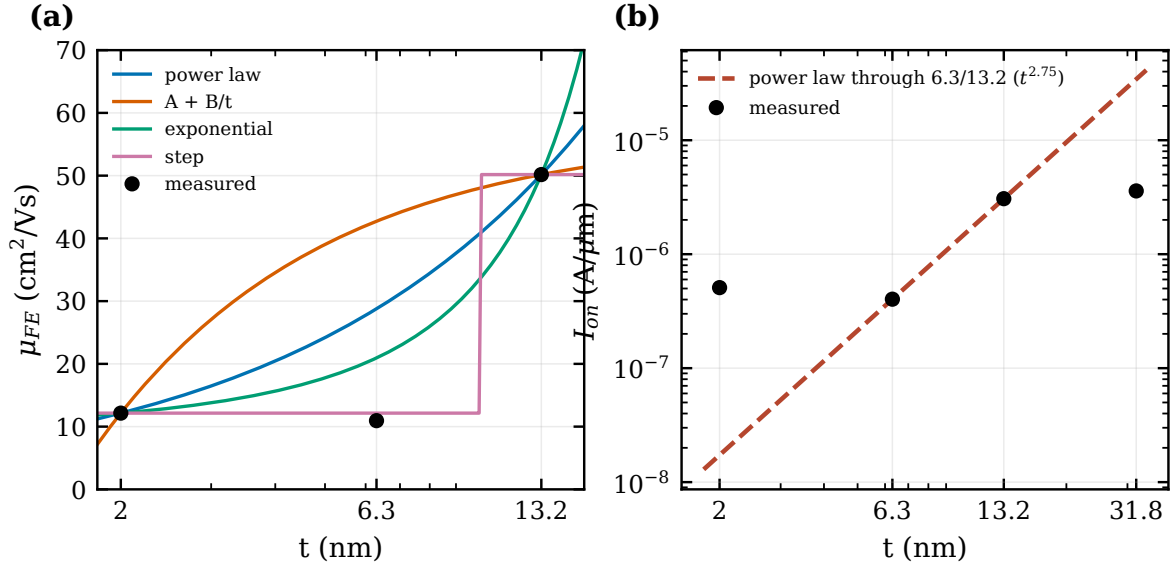


Figure 5.3. Leave-one-out tests of thickness laws for the measured $\mu_{FE}(t)$ and $I_{on}(t)$. Power law, $A + B/t$, exponential and step forms are fitted through 2 and 13.2 nm and used to predict 6.3 nm (measured $\mu_{FE} = 10.95$ cm²/V s; predictions 28.78, 42.73, 20.94 and 12.15). The panels also show the extrapolation to 31.8 nm. An I_{on} power law through 6.3 and 13.2 nm (exponent 2.75) predicts 3.43×10^{-5} A/μm against 3.59×10^{-6} measured. Note the non-monotonic data: no monotonic law passes through all three films.

(crystallinity, grain boundaries, percolation), which S2 rates **plausible, unverified**. It would be tested by a temperature series and GIXRD, and by more thicknesses between 6.3 and 13.2 nm.

Summary.

- Mobility is fitted, not derived. $\mu_{band} = 17.69 / 11.58 / 61.60$ cm²/V s at DOS 384/192 (run_0038, run_0039 $\times 1.015074$, run_0040 $\times 1.020113$), replacing 18.13 / 11.69 / 61.9 at 96/48. The fitted μ_{band} carries 109 % of the 6.3 $\rightarrow$ 13.2 nm mobility step. The roughness factor and the trap partition each carry about 3 % (CAL).
- I_D is exactly linear in a uniform constant mobility: run_0029/run_0038 = 1.024879 against 1.024873. This makes the one-step calibration, the rescaled 6.3 / 13.2 nm runs and both temperature variants exact (NUM, demonstrated).
- The paper's 5.1 and 27.4 cm²/V s are the saturation formula applied to the same $V_D = 0.7$ V curves (5.09 / 27.40), not linear-regime μ_{FE} (12.15 / 50.17). This is demonstrated provenance evidence, and it predicts 3.86 cm²/V s at 6.3 nm.
- Contacts are untested in V1. The S/D are ideal Ohmic. run_0027 is malformed and retracted (Section 7.4). R_{sd} is not identifiable from one curve ($\theta_{net} < 0$, α - θ correlation +1.00). A contact explanation of the thickness trend is rejected.
- The elevated- T runs used ATLAS's default $tmu = 1.5$. Both variants are exact, with P-MTR multipliers 1.30441 (358.15 K) and 1.19669 (338.15 K). Their activation energies differ by exactly 42.3 meV. A direct check, run_0054 (2 nm, 358.15 K, $tmun = 0$ declared explicitly), reproduces run_0045 $\times 1.30441$ to within 0.11 % at every gate voltage, which confirms the rescaling numerically. All temperature results are registered predictions (PRED).
- SS_{min} is retired: 73.2 vs 83.1 mV/dec at identical I_{on} (run_0012 vs run_0038). Fixed-current SS

reproduces the 10^{-11} – 10^{-10} A/ μ m slope within 2.3 mV/dec in all films (13.2 nm floor-subtracted).

- DOS 384/192 is converged where checked (2 nm, 300 K: order 2.03, residual +0.16 % I_{on}). The 96/48 offsets of +2.48 / +0.95 / +0.50 % scale with N_t .
- No thickness power law is identifiable from three non-monotonic films. The leave-one-out misses are $\times 2.63$ (μ_{FE}) and $\times 3.76$ (I_{on}), and the I_{on} law overshoots 31.8 nm by $\times 9.6$. No functional form is claimed.

With every physical input derived, the next chapter documents how the model is turned into ATLAS decks, how each launch is validated before it is scored, and how far the numerical results are converged.

Chapter 6

Numerical implementation in Silvaco ATLAS

The physics of Chapters 4 and 5 reaches the device only through an input deck, a mesh, a nonlinear solver and a runner that decides whether a result is accepted. If set carelessly, each of these elements can change the drain current by more than the calibration residual, and each could in principle produce artificial agreement with the data, for instance by replacing numerical zeros with a floor. This chapter documents the implementation so that every simulated number of the report can be traced to a statement of the manual and to a convergence check. It describes the content of a final deck (Section 6.1); the set-up of the mesh, the solver and the gate sweep and the reasons for it (Sections 6.2 to 6.4); the validation of each run before it is scored (Section 6.5); the magnitude of the numerical errors (Section 6.6); and the execution of the 2026-09-25 campaigns (Section 6.7). All simulations use Silvaco ATLAS 5.28.1.R, run in batch mode through DeckBuild on a single 32-bit licence. Manual pages are the printed pages of the ATLAS user's manual [22].

6.1 Anatomy of a final deck

Every deck is generated by `scripts/build_iwo_decks.py` from one material model (`config/iwo_material_model`), one geometry file (`config/geometry.yaml`) and one solver file (`config/solver.yaml`). A run differs from the reference configuration only by the dotted-path overrides recorded in its catalogue entry (Table A.2). The description below uses the final 2 nm deck `run_0038` (DOS levels 384/192, $\mu_{\text{band}} = 17.69 \text{ cm}^2/(\text{V s})$), which is reproduced in full in Chapter C. Line numbers refer to that listing.

6.1.1 Provenance header (lines 1–31)

```
1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 2.0 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
   divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
   numerical.
8 #=====
```

```

9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.3533 eV | DFT_PURE_IN203_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.4033 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 3.9467 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2573 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN203_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 3.2744e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.0000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 2.0000e+19 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 5.0000e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5002e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN203_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 17.69 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.7336 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 12.9772 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVEs (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====

```

The header is written by the generator and records, for every parameter, its value at this thickness, its provenance label, its source and the formula that produced it. At 2 nm it states: $E_g = 3.05 + 0.3533 = 3.4033$ eV; $\chi = 4.3 - 0.3533 = 3.9467$ eV (the confinement shift is assigned entirely to the conduction band); $m^* = 0.2573 m_0$ and hence $N_c = 3.2744 \times 10^{18} \text{ cm}^{-3}$; $\epsilon_{\text{IWO}} = 9.3$; $N_t = 2 \times 10^{19} \text{ cm}^{-3}$ and $W_{\text{TA}} = 0.04$ eV, so that $N_{\text{TA}} = N_t/W_{\text{TA}} = 5 \times 10^{20} \text{ cm}^{-3} \text{ eV}^{-1}$; $N_{\text{d,eff}} = 2.50025 \times 10^{17} \text{ cm}^{-3}$; the interface sheet $3 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$ at $E_c - 0.3$ eV; $Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$ ($\Delta V_{\text{fb}} = -qQ_f/C_{\text{ox}} = -0.3095$ V); $\mu_{\text{band}} = 17.69$ and the roughness factor $(1 - 0.287/2)^2 = 0.7336$, which give $\text{MUN} = 12.9772 \text{ cm}^2/(\text{V s})$. Two remarks on the header are necessary. First, the comment in lines 3–4 still describes the reduced electrical deck, whereas the body of the deck (lines 53–65) builds the physical structure; the comment is an outdated generator string and has no effect on the solution. Second, the “not inserted natively” remark on line 27 refers to the gate-dependent (Ando-type) mobility roll-off, which is not modelled, and not to MUN itself; MUN is set on the material statement (line 72). The derivation of every value is given in Chapters 4 and 5.

6.1.2 Mesh (lines 32–52)

```
1 go atlas
2 mesh width=1.0
3 x.mesh loc=0 spac=0.3
4 x.mesh loc=1.8 spac=0.05
5 x.mesh loc=2 spac=0.01
6 x.mesh loc=2.2 spac=0.05
7 x.mesh loc=3 spac=0.2
8 x.mesh loc=12 spac=0.7
9 x.mesh loc=21 spac=0.2
10 x.mesh loc=21.8 spac=0.05
11 x.mesh loc=22 spac=0.01
12 x.mesh loc=22.2 spac=0.05
13 x.mesh loc=24 spac=0.3
14 y.mesh loc=-0.072 spac=0.02
15 y.mesh loc=-0.002 spac=0.00025
16 y.mesh loc=-0.00025 spac=6.25e-05
17 y.mesh loc=-0.000125 spac=6.25e-05
18 y.mesh loc=0 spac=6.25e-05
19 y.mesh loc=0.002 spac=0.00025
20 y.mesh loc=0.017 spac=0.001
21 y.mesh loc=0.067 spac=0.02
```

`mesh width=1.0` sets the scale factor for the dimension that is not simulated, which is applied to all run-time and log-file outputs [22, p. 1376]; with the default of 1 μm [22, p. 1374] the logged currents are in $\text{A}/\mu\text{m}$. Each `x.mesh/y.mesh` line places one mesh line at `loc` with local spacing `spac`, both in micrometres [22, pp. 633, 1117]; the MESH statement itself is described on pp. 1373–1376 of the manual. The design of the mesh is discussed in Section 6.2.

6.1.3 Regions and electrodes (lines 53–65)

```
1 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.002 y.max=-0.00025
2 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
3 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
  nm S/D, air above the channel.
4 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
  CONTACT (4.70 eV, ASSUMED).
5 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
6 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
7 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
8 region num=7 material=Air x.min=0 x.max=24 y.min=-0.072 y.max=-0.002
9 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
10 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
  ASSUMED, no TLM data);
11 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
  electrical equivalence.
12 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.072 y.max=-0.002
13 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.072 y.max=-0.002
```

Regions are rectangles bounded by `x.min`, `x.max`, `y.min` and `y.max` [22, p. 1559]. The IWO film is divided into region 1 (bulk, $-2 \leq y \leq -0.25 \text{ nm}$) and region 2, the 0.25 nm layer adjacent to the Al_2O_3 that carries the regularized interface DOS (Section 4.12). Region 5 is not used. Electrodes are named and positioned with `name` and the bounding coordinates [22, pp. 1207–1209]. The `material=Palladium` parameter of the source and drain electrodes only sets the material shown in

the plots and its thermal and optical properties; it does not apply an electrical property such as a work function, which is set on the CONTACT statement [22, p. 1208]. The gate is the Conductor volume (region 6), with the electrode defined on the same rectangle.

6.1.4 Background donors (lines 66–67)

```
1 doping uniform n.type conc=2.50025e+17 region=1
2 doping uniform n.type conc=2.50025e+17 region=2
```

doping uniform n.type conc= places a constant donor density in the named region; UNIFORM requires the type and the concentration and by default fills the whole region [22, p. 1189]. The value is the fitted law $N_{d,\text{eff}}(t) = 2.5 \times 10^{17} [1 + (t/20)^4] \text{ cm}^{-3}$ evaluated at $t = 2 \text{ nm}$ (Section 4.14).

6.1.5 Material parameters (lines 68–78)

```
1 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
2 material material=IWO user.group=semiconductor user.default=silicon \
3   eg300=3.40331 affinity=3.94669 permittivity=9.3 \
4   nc300=3.27444e+18 nv300=1e+19 \
5   mun=12.9772 mup=0.01 \
6   taun0=1e-06 tau0=1e-06
7 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
   MEASURED_TARGET_DEVICE (SI Fig.S1)).
8 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
   values.
9 material region=3 permittivity=9.0
10 material region=4 permittivity=19.57
11 contact name=gate workfunction=4.7
```

IWO is a user-defined semiconductor (user.group=semiconductor) whose unspecified parameters are taken from the material named by user.default [22, p. 1366]. Every parameter that is relevant for an electrons-only DC simulation (eg300, affinity, permittivity, nc300, mun) is overridden explicitly. The MATERIAL statement sets band-structure and model parameters for a material or a region [22, p. 1336]. Lines 76–77 set the measured or assumed permittivities of the two named oxides, because the ATLAS defaults for Al_2O_3 and HfO_2 are not the values of this stack. The TiN work function 4.7 eV is set on the gate CONTACT (line 78). A known consequence of user.default=silicon, namely that the temperature exponent of the constant mobility is inherited from silicon, is discussed in Section 6.3.

6.1.6 Physical models (line 79)

```
1 models fermi srh temp=300 print
```

fermi selects Fermi-Dirac statistics [22, pp. 108–109], which are required because the channel is degenerate in accumulation and N_c is small. srh adds a Shockley-Read-Hall background, which plays no role in the unipolar current; temp=300 sets the lattice temperature; and print echoes the model and electrode settings to the run output (the MODELS parameter list is on pp. 1433 and 1469–1477). The physical justification is given in Chapter 4.

6.1.7 Defect DOS (lines 80–97)

```

1  # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
   ^-3/eV;
2  #   EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
3  defects region=1 continuous numa=384 numd=192 \
4    nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
5    nga=5e+16 ega=0.6 wga=0.15 \
6    ngd=0 egd=3.30331 wgd=0.1 \
7    sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
8    siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
9    afile=acceptor_r1.dat dfile=donor_r1.dat
10 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
   ^-3/eV;
11 #   EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
   regularized: NGA = sheet peak / layer thickness.
12 defects region=2 continuous numa=384 numd=192 \
13    nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
14    nga=1.16757e+19 ega=0.301554 wga=0.123975 \
15    ngd=0 egd=3.30331 wgd=0.1 \
16    sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
17    siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
18    afile=acceptor_r2.dat dfile=donor_r2.dat

```

The DEFECTS statement describes the density of states in the gap as up to four distributions, namely an exponential tail and a Gaussian for each of the acceptor-like and donor-like states [22, p. 1169]. `continuous` selects the continuous defect integral model [22, p. 1169]; `nta` is the acceptor-like tail density at the conduction-band edge and `nga` the total density of a Gaussian acceptor band [22, p. 1171]; `wta` is the characteristic decay energy of the tail [22, p. 1173]; and `afile/dfile` store the resulting DOS versus energy [22, pp. 1169–1170]. `numa/numd` set the number of energy levels of the numerical integration. The manual default is 12 for both [22, p. 1168], far below the 96/48 used for the calibration runs and the 384/192 of the final runs (Sections 5.11 and 6.6). All capture cross sections are set to 10^{-15} cm^2 (ASSUMED); the defaults are 10^{-16} or 10^{-14} , depending on the carrier and state type [22, pp. 1168–1169]. In a DC unipolar simulation the occupancy of acceptor-like states is set by the electron quasi-Fermi level and is insensitive to the cross sections. In addition to the bulk tail, region 2 carries the interface Gaussian converted into a volume density. The generator writes $N_{\text{GA}} = 1.16757 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$, $E_{\text{GA}} = 0.3016 \text{ eV}$ and $W_{\text{GA}} = 0.1240 \text{ eV}$, which are close to the nominal sheet peak divided by the layer thickness ($3 \times 10^{11} / 2.5 \times 10^{-8} \text{ cm} = 1.2 \times 10^{19}$) and to the nominal centre and width (0.3 and 0.12 eV). The deck comments do not state the reason for the small differences, which are produced by the generator (Chapter B). Halving the layer at constant sheet density changes I_D by at most 0.009 dec (check E, Table 6.2).

6.1.8 Fixed interface charge (line 98)

```

1  interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001

```

`qf` places a fixed sheet charge, in units of elementary charges per cm^2 , at the interface inside the bounding box; a positive value denotes a positive charge [22, p. 1260]. By default the statement applies to semiconductor-insulator interfaces [22, p. 1264], so the thin box around $y = 0$ selects the IWO/ Al_2O_3 interface. The same mechanism with a box on the IWO/air surface implements the back-surface charge of hypothesis A3 (run_0030; see Chapter C).

6.1.9 Numerical method (lines 99–101 and 117–119)

```
1 # Strict acceptance (XANDRNORM, cr.toler 1e-17) up to Vg = 0.7 V; default acceptance above.
2 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
3   cr.toler=1e-17 xandrnorm carriers=1 electrons
```

The METHOD statement sets the solution technique and the tolerances for the subsequent SOLVE statements [22, p. 1384]. The two METHOD statements of the deck and their tolerances are discussed in Section 6.3.

6.1.10 Outputs and probes (lines 102–106)

```
1 output con.band val.band e.mobility
2 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
3 probe name=chan_mob x=12 y=-0.001 n.mob dir=0
4 probe name=chan_n x=12 y=-0.001 n.conc
5 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.002 y.max=-0.00025
```

output adds the band edges and the electron mobility to the saved structures. Three probes record, as functions of the gate bias, the electron mobility and density at the channel centre ($x = 12 \mu\text{m}$, $y = -1 \text{ nm}$) and the electron density averaged over the bulk region of the channel. Probe values are written to the log file together with the terminal currents [22, p. 1299].

6.1.11 Initial solution and bias ramp (lines 107–111)

```
1 solve init
2 save outf=equilibrium.str
3 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
4 solve vstep=-0.1 vfinal=-3 name=gate
5 solve vstep=0.1 vfinal=0.7 name=drain
```

solve init computes the zero-bias solution that serves as the initial guess of all later solutions; it is solved in zero-carrier mode [22, p. 1598]. The equilibrium structure is saved for the band diagrams. The gate is then ramped to -3 V in -0.1 V steps and the drain to 0.7 V in 0.1 V steps with stepped SOLVE statements (vstep, vfinal, name; [22, pp. 1593, 1609]). No point of the ramp is logged.

6.1.12 Logged transfer sweep (lines 112–123)

```
1 log outf=transfer.log
2 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
3 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
4 solve vstep=0.05 vfinal=0.7 name=gate
5 save outf=near_vth.str
6 # On-state: default convergence acceptance.
7 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
8   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
9 solve vstep=0.05 vfinal=1.5 name=gate
10 solve vstep=0.1 vfinal=3 name=gate
11 log off
12 save outf=final.str
```

log outf= opens the log file, in which every solution of later SOLVE statements is stored until log off [22, pp. 1299, 1301]. The sweep runs in four stepped segments, with a change of the

convergence acceptance between the second and the third segment (Sections 6.3 and 6.4). The structure near threshold (`near_vth.str`) and in the on-state (`final.str`) is saved for the band diagrams of Chapter 4.

6.1.13 Extraction of curves and profiles (lines 124–147)

```
1 output con.band val.band qfn e.mobility
2 extract init infile="transfer.log"
3 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
4 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
5 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
6 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
7 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
8 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
9 extract init infile="equilibrium.str"
10 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_eq.dat"
11 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_eq.dat"
12 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
13 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"
14 extract init infile="near_vth.str"
15 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"
16 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
17 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
18 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_vth.dat"
19 extract init infile="final.str"
20 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
21 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
22 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
23 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
24 quit
```

The DeckBuild `extract` statements [24] write the drain, gate and source currents versus V_G (`idvg.dat`, `igvg.dat`, `isvg.dat`), which are used for scoring and for the KCL check, the probe curves, and vertical cuts at $x = 12 \mu\text{m}$ of the conduction- and valence-band edges, the electron density and the electron quasi-Fermi level in the three saved states. No quantity is post-processed inside the deck.

6.2 Mesh

Lateral mesh. The x mesh (`config/geometry.yaml`) is dense where the potential varies laterally, at the contact edges, and coarse in the channel. The spacing is $0.3 \mu\text{m}$ under the outer contacts, $0.05 \mu\text{m}$ at 1.8 and $2.2 \mu\text{m}$, $0.01 \mu\text{m}$ at the source edge ($x = 2 \mu\text{m}$) and the drain edge ($x = 22 \mu\text{m}$), $0.2 \mu\text{m}$ at 3 and $21 \mu\text{m}$, and $0.7 \mu\text{m}$ at the channel centre. Because $L = 20 \mu\text{m}$ is three orders of

Table 6.1. Statements of the final deck and the manual pages that define them (ATLAS 5.28.1.R user's manual, printed pages).

Statement	Role in the deck	Manual pages
MESH, X.MESH, Y.MESH REGION	domain, width scale factor, mesh lines IWO bulk, IWO interface layer, Al ₂ O ₃ , HfO ₂ , Conductor, Air	633, 1373–1376 1553–1560
ELECTRODE	gate volume, Pd source/drain volumes	1207–1211
DOPING UNIFORM	$N_{d,eff}$ in regions 1 and 2	1188–1189
MATERIAL	user-defined IWO; oxide permittivities	1336, 1355, 1366
CONTACT	gate work function; S/D Ohmic by omission	1145, 1148
MODELS	Fermi-Dirac, SRH, temperature, print	108–109, 1433, 1469–1477
DEFECTS	continuous tail + Gaussian DOS, NUMA/NUMD	1168–1174
INTERFACE	fixed charge Q_f	1258–1264
METHOD	Newton, trap, tolerances, XANDRNORM, carriers	1383–1389, 1395
SOLVE	init, stepped bias ramps	1593–1611
LOG	terminal characteristics and probes	1299–1305

magnitude larger than the film thickness, the channel is a gradual channel, and the coarse spacing at the centre is adequate (check A below).

Vertical mesh. The vertical mesh resolves the film, the interface and the oxides separately. In the IWO the bulk spacing is $\min(1 \text{ nm}, t/8)$ (0.25 nm at 2 nm, i.e. eight intervals), refined to 0.0625 nm in the 0.25 nm interface layer (four intervals). The spacing is 0.25 nm in the Al₂O₃, 1 nm in the HfO₂ and 20 nm in the gate volume and in the air/Pd region. Thicker films are graded from the interface to keep the node count within the memory limit of the 32-bit ATLAS executable (about 30 000 nodes). The 13.2 nm deck with all spacings multiplied by 0.7 has 24 255 nodes, a uniform refinement by 0.5 (about 40 000 nodes) exceeds the limit, and the 31.8 nm decks use 16 500–18 000 nodes graded from 1 to 0.0625 nm.

Interface regularization layer. The interface DOS is represented as a volume DOS in the 0.25 nm region 2 (Section 4.12). The layer is a numerical construct: its thickness has no physical meaning, and the result must not depend on it. Halving it to 0.125 nm at constant sheet density changes I_D by at most 0.009 dec and $I_D(3 \text{ V})$ by -0.06% (V0 run_0017 vs V0 run_0013; check E).

Refinement tests. The mesh was tested by multiplying all spacings by a common factor (`geometry.mesh.spacing_` $\times 0.5$ at 2 nm (V0; 28 203 nodes), $\times 0.7$ at 13.2 nm (V0 and V1: run_0028 vs run_0013) and $\times 0.7$ at 6.3 nm on the tuned configuration (run_0036 vs run_0015). All tests pass with changes below 0.005 dec (Table 6.2). The first V1 attempt at 13.2 nm, run_0018, timed out at the default limit of 900 s and was repeated with 1500 s as run_0028 (1068 s).

6.3 Solver configuration

Equations and carriers. Only the Poisson equation and the electron continuity equation are solved (carriers=1 electrons on METHOD, [22, p. 1386]; the MODELS parameter for the number of carriers is on p. 1475). Holes are numerically absent in a 3.05 eV n-type oxide. In every earlier V0 and

Astra run the hole current was 10^{-38} to 10^{-55} A, and the Astra comparison of electrons-only with two-carrier solutions at 2 nm gave identical I_D at $V_G = 0, 0.5, 1$ and 3 V (check I). Transport is described by classical drift-diffusion with Fermi-Dirac statistics and a spatially uniform constant mobility MUN. Quantum confinement enters only through the thickness laws of E_g , χ and m^* (Section 4.6) and not through a Schrödinger solution.

Nonlinear iteration. The fully coupled Newton method is used together with the trap procedure (`trap`, enabled by default, [22, p. 1383]): if a bias step diverges, the step is reduced and retried up to `maxtraps` times, with a maximum of 10 [22, pp. 1389, 1395]. `itlimit=80` bounds the number of Newton loops per bias point [22, p. 1389]; `climit=1e-6` and `ir.tol=1e-19` tighten the carrier-concentration limit and the absolute current convergence criterion (`IR.TOL`: [22, p. 1386]).

Two-level acceptance. The acceptance of a bias point is defined in two stages (`config/solver.yaml`, `strict_acceptance`). The strict stage applies from -3 V up to the gate voltage at which the measured current reaches 10^{-9} A/ μ m, which is 0.7 V in the 2 nm deck. It uses `cr.toler=1e-17` with `xandrnorm`. `CR.TOLER` is the absolute tolerance of the continuity equation [22, p. 1386], and `XANDRNORM` accepts a point only if both the update (X) and the residual (RHS) norms are satisfied [22, p. 1387] (default: false, [22, p. 1383]). The strict tolerance is $\max(10^{-17}$ A, $10^{-5} \times$ the smallest measured active current). Without it the solver accepted residual glitches of order 10^{-15} A/ μ m in depletion (`V0 run_0006`), while a tolerance of 10^{-19} never converges because the residual floor is $10^{-17.7}$ A/ μ m (check F). Above that point the default stage applies, with `cr.toler=5e-18` and without `XANDRNORM` (`^xandrnorm`). In the on-state the residual floor scales with the current, and the strict setting would reject converged points. Both levels give the same on-state current (check F); the switch only removes spurious sub- 10^{-15} glitches in deep depletion.

DOS discretization. The continuous DOS integral is evaluated on NUMA/NUMD energy levels. The calibration runs use 96/48. The convergence test (Section 6.6) showed that 96/48 under-resolves the on-current by 2–3 % at 2 nm, and the final runs `run_0038–run_0052` therefore use 384/192 (except the B0 test `run_0037`, which was pre-registered at 96/48).

Mobility. MUN is a constant, uniform mobility. ATLAS 5.28.1.R ignores MOBILITY updates between SOLVE statements (`V0 run_0008`), so a gate-dependent mobility cannot be imposed by resetting MUN during the sweep; the trapped/free partition of the field-effect mobility is produced self-consistently by the DOS occupancy (Section 5.3). Because the mobility is constant and uniform in an isothermal simulation, I_D is exactly proportional to MUN. Raising μ_{band} from 16.8 to 18.13 $\text{cm}^2/(\text{V s})$ (+7.9 %) raises I_{on} from 4.717×10^{-7} to 5.090×10^{-7} A/ μ m (+7.9 %; `run_0003` $\rightarrow$ `run_0008`). This linearity is used to rescale curves to a slightly different μ_{band} without a new launch (Section 6.6).

Temperature. The constant-mobility model contains the lattice-temperature law $\mu = \text{MUN} (T_L/300)^{-\text{TMUN}}$ with the default `TMUN = 1.5` [22, p. 179]; the same default is tabulated for silicon [22, p. 1672]. The decks do not set TMUN, so in the elevated-temperature runs `run_0044–run_0047` ATLAS applied $(T/300)^{-1.5}$ to the band mobility (verified in the run output of `run_0045`), although the plan

declared a temperature-independent band mobility. Both interpretations are reported as exact variants in Chapter 10 (CORR; Section 5.8).

6.4 Stepped and pointwise gate sweeps

Two sweep forms were used (`config/solver.yaml`, `sweep`):

Pointwise (runs `run_0001`–`run_0009`): one SOLVE statement per 0.05 V point from -3 to 3 V, giving 121 logged points that match the measured grid point by point.

Stepped (every run from `run_0012` on, following the user’s directive of 2026-09-21): stepped SOLVE blocks with `vstep/vfinal` and segments (end voltage, step) = $(-1, 0.2)$, $(1.5, 0.05)$, $(3, 0.1)$ from the start at -3 V. The grid is coarse in deep depletion, 0.05 V around turn-on, where $V_{th,cc}$ and the swing are extracted, and 0.1 V in the on-state. This gives 76 logged points: 11 from -3 to -1 V, 50 from -0.95 to 1.5 V and 15 from 1.6 to 3 V. The runs are scored at the solved points only.

The two forms are electrically equivalent: `run_0012` (stepped, physical structure) and `run_0008` (pointwise, reduced structure) agree to better than 0.1 % at all compared biases (check M). Earlier, the V0 package showed that solving every 0.1 V instead of every 0.05 V introduces an interpolation error below 0.005 dec on the smooth parts of the curve (check D).

Consequence for SS_{min} . On this non-uniform grid the minimum local swing depends on where the grid points fall relative to the floor-gated search window. SS_{min} is therefore not used, and all swings are quoted on fixed current windows (Section 5.10).

Output characteristics. For the I_D – V_D predictions (`run_0041`–`run_0043`) the runner writes, after the transfer sweep, one drain sweep per gate voltage $V_G = 1, 1.5, 2, 2.5$ and 3 V. Each sweep starts at 0 and uses the default segments of 0.05 V up to 0.5 V and 0.1 V up to 3 V (option `-output-vgs` of `scripts/run_atlas.py`); the deck of `run_0041` is listed in Chapter C.

6.5 Per-run validation

Every launch goes through `scripts/run_atlas.py`, which never substitutes a Python model for the solver. After the run, the script checks the exported currents and only then scores the curve against the measurement. The outcome is recorded in `results/RUN_INDEX.csv` and in the `execution.json` file of the run. The validation comprises the following elements.

- *Status.* `REAL_ATLAS_SCORED` (transfer curve exported, validated and scored), `REAL_ATLAS_PREDICTION` (I_D – V_D families, no measurement to score against), `NO_EXPORT` (solver aborted before a current was written; `run_0010`), `NO_EXECUTION_JSON` (never executed; `run_0011`) and `TIMEOUT` (`run_0018`). Timeouts are 900 s by default and 1500 s for thick films.
- *KCL.* The maximum of $|I_D + I_S + I_G|$ over the sweep must remain below $\max(10^{-17}$ A, $0.1 \times \text{floor}$) plus 1 % of the current. The floor was originally the minimum measured current of the device. For `run_0012`, one deep-depletion point ($V_G = -1.2$ V, $I_D \approx 10^{-19}$ A/ μm) had a residual of

1.0×10^{-16} A, above the original limit of 1.1×10^{-17} A derived from the single 2 nm outlier at -3 V (Section 3.3). The floor was therefore redefined as the median off-band current (limit 4.6×10^{-16} A), and the run was re-scored without a new launch (`-rescore`; the original `execution.json` is kept). The residual is 2 % of the measured floor and four decades below any scored current (check G'). The 2 nm runs lie at the absolute residual floor of the solver (10^{-17} to 10^{-16} A in Table A.2).

- *Drain sign.* The drain current must be positive (electrons flowing from source to drain at $V_D > 0$).
- *Native zeros.* Below the strict-acceptance floor the solver returns $I_D = 0$ or $\pm 10^{-19}$ A/ μm . Such points are counted and excluded from logarithmic residuals; they are never replaced by a floor (logging epsilon 10^{-40} A/ μm). The measured off-state floors are therefore a limitation of the model, not a numerical artefact.
- *Score.* The fit residual is the RMS of $\log_{10} I_D^{\text{sim}} - \log_{10} I_D^{\text{meas}}$ over the active region (measured current above five times the off-band median; all points for $t \geq 20$ nm), together with the metric differences of Section 5.10.

6.6 Convergence evidence

A check passes if the active-region $\log_{10}$ RMSE changes by less than 0.01 dec, $|\Delta V_{\text{th,cc}}| < 10$ mV, $|\Delta SS| < 3$ mV/dec and $|\Delta I_{\text{on}}| < 1$ %, and if the off-current remains at native zero (thin films) or changes by less than 5 % (31.8 nm). Two sources of evidence exist. The first consists of checks of the V0 package, which shares the generator lineage, mesh, DOS discretization, interface regularization and solver controls with V1; the second consists of V1 runs that repeat the most important checks on the final decks. Table 6.2 collects both (`docs/NUMERICAL_CONVERGENCE.md`; metric values of V1 runs from Table A.2).

Table 6.2. Numerical convergence and consistency checks. Δ values are refined minus reference.

Check	Evidence	Result	Verdict
A. Lateral mesh $\times 0.5$, 2 nm (28 203 nodes)	V0 run_0015 vs V0 run_0013	max $\Delta \log_{10} I_D$ 0.0046 dec, RMS 0.0013; $I_D(3 \text{ V}) -0.02$ %	passed
A. Mesh $\times 0.7$, 13.2 nm (19 965 nodes)	V0 run_0026 vs V0 run_0024	max 0.0040 dec; $I_D(3 \text{ V}) -0.007$ %	passed
A'. V1 final deck, 13.2 nm, all spacings $\times 0.7$ (24 255 nodes)	run_0028 vs run_0013	max 0.0032 dec, RMS 0.0008; $V_{\text{th,cc}} -0.4$ mV; $SS_{[10^{-10}, 10^{-9}]}$ -0.03 mV/dec; $I_{\text{on}} -0.01$ %	passed
L. V1 tuned deck, 6.3 nm, spacings $\times 0.7$	run_0036 vs run_0015	active RMSE 0.0625 vs 0.0626 dec; $V_{\text{th,cc}} -0.3$ mV; $SS_{[10^{-10}, 10^{-9}]}$ $+0.02$ mV/dec; $I_{\text{on}} -0.01$ %	passed
B. Vertical IWO mesh at 2 nm: 0.25 nm + 0.0625 nm in the interface layer	V0 (part of A)	included in A	passed

Check	Evidence	Result	Verdict
C. DOS levels 96/48 $\rightarrow$ 192/96, 2 nm (V0)	V0 run_0016 vs V0 run_0013	max 0.0056 dec; $I_D(3\text{ V}) +1.1\%$	passed (marginal on I_{on})
C'. V1 final deck, 2 nm, DOS 96/48 $\rightarrow$ 192/96 $\rightarrow$ 384/192	run_0019, run_0029 vs run_0012	$I_{on} +2.0\%$ and $+2.5\%$; max 0.0125 dec; $V_{th,cc} +0.9 / +1.1\text{ mV}$; $SS_{[10^{-10}, 10^{-9}]} +0.7 / +0.8\text{ mV/dec}$	failed on I_{on} (1% criterion)
C''. DOS offset vs thickness (recalibration at 384/192)	run_0038, run_0039, run_0040	I_{on} at 384/192 relative to 96/48: $+2.5 / +1.0 / +0.5\%$ at 2.0 / 6.3 / 13.2 nm	resolved by recalibration
D. Gate step: solve every 0.1 V vs every 0.05 V	V0 run_0032 vs full sweeps	interpolation error $< 0.005\text{ dec}$ on smooth regions	passed
E. Interface layer 0.25 $\rightarrow$ 0.125 nm (sheet preserved)	V0 run_0017 vs V0 run_0013	max 0.009 dec; $I_D(3\text{ V}) -0.06\%$	passed
F. Continuity tolerance: 10^{-19} (never converges), $10^{-17} + \text{XANDRNORM}$, default 5×10^{-18}	V0 run_0006, 0009, 0010, 0011	10^{-15} glitches removed; on-state unaffected	passed (documented choice)
G. Terminal-current KCL, every V1 run	execution.json	e.g. run_0001 1.0×10^{-17} , run_0002 3.4×10^{-17} , run_0003 $6.9 \times 10^{-18}\text{ A}$; 2 nm runs at the 10^{-17} A solver floor	passed per run
G'. KCL rule refinement	run_0012	one point at -1.2 V with $1.0 \times 10^{-16}\text{ A}$; limit redefined on the median floor ($4.6 \times 10^{-16}\text{ A}$); re-scored without relaunch	passed after documented rule change
H. Repeated identical deck	V0 run_0012 vs run_0013 family; Astra retry	identical currents to printed precision	passed
I. Electrons-only vs two carriers, 2 nm	Astra rebuild (contact both vs electron)	identical I_D at $V_G = 0, 0.5, 1, 3\text{ V}$; hole current 10^{-38} – 10^{-55} A	passed
J. Width convention	V0 run_0007 (WIDTH=2) vs V0 run_0005	raw currents $\times 2.000000$; $A/\mu\text{m}$ identical	passed
M. Physical structure + stepped sweep (76 points) vs reduced boundaries + pointwise (121 points), 2 nm	run_0012 vs run_0008	I_D at 0.5 / 0.7 / 1 / 2 / 3 V: $+0.08 / +0.05 / +0.01 / -0.003 / -0.002\%$	passed
Linearity of I_D in μ_{band}	run_0003 run_0008	$\rightarrow \mu_{band} +7.9\%$ gives $I_{on} +7.9\%$	exact
SS_{min} robustness	run_0038 run_0012	I_{on} equal within 0.01%, but SS_{min} 83.1 vs 73.2 mV/dec	SS_{min} failed (metric withdrawn)
K. 31.8 nm mesh (graded, 16.5–18k nodes)	not refined (memory, time)	device excluded by the user	closed

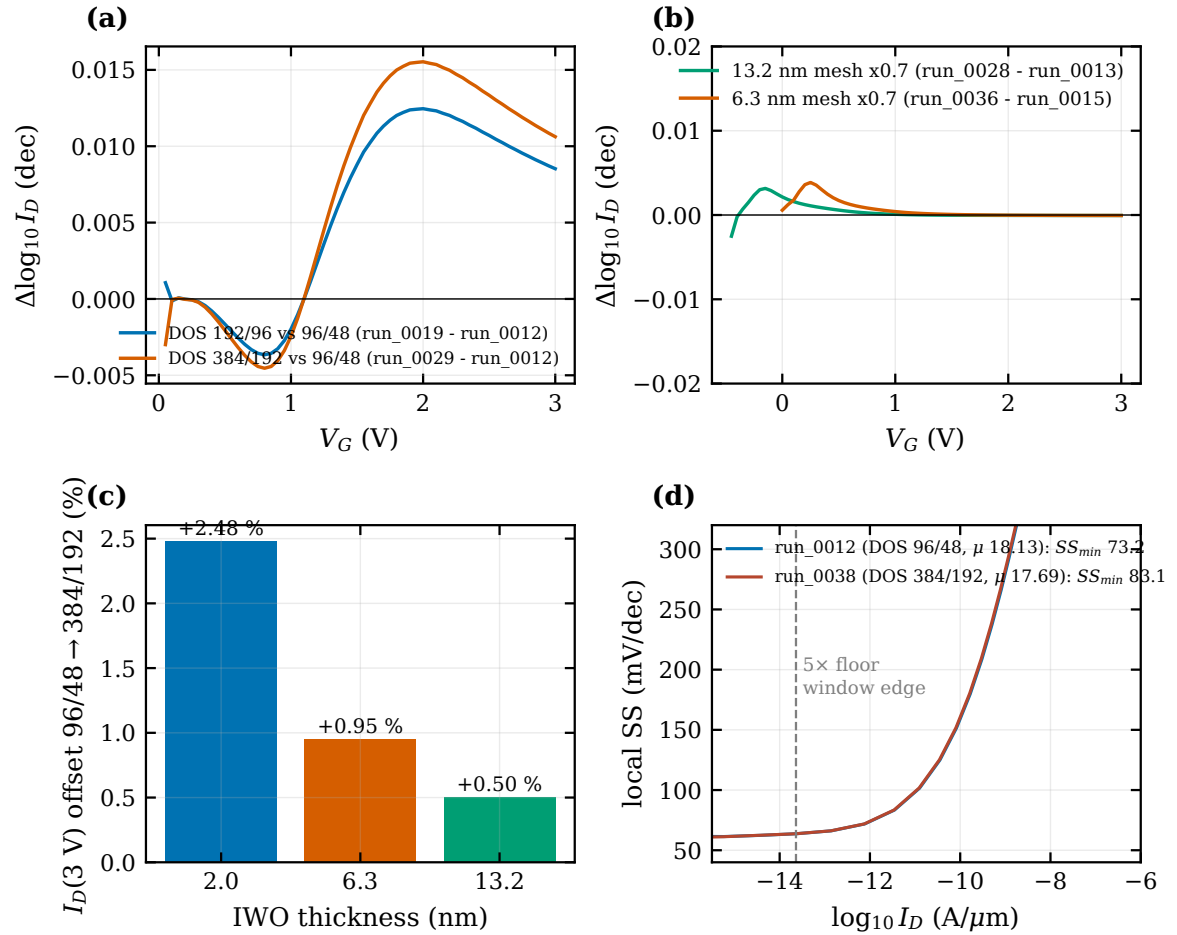


Figure 6.1. Numerical convergence of the final decks. (a) DOS discretization at 2 nm: $\Delta \log_{10} I_D$ versus V_G for 192/96 (run_0019) and 384/192 (run_0029) relative to 96/48 (run_0012). (b) Mesh refinement by 0.7 at 13.2 nm (run_0028 vs run_0013) and at 6.3 nm (run_0036 vs run_0015). (c) I_{on} at 384/192 relative to 96/48 versus thickness (+2.5, +1.0, +0.5 %; S6 report, B1–B3). (d) The $SS_{\min}$ artefact: local swing versus $\log_{10} I_D$ for run_0012 and run_0038, with the five-times-floor window marked; the minimum occurs at currents near the artefact threshold 2.30×10^{-14} A/ μm (S6), where the two runs differ although their curves are otherwise identical.

Discussion. Where checked, the choices of mesh, sweep form, structure representation, interface regularization, tolerance, width and carriers change the drain current by less than 0.01 dec, well below the calibration residuals (0.043, 0.059 and 0.081 dec for run_0038, run_0039 and run_0040). The only failed check is the DOS discretization at 96/48, which under-resolves the tail integral and underestimates the on-current: the 384/192 discretization gives +2.5 % more I_{on} at 2 nm, +1.0 % at 6.3 nm and +0.5 % at 13.2 nm. Because the offset depends on thickness, it was removed by recalibrating μ_{band} at 384/192 rather than by a common correction. The recalibrated values are $17.69 \text{ cm}^2/(\text{V s})$ at 2 nm (run_0038, I_{on} error 0.00 %), 11.58 at 6.3 nm (run at 11.41, run_0039, I_{on} error -1.48 %, curve rescaled by $\times 1.015074$) and 61.60 at 13.2 nm (run at 60.39, run_0040, I_{on} error -1.97 %, rescaled by $\times 1.020113$). The rescaling is exact because I_{D} is linear in MUN . The number of DOS levels also shifts $V_{\text{th,cc}}$ by only 1 mV and the fixed-current swing by less than 1 mV/dec, so no conclusion on threshold or swing depends on it. SS_{min} , in contrast, changed by 10 mV/dec between run_0012 and run_0038 with an unchanged curve and is withdrawn as a metric (CORR; Section 5.10).

6.7 The 2026-09-25 campaign machinery

The runs of 2026-09-25 (run_0030–run_0052) were executed as three campaigns defined in JSON plans (config/campaign_A_6p3_hypotheses.json, config/campaign_B_recal_predictions.json, config/campaign_C_6p3_nearEc_band.json; listed in Chapter C) and run sequentially by scripts/run_campaign.py. Figure 6.2 shows the execution chain.

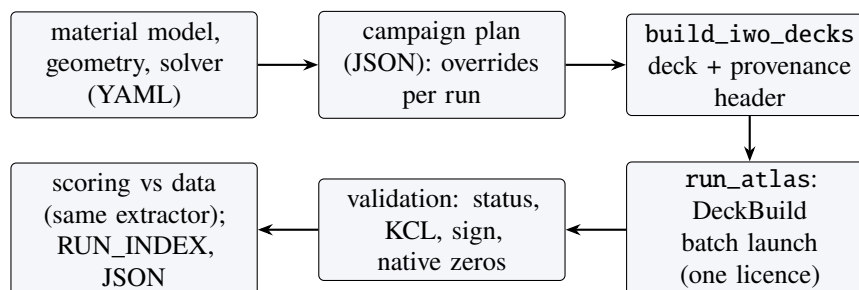


Figure 6.2. Execution chain of a V1 launch. A campaign plan lists dotted-path overrides of the configuration; each entry is turned into a deck by the generator, launched once in the real solver, validated and scored. No step replaces the solver by a surrogate.

Generator. scripts/build_iwo_decks.py evaluates the thickness laws of scripts/material_laws.py at the requested thickness, applies the overrides (-override path=value, e.g. traps.fixed_interface_charge_0) and writes the deck with its provenance header. It supports the two structure forms (-structure full|reduced) and the two sweep forms (-sweep stepped|pointwise).

Runner. scripts/run_atlas.py launches one deck with the verified batch recipe deckbuild.exe -run <deck> -outfile <stdout> in the run directory, enforces the timeout, validates and scores the result, writes execution.json, comparison.csv and idvg.dat, and appends one row to results/RUN_INDEX.csv. The option -rescore <run dir> repeats validation and scoring from the exported currents without a launch, and -output-vgs adds the $I_{\text{D}}-V_{\text{D}}$ families.

Campaign driver. `scripts/run_campaign.py` reads a plan, launches its entries one after another (the single 32-bit licence does not permit parallel runs, and specialist agents were not allowed to launch the solver), and writes the per-run metrics into `results/campaigns/<name>.json`. Campaign A (`run_0030–run_0036`) tested one mechanism at a time for the 6.3 nm film relative to `run_0014`, together with the mesh check L. Campaign B (`run_0037–run_0051`) ran the pre-registered B0 re-partition test, the DOS 384/192 recalibration, the I_D – V_D and temperature predictions, and the work-function and effective-mass isolation tests. Campaign C (`run_0052`) tested the near- E_c acceptor band proposed by specialist S4. The physics of these campaigns is discussed in Chapters 7, 8 and 10.

Budget and cost. The launch budget (`runner.max_launches`) was raised from 30 to 52 on 2026-09-25 at the user’s request and is exhausted (52/52). Of the 52 recorded launches, 49 produced scored currents (`run_0010` aborted, `run_0011` was never executed and `run_0018` timed out; Section A.1). Wall-clock times of single launches range from 125 s (`run_0012`) to 1926 s (`run_0043`, I_D – V_D at 13.2 nm); a final 2 nm transfer run at DOS 384/192 took 436 s (`run_0038`) and the corresponding 13.2 nm run 1122 s (`run_0040`).

Summary. Every simulated curve is obtained from a real ATLAS 5.28.1.R solution of a generated deck whose statements are documented against the manual (NUM). Where checked, the mesh ($\times 0.5$ at 2 nm on the V0 deck of the same generator, not repeated on the V1 2 nm deck; $\times 0.7$ at 6.3 and 13.2 nm), the sweep form, the structure representation, the interface regularization, the tolerances and the width convention change I_D by less than 0.01 dec. The DOS discretization 96/48 fails the 1 % criterion on I_{on} (+2.5/+1.0/+0.5 % at 2.0/6.3/13.2 nm); this error was removed by recalibrating μ_{band} at 384/192 (`run_0038–run_0040`). SS_{min} is not stable with respect to the grid and is withdrawn (Section 5.10); fixed-current swings are used instead. Native zeros are never replaced by a floor, so the measured off-state plateaus remain a limitation of the model.

With the numerics checked as far as the launch budget allowed, the next chapter uses these runs to calibrate the model on the 2 and 13.2 nm curves and shows, one parameter at a time, how each input controls the simulated transfer curve.

Chapter 7

Calibration and how each parameter controls the transfer curve

The derivation chapters (Chapters 4 and 5) fixed most inputs of the ATLAS model from the literature, from first-principles proxies and from physical laws in the thickness t . A small number of quantities cannot be fixed in this way and must be fitted to the measured transfer curves of Chapter 3. This chapter documents that fitting in full. It addresses four points: (i) the order in which the free quantities were fitted and the reason why this order gives nearly orthogonal adjustments; (ii) the contribution of every stage run, including runs that did not fit or did not finish; (iii) the final converged calibrated state and the features of the measured curves that it reproduces or misses; and (iv) the action of each model parameter on the transfer curve, as measured by one-at-a-time ATLAS runs rather than estimated analytically. Point (iv) forms the basis of the identifiability discussion in Section 9.7. Several parameters act on one transfer curve in exactly the same way, so that the calibrated set is one member of a family of equivalent calibrations and not a unique extraction. Throughout the chapter the model is regarded as *calibrated*, not validated: the out-of-sample test of the shared parameters, the 6.3 nm threshold, failed, and the registered 6.3 nm hypothesis tests B0 and C1 failed or partly failed (Chapter 8).

Metrics follow Section 5.10. In accordance with the corrections of the 2026-09-25 synthesis (analysis_2026-09-25/SYNTHESIS.md, section K), the windowed minimum slope $SS_{\min}$ is not used as evidence in this chapter. It depends on where a 5-point window falls on the 0.05 V grid and can move by up to 10 mV/dec without any physical change (S6 report, section 1.4). Subthreshold slopes are therefore reported at fixed current, $SS_{[10^{-11}, 10^{-10}]}$ and $SS_{[10^{-10}, 10^{-9}]}$ (in A/ μm). They are given together with SS_{cc} , the linear-extrapolation threshold $V_{\text{th,lin}}$, the turn-on delay (“gap”) $V_{\text{th,lin}} - V_{\text{th,cc}}$, the transconductance $g_{\text{m,max}}$, μ_{FE} and I_{on} at $V_{\text{G}} = 3 \text{ V}$.

7.1 Staged calibration strategy

7.1.1 What is fitted and what is not

The calibration was designed so that as few numbers as possible are adjusted and so that each adjusted number has a single dominant effect on the curve. The complete set of fitted quantities is given here as corrected in SYNTHESIS section K; the earlier statement that there is “one fitted electrostatic

value” is withdrawn. The set comprises one shared front fixed charge $Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$ for the 2 and 13.2 nm films (Section 4.13); a device-specific $Q_f = 8.7 \times 10^{10} \text{ cm}^{-2}$ for the 6.3 nm film, introduced only after the shared value had failed there (Chapter 8); one band mobility μ_{band} per film, that is, three numbers (Section 5.1); the tail-state anchor $N_t(2 \text{ nm}) = 2 \times 10^{19} \text{ cm}^{-3}$ and the tail width $W_{\text{TA}} = 40 \text{ meV}$ (Section 4.10); and the three constants of the effective donor law $N_{\text{d,eff}}(t)$, inherited from the V0 fit (Section 4.14).

The following quantities are assumed, that is, neither fitted nor measured on these devices: the gate work function, the electron affinity χ , the interface acceptor sheet D_{it} , the deep Gaussian, $\varepsilon(\text{Al}_2\text{O}_3)$ and the confinement law $\Delta E_c(t)$ with its effective-mass companion $m^*(t)$ (Sections 4.4, 4.5, 4.7, 4.11, 4.12 and 4.16). The anchor films are 2.0 and 13.2 nm. The 6.3 nm film was held out for a shared-parameter threshold test. The 31.8 nm film lies outside the scope of the model and is reported only as a documented failure (SYNTHESIS H.2, item 2.5).

7.1.2 Why the order gives near-orthogonal adjustments

Two exact properties of the drift-diffusion model with a uniform, constant mobility make the procedure well conditioned; both are demonstrated in Section 7.4.

1. A front fixed sheet charge produces a rigid shift of the whole transfer curve, $\Delta V_G = -q\Delta Q_f/C_{\text{ox}}$ (Equation (4.12)). With $C_{\text{ox}} = 8.955 \times 10^{-7} \text{ F cm}^{-2}$ this shift is -0.3095 V for $\Delta Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$, and ATLAS reproduces it to within 0.3 mV (run_0001→run_0003: -0.3092 V). The shift is constant to within 0.6 mV from 10^{-14} to $3 \times 10^{-7} \text{ A}/\mu\text{m}$ (SYNTHESIS D2).
2. At fixed electrostatics the drain current is exactly proportional to a uniform constant mobility (Equation (5.8)). The current ratio run_0008/run_0003 is 1.079171 against $18.13/16.8 = 1.079167$, and run_0029/run_0038 is 1.024879 against 1.024873 (S6 report, section 1.1).

The procedure was therefore as follows. (1) The literature/DFT laws were run with $Q_f = 0$ and the V0 mobilities (baseline laws). (2) The shared Q_f was fitted on the threshold, which leaves the slopes unchanged. (3) μ_{band} was fitted on I_{on} by exact linear rescaling, which moves $V_{\text{th,cc}}$ by less than 10 mV, since the constant-current definition couples the mobility into the threshold (Section 9.6). (4) The held-out 6.3 nm device was predicted with the shared parameters. (5) All runs were repeated on the physical structure with the stepped sweep (Sections 6.1 and 6.4). (6) The trap-DOS discretization was refined from 96/48 to 384/192 levels, and μ_{band} was recalibrated by exact rescaling. The subthreshold parameters $N_t(2 \text{ nm})$ and W_{TA} were fixed before stage (1) from the V0 fit; their origin is documented in Section 4.10. They were not re-fitted during these stages, and their leverage is measured in Section 7.4. Figure 7.1 summarises the procedure.

7.1.3 Goodness-of-fit measures

The “active” root-mean-square error of $\log_{10} I_D$ is evaluated where the measured current exceeds five times the off-band median floor (MANIFEST, calibrated_overlays). It is a useful summary for each film but is *not* comparable across films, because the active spans are $6.57 / 5.25 / 4.59$ decades at $2 / 6.3 / 13.2 \text{ nm}$ (SYNTHESIS H.2, item 2.11). The shape metrics listed in the introduction to this chapter are therefore always reported alongside it. The measured 13.2 nm window 10^{-11} – $10^{-10} \text{ A}/\mu\text{m}$ lies only 1.5 times above its floor ($6.6 \times 10^{-12} \text{ A}/\mu\text{m}$). Its slope is 136.1 mV/dec before

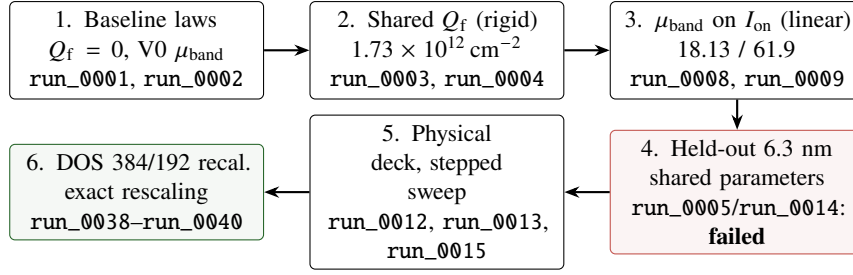


Figure 7.1. Staged calibration flow. Each stage adjusts one class of quantity whose action on the transfer curve is known exactly: a rigid shift for Q_f (stage 2) and a proportional current scaling for μ_{band} (stages 3 and 6). Stage 4 is the only out-of-sample test and failed for the threshold (Chapter 8); the 6.3 nm device was then given its own Q_f and μ_{band} (device-specific tuning, run_0007, run_0015, run_0039).

Table 7.1. Every run of the calibration chain (stages 1–6) with its robust metrics. $\Delta I_D(3\text{ V})$ is the simulated minus measured current at $V_G = 3\text{ V}$ in per cent. RMSE is the active log-current error (dec). $SS_{[-11, -10]}$ and $SS_{[-10, -9]}$ are the fixed-current slopes between 10^{-11} and 10^{-10} and between 10^{-10} and $10^{-9}\text{ A}/\mu\text{m}$. Stage numbers follow Figure 7.1; the stage labels in the run names are quoted. Units: V, mV/dec, $\text{cm}^2/\text{V s}$. The measured rows are from data/experimental_clean.csv; for 13.2 nm the slopes are raw (floor-subtracted: 91.8 and 118.8 mV/dec).

run	t	stage	change vs previous	$V_{\text{th,cc}}$	$V_{\text{th,lin}}$	$SS_{[-11, -10]}$	$SS_{[-10, -9]}$	μ_{FE}	$\Delta I_D(3\text{ V})$	RMSE
measured	2.0	data	–	0.662	1.663	121.2	194.8	12.15	–	–
measured	6.3	data	–	0.644	1.820	124.5	210.0	10.95	–	–
measured	13.2	data	–	0.083	1.044	136.1	121.6	50.17	–	–
run_0001	2.0	1 baseline laws	$Q_f = 0, \mu_{\text{band}} = 16.8$	0.995	1.838	125.3	216.4	10.19	–27.1	3.928
run_0002	13.2	1 baseline laws	$Q_f = 0, \mu_{\text{band}} = 57.6$	0.369	1.285	92.4	145.7	44.97	–21.3	1.030
run_0003	2.0	2 (“stage3”)	$Q_f = 1.73 \times 10^{12}$	0.686	1.561	125.0	216.7	10.46	–7.3	0.056
run_0004	13.2	2 (“stage3”)	$Q_f = 1.73 \times 10^{12}$	0.059	1.001	92.0	145.7	45.60	–7.0	0.086
run_0005	6.3	4 (“stage5”)	shared Q_f , held out	0.350	1.215	122.3	211.4	9.14	+26.8	0.678
run_0006	31.8	4 (“stage5”)	shared Q_f	–2.710	–0.168	84.9	140.2	39.90	–5.8	1.050
run_0007	6.3	4b (“stage5b”)	device $Q_f = 8.7 \times 10^{10}$	0.644	1.474	122.8	211.5	8.95	+6.1	0.082
run_0008	2.0	3 (“stage6”)	$\mu_{\text{band}} = 18.13$	0.676	1.561	122.5	212.9	11.29	+0.01	0.037
run_0009	13.2	3 (“stage6”)	$\mu_{\text{band}} = 61.9$	0.053	1.001	90.8	143.1	49.00	–0.02	0.081
run_0010	31.8	hypothesis	back donor sheet, $Q_f = 1.22 \times 10^{13}$	aborted in SOLVE INIT; no currents (NO_EXPORT)						
run_0011	6.3	3 (“stage6”)	$\mu_{\text{band}} = 11.69$	not executed; stopped with the chain (no execution.json)						
run_0012	2.0	5 physical deck	stepped sweep, DOS 96/48	0.676	1.559	122.5	212.9	11.27	+0.01	0.037
run_0013	13.2	5 physical deck	stepped sweep, DOS 96/48	0.053	0.999	90.8	143.1	48.95	–0.02	0.081
run_0014	6.3	5 validation	shared parameters, $\mu_{\text{band}} = 12.4$	0.350	1.212	122.3	211.4	9.13	+26.8	0.678
run_0015	6.3	5 tuned	$Q_f = 8.7 \times 10^{10}, \mu_{\text{band}} = 11.69$	0.651	1.471	124.1	214.4	8.42	–0.01	0.063
run_0038	2.0	6 recal. B1	DOS 384/192, $\mu_{\text{band}} = 17.69$	0.680	1.537	123.5	215.0	11.10	+0.00	0.043
run_0039	6.3	6 recal. B2	DOS 384/192, $\mu_{\text{band}} = 11.41$	0.655	1.466	124.8	216.5	8.27	–1.48	0.059
run_0040	13.2	6 recal. B3	DOS 384/192, $\mu_{\text{band}} = 60.39$	0.055	0.995	91.2	144.1	47.89	–1.97	0.081

and 91.8 mV/dec after floor subtraction, and both values are given (S6 report, section 1.4). The floor subtraction is an assumption, since the origin of the floor has not been determined (Section 9.4).

7.2 Calibration history: every stage run and what it taught

Table 7.1 lists every run of the calibration chain, including runs that did not fit, did not finish or were superseded. All numbers are taken from digest/runs.csv, which was extracted from the execution.json of each run with the unchanged extractor. Figure 7.2 shows the curves of the two anchor films through the stages.

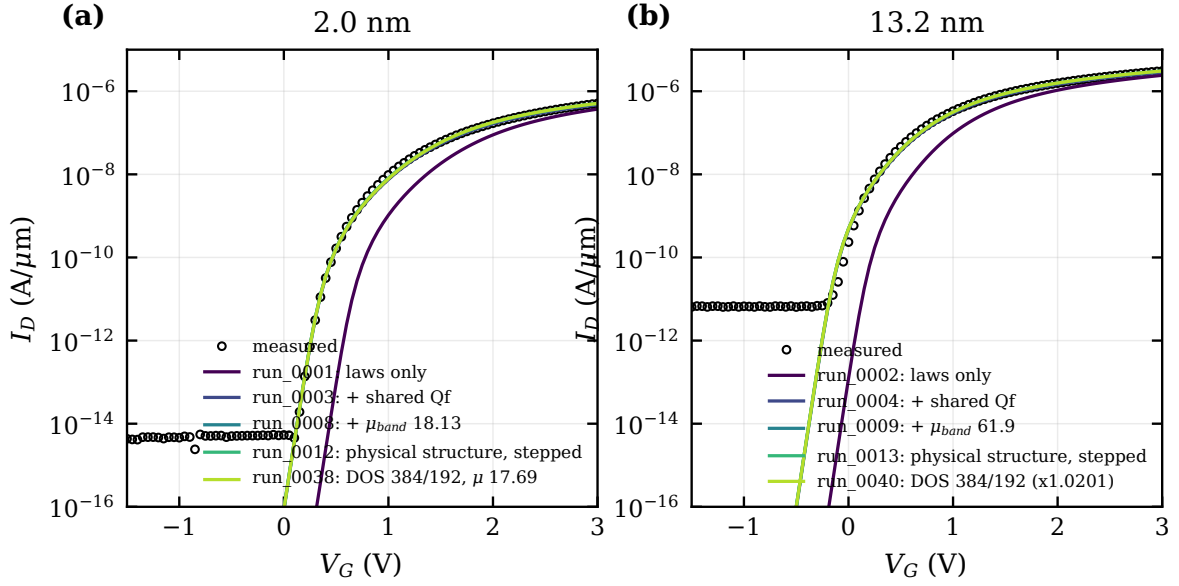


Figure 7.2. Staged calibration of the two anchor films, $\log_{10} I_D$ versus V_G at $V_D = 0.7$ V against the measured curves. 2.0 nm: run_0001 (baseline laws, $Q_f = 0$) → run_0003 (shared $Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$) → run_0008 ($\mu_{\text{band}} = 18.13$) → run_0012 (physical deck, stepped sweep) → run_0038 (DOS 384/192, $\mu_{\text{band}} = 17.69$). 13.2 nm: run_0002 → run_0004 → run_0009 → run_0013 → run_0040. The Q_f step is a pure horizontal translation, the mobility step a pure vertical scaling of the above-threshold current, and the last two steps are not visible at this scale. Metrics are in Table 7.1.

7.2.1 Stage 1: baseline laws (run_0001, run_0002)

With the literature/DFT laws, $Q_f = 0$ and the V0 band mobilities, the 2 nm threshold is 0.333 V too late (run_0001: $V_{\text{th,cc}} = 0.995$ V against 0.662 V), and the 13.2 nm threshold is 0.287 V too late (run_0002: 0.369 against 0.083 V). These law-only offsets differ by only 47 mV between the two anchors (SYNTHESIS K, FINAL_STATUS Conclusion 1), and this is the reason why a single shared charge could be tried. Before any electrostatic fit, the fixed-current subthreshold slope $SS_{[10^{-11}, 10^{-10}]}$ is already close to the data at 2 nm (125.3 against 121.2 mV/dec) and, at 13.2 nm, close to the floor-subtracted value (92.4 against 91.8 mV/dec). This slope is set by the tail parameters, which were fixed beforehand. The active RMSE (3.93 and 1.03 dec) is dominated by the threshold offset.

7.2.2 Stage 2: shared fixed charge (run_0003, run_0004)

The charge $Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$ moves both curves rigidly by -0.309 and -0.310 V, in agreement with the analytic value to within 1 mV, and leaves both fixed-current slopes unchanged to within 0.4 mV/dec. The residual threshold errors are +24 mV (2 nm) and -23 mV (13.2 nm), and the RMSE falls to 0.056 and 0.086 dec. $V_{\text{th,lin}}$ moves less than $V_{\text{th,cc}}$ (-0.277 and -0.284 V) because $V_{\text{th,lin}}$ is extracted at the transconductance maximum, which at 2 nm lies at the end of the sweep (3 V). A rigid shift of the curve then changes the part of the curve that is sampled at the sweep end (SYNTHESIS H.2, item 2.13). I_{on} changes by +27.1 and +18.2 % because it is read at fixed V_G .

7.2.3 Stage 4: the held-out 6.3 nm device (run_0005, run_0014) and the tuned alternative (run_0007, run_0015)

With the shared parameters and $\mu_{\text{band}} = 12.4$, the 6.3 nm threshold is predicted at 0.350 V against 0.644 V measured. This is a miss of -0.294 V, with $I_D(3\text{ V})$ 26.8 % too high and an RMSE of 0.678 dec. The reduced-deck run (run_0005) and the physical-deck repeat (run_0014) agree to within 0.1 mV. Because $\mu_{\text{band}} = 12.4$ is the V0 fit of the same curve, the test is out-of-sample for the threshold only (SYNTHESIS D4). About three quarters (log share) of the +26.8 % current excess follows from the threshold miss and one quarter from the choice of mobility (SYNTHESIS K). A device-specific $Q_f = 8.7 \times 10^{10} \text{ cm}^{-2}$, which is $1.64 \times 10^{12} \text{ cm}^{-2}$ less positive than the shared value, removes the threshold miss exactly (run_0007: +0.294 V, $V_{\text{th,cc}} = 0.644$ V). Chapter 8 shows that this tuning does not repair the shape of the on-state.

7.2.4 Stage 4, 31.8 nm: a documented failure (run_0006, run_0010)

The shared parameters give $V_{\text{th,cc}} = -2.71$ V at 31.8 nm (run_0006). The measured device is not turned off within the sweep, and the model with these laws is not applicable there. A hypothesis of a back-surface donor sheet for 31.8 nm (run_0010) aborted in the initial solution and produced no currents. The 31.8 nm film is not used as support for any claim (SYNTHESIS H.2, item 2.5).

7.2.5 Stage 3: band mobility (run_0008, run_0009; run_0011 not executed)

μ_{band} was raised from 16.8 to 18.13 (2 nm) and from 57.6 to 61.9 $\text{cm}^2/\text{V s}$ (13.2 nm), after which $I_D(3\text{ V})$ agrees to within 0.02 %. The threshold moves by -9 and -6 mV and the fixed-current slopes by -1.2 to -3.8 mV/dec. A change of mobility is thus not perfectly orthogonal to the threshold, because the constant-current threshold and the free-carrier contribution to the slope depend on the current scale (Section 9.6). The planned 6.3 nm mobility run run_0011 was stopped together with the chain and never executed; its setting was carried over into run_0015.

7.2.6 Stage 5: physical structure and stepped sweep (run_0012–run_0015)

The full physical deck (stack, overlaps and air region as described in Section 6.1) with the stepped V_G sweep reproduces the reduced, pointwise stage-3 runs to within -0.07 and -0.09 mV in $V_{\text{th,cc}}$ and 0.002 % in I_{on} (run_0008→run_0012, run_0009→run_0013; tables/SENSITIVITY_RESULTS.md). The physical deck and the reduced deck are therefore equivalent for the transfer metrics (SYNTHESIS E). Runs run_0014 and run_0015 repeat the shared and tuned 6.3 nm cases on the physical deck.

7.2.7 Stage 6: DOS refinement and exact recalibration (run_0038–run_0040)

Refinement of the tail-DOS discretization changes the current in a manner that depends on the gate voltage. At 2 nm, I_D changes by -0.95 / $+2.56$ / $+3.64$ / $+2.48$ % at $V_G = 0.7$ / 1.5 / 2 / 3 V (run_0012→run_0029), and $V_{\text{th,lin}}$ changes by -22 mV. The refinement series 96/48 → 192/96 → 384/192 at 2 nm (run_0012, run_0019, run_0029) changes I_{on} by $+1.98$ % and then by $+0.48$ %, which corresponds to an observed order of 2.03 with a Richardson residual of $+0.16$ % (S6 report, section 1.2; Section 6.6, Figure 6.1). The offset at 3 V scales with thickness as $1 : 0.38 : 0.20$, close

Table 7.2. Final converged calibration (DOS 384/192): measured versus simulated metrics. Simulated values are `run_0038` ($\times 0.999983$), `run_0039` ($\times 1.015074$) and `run_0040` ($\times 1.020113$), i.e. exact rescalings to $\mu_{\text{band}} = 17.6897 / 11.582 / 61.6046 \text{ cm}^2/\text{V s}$ (S6 report, section 1.6). *Measured 13.2 nm floor ($6.6 \times 10^{-12} \text{ A}/\mu\text{m}$) subtracted; raw values 136.1 ($SS_{[-11, -10]}$) and 160.0 mV/dec (SS_{cc}). Active RMSE from figures/MANIFEST.md. $\text{gap} = V_{\text{th,lin}} - V_{\text{th,cc}}$; $\text{gap7} = V(10^{-7} \text{ A}/\mu\text{m}) - V_{\text{th,cc}}$.

metric	2.0 nm		6.3 nm (device-tuned)		13.2 nm	
	meas.	<code>run_0038</code>	meas.	<code>run_0039</code>	meas.	<code>run_0040</code>
$Q_f \text{ (cm}^{-2}\text{)}$	–	1.73×10^{12}	–	8.7×10^{10}	–	1.73×10^{12}
$\mu_{\text{band}} \text{ (cm}^2/\text{V s)}$	–	17.69	–	11.58	–	61.60
$V_{\text{th,cc}} \text{ (V)}$	0.662	0.680	0.644	0.653	0.083	0.054
$SS_{[10^{-11}, 10^{-10}]} \text{ (mV/dec)}$	121.2	123.5	124.5	124.5	91.8*	90.9
$SS_{[10^{-10}, 10^{-9}]} \text{ (mV/dec)}$	194.8	215.0	210.0	215.6	118.8*	143.4
$SS_{\text{cc}} \text{ (mV/dec)}$	269.9	290.9	295.4	289.7	158.4*	193.0
$V_{\text{th,lin}} \text{ (V)}$	1.663	1.537	1.820	1.466	1.044	0.995
$\text{gap} \text{ (V)}$	1.001	0.856	1.176	0.814	0.962	0.941
$\text{gap7} \text{ (V)}$	1.051	1.024	1.224	1.085	0.571	0.630
$g_{\text{m,max}} \text{ (A/V}/\mu\text{m)}$	3.807×10^{-7}	3.478×10^{-7}	3.431×10^{-7}	2.631×10^{-7}	1.572×10^{-6}	1.531×10^{-6}
$\mu_{\text{FE}} \text{ (cm}^2/\text{V s)}$	12.15	11.10	10.95	8.39	50.17	48.85
$I_{\text{on}}/g_{\text{m,max}} \text{ (V)}$	1.337	1.463	1.176	1.534	1.953	2.005
$I_{\text{on}} \text{ (A}/\mu\text{m)}$	5.09×10^{-7}	0.00 %	4.03×10^{-7}	0.00 %	3.07×10^{-6}	0.00 %
active RMSE (dec)	–	0.0429	–	0.0630	–	0.0813

to the ratio of $N_t(t)$ (1 : 0.43 : 0.25); it is therefore a quadrature error of the discretized tail. Because I_D is exactly linear in μ_{band} , the three films were recalibrated at 384/192 by rescaling: `run_0038` at $\mu_{\text{band}} = 17.69$ (Ion error 0.00 %; exact value 17.6897), `run_0039` at 11.41 rescaled by 1.015074 to 11.582, and `run_0040` at 60.39 rescaled by 1.020113 to 61.6046 $\text{cm}^2/\text{V s}$. Path independence was verified: the ID-VD runs at $V_D = 0.7 \text{ V}$ agree with the rescaled transfer runs at $V_G = 1\text{--}3 \text{ V}$ to 0.000 % (S6 report, section 1.1). The earlier DOS 96/48 set (`run_0012`–`run_0015`) is superseded.

7.3 Final converged calibration at DOS 384/192

Table 7.2 gives the final calibrated state, and Figure 7.3 overlays it on the data. This is a *calibration* (evidence class CAL): every row is fitted to, or evaluated on, the curve with which it is compared.

Features that are reproduced. $V_{\text{th,cc}}$ is reproduced to within +19 / +9 / –29 mV, and the deep-subthreshold slope $SS_{[10^{-11}, 10^{-10}]}$ to within 2.3 mV/dec in all three films (13.2 nm floor-subtracted). I_{on} agrees by construction. At 13.2 nm the shape of the on-state is also reproduced (gap within 21 mV, $g_{\text{m,max}}$ within 2.6 %).

Features that are missed. Three deviations remain. First, in the window 10^{-10} – $10^{-9} \text{ A}/\mu\text{m}$ the model is too soft by +20 / +6 / +25 mV/dec. This is the same deficit that the trap-DOS inversion attributes to under-represented states at $E_c - 0.15$ to -0.20 eV (Section 9.4). Second, the shape of the on-state is missed at 6.3 nm (gap –0.362 V, $g_{\text{m,max}}$ –23.3 %) and, with the same sign but a smaller magnitude, at 2 nm (gap –0.145 V, $g_{\text{m,max}}$ –8.6 %) (SYNTHESIS D5). Third, $V_{\text{th,lin}}$ at 13.2 nm and 2 nm carries a bias of –11.7 and –5.4 mV from the 0.1 V solver grid above 1.5 V (S6 report, section 1.5); this bias is included in the numbers above.

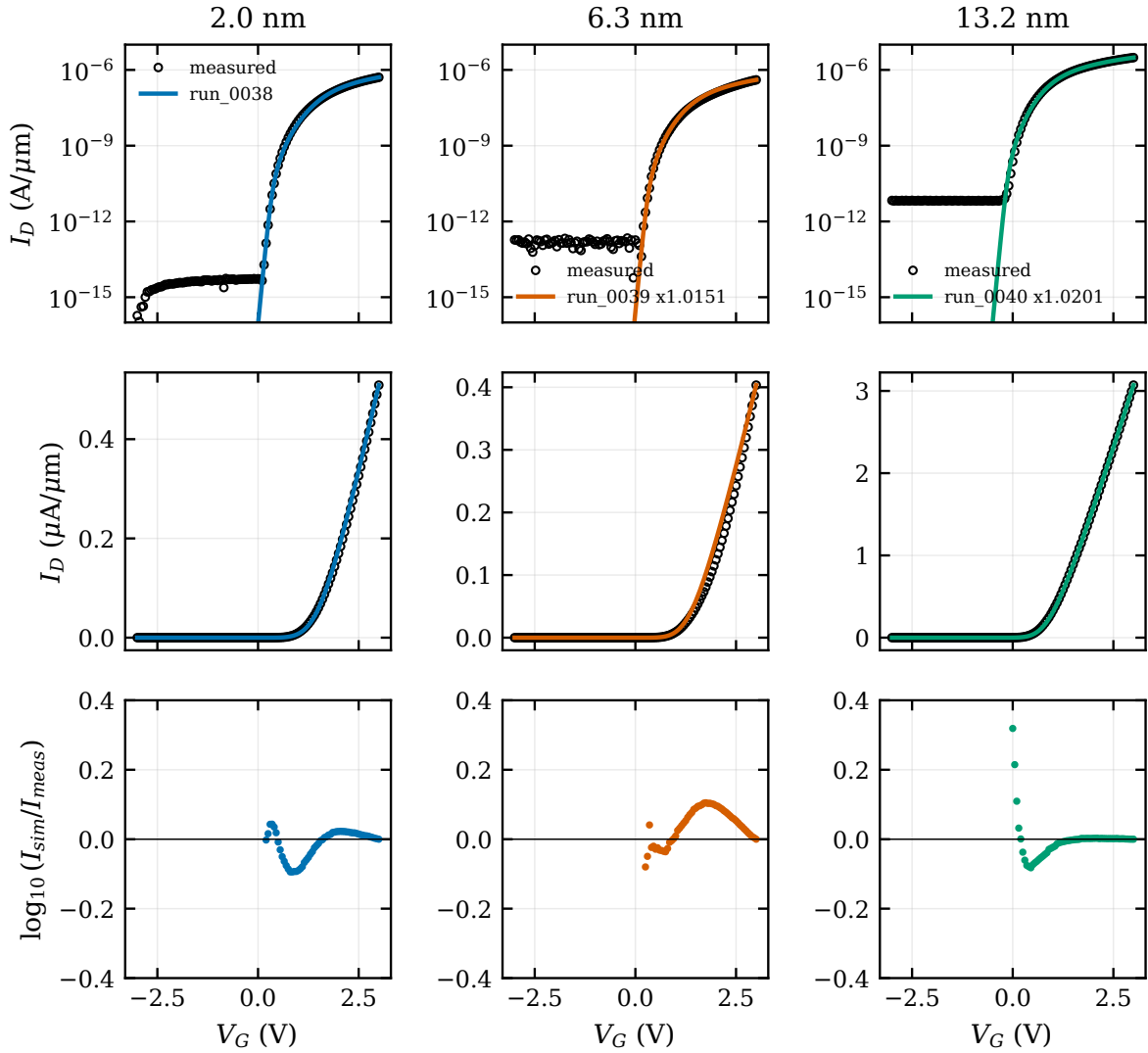


Figure 7.3. Final converged calibration against the measured transfer curves ($V_D = 0.7$ V): run_0038 (2.0 nm, $\times 0.999983$), run_0039 $\times 1.015074$ (6.3 nm) and run_0040 $\times 1.020113$ (13.2 nm). Row 1: $\log_{10} I_D$; row 2: linear I_D ; row 3: log residual. Simulated metrics: $V_{th,cc}$ 0.680 / 0.653 / 0.054 V, $V_{th,lin}$ 1.537 / 1.466 / 0.995 V, μ_{FE} 11.10 / 8.39 / 48.85 $\text{cm}^2/\text{V s}$, I_{on} 5.09×10^{-7} / 4.034×10^{-7} / 3.071×10^{-6} A/ μm ; active RMSE (measured current above five times the off-band median) 0.0429 / 0.0630 / 0.0813 dec. The late, steep 6.3 nm turn-on is not reproduced by the model (Section 8.3), and a smaller on-state residual of the same sign appears at 2 nm.

Numerical uncertainty of the final state. The numerical uncertainties, taken from S6 report, section 1.5, are ± 1.5 mV for $V_{th,cc}$ and $V(I)$, ± 1 mV/dec for the fixed-current SS, ± 2 mV plus the grid bias stated above for $V_{th,lin}$, ± 0.7 % for $g_{m,max}$ and μ_{FE} , and ± 0.2 % for I_{on} . A mesh refinement by a factor of 0.7 changes $V_{th,cc}$ by -0.32 mV at 6.3 nm (run_0036 vs run_0015) and by -0.36 mV at 13.2 nm (run_0028 vs run_0013); the maximum KCL error over the transfer runs is 2.5×10^{-16} A. These uncertainties are one to two orders of magnitude smaller than every model-data difference discussed in this report. Convergence is established for 300 K transfer metrics only. The V1 2 nm mesh, the DOS order at 6.3 and 13.2 nm, and the elevated-temperature and ID-VD runs have not been demonstrated (SYNTHESIS E and F; Section 6.6).

Status. The two anchor films are *calibrated* (CAL). The 6.3 nm row is a device-specific calibration with its own Q_f and μ_{band} , and it does not constitute evidence that the shared laws describe that device. The recalibrated band mobilities are 17.69 / 11.58 / 61.60 $\text{cm}^2/\text{V s}$; the non-monotonic value at 6.3 nm is discussed in Section 9.2.

7.4 How each parameter controls the transfer curve

Every parameter effect in this section is *measured in ATLAS* from pairs of executed runs that differ in exactly one input or in one declared package of inputs; none is estimated analytically. Three kinds of pair are used: stage pairs from the calibration chain; one-at-a-time perturbations of the final DOS 96/48 2 nm run run_0012 (run_0016, run_0020–run_0026); and isolation runs at DOS 384/192 relative to run_0038 and run_0040 (run_0048–run_0051; Section 7.5). The 6.3 nm hypothesis runs (run_0030–run_0037, run_0052) are also single-mechanism variants; they are analysed in Chapter 8.

7.4.1 Complete table of perturbation and counterfactual runs

Table 7.3 lists every pair. Differences are given as perturbed minus baseline and are computed from `digest/runs.csv`.

Table 7.3. Every one-at-a-time, stage-pair and counterfactual run, with its measured effect. Δ = perturbed – baseline. ΔI_{on} at $V_G = 3$ V. RMSE: active log error of baseline $\rightarrow$ perturbed (dec). Action: R = rigid shift; S = subthreshold slope; G = turn-on delay (gap); I = current scale; – = negligible.

t	change	runs	$\Delta V_{th,cc}$ (V)	$\Delta V_{th,lin}$ (V)	$\Delta SS_{[-11, -10]}$ (mV/dec)	$\Delta SS_{[-10, -9]}$ (mV/dec)	ΔI_{on} (%)	RMSE (dec)	action
<i>Stage pairs</i>									
2.0	$Q_f 0 \rightarrow 1.73 \times 10^{12} \text{ cm}^{-2}$	run_0001→run_0003	-0.309	-0.277	-0.4	+0.4	+27.1	3.928 $\rightarrow$ 0.056	R
13.2	$Q_f 0 \rightarrow 1.73 \times 10^{12} \text{ cm}^{-2}$	run_0002→run_0004	-0.310	-0.284	-0.4	0.0	+18.2	1.030 $\rightarrow$ 0.086	R
6.3	$Q_f 1.73 \times 10^{12} \rightarrow 8.7 \times 10^{10}$	run_0005→run_0007	+0.294	+0.260	+0.5	+0.1	-16.3	0.678 $\rightarrow$ 0.082	R
2.0	$\mu_{band} 16.8 \rightarrow 18.13$	run_0003→run_0008	-0.009	0.000	-2.5	-3.8	+7.9	0.056 $\rightarrow$ 0.037	I
13.2	$\mu_{band} 57.6 \rightarrow 61.9$	run_0004→run_0009	-0.006	0.000	-1.2	-2.5	+7.5	0.086 $\rightarrow$ 0.081	I
2.0	reduced $\rightarrow$ physical deck, stepped	run_0008→run_0012	-0.0001	-0.002	0.0	0.0	-0.002	0.037 $\rightarrow$ 0.037	–
13.2	reduced $\rightarrow$ physical deck, stepped	run_0009→run_0013	-0.0001	-0.002	0.0	0.0	+0.000	0.081 $\rightarrow$ 0.081	–
<i>One-at-a-time at 2.0 nm, baseline run_0012 (DOS 96/48)</i>									
2.0	$\chi -0.1$ eV (4.20)	run_0012→run_0020	+0.100	+0.090	0.0	0.0	-6.9	0.037 $\rightarrow$ 0.463	R
2.0	$\chi +0.1$ eV (4.40)	run_0012→run_0021	-0.100	-0.091	0.0	0.0	+7.0	0.037 $\rightarrow$ 0.361	R
2.0	N_d anchor $\times 2$ ($N_{d0} = 5 \times 10^{17}$)	run_0012→run_0022	-0.009	-0.008	-1.0	+0.7	+0.7	0.037 $\rightarrow$ 0.054	–

t change	runs	$\Delta V_{th,cc}$	$\Delta V_{th,lin}$	$\Delta SS_{[-11,-10]}$	$\Delta SS_{[-10,-9]}$	ΔI_{on}	RMSE	action
2.0 N_t anchor $\times 1.5$ ($3 \times 10^{19} \text{ cm}^{-3}$)	run_0012→run_0023	+0.082	+0.283	+21.8	+49.0	-25.1	0.037 → 0.263	S, G, I
2.0 W_{TA} +5 meV (45 meV)	run_0012→run_0024	+0.020	-0.002	+7.5	+4.6	+0.4	0.037 → 0.063	S
2.0 $D_{it} \times 3$ ($9 \times 10^{11} \text{ cm}^{-2} \text{ eV}^{-1}$)	run_0012→run_0025	+0.025	+0.024	+1.4	+0.1	-1.8	0.037 → 0.098	R (small)
2.0 $\varepsilon_{IWO} \rightarrow 10.55$	run_0012→run_0026	-0.001	-0.001	-0.4	-0.4	+0.7	0.037 → 0.037	–
2.0 overlap 2 → 4 μm	run_0012→run_0027	retracted: malformed deck (drain collapsed; see text)						
2.0 overlap 2 → 4 μm (corrected mesh)	run_0038→run_0055	$\Delta V_{th,cc} = 0.000 \text{ V}$, $\Delta I_{on} = -0.006 \%$, identical SS (DOS 384/192)						
2.0 confinement off (ΔE_c and $m^*(t)$ law)	run_0012→run_0016	-0.316	-0.299	+9.6	+15.3	+21.4	0.037 → 0.846	R (+S)
13.2 confinement off	run_0013→run_0017	-0.023	-0.020	+1.2	+1.1	+0.9	0.081 → 0.114	R
<i>Isolation at DOS 384/192, baselines run_0038 (2.0 nm) and run_0040 (13.2 nm)</i>								
2.0 gate WF +0.1 eV (4.8 eV)	run_0038→run_0048	+0.100	+0.092	0.0	0.0	-6.8	0.043 → 0.470	R
13.2 gate WF +0.1 eV (4.8 eV)	run_0040→run_0049	+0.100	+0.092	0.0	0.0	-5.0	0.081 → 0.221	R
2.0 $m^* \times 1.3$	run_0038→run_0050	-0.044	-0.043	-10.2	-18.0	+4.1	0.043 → 0.087	S
13.2 $m^* \times 1.3$	run_0040→run_0051	-0.029	-0.061	-4.7	-10.5	+5.8	0.081 → 0.122	S

Two remarks apply to the reading of the table. First, parameters that shift the curve exactly rigidly (Q_f , χ , WF) nevertheless change $V_{th,lin}$ by only 0.090–0.092 V per 0.100 V of $V_{th,cc}$. This is an extraction effect of the 0.1 V solver grid above 1.5 V and of the sweep end, not a physical effect (S6 report, section 1.5): the maximum $\Delta \log_{10} I_D$ between the WF-shifted curve and the translated baseline curve is 1.6×10^{-5} (2 nm) and 4.1×10^{-6} (13.2 nm). Second, the SS_{min} column of the earlier tables/SENSITIVITY_RESULTS.md (for example “–9.89” for the mobility step and “+10.7 / –12.7” for confinement off) is retired as a window-phase artefact (Section 5.10). On fixed-current windows these changes amount to at most 2–4 mV/dec (SYNTHESIS K).

The retracted overlap run (run_0027). Run run_0027 was intended to test a 4 μm source/drain overlap (override geometry.contact_overlap_um.value=4.0). Its deck extended the regions and electrodes to $x = 28 \mu\text{m}$ but left x.mesh ending at 24 μm . ATLAS therefore shortened the gate to 0–24 μm and the source to 0–3.905 μm , and it collapsed the drain to a zero-width line at $x = 24 \mu\text{m}$ (15 nodes) that touches the IWO only at a corner. The file deckbuild.out reports “Electrode shortened” three times (S5 report, section 2; SYNTHESIS D11). The run converged and was scored, but its apparent $\Delta I_{on} = -2.15 \%$ reflects a slightly different L (20.1 μm) and a point drain, not an effect of the overlap. Three statements are retracted (S6 report; finding of specialist S5, CORR): the “OAT overlap_4um” row of tables/SENSITIVITY_RESULTS (–2.17 % I_{on}), the statement that contact geometry is “irrelevant”, and the earlier claim that a 4 μm overlap changes the current by less than 2 %. The run is kept in the catalogue (Chapter A) and is excluded from every figure and table of sensitivities. Contact geometry and contact resistance were *not tested* in this study (Section 9.5).

7.4.2 Small multiples and difference curves

Figures 7.4 to 7.6 show the same pairs as curves.

7.4.3 The control matrix

Table 7.4 condenses Table 7.3 into the leverage of each parameter on each feature of the curve and names the parameters with which it is degenerate on a single curve. It forms the empirical basis of Table 9.2.

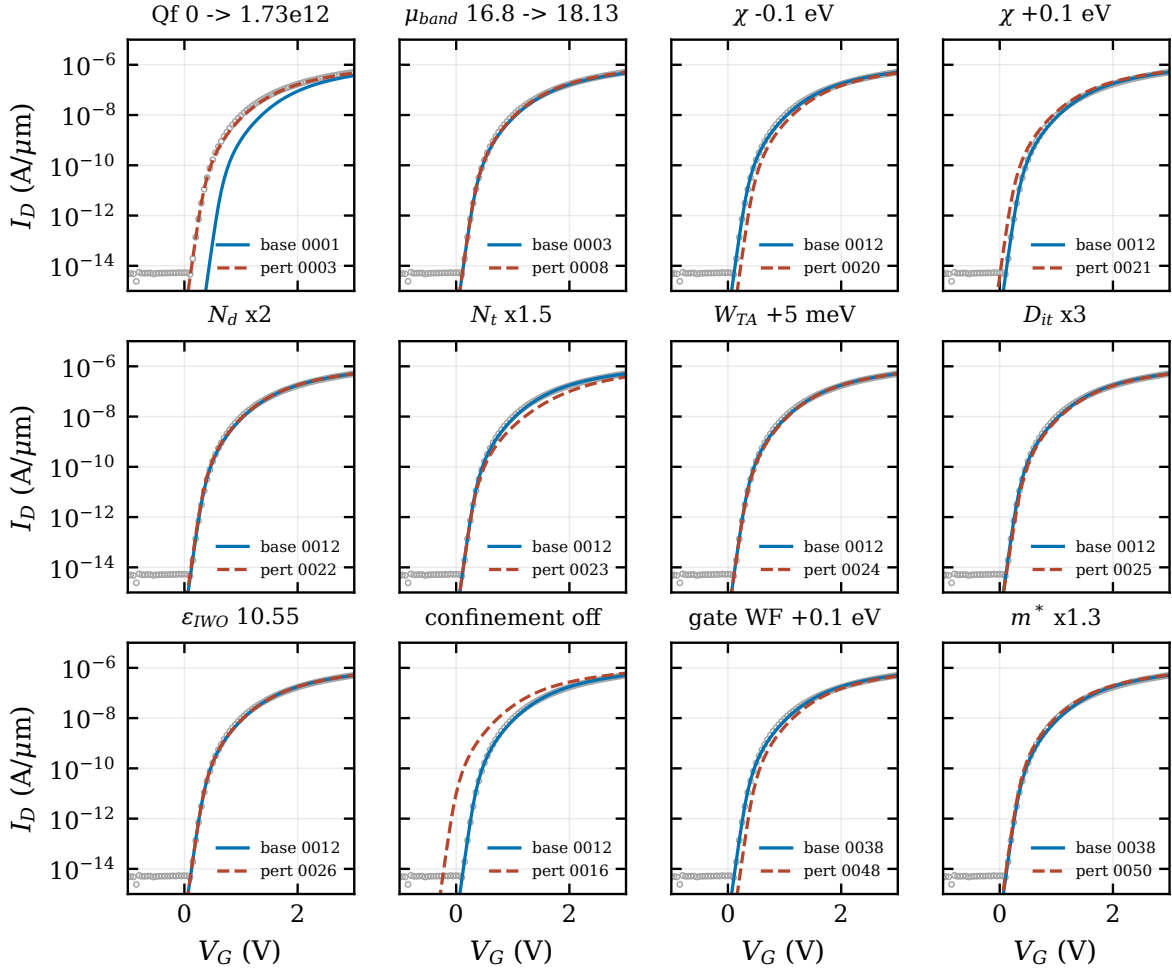


Figure 7.4. Effect of each parameter on the 2.0 nm transfer curve ($\log_{10} I_D$); each panel shows a perturbation against its baseline and the measured curve (grey circles). Q_f $0 \rightarrow 1.73 \times 10^{12} \text{ cm}^{-2}$ (run_0001 $\rightarrow$ run_0003); μ_{band} $16.8 \rightarrow 18.13$ (run_0003 $\rightarrow$ run_0008); χ $-0.1 / +0.1 \text{ eV}$ (run_0020 / run_0021 vs run_0012); $N_d \times 2$ (run_0022); $N_t \times 1.5$ (run_0023); $W_{TA} + 5 \text{ meV}$ (run_0024); $D_{it} \times 3$ (run_0025); $\epsilon_{IWO} = 10.55$ (run_0026); confinement off (run_0016); gate WF $+0.1 \text{ eV}$ (run_0048 vs run_0038); $m^* \times 1.3$ (run_0050 vs run_0038). run_0027 (overlap) is excluded as malformed. Four very different physical quantities (Q_f , χ , WF and, to a good approximation, confinement) produce the same horizontal translation, whereas only N_t changes the turn-on delay.

Table 7.4. Control matrix: measured leverage of each parameter on each transfer-curve feature (2.0 nm unless stated; from Table 7.3). “gap” is $\Delta(V_{\text{th,lin}} - V_{\text{th,cc}})$. Entries below 5 mV, 1 mV/dec or 1 % are marked “0”.

parameter (step)	$V_{\text{th,cc}}$	deep SS	mid SS	gap	I_{on} at 3 V	degenerate on one curve with
Q_f (10^{12} cm^{-2})	-0.179 V (exact)	0	0	0 ^a	via overdrive	WF, χ , ΔE_c , filled D_{it} , $N_d t$
WF, χ ($\pm 0.1 \text{ eV}$)	$\pm 0.100 \text{ V}$ (exact)	0	0	0 ^a	$\mp 7 \%$	Q_f , ΔE_c
confinement package	-0.316 V (2 nm), -0.023 V (13.2 nm)	+9.6	+15.3	+0.017	+21 %	Q_f (rigid part); m^* , N_t (residual)
μ_{band} (+7.9 %)	-0.009 V	-2.5	-3.8	+0.009	+7.9 % (exact)	N_t ($\times 1.5 \equiv \mu \times 1.33$ on I_{on}), R_{sd}
N_t ($\times 1.5$)	+0.082 V	+21.8	+49.0	+0.201	-25 %	μ_{band} (I_{on}); m^* (small steps)
W_{TA} (+5 meV)	+0.020 V	+7.5	+4.6	-0.021	0	D_{it} , deep states
D_{it} ($\times 3$)	+0.025 V	+1.4	0	0	-1.8 %	Q_f (when filled)
m^* ($\times 1.3$)	-0.044 V	-10.2	-18.0	0	+4.1 %	confinement residual, N_t
N_d ($\times 2$)	-0.009 V	-1.0	0	0	0	Q_f
ϵ_{IWO} (10.55)	0	0	0	0	0	—

^a Apparent $V_{\text{th,lin}}$ changes of 8–10 mV per 0.1 V are solver-grid/sweep-end effects (see text).

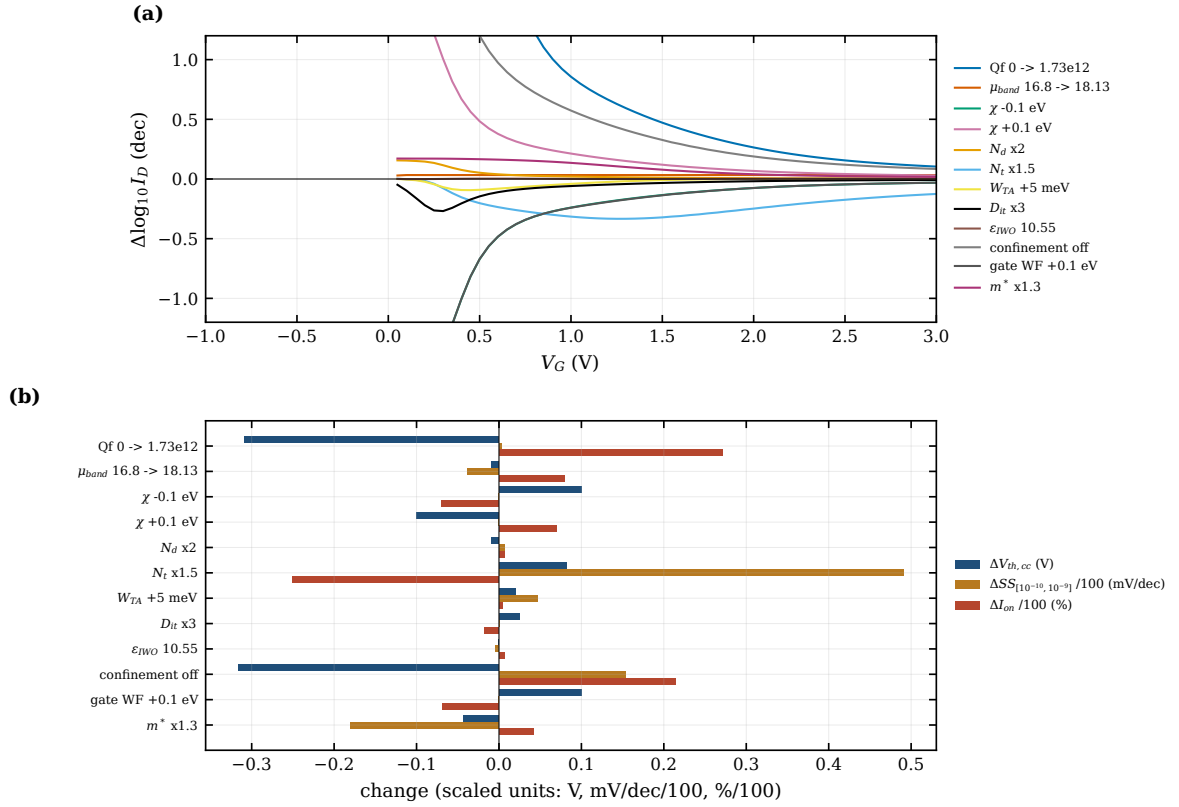


Figure 7.5. (a) $\Delta \log_{10} I_D(V_G) = \text{perturbed} - \text{baseline}$ for every one-at-a-time case at 2.0 nm. (b) Bars of $\Delta V_{th,cc}$ (V), $\Delta SS_{[10^{-10}, 10^{-9}]}$ (mV/dec) and ΔI_{on} (%): $Q_f 0 \rightarrow 1.73 \times 10^{12}$: -0.309, +0.4, +27.1; $\mu_{band} 16.8 \rightarrow 18.13$: -0.009, -3.8, +7.9; $\chi -0.1$ eV: +0.100, 0.0, -6.9; $\chi +0.1$ eV: -0.100, 0.0, +7.0; $N_d \times 2$: -0.009, +0.7, +0.7; $N_t \times 1.5$: +0.082, +49.0, -25.1; $W_{TA} +5$ meV: +0.020, +4.6, +0.4; $D_{it} \times 3$: +0.025, +0.1, -1.8; $\epsilon_{IWO} 10.55$: -0.001, -0.4, +0.7; confinement off: -0.316, +15.3, +21.4; gate WF +0.1 eV: +0.100, 0.0, -6.8; $m^* \times 1.3$: -0.044, -18.0, +4.1. Rigid parameters give flat difference curves in the subthreshold region and a decaying difference above threshold (because I_{on} is read at fixed V_G).

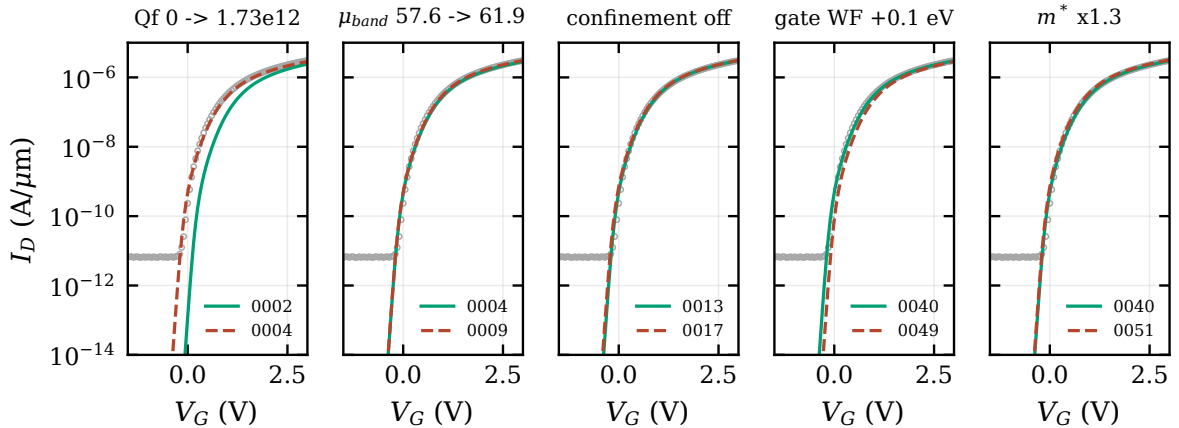


Figure 7.6. Parameter control at 13.2 nm: $Q_f 0 \rightarrow 1.73 \times 10^{12} \text{ cm}^{-2}$ (run_0002→run_0004), $\mu_{band} 57.6 \rightarrow 61.9$ (run_0004→run_0009), confinement off (run_0017 vs run_0013), gate WF +0.1 eV (run_0049 vs run_0040) and $m^* \times 1.3$ (run_0051 vs run_0040), against the measured 13.2 nm curve. Switching confinement off changes the 13.2 nm threshold by only -23 mV, against -316 mV at 2 nm.

The parameters fall into four groups. The first group consists of the rigid-shift parameters: Q_f , WF, χ , a filled D_{it} , $qN_d t/C_{ox}$ and the pure ΔE_c term. They amount to one number per curve. The pure $\Delta E_c(2\text{ nm}) = 0.353\text{ eV}$ is equivalent to $1.97 \times 10^{12}\text{ cm}^{-2}$ of fixed charge (SYNTHESIS D2).

The second group is the current-scale parameter μ_{band} . The current is exactly proportional to it, with a second-order coupling into $V_{th,cc}$ and into the fixed-current slope through the constant-current definition and the free-carrier capacitance.

The third group comprises the slope parameters: W_{TA} , m^* (through N_c) and the deep states. W_{TA} acts mainly on the deep window, whereas m^* acts on both windows and on the threshold. The m^* threshold shift equals the local slope times $\log_{10}(1.3^{1.5})$, since N_c changes the partition between free and trapped charge at fixed current (S6 report, section 2).

The fourth group contains only N_t , which is the only parameter that changes the turn-on delay. $N_t \times 1.5$ opens the gap by 0.20 V, but at the cost of +49 mV/dec in the mid window and +69 mV/dec in SS_{cc} (289.1 $\rightarrow$ 357.8, run_0012 $\rightarrow$ run_0023). This ratio, about 2.9 mV of gap per mV/dec of SS_{cc} , lies on the same trade-off line that limits every attempt to repair the 6.3 nm on-state (Section 8.3).

Donors and permittivity. The 2 nm result is insensitive to the donor law ($N_d \times 2$: -9 mV, +0.7 %) and to ϵ_{IWO} (10.55: -1 mV, +0.7 %) (run_0022, run_0026). The permittivity is therefore irrelevant at 2 nm, and the donor density cannot be identified there (0.036 V per 10^{18} cm^{-3} according to S4).

Evidence class. All entries are NUM: they are converged, model-internal statements on the response of ATLAS. They carry no information on which parameter values are physically correct.

7.5 Isolation tests of the gate work function and the effective mass

The package documents originally listed the work-function and effective-mass cases as “not run” (tables/SENSITIVITY_RESULTS.md); that statement is superseded. Runs run_0048–run_0051 were performed at the final DOS 384/192 with the same mobilities as their baselines run_0038 (2 nm) and run_0040 (13.2 nm), so that each difference isolates a single parameter.

7.5.1 Gate work function +0.1 eV (run_0048, run_0049)

The work function was raised from 4.7 to 4.8 eV (Section 4.16). Both films shift by exactly +0.100 V in $V_{th,cc}$, and all three fixed-current slopes remain unchanged to 0.0 mV/dec. The curves are exact translations: the maximum $\Delta \log_{10} I_D$ after a 0.100 V translation is 1.6×10^{-5} at 2 nm and 4.1×10^{-6} at 13.2 nm (S6 report, section 2). I_{on} decreases by 6.8 % at 2 nm and by 5.0 % at 13.2 nm relative to the baselines with the same mobility. The value of -6.9 % at 13.2 nm given earlier in the evidence brief was computed relative to the measurement and is corrected here (SYNTHESIS A.3). The different changes of I_{on} do not indicate a thickness-dependent effect of the work function. I_{on} is read at a fixed V_G of 3 V, and $I_{on}/g_{m,max}$ is 1.46 V at 2 nm against 2.00 V at 13.2 nm, so that the same translation removes a different fraction of the current. The thickness step is unchanged: $\Delta V_{th,cc}(2 \rightarrow 13.2)$ is -0.6251 V with and without the shift, and $I_{on}(13.2)/I_{on}(2)$ changes from 5.914 to 6.031 (+2.0 %), a change that a refit of Q_f removes.

Consequence. WF, χ and Q_f act as one number per curve and are independent of thickness (NUM, demonstrated). A thickness-dependent interface dipole or charge would therefore be indistinguishable from $\Delta E_c(t)$ on ID-VG, because the Q_f shift is constant over five decades of current.

7.5.2 Effective mass $\times 1.3$ (run_0050, run_0051)

The bulk effective mass was raised from 0.208 to 0.2704 m_0 , and the thickness coefficient was scaled by the same factor (0.131 to 0.17043); the whole $m^*(t)$ law was thus multiplied by 1.3 (Section 4.7). In V1, m^* enters only through the effective density of states of the conduction band, $N_c \propto m^{*3/2}$, and not through the confinement law. The effect is not a rigid shift (0.17 dec deviation from a translation), as the following values show.

film (runs)	$\Delta V_{th,cc}$ (V)	$\Delta SS_{[-11, -10]}$	$\Delta SS_{[-10, -9]}$	ΔSS_{cc}	$\Delta V_{th,lin}$ (V)	ΔI_{on}
2.0 nm (run_0038→run_0050)	−0.044	−10.2	−18.0	−17.7	−0.043	+4.1 %
13.2 nm (run_0040→run_0051)	−0.029	−4.7	−10.5	−13.2	−0.061	+5.8 %

A larger N_c places a larger fraction of the induced charge in free electrons at a given quasi-Fermi level, so that a given current is reached with less trapped charge. The threshold therefore moves earlier by the local slope times $\log_{10}(1.3^{1.5}) = 0.171$ dec, and the slopes become steeper. With respect to the thickness dependence, $m^* \times 1.3$ changes the 2 $\rightarrow$ 13.2 nm threshold step by +14.9 mV (2.4 % of the step), the I_{on} ratio by +1.6 % and the SS_{cc} difference from −97.1 to −92.7 mV/dec (S6 report, section 2). The effect is thus of second order for the thickness trend (NUM).

7.5.3 What the isolation runs imply for confinement

The V1 confinement package ($\Delta E_c(t)$ together with $m^*(t) \rightarrow N_c$) is only approximately rigid. At 2 nm its removal shifts the curve by 0.345 V at 10^{-14} , 0.302 V at 10^{-8} and 0.311 V at 3×10^{-7} A/ μ m, and it changes SS_{cc} from 289.1 to 303.7 mV/dec (run_0012 vs run_0016). Beyond a rigid shift, its only signature on the same curve is therefore +38.8 mV of non-rigidity between 10^{-11} and 10^{-8} A/ μ m and +14.6 mV/dec in SS_{cc} . Log-linear scaling of the isolation run run_0050 and of the tail run run_0023 shows that this signature is nearly reproduced by $m^* \times 0.8$ ($SS_{cc} + 14.6$ mV/dec, +37.5 mV) or by $N_t \times 1.09$ (+14.6 mV/dec, +34 mV) (SYNTHESIS D2 and [SYN-d]). The degeneracy is with m^* and N_t , not with W_{TA} : on SS_{cc} , $W_{TA} + 5$ meV gives only +0.8 mV/dec and $D_{it} \times 3$ only +0.5 mV/dec. Confinement therefore cannot be identified from one transfer curve per thickness; the consequences are developed in Section 9.3.

Summary.

- The calibration fits few numbers, in an order that exploits two exact model properties: Q_f acts as a rigid shift (−0.309 V for 1.73×10^{12} cm $^{-2}$, analytic −0.3095 V; run_0001→run_0003) and I_D is exactly proportional to μ_{band} (to 6×10^{-6}).
- Fitted: shared Q_f ; the 6.3 nm Q_f ; μ_{band} for each film (17.69 / 11.58 / 61.60 cm 2 /V s at DOS 384/192); the N_t anchor and W_{TA} ; the $N_{d,eff}$ constants. Assumed: WF, χ , D_{it} , deep states, $\varepsilon(\text{Al}_2\text{O}_3)$, $\Delta E_c(t)$, $m^*(t)$.

- Final state (run_0038–run_0040, CAL): $V_{th,cc}$ within +19 / +9 / –29 mV, deep-subthreshold slope within 2.3 mV/dec, I_{on} exact; mid-window slope too soft by +20 / +6 / +25 mV/dec; the 6.3 nm on-state is missed (gap –0.36 V, $g_{m,max}$ –23 %).
- One-at-a-time runs show four classes of action: rigid (Q_f , WF, χ , ΔE_c), current scale (μ_{band}), slope (W_{TA} , m^*) and turn-on delay (N_t only, at a large slope cost). Donors and ϵ_{IWO} are irrelevant at 2 nm.
- run_0027 (overlap) is retracted as malformed. The corrected repetition run_0055 (overlap 4 μ m, mesh following the contacts; the first attempt run_0053 exceeded the 900 s time limit) changes I_{on} by –0.006 % and leaves $V_{th,cc}$ and the subthreshold slope unchanged, so with ideal Ohmic contacts the assumed overlap does not influence the results. SS_{min} is retired (Section 5.10).
- The calibrated set is one member of a family of equivalent calibrations, not a unique parameter extraction; the model is calibrated, not validated.

The only film that was not used for calibration, 6.3 nm, is the film for which the shared-parameter model fails. The next chapter decomposes that failure and tests every hypothesis proposed for it.

Chapter 8

The 6.3 nm discrepancy: a controlled hypothesis study

The 6.3 nm device is the only film that was held out of the calibration, and it is the film for which the shared-parameter model fails. In the measurement it behaves almost like the 2 nm device ($V_{th,cc}$ 0.644 against 0.662 V, μ_{FE} 10.95 against 12.15 $\text{cm}^2/\text{V s}$; Table 3.2), whereas the thickness laws of the model, anchored on 2 and 13.2 nm, place it much closer to the thick film. This chapter treats the discrepancy as a controlled experiment. It determines the size of the miss and whether the miss has more than one component. It examines which physical mechanisms, implemented one at a time in ATLAS with all other inputs held fixed, can close the miss and at what cost to the rest of the curve. Finally, it reports the outcome of two predictions that were registered before their runs finished. The result is a demonstrated *decomposition* of the discrepancy into a rigid threshold part and an on-state shape part, a set of quantitative exclusions, and no assigned cause. Every simulated number carries its run ID. Unless stated otherwise, the metrics are taken from `digest/runs.csv` and from the metric table of the S6 report (section 3.1).

8.1 The problem: a failed shared-parameter test and a tuned fit that still misses

8.1.1 The measured 6.3 nm curve

The measured 6.3 nm device has $V_{th,cc} = 0.644$ V, $V_{th,lin} = 1.820$ V, a turn-on delay (gap) $V_{th,lin} - V_{th,cc} = 1.176$ V, $V(10^{-7} \text{ A}/\mu\text{m}) - V_{th,cc}(\text{gap}) = 1.224$ V, fixed-current slopes of 124.5 (10^{-11} – $10^{-10} \text{ A}/\mu\text{m}$) and 210.0 mV/dec (10^{-10} – $10^{-9} \text{ A}/\mu\text{m}$), $SS_{cc} = 295.4$ mV/dec, $g_{m,max} = 3.431 \times 10^{-7} \text{ A/V}/\mu\text{m}$ ($\mu_{FE} = 10.95 \text{ cm}^2/\text{V s}$) and $I_{on} = 4.03 \times 10^{-7} \text{ A}/\mu\text{m}$ at $V_G = 3$ V, $V_D = 0.7$ V (S6 report, section 3.1; Figure 3.2). After passing threshold, the current rises late and then steeply: $I_{on}/g_{m,max} = 1.176$ V, which is smaller than at 2 nm (1.337 V) and at 13.2 nm (1.953 V). The data come from a single device ($N = 1$); device spread and sweep history are not recorded (Section 3.5).

Table 8.1. Horizontal offset $V_G^{\text{meas}}(I) - V_G^{\text{sim}}(I)$ (V) at fixed currents ($\text{A}/\mu\text{m}$), from SYNTHESIS A.2 [SYN-a]. A rigid error is a constant row; a shape error is a row that changes with current.

run	10^{-11}	10^{-10}	10^{-9}	10^{-8}	3×10^{-8}	10^{-7}	2×10^{-7}	3×10^{-7}
6.3 nm run_0014 (shared)	+0.293	+0.296	+0.294	+0.316	+0.358	+0.455	+0.484	+0.449
6.3 nm run_0015 (tuned Q_f)	-0.003	-0.003	-0.007	+0.011	+0.047	+0.132	+0.141	+0.087
6.3 nm A6 run_0034 (no confinement)	+0.063	+0.062	+0.057	+0.077	+0.117	+0.213	+0.239	+0.201
6.3 nm C1 run_0052	+0.010	+0.009	-0.004	-0.009	+0.024	+0.134	+0.201	+0.212
6.3 nm B0 run_0037	+0.142	+0.094	-0.010	-0.138	-0.167	-0.125	-0.088	-0.084
2.0 nm B1 run_0038	+0.004	+0.002	-0.019	-0.040	-0.029	+0.008	+0.034	+0.037
13.2 nm B3 run_0040	+0.006	+0.051	+0.029	-0.015	-0.029	-0.029	-0.026	-0.020

8.1.2 The shared-parameter test (run_0014)

With the thickness laws, the shared $Q_f = 1.73 \times 10^{12} \text{ cm}^{-2}$ fitted on 2 and 13.2 nm, and $\mu_{\text{band}} = 12.4 \text{ cm}^2/\text{V s}$, the model predicts $V_{\text{th,cc}} = 0.350 \text{ V}$, a miss of -0.294 V (run_0014; identical to the reduced-deck run run_0005 to within 0.1 mV). The fixed-current slope is reproduced ($SS_{[10^{-11}, 10^{-10}]}$ -2.2 mV/dec), but the on-state is not: the gap error is -0.314 V , the $g_{\text{m,max}}$ error -16.6% and the I_{on} error $+26.8 \%$, and the active RMSE is 0.678 dec. Because $\mu_{\text{band}} = 12.4$ is the V0 fit of the same curve (config/iwo_material_model.yaml, line 105), the test is out-of-sample for the threshold only, and it *failed* (OOS, SYNTHESIS D4). The earlier description of run_0014 as “validation, nothing tuned” and as “reproduced in shape” is withdrawn (SYNTHESIS K).

8.1.3 The tuned fit (run_0015, run_0039)

Assigning the device its own $Q_f = 8.7 \times 10^{10} \text{ cm}^{-2}$ ($1.64 \times 10^{12} \text{ cm}^{-2}$ less positive, $+0.294 \text{ V}$) and $\mu_{\text{band}} = 11.69$ removes the threshold miss ($+7 \text{ mV}$) and matches I_{on} (run_0015). At DOS 384/192 the same holds with $\mu_{\text{band}} = 11.582$ (run_0039 $\times 1.015074$). The on-state, however, remains wrong: the gap error is -0.356 (run_0015) and -0.362 V (run_0039), the $V_{\text{th,lin}}$ error -0.349 V , and the $g_{\text{m,max}}$ error -23.1 and -23.3% . A single electrostatic number thus removes the threshold part but not the shape.

8.1.4 Decomposition into a rigid part and a shape part

The decomposition uses the horizontal offset $V_G^{\text{meas}}(I) - V_G^{\text{sim}}(I)$ at fixed currents (Table 8.1; SYNTHESIS A.2, check [SYN-a]). A positive value means that the measured curve reaches the given current later.

Three observations follow from the table. The first concerns the rigid part. In run_0014 the offset is 0.294 V and constant to within 3 mV from 10^{-11} to $10^{-9} \text{ A}/\mu\text{m}$, which is an exact fixed-charge signature equivalent to $1.64 \times 10^{12} \text{ cm}^{-2}$ ($q/C_{\text{ox}} = 0.179 \text{ V per } 10^{12} \text{ cm}^{-2}$). Its size depends on the confinement law: without the law (A6, run_0034) it is 0.057 V, equivalent to $0.32 \times 10^{12} \text{ cm}^{-2}$. The second observation concerns the shape part. The offset minus its value at $10^{-9} \text{ A}/\mu\text{m}$ is, at $10^{-7} / 2 \times 10^{-7} \text{ A}/\mu\text{m}$, $+0.161 / +0.190 \text{ V}$ (run_0014), $+0.139 / +0.148 \text{ V}$ (run_0015) and $+0.156 / +0.182 \text{ V}$ (A6). The shape part is therefore independent of the confinement law and of Q_f . The third observation is a comparison with the other films: the residual swing of the final calibration is 0.07–0.08 V at 2 and 13.2 nm, against about 0.15 V at 6.3 nm, where it has a “late, then steep” form.

The rigid part is what a shared-parameter model must explain on the threshold, and the shape part is what any model of the 6.3 nm device must explain, whatever value of Q_f it uses. Both are demonstrated decompositions of the data relative to the model (NUM on DATA). Neither is assigned to a cause in this chapter.

8.2 Campaign A: one mechanism at a time

8.2.1 Design

Campaign A implemented six physically motivated modifications of the shared-parameter 6.3 nm deck (`run_0014`), each with all other inputs unchanged, together with a mesh check (A7). The design file (config A) was written after the `run_0014` miss was known. The campaign is therefore a set of post-hoc hypothesis tests, and A6 in particular is classed as POST, not as pre-registered (SYNTHESIS A.3 and H.2, item 2.15).

A1, thickness error (`run_0031`). Film thickness of 5.3 nm instead of 6.3 nm in the geometry and in every thickness law, $\mu_{\text{band}} = 12.4$. This case tests whether a 1 nm metrology error explains the offset through $\Delta E_c(t)$, $N_t(t)$ and $N_{d,\text{eff}}(t)$.

A2, fewer donors (`run_0032`). $N_{d0} = 10^{16} \text{ cm}^{-3}$ (effectively no donors). This case tests whether the donor term $qN_d t / C_{\text{ox}}$ explains the offset; the analytic bound is $\leq 0.03 \text{ V}$ (Section 4.14).

A3, back-surface charge (`run_0030`). A negative fixed sheet $Q_b = -1.64 \times 10^{12} \text{ cm}^{-2}$ at the air-exposed back surface, corresponding to the full rigid miss (Section 4.15).

A4, front interface traps (`run_0033`). Interface acceptor sheet $\times 5$ ($1.5 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$) (Section 4.12).

A5, combination (`run_0035`). $t = 5.8 \text{ nm}$, $N_{d0} = 10^{16} \text{ cm}^{-3}$, $Q_b = -1.0 \times 10^{12} \text{ cm}^{-2}$, $\mu_{\text{band}} = 12.4$: a “plausible combination” of small effects.

A6, no confinement (`run_0034`). ΔE_c and $m^*(t)$ laws switched off, with $Q_f = 4.76 \times 10^{10} \text{ cm}^{-2}$ taken from a no-confinement fit of the 2 nm device.

A7, mesh check (`run_0036`). The tuned deck with mesh spacing $\times 0.7$, a numerical duplicate of `run_0015` ($\Delta V_{\text{th,cc}} = -0.32 \text{ mV}$, RMSE 0.0625 vs 0.0626 dec; NUM, PASS).

Two further hypotheses were tested after campaign A, with predictions registered before the runs finished (Section 8.3): B0, a re-partition of tail density and mobility (`run_0037`), and C1, a narrow acceptor band just below E_c (`run_0052`).

8.2.2 Results

Table 8.2 gives every 6.3 nm variant in terms of the shape metrics, and Figures 8.1 and 8.2 show the curves and the errors.

Table 8.2. All 6.3 nm variants: model minus measured (measured row absolute), from S6 report, section 3.1. run_0014, run_0015, A1–A6 and B0 use DOS 96/48; B2 and C1 use 384/192 (the DOS effect on these metrics is ≤ 1 mV, 0.6 mV/dec, 1 %). $SS_{[-11, -10]}$, $SS_{[-10, -9]}$: fixed-current slopes between the stated decades of $A/\mu\text{m}$. $g_{m,\text{max}}$ and I_{on} in per cent of the measured value. The last column gives what the mechanism would need to close the rigid miss, or the outcome of its registered test.

case	run	$V_{\text{th,cc}}$ (V)	$SS_{[-11, -10]}$ (mV/dec)	$SS_{[-10, -9]}$ (mV/dec)	SS_{cc} (mV/dec)	gap (V)	gap7 (V)	$g_{m,\text{max}}$ (%)	I_{on} (%)	RMSE (dec)	requirement / verdict
measured	–	0.644	124.5	210.0	295.4	1.176	1.224	3.431×10^{-7}	4.03×10^{-7}	–	–
shared laws	run_0014	–0.294	–2.2	+1.4	–10.5	–0.314	–0.161	–16.6	+26.8	0.678	OOS threshold test: fa
tuned Q_f, μ	run_0015	+0.007	–0.4	+4.4	–6.6	–0.356	–0.137	–23.1	0.0	0.063	CAL; shape missed
B2 (tuned, 384/192)	run_0039	+0.009	0.0	+5.6	–5.7	–0.362	–0.139	–23.3	0.0	0.059 [†]	CAL; shape missed
A1 $t = 5.3$ nm	run_0031	–0.258	0.0	+5.0	–4.9	–0.297	–0.137	–18.1	+20.8	0.613	needs $t \approx 2.1$ nm; reje
A2 $N_{d0} = 10^{16}$	run_0032	–0.264	–2.0	–1.3	–14.7	–0.329	–0.168	–17.5	+24.4	0.634	+0.030 V max; rejecte
A3 back -1.64×10^{12}	run_0030	–0.017	–21.8	–43.2	–64.1	–0.452	–0.267	–18.4	+14.4	0.237	SS broken; rejected as
A4 $D_{it} \times 5$	run_0033	–0.243	+2.8	+3.5	–9.1	–0.319	–0.160	–16.9	+23.1	0.595	needs $\times 23$ –24; rejecte
A5 combination	run_0035	–0.079	–14.9	–28.4	–45.9	–0.405	–0.221	–19.6	+13.9	0.246	73 % of miss, SS –46;
A6 no confinement	run_0034	–0.057	+0.6	+4.8	–7.7	–0.337	–0.154	–18.4	+9.2	0.211	POST; threshold withi
B0 re-partition	run_0037	+0.010	+48.8	+103.8	+115.5	+0.026	+0.115	–5.0	–7.6	0.270	OOS: on-state met, SS
C1 near- E_c band	run_0052	+0.004	+1.4	+12.9	+9.0	–0.237	–0.137	–3.5	+16.0	0.081	OOS: partly failed

[†] At $\mu_{\text{band}} = 11.41$ before rescaling; 0.0630 dec after the exact rescaling (figures/MANIFEST.md).

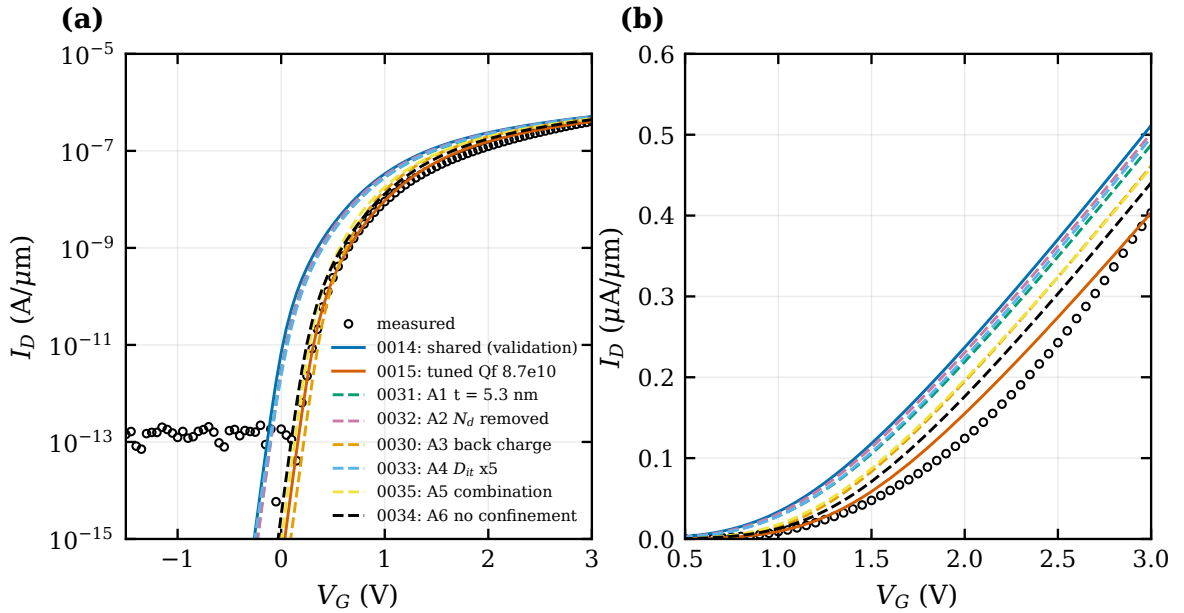


Figure 8.1. The 6.3 nm device against every campaign-A variant (DOS 96/48): measured curve versus run_0014 (shared-parameter test), run_0015 (tuned Q_f and μ_{band}), A1 run_0031 ($t = 5.3$ nm), A2 run_0032 (donors removed), A3 run_0030 (back charge $-1.64 \times 10^{12} \text{ cm}^{-2}$), A4 run_0033 ($D_{it} \times 5$), A5 run_0035 (combination) and A6 run_0034 (no confinement): $\log_{10} I_D$ and a linear on-state panel. Every variant that moves the threshold to the data leaves the linear on-state below and to the left of the measurement.

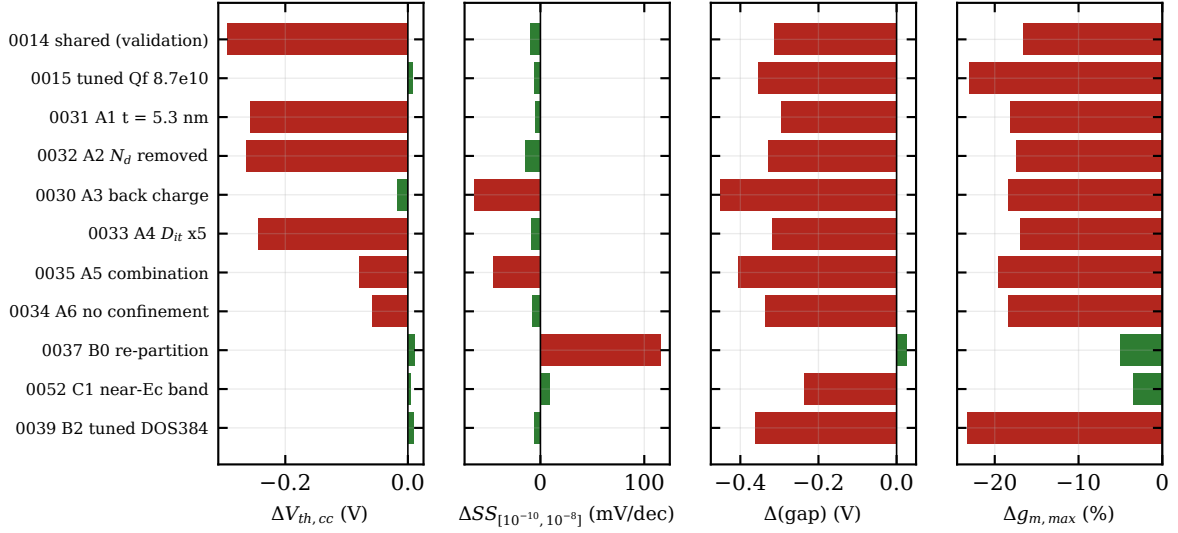


Figure 8.2. Errors of every 6.3 nm hypothesis relative to the measurement ($V_{th,cc}$ 0.644 V, SS_{cc} 295.4 mV/dec, gap 1.176 V, $g_{m,max}$ 3.431×10^{-7} A/V/ μm); green marks agreement within 0.05 V / 20 mV/dec / 0.1 V / 8 %. $V_{th,cc}$, SS_{cc} , gap and $g_{m,max}$ errors: run_0014 -0.294 V, -10.4 , -0.314 V, -16.6 %; run_0015 $+0.007$ V, -6.5 , -0.356 V, -23.1 %; A1 run_0031 -0.258 V, -4.9 , -0.297 V, -18.1 %; A2 run_0032 -0.264 V, -14.7 , -0.329 V, -17.5 %; A3 run_0030 -0.017 V, -64.1 , -0.452 V, -18.4 %; A4 run_0033 -0.243 V, -9.1 , -0.319 V, -16.9 %; A5 run_0035 -0.079 V, -45.9 , -0.405 V, -19.6 %; A6 run_0034 -0.057 V, -7.7 , -0.337 V, -18.4 %; B0 run_0037 $+0.010$ V, $+115.6$, $+0.026$ V, -5.0 %; C1 run_0052 $+0.004$ V, $+9.1$, -0.237 V, -3.5 %; B2 run_0039 $+0.009$ V, -5.7 , -0.362 V, -23.3 %. No case is green in all four metrics.

8.2.3 Mechanism-by-mechanism reading

A1, thickness error: rejected (model-conditional). A film thinner by 1 nm moves $V_{th,cc}$ by only $+0.036$ V (0.386 against 0.350 V), which is 12 % of the required $+0.294$ V. Closing the rigid miss through $\Delta E_c(t)$ would require $t \approx 2.1$ nm, that is, a metrology error of 67 %. The gap error remains at -0.297 V and the $g_{m,max}$ error at -18 %. With a physical, weaker $\Delta E_c(t)$ beyond 3 nm the gain would be even smaller (Section 9.3). NUM.

A2, fewer donors: rejected. Removal of the donors gives $+0.030$ V (analytic bound 0.028 V), about ten times too little. NUM.

A3, back-surface negative charge: rejected as the sole cause (model-conditional). A back sheet of $-1.64 \times 10^{12} \text{ cm}^{-2}$ matches the threshold (-17 mV). This shows that at threshold the back surface has nearly the same lever arm as the front (0.96–1.2 times, not the 1.7–2.4 times of a naive Qt/ϵ_s estimate; S1 verdicts). However, the back charge destroys the agreement in the subthreshold region (SS_{cc} -64.1 mV/dec, with changes of -22 and -43 mV/dec in the two fixed-current windows), closes the gap further (0.724 V against 1.176 V) and increases the offset at 10^{-7} A/ μm to $+0.285$ V. NUM.

A4, front interface traps: rejected at plausible densities. $D_{it} \times 5$ gives $+0.051$ V with the gap unchanged (0.857 against 0.862 V). Closing the miss would require about $\times 23$ – 24 ($\approx 7 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$) at 6.3 nm only, on a gate stack that is shared with the other films, and the slope would degrade accordingly. The discriminator registered in config A (traps shift the threshold *and* degrade the slope) was confirmed qualitatively. NUM.

A5, combination: rejected. A plausible combination covers 0.215 of 0.294 V (73 %) but costs -45.9 mV/dec in SS_{cc} , because its back-charge component carries the slope penalty of A3. NUM.

A6, no confinement: the threshold miss is reduced but not explained. Without the confinement law, and with Q_f taken from a no-confinement fit of the 2 nm film, the 6.3 nm threshold is predicted to within -0.057 V (RMSE 0.21 against 0.68 dec). This statement concerns the threshold only, since $\mu_{band} = 12.4$ is not held out, and the on-state fails as in every other run (gap -0.337 V). A6 was designed after the run_0014 miss was known and is classed as POST. It does not show that the data prefer the absence of confinement. Without the law it is the 13.2 nm device that requires its own charge (1.43×10^{12} against 0.047×10^{12} cm $^{-2}$); the no-confinement 13.2 nm threshold is 0.331 V, a miss of $+0.248$ V, obtained from the exact rigid identity applied to run_0017. Thresholds with a single offset do not discriminate between the two readings (Section 9.3).

Any rigid charge as the whole miss: rejected. In the tuned run the offset still varies from -0.007 to $+0.141$ V between 10^{-9} and 2×10^{-7} A/ μ m (Table 8.1), and no rigid term can remove this variation. Eight charge or electrostatic variants (run_0014, run_0015, run_0030–run_0035) give gaps of 0.72–0.88 V against 1.176 V measured (SYNTHESIS D5), and they change $g_{m,max}$ by at most 2 % relative to each other.

8.3 The on-state shape problem and two pre-registered tests

8.3.1 What must be reproduced

The measured 6.3 nm curve reaches 10^{-7} A/ μ m 1.224 V after $V_{th,cc}$ and then rises with $I_{on}/g_{m,max} = 1.176$ V. Every simulation whose fixed-current slopes lie within about 15 mV/dec of the data in every window has gap7 ≤ 1.09 V and $I_{on}/g_{m,max} \geq 1.41$ V (S6 report, section 3.3). The feature requires additional charge that is trapped between $V_{th,cc}$ and $V_{th,lin}$ (about 2×10^{12} cm $^{-2}$ according to the S4 surrogate) and then released into a steeper rise, without additional charge in the subthreshold region. Two trap-side remedies that could plausibly produce this behaviour were proposed by the specialist analyses and registered, with file timestamps, before the corresponding ATLAS runs finished.

8.3.2 B0: re-partition of tail density and band mobility (run_0037)

The S2 analysis proposed that the non-monotonic μ_{band} (lower at 6.3 than at 2 nm) is an artefact of the imposed $N_t(t)$ law and that a partition consistent with the shape would be $\mu_{band} = 19.3 / 19.6 / 62.7$ cm 2 /V s with $N_t = 2.4 \times 10^{19} / 2.7 \times 10^{19} / 4.8 \times 10^{18}$ cm $^{-3}$. B0 implements this proposal at 6.3 nm: the exponent of the N_t law is set to 0 with $N_t = 2.66 \times 10^{19}$ cm $^{-3}$, $\mu_{band} = 19.6$ and $Q_f = 9.8 \times 10^{11}$ cm $^{-2}$. The S2 prediction was written at 18:59:35Z, config B at 19:06:27Z, and run_0037 ended at 19:21:30Z (UTC file times; S6 report, section 3.2).

Outcome. The on-state predictions passed: $V_{th,cc}$ $0.64 \pm 0.03 \rightarrow 0.654$ V, $V_{th,lin}$ $1.82\text{--}1.90 \rightarrow 1.856$ V, I_{on} -5 to -10 % $\rightarrow -7.6$ %, and μ_{FE} $10.5\text{--}11.0 \rightarrow 10.40$ (1 % low). The subthreshold prediction failed by a wide margin: SS_{cc} is 410.9 against 295.4 mV/dec measured (+115.5), and the error in $SS_{[10^{-11}, 10^{-10}]}$ is +48.8 mV/dec. Rescaling the mobility so that I_{on} matches (21.2 cm 2 /V s) gives $g_{m,max}$ +2.8 % and gap +0.026 V. B0 thus reproduces the whole on-state, but only by adding

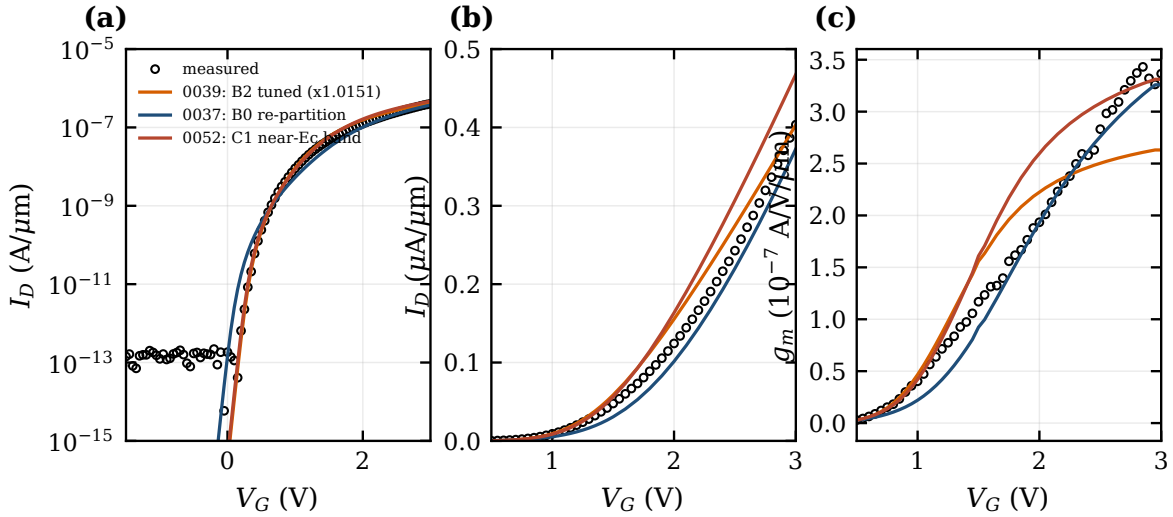


Figure 8.3. The 6.3 nm on-state shape: measured curve against B2 (run_0039, rescaled by 11.581987/11.41), B0 (run_0037, tail re-partition) and C1 (run_0052, near- E_c acceptor band). (a) $\log_{10} I_D$; (b) linear I_D ; (c) $g_m(V_G)$ from the numerical gradient on the 0.05 V grid. B0 reproduces the on-state (b, c) but not the subthreshold region (a); C1 retains the subthreshold behaviour and recovers only part of the delayed, steep rise; B2 rises too early and too slowly.

116 mV/dec to SS_{cc} . The re-partition is therefore rejected as the explanation (OOS, failed). A competing blind calculation by S1 for the same configuration (SS_{cc} 418 $\pm$ 10 mV/dec) agreed with the outcome. Its code, however, was saved during the run (19:18:00Z) and its output after the run had ended, so it is classed as POST (SYNTHESIS A.3).

8.3.3 C1: a narrow acceptor band just below E_c (run_0052)

The S4 analysis proposed states concentrated within 1–2 $k_B T$ of E_c together with a higher mobility. C1 adds a Gaussian acceptor band $N_{GA} = 9.0 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$ at $E_{GA} = E_c - 0.03 \text{ eV}$ with $W_{GA} = 0.02 \text{ eV}$ (“2e12” in the run name), keeps $Q_f = 8.7 \times 10^{10} \text{ cm}^{-2}$ and sets $\mu_{band} = 15.4$ (DOS 384/192). The S4 prediction was written at 19:21:48Z, config C at 19:35:23Z, and run_0052 ended at 22:47:22Z.

Outcome (relative to run_0015). The gap was predicted to change by +0.28 V and changed by +0.119 V (**failed**); $g_{m,max}$ was predicted to rise by +24 % and rose by +25.5 % (**passed**); SS_{cc} was predicted to change by +23 (+20 to +30) mV/dec and changed by +15.6 mV/dec (close); and I_{on} was predicted to change by ± 3 % and changed by +16.0 % (**failed**). The stated falsifiers ($g_{m,max}$ not rising, or $SS_{cc} > 330$ mV/dec) were not triggered. At matched I_{on} ($\mu_{band} = 13.28$) the $g_{m,max}$ error returns to –16.8 %, with the gap error unchanged at –0.237 V. C1 therefore recovers about one third of the gap miss (33 % of 0.356 V) and 28 % of the $g_{m,max}$ miss at a moderate cost in slope ($SS_{cc} + 15.6$ mV/dec). The verdict is partly failed (SYNTHESIS section D): a near- E_c band is at most a partial contributor (OOS). The DOS resolution of the 20 meV band was not checked, since a repeat with 768/384 levels was not run.

8.3.4 Registered predictions and outcomes

Table 8.3 collects every prediction made for the 6.3 nm study together with its registration status. It follows section 3.2 of the S6 report, with the classification corrections of SYNTHESIS A.3.

Table 8.3. Predictions for the 6.3 nm study and their outcomes. OOS: prediction file time-stamped before the run ended. POST: designed or evaluated after the outcome was known.

#	prediction (source)	predicted → obtained	outcome	class
P1	S2 for B0 (run_0037)	$V_{th,cc} 0.64 \pm 0.03 \rightarrow 0.654$; $V_{th,lin} 1.82 \rightarrow 1.90$; $I_{on} -5 \text{ to } -10 \% \rightarrow -7.6 \%$; $\mu_{FE} 10.5 \rightarrow 11.0$; slope +3 to +15 over run_0015 → $SS_{[-11, -10]} +49$	on-state passed, sub-threshold failed: re-partition rejected	OOS
P2	S1 1-D code for B0	$SS_{cc} 418 \pm 10 \rightarrow 410.9$; $V_{th,cc} 0.656 \rightarrow 0.654$; $g_{m,max} 3.31 \rightarrow 3.26 \times 10^{-7}$; $V_{th,lin} 1.889 \rightarrow 1.856$	agreed	POST
P3	S4 for C1 (run_0052), vs run_0015	gap +0.28 → +0.119; $g_{m,max} +24 \rightarrow +25.5 \%$; $SS_{cc} +23 \rightarrow +15.6$; $I_{on} \pm 3 \rightarrow +16.0 \%$	partly failed	OOS
P3'	S1: gap cannot open without SS_{cc} rising well above 295	gap +0.12 of +0.36 needed at $SS_{cc} 304$	consistent	OOS
P4	A6 (run_0034) held-out threshold	$V_{th,cc} -57 \text{ mV}$; slope within 8 mV/dec; gap -0.34; $I_{on} +9 \%$	threshold within 0.1 V; on-state failed	POST
P5	A2 analytic bound	$\leq +0.03 \rightarrow +0.030 \text{ V}$	passed	OOS
P6	A4 discriminator	traps shift $V_{th,cc}$ and degrade slope → +0.051 V, $SS_{[-11, -10]} +5.0$	passed (qualitative)	OOS
P8	S3: SP/BQP moves $V_G (n_s = 10^{12} \text{ cm}^{-2})$ by +0.15–0.19 V at 6.3/13.2 nm	not run (launch budget)	untested	PRED

8.3.5 The trade-off line

Every trap-side change moves the model along one line in the plane of gap against SS_{cc} (S6 report, section 3.3). From B2 to C1 the gap opens by +0.125 V at +15 mV/dec (8.5 mV of gap per mV/dec), and from B2 to B0 by +0.388 V at +121 mV/dec (3.2 mV per mV/dec). The 2 nm tail run lies on the same kind of line: $N_t \times 1.5$ opens the gap by +0.201 V at +68.7 mV/dec in SS_{cc} (run_0012→run_0023; about 2.9 mV per mV/dec; Table 7.3). Even at the efficiency of C1, the missing +0.36 V would cost $SS_{cc} \approx 332 \text{ mV/dec}$ against 295 mV/dec measured. Within the V1 model form (equilibrium tails and a constant, density-independent mobility), the on-state and the subthreshold region of the 6.3 nm device cannot be fitted simultaneously.

8.4 Verdict: what is and is not understood

Demonstrated (NUM on DATA). The departure of the 6.3 nm device from the shared calibration consists of two separable parts. The first is a rigid threshold offset (0.294 V with the confinement law, $\equiv 1.64 \times 10^{12} \text{ cm}^{-2}$; 0.057 V without it). The second is an on-state shape part that depends on gate voltage (turn-on delay 0.31–0.36 V too small, $g_{m,max}$ 17–23 % too low) and is independent of Q_f and of the confinement law.

Rejected, with the quantitative reason (NUM, model-conditional). The following explanations are rejected: thickness error (+0.036 V per nm; would need $t \approx 2.1 \text{ nm}$), donor reduction (+0.030 V), front D_{it} (needs $\times 23$ –24), back-surface charge as the sole cause ($SS_{cc} -64 \text{ mV/dec}$), the plausible

combination (73 % of the offset, $SS_{cc} -46$), the S2 re-partition B0 ($SS_{cc} +116$), a near- E_c band as the full explanation (C1: 33 % of the gap, $I_{on} +16$ %), and any rigid charge as the whole miss.

Not determined. Device-to-device spread, sweep history and bias stress during the sweep, the measurement ambient, and a thickness error beyond the model-conditional bound have not been determined. Extrapolation of the positive-bias-stress shifts of the same device family [1, p. 9] to a -3 to $+3$ V sweep gives 0.003 – 0.24 V (S4 report, section 6). The upper end reaches the order of the rigid offset, so that sweep history is a possible but unquantified confounder (Section 9.6). With one device per thickness, the rigid offset cannot be distinguished from process variation. Separating a film property from spread at the size of the offset requires $n \approx 15.7\sigma^2/\Delta^2$ replicates per thickness, in practice 3–5 devices with dual sweeps and recorded dwell (SYNTHESIS Q4 and F, Tier 4).

Ranked model-form causes of the shape part (plausible, unverified).

1. *Constant, density-independent mobility.* The feature requires a current that rises faster than the free charge above threshold without additional subthreshold charge. This is the signature of a carrier-density-dependent (percolation or power-law) mobility, which would also lower the slope. The ATLAS version used ignores MOBILITY statements between SOLVE statements (a verified runner fact), so that V1 cannot emulate such a mobility within a sweep (SYNTHESIS C5).
2. *Rigid equilibrium tails,* without above-threshold trapping or sweep history.
3. *Gate-field quantization with local tails.* S3 estimates $+0.15$ – 0.19 V in $V_G(n_s = 10^{12} \text{ cm}^{-2})$ for $t \geq 6.3$ nm, but this effect should also appear at 13.2 nm, where the classical model already fits (P8, untested).

The miss is largest at 6.3 nm, smaller at 2 nm (gap -0.145 V, $g_{m,max} -8.6$ %) and negligible at 13.2 nm (-0.021 V, -2.6 %). This ordering is consistent with a distinction between disordered thin films and ordered thick films (Section 9.2), but it is a correlation over three devices, not a demonstration.

Falsification tests. Hypothesis C5 (density-dependent mobility) is falsified if a density-dependent mobility calibrated on 2 and 13.2 nm leaves the 6.3 nm gap miss above 0.1 V at $SS_{cc} \leq 330$ mV/dec, or if dual sweeps show the feature to be hysteretic. Hypothesis C4 (device-specific offset) is supported if replicates give $\sigma(V_{th}) < 0.1$ V with a mean at least 0.2 V away from the shared law (SYNTHESIS C). The registered predictions of Chapter 10 do not test the 6.3 nm shape directly.

Summary.

- The shared-parameter model misses the held-out 6.3 nm threshold by -0.294 V (run_0014): a failed out-of-sample test of the threshold only.
- The discrepancy is a rigid part ($0.294 \text{ V} \equiv 1.64 \times 10^{12} \text{ cm}^{-2}$ with the confinement law; 0.057 V without) plus an on-state shape part ($+0.14$ to $+0.19$ V of extra delay between 10^{-9} and $2 \times 10^{-7} \text{ A}/\mu\text{m}$; gap 0.31 – 0.36 V too small; $g_{m,max}$ 17–23 % low), independent of Q_f and confinement.

- Thickness error, donors, front D_{it} , back charge, a plausible combination, the S2 re-partition (B0, $SS_{cc} + 116$ mV/dec) and a near- E_c band (C1, 33 % of the gap) are excluded as the full explanation.
- Within V1, gap and subthreshold slope trade along one line (3.2–8.5 mV of gap per mV/dec); the shape part remains unexplained, and its most plausible cause, the constant, density-independent mobility, is untested.
- With one device per thickness, the rigid part cannot be assigned: device spread and sweep history are not determined.

The 6.3 nm study excludes several single mechanisms but assigns none. The next chapter widens the view to all five mechanism families and examines, for the whole thickness series, which of them the data require, identify or exclude.

Chapter 9

Mechanism analysis of the thickness dependence

The calibration of Chapter 7 and the hypothesis tests of Chapter 8 show that the V1 model can be made to reproduce the three transfer curves and where it fails when parameters are shared. They do not by themselves explain why the measured devices behave as they do. The measured pattern is unusual. In the single devices available, the 2.0 nm and 6.3 nm transistors are electrically similar ($V_{th,cc} = 0.662$ and 0.644 V; $\mu_{FE} = 12.15$ and 10.95 cm²/V s), whereas the 13.2 nm device differs strongly ($V_{th,cc} = 0.083$ V, $\mu_{FE} = 50.17$ cm²/V s). Since $\mu_{FE}(6.3) < \mu_{FE}(2)$, no monotonic thickness law passes through all three points (SYNTHESIS D1, DATA).

This chapter examines, one mechanism class at a time, which physical effects can produce this pattern: electrostatics and the threshold charge budget (Section 9.1), transport and the mobility step (Section 9.2), quantum confinement (Section 9.3), traps, donors and off-state floors (Section 9.4) and contacts (Section 9.5). For each mechanism the following are given: (i) the expected dependence on the channel thickness t , (ii) its expected magnitude, (iii) a prediction that distinguishes it from its competitors, (iv) the comparison with the measured data and the ATLAS runs, (v) the evidence for and against, and (vi) the evidence class in the vocabulary of ?? (DATA, NUM, CAL, OOS, POST, PRED, SURR, THEORY, LIT, CORR). Section 9.6 then describes the couplings that make the mechanisms difficult to separate, Section 9.7 states which parameter combinations one ID–VG curve can and cannot fix, and Section 9.8 collects the verdicts in Table 9.3.

Three limitations apply to every statement below. First, each thickness is represented by one device ($N = 1$), and device-to-device spread, sweep history and ambient were not recorded. Second, many quantitative statements rest on one-dimensional specialist surrogates (SURR) that were checked against ATLAS but not independently re-run in ATLAS. Third, the verdict “rejected” is usually model-conditional: it holds within the V1 density of states (DOS) and electrostatics described in Chapters 4 and 5. The source reports are `analysis_2026-09-25/S1_electrostatics_6p3nm.md` (S1), `S2_transport_mobility_powerlaw.md` (S2), `S3_quantum_confinement.md` (S3), `S4_defects_traps.md` (S4), `S5_contacts_experimental_validation.md` (S5) and the synthesis `analysis_2026-09-25/SYNTHESIS.r` which resolves their disagreements from primary files and takes precedence where they differ.

Table 9.1. Threshold charge budget at $V_{\text{th,cc}}$ (volts; positive terms raise $V_{\text{th,cc}}$). Values from the S1 1-D surrogate (S1 section 1.1), which matches ATLAS within 3 mV. Sheet densities of filled tail states and free electrons in parentheses (cm^{-2}). “Status” is the provenance of the input that sets the term.

term	2.0 nm run_0012	6.3 nm run_0014 (shared)	6.3 nm run_0015 (tuned)	13.2 nm run_0013	status
WF–	+0.400	+0.400	+0.400	+0.400	assumed; degenerate with Q_f
χ^{bulk} confinement	+0.353	+0.072	+0.072	+0.026	DFT proxy, extrapolated
ΔE_c $E_{Fn} - E_c$ at the front front	–0.038	–0.039	–0.037	–0.111	model
Q_f qN_{dt}/C_{ox}	–0.310	–0.310	–0.016	–0.310	fitted
filled tail interface	+0.009 +0.256 (1.43e12)	–0.028 +0.226 (1.27e12)	–0.028 +0.231 (1.29e12)	–0.070 +0.098 (5.5e11)	fitted law fitted N_t , W_{TA}
traps	+0.012	+0.012	+0.012	+0.012	assumed D_{it}
free elec- trons deep	+0.014 (7.6e10)	+0.016 (9.1e10)	+0.017	+0.005 (2.6e10)	model
Gaus- sian	0.000	+0.001	+0.001	+0.003	assumed
$V_{\text{th,cc}}$ 1- D / AT- LAS / mea- sured	0.679/0.676/0.662	0.351/0.350/0.644	0.652/0.651/0.644	0.052/0.053/0.083	

9.1 Electrostatics and the threshold charge budget

9.1.1 What the threshold measures

The constant-current threshold $V_{\text{th,cc}}$ (Section 5.10) is the gate voltage at which the source-end sheet density reaches the value that gives the reference current. At that bias, Gauss’s law splits $V_{\text{th,cc}}$ into a sum of charge and work-function terms (Section 4.18): the metal–semiconductor alignment $\text{WF} - \chi$, the confinement shift ΔE_c , the quasi-Fermi level position $E_{Fn} - E_c$ that sets n_s , the fixed front charge Q_f , the donor sheet qN_{dt}/C_{ox} , the filled acceptor tail, the filled interface traps, the free electrons and any deep Gaussian states. The S1 analysis implemented this budget in a 1-D Poisson surrogate that reproduces the ATLAS $V_{\text{th,cc}}$ within 3 mV in every film (S1 section 1.1; runs run_0012, run_0014, run_0015, run_0013). Table 9.1 reproduces the terms and Figure 9.1 shows them as grouped bars.

9.1.2 Expected thickness dependence of each term

The terms scale differently with t , and this difference is the basis of every discriminating experiment proposed in Section 9.7. Front sheet charges (Q_f , filled D_{it}) enter as Q/C_{ox} and are independent

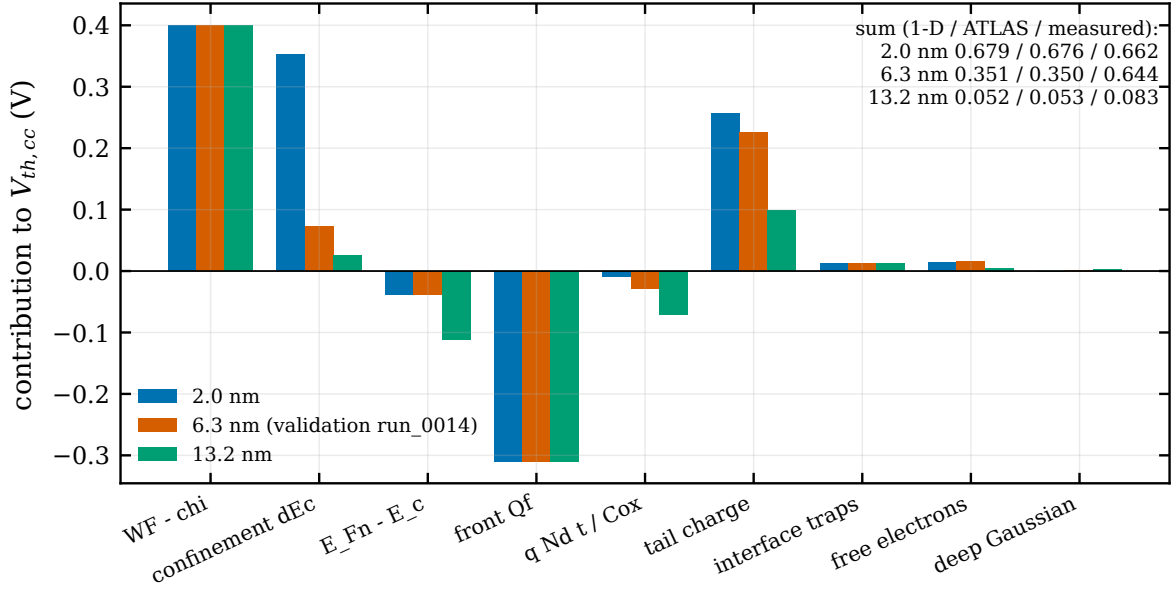


Figure 9.1. Threshold charge budget per film. Grouped bars give the Gauss-law terms at $V_{th,cc}$ from the S1 1-D surrogate (ΔE_c , filled tail, $qN_d t / C_{ox}$, interface traps, Q_f , $E_{Fn} - E_c$), which agrees with ATLAS within 3 mV for runs run_0012 (2 nm), run_0014 (6.3 nm, shared parameters) and run_0013 (13.2 nm); markers give the measured $V_{th,cc}$. The 0.281 V fall of ΔE_c from 2 to 6.3 nm is not compensated by any other term in the shared-parameter model, which is why run_0014 misses the measured 6.3 nm threshold by -0.294 V.

of t (t^0). Volume charges (donors, bulk tail states) give a sheet proportional to t plus a lever-arm correction proportional to t^2 / ϵ_{IWO} ; with the C_{ox} of the HfO_2/Al_2O_3 stack the t^2 part is small below 13.2 nm (S1). A back-surface sheet acts through the lever arm $1 + t C_{ox} / \epsilon_s$ (Section 4.15). S1 found an effective back-referenced lever of 0.96–1.2 times the front value at threshold, not the 1.7–2.4 needed to make a thin-film-specific back charge behave differently (S1 verdicts, rejected). The confinement shift $\Delta E_c(t)$ falls steeply with t ; for an infinite well the scaling is t^{-2} , while the V1 law uses $0.9205 t^{-1.3815}$ (Section 4.5). Finally, the CC definition converts mobility into threshold: a film with a larger mobility reaches the reference current at a lower n_s and therefore at a lower $V_{th,cc}$.

9.1.3 Magnitudes and comparison with the data

The 2 versus 13.2 nm step. The measured step is $V_{th,cc}(2) - V_{th,cc}(13.2) = 0.579$ V. In the calibrated model the confinement term accounts for $0.3156 - 0.0234 = 0.292$ V of it, i.e. 50.4 % (SYNTHESIS D6, NUM). The CC-definition coupling also contributes: giving the 13.2 nm curve the 2 nm mobility moves $V_{th,cc}$ by $+0.108$ to $+0.115$ V, 19–20 % of the step (SYNTHESIS D6, DATA/NUM; S1 quotes $+0.11$ to 0.14 V, strongly supported). The tail term changes by $0.256 - 0.098 = 0.158$ V between the two films and the donor sheet by 0.061 V (Table 9.1); both are set by fitted laws ($N_t(t)$ and $N_{d,eff}(t)$) and are therefore CAL.

The 6.3 nm threshold. With all laws shared, run_0014 gives $V_{th,cc} = 0.350$ V against 0.644 V measured, a miss of -0.294 V (SYNTHESIS D4). Because the value $\mu_{band} = 12.4 \text{ cm}^2/\text{Vs}$ used in run_0014 is the V0 fit of the same curve, this is a failed out-of-sample test of the threshold only (OOS, failed). S1 attributes 0.281 of the 0.328 V model-internal difference between 2 and 6.3 nm (86 %) to the fall of ΔE_c ; SYNTHESIS corrects this share to 77–86 %, depending on the decomposition.

Tested electrostatic explanations of the 6.3 nm miss. Campaign A (Chapter 8) tested the physical charge candidates in ATLAS. The quantitative inconsistencies, taken from SYNTHESIS section D, are listed below.

- thickness error, A1 (run_0031, $t = 5.3$ nm): +0.036 V of the required +0.294 V (12 %); closing the miss through ΔE_c needs $t \approx 2.1$ nm, a 67 % error (rejected, model-conditional);
- fewer donors, A2 (run_0032): +0.030 V, a factor of ten short (rejected);
- back-surface charge, A3 (run_0030): matches $V_{th,cc}$ (−0.017 V) but SS_{cc} is off by −64.1 mV/dec and the gap closes to 0.724 V (rejected as the sole cause, model-conditional);
- front $D_{it} \times 5$, A4 (run_0033): +0.051 V with the gap unchanged; the full miss would need $\times 23\text{--}24$ ($\approx 7 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$) at 6.3 nm only on a shared gate stack (rejected at plausible D_{it});
- combination A5 (run_0035): 0.215 of 0.294 V (73 %) with SS_{cc} −45.9 mV/dec (rejected);
- a uniform donor density or neutral layer as the 6.3 → 13.2 step: the steepest possible slope ratio is 2.35 against $\approx 9\text{--}20$ measured; N_d fitted to 6.3 → 13.2 ($2.2 \times 10^{18} \text{ cm}^{-3}$) predicts −0.25 V for 2 → 6.3 against −0.018 V measured (S1, S4; SURR+DATA, rejected).

The only change that reproduces the 6.3 nm threshold is a film-specific front charge: run_0015 uses $Q_f = 8.7 \times 10^{10} \text{ cm}^{-2}$ instead of the shared $1.73 \times 10^{12} \text{ cm}^{-2}$, a difference of $1.64 \times 10^{12} \text{ cm}^{-2}$ (0.294 V). This is a calibration (CAL), not an explanation.

Small terms. Two budget terms are shown to be small: the donor sheet $qN_d t / C_{ox} = 0.009/0.028/0.070$ V (run_0022, A2) and the interface-trap charge 0.012 V, independent of thickness (run_0025, A4) (S1 verdicts, NUM).

9.1.4 Distinguishing predictions and verdict

Front sheets scale as t^0 , volume charges as t and t^2 , and a back sheet as t/ϵ_s . A quasi-static C–V flat-band voltage $V_{FB}(t)$ measured for each thickness therefore separates them, and Kelvin-probe or UPS measurements fix WF and χ (SYNTHESIS Q4). A confinement shift, in contrast, is a band-edge property and changes the optical gap. None of these measurements is available for the present films.

Summary. The threshold budget is fully accounted for within the model, but its thickness dependence is carried by fitted or assumed terms. The 6.3 nm miss of −0.294 V (run_0014) is rigid between 10^{-11} and $10^{-9} \text{ A}/\mu\text{m}$ and equivalent to $1.64 \times 10^{12} \text{ cm}^{-2}$. Every tested physical charge (A1–A5) fails in magnitude or in its SS signature (NUM, model-conditional). The origin of the rigid part is not determined, and device spread and sweep history are not excluded.

9.2 Transport and the mobility step

9.2.1 The observation

The linear field-effect mobility (Section 5.5, Figure 5.1) is 12.15, 10.95 and $50.17 \text{ cm}^2/\text{V s}$ for 2, 6.3 and 13.2 nm. It is flat or slightly falling up to 6.3 nm and then rises by a factor of 4.58 to 13.2 nm (SYNTHESIS D1, DATA). The published mobilities of 5.1 and $27.4 \text{ cm}^2/\text{V s}$ are reproduced as 5.09

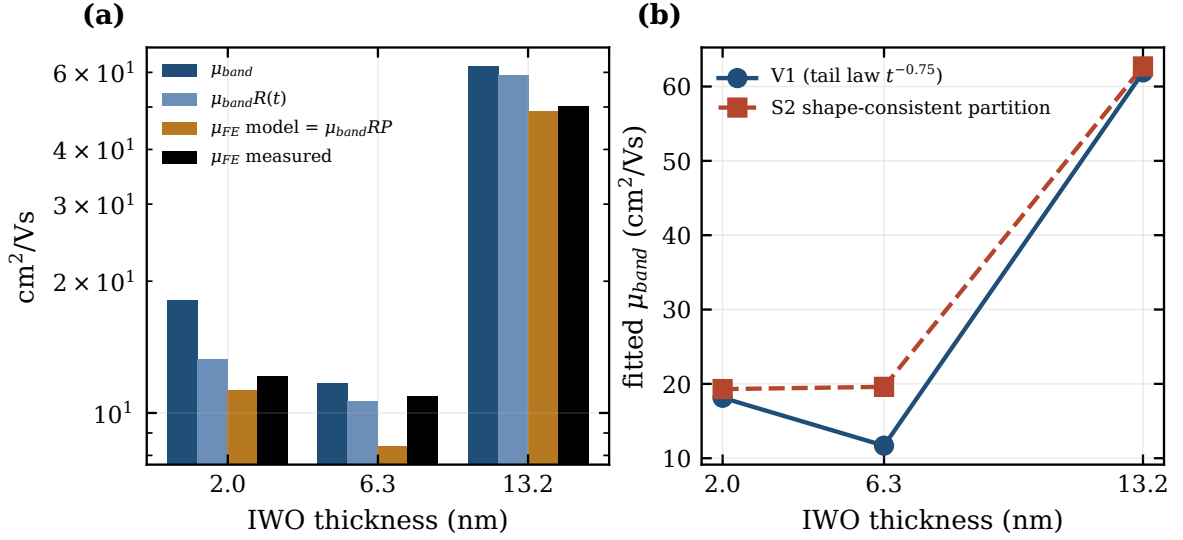


Figure 9.2. Partition of the apparent mobility, $\mu_{FE} = \mu_{band} \times R(t) \times P$ (Equation (9.1)), per film. Left: the V1 calibration (runs `run_0012`, `run_0015`, `run_0013`); right: the S2 shape-consistent partition from the 1-D surrogate, $\mu_{band} = 19.3/19.6/62.7 \text{ cm}^2/\text{Vs}$ with $N_t = 2.4\text{e}19/2.7\text{e}19/4.8\text{e}18 \text{ cm}^{-3}$. In both partitions the roughness factor and the partition factor change by only a few per cent between 6.3 and 13.2 nm, so the $\times 4.6$ step lies in μ_{band} .

and $27.40 \text{ cm}^2/\text{Vs}$ when the saturation formula is applied to the same curves at $V_D = 0.7 \text{ V}$, outside the regime of that formula; the same procedure predicts $3.86 \text{ cm}^2/\text{Vs}$ at 6.3 nm (SYNTHESIS D12; S2 and S5 verdicts, demonstrated, provenance evidence).

9.2.2 Decomposition of the apparent mobility

In the V1 model the apparent mobility is the product

$$\mu_{FE} = \mu_{band} \times R(t) \times P, \quad R(t) = \left(1 - \frac{\Delta_{sr}}{t}\right)^2, \quad (9.1)$$

where μ_{band} is the fitted band mobility (Section 5.1), $R(t)$ the roughness factor (Section 5.2) and P the free/trapped partition factor at $g_{m,max}$ (Section 5.3). S2 evaluated Equation (9.1) for the calibrated runs (`run_0012`, `run_0015`, `run_0013`) and decomposed $\ln(50.17/10.95)$, the $6.3 \rightarrow 13.2$ step, as follows: 109 % is carried by the fitted μ_{band} , 3 % by the roughness factor, 3 % by the partition factor and -16% is model residual (S2, CAL). The $N_t(t)$ law changes P at $g_{m,max}$ only between 0.79 and 0.85 across all films ($0.791 \rightarrow 0.826$ for $6.3 \rightarrow 13.2$; SYNTHESIS B1). In the recalibrated DOS-384 state the band mobilities are $17.69/11.58/61.60 \text{ cm}^2/\text{Vs}$ (`run_0038`, `run_0039`, `run_0040`; SYNTHESIS D8).

9.2.3 Candidate mechanisms

Trap-limited partition (Januar-type). A denser tail in thinner films is expected to lower the free fraction. The partition factor P , however, changes by only $+4 \%$ between 6.3 and 13.2 nm, and a $\times 4.6$ step would require $N_t \times 6.7 \times 10^3$ (S2). This mechanism is therefore rejected as the cause of the step (SURR/CAL).

Surface roughness. The roughness factor $R(t) = (1 - \Delta_{\text{sr}}/t)^2$ rises monotonically with t . Between 6.3 and 13.2 nm it changes by a factor of 1.051, whereas a factor of 4.58 is needed (3 % of the logarithm). Roughness is therefore rejected as the cause of the step (CAL). The thickness-fluctuation form ($\propto t^6$) is rejected independently: $\mu(6.3) = 17 \text{ cm}^2/\text{V s}$ would imply $\mu(2) = 0.017 \text{ cm}^2/\text{V s}$, a factor of ≈ 700 below the extracted value (S2, SURR).

Coulomb scattering. Front Q_{f} , back-surface and trapped charge give μ_{C} between 283 and 515 $\text{cm}^2/\text{V s}$ (front) and $4.7 \times 10^4 \text{ cm}^2/\text{V s}$ (back, 6.3 nm). All of these are monotonic in t and too weak by orders of magnitude (S2; rejected as the step). Remote phonons are independent of t to first order (S2; rejected as the step).

Contacts, electrostatics and quantum capacitance. Campaign A changes g_{m} by at most 2 %, and the effective gate capacitance $C_{\text{eff}}/C_{\text{ox}}$ changes by ≤ 1 % at 2 nm and by 3–5 % at 6.3/13.2 nm (S3, SURR). Contacts are treated in Section 9.5 and cannot create the trend. Taken together, these results make the statement “the $\times 4.6$ rise is a change in channel transport” strongly supported within the model (SYNTHESIS B1).

A structural transition between 6.3 and 13.2 nm. Such a transition is expected to produce a step, not a continuous law, in μ_{band} at the thickness where crystallinity, grain coalescence or a percolation threshold changes. S2 estimates that a grain-boundary barrier difference of 39 meV is sufficient. In favour of this reading, $V_{\text{th,cc}}$, μ_{FE} , SS_{cc} and the off-floor change together between 6.3 and 13.2 nm (CORR), and the target paper states that its ultrathin films lack long-range crystallinity (Section 2.2). Against it, there are no structural data at 6.3 or 13.2 nm, the Kim et al. study of IWO films from a different process (Section 2.7) shows no step at 10–30 nm, and $N = 1$. The distinguishing measurements are the GIXRD/TEM grain size, the gated Hall mobility at 6.3 and 13.2 nm, and the temperature dependence of $\mu_{\text{FE}}(13.2)/\mu_{\text{FE}}(2)$, which is registered in Section 10.2. The verdict is plausible, unverified; this is the working hypothesis (SYNTHESIS C1). The value “ $\mu_{\text{FE}} = 49.2$ ” at 31.8 nm that is sometimes quoted as support is not a mobility (g_{m} has maxima at -0.30 , 0.85 , 1.15 and 1.70 V) and is not used as evidence.

Power laws in thickness. Leave-one-out fits through 2 and 13.2 nm predict $\mu_{\text{FE}}(6.3) = 28.8$ (power law), $42.7 (A + B/t)$ and 20.9 (exponential) against $10.95 \text{ cm}^2/\text{V s}$ measured. The I_{on} power law with exponent 2.75 predicts $3.43 \times 10^{-5} \text{ A}/\mu\text{m}$ at 31.8 nm against $3.59 \times 10^{-6} \text{ A}/\mu\text{m}$ measured (Figure 5.3; S2, DATA). Power laws are therefore rejected. The “step” form fits three points by construction, so its leave-one-out success is circular (SYNTHESIS Q3).

Artefact status of the non-monotonic μ_{band} . The calibrated relation $\mu_{\text{band}}(6.3) < \mu_{\text{band}}(2)$ is, with strong support, an artefact of the $N_{\text{t}}(t)$ partition (S2). The 6.3 nm on-state is not reproduced (gap -0.36 V , $g_{\text{m}} -23$ %), and the S2 shape-consistent partition (19.3/19.6/62.7, Figure 9.2) is flat up to 6.3 nm. Its ATLAS implementation, run B0 (run_0037), met the on-state but failed the subthreshold ($SS_{\text{cc}} 410.9$ vs 295.4 mV/dec), so that partition is rejected in its tested form (SYNTHESIS D, OOS, failed).

Summary. Within the calibrated model, the $\times 4.6$ rise of the apparent mobility between 6.3 and 13.2 nm cannot be attributed to contacts, roughness, electrostatics or tail filling, since 109 % of $\ln(\times 4.58)$ is in the fitted μ_{band} (strongly supported, model-conditional). Its physical origin is not determined; a structural transition is the plausible, unverified working hypothesis.

9.3 Quantum confinement

9.3.1 Expected dependence and magnitude

Geometric confinement of the electrons in a film of thickness t raises the lowest subband above the bulk conduction-band edge. For an infinite well the shift scales with t^{-2} and depends on m^* ; a finite barrier, non-parabolicity and the gate field reduce it (Section 4.6, Figure 4.1). The V1 model implements $\Delta E_c = 0.9205 t^{-1.3815}$ eV, i.e. 0.353 eV at 2 nm, 0.072 at 6.3 and 0.026 eV at 13.2 nm (Table 9.1), together with the $m^*(t)$ law that enters N_c (Section 4.7). S3 solved a 1-D Schrödinger–Poisson problem anchored on PBE slab calculations of pure In_2O_3 (Figure 4.2): the subthreshold-equivalent shift at 2 nm is 0.26–0.30 V for $m^* = 0.18$, bracket 0.14–0.38 V over effective mass and barrier treatment (S3 verdicts, strongly supported as THEORY; not measured on IWO). Image-charge (dielectric) confinement adds an estimated +0.02 to 0.03 eV at 2 nm (S3, plausible, unverified).

Above $t \approx 2.98$ nm the V1 law exceeds the infinite-well (effective-mass) upper bound: by $\times 1.59$ at 6.3 nm and $\times 2.51$ at 13.2 nm (SYNTHESIS Q3). Relative to the DFT-anchored effective-width estimate of S3 the same overshoot is $\times 1.45$ and $\times 2.15$ (Table 4.1); the two pairs differ only in their reference. The $t^{-1.38}$ exponent is a local three-point slope and is rejected as physics beyond ≈ 3 nm; its impact on V_{th} is at most 35 mV.

9.3.2 What one ID–VG curve can see

The pure band-edge shift is exactly rigid. A shift of E_c moves the whole transfer curve along V_G in the same way as a change of WF, χ or Q_f : WF +0.1 eV gives +0.1000 V with SS unchanged (run_0048, run_0049), Q_f is rigid within 0.6 mV over 10^{-14} to 3×10^{-7} A/ μm (run_0001 $\rightarrow$ run_0003, run_0002 $\rightarrow$ run_0004, run_0005 $\rightarrow$ run_0007), and $\Delta E_c(2 \text{ nm}) = 0.353$ eV is equivalent to $1.97 \times 10^{12} \text{ cm}^{-2}$ of fixed charge (SYNTHESIS D2, NUM).

The confinement package is only near-rigid, and its signature is not unique. Switching the V1 package (ΔE_c plus $m^* \rightarrow N_c$) off at 2 nm (run_0012 vs run_0016) shifts the curve by 0.345 V at 10^{-14} , 0.302 V at 10^{-8} and 0.311 V at 3×10^{-7} A/ μm , and SS_{cc} is 289.1 vs 303.7 mV/dec. The non-rigid part (+38.8 mV between 10^{-11} and 10^{-8} A/ μm ; $SS_{\text{cc}} + 14.6$ mV/dec) is nearly reproduced by $m^* \times 0.8$ or by $N_t \times 1.09$ (log-linear scaling of run_0050 and run_0023; SYNTHESIS D2 and [SYN-d]). On the same metric, $W_{\text{TA}} + 5$ meV moves SS_{cc} by only +0.8 and $D_{\text{it}} \times 3$ by +0.5 mV/dec: the degeneracy is with m^* and N_t , not with W_{TA} (SYNTHESIS A.3).

The subthreshold-slope argument is an artefact. The earlier claim that removing confinement moves SS toward the data rests on SS_{min} , which is 151.3 vs 138.6 mV/dec for run_0013 vs run_0017, whereas the fixed-current SS is 90.8 vs 92.0 mV/dec (SYNTHESIS D7). The SS_{min} window is set

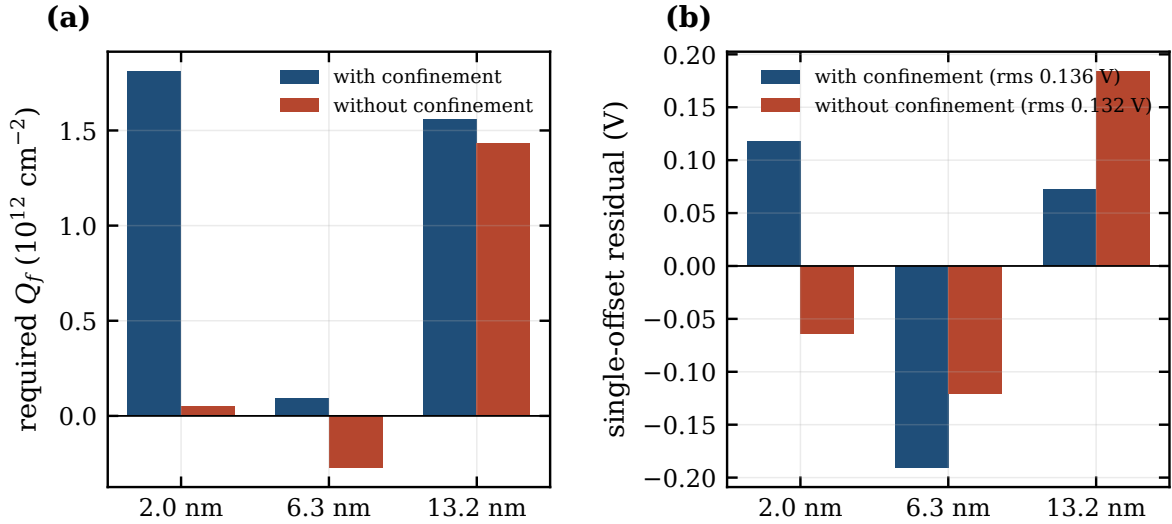


Figure 9.3. Symmetric calibration of the thresholds with and without the confinement law. (a) Front charge Q_f required by each film to reproduce its measured $V_{th,cc}$: with confinement 1.81/0.09/1.56 and without 0.048/−0.27/1.43 (units of 10^{12} cm^{-2} ; 2/6.3/13.2 nm). (b) Residuals of the best single shared Q_f : with confinement $Q_f = 1.153 \times 10^{12} \text{ cm}^{-2}$, residuals +0.117/−0.19/+0.073 V (rms 0.1358 V); without, $0.403 \times 10^{12} \text{ cm}^{-2}$, residuals −0.063/−0.12/+0.184 V (rms 0.1320 V) (figure data from S1 key numbers / SYNTHESIS D3; SYNTHESIS quotes 0.1363 and 0.1325 V from its own re-derivation). The two readings fit equally well, and each needs one film with an anomalous charge: 6.3 nm with confinement and 13.2 nm without.

by a current of $\approx 2.3 \times 10^{-14} \text{ A}/\mu\text{m}$ at $V_G = 0.15 \text{ V}$, and SS_{min} is therefore not a reliable metric (Section 5.10); S3 obtains 76–82 and 136–137 mV/dec from resampling alone (S3 verdicts, rejected as evidence).

9.3.3 Symmetric calibration: with and without confinement

The most direct test is to fit the three thresholds with one shared offset, once with the confinement law and once without it, and to compare the residuals. SYNTHESIS D3 re-derived this comparison from the `execution.json` files and the exact Q_f identity (NUM on DATA). The one-offset rms is 0.1363 V with confinement and 0.1325 V without; the leave-one-out values are 0.2045 and 0.1988 V. The preferred reading changes with the anchor: anchored on 2 nm the values are 0.2205 vs 0.1800 V, anchored on 13.2 nm they are 0.1897 vs 0.2782 V. Each reading needs one film-specific charge. With confinement, the 6.3 nm film needs $Q_f = 0.086 \times 10^{12} \text{ cm}^{-2}$ instead of the shared 1.73×10^{12} ($1.64 \times 10^{12} \text{ cm}^{-2}$ less positive, 0.294 V). Without confinement, the 13.2 nm film needs 1.434×10^{12} instead of the $0.047 \times 10^{12} \text{ cm}^{-2}$ fitted on 2 nm (1.39×10^{12} more positive, 0.248 V). Figure 9.3 shows the required Q_f per film in both readings and the single-offset residuals.

9.3.4 Comparison with the data and verdict

Necessity. Confinement is not necessary (demonstrated), because the fit quality is equal with and without it (D3). The A6 run (`run_0034`, no confinement, Q_f from 2 nm) misses the 6.3 nm threshold by only −0.057 V. However, configuration A was written after the `run_0014` miss was known, so this result is POST and not a held-out test (SYNTHESIS A.3). The S1 verdict “favoured” and the S3 verdict “mildly disfavoured” are therefore both overstated; the thresholds are neutral.

The pattern $2 \approx 6.3 \gg 13.2$. Confinement does not explain this pattern (rejected, S3). The offsets needed are +0.30/+0.36/+0.05 V, whereas confinement at 6.3 nm contributes at most 0.05 V.

Identifiability. Confinement is not identifiable (rejected as identifiable, S3): on one curve, 0.301–0.315 V of shift is equivalent to $1.68\text{--}1.76 \times 10^{12} \text{ cm}^{-2}$ of fixed charge.

Physical expectation. A shift of 0.26–0.30 V at 2 nm is expected from theory (THEORY). If this shift is present, the 2 nm film must carry $0.8\text{--}2.1 \times 10^{12} \text{ cm}^{-2}$ more positive charge than the 6.3 nm film, or the 6.3 nm device must carry a coincident offset of that size (SYNTHESIS Q1).

Tail-energy reference. Whether the tail states follow the confined or the bulk edge changes the 2 nm onset by 0.26 vs 0.91 V in the S3 surrogate. This is the dominant model-form uncertainty at 2 nm and is not determined.

Field quantization at 6.3 and 13.2 nm. The gate field produces a triangular-well subband of $\approx 0.5 \text{ eV}$ at 3 V and lowers C_{eff} by 3–5 %. With local tails, $V_G(n_s = 10^{12} \text{ cm}^{-2})$ rises by 0.15–0.19 V (S3: existence strongly supported, device impact plausible, unverified). Because the model already fits the 13.2 nm film, this effect cannot be the sole cause of the 6.3 nm shape miss.

Distinguishing predictions. The optical gap should widen by +0.2 to 0.45 eV at 2 nm relative to thick films ($< 0.02 \text{ eV}$ at 13.2 nm); the confinement contribution to V_{th} changes by only -6 mV between 300 and 358 K, whereas tail filling produces the registered $V_{\text{th,lin}}$ shifts of Section 10.2 (SYNTHESIS Q1). Falsified if the optical-gap shift between 2 and 13.2 nm is below 0.1 eV (SYNTHESIS C3).

Summary. A shift of about 0.3 eV at 2 nm is theoretically expected (THEORY, bracket 0.14–0.38 V), but the transfer data neither require nor identify it. On one ID–VG curve it is equivalent to $\approx 1.7 \times 10^{12} \text{ cm}^{-2}$ of fixed charge, and one shared offset fits the three thresholds equally well with or without it (rms 0.136 vs 0.133 V; NUM on DATA). It does not explain the similarity of the 2 nm and 6.3 nm thresholds.

9.4 Traps, donors and off-state floors

9.4.1 Near-conduction-band acceptor tail

The V1 model uses an exponential acceptor tail $N_{\text{TA}} \exp[(E - E_c)/W_{\text{TA}}]$ with $N_t(t) = 2 \times 10^{19} (2/t)^{0.75} \text{ cm}^{-3}$ and a shared $W_{\text{TA}} = 40 \text{ meV}$ (Section 4.10). The filling of this tail sets both the threshold (0.256/0.226/0.098 V of the budget, Table 9.1) and the subthreshold slope. One shared W_{TA} reproduces the measured $SS_{[10^{-11}, 10^{-10}]}$ within 2.3 mV/dec in all films (121.2/123.5, 124.5/124.5, 91.8/90.9 mV/dec, measured/simulated; the 13.2 nm value is floor-subtracted, raw 136.1; SYNTHESIS D8). Ideal-contact ATLAS runs reproduce the supralinear exponents $\alpha_H = 1.59/2.14/1.29$ and the negative θ through trap filling alone (S5). A trap-DOS inversion validated on ATLAS curves (Figure 4.4) gives measured/ATLAS DOS ratios of 0.81–1.46 near E_c (S4; conditional on the mobility, because the energy axis moves by $k_B T \ln$ of the mobility ratio). The tail-filling mechanism is therefore strongly supported (SYNTHESIS B2; CAL + inversion).

The specific V1 DOS form, however, is not supported. The model is too soft in the 10^{-10} to 10^{-9} A/ μ m window by +20/+6/+25 mV/dec (SYNTHESIS D8), and deeper states near $E_c - 0.15$ to $E_c - 0.2$ eV are underestimated by $\times 2.2$ – 2.8 at 2 nm and $\times 2.6$ at 6.3 nm (S4, strongly supported).

The $N_t(t)$ law. This law has no identified physical origin. It is a two-point interpolation whose exponent comes from one qualitative ratio. The positive-bias-stress devices of the same paper imply an exponent of 1.71 instead of 0.75, and a surface + bulk form ($D_s \approx 8.5 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$ plus $g_b \approx 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$ at $E_c - 0.1$ eV, sheet $\propto t^{0.38}$) differs by only 8 % at 6.3 nm (S4; SYNTHESIS Q3). The law is kept as an interpolation and rejected as physics. Multi-frequency C–V measured for each thickness would separate surface and bulk trap densities (SYNTHESIS C6).

Reading $SS_{cc}(t)$ as a trap trend is biased. The free-carrier capacitance puts ≈ 60 mV/dec of mobility into the thin-film SS_{cc} : a trap-free surrogate gives $SS_{cc} = 122$ at $\mu = 13$ vs 75 mV/dec at $\mu = 59 \text{ cm}^2/\text{V s}$ (S4, rejected as a naive trap reading, SURR).

States close to E_c and the 6.3 nm shape. S4 proposed $2 \times 10^{12} \text{ cm}^{-2}$ at $E_c - 0.03$ eV together with a higher mobility, with a predicted gap change of +0.28 V, $g_m +24$ % and $SS_{cc} +23$ mV/dec. The ATLAS test C1 (run_0052), registered at 19:21Z before its outcome was known, gave a gap gain of +0.119 V (33 % of the 0.356 V miss), $I_{on} +16.0$ % against a ± 3 % requirement and $g_m -16.8$ % at matched I_{on} (SYNTHESIS D, OOS, partly failed; Section 8.3). The near- E_c band remains a plausible partial contributor; the DOS resolution of the 20 meV band was not checked.

9.4.2 Donors

A uniform effective donor density contributes $qN_d t / C_{ox}$ to the budget, a term linear in t . Its magnitude is 0.009/0.028/0.070 V (Table 9.1), and doubling N_d at 2 nm moves $V_{th,cc}$ by only -0.009 V (run_0022, Figure 7.5). A low effective density ($\approx 2.5 \times 10^{17} \text{ cm}^{-3}$, compensated) is consistent with the 4.3 – $4.5 \times 10^{17} \text{ cm}^{-3}$ reported for related IWO films (Section 2.7); it is not identifiable from these curves (S4, plausible, unverified). A thickness-independent donor density cannot explain the 13.2 nm drop (it needs $2.2 \times 10^{18} \text{ cm}^{-3}$ and then misses $2 \rightarrow 6.3$ by 0.23 V; rejected). A donor or positive-charge step ($\approx 2.5 \times 10^{17}$ at ≤ 6.3 nm to $\approx 10^{18} \text{ cm}^{-3}$ at 13.2 nm) is what the no-confinement reading requires ($+1.39 \times 10^{12} \text{ cm}^{-2}$, 0.248 V); it is degenerate with ΔE_c on $V_{th}(t)$ (SYNTHESIS C2, plausible, unverified). Falsified if C–V gives $V_{FB}(t)$ flat within 50 mV across the three films.

The fixed charge Q_f . The fitted $1.73 \times 10^{12} \text{ cm}^{-2}$ is electrostatically indistinguishable from an ionized oxygen-vacancy layer of $\approx 1.7 \times 10^{19} \text{ cm}^{-3}$ over 1 nm (S4), but it also absorbs any error of the alignment: ± 0.2 eV of work function equals $\pm 1.12 \times 10^{12} \text{ cm}^{-2}$ (SYNTHESIS C9). Its physical identity is plausible, unverified.

9.4.3 Off-state floors

Below threshold all three curves flatten to gate-independent plateaus (Figure 3.2). The median plateau between -2 and -0.5 V is 4.61×10^{-15} , 1.53×10^{-13} and 6.60×10^{-12} A/ μ m for 2, 6.3 and 13.2 nm, which corresponds to a factor of 1432 from 2 to 13.2 nm, with a log–log slope of 3.78 over the

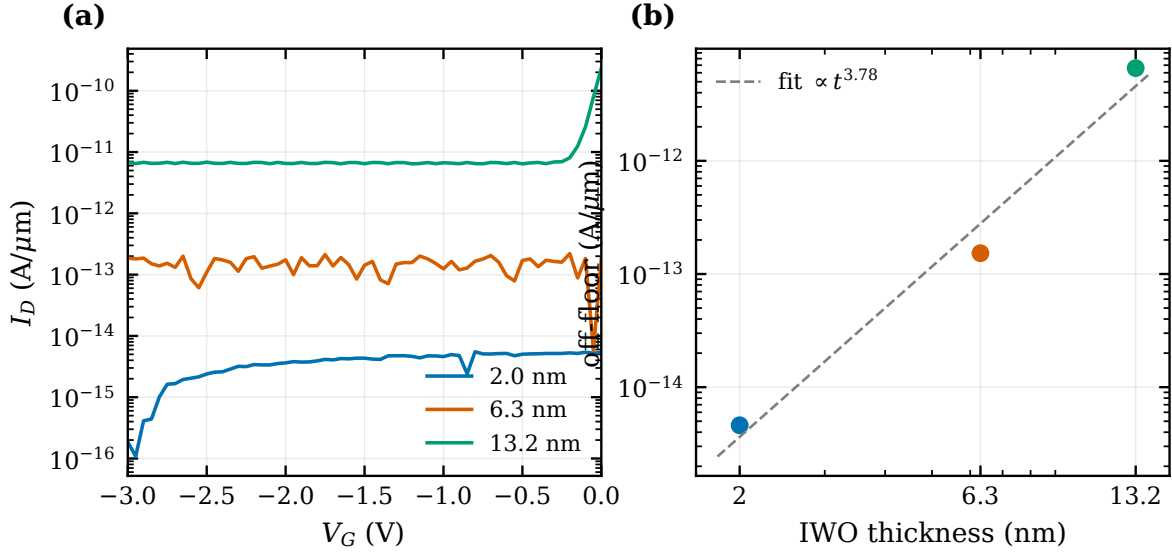


Figure 9.4. Off-state floors. (a) Measured I_D between $V_G = -3$ and 0 V for 2, 6.3 and 13.2 nm at $V_D = 0.7$ V: flat plateaus. (b) Plateau (median over -2 to -0.5 V) versus thickness: 4.61×10^{-15} , 1.53×10^{-13} , 6.60×10^{-12} A/ μm ; ratio 1432 between 13.2 and 2 nm; log-log slope 3.78 (S4 quotes 3.85), with a $t^{3.85}$ guide. The floor rises by three orders of magnitude on an identical gate stack.

three films (figures/MANIFEST.md; S4 quotes $t^{3.85}$) (DATA). Figure 9.4 shows the plateaus and the thickness scaling.

Three candidate mechanisms were considered. Gate leakage is expected to depend on the gate-drain voltage and not on t . The 13.2 nm floor, however, is flat within $\times 1.06$ while V_{gd} changes from 1.2 to 3.7 V; the 6.3 nm floor scatters by $\pm 45\%$ (max/min 3.0–3.5) with no trend (-0.011 dec/V); and the $\times 1432$ rise occurs on an identical stack. Gate leakage is therefore rejected as the dominant floor at 6.3 and 13.2 nm (SYNTHESIS B4 and A.3, DATA; S4 had described the floors as flat to $\pm 3\%$, which SYNTHESIS corrects for 6.3 nm). Thermal (SRH) generation falls 13–17 decades short (calculation from the Astra audit; rejected). The third candidate is an ungated parallel IWO path. The corresponding sheet conductances, 1.3×10^{-13} , 4.4×10^{-12} and 1.9×10^{-10} S/sq, scale as $t^{3.85}$, and the order of magnitude fits (S4, plausible, unverified). If this reading is correct, the $\times 1432$ rise requires $E_{F0} - E_c$ to move by only ≈ 0.10 eV between 2 and 13.2 nm, compared with the 0.33 eV extrapolated ΔE_c difference, so the floor would bound confinement and donor compensation jointly (SYNTHESIS Q5.6, conditional). An instrument floor or an unpatterned film is not excluded. The path hypothesis is falsified if $I_G \approx I_D$ in the off state or if the floor shows no $1/L$ scaling on an L-series (SYNTHESIS C8).

The floor also biases the extraction: the 13.2 nm fixed-current SS is 136.1 raw vs 91.8 mV/dec floor-subtracted, and the $SS_{\min}$ window depends on it (SYNTHESIS Q5.9).

9.4.4 Sweep history and bias stress

Positive-bias stress in the target paper roughly doubles N_t at 2 nm. Extrapolated to the measurement sweep, this gives a threshold shift between 0.003 and 0.24 V (S4 section 6), and the upper end is of the same order as the 0.29 V offset at 6.3 nm. Sweep direction, rate and dwell were not recorded, so this mechanism is not determined (SYNTHESIS C4).

Summary. Subthreshold conduction in all films is governed by the filling of a near- E_c exponential acceptor tail (strongly supported). The V1 DOS form, however, is not supported (too soft in 10^{-10} – 10^{-9} A/ μ m; deep states underestimated $\times 2.2$ – 2.8), and the $N_t(t)$ law has no physical origin. Donors are a small, non-identifiable term. The floors rise $\times 1432$ ($t^{3.78}$ to $t^{3.85}$) and are due neither to gate leakage nor to thermal generation; their origin is not determined.

9.5 Contacts

The V1 deck uses ideal Ohmic Pd source/drain contacts (Section 5.6). A contact resistance R_{sd} would lower the apparent mobility and could, in principle, create a thickness trend if it depended on t or if the contact fraction of the total resistance did.

Thickness-independent R_{sd} as the cause of the trend. The required resistance is $R_{sd}W \approx 1.15 \times 10^6 \Omega \mu\text{m}$ at 2 nm, i.e. $\rho_c \approx 0.26 \Omega \text{cm}^2$ and a transfer length $L_T \approx 45 \mu\text{m}$. Such a resistance would drop 83 % of V_D and is ≈ 5700 times the contact resistance of related metal/ In_2O_3 contacts (Section 2.5; S5). In addition, the mobility-degradation parameter θ_{net} is $-0.10/-0.12/-0.046 \text{V}^{-1}$ in all films, which is the wrong sign for series resistance. Finally, at fixed ρ_c the contact fraction scales as $R_{sh}^{-1/2}$, so a thickness-independent contact penalises the thick film, which is opposite to the observation. This mechanism is therefore rejected (SYNTHESIS B3 and D; CORR + LIT, related material).

R_{sd} as a minor contributor. Within the calibrated trap model, $R_{sd} \lesssim 5 \times 10^4 \Omega \mu\text{m}$ (≤ 4 %) at 2 nm and $\lesssim 1\text{--}2 \times 10^4 \Omega \mu\text{m}$ (4–9 %) at 13.2 nm (S5). These are calibration-class bounds, so a minor contribution is not determined.

Confinement-raised Pd/IWO barrier at 2 nm. Such a barrier would need $\Delta\phi \geq 0.20 \text{eV}$, and $\Delta E_c(6.3) = 0.07 \text{eV}$ cannot give $I_{\text{on}}(6.3) = 0.79 I_{\text{on}}(2)$. The barrier is plausible but unverified at 2 nm, and it is rejected as the cause of the 2/6.3 vs 13.2 step (S5; CORR).

Geometric effects. Vertical access through the film gives $\rho t \leq 5 \times 10^{-7} \Omega \text{cm}^2$ (< 1 % of R_{on}) and grows with t (rejected). Current crowding in the 2 nm film gives $L_T \leq 0.35 \mu\text{m}$ at $\rho_c \leq 10^{-4} \Omega \text{cm}^2$ (rejected) (S5, order-of-magnitude estimates). The V0 31.8 nm series resistance ($9.2 \times 10^4 \Omega \mu\text{m}$), if carried over to 13.2 nm, would give -27 % I_{on} and $\theta = +0.17 \text{V}^{-1}$, against -0.046V^{-1} observed (rejected within the model).

Extraction methods. The Y-function (Section 5.7, Figure 5.2) is invalid for these films: in ATLAS method checks with ideal contacts, α and θ correlate with $r = +1.00$ and $\mu_{0,Y}/\mu_{\text{band}} = 0.53\text{--}0.68$ (S5, not determined). run_0027, originally intended as an overlap sensitivity test, is malformed and retracted, so the contact geometry was never tested (Section 7.4).

The distinguishing measurements are a transfer-length-method or channel-length series, low- V_D output characteristics ($V_D = 0\text{--}0.2 \text{V}$) and the registered ID–VD ratios of Section 10.1. An S-shape or $I_D(0.1)/I_D(0.05) > 2.0$, especially at 2 nm only, would indicate non-Ohmic injection.

Summary. Contacts do not create the thickness trend. The required resistance would drop 83 % of V_D , θ_{net} has the wrong sign in all films, and a thickness-independent contact would penalise the thick film (strongly supported; CORR + LIT). A minor R_{sd} contribution is bounded only within the calibrated model; contact geometry and resistance were not tested in ATLAS.

9.6 Couplings between mechanisms

The mechanisms discussed above are not independent. Several of them act on the same observable, and some change the way in which another is extracted. The following couplings, collected in SYNTHESIS Q5, explain why single-mechanism conclusions drawn from one curve per film are fragile.

1. *Mobility* $\rightarrow$ *threshold*. The constant-current definition converts mobility into threshold: +0.108 to +0.115 V of the 2 vs 13.2 nm step (Section 9.1; SYNTHESIS D6).
2. *Mobility* $\rightarrow$ SS_{cc} . The free-carrier capacitance puts ≈ 60 mV/dec of mobility into the thin-film SS_{cc} ; $SS_{\text{cc}}(t)$ read as a trap trend is biased (S4, SURR).
3. *Confinement* $\leftrightarrow$ *tail filling*. The energy reference of the tail under confinement changes the 2 nm onset by 0.65 V (0.26 vs 0.91 V; S3, SURR); this is the dominant model-form uncertainty at 2 nm.
4. *Confinement* $\leftrightarrow$ $N_c(m^*) \leftrightarrow SS$. $m^* \times 1.3$ gives -44 mV and -17.7 mV/dec at 2 nm (run_0050); confinement also raises the Pd/IWO barrier at 2 nm (S5).
5. *Tail density* $\leftrightarrow$ *mobility* $\leftrightarrow$ *gap* $\leftrightarrow$ SS . B0 (run_0037) and C1 (run_0052) lie on one trade-off line of 3.2–8.5 mV of gap per mV/dec of SS_{cc} (S6 section 3.3; Section 8.3).
6. *Donors* / $E_{F0} \leftrightarrow$ *confinement* $\leftrightarrow$ *floor*. If the floor is an ungated IWO path, it bounds confinement and donor compensation jointly (Section 9.4; conditional).
7. $R_{\text{sd}} \leftrightarrow \alpha \leftrightarrow \theta$. Perfectly correlated ($r = +1.00$) in the Y-function analysis (S5).
8. *Bias stress* $\leftrightarrow$ N_t and V_{th} . Stress doubles N_t at 2 nm in the target paper, so sweep history can alter both the offset and the shape (Section 9.4).
9. *Floor* $\leftrightarrow$ *SS extraction*. The SS_{min} window and the 13.2 nm fixed-current SS (136.1 raw vs 91.8 mV/dec) depend on the floor.
10. *Numerics* $\leftrightarrow$ *thickness trend*. The DOS quadrature error of I_{on} scales roughly with $N_t(t)$ (+2.48/+0.95/+0.50 % at equal mobility; SYNTHESIS D9). It is removed by recalibration (run_0038–run_0040), but it shows that numerics can mimic a thickness trend (Section 6.6).

The 6.3 nm discrepancy illustrates these couplings. SYNTHESIS A.2 splits the horizontal offset $V_G^{\text{meas}}(I) - V_G^{\text{sim}}(I)$ into a rigid part (0.294 V with confinement, constant within 3 mV from 10^{-11} to 10^{-9} A/ μm in run_0014; 0.057 V without it, run_0034) and a shape part (offset minus the offset at 10^{-9} A/ μm) of +0.161/+0.190 V (run_0014), +0.139/+0.148 V (run_0015) and +0.156/+0.182 V (run_0034) at $10^{-7}/2 \times 10^{-7}$ A/ μm . The shape part is therefore independent of both the confinement law and Q_f ; at 2 and 13.2 nm the residual swing is only 0.07–0.08 V. The rigid part is an electrostatic question (Section 9.1). The shape part remains unexplained; the constant, density-independent

Table 9.2. Degenerate parameter sets on one ID–VG curve per film, the observable they share, and the independent experiment that separates them (SYNTHESIS Q4).

degenerate set	shared observable	experiment that breaks the degeneracy
WF, χ , Q_f , pure ΔE_c , filled D_{it} , $N_d t$ (rigid)	V_{th} offset	C–V $V_{FB}(t)$ (front sheet t^0 ; volume t , t^2 ; back sheet t/ϵ_s); KPFM/UPS for WF and χ
$\Delta E_c(t)$ vs a t -dependent fixed charge	$V_{th}(t)$	optical gap / IPES for $E_c(t)$ with C–V; $V_{th}(T)$
μ_{band} vs N_t ($N_t \times 1.5 \equiv \mu \times 1.33$)	I_{on}	gap and g_m (partial); split C–V $n_{total}(V_G)$; gated Hall
N_t , W_{TA} , D_{it} , deep states, near- E_c band, back acceptors	SS, gap	temperature series (subthreshold E_a gives $E_F - E_c$ without μ); multi-frequency C–V; passivation; more thicknesses
R_{sd} , tail exponent α , θ ($r = +1.00$)	g_m roll-off	TLM or L-series; low- V_D ID–VD
μ_{band} lumping R_{sd} , C_{eff}/C_{ox} (≈ 0.87), m^*	g_m	Hall; TLM; C–V
Drude term (-18% at 2 nm), roughness		
6.3 nm offset: film property vs device spread vs sweep history	$V_{th}(6.3)$	3–5 replicates ($n \approx 15.7\sigma^2/\Delta^2$); dual sweeps with dwell recorded
charge location (front / back / bulk)	second-order SS, gap	$V_{FB}(t)$; vacuum vs air; passivation
tail energy reference under confinement (0.26 vs 0.91 V at 2 nm)	2 nm onset	split C–V + Hall at 2 nm; temperature series
band vs percolation vs phonon transport	$\mu_{FE}(T)$	temperature series (Section 10.2)

mobility of the model is the leading candidate (SYNTHESIS C5, untested), with above-threshold trapping or sweep history as alternatives.

9.7 Identifiability

One transfer curve per film constrains a small number of parameter combinations, not the individual parameters. The one-at-a-time runs of Section 7.4 (Figure 7.5) give the following leverages at 2 nm: WF +0.1 eV shifts $V_{th,cc}$ by +0.100 V with $\Delta SS = 0$ (run_0048); $\chi \pm 0.1$ eV by ± 0.100 V (run_0020, run_0021); $N_t \times 1.5$ by +0.082 V, +49.0 mV/dec and -25.1% I_{on} (run_0023); $W_{TA} +5$ meV by +0.020 V and +4.6 mV/dec (run_0024); $D_{it} \times 3$ by +0.025 V (run_0025); $N_d \times 2$ by -0.009 V (run_0022); $\epsilon_{IWO} = 10.55$ by -0.001 V (run_0026); $m^* \times 1.3$ by -0.044 V and -18.0 mV/dec (run_0050); switching confinement off by -0.316 V, +15.3 mV/dec and +21.4% I_{on} (run_0016) (SS in the 10^{-10} – 10^{-9} A/ μ m window; figures/MANIFEST.md). Table 9.2 lists the degenerate sets and the experiments that break each degeneracy, and Figure 9.5 summarises them. This assessment rests on one-at-a-time perturbations only. A Jacobian or profile analysis over the fitted metrics has not been made (SYNTHESIS F.1, item 0.4), so the fitted parameter set is one member of a family of equivalent calibrations.

9.8 Verdicts

Table 9.3 collects the verdict for every mechanism discussed in this chapter, together with its evidence class and the main supporting number. The verdicts follow SYNTHESIS sections B–D, which resolve the disagreements between the specialist reports; where a specialist verdict was corrected, the table gives the corrected verdict.

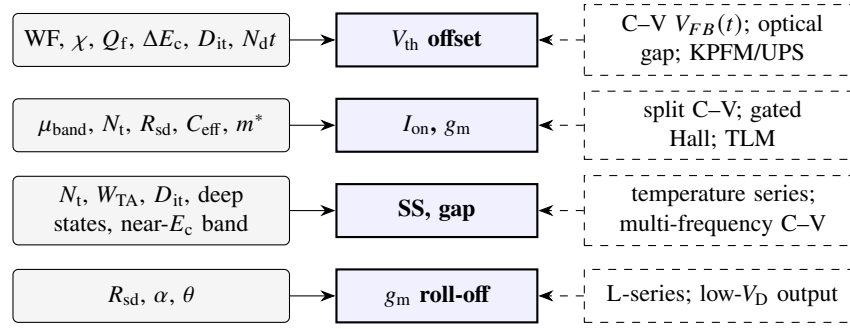


Figure 9.5. Identifiability map. Left: parameter sets that act on the same observable of one ID–VG curve (solid arrows); right: the independent measurements that separate them (dashed arrows). Summary of Table 9.2 (SYNTHESIS Q4).

Table 9.3. Mechanism verdicts. Classes: DATA, NUM, CAL, OOS, POST, SURR, THEORY, LIT, CORR. “Model-conditional” means within the V1 DOS and electrostatics.

mechanism / claim	verdict	class	key number (source)
<i>Electrostatics (Section 9.1)</i>			
6.3 nm threshold miss with shared laws	demonstrated (failed OOS)	OOS	−0.294 V (run_0014)
Tuned 6.3 nm-specific front charge	calibration only	CAL	$1.64 \times 10^{12} \text{ cm}^{-2}$ (run_0015)
Thickness error (A1)	rejected, model-cond.	NUM	+0.036 of +0.294 V (run_0031)
Fewer donors (A2)	rejected	NUM	+0.030 V (run_0032)
Back-surface charge as sole cause (A3)	rejected, model-cond.	NUM	$SS_{cc} -64.1 \text{ mV/dec}$ (run_0030)
Front D_{it} (A4)	rejected at plausible D_{it}	NUM	$\times 5$ gives +0.051 V; needs $\times 23$ –24 (run_0033)
Combination (A5)	rejected	NUM	73 % of miss, $SS_{cc} -45.9$ (run_0035)
Uniform donors / neutral layer as 6.3 → 13.2 step	rejected	SURR+DATA	slope ratio ≤ 2.35 vs 9–20 (S1)
CC-definition (mobility) share of the $V_{th,cc}$ step	strongly supported	DATA/NUM	+0.108 to +0.115 V of 0.579 V
qN_{dt}/C_{ox} ; interface-trap charge	demonstrated small	NUM	$\leq 0.070 \text{ V}$; 0.012 V (run_0022, run_0025)
Sweep history / device spread	not determined	–	0.003–0.24 V from PBS (S4)
<i>Transport (Section 9.2)</i>			
$\times 4.6 \mu_{FE}$ rise is a change in channel transport	strongly supported, model-cond.	CAL/SURR	109 % of $\ln(\times 4.58)$ in μ_{band} (S2)
Trap partition as the step	rejected	SURR	needs $N_t \times 6.7 \times 10^3$
Roughness factor as the step	rejected	CAL	$\times 1.051$ vs $\times 4.58$
Thickness-fluctuation scattering (t^6)	rejected	SURR	$\times 700$ contradiction at 2 nm
Coulomb (front / back / trapped)	rejected as the step	SURR	back $\times 2700$ short at 6.3 nm
S2 re-partition (B0)	rejected	OOS, failed	$SS_{cc} 410.9$ vs 295.4 (run_0037)
Structural transition 6.3–13.2 nm	plausible, unverified (working hypothesis)	CORR	39 meV barrier suffices (S2)
Power laws in t (μ_{FE} , I_{on} , V_{th})	rejected	DATA	LOO miss $\times 2.63$ / $\times 3.76$

mechanism / claim	verdict	class	key number (source)
Paper μ = saturation formula	demonstrated (provenance)	DATA	5.09 / 27.40 cm ² /V s
<i>Confinement (Section 9.3)</i>			
Confinement at 2 nm	expected (theory); not identified from ID–VG	THEORY	0.26–0.30 V (0.14–0.38) (S3)
Thresholds require or refute confinement	rejected	NUM on DATA	rms 0.136 vs 0.133 V (D3)
Confinement explains $2 \approx 6.3 \gg 13.2$	rejected	CORR/CAL	≤ 0.05 V at 6.3 nm vs 0.36 V needed
ΔE_c identifiable from one ID–VG	rejected	NUM	0.353 eV $\equiv 1.97 \times 10^{12}$ cm ⁻²
$\Delta E_c \propto t^{-1.38}$ beyond 3 nm as physics	rejected	THEORY	$\times 1.59 / \times 2.51$ above the infinite-well bound ($\times 1.45 / \times 2.15$ vs S3 effective width)
“Removing confinement improves SS”	rejected (artefact)	NUM	fixed-current SS 90.8 vs 92.0 (run_0013/run_0017)
Tail-energy reference under confinement	not determined	SURR	0.26 vs 0.91 V onset at 2 nm
Gate-field quantization (device impact)	plausible, unverified	SURR	+0.15–0.19 V in V_G (10^{12})
<i>Traps, donors, floors (Section 9.4)</i>			
Near- E_c exponential tail governs subthreshold	strongly supported	CAL+inversion	SS within 2.3 mV/dec; DOS ratio 0.81–1.17 (2, 13.2 nm), 0.90–1.46 (6.3 nm)
V1 DOS form	not supported	NUM	too soft +20/+6/+25 mV/dec; deep $\times 2.2$ –2.8
$N_t \propto t^{-0.75}$ as physics	rejected (kept as interpolation)	CAL	exponents 0.75 vs 1.71
Surface + bulk trap DOS	plausible, unverified	CORR	$D_s \approx 8.5 \times 10^{12}$, $g_b \approx 10^{19}$
Naive $SS_{cc}(t)$ trap reading	rejected	SURR	122 vs 75 mV/dec trap-free
Near- E_c band (C1) as full 6.3 nm explanation	rejected; partial contributor plausible	OOS, partly failed	gap +0.119 vs +0.28 V (run_0052)
Donor step at 13.2 nm	plausible, unverified	CORR	$+1.39 \times 10^{12}$ cm ⁻² needed
Q_f = ionized V_O layer	plausible, unverified	–	1.73×10^{12} cm ⁻²
Floors = gate leakage (6.3, 13.2 nm)	rejected	DATA	$\leq \times 1.06$ over V_{gd} 1.2–3.7 V
Floors = thermal generation	rejected	calculation	13–17 decades short
Floors = ungated IWO path	plausible, unverified	CORR	$\times 1432$, slope 3.78 (3.85 in S4)
<i>Contacts (Section 9.5)</i>			
Thickness-independent R_{sd} causes the trend	rejected	CORR+LIT	$\rho_c \approx 0.26 \Omega \text{ cm}^2$; $\theta_{net} < 0$
R_{sd} as minor contributor	not determined	CAL	$\lesssim 5 \times 10^4 / \lesssim 1\text{--}2 \times 10^4 \Omega \mu\text{m}$
Confinement-raised Pd/IWO barrier as the step	rejected	CORR	needs $\Delta\phi \geq 0.20$ eV
Vertical access; current crowding	rejected	estimate	$< 1\%$ of R_{on} ; $L_T \leq 0.35 \mu\text{m}$
Y-function R_{sd} / intrinsic μ	not determined (method invalid)	NUM	$r(\alpha, \theta) = +1.00$
run_0027 as overlap test	retracted (malformed)	audit	“Electrode shortened” $\times 3$

Summary. In these single devices and within the calibrated model, the dominant thickness effect is

a change in channel transport and charge between 6.3 and 13.2 nm, on top of trap-limited conduction through a near- E_c exponential tail present in all films. Contacts, roughness, electrostatics and tail filling cannot produce the $\times 4.6$ mobility step within the calibrated model (strongly supported, model-conditional). Its physical origin is not determined, with a structural transition as the plausible, unverified working hypothesis. Quantum confinement of ≈ 0.3 eV at 2 nm is theoretically expected but neither required nor identified by the data. The 6.3 nm discrepancy has a rigid part that cannot be assigned with $N = 1$ and an on-state shape part that no tested V1 change reproduces and that remains unexplained. Every mechanism verdict rests on one device per thickness; the experiments of Table 9.2 and the registered predictions of Chapter 10 are required before any mechanism claim can move beyond “plausible”.

Since the present data cannot decide between the remaining readings, the next chapter registers what the calibrated model predicts for measurements it has not seen, together with the rules by which those predictions will be judged.

Chapter 10

Registered predictions

A model that has only been fitted to the curves it describes is calibrated, not validated (Chapter 7). The out-of-sample tests available within the present data set failed or partly failed: the shared-parameter 6.3 nm threshold of `run_0014` missed by -0.294 V, and the registered 6.3 nm hypothesis tests B0 (`run_0037`) and C1 (`run_0052`) did not succeed (Chapter 8). The only remaining route to a validation claim is therefore prospective. Quantities that were not used in calibration and were not measured when the model was frozen must be computed, recorded with a timestamp and cryptographic hashes, and then compared with data obtained later. This chapter documents that register. It describes which output-characteristic (ID-VD) and elevated-temperature transfer quantities the calibrated V1 model predicts (Sections 10.1 and 10.2), which physical assumption each prediction tests, and which rules decide, before any comparison, whether an outcome confirms or falsifies the model (Section 10.3).

Registration. The register `analysis_2026-09-25/PREDICTIONS_REGISTER.md` was written at 2026-09-24T23:24:53Z by specialist S6 from the real ATLAS runs `run_0038`–`run_0047` (S6 made no ATLAS launch). None of the predicted quantities was used in calibration, and no measurement of them was available to the project at the time of registration. Temperature-dependent transfer curves of a 2 nm device at 65 and 85 °C exist in the supplementary information of the target paper [1, SI Fig. S12, referred to on p. 9 and captioned on p. 11]; they are not part of the package and form the first intended test (Section 11.3). The file list and SHA-256 hashes of all 44 registered files are reproduced in Chapter D; the synthesis re-computed all 44 hashes and found them unchanged (`scratch/SYNTHESIS/syn_checks_out.txt`, check [SYN-c]).

Model state that is registered. The registered model is V1 (`config/iwo_material_model.yaml`) in the physical structure, with classical electrons-only drift-diffusion with Fermi statistics, acceptor tail N_{TA}/W_{TA} ($W_{TA} = 0.040$ eV), deep and interface Gaussians, fixed front charge Q_f , uniform $N_{d,eff}(t)$, confinement law $\Delta E_c = 0.9205 t^{-1.3815}$ eV, $N_c(m^*(t))$, constant spatially uniform band mobility, ideal Ohmic Pd contacts, DOS discretization 384/192 levels. Calibrated values (300 K, $V_D = 0.7$ V, one ID-VG per film): $\mu_{band} = 17.6897, 11.582$ and 61.6046 $\text{cm}^2/\text{V s}$ for 2, 6.3 and 13.2 nm; $Q_f = 1.73 \times 10^{12} \text{cm}^{-2}$ (2 and 13.2 nm, shared) and $8.7 \times 10^{10} \text{cm}^{-2}$ (6.3 nm, device-specific “tuned” configuration). The 300 K reference curves are `run_0038` ($\times 0.999983$), `run_0039` ($\times 1.015074$) and `run_0040` ($\times 1.020113$); these factors are exact because I_D is linear in a uniform constant mobility (verified by the `run_0029/run_0038` ratio 1.024879 against 1.024873 expected

over 43 points; Section 6.5). The ID-VD runs reproduce the rescaled transfer curves at $V_D = 0.7$ V to better than 0.001 % at all 15 (film, V_G) points.

Inherited residuals. The predictions inherit the calibration residuals of Section 7.3: at 2 nm $g_{m,\max}$ -8.6 % and $V_{th,lin}$ -0.127 V; at 6.3 nm $g_{m,\max}$ -23 % and $V_{th,lin}$ -0.354 V (the on-state shape is not reproduced, Section 8.3; the 6.3 nm predictions are therefore of lower confidence); at 13.2 nm $g_{m,\max}$ -2.6 %. For this reason the primary registered quantities are changes (voltage shifts, ratios, activation energies) to be compared with changes measured on the same device relative to its own 300 K curve. Absolute values carry the 300 K calibration residual.

Extraction definitions. All metrics are extracted with the unchanged `scripts/extract_metrics.py` (SHA-256 `a7d0b236...`): $V_{th,cc}$ at $I_D = 10^{-9}$ A/ μ m (log-linear), $V_{th,lin}$ by secant g_m on the 0.05 V grid and linear extrapolation at $g_{m,\max}$, $\mu_{FE} = g_{m,\max}L/(WC_{ox}V_D)$ with $L = 20$ μ m, $C_{ox} = 8.955 \times 10^{-7}$ F/cm², $V_D = 0.7$ V, and $I_{on} = I_D(3 \text{ V})$. Simulated curves are first interpolated onto the 0.05 V measurement grid. Fixed-current quantities added by S6: $V(I)$, the gate voltage at $I_D = I$ (log-linear), and $SS_{[I_1, I_2]} = [V(I_2) - V(I_1)]/\log_{10}(I_2/I_1)$ for the windows $10^{-11}..10^{-10}$, $10^{-10}..10^{-9}$ and $10^{-10}..10^{-8}$ A/ μ m. SS_{min} is deliberately not registered (window-phase artefact, Section 5.10). The apparent activation energy is $E_a(V_G) = -k_B \partial \ln I_D / \partial (1/T)$ at fixed V_G and $V_D = 0.7$ V, by least squares over the available temperatures.

10.1 Output characteristics (ID-VD) at 300 K

Runs `run_0041` (B8, 2 nm), `run_0042` (B9, 6.3 nm tuned) and `run_0043` (B10, 13.2 nm) sweep V_D from 0 to 0.5 V in 0.05 V steps and from 0.5 to 3 V in 0.1 V steps at $V_G = 1, 1.5, 2, 2.5$ and 3 V (Figure 10.1). Because the calibration fixes only the $V_D = 0.7$ V column, the genuine prediction content is the shape of each output curve normalised to that calibrated point: the low- V_D linearity, the saturation voltage, the output conductance and the film ratios at low V_D . These test auxiliary assumptions (ideal Ohmic injection, negligible R_{sd} , constant mobility, no parallel path); they do not by themselves test the thickness mechanism (REVIEW objection U7, conceded in `SYNTHESIS.md` §G.2).

Low-drain linearity. For an Ohmic contact and gradual-channel transport, $I_D(0.1)/I_D(0.05) = 2[1 - 0.05/V_{ov}]/[1 - 0.025/V_{ov}] < 2$. Every registered value lies between 1.80 and 1.98, with negative curvature $\partial^2 I_D / \partial V_D^2$ at $V_D \rightarrow 0$ (from -1.48×10^{-7} to -2.21×10^{-6} A/ μ m/V²; register §4.2). No S-shape appears in any film. A measured ratio above 2.0, or an upward curvature below about 0.2 V, in particular if it occurs at 2 nm only, would falsify ideal Ohmic Pd/IWO injection; specialist S5 identified a confinement-raised injection barrier at 2 nm as the physical alternative (mechanism M2, Section 9.5).

Series resistance. The normalised ratio $I_D(0.1)/I_D(0.7)$ at $V_G = 3$ V is 0.172, 0.171 and 0.164 for 2, 6.3 and 13.2 nm. Source/drain resistance compresses the low- V_D end. A measured value that is lower than predicted by more than the repeatability, with a deficit that grows with I_D , indicates R_{sd} , and the thickness ordering of the deficit locates it.

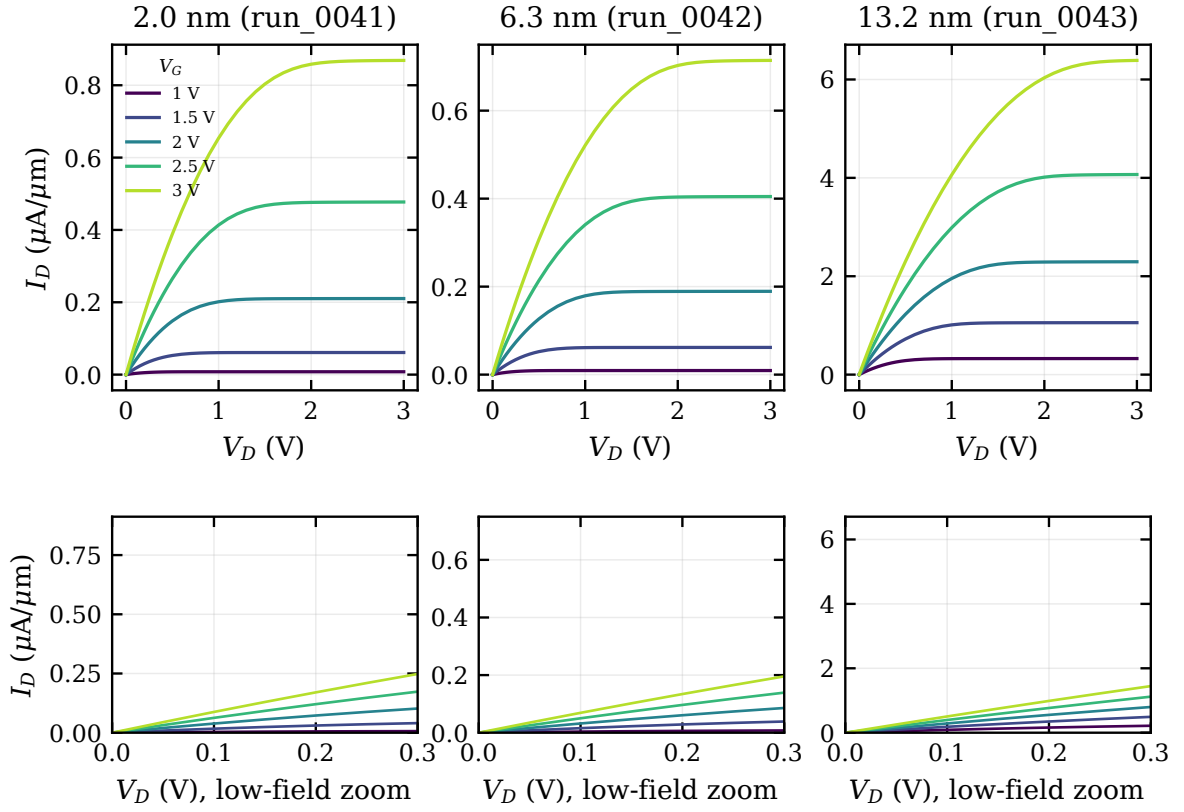


Figure 10.1. Registered ID-VD predictions of the calibrated V1 model at 300 K, run_0041 (2 nm), run_0042 (6.3 nm, tuned) and run_0043 (13.2 nm), recalibrated DOS 384/192, at $V_G = 1, 1.5, 2, 2.5$ and 3 V. Top row: full V_D range 0–3 V; bottom row: low-field zoom $V_D = 0\text{--}0.3$ V. All films show a strictly sub-linear low- V_D curvature (no S-shape) and a flat saturation (output conductance < 0.7 % of g_{d0}). $I_D(V_G = 3, V_D = 3 \text{ V}) = 8.687 \times 10^{-7}, 7.142 \times 10^{-7}$ and $6.388 \times 10^{-6} \text{ A}/\mu\text{m}$ (figures/MANIFEST.md).

Table 10.1. Registered ID-VD predictions (300 K). Drain current in $\text{A}/\mu\text{m}$ and shape metrics normalised to the calibrated $V_D = 0.7 \text{ V}$ point. $V_{ov} = V_G - V_{th,lin}$ of the 300 K model. $V_{d,sat}$ is the drain bias at which g_d falls to 10 % of g_{d0} (quadratic fit 0–0.1 V). Runs run_0041–run_0043; PREDICTIONS_REGISTER.md §4.1–4.2.

film	V_G (V)	V_{ov} (V)	$I_D(0.1)$	$I_D(0.7)$	$I_D(3)$	$\frac{I_D(0.1)}{I_D(0.05)}$	$\frac{I_D(0.1)}{I_D(0.7)}$	$V_{d,sat}$ (V)	$\frac{g_d(3)}{g_{d0}}$
2 nm	1.0	−0.54	3.34e-9	7.86e-9	7.888e-9	1.8011	0.4249	0.43	0.01 %
	1.5	−0.04	1.699e-8	5.841e-8	6.111e-8	1.8941	0.2909	0.68	0.02 %
	2.0	+0.46	3.86e-8	1.786e-7	2.103e-7	1.9414	0.2161	1.01	0.04 %
	2.5	+0.96	6.27e-8	3.376e-7	4.772e-7	1.9614	0.1857	1.40	0.08 %
	3.0	+1.46	8.768e-8	5.09e-7	8.687e-7	1.9716	0.1723	1.80	0.17 %
6.3 nm	1.0	−0.47	3.836e-9	9.214e-9	9.254e-9	1.8126	0.4163	0.44	0.01 %
	1.5	+0.03	1.575e-8	5.859e-8	6.181e-8	1.9103	0.2688	0.73	0.03 %
	2.0	+0.53	3.202e-8	1.553e-7	1.893e-7	1.9472	0.2061	1.09	0.05 %
	2.5	+1.03	5.005e-8	2.745e-7	4.048e-7	1.9636	0.1823	1.49	0.09 %
	3.0	+1.53	6.9e-8	4.034e-7	7.142e-7	1.9725	0.1710	1.90	0.19 %
13.2 nm	1.0	+0.01	8.863e-8	3.142e-7	3.263e-7	1.9057	0.2821	0.68	0.03 %
	1.5	+0.51	1.838e-7	8.85e-7	1.056e-6	1.9468	0.2077	1.05	0.05 %
	2.0	+1.01	2.872e-7	1.576e-6	2.294e-6	1.9639	0.1822	1.45	0.09 %
	2.5	+1.51	3.946e-7	2.312e-6	4.068e-6	1.9728	0.1707	1.87	0.20 %
	3.0	+2.01	5.044e-7	3.071e-6	6.388e-6	1.9783	0.1643	2.30	0.68 %

Saturation and output conductance. With constant mobility and zero R_{sd} , $V_{d,sat}$ rises from 0.43 to 1.80 V (2 nm), 0.44 to 1.90 V (6.3 nm) and 0.68 to 2.30 V (13.2 nm) for V_G from 1 to 3 V. The output conductance at 3 V is 0.01–0.19 % of g_{d0} for 2 and 6.3 nm and 0.03–0.68 % for 13.2 nm ($L = 20 \mu\text{m}$, no channel-length modulation, no parallel path in the model). Quasi-saturation more than about 0.3 V earlier would indicate field-dependent mobility or R_{sd} ; an output conductance of several percent, or one rising with t , would indicate an ungated or back-channel path (compare the off-floor analysis in Section 9.4).

Film ratios. The ratio $I_D(13.2)/I_D(2)$ at $V_D = 0.1$ V is 26.5, 10.8, 7.44, 6.29 and 5.75 for $V_G = 1, 1.5, 2, 2.5$ and 3 V (register §4.3; at $V_D = 3$ V the corresponding values are 41.4, 17.3, 10.9, 8.52 and 7.35). The ratio $I_D(6.3)/I_D(2)$ at 0.1 V is 1.149, 0.927, 0.829, 0.798 and 0.787. A low- V_D ratio that departs from the $V_D = 0.7$ V ratio beyond this predicted V_D dependence falsifies a thickness-independent (zero) contact resistance.

Numerical uncertainty. $V_{d,sat}$ carries ± 0.05 V (0.1 V grid); $g_d(3 \text{ V})$ at $V_G = 1$ V is set by current differences of about 5×10^{-13} A/ μm per 0.1 V and carries up to about 5 % numerical uncertainty (KCL residual up to 10^{-14} A in run_0041, run_0042). No DOS or mesh check was performed for ID-VD (Section 6.6; checklist item 2.3 in Section 11.3).

10.2 Elevated-temperature transfer characteristics

Runs run_0044 (B11, 2 nm, 338.15 K), run_0045 (B12, 2 nm, 358.15 K), run_0046 (B13, 6.3 nm, 358.15 K) and run_0047 (B14, 13.2 nm, 358.15 K) repeat the calibrated transfer sweep at $V_D = 0.7$ V. The declared temperature assumptions of campaign B were: $N_c, N_v \propto T^{1.5}$ (ATLAS); E_g temperature-independent (egalp = 0); N_{TA}, W_{TA}, D_{it} , deep states, Q_f and $N_{d,eff}$ temperature-independent; trap occupancy at the lattice temperature (isothermal, no self-heating); and a temperature-independent band mobility.

Limitation. ATLAS did not honour the last assumption: it applied its silicon default $\text{tmu} = 1.5$ to the constant mobility, so the runs as simulated (P-phonon, $\mu_{band} \propto T^{-1.5}$) and the same currents multiplied by the exact factors $(T/300)^{+1.5}$ (P-MTR, temperature-independent band mobility) are registered as two exact variants, and the as-simulated runs are never quoted as “temperature-independent mobility”; the evidence, the factors and the register’s prose typo are given in Section 5.8.

A measured result between the variants is read through an effective exponent g defined by $\mu_{band} \propto T^{-g}$,

$$g = -\frac{\ln(R_{meas}/R_{MTR})}{\ln(T/300 \text{ K})}, \quad (10.1)$$

where R is the measured and the P-MTR-predicted ratio of the same current quantity (I_{on} or μ_{FE}) at T and at 300 K; $g = 0$ is P-MTR and $g = 1.5$ is P-phonon.

The 300 K references are (register §5.1): $V_{th,cc} = 0.6804, 0.6529$ and 0.0536 V; $SS_{[10^{-10}, 10^{-9}]} = 215.0, 215.6$ and 143.4 mV/dec; $V_{th,lin} = 1.5365, 1.4664$ and 0.9946 V; $\mu_{FE} = 11.096, 8.393$ and 48.851 $\text{cm}^2/\text{V s}$; $I_{on} = 5.09 \times 10^{-7}, 4.034 \times 10^{-7}$ and 3.071×10^{-6} A/ μm for 2, 6.3 and 13.2 nm.

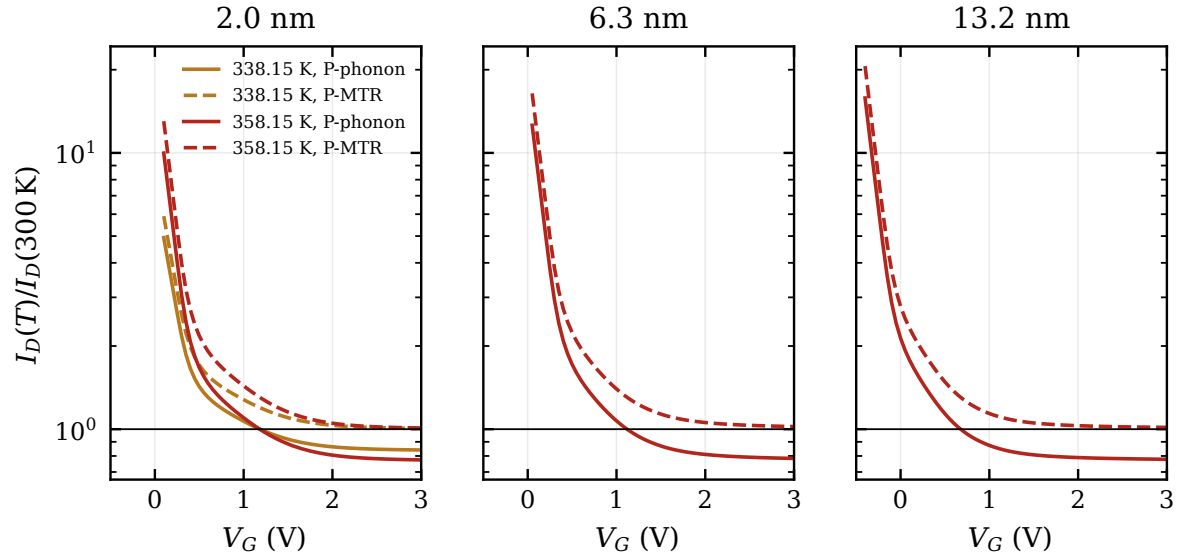


Figure 10.2. Predicted ratio $I_D(T)/I_D(300\text{ K})$ versus V_G at $V_D = 0.7\text{ V}$ for 2 nm at 338.15 K (run_0044) and 358.15 K (run_0045), 6.3 nm at 358.15 K (run_0046) and 13.2 nm at 358.15 K (run_0047), relative to the 300 K references run_0038, run_0039 ($\times 1.015074$) and run_0040 ($\times 1.020113$). Solid: P-phonon (as simulated, ATLAS $\text{tmu} = 1.5$). Dashed: P-MTR ($\times (T/300)^{1.5}$). At $V_G = 3\text{ V}$ the P-phonon/P-MTR ratios are 0.841/1.006 (2 nm, 338 K), 0.774/1.010 (2 nm, 358 K), 0.784/1.022 (6.3 nm) and 0.778/1.015 (13.2 nm); at 0.5 V the P-MTR ratio is 1.71, 2.18, 2.24 and 1.48 (figures/MANIFEST.md).

Table 10.2. Registered elevated-temperature predictions ($V_D = 0.7\text{ V}$): changes relative to each film's own 300 K reference (2 nm run_0038; 6.3 nm run_0039 $\times 1.015074$; 13.2 nm run_0040 $\times 1.020113$). Shifts in mV; fixed-current SS in mV/dec with its change in brackets. Runs run_0044–run_0047; PREDICTIONS_REGISTER.md §5.2.

film	$T\text{ (K)}$	variant	$\Delta V_{\text{th,cc}}$	$\Delta V(10^{-11})$	$\Delta V(10^{-7})$	$SS_{[10^{-10}, 10^{-9}]}$	$\Delta V_{\text{th,lin}}$	μ_{FE} ratio	$\Delta I_{\text{on}}\text{ (}\%\text{)}$
2 nm	338.15	P-phonon	−27.9	−28.9	+56.4	216.0 (+1.0)	−17.2	0.8312	−15.91
	338.15	P-MTR	−49.3	−36.9	−27.1	206.3 (−8.6)	−17.2	0.9947	+0.63
	358.15	P-phonon	−40.5	−44.0	+85.4	216.9 (+2.0)	−26.5	0.7604	−22.58
	358.15	P-MTR	−73.1	−54.6	−41.6	202.8 (−12.1)	−26.5	0.9919	+0.99
6.3 nm	358.15	P-phonon	−46.7	−48.3	+107.8	217.9 (+2.4)	−34.5	0.7664	−21.63
	358.15	P-MTR	−78.9	−60.2	−45.4	203.1 (−12.5)	−34.5	0.9997	+2.22
13.2 nm	358.15	P-phonon	−56.6	−55.4	+0.9	140.1 (−3.3)	−30.1	0.7668	−22.17
	358.15	P-MTR	−76.3	−65.2	−64.2	132.9 (−10.6)	−30.1	1.0002	+1.52

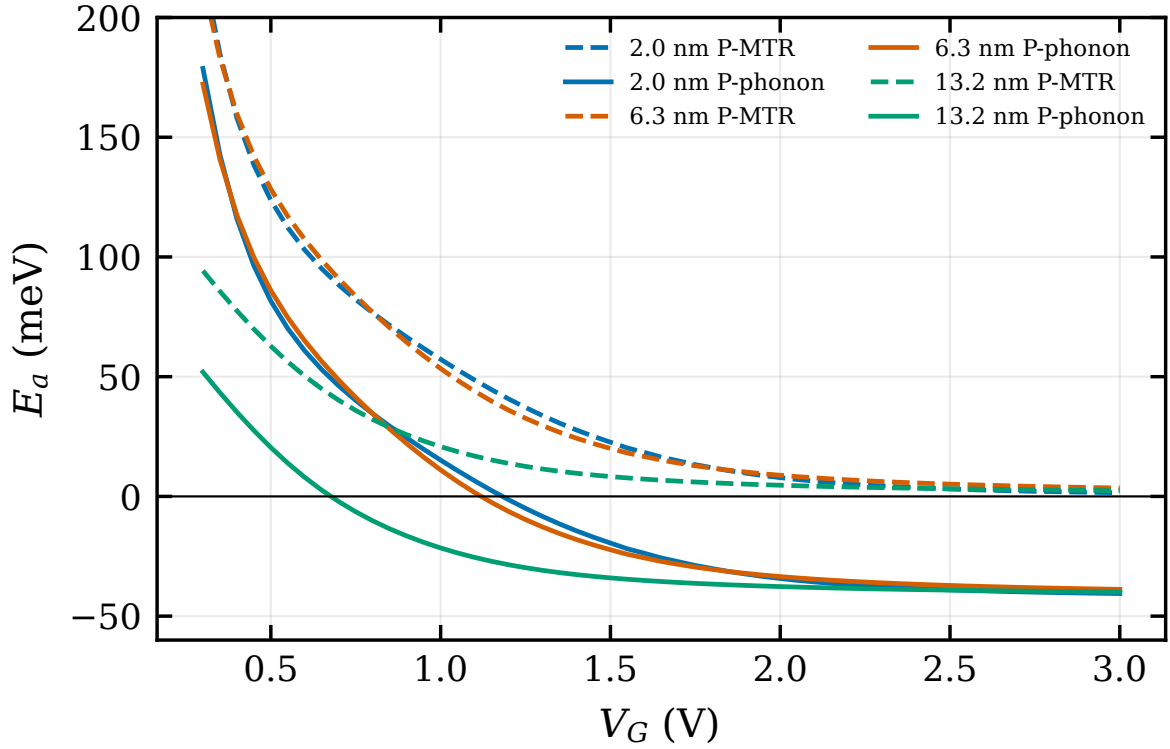


Figure 10.3. Predicted apparent activation energy $E_a(V_G)$ of I_D at $V_D = 0.7$ V for both variants and the three films, least-squares Arrhenius fit over 300/338.15/358.15 K (2 nm; run_0038, run_0044, run_0045) and 300/358.15 K (6.3 nm run_0039, run_0046; 13.2 nm run_0040, run_0047). At $V_G = 1$ V: P-MTR 57.3/53.4/20.8 meV, P-phonon 15.1/11.1/−21.5 meV for 2/6.3/13.2 nm (figures/MANIFEST.md). The two variants differ by the constant τ_{mu} term $1.5 k_B T_{eff} = 42.3$ meV.

Three features of Table 10.2 are relevant to the test design. First, $\Delta V_{th,lin}$ is identical in the two variants (−17.2 and −26.5 mV at 2 nm, −34.5 mV at 6.3 nm, −30.1 mV at 13.2 nm), because a common current factor does not move the linear-extrapolation intercept. It is therefore the most selective test of the equilibrium tail-filling electrostatics, independent of the mobility assumption. Second, $\Delta V_{th,cc}$ differs between variants, because the constant-current definition converts a current factor into a voltage (the same coupling as SYNTHESIS D6). Third, the thickness ratio $\mu_{FE}(13.2)/\mu_{FE}(2)$ changes from 4.403 (300 K) to 4.440 (358.15 K) in both variants (+0.8 %), and $I_{on}(13.2)/I_{on}(2)$ from 6.033 to 6.065. The model, which uses one transport mechanism in all films, thus predicts that the thickness contrast does not decrease with temperature.

For 2 nm, the bracketed pair values of the register (e.g. P-MTR at 0.5 V: +122.5 and +127.1 meV

Table 10.3. Registered apparent activation energy of I_D at fixed V_G (meV; $V_D = 0.7$ V). 2 nm: fit over three temperatures (the two pair values in brackets); 6.3 and 13.2 nm: 300/358.15 K. *Below $5\times$ the measured 300 K off-floor of that film: not measurable on the existing devices. PREDICTIONS_REGISTER.md §5.3.

film	variant	0 V	0.5 V	1.0 V	1.5 V	2.0 V	3.0 V
2 nm	P-phonon	n/a	+81.6	+15.2	−19.5	−34.3	−40.6
	P-MTR	n/a	+123.7	+57.3	+22.6	+7.8	+1.5
6.3 nm	P-phonon	+452.0*	+86.0	+11.1	−22.3	−33.5	−38.8
	P-MTR	+494.3*	+128.3	+53.4	+20.0	+8.8	+3.5
13.2 nm	P-phonon	+120.8	+20.4	−21.5	−34.0	−37.7	−39.9
	P-MTR	+163.1	+62.7	+20.8	+8.3	+4.7	+2.4

for 300–338 and 300–358 K) differ from the three-point fit by at most a few meV, so a single Arrhenius slope describes the model in this range. At $V_G = 0$ the 2 nm current is below 10^{-16} A/ μm and no value is registered.

Cross-check against the analytic MTR surrogate. Specialist S2 had registered, at 2026-09-24T18:57Z, a one-dimensional MTR surrogate expectation for 300→350 K with temperature-independent mobility. Interpolated linearly in T , ATLAS P-MTR gives $\Delta V_{\text{th,cc}} = -63.4$ (surrogate -64) and -65.6 (-66) mV, μ_{FE} ratios 0.9930 (0.993) and 1.0002 (0.999), and I_{on} ratios 1.0084 (1.004) and 1.0131 (1.011) at 2 and 13.2 nm; the largest difference is $\Delta V_{\text{th,lin}}$ at 2 nm (-22.7 against -15 mV), within the solver-grid bias of $V_{\text{th,lin}}$. The P-MTR activation energies agree with the surrogate to within 7 meV at every V_G (register §5.4). This is a model-versus-surrogate numerical consistency test, not an experimental validation (SURR against NUM).

Numerical uncertainty. I_D , I_{on} and μ_{FE} ratios: DOS residual beyond 384/192 of $+0.16\%$ at 2 nm (Richardson, observed order 2.0 from `run_0012`, `run_0019`, `run_0029`), smaller at 6.3 and 13.2 nm; mesh $\times 0.7 \leq 0.013\%$ (`run_0036`, `run_0028`); linear rescale exact to 10^{-5} . $V_{\text{th,cc}}$ and $V(I)$: ± 1 mV; fixed-current SS: ± 1 mV/dec; $V_{\text{th,lin}}$: grid-limited, -5 to -12 mV bias relative to a 0.05 V measurement grid (0.1 V solver steps above 1.5 V); $g_{\text{m,max}}$: -0.4 to -0.7% ; E_a : below 0.5 meV. DOS convergence was verified at 300 K only, not at 338 or 358 K. Physical and model-form uncertainty is not included in these numbers.

10.3 Falsification criteria and prospective status

Table 10.4 lists, for each registered prediction, the outcome that falsifies it and the assumption that would then be falsified. Table 10.5 gives the decision rules that the synthesis attached to the register before any comparison data were requested. The symbol σ_{rep} denotes the repeatability of the same device; it is not determined and must be measured before the comparison is reported.

Only the test of the μ_{FE} ratio versus T and the ordering of E_a (3 V) bear on the thickness mechanism; the ID-VD items test auxiliary assumptions (conceded, REVIEW U7).

Rules that preserve prospective status. The register remains prospective only if the following rules are observed (SYNTHESIS.md §G.3):

1. *Hashes.* All 44 files of register §10 are unchanged (check [SYN-c]); the register's own SHA-256 is 836ead1f731745f95e339dedff351c2cc4dae5c6cdd3979d5c992e8e3030a681 (Chapter D).
2. *External timestamp.* The register, `predictions_canonical.csv`, the synthesis and their hashes are deposited with an external timestamp (e-mail to the supervisor and an OSF/Zenodo or repository commit) before the S12 data are requested or opened. Until then the only timestamp is internal (2026-09-24T23:24:53Z); the limitation sentence is: “Predictions were registered internally with SHA-256 hashes; no third-party timestamp exists.”
3. *No edits.* Neither the register nor the CSV is edited. The prose multiplier error (1.19665/1.30426 for the exact 1.19669/1.30441) goes in a separately timestamped addendum.

Table 10.4. What each registered outcome would falsify (PREDICTIONS_REGISTER.md §6).

prediction	model value(s)	falsifying outcome	assumption falsified
Low- V_D linearity	$I_D(0.1)/I_D(0.05) = 1.80\text{--}1.98$; negative curvature	ratio > 2.0 or S-shape below ≈ 0.2 V, especially at 2 nm only	ideal Ohmic Pd/TWO injection; μ_{band} lumping
$I_D(0.1)/I_D(0.7)$ at 3 V	0.172 / 0.171 / 0.164	lower by more than repeatability, growing with I_D	negligible R_{sd}
Saturation	$V_{\text{d,sat}}$ 0.43–1.80, 0.44–1.90, 0.68–2.30 V	quasi-saturation > 0.3 V earlier	constant mobility, zero R_{sd}
Output conductance	0.01–0.19 % (2, 6.3 nm), 0.03–0.68 % (13.2 nm) of g_{d0}	several % or rising with t	no parallel/ungated path
Film ratio at low V_D	$I_D(13.2)/I_D(2) = 26.5 / 10.8 / 7.44 / 6.29 / 5.75$	departure from the predicted V_D dependence	zero contact resistance
ΔI_{on} (2 nm, 358 K)	P-MTR +0.99 %, P-phonon –22.6 %	$> +5$ % ($g < -0.2$) or near –22 %	T-independent band mobility (activated/percolation or phonon-like transport)
$\mu_{\text{FE}}(13.2)/\mu_{\text{FE}}(2)$	4.40 $\rightarrow$ 4.44 at 358 K	shrinks toward $\approx 2.7\text{--}3.3$	common band-transport mechanism in all films
$\Delta V_{\text{th,cc}}$ (358 K)	P-MTR –73/–79/–76; P-phonon –41/–47/–57 mV	outside –35 to –85 mV, or thickness spread $\gg 20$ mV	equilibrium tail filling as only T-dependent charge
Subthreshold E_a	P-MTR 124/128/63 (0.5 V), 57/53/21 meV (1 V)	ordering $E_a(2) \approx E_a(6.3) > E_a(13.2)$ violated by > 10 meV	tail/ E_F alignment ordering
On-state E_a (3 V)	P-MTR +1.5/+3.5/+2.4; P-phonon –40.6/–38.8/–39.9 meV	thin-film value exceeding 13.2 nm by > 10 meV	thickness-independent transport mechanism
SS (358 K)	$SS_{[10^{-10}, 10^{-9}]}$ P-MTR –12.1/–12.5/–10.6 mV/dec	increase > 20 mV/dec	T-independent W_{TA} , D_{it} ; equilibrium trapping

Table 10.5. Decision rules fixed before any comparison (SYNTHESIS.md §G.2).

test	outcome and meaning
ΔI_{on} (2 nm, 358 K)	$\geq +5$ %: activated band mobility or percolation; both variants and constant mobility falsified. Between -5 and $+5$ %: P-MTR consistent. ≤ -15 %: phonon-like; P-MTR falsified. Otherwise report g from Equation (10.1); neither variant confirmed.
$\Delta V_{\text{th,lin}}$ at 338/358 K (–17.2/–26.5 mV, both variants)	pass if $ \text{meas} - \text{pred} \leq \max(15 \text{ mV}, 2\sigma_{\text{rep}})$; the 15 mV exceeds the -5 to -12 mV grid bias. A fail falsifies the equilibrium tail-filling electrostatics.
$\Delta V_{\text{th,cc}}$ (358 K)	inside -35 to -85 mV: tail filling sufficient; outside: additional temperature-dependent charge.
$\Delta SS_{[10^{-10}, 10^{-9}]}$ at 358 K	increase > 20 mV/dec: temperature-dependent tail/ D_{it} or non-equilibrium trapping.
$\mu_{\text{FE}}(13.2)/\mu_{\text{FE}}(2)$ vs T	constant within ± 5 %: common mechanism; falling toward ≈ 3.3 : supports the structural-transition hypothesis C1 (Section 14.3).
Low- V_D output	$I_D(0.1)/I_D(0.05) > 2.0$ or S-shape (especially 2 nm only): non-Ohmic injection.
$I_D(0.1)/I_D(0.7)$ at $V_G = 3$ V	lower than predicted by more than σ_{rep} and growing with I_D : R_{sd} present; the film ordering of the deficit locates it.
μ_{sat} (6.3 nm) in the target paper's Fig. 5a	$3.86 \text{ cm}^2/\text{V s}$ within the digitisation error confirms the provenance chain (Section 5.5).

4. *No refit before comparison.* μ_{band} , Q_{f} , N_{t} and W_{TA} are not refitted before the comparison is written; the unchanged extractor is used. If S12 exists only as figures, it is digitised before the predictions are viewed and the digitisation error is recorded.
5. *Re-checks beside, never instead of, the registered numbers.* Tier-2 re-checks (explicit TMUN, DOS at 358 K, ID-VD mesh; Section 11.3) are reported next to the registered numbers; a shift beyond the registered numerical uncertainty is reported as a numerical flaw of that prediction.
6. *Device identity.* Whether the S12 device is the workbook 2 nm device is recorded (currently not determined). Only changes relative to that device's own 300 K curve are compared.

Any later change to the registered files, the extraction definitions or the model state invalidates the prospective status; a revised register must be a new, separately timestamped file.

Summary.

- ID-VD (run_0041–run_0043): strictly sub-linear low- V_{D} response ($I_{\text{D}}(0.1)/I_{\text{D}}(0.05) = 1.80\text{--}1.98$), $I_{\text{D}}(0.1)/I_{\text{D}}(0.7) = 0.172/0.171/0.164$ at $V_{\text{G}} = 3\text{ V}$, $V_{\text{d,sat}}$ up to 1.80/1.90/2.30 V and output conductance $\leq 0.68\%$ of g_{d0} . These test contact and parasitic-path assumptions, not the thickness mechanism.
- Temperature (run_0044–run_0047): two exact variants because ATLAS applied $\text{tmu} = 1.5$; factors $(338.15/300)^{1.5} = 1.19669$ and $(358.15/300)^{1.5} = 1.30441$. At 358 K, $\Delta I_{\text{on}}(2\text{ nm}) = +0.99\%$ (P-MTR) or -22.58% (P-phonon); $\Delta V_{\text{th,lin}} = -26.5\text{ mV}$ in both; $\mu_{\text{FE}}(13.2)/\mu_{\text{FE}}(2) 4.40 \rightarrow 4.44$.
- Status: PRED (registered 2026-09-24T23:24:53Z, 44/44 hashes unchanged), untested; no third-party timestamp yet.

The registered predictions provide the route to validation, but none of them has yet been tested. The next chapter states the validation status of every model claim and the work required, in order of dependency, before submission.

Chapter 11

Validation status, peer-review assessment and required work

A calibrated model reproduces the curves to which it was fitted, whereas a validated model predicts curves it has not seen. This chapter applies that distinction to every model claim of the report (Section 11.1) and answers the eight questions of the internal adversarial review with the evidence of Chapters 7 to 10 (Section 11.2). It then lists the minimum work required before submission in the order of scientific dependency, together with the limitation sentence to be printed if an item cannot be done (Section 11.3), and identifies the topics that belong in a second study (Section 11.4). The sources are analysis_2026-09-25/SYNTHESIS.md §E, F, H and I and analysis_2026-09-25/REVIEW.md.

11.1 Calibration versus validation status of every model claim

Table 11.1 contains no entry of class “validated”. The strongest positive statements are numerical (NUM) or rest on the measured data themselves (DATA), and every genuinely out-of-sample test run so far (run_0014, B0 run_0037, C1 run_0052) failed or partly failed. Two tests that were described as pre-registered in specialist reports (A6 and the B0 value of SS_{cc} given by specialist S1) are post hoc. The S1 code was saved at 19:18:00Z, during run_0037 (19:16–19:21Z), and configuration A was written days after the run_0014 miss was known.

11.2 Adversarial review: objections and responses

An independent referee pass (REVIEW.md) posed eight required questions. Table 11.2 gives the answer supported by the evidence of this report. In five of the eight cases the objection is conceded or agreed.

Further objections. The review raised sixteen further points (REVIEW §2) and lists of overstatements (O1–O9) and understatements (U1–U7); all are accepted. The most consequential points are the following. The drain bias and current normalisation come from the device schematic (the μ_{sat} reproduction fixes $L/(WC_{ox})$ and the $A/\mu m$ normalisation jointly, but not V_D). Since $N = 1$, no film-specific mechanism is claimed, and sweep history remains an open confounder of offset and shape. The 31.8 nm curve is a documented model failure ($I_D(-3 V) = 3.0 \times 10^{-7} A/\mu m$) and is never used as support. The origin of the off-floor is not determined, and the 13.2 nm SS is given both raw

Table 11.1. Status of every major model claim (SYNTHESIS.md §E). Evidence classes as in ?? .

claim	status	evidence
2 and 13.2 nm fits ($V_{th,cc} +19/-29$ mV; I_{on} 0.00 %)	CAL	run_0038, run_0040
6.3 nm tuned fit	CAL (device-specific Q_f , μ)	run_0015, run_0039
Shared laws predict the 6.3 nm threshold	OOS, failed (-0.294 V)	run_0014
6.3 nm “reproduced in shape”	SS yes; on-state no (gap -0.314 V, $g_m -16.6$ %)	Section 8.3
Confinement accounts for the 2 vs 13.2 nm step	not supported : realises 50.4 %; data neutral	Section 14.1
No confinement + Q_f from 2 nm predicts 6.3 nm	POST, threshold only (-0.057 V)	A6, run_0034
Subthreshold follows $N_t(t)$, W_{TA} , D_{it} “without tuning”	CAL (N_t anchor and W_{TA} fitted)	Section 7.3
μ_{band} per film; no mobility law	CAL	Section 5.1
Q_f rigid; $WF/\chi/Q_f$ one number; I_D linear in μ ;	NUM	run_0048–run_0051;
m^* second-order		run_0029/run_0038
300 K transfer convergence (where checked)	NUM	run_0019, run_0029, run_0036, run_0028
V1 2 nm mesh; DOS order at 6.3/13.2 nm; T and ID-VD convergence	not demonstrated	Section 6.6
Physical deck $\equiv$ reduced deck; donors, ε irrelevant at 2 nm	NUM	run_0008/run_0012, run_0009/run_0013, run_0022, run_0026
Contact geometry irrelevant	retracted	run_0027 malformed
A1–A5 rejected as sole cause	NUM, model-conditional	run_0030–run_0035
B0 re-partition	OOS, failed ($SS_{cc} +115.6$)	run_0037
C1 near- E_c band	OOS: passed on g_m , SS; failed on gap, I_{on}	run_0052
R_{sd} cannot create the trend	CORR + LIT; bounds CAL	Section 9.5
Paper μ = saturation formula	DATA (provenance)	Section 5.5
Structural step	CORR	hypothesis C1
Confinement 0.26–0.30 V at 2 nm (expected; not identified from ID-VG)	THEORY	Section 4.6
MTR surrogate vs ATLAS P-MTR	numerical cross-check	Section 10.2
ID-VD and 338/358 K predictions	PRED	run_0041–run_0047
Ideal Ohmic contacts, constant mobility, T-independent tails	assumed	register §6

Table 11.2. The eight required review questions and the strongest defensible response (SYNTHESIS.md §H.1).

question	response and evidence
(a) Are the parameters unique?	Conceded. WF, χ , Q_f and ΔE_c are one number (run_0048, run_0049, run_0001→run_0003); μ_{band} trades with N_t ($N_t \times 1.5 \equiv \mu \times 1.33$), R_{sd} with the tail exponent. The report speaks of a “calibrated, non-unique parameter set”; checklist item 0.4 identifies the combinations; C-V, split C-V/Hall and TLM pin three of them (Section 9.7).
(b) Validated or calibrated?	Conceded: calibrated. run_0014 is a failed out-of-sample threshold test; A6 is post hoc; B0 failed and C1 partly failed; the μ_{sat} reproduction is provenance evidence. The first validation is the register against the SI temperature data (Chapter 10).
(c) Could the power law be accidental?	Conceded. All thickness laws are two-point laws or extrapolations; the leave-one-out miss of a power law in μ_{FE} is $\times 2.63$ (exponent 0.7516). The step form is equally unsupported; no functional form in t is claimed (Section 5.12).
(d) Is the 6.3 nm miss understood?	Partly. The decomposition is demonstrated: rigid 0.294 V plus a shape part of +0.16 to +0.19 V at 10^{-7} – 2×10^{-7} A/ μm that is independent of confinement (run_0014, run_0015, run_0034). Eight electrostatic variants, B0 and C1 are excluded quantitatively; the cause is not assigned and the shape part remains unexplained (Section 8.4).
(e) Converged?	Agreed: partially. Converged where checked for 300 K transfer (DOS order 2.03 at 2 nm, residual +0.16 %; mesh $\times 0.7$ at 6.3 and 13.2 nm). Open: V1 2 nm mesh, DOS order at 6.3/13.2 nm, $T \neq 300$ K and ID-VD checks, the -5 to -12 mV $V_{\text{th,lin}}$ grid bias (Tiers 2–3).
(f) Is confinement demonstrated?	Conceded: assumed and theoretically expected, not identifiable. The residual signature of the confinement package is degenerate with m^* ($\times 0.8$) and N_t ($\times 1.09$), not with W_{TA} as the referee argued (on SS_{cc} , W_{TA} +5 meV moves +0.8 and $D_{\text{it}} \times 3$ +0.5 mV/dec against +14.6 for the package). “It was tested” is deleted.
(g) Are contacts and interfaces separated?	Agreed: partially. The trend argument stands (Section 14.2, S3). R_{sd} magnitude, charge location and interface vs bulk traps are not separated (Tiers 6–7).
(h) Is the preferred mechanism falsifiable?	Answered. Hypothesis C1 and the auxiliaries C3, C4, C5 and C10 have pre-declared falsifiers (Table 14.1); the ID-VD register items test auxiliary assumptions only.

(136.1) and floor-subtracted (91.8 mV/dec). The value $t_{mu} = 1.5$ is a configuration-control failure that is handled by two exact variants. `run_0027` is retracted and SS_{min} is retired. The log-RMSE is not comparable across films (active spans 6.57/5.25/4.59 decades), and the constant-current threshold is confounded by mobility. $\mu_{FE}(2\text{ nm})$ is a lower bound because $g_{m,max}$ falls at the last sweep point. Finally, a value of $\varepsilon(\text{Al}_2\text{O}_3)$ between 7 and 9 moves the EOT by about 3 %, and every q/C_{ox} charge with it.

Further expected criticisms. A first expected criticism is that three curves and about ten fitted numbers teach little. The contribution of the study, however, lies in the degeneracy map, the anomaly decomposition, the quantitative exclusions and the registered predictions, not in the parameter values. A second concerns the continuity of a 2 nm film; this is not determined, and the AFM roughness is higher at 2 nm than at 10 nm [1, p. 8]. A third concerns the validity of drift-diffusion at 2 nm. For trap-free electrostatics, Schrodinger-Poisson agrees within 11 mV and C_{eff} within 1 % (SURR); with traps, the tail energy reference (0.26 vs 0.91 V at 2 nm) is a stated model-form uncertainty. A fourth concerns the absence of a density-dependent mobility. This ATLAS version ignores MOBILITY updates between SOLVE statements, and no IWO-calibrated model exists locally, so C5 remains plausible and is deferred. A fifth is that the temperature data come from another device. Device identity is not determined, and only changes relative to that device's own 300 K curve are compared.

11.3 Minimum work before submission

Completed in this study. The following work was completed: the 6.3 nm hypothesis study (A1–A6 `run_0030`–`run_0035`, B0 `run_0037`, C1 `run_0052`), in which no tested mechanism explains the miss; the DOS 384/192 recalibration (`run_0038`–`run_0040`; refinement series `run_0019`, `run_0029`); the 6.3 nm mesh check (A7 `run_0036`, pass); the WF/m^* isolation (`run_0048`–`run_0051`); the ID-VD predictions (`run_0041`–`run_0043`, registered); and the temperature predictions (`run_0044`–`run_0047`, registered in two variants). The launch budget of 52 is exhausted (52 recorded launches, of which 49 produced scored currents; Section A.1). The following items were not done: explicit TMUN, Schrodinger-Poisson/BQP, a corrected overlap run, the V1 2 nm mesh check, the DOS order at 6.3 and 13.2 nm, DOS/mesh checks at $T \neq 300\text{ K}$ and for ID-VD, and the C1 DOS resolution. Together these require about 10–11 launches.

Checklist. The items are ordered by scientific dependency and labelled ANALYSIS, ATLAS or LAB. The italic sentence is the limitation to be printed if the item cannot be done.

- Tier 0 (analysis; no dependency).**
- 0.1 [ANALYSIS] Correct the package documents (Section 14.6): retract `run_0027`, re-tabulate with fixed-current metrics instead of SS_{min} , give the 13.2 nm SS raw and floor-subtracted. Mandatory; no limitation sentence is acceptable. *Done in this report.*
 - 0.2 [ANALYSIS] Freeze the register externally before any comparison data are requested. *“Predictions were registered internally with SHA-256 hashes; no third-party timestamp exists.”*

- 0.3 [ANALYSIS] Audit ATLAS defaults from one 300 K and one 358 K deckbuild.out (tmu 1.5, vsat 9.78×10^6 cm/s, SRH/Auger, lattice constant). *“Unset ATLAS material parameters took silicon defaults; the constant-mobility temperature exponent (1.5) was active in the elevated-temperature runs and is reported as a separate prediction variant.”*
- 0.4 [ANALYSIS] Identifiability: Jacobian or profile over the fitted metrics from the one-at-a-time pairs (2 nm: run_0012, run_0020–run_0026, run_0048, run_0050; 13.2 nm: run_0049, run_0051). *“Identifiability was assessed only by one-at-a-time perturbations; the fitted parameter set is one member of a family of equivalent calibrations.”*

Tier 1 (records; everything below depends on these).

- 1.1 [LAB/records] V_D , current normalisation, sweep direction, rate and dwell, ranges, temperature, ambient, device IDs, I_G/I_S , thickness method. *“Drain bias (0.7 V), current normalisation and geometry were taken from the device schematic; sweep direction, hysteresis, temperature and gate current were not recorded; one device per thickness was measured.”*
- 1.2 [LAB/records] Target-paper SI: Fig. S12 (2 nm at 338/358 K with its own 300 K curve), Fig. 5a μ_{sat} (6.3 nm), MIM/MISM C-V, ID-VD S7b, PBS of Fig. 6; establish device identity. Needs 0.2 and 2.1–2.2 first. *“No measurement independent of the calibration curves was available; the model is calibrated, not validated, and its registered predictions are untested.”*

Tier 2 (ATLAS; before a registered prediction is compared). 2.1 explicit TMUN = 0 re-run of B12 (expected: P-MTR within $\pm 0.2\%$); 2.2 DOS check at 358 K; 2.3 ID-VD mesh $\times 0.7$ near the drain. One launch each. *“Elevated-temperature predictions were computed with the ATLAS default constant-mobility exponent ($T^{1.5}$) and, by exact rescaling, for a temperature-independent band mobility; discretization convergence was verified at 300 K for transfer curves only.”*

Tier 3 (ATLAS numerical closure). 3.1 V_1 2 nm mesh $\times 0.7$; 3.2 DOS 192/96 at 6.3 and 13.2 nm (two launches); 3.3 C1 at 768/384 levels; 3.4 corrected overlap run (x.mesh beyond the electrodes, reject any log with “Electrode shortened”). *“Mesh convergence of the 2 nm deck is inherited from an earlier deck of the same generator; discretization order was established only at 2 nm; contact geometry and resistance were not tested.”*

Tier 4 (replicates; before any film-specific claim). 4.1 [LAB] 3–5 devices per thickness, same process run, dual sweeps with rate and dwell recorded; decides C4. *“Each thickness is represented by a single device; device-to-device variation was not quantified, so the 6.3 nm threshold offset cannot be distinguished from process variation or sweep history.”*

Tier 5 (thickness metrology; before any t -law). 5.1 [LAB] XRR or ellipsometry, cross-sectional TEM, XPS/RBS for W. *“Channel thicknesses are nominal, without measured uncertainty; ‘ $\sim 2\%$ W’ is undefined; all thickness laws are evaluated at nominal t .”*

Tier 6 (contacts and parasitic paths; needs 1, 5). 6.1 [LAB] TLM or L-series; low- V_D output (0–0.2 V); transfer at $V_D = 0.05$ –0.1 V; I_G , I_S recorded; mesa vs unpatterned devices. Tests the registered ID-VD shape and the floor path. *“Contacts were not characterised; extracted mobilities include any source/drain resistance, which is not identifiable from one transfer curve*

Table 11.3. Topics deferred to a second study and the reason (SYNTHESIS.md §I).

topic	why not now
QE/DFT for W:In ₂ O ₃ (W substitution, surfaces, tail localisation vs t) 31.8 nm and thicker films	no QE installation; even a correct ΔE_c is not identifiable from these data; useful only with an optical gap vs t needs a different mechanism ($N_d \geq 3 \times 10^{18} \text{ cm}^{-3}$ or a back donor sheet); reported here only as a failure
Schrodinger-Poisson/BQP with traps Density-dependent/percolation mobility	changes the model form; only the single check 8.1 belongs here leading candidate for the shape miss (C5); needs T and split C-V/Hall data; cannot be emulated within a sweep in this ATLAS version
Off-state floor physics Bias-stress and hysteresis dynamics	needs I_G/I_S , an L-series and mesa devices needs dual sweeps and PBS/NBS on replicates
Denser thickness series; functional form in t Contact physics with measured ρ_c	needs Tiers 4–5; a step and a steep sigmoid cannot be separated with three points needs TLM (6.1)

at one channel length; off-state currents are not attributed to the channel.”

Tier 7 (electrostatics; needs 5, 6). 7.1 [LAB] quasi-static, multi-frequency and split C-V per thickness ($V_{FB}(t)$, $n_{\text{total}}(V_G)$, charge-referenced threshold); 7.2 passivated vs air-exposed. *“Flat-band voltage and the free/trapped partition were not measured; work function, electron affinity, fixed charge and any confinement shift enter as one fitted offset per curve, and the charge location is not identified.”*

Tier 8 (band edge and confinement; needs 5, 7). 8.1 [ATLAS] Schrodinger-Poisson/BQP with the V1 DOS at the three thicknesses (register before 8.2; two to three launches); 8.2 [LAB] optical gap vs t , UPS/XPS, IPES. *“The confinement shift is an assumed input taken from PBE calculations on pure In₂O₃ slabs of 0.95–1.98 nm; it was not measured on IWO and cannot be distinguished from a fixed-charge offset of about $1.7 \times 10^{12} \text{ cm}^{-2}$ with the present data.”*

Tier 9 (transport mechanism; needs 6, 7). 9.1 [LAB] temperature series (≥ 3 –4 temperatures, 300–360 K) on all films; 9.2 [LAB] gated Hall; GIXRD/HRTEM/SEM grain size at 6.3 and 13.2 nm. *“Only room-temperature data were used; band mobility and tail-state density form a calibrated pair; the origin of the mobility increase between 6.3 and 13.2 nm is not determined.”*

Tier 10 (functional form in t ; needs all above). 10.1 [LAB] at least two more thicknesses (3–5 and 8–11 nm), predicted blind with the frozen model. *“No functional form in thickness is identified; the thickness laws are interpolations between two anchors.”*

Submission thresholds. A calibration or methods paper without a mechanism claim requires Tier 0, items 1.1 and 1.2 with the S12 comparison, and Tiers 2–3 (about 8 launches). A mechanism paper additionally requires Tiers 4, 5, 7 and 8.2.

11.4 What to leave for a second paper

Summary.

- No model claim is validated; every out-of-sample test so far (run_0014, run_0037, run_0052) failed or partly failed. The fits are calibrations of a non-unique parameter set.

- Of the eight review questions, (a), (b), (c), (f) are conceded, (e), (g) agreed as partial, (d) partly answered, (h) answered with pre-declared falsifiers.
- The first validation test is the registered temperature prediction against the existing 2 nm SI data, after external timestamping and the three Tier-2 launches. A methods paper needs Tiers 0–3 (about 8 launches); a mechanism paper additionally Tiers 4, 5, 7 and 8.2.

The final chapter collects the conclusions of the study, ordered by strength of evidence, together with the corrections that they imply for the earlier package documents.

Chapter 12

First-principles inputs: density-functional calculations of In_2O_3 and IWO films

The device model of the previous chapters needs to know how the electronic properties of an IWO film change when the film becomes only one or two nanometres thick: by how much the band gap opens, how the opening is shared between the conduction and the valence band, and how heavy the electrons become. It also needs to know what a tungsten atom does inside indium oxide. None of these quantities has been measured for the target films. Until this chapter they were taken over from published calculations on pure In_2O_3 and carried the provenance label `DFT_PURE_IN2O3_PROXY` with an assigned uncertainty of $\pm 50\%$ (Sections 4.5 and 4.7).

This chapter describes a first-principles study carried out to replace those proxies by values computed in one consistent setup, and to add what the literature does not give for IWO. Because the method is far from the device physics of the other chapters, the chapter is written so that it can be read without prior knowledge of density-functional theory (DFT). It proceeds in the order in which the work was done: why the calculations are needed (Section 12.1); the vocabulary (Section 12.2); how a DFT calculation works (Section 12.3); how the material is defined for the computer (Section 12.4); every calculation with its settings and result (Section 12.5); how the calculations were run on the CERN batch system and on GPUs (Section 12.6); how the numbers and figures are obtained from the output files (Section 12.7); and what changes in the device model (Section 12.9). Each idea is first explained in plain words and then stated precisely. All numbers are read from the output files of the calculations by scripts; the dated log of every value is `qe_workflow/cern_htcondor/results/RESULTS_LOG.md`.

Method references that are not bundled with the package (marked by the absence of a page number) are cited for the method as a whole; no page-specific statement is taken from them. Statements about Quantum ESPRESSO, Lin et al. and Si et al. carry the printed page of the bundled text.

12.1 Why first-principles calculations are needed

In plain words. A transistor channel that is only 2 nm thick is a few atomic layers of oxide. In such a thin film the electrons are squeezed between the two surfaces, and quantum mechanics then raises the

lowest energy they can have (the band gap opens) and changes how easily they move (their effective mass grows). The device simulator cannot work this out by itself; it must be told. The numbers can come from measurements on the films, which do not exist, or from a calculation that starts from the atoms alone. Such a calculation is called “first-principles” or “ab initio”, because it uses no adjustable parameters fitted to the material, only the laws of quantum mechanics and the positions of the atoms.

The thickness dependence enters the device model through two laws, the confinement shift of the conduction-band edge $\Delta E_c(t) = 0.9205 t^{-1.3815}$ eV (t in nm, Equation (4.3)), fitted to the slab-minus-bulk gaps of Lin et al. [2, p. 21542] and the conduction-band shift of Si et al. [4, p. 504], and an increment of the effective mass based on the slab masses of Lin et al. [2, p. 21540]. Three weaknesses motivated new calculations:

1. The proxies describe pure In_2O_3 ; the channel is W-doped, and the role of W was unknown.
2. The published values come from different codes and settings, and their uncertainty could only be assigned ($\pm 50\%$), not estimated.
3. The semi-local functional used in all of them is known to fail for band gaps (Section 12.3.3); whether it is adequate for the *change* of the gap with thickness had not been tested.

The study therefore asks five questions:

Q1 Does an independent calculation reproduce the published confinement of a 1 nm film?

Q2 What are the confinement shift and the mass at 2 nm, the thinnest device?

Q3 How is the gap opening shared between conduction and valence band (ΔE_c versus ΔE_v)?

Q4 Is the semi-local functional adequate for the shift, as opposed to the gap itself?

Q5 Where does W sit in the lattice, which valence does it take, and what happens to its electrons?

Figure 12.1 shows how the answers travel from the atoms to the device model.

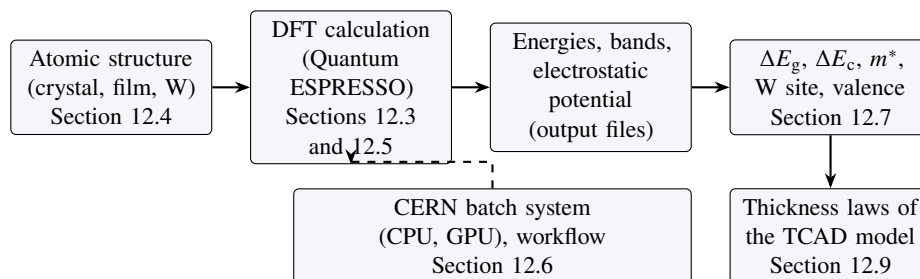


Figure 12.1. From atoms to device inputs: the path of this chapter. The structure is defined, the DFT calculation produces energies, bands and potentials, analysis scripts turn them into the quantities the device model uses, and these replace the literature proxies of the thickness laws.

What the calculations can and cannot deliver must be stated at the start. They describe ideal crystalline films with hydrogen-terminated surfaces in vacuum, whereas the device channels are sputtered, largely amorphous or nanocrystalline films on Al_2O_3 with an air-exposed top. The results are therefore *computed* inputs that describe trends and give physical grounding; they are not *validated* against measurements of the target films (Section 12.10).

12.2 Terms used in this chapter

Table 12.1 explains every technical term of the chapter in plain words, roughly in the order in which the terms appear. Table 12.2 lists the units: DFT codes work in “atomic” units (Rydberg, bohr), the device model in eV and nm.

Table 12.1. Glossary of the terms used in this chapter.

Term	Meaning in plain words	Section
<i>Physics of electrons in solids</i>		
Band, band structure	In a crystal the allowed electron energies form continuous ranges called bands. The band structure is the plot of these energies against the electron’s wave vector k .	12.3.4
Valence band (VB), conduction band (CB)	The highest band filled with electrons, and the lowest empty band above it. Current in an n-type oxide is carried by electrons in the CB.	12.3.4
VBM, CBM	Valence-band maximum and conduction-band minimum: the top of the VB and the bottom of the CB.	12.3.10
Band gap E_g	The energy distance CBM – VBM. “Fundamental” gap: smallest distance anywhere in k ; “direct” gap: distance at one k (here Γ).	12.3.10
Confinement shift $\Delta E_g(t)$	How much the gap of a film of thickness t exceeds the gap of the thick (bulk) crystal, because the film squeezes the electrons.	12.3.10
$\Delta E_c, \Delta E_v$	The parts of ΔE_g that come from the CB moving up and the VB moving down.	12.3.10
Effective mass m^*	How heavy an electron appears when it moves in the crystal; read from the curvature of the CB near its minimum. Given in units of the free-electron mass m_0 .	12.3.10
Non-parabolicity α	How much the CB deviates from a parabola away from its minimum (the mass grows with energy).	12.3.10
Wave vector k , Γ	A label of an electron state in a crystal (its momentum divided by $\hbar$). Γ is $k = 0$.	12.3.4
Brillouin zone (BZ)	The set of all distinct wave vectors of a crystal. Calculations sum over points of this zone.	12.3.4
k -point grid	The finite set of k points used for those sums, e.g. $3 \times 3 \times 3$. Denser grids are more accurate and more expensive.	12.3.4
Fermi level E_F	The energy up to which states are filled. In a metal it lies inside a band.	12.3.8
Density of states (DOS), projected DOS (PDOS)	How many electron states exist per energy interval; the PDOS splits this into contributions of each atom and orbital (e.g. the W 5d orbitals).	12.3.10
Vacuum level E_{vac}	The energy of an electron at rest just outside the material; the common reference that lets band edges of different films be compared.	12.3.10
Electron affinity (EA), ionization potential (IP)	$E_{\text{vac}} - \text{CBM}$ and $E_{\text{vac}} - \text{VBM}$: the depths of the band edges below the vacuum level.	12.3.10

Table 12.1, continued

Term	Meaning in plain words	Section
Spin polarization, magnetic moment	Whether spin-up and spin-down electrons are distributed differently; the difference, in Bohr magnetons (μ_B), is the magnetic moment.	12.5.9
<i>The DFT method</i>		
Density-functional theory (DFT)	A way to compute the ground state of many interacting electrons from the electron density alone, instead of the many-electron wave function.	12.3.2
Electron density $n(\mathbf{r})$	The number of electrons per volume at each point in space.	12.3.2
Kohn–Sham equations, orbitals, eigenvalues	The one-electron equations that DFT actually solves. Their solutions (orbitals) and energies (eigenvalues) are used as the band structure.	12.3.2
Exchange–correlation (xc) functional	The one approximate ingredient of DFT: the quantum-mechanical part of the electron–electron interaction.	12.3.3
PBE	The semi-local (generalized-gradient) xc functional used for all main results. Good for structures and energy differences, but gives band gaps that are far too small.	12.3.3
Hybrid functional, HSE06	An xc functional that mixes in 25 % of exact (Fock) exchange at short range. Gives much better gaps, at much higher cost. Used here as a check.	12.3.3
Plane waves, cutoff (ecutwfc, ecutrho)	The orbitals are written as sums of plane waves; the cutoff is the highest kinetic energy kept. A higher cutoff means more plane waves and more accuracy.	12.3.5
Pseudopotential, PAW, norm-conserving (NC)	Replacing the tightly bound core electrons and the steep nuclear potential by a smooth effective potential, so that few plane waves suffice. PAW (projector augmented wave) and NC are two kinds.	12.3.6
Self-consistent field (SCF)	The iterative loop: guess the density, compute the potential, solve for the orbitals, build a new density, repeat until input and output agree.	12.3.7
Mixing, mixing_beta, local-TF	How the next input density is formed from previous ones; a small mixing_beta is a cautious step; local-TF is a variant designed for slabs.	12.3.7
Convergence threshold	The tolerance at which an iteration is declared finished (e.g. conv_thr for the SCF).	12.3.7
Occupations: fixed, smearing	Insulators: every band either full or empty. Metals: occupations smeared over a small energy width so that sums over k converge.	12.3.8
Total energy, force, stress, pressure	The ground-state energy of the whole cell; its derivatives with respect to atom positions (forces) and cell shape (stress; pressure is its mean).	12.3.9
Relaxation (relax, vc-relax), BFGS	Moving the atoms (and in vc-relax also the cell) step by step downhill in energy until the forces vanish. BFGS is the step algorithm.	12.3.9
Non-SCF (“bands”) calculation	A run that keeps the converged density fixed and only computes orbitals at chosen k points, e.g. along lines for a band-structure plot.	12.3.10

Table 12.1, continued

Term	Meaning in plain words	Section
Planar / macroscopic average	The electrostatic potential averaged over each plane parallel to the film, and then over one atomic-layer period, to remove atomic wiggles.	12.3.10
<i>Describing the material</i>		
Unit cell, conventional / primitive cell, supercell	The repeating box of a crystal. The conventional cubic cell of In_2O_3 has 80 atoms; the smallest (primitive) cell has 40. A supercell is several cells stacked.	12.4.1
Bixbyite, space group $\text{Ia}\bar{3}$	The crystal structure of In_2O_3 and its symmetry group (number 206).	12.4.1
Wyckoff site (8b, 24d, 48e)	A class of symmetry-equivalent positions in the cell; the number is how many there are. In In_2O_3 indium occupies 8b and 24d, oxygen 48e.	12.4.1
Symmetry operation	A rotation, reflection or shift that maps the crystal onto itself; their number is a quick check that a structure is correct.	12.4.1
Slab, vacuum layer	A film model: a few atomic layers, repeated periodically with empty space (vacuum) between the copies so that they do not interact.	12.4.2
Passivation	Saturating the broken bonds of the surface oxygen atoms with hydrogen so that no spurious surface states appear in the gap.	12.4.2
Surface dipole	An electric dipole across the film when its two surfaces differ; it would tilt the vacuum potential. The slabs here have equivalent surfaces and no dipole.	12.4.2
Substitution, cation fraction	Replacing one In atom by W; the cation fraction $W/(W+\text{In})$ is the W content.	12.4.3
<i>Computing</i>		
Quantum ESPRESSO (QE), <code>pw.x</code> , <code>pp.x</code> , <code>projwfc.x</code>	The open-source DFT package used; <code>pw.x</code> solves the equations, <code>pp.x</code> extracts potentials, <code>projwfc.x</code> computes the PDOS.	12.5
MPI rank, pool (-nk)	The parallel processes of one calculation, and groups of them that each handle a subset of the k points.	12.6.2
CPU, GPU, core-hour	Processor types; GPUs (graphics processors) run QE several times faster for large cells. A core-hour is one CPU core used for one hour.	12.6.3
HTCondor, job, flavour, wall time	CERN's batch system; a job is one queued calculation; the flavour sets its maximum run time (wall time).	12.6.1
lxplus, EOS, Kerberos ticket	CERN's login computers, its disk storage, and the time-limited login credential.	12.6.1
Container (Apptainer)	A packaged software environment; used to run NVIDIA's GPU build of QE.	12.6.3
Checkpoint, continuation	A saved copy of a job's results after each step, and a resubmission that restarts an unfinished relaxation from its last geometry.	12.6.4

Table 12.2. Units used by the DFT code and their conversion (CODATA 2018 values, as in the analysis scripts).

Unit	Value	Used for
Rydberg (Ry)	1 Ry = 13.605693 eV	energies, cutoffs, convergence thresholds
bohr	1 bohr = 0.529177 Å	lengths, forces (Ry/bohr)
kbar	1 kbar = 0.1 GPa	stress and pressure
$\hbar^2/2m_0$	3.80998 eV Å ²	effective-mass fits
μ_B	Bohr magneton	magnetic moments

12.3 How a density-functional calculation works

This section explains the method from the beginning. The textbook treatment is Martin [25]; the implementation used is Quantum ESPRESSO [14, 26].

12.3.1 The problem: many interacting electrons

In plain words. All properties of a solid follow from its electrons, which move around the much heavier nuclei and repel each other. Quantum mechanics describes them exactly by one equation, but that equation couples every electron to every other one. For the 176-atom film of this study, which has 1328 electrons, it cannot be solved directly. DFT is the standard way around this.

Because nuclei are thousands of times heavier than electrons, the electrons can be treated as moving in the field of fixed nuclei at positions $\{\mathbf{R}_I\}$ [27]. The electrons then obey

$$\left[\sum_i -\frac{\hbar^2}{2m_0} \nabla_i^2 + \sum_i V_{\text{ext}}(\mathbf{r}_i) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} \right] \Psi = E \Psi, \quad (12.1)$$

where V_{ext} is the attraction of the nuclei and $\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$ depends on all N electron coordinates at once. Storing Ψ on even a coarse grid is impossible for N in the hundreds.

12.3.2 The density instead of the wave function: Kohn–Sham equations

In plain words. DFT rests on a theorem: the ground-state energy is fully determined by the electron density $n(\mathbf{r})$, a function of only three coordinates. Kohn and Sham then showed how to get that density by solving one-electron equations: each electron moves in an average potential created by the nuclei and by all the other electrons. Everything that is hard about the interaction is collected in one term, the exchange–correlation energy, which has to be approximated.

Hohenberg and Kohn [28] proved that the ground-state energy is a functional $E[n]$ of the density and is minimal for the true density. Kohn and Sham [29] wrote the density as that of fictitious non-interacting electrons in orbitals ψ_i ,

$$n(\mathbf{r}) = \sum_i f_i |\psi_i(\mathbf{r})|^2, \quad (12.2)$$

with occupations f_i , and obtained the Kohn–Sham equations

$$\left[-\frac{\hbar^2}{2m_0} \nabla^2 + V_{\text{ext}}(\mathbf{r}) + V_H[n](\mathbf{r}) + V_{\text{xc}}[n](\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r}). \quad (12.3)$$

V_H is the classical (Hartree) repulsion of the density with itself and $V_{xc} = \delta E_{xc}/\delta n$ the exchange–correlation potential. The total energy is

$$E[n] = T_s[n] + \int V_{\text{ext}} n \, d\mathbf{r} + E_H[n] + E_{xc}[n] + E_{II}, \quad (12.4)$$

with T_s the kinetic energy of the orbitals and E_{II} the repulsion between nuclei. Because V_H and V_{xc} depend on n , which depends on the orbitals, Equation (12.3) must be solved self-consistently (Section 12.3.7). The eigenvalues ε_i are used as the band energies; strictly they are auxiliary quantities, which is the origin of the band-gap problem discussed next.

12.3.3 The approximation: exchange–correlation functionals

In plain words. The exchange–correlation term is the only approximation in DFT, and the choice of approximation decides what the calculation gets right. The common choice (PBE) gives good crystal structures and energy differences, but it places the conduction band too low: band gaps come out about half or less of the measured value. More expensive “hybrid” functionals (HSE06) correct most of this. This study uses PBE for everything and checks with HSE06 that the *change* of the gap with film thickness is reliable, even though the gap itself is not.

All main calculations use the generalized-gradient approximation of Perdew et al. [30] (PBE), in which E_{xc} depends on n and $|\nabla n|$ at each point. Semi-local functionals underestimate band gaps, partly because the Kohn–Sham gap lacks the discontinuity of the exact exchange–correlation potential at integer electron number [31, 32]. For In_2O_3 the PBE gap of Lin et al. [2, p. 21542] is 0.94 eV, against experimental values of 2.7–3.25 eV quoted on the same page; the device model uses 3.05 eV. Lattice constants are typically overestimated by 1–2 % with PBE (Section 12.5.4).

The hybrid functional HSE06 [33–35] splits the Coulomb interaction into a short-range and a long-range part with a screening parameter ω and replaces a quarter of the short-range PBE exchange by exact (Fock) exchange:

$$E_{xc}^{\text{HSE}} = \frac{1}{4} E_x^{\text{HF,SR}}(\omega) + \frac{3}{4} E_x^{\text{PBE,SR}}(\omega) + E_x^{\text{PBE,LR}}(\omega) + E_c^{\text{PBE}}. \quad (12.5)$$

The output of the HSE06 runs confirms the exact-exchange fraction of 0.25. Exact exchange couples every occupied orbital at $\mathbf{k}$ with every orbital at $\mathbf{k}' = \mathbf{k} + \mathbf{q}$; Quantum ESPRESSO evaluates these terms with fast Fourier transforms, at a cost at least an order of magnitude above a semi-local calculation, and allows the sum over $\mathbf{q}$ to run over a coarser auxiliary grid whose adequacy must be checked [14, pp. 16–17]. The implementation in the versions used applies the exchange operator in the adaptively compressed form of Lin [36].

12.3.4 Crystals: Bloch states, wave vectors and k-point sampling

In plain words. A crystal repeats itself, so the computer only needs one repeating box (the unit cell). The price is that each electron state carries a label k , its wave vector, and the properties of the crystal are averages over all k in a finite region called the Brillouin zone. In practice the average is taken over a grid of k points. Plotting the energies against k along lines through this zone gives the familiar band-structure diagram.

In a periodic potential every orbital can be written as a plane wave times a cell-periodic function [37], $\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$, with band index n and wave vector $\mathbf{k}$ in the first Brillouin zone. Densities and energies become averages over $\mathbf{k}$, evaluated on Monkhorst–Pack grids [38]: an $N_1 \times N_2 \times N_3$ grid samples the zone uniformly; for slabs the grid is $3 \times 3 \times 1$, because there is no dispersion across the vacuum. Quantum ESPRESSO detects the crystal symmetries and restricts the sum to the irreducible wedge of the zone with appropriate weights, then symmetrizes the density, forces and stress [14, pp. 7, 16]. The bulk primitive cell has 4 irreducible points at $3 \times 3 \times 3$ and 8 at $4 \times 4 \times 4$; the 1 nm slab has 4 at $3 \times 3 \times 1$ (output files of the runs).

12.3.5 Plane waves and the cutoff energy

In plain words. To put the orbitals into a computer, each one is written as a sum of simple waves of different wavelengths. Short wavelengths are needed to describe sharp features; the cutoff energy decides how short the shortest wave is. A higher cutoff gives a more accurate answer and costs more. The cutoff is therefore chosen by testing: it is raised until the results stop changing (Section 12.5.3).

The periodic part of each orbital is expanded in plane waves with reciprocal-lattice vectors $\mathbf{G}$,

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} c_{n,\mathbf{k}+\mathbf{G}} e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}, \quad \frac{\hbar^2}{2m_0} |\mathbf{k} + \mathbf{G}|^2 \leq E_{\text{cut}}, \quad (12.6)$$

where E_{cut} is `ecutwfc`. In Rydberg units $\hbar^2/2m_0 = 1$, so the condition reads $|\mathbf{k} + \mathbf{G}|^2 \leq E_{\text{cut}}$ in bohr^{-2} , and the number of plane waves is close to the number of lattice points in a sphere of radius $\sqrt{E_{\text{cut}}}$,

$$N_{\text{PW}} \approx \frac{\Omega E_{\text{cut}}^{3/2}}{6\pi^2}, \quad (12.7)$$

with Ω the cell volume in bohr^3 . For the relaxed bulk primitive cell ($\Omega = 3693.6 \text{ bohr}^3$) and 71 Ry this gives 37 315, against 37 359 plane waves at Γ in the output; for the relaxed 1 nm slab ($\Omega = 25 140 \text{ bohr}^3$) it gives 253 982, against 253 973. The density, a product of orbitals, needs a higher cutoff `ecutrho`: for PAW and ultrasoft data sets typically 6–12 times `ecutwfc`, for norm-conserving pseudopotentials 4 times [14, p. 7]. This work uses 8 (568 Ry) with PAW and 4 (320 Ry) with the norm-conserving set of the hybrid check. The density of the 1 nm slab is then represented by 5 746 731 plane waves on a $150 \times 150 \times 500$ fast-Fourier-transform grid.

12.3.6 Pseudopotentials and the projector augmented-wave method

In plain words. Near a nucleus the orbitals oscillate rapidly, which would need an enormous number of plane waves. The core electrons there do not take part in bonding. Pseudopotentials replace the nucleus and its core electrons by a smooth effective potential that acts on the outer (valence) electrons only. The PAW method does this while keeping a way to reconstruct the true orbitals near the nucleus, so it is accurate at a moderate cutoff.

The atomic cores are described by the projector augmented-wave (PAW) method [39, 40] with the `pslibrary 1.0.0` data sets [41]; Quantum ESPRESSO supports norm-conserving, ultrasoft and PAW descriptions [14, pp. 5, 7]. Table 12.3 lists the data sets. The valence configuration decides

how many electrons the calculation contains and therefore how many bands are occupied. For the hybrid-functional check, optimized norm-conserving Vanderbilt pseudopotentials [42] of the SG15 library [43] were used instead (Section 12.5.8).

Table 12.3. Pseudopotential data sets (PAW, pslibrary 1.0.0). Valence charge and suggested cutoffs from the file headers; the calculations use 71/568 Ry, above every suggestion.

Element	File	Valence electrons	Suggested cutoffs (Ry)
In	In.pbe-dn-kjpaw_psl.1.0.0.UPF	13 ($4d^{10}5s^25p^1$)	41 / 212
O	O.pbe-n-kjpaw_psl.1.0.0.UPF	6 ($2s^22p^4$)	47 / 323
W	W.pbe-spn-kjpaw_psl.1.0.0.UPF	14 ($5s^25p^65d^46s^2$)	50 / 475
H	H.pbe-kjpaw_psl.1.0.0.UPF	1	46 / 221

Counting electrons is a useful check of every structure. The bulk primitive cell $\text{In}_{16}\text{O}_{24}$ has $16 \times 13 + 24 \times 6 = 352$ electrons and therefore 176 occupied bands; the 1 nm slab $\text{In}_{24}\text{O}_{48}\text{H}_{24}$ has 624 (312 occupied bands); the 2 nm slab $\text{In}_{56}\text{O}_{96}\text{H}_{24}$ has 1328 (664). A W atom carries one electron more than In in these data sets (14 against 13), but the chemically relevant count is different: In gives three electrons to oxygen (In^{3+}), whereas W in its highest oxidation state gives six (W^{6+}), so each W substituted for In adds three electrons that oxygen cannot take. The doped 80-atom cell has 705 electrons, an odd number, and is a metal in the calculation.

12.3.7 The self-consistent field loop

In plain words. The potential in the Kohn–Sham equation depends on the density, and the density depends on the solutions. The calculation therefore guesses a density, computes the potential, solves the equations, builds the new density, and repeats. Each round is one “SCF iteration”. To avoid oscillations, the new guess is a careful blend of the previous ones (“mixing”). The loop stops when the density no longer changes by more than a tiny tolerance.

Figure 12.2 shows the loop. Self-consistency is recast as a fixed-point problem $x = F[x]$ for the Fourier components of the density, and Quantum ESPRESSO solves it with the modified Broyden method of Johnson [44]: the next input density is the optimal combination of the inputs of a few previous iterations, plus a step along the residual [14, pp. 7, 13–14]. The step can be preconditioned by an approximate model of electronic screening: simple, Thomas–Fermi, or local Thomas–Fermi (local-TF) mixing [14, p. 14] [45]. Local-TF accounts for the strongly inhomogeneous density of a slab with vacuum, and is used for all slabs here; `mixing_beta` sets the size of the step (0.4 bulk, 0.2 slabs, 0.3 doped bulk).

The convergence measure is an estimate of the error of the total energy built from the density residual, $\|\rho^{\text{out}} - \rho^{\text{in}}\|^2 \propto \sum_{\mathbf{G}} |\Delta\rho(\mathbf{G})|^2 / G^2$ [14, p. 14], printed as “estimated scf accuracy”. The loop stops when it falls below `conv_thr` = 10^{-9} Ry for every calculation of this study. Within each iteration the Kohn–Sham equations are solved iteratively for the lowest bands only, by the block Davidson method or by band-by-band conjugate-gradient minimization [14, pp. 14–15] [46, 47]. Davidson is faster; conjugate gradient (CG) is slower but more robust (Section 12.5.5).

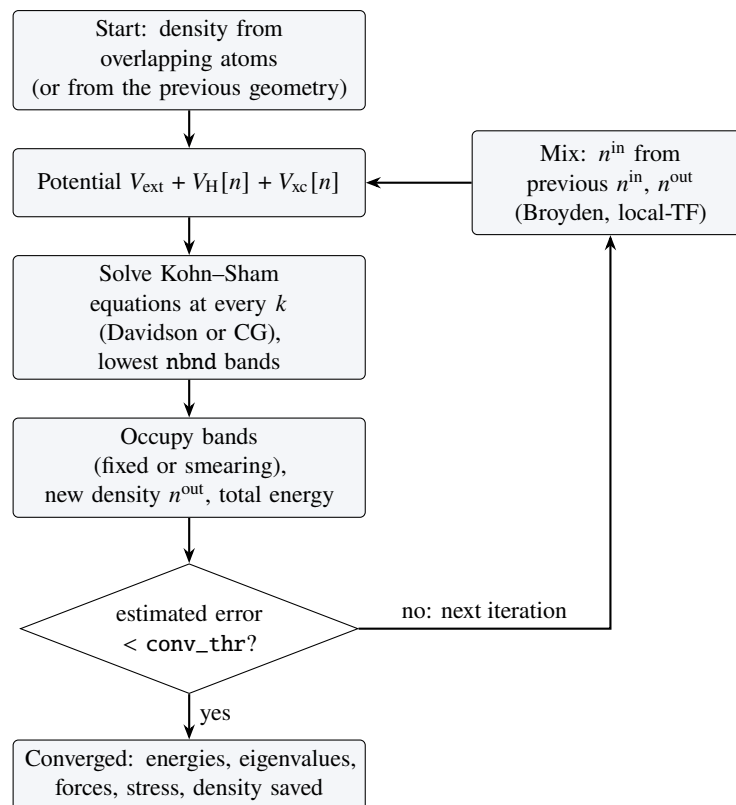


Figure 12.2. The self-consistent field (SCF) loop of a Kohn–Sham calculation, as implemented in Quantum ESPRESSO [14, pp. 13–15]. In this study the loop stops at an estimated energy error of 10^{-9} Ry; it takes 25 iterations for the bulk cell and 47 for the relaxed 1 nm slab.

12.3.8 Insulators and metals: occupations

In plain words. In an insulator every band is either completely full or completely empty, and the program simply fills the lowest ones. In a metal a band is partly filled, and the boundary between filled and empty states (the Fermi level) cuts through it. Sums over a finite k grid then converge badly, so the occupations are smoothed over a small energy window.

Pure In_2O_3 cells and slabs are insulators and use fixed occupations. The W-doped cells and slabs have three extra electrons per W in the conduction band and are metallic in the calculation; they use the cold smearing of Marzari et al. [48] with a width of 0.01 Ry, one of the smearing schemes of Quantum ESPRESSO [14, p. 7]. The Fermi level E_F is then determined by the electron count.

12.3.9 Forces, stress and relaxation

In plain words. Once the electrons are in their ground state, the program can compute the force on every atom and the stress on the cell. If the forces are not zero, the atoms are not where they want to be. A relaxation moves them step by step downhill in energy, recomputing the electrons at each step, until the forces are negligible. In a variable-cell relaxation the size and shape of the box are adjusted as well.

Forces are the derivatives of the total energy with respect to the atomic positions, obtained from the ground state by the Hellmann–Feynman theorem [49], and the stress tensor follows from the

derivative with respect to a strain of the cell [50]; the pressure is minus one third of its trace. Quantum ESPRESSO computes both and uses them for structural optimizations of atomic coordinates and cell [14, p. 5]. The optimization uses the Broyden–Fletcher–Goldfarb–Shanno (BFGS) quasi-Newton algorithm [14, p. 7]: each step moves along $-\mathbf{H}^{-1}\nabla E$, where the inverse Hessian $\mathbf{H}^{-1}$ is built up from the change of the gradients between steps. Variable-cell runs (`vc-relax`) use the invariant formulation of Wentzcovitch [51]. For slabs only the in-plane cell vectors are allowed to change (`cell_dofree = '2Dxy'`), so that the vacuum stays fixed. A relaxation ends when the energy change between steps and the largest force component fall below `etot_conv_thr` and `forc_conv_thr`, and, for `vc-relax`, the pressure below `press_conv_thr` (values in Table 12.6). At the end, `pw.x` repeats the SCF with the plane-wave basis of the final cell, which removes the error of a basis set that was defined for the starting cell (Pulay stress).

12.3.10 What is read from a calculation

In plain words. A finished calculation gives a list of energies at each k point. The highest filled and lowest empty ones are the band edges, and their difference is the gap. The curvature of the conduction band near its minimum gives the effective mass. The average electrostatic potential far out in the vacuum is the vacuum level, which serves as a common zero, so that band edges of different films can be compared. Figure 12.3 shows how these quantities are defined.

Band edges and gaps. `pw.x` prints the highest occupied and lowest unoccupied eigenvalue of every SCF; their difference is the Kohn–Sham gap on the k grid (which contains Γ). For the bulk the fundamental gap is taken from a separate non-self-consistent (“bands”) run along a path through the Brillouin zone, which uses the converged density and computes orbitals at the requested k points only.

Effective mass. Near the conduction-band minimum at Γ the band is fitted with a parabola,

$$E(k) = E_{\text{CBM}} + \frac{\hbar^2 k^2}{2m^*}, \quad \frac{m^*}{m_0} = \frac{\hbar^2/2m_0}{c_2} = \frac{3.80998 \text{ eV } \text{\AA}^2}{c_2}, \quad (12.8)$$

where c_2 is the slope of $E - E_{\text{CBM}}$ against k^2 in a least-squares fit over $|k| \leq 0.05 \text{ \AA}^{-1}$. Over a wider range the band is described by the non-parabolic (Kane-type) dispersion [52]

$$E(1 + \alpha E) = \frac{\hbar^2 k^2}{2m^*}, \quad (12.9)$$

which is linear in the two unknowns $1/m^*$ and α for known E and k and is solved by linear least squares.

Density of states and its projections. `projwfc.x` projects the orbitals on atomic orbitals (Löwdin projections, 53; 14, p. 10) and gives the density of states resolved by atom and orbital, here used for the W 5d share of the states near E_F .

Vacuum level, EA and IP. `pp.x` computes the electrostatic potential felt by an electron (bare ionic plus Hartree; `plot_num = 11`), and `average.x` forms its planar average over each plane parallel to the

film,

$$\bar{V}(z) = \frac{1}{A} \iint_A V(x, y, z) dx dy, \quad (12.10)$$

and the macroscopic average $\bar{\bar{V}}(z)$, a further running average over one structural period L that removes the atomic oscillations [54][14, p. 10]; here $L = a/4$, the distance between cation layers along [001]. In the middle of the vacuum the potential is flat and defines E_{vac} . Then

$$EA = E_{\text{vac}} - E_{\text{CBM}}, \quad IP = E_{\text{vac}} - E_{\text{VBM}}, \quad E_g = IP - EA. \quad (12.11)$$

Because both surfaces of the slabs are equivalent by symmetry, the potential is the same in the vacuum on both sides and no dipole correction [55] is needed. The confinement shift and its partition between two films t_1 and t_2 follow as

$$\begin{aligned} \Delta E_g(t) &= E_g(t) - E_g^{\text{bulk}}, \\ \Delta E_c(t_1) - \Delta E_c(t_2) &= EA(t_2) - EA(t_1), \quad \Delta E_v(t_1) - \Delta E_v(t_2) = IP(t_1) - IP(t_2). \end{aligned} \quad (12.12)$$

The bulk crystal has no vacuum, so its band edges cannot be placed on the vacuum scale directly. The partition relative to bulk requires aligning the bulk band edges, measured from the macroscopic average potential of a bulk calculation, with the macroscopic average inside the slab (the two-step method of 54, 56). This step has not yet been done (Section 12.9).

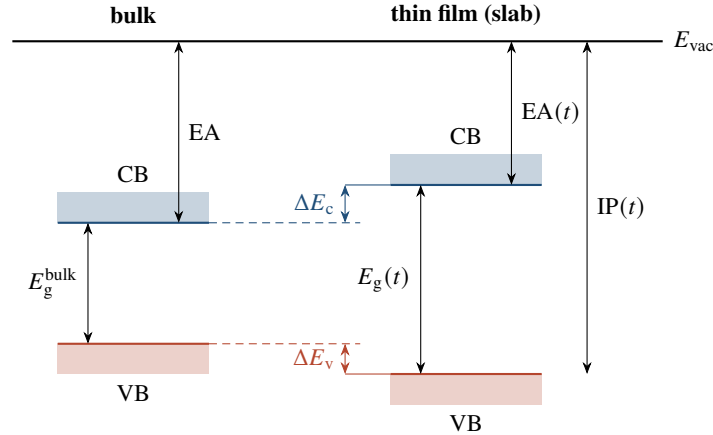


Figure 12.3. Definitions used throughout the chapter (schematic, not to scale). Confinement in a thin film raises the conduction-band minimum by ΔE_c and lowers the valence-band maximum by ΔE_v , so the gap opens by $\Delta E_g = \Delta E_c + \Delta E_v$. Measured from the vacuum level, the electron affinity (EA) and ionization potential (IP) of films of different thickness give the partition, Equation (12.12). The device model so far assumes $\Delta E_g = \Delta E_c$.

12.4 Defining the material

In plain words. Before anything can be computed, the computer needs a precise list of atoms: the box (cell) and the position of every atom in it. For bulk In_2O_3 this comes from the measured crystal structure. A thin film is modelled as a slab: a few layers of the crystal, cut parallel to a crystal plane, with the broken bonds at the surfaces closed by hydrogen, and with empty space above and below. Doping is modelled by replacing one indium atom by tungsten. Every structure is checked

automatically before it is used, because a single wrong coordinate gives meaningless results.

All structures were built with the Atomic Simulation Environment [57] and checked with spglib [58] by three scripts of the package: `build_structure.py` for the bulk crystal, and `build_structures.py` and `build_iwo_slab.py` for the films and the doped cells. Table 12.4 summarizes them.

12.4.1 Bulk In_2O_3 : the bixbyite crystal

Cubic In_2O_3 has the bixbyite structure, space group $\text{Ia}\bar{3}$ (No. 206). The conventional cubic cell, with $a = 10.117 \text{ \AA}$, contains 80 atoms on three Wyckoff sites [59]: In1 on 8b at $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, In2 on 24d at $(u, 0, \frac{1}{4})$ with $u = -0.0336$, and O on 48e at $(0.3905, 0.1529, 0.3832)$. Each In is surrounded by six O. The 8b site has the higher symmetry ($\bar{3}$, six equal In–O bonds); the 24d site has only a twofold axis and three pairs of different bonds. Because the cubic cell is body-centred, the same crystal can be described by a primitive cell of 40 atoms (16 In, 24 O), which halves the cost and is used for all bulk calculations.

The structure was generated from these crystallographic data and verified before use: space group 206 with 48 symmetry operations in the conventional cell and 24 in the primitive cell, 80 and 40 atoms, In–O bond lengths of 2.123, 2.191 and 2.208 $\text{\AA}$. The verification was introduced after a recorded failure: an inherited 80-atom input that had never been checked gave only two symmetry operations, a total force of 2.86 Ry/bohr, a pressure of 765 kbar and a negative “gap”. A second defect followed when spglib’s primitive coordinates were combined with Quantum ESPRESSO’s built-in body-centred lattice (`ibrav = 3`), whose basis vectors differ; since then all inputs give the cell vectors explicitly (`ibrav = 0`), and pw.x reports the expected 24 operations, 18 of them with a fractional translation (PROTOCOL.md, section 7a).

12.4.2 Thin films: the slab model

In plain words. A plane-wave code always works with a repeating box. To describe a film, the box contains the film plus a thick layer of empty space, so that the film does not feel its periodic copies above and below (Figure 12.4). The surface oxygen atoms lose bonds when the film is cut; one hydrogen atom on each of them restores a closed electronic shell, as in a hydroxylated oxide surface.

The slabs follow the recipe of Lin et al. [2, p. 21542], which was chosen so that the results can be compared with theirs directly:

1. take the conventional cell (for the 1 nm film) or a $1 \times 1 \times 2$ supercell stacked along [001] (for the 2 nm film) of the relaxed bulk crystal;
2. remove the In layer at the cell boundary (8 cations; the layer is corrugated by about $\pm 0.35 \text{ \AA}$, so all cations within $0.6 \text{ \AA}$ of the boundary are removed);
3. put one H on every O that lost In neighbours, $0.97 \text{ \AA}$ from the O, along the mean direction of the two removed In–O bonds;
4. add $25 \text{ \AA}$ of vacuum and centre the slab in the cell.

The script checks the result: every one of the 24 surface O atoms has lost exactly two In neighbours; 12 H atoms sit on each surface; the electron count closes, $3n_{\text{In}} + n_{\text{H}} = 2n_{\text{O}}$ (In^{3+} , H^+ , O^{2-}), so the slab should be an insulator; no non-bonded atoms come closer than 2.61 Å; and the space group is Pcca (8 operations), which contains an operation that maps the top surface onto the bottom one. The slab therefore has no dipole. Applied to the PBE-relaxed bulk ($a = 10.306$ Å), the recipe gives $\text{In}_{24}\text{O}_{48}\text{H}_{24}$ (96 atoms, 9.875 Å from the lowest to the highest H, 8.03 Å between the outermost O layers) and $\text{In}_{56}\text{O}_{96}\text{H}_{24}$ (176 atoms, 20.18 Å and 18.34 Å). The films are called “1 nm” and “2 nm”; Lin et al. quote 0.95 and 1.98 nm for their versions [2, p. 21540]. After relaxation the 1 nm slab measures 9.30 Å from H to H and 7.88 Å between the outer O layers (Figure 12.5). For the vacuum test the 1 nm slab was also built with 15 and 35 Å.

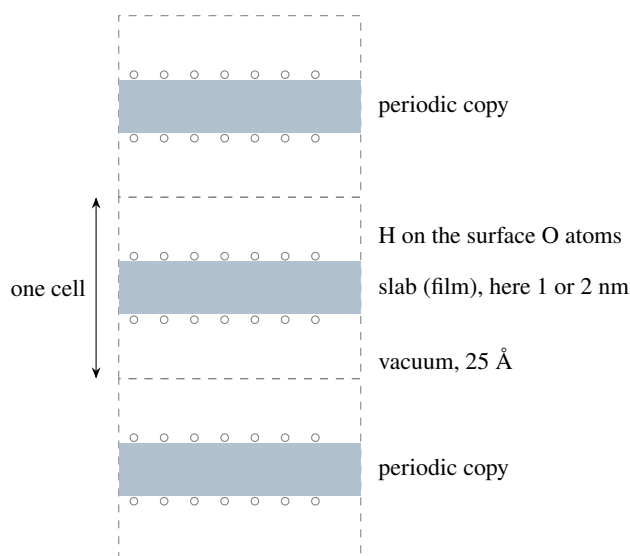


Figure 12.4. The slab model (schematic). The calculation repeats the cell (dashed) in all directions; along z the film alternates with 25 Å of vacuum, which separates the periodic copies. Hydrogen atoms (open circles) terminate the surface oxygen atoms.

12.4.3 Tungsten doping

In plain words. To learn where tungsten goes, one In atom of the 80-atom cell is replaced by W, once on each of the two kinds of In site. Comparing the energies of the two relaxed cells shows which site W prefers. One W in 32 cations is 3.1 %, the same order as the “about 2 % W” of the target films.

In the conventional 80-atom cell, one In was replaced by W on an 8b site and, in a second cell, on a 24d site, giving $\text{In}_{31}\text{WO}_{48}$ with $W/(W+\text{In}) = 1/32 = 3.1$ %. Substitution lowers the symmetry; the number of remaining operations is 48 divided by the multiplicity of the site, which the build script checks: 6 operations (space group $R\bar{3}$) for W on 8b, 2 (P2) for W on 24d. Relative to In^{3+} , W^{6+} releases three electrons; in this cell of 1095 Å³ that corresponds to about $2.8 \times 10^{21} \text{ cm}^{-3}$, far above the carrier density of the transistor channel. The cell therefore describes the fully ionized, degenerate limit, and band-edge quantities are read from the dispersion and the projected states, not from the position of E_F alone.

For the doped film, one In of the central cation layer of the relaxed 1 nm slab was replaced by W (build_iwo_slab.py). In the conventional-cell slabs this layer holds only 24d cations, the site



Atoms are drawn back to front; the arrows give the distance between the outermost H atoms.

preferred in bulk (Section 12.5.9). The result is $\text{In}_{23}\text{WO}_{48}\text{H}_{24}$ ($\text{W}/(\text{W}+\text{In}) = 4.2\%$), with W exactly at the slab centre; the space group is P2, and its twofold operation still maps z to $-z$, so the slab has no dipole. Three electrons in a 1 nm slab of $10.34^2 \text{ \AA}^2$ correspond to about $2.8 \times 10^{14} \text{ cm}^{-2}$.

Table 12.4. Structures of the study. “Electrons” counts valence electrons of the data sets (Table 12.3); ops: symmetry operations found by spglib.

Name	Composition, cell	Atoms	Electrons	Symmetry	Purpose
bulk (primitive)	$\text{In}_{16}\text{O}_{24}$, bcc primitive cell	40	352	$\text{Ia}\bar{3}$, 24 ops	convergence, relaxation, bands, mass
bulk_conv	$\text{In}_{32}\text{O}_{48}$, cubic, $a = 10.306 \text{ \AA}$	80	704	$\text{Ia}\bar{3}$, 48 ops	host of all other structures
iwo_W8b	$\text{In}_{31}\text{WO}_{48}$, W on 8b	80	705	$\text{R}\bar{3}$, 6 ops	W site, valence, states
iwo_W24d	$\text{In}_{31}\text{WO}_{48}$, W on 24d	80	705	P2, 2 ops	W site, valence, states
slab1_v15/v25/v35	$\text{In}_{24}\text{O}_{48}\text{H}_{24}$, 15/25/35 $\text{\AA}$ vacuum	96	624	Pcca, 8 ops	vacuum test
slab1 (relaxed)	$\text{In}_{24}\text{O}_{48}\text{H}_{24}$, $10.336 \times 10.335 \times 34.87 \text{ \AA}$	96	624	Pcca, 8 ops	1 nm confinement
slab2_v25	$\text{In}_{56}\text{O}_{96}\text{H}_{24}$, $c = 45.18 \text{ \AA}$	176	1328	Pcca, 8 ops	2 nm confinement
slab1r_W24d	$\text{In}_{23}\text{WO}_{48}\text{H}_{24}$	96	625	P2, 2 ops	W in a 1 nm film
slab2r_W24d	$\text{In}_{55}\text{WO}_{96}\text{H}_{24}$	176	1329	–	W in a 2 nm film (submitted 4 Oct.)

12.5 The calculations, step by step

In plain words. The work was done in stages, each building on the previous one (Figure 12.6). First the numerical settings were tested on the bulk crystal; then the crystal was relaxed and its bands computed; the relaxed crystal became the starting point for every film and doped cell. Before each stage the criteria for accepting its result were written down, so that a result could not be accepted just because it looked good.

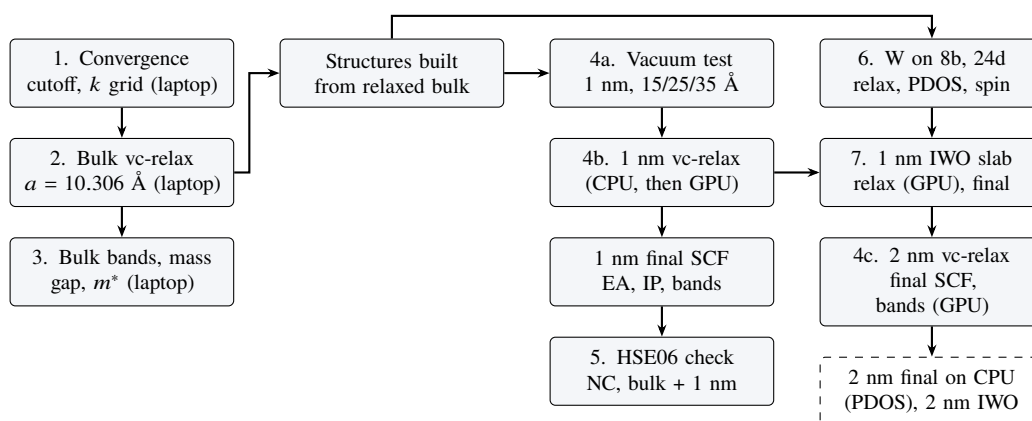


Figure 12.6. Stages of the study and their dependencies. Solid: finished; dashed: running or queued at the time of writing. Stages 1–3 ran on the author’s laptop, all others on the CERN batch system.

12.5.1 Protocol and acceptance criteria

The criteria were fixed in two protocols written before the corresponding results existed (PROTOCOL.md of the bulk stages, 2026-09-26, and PROTOCOL_CERN.md of the CERN stages, 2026-09-27, both in `qe_workflow`); every later deviation is recorded there with its reason. Table 12.5 collects them.

Table 12.5. Acceptance criteria, fixed before the results, and their outcome.

Item	Criterion	Outcome
Cutoff, k grid	a further increase changes the energy by < 1 mRy/atom, the pressure by < 1 kbar and the gap by < 10 meV	met at 71 Ry and $2 \times 2 \times 2$; $3 \times 3 \times 3$ used
Bulk relaxation	forces $< 10^{-4}$ Ry/bohr, pressure < 0.2 kbar; lattice constant within 3 % of experiment	met; +1.87 %
k grid of doped cells	$3 \times 3 \times 3$ vs $4 \times 4 \times 4$: energy < 1 mRy/atom, site energy < 10 meV	met (0.3 meV)
Vacuum	EA and IP change by < 10 meV from 15 to 25 and 25 to 35 Å	met (≤ 0.1 meV)
Reproduction	ΔE_g of the 1 and 2 nm slabs within 0.1 eV of Lin et al.	met (0.04 and 0.005 eV)
Functional	$ \Delta E_g^{\text{HSE}} / \Delta E_g^{\text{PBE}} - 1 < 10\%$	met (6 %; 11 % worst case)
Magnetism	$ M < 0.05 \mu_B$ and spin energy gain < 1 mRy	not met; preliminary
Status	proxies replaced only after the k and vacuum tests; “computed”, not “validated”	applied

12.5.2 Anatomy of an input file

In plain words. A calculation is described by a short text file. It says what to compute (an SCF, a relaxation, bands), how accurately (cutoffs, thresholds), with which atoms (pseudopotentials, cell, positions) and on which k points. All input files of this study are written by scripts, never by hand, so that every job uses exactly the intended settings.

Listing 12.1 shows the input of the final SCF of the relaxed 1 nm slab, as written by the job generator `make_jobs.py`; Table 12.6 lists the settings of every type of calculation.

Listing 12.1. Input of the final SCF of the relaxed 1 nm slab (`slab1r_final_cpu/scf.in`); the 96 atomic positions are shortened to three.

```

&CONTROL
  calculation = 'scf', prefix = 'slab', outdir = './tmp', pseudo_dir = './'
  verbosity = 'high', tprnfor = .true., tstress = .true., max_seconds = 1.0d8
/
&SYSTEM
  ibrav = 0, nat = 96, ntyp = 3, ecutwfc = 71, ecutrho = 568
  occupations = 'fixed'
  nbnd = 328
/
&ELECTRONS
  conv_thr = 1.0d-9, mixing_beta = 0.2, electron_maxstep = 150, mixing_mode = 'local-TF'
/
ATOMIC_SPECIES
  In 114.818 In.pbe-dn-kjpaw_psl.1.0.0.UPF
  O 15.999 O.pbe-n-kjpaw_psl.1.0.0.UPF
  H 1.008 H.pbe-kjpaw_psl.1.0.0.UPF
CELL_PARAMETERS angstrom
  10.3364310600 0.0000000000 0.0000000000
  0.0000000000 10.3345042100 0.0000000000
  0.0000000000 0.0000000000 34.8745389300
ATOMIC_POSITIONS crystal
  In 0.2477884238 0.2524784941 0.4261425288
  ...
  H 0.6367869492 0.3539040563 0.3667720191
K_POINTS automatic
  3 3 1 0 0 0

```

Line by line: `calculation` selects the task; `prefix` and `outdir` name the directory where the density and orbitals are saved (later read by `pp.x` and by the bands run); `tprnfor` and `tstress`

request forces and stress; `max_seconds` is overwritten by the job driver so that `pw.x` stops cleanly before the batch time limit (Section 12.6.2). `ibrav = 0` means that the cell is given explicitly in `CELL_PARAMETERS`; `nat` and `ntyp` count atoms and species; `ecutwfc` and `ecutrho` are the cutoffs of Section 12.3.5; `nbnd` is the number of bands computed, here the 312 occupied plus 16 empty ones (the empty ones give the conduction-band minimum). The `&ELECTRONS` block sets the SCF threshold, the mixing and the maximum number of SCF iterations. `ATOMIC_SPECIES` gives each element its mass (IUPAC standard atomic weight) and data set; `ATOMIC_POSITIONS crystal` gives fractional coordinates; `K_POINTS automatic 3 3 1 0 0 0` requests an unshifted $3 \times 3 \times 1$ Monkhorst–Pack grid.

Table 12.6. Settings per type of calculation (`make_jobs.py`; bulk stages 1–3: `make_inputs.py`, `make_stage3.py`). All use PBE, PAW, 71/568 Ry and `conv_thr = 10-9` Ry, except the hybrid check.

Type	Occupations, mixing	k points	Other settings
Bulk SCF / vc-relax	fixed; $\beta = 0.4$	$2^3, 3^3, 4^3$	<code>nbnd</code> 190; relax: forces 10^{-4} Ry/bohr, energy 10^{-5} Ry, pressure 0.2 kbar
Bulk bands	fixed; CG diagonalization	path (80), Γ set (31)	non-SCF on the vc-relax density
Slab SCF	fixed; $\beta = 0.2$, local-TF; max. 150 iterations	$3 \times 3 \times 1$	<code>nbnd</code> = occupied + 16
Slab vc-relax	as slab SCF	$3 \times 3 \times 1$	BFGS; forces 10^{-3} Ry/bohr, energy 10^{-4} Ry; <code>cell_dofree</code> = '2Dxy', pressure 0.5 kbar; <code>nbnd</code> = occupied + 8
Slab bands	Davidson	Γ + 10 along [100], [110]	$ k \leq 0.15 \text{ \AA}^{-1}$
Doped bulk	MV smearing 0.01 Ry; $\beta = 0.3$; max. 100 iterations	relax 3^3 , SCF 4^3	PDOS; bands near Γ ; spin-polarized SCF 3^3 ($\beta = 0.2$, starting moment 0.5 on W)
Doped slab	MV smearing 0.01 Ry; $\beta = 0.2$, local-TF	$3 \times 3 \times 1$	relax at fixed cell; <code>nbnd</code> = occupied + 24; PDOS
Hybrid check	fixed; $\beta = 0.3$ (local-TF for the slab)	$3^3 / 3 \times 3 \times 1$	SG15 NC, 80/320 Ry; <code>input_dft</code> = 'hse', EXX q grid 1 (bulk also 3), <code>ecutfock</code> 160 Ry

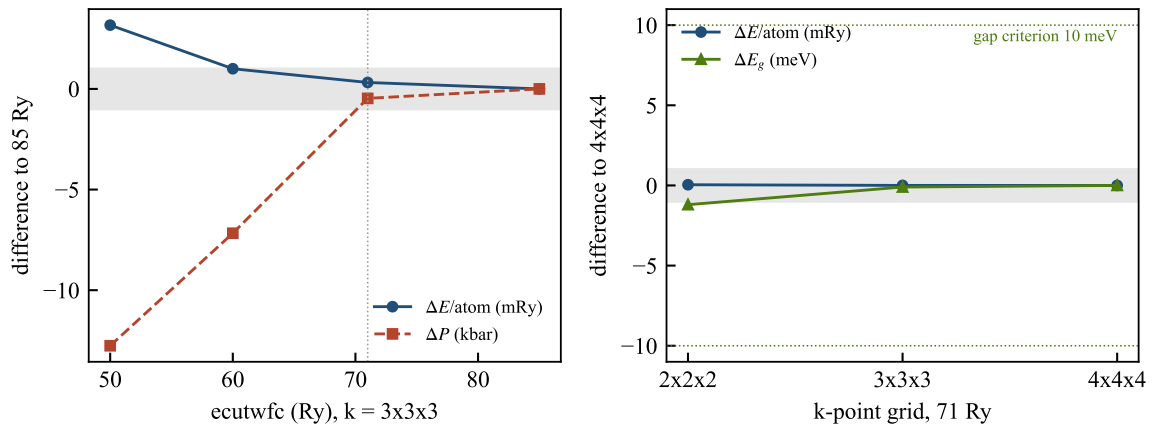
12.5.3 Stage 1: convergence of the numerical settings

In plain words. A calculation is only meaningful if its answer does not depend on the numerical settings. The cutoff and the k grid were therefore raised step by step until energy, pressure and gap no longer changed by more than the criteria.

Table 12.7 and Figure 12.7 summarize the tests on the 40-atom primitive cell at the experimental lattice constant (laptop, 10 MPI processes; one SCF at 71 Ry and $3 \times 3 \times 3$ took 12 min). A cutoff of 71 Ry is the lowest that meets all three criteria against 85 Ry (0.32 mRy/atom, 0.47 kbar, 0.5 meV); 60 Ry fails the energy and pressure criteria. The $2 \times 2 \times 2$ grid already meets the criteria against $4 \times 4 \times 4$; the $3 \times 3 \times 3$ grid was nevertheless used for the relaxation (a recorded deviation in favour of accuracy, at almost no cost).

Table 12.7. Convergence of bulk In_2O_3 at $a = 10.117 \text{ \AA}$ (40-atom primitive cell, PAW, PBE). Energies relative to the best-converged run of each series.

Run	E (Ry)	$\Delta E/\text{atom}$ (mRy)	P (kbar)	KS gap (eV)
50 Ry, $3\times 3\times 3$	-7573.24296524	3.17	84.28	1.1897
60 Ry, $3\times 3\times 3$	-7573.32959316	1.00	89.87	1.1894
71 Ry, $3\times 3\times 3$	-7573.35691861	0.32	96.57	1.1897
85 Ry, $3\times 3\times 3$	-7573.36970781	0	97.04	1.1902
71 Ry, $2\times 2\times 2$	-7573.35524852	+0.044 vs $4\times 4\times 4$	96.27	1.1886
71 Ry, $4\times 4\times 4$	-7573.35701116	reference	96.61	1.1898

**Figure 12.7.** Convergence of total energy, pressure and Kohn–Sham gap of bulk In_2O_3 with the plane-wave cutoff (left, $3\times 3\times 3$) and the k -point grid (right, 71 Ry). The grey band marks the energy criterion of 1 mRy per atom; the dotted lines on the right mark the gap criterion of 10 meV.

12.5.4 Stage 2: the relaxed bulk crystal

At the experimental lattice constant PBE gives a pressure of +97 kbar: the crystal “wants” to expand, the expected behaviour of this functional. The variable-cell relaxation (71 Ry, $3\times 3\times 3$, BFGS, 1 h 47 min on the laptop) converged in 11 BFGS steps to a primitive volume of $547.333 \text{ \AA}^3$, i.e. $a = 10.306 \text{ \AA}$, 1.87 % above the experimental $10.117 \text{ \AA}$ [59] and within 0.06 % of the PBE value of $10.30 \text{ \AA}$ reported by Lin et al. [2, p. 21542]. The final SCF with the plane-wave basis of the new cell gives $P = 0.41 \text{ kbar}$, inside the 1 kbar criterion. Space group $\text{Ia}\bar{3}$ is preserved; the In–O bonds become $2.215 \text{ \AA}$ (8b) and $2.167/2.240/2.256 \text{ \AA}$ (24d). All later structures are built from this relaxed cell.

12.5.5 Stage 3: bulk band structure and effective mass

Two non-SCF runs use the density of the relaxed crystal: one along the high-symmetry path $\Gamma\text{--H--N--}\Gamma\text{--P--H|P--N}$ of the body-centred cubic Brillouin zone [60] (80 k points, about 15 per $2\pi/a$; 11 h 16 min), and one on Γ plus 10 points along each of $[100]$, $[110]$ and $[111]$ up to $|k| = 0.15 \text{ \AA}^{-1}$ (31 points; 4 h 21 min). The first attempt with Davidson diagonalization aborted after 9 of 16 k points with unconverged eigenvalues; both runs therefore use conjugate-gradient diagonalization.

Figure 12.8 shows the bands. The conduction-band minimum lies at Γ ; the valence band is very flat, with its maximum on $\Gamma\text{--H}$ at $k = (0, 0, 0.196) 2\pi/a$, only 4.4 meV above the highest valence state at Γ . The fundamental gap is 0.889 eV and the direct gap at Γ 0.893 eV, against 0.94 eV in Lin et al. [2, p. 21542]; the bulk gap of 3.05 eV adopted in the device model is, as expected, strongly underestimated by PBE (Section 12.3.3). The bulk reference for all confinement shifts is the fundamental gap, 0.8887 eV.

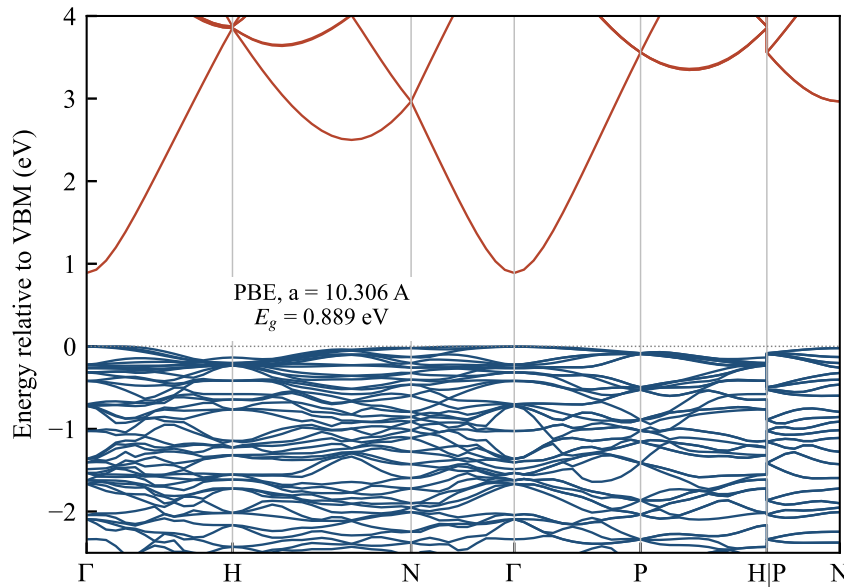


Figure 12.8. PBE band structure of bulk In_2O_3 at the relaxed lattice constant, energies relative to the valence-band maximum. Occupied bands blue, empty bands red. Path after Setyawan and Curtarolo [60].

Figure 12.9(a) shows how the mass is obtained. Within $|k| \leq 0.05 \text{ \AA}^{-1}$ the parabolic fit (Equation (12.8)) gives $m^* = 0.1588$, 0.1587 and $0.1587 m_0$ along $[100]$, $[110]$ and $[111]$: an isotropic $m^* = 0.159 m_0$, as expected for the s-like conduction-band minimum of a cubic crystal. Further

from Γ the points fall below the parabola: the band is non-parabolic. The Kane fit (Equation (12.9)) over $|k| \leq 0.153 \text{ \AA}^{-1}$ gives $\alpha = 0.63, 0.57$ and 0.55 eV^{-1} . For comparison, Lin et al. [2, p. 21540] report $0.17 m_0$ with PBE, and the device model uses the measured $0.208 m_0$ [7, p. 225102-12]. PBE underestimates the mass because the gap is too small (the conduction-band curvature is coupled to the gap); as for the gap, only differences between films and bulk computed in the same setup are transferred to the device model.

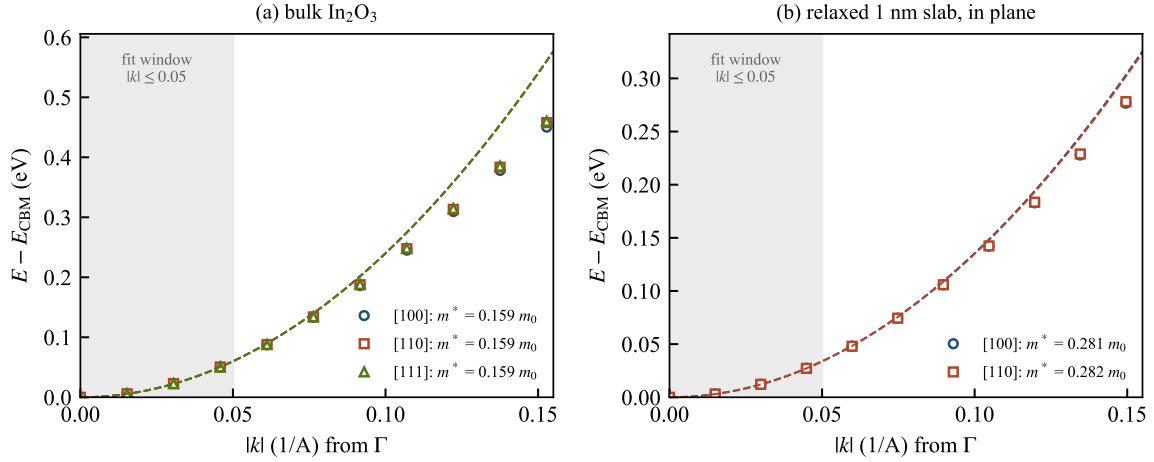


Figure 12.9. How the effective mass is obtained. Symbols: conduction-band energies from the non-SCF runs, relative to the minimum at Γ ; dashed: parabola from the fit over $|k| \leq 0.05 \text{ \AA}^{-1}$ (shaded). (a) Bulk, three directions; (b) relaxed 1 nm slab, two in-plane directions. The departure from the parabola at larger k is the non-parabolicity.

12.5.6 Stage 4a: vacuum thickness

In plain words. The vacuum layer must be thick enough that the film does not “see” its periodic copies. This was tested by computing the same film with 15, 25 and $35 \text{ \AA}$ of vacuum: if EA and IP do not change, the vacuum is sufficient.

The unrelaxed 1 nm slab was computed with the three vacuum thicknesses (16 CPUs each; the $35 \text{ \AA}$ SCF took 1 h 48 min and 77 iterations). EA and IP change by at most 0.1 meV and the vacuum plateau of the potential is flat to 0.01 meV (Table 12.8), far inside the 10 meV criterion; the symmetric, dipole-free slab converges fast. The $25 \text{ \AA}$ of Lin et al. [2, p. 21542] is therefore sufficient and is used for all slabs.

Table 12.8. Vacuum convergence of the unrelaxed 1 nm slab ($\text{In}_{24}\text{O}_{48}\text{H}_{24}$).

Vacuum	Gap (eV)	EA (eV)	IP (eV)	SCF iterations
15 $\text{\AA}$	0.9586	3.4331	4.3917	46
25 $\text{\AA}$	0.9586	3.4332	4.3918	64
35 $\text{\AA}$	0.9586	3.4332	4.3918	77

12.5.7 Stage 4b: relaxation and confinement of the 1 nm film

In plain words. A freshly cut film is not in equilibrium: the surface atoms and the added hydrogens are pulled into new positions. The relaxation lets them move. For the 1 nm film this changed the result completely: the gap of the unrelaxed film was only 0.07 eV above the bulk gap, the relaxed one 0.90 eV.

The relaxation (all atoms and the in-plane cell) started on 16 CPU cores; after 13 BFGS steps and the successful GPU benchmark (Section 12.6.3) it was continued on one A100 GPU from the latest geometry, where it converged after 54 further steps (15 h 31 min). Figure 12.10 shows the 72 SCF cycles: the total energy fell by 7.05 eV (73 meV per atom), the residual force by a factor of 40, and the gap rose from 0.959 to 1.789 eV, most of it in the first ten steps. The final in-plane cell is 10.336×10.335 Å (bulk 10.306; Lin et al. 10.26 and 10.36 Å, 2, p. 21542). The total residual force of 0.0061 Ry/bohr is the norm over all 96 atoms; BFGS ended with the criteria as Quantum ESPRESSO applies them.

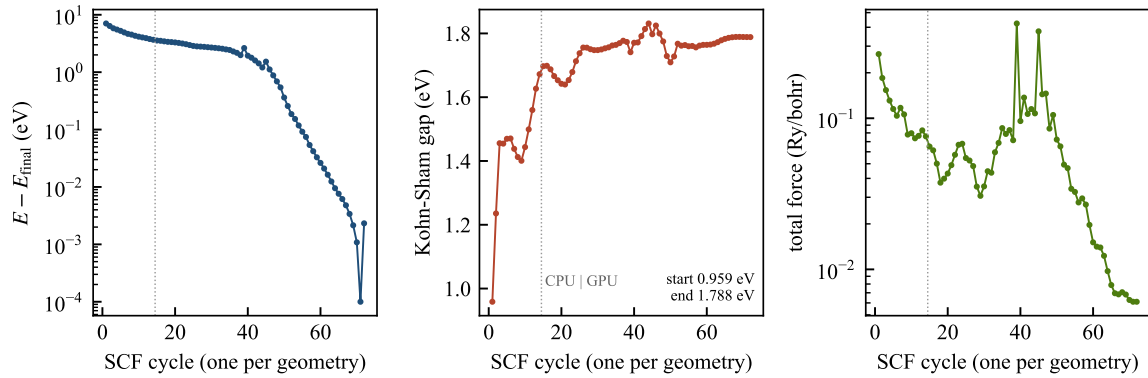


Figure 12.10. Relaxation of the 1 nm slab: total energy relative to the final value, Kohn–Sham gap and total force at every SCF cycle (one per geometry). Dotted: continuation of the CPU run on a GPU from its latest geometry. The gap of the relaxed film is the one used.

The confinement shift is

$$\Delta E_g(0.99 \text{ nm}) = 1.7885 - 0.8887 = 0.900 \text{ eV}, \quad (12.13)$$

against +0.94 eV for the 0.95 nm slab of Lin et al. [2, p. 21542]. The in-plane mass is $0.281 m_0$ along [100] and $0.282 m_0$ along [110] (Figure 12.9b), an increment of $+0.122 m_0$ over the bulk value, against $+0.13 m_0$ in Lin et al. [2, p. 21540]. Both agree within the pre-registered tolerance: the published confinement of a 1 nm film is reproduced independently, with a different pseudopotential type and cutoff (PAW at 71 Ry versus norm-conserving at 140 Ry, 2, p. 21542).

In plain words. Figure 12.11 puts every quantity that both studies computed side by side. A point on the vertical line would mean perfect agreement; the grey band is $\pm 5\%$. All our energies and masses lie a few per cent below theirs, and our lattice constants agree to 0.2 %.

The lattice constants agree within 0.2 %. Gaps and masses are uniformly 4–7 % lower than those of Lin et al. [2, pp. 21540, 21542]: bulk gap -5.5% , 1 nm gap -4.9% , bulk mass -6.6% , 1 nm mass -6.3% . The quantities transferred to the device model, the differences between film and bulk, show

the same small offset (ΔE_g -4.3% , Δm^* -5.8%), well inside the pre-registered tolerance of 0.1 eV for the shift. An offset of the same sign and size for every quantity points to a systematic difference between the two numerical setups (pseudopotential type and cutoff) rather than to a difference between the films.

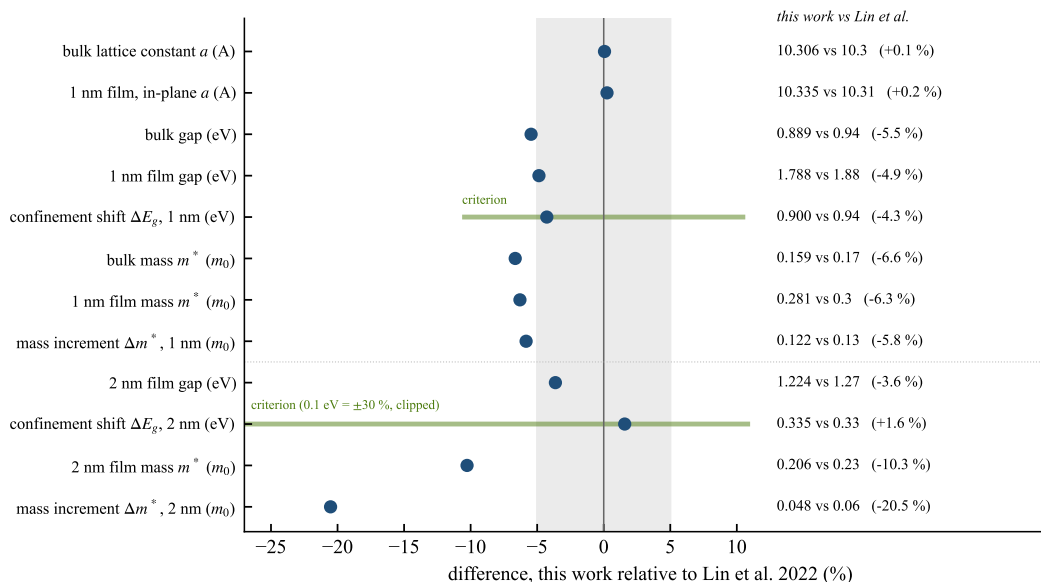


Figure 12.11. This work against the PBE calculations of Lin et al. [2, pp. 21540, 21542], which used the same slab recipe: relative difference of each quantity (dots) with both values written out. Grey: $\pm 5\%$; green: the pre-registered tolerance of the confinement shift (± 0.1 eV; at 2 nm this is $\pm 30\%$ and the bar is clipped). For the in-plane lattice constant of the 1 nm film, Lin et al. give 10.26 and 10.36 Å; their mean is compared with ours. Below the dotted line: the 2 nm film (Section 12.5.11); no criterion was set for the masses. The increments of Lin et al. are differences of their rounded values ($1.27 - 0.94$ eV and $0.23 - 0.17 m_0$).

In plain words. Figure 12.12 shows the allowed electron energies at the centre of the Brillouin zone (Γ) as horizontal lines, for the bulk crystal and for the 1 nm film before and after relaxation. Below zero are the filled (valence) levels, above the gap the empty (conduction) ones. Two things change in the film: the gap opens, and new, separate conduction levels appear above the lowest one. They are the signature of confinement: an electron that can move freely through the crystal can only take certain energies when it is trapped between two surfaces, like a wave on a string of fixed length.

At Γ the gap is 0.893 eV in bulk, 1.015 eV in the film as built (whose valence-band maximum lies away from Γ , so its fundamental gap is 0.959 eV) and 1.788 eV after relaxation. In the bulk crystal the next conduction level at Γ lies 4.28 eV above the minimum; in the film there are further levels 1.57 and 2.43 eV (as built) and 1.50 and 2.45 eV (relaxed) above the lowest one. This is consistent with the conduction band splitting into separate sub-bands when the motion across the film is confined. The valence levels are denser in the film because its cell contains more atoms.

The pp.x step of the GPU job failed to read the density written by the GPU build (Section 12.6.3), so the final geometry was recomputed on 16 CPU cores (SCF 2 h 19 min, 47 iterations; bands 4 h 38 min), which reproduced the gap and mass to the last printed digit and gave the potential of Figure 12.13: EA = 3.93 eV and IP = 5.72 eV (PBE). Figure 12.15 places the result next to the literature points and the thickness laws of the device model.

In plain words. Figure 12.14 draws the band edges of three 1 nm films on one common scale: the

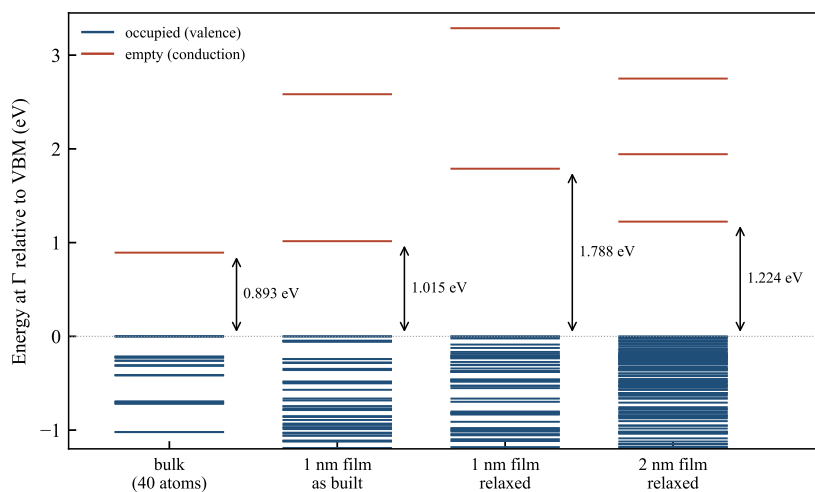


Figure 12.12. Kohn–Sham levels at Γ near the gap, relative to the highest occupied level at Γ : bulk In_2O_3 (40-atom cell), the 1 nm film as built, and the relaxed 1 nm and 2 nm films (PBE). Blue: occupied; red: empty. Arrows: gap at Γ .

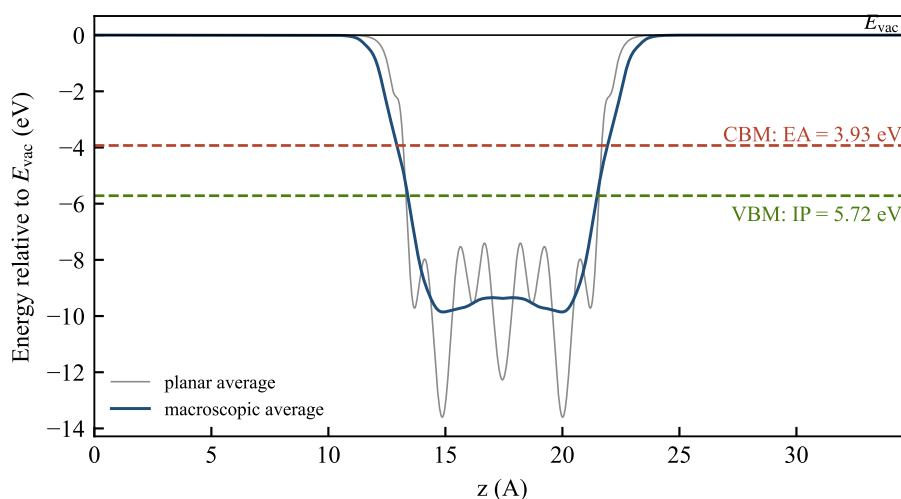


Figure 12.13. Planar (grey) and macroscopic (blue) averages of the electrostatic potential of the relaxed 1 nm slab along the film normal, relative to the vacuum level, with the conduction- and valence-band edges (PBE). The flat regions near $z = 0$ and $z = c$ are the vacuum.

energy of an electron at rest outside the film (the vacuum level) is zero, and every edge is placed below it. This is the computed version of the schematic Figure 12.3.

On the vacuum scale, relaxation lowers the conduction-band minimum of the 1 nm film from -3.43 to -3.93 eV and the valence-band maximum from -4.39 to -5.72 eV. Both edges move down, by different amounts. Besides the film interior, relaxation changes the surface O–H bonds, and any change of their electric dipole shifts the potential step between film and vacuum; absolute EA and IP of a single film therefore contain a contribution of the surface termination, which this calculation does not separate. Only differences between films with the same termination (here the relaxed 1 and 2 nm films) isolate the confinement shift of each band edge (Equation (12.12)). The third column is discussed in Section 12.5.10.

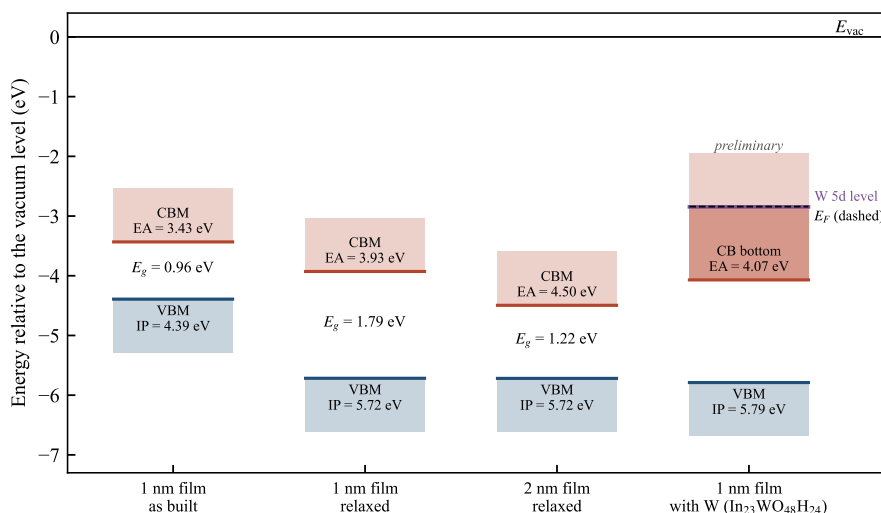


Figure 12.14. Band edges of the films on the vacuum scale, computed from the SCF eigenvalues and the planar-averaged potential (PBE): 1 nm film as built and relaxed, relaxed 2 nm film, and 1 nm film with one W on the central 24d site. For the pure films, the band edges are taken over the whole k grid, as in Table 12.8. For the W-doped film, the levels at Γ are shown: the top valence level, the bottom of the host conduction band (red; dark: filled up to E_F), the W-5d-derived level (purple) and E_F (dashed). That column is preliminary.

12.5.8 Stage 5: is PBE adequate for the shift? The hybrid-functional check

In plain words. PBE puts the gap far too low. If it also got the *change* of the gap wrong, the confinement values would be useless. The check repeats the bulk and the 1 nm film with the much more accurate HSE06 functional and compares the two shifts. They differ by only 6 %.

The check uses HSE06 single points at the PBE geometries. It was first run with PAW; after more than 4.5 h on an H100 GPU at 0 % utilization (and more than 5.5 h on 16 CPU cores before) the first exact-exchange step had not finished, the PAW exact exchange of the GPU build evidently running on the host. Following a recorded deviation the check was moved to SG15 norm-conserving pseudopotentials [42, 43] (In, O, H at 80 Ry, a cutoff verified against 100 Ry: gap change 0.1 meV), with PBE and HSE06 computed in the same setup and the ratio of the shifts transferred. The exact-exchange cutoff `ecut_fock` was set to twice the wave-function cutoff after the default ran the GPU out of memory for the slab (the output reports 40.6 GB for the exchange arrays at this setting); for the bulk this changes the HSE06 gap by less than 0.1 meV.

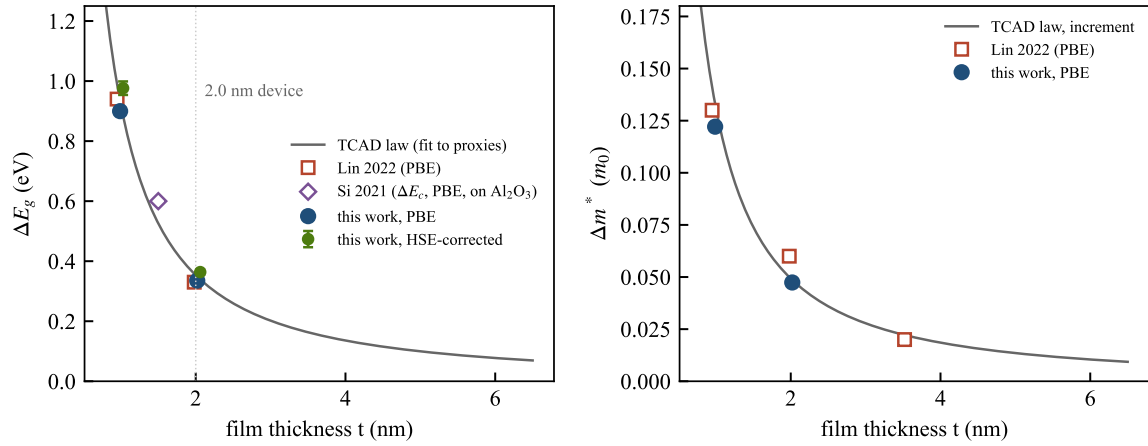


Figure 12.15. Confinement shift (left) and mass increment (right) versus film thickness: thickness laws of the device model, PBE slab values of Lin et al. [2, pp. 21540, 21542], the conduction-band shift of Si et al. [4, p. 504] (corundum In_2O_3 on Al_2O_3 , PBE), and this work (PBE, and the HSE06-corrected range of Section 12.5.8, drawn slightly offset in thickness for visibility). Thickness of this work: H–H distance of the slab as built (0.99 and 2.02 nm). The HSE06 correction measured at 1 nm is applied unchanged at 2 nm (an assumption, not a calculation).

A hybrid calculation is a loop within a loop: pw.x first converges a PBE SCF (34 iterations for the slab), then repeatedly fixes the exchange operator, converges the density in it (23, 14, 9, 5, 3, 3, 1 iterations), and updates the operator, until the estimated exchange error falls from 0.013 Ry to 1.4×10^{-9} Ry. The slab took 4.0 h and the bulk 12 min on one H100 NVL GPU.

Table 12.9. PBE versus HSE06 (norm-conserving, same geometries): Kohn–Sham gaps at Γ and the confinement shift.

	Bulk gap (eV)	1 nm slab gap (eV)	ΔE_g (eV)
PBE	0.9204	1.8068	0.8864
HSE06 ($q = 1$)	2.1080	3.0475	0.9395
Ratio HSE06/PBE			1.060

HSE06 raises both gaps by about 1.2 eV but changes the shift by only 6 % (Table 12.9 and Figure 12.16): the ratio 1.060 is inside the pre-registered 10 % criterion, so the PBE confinement is accepted. The reduced exact-exchange q grid is the dominant numerical uncertainty: with $q = 3 \times 3 \times 3$ the bulk HSE06 gap is 42 meV lower (2.0662 eV); if the slab gap did not shift by the same amount, the ratio would be 1.107, so it lies between 1.06 and 1.11. The norm-conserving PBE shift (0.886 eV) agrees with the PAW value (0.900 eV) within 1.6 %, which justifies applying the ratio to the PAW result: the HSE-corrected 1 nm shift is 0.95–1.00 eV. The semi-local functional, inadequate for the absolute gap, is adequate for the confinement *shift* to within about 10 %, which supports the slab-minus-bulk construction of the device model (Section 4.6) with evidence rather than assumption.

12.5.9 Stage 6: tungsten in bulk In_2O_3

In plain words. Three questions were asked about W: which of the two In sites it takes, which charge state (valence) it has, and whether its extra electrons become free carriers. The site follows from

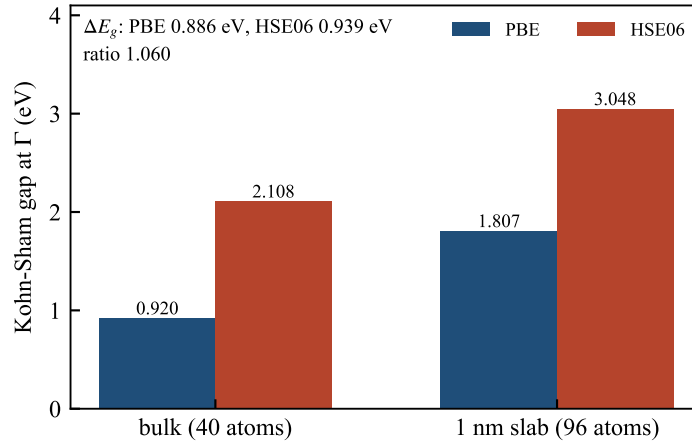


Figure 12.16. Kohn–Sham gaps at Γ of bulk In_2O_3 and of the relaxed 1 nm slab with PBE and HSE06 (norm-conserving, same geometries) and the resulting confinement shifts.

comparing energies, the valence from the length of the W–O bonds, and the fate of the electrons from where their states sit in energy and on which atoms.

Each doped cell was relaxed at the PBE lattice constant ($3\times 3\times 3$, 32 CPU cores; 10 and 24 h), followed by an SCF on $4\times 4\times 4$ (30 irreducible points for W on 24d), a PDOS, bands near Γ and a spin-polarized SCF (Table 12.6). Table 12.10 gives the results.

Table 12.10. W in the 80-atom In_2O_3 cell ($\text{In}_{31}\text{WO}_{48}$, 3.1 % W).

Quantity	W on 8b	W on 24d
E , relaxed, $3\times 3\times 3$ (Ry)	−15496.3476466	−15496.3666037
E , SCF $4\times 4\times 4$ (Ry)	−15496.34766084	−15496.36660308
W–O bonds (Å)	6×1.972	1.892, 1.902, 2.152 (each $\times 2$; mean 1.982)
E_F (eV), $4\times 4\times 4$	10.758	10.794
Spin-polarized: moment (μ_B /cell);	0.74; −1.67 mRy	0.62; −1.43 mRy
$E_{\text{spin}} - E$		
W 5d weight of the three bands at and above E_F (first k point)	0.50, 0.50, 0.58	0.55, 0.48, 0.30

Site. The energy difference

$$E(8b) - E(24d) = +0.258 \text{ eV } (3 \times 3 \times 3), \quad +0.2577 \text{ eV } (4 \times 4 \times 4), \quad (12.14)$$

is converged to 0.3 meV against the 10 meV criterion: W prefers the 24d site. With the multiplicity ratio of three, the equilibrium occupation ratio is $N_{24d}/N_{8b} = 3 \exp(\Delta E/k_B T)$, about 430:1 at 600 K.

Valence. The W–O bonds are about 0.2 Å shorter than In–O. Bond lengths expected from the sums of the Shannon radii [61] (O^{2-} 1.38 Å, six-coordinated cations) are 1.98 Å for W^{6+} , 2.00 for W^{5+} , 2.04 for W^{4+} and 2.18 for In^{3+} ; the computed means of 1.97–1.98 Å identify W^{6+} . Relative to In^{3+} , each W can therefore release three electrons.

The electrons of W (preliminary). The projected densities of states (Figure 12.17) show that the states at and just above E_F carry about half of their weight on W 5d orbitals (Table 12.10), while the valence-band top carries 1.5 %. A spin-polarized calculation finds weak moments of 0.6–0.7 μ_B per cell, 1.4–1.7 mRy below the non-magnetic state; this does not meet the pre-registered non-magnetic criterion and is treated as preliminary, because coarse sampling of a degenerate electron gas can produce spurious moments. At the first k point the host conduction band (band 352, 67 % In 5s) lies 1.76 eV below E_F , and E_F lies in the W-5d-derived band (band 353, 54 % W 5d). Within PBE and for one W per cell (3.1 % of the cations, 5.5×10^{21} donated electrons per cm^3), the donated electrons thus fill the host conduction band up to a W-5d-derived level that is resonant in the conduction band, about 1.8 eV above its bottom, and partly occupy that level. How the three electrons divide between the two is not determined from the available data.

12.5.10 Stage 7: tungsten in the 1 nm film

The W-doped 1 nm slab (Section 12.4.3) was relaxed at the cell of the pure relaxed slab on one H100 NVL GPU (27 BFGS steps, 4 h 29 min; $E = -12238.28910896$ Ry); the final SCF, potential and PDOS ran on 16 CPU cores (2 h 55 min), and its bands run aborted, so no mass is available. The W–O bonds are 1.880, 1.907 and 2.212 Å (each twice).

Which band is which. The doped slab has one occupied band *fewer* below the gap than the pure slab: the In data set treats the filled 4d shell as valence (five bands per In), the W data set the 5s and 5p shells (four bands), so replacing one In by W removes one band. The projections at Γ confirm this: band 311 is the top of the O 2p valence band (–2.366 eV), band 312 (–0.647 eV) is the bottom of the host conduction band (In 5s 50 %, O 2s 21 %), band 313 (0.576 eV) carries 56 % W 5d weight and band 314 (0.766 eV) is the next host conduction level. A first version of this analysis took band 312 as the valence-band top, as in the pure slab; it therefore reported a W level 0.19 eV *below* the conduction band and an unexplained upward shift of 1.1–1.65 eV of all levels on the vacuum scale. Both were artefacts of the band count and were corrected on 5 October 2026.

Result (PBE, one W position, preliminary). E_F lies 1.23 eV above the bottom of the host conduction band and 5 meV above band 313. The host band is filled: band 312 lies below E_F at every k point and holds two of the three donated electrons; about one electron occupies the W-5d-derived level, which is resonant in the conduction band, 1.22 eV above its bottom. The gap between the valence-band top and the host conduction-band bottom is 1.719 eV, 0.069 eV less than in the pure film. On the vacuum scale (Figure 12.14, fourth column) the doped film has EA = 4.07 eV and IP = 5.79 eV, 0.14 and 0.07 eV larger than the pure film: the levels of the two films agree within 0.14 eV.

Consequence for the device model (inference). Full activation of about 2 % W would give of the order of 10^{21} cm^{-3} electrons, orders of magnitude more than the channels of transistors that switch off can carry. In both cells the W-5d-derived level is resonant in the conduction band, 1.2 eV (1 nm film) to 1.8 eV (bulk) above its bottom, and takes up electrons only because these cells fill the band up to it (about 3×10^{21} and 5.5×10^{21} donated electrons per cm^3 ; in the 1 nm film, about one of the three electrons). The first result for the 2 nm film with 1.8 % W (last SCF of its relaxation; projections

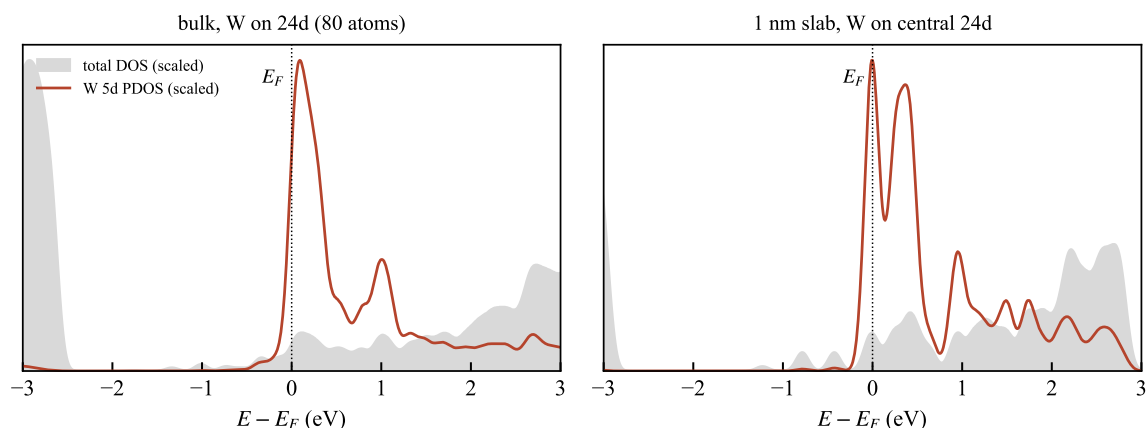


Figure 12.17. Total density of states (grey, scaled) and W 5d projected density of states (red, scaled) relative to E_F : bulk 80-atom cell with W on 24d (left) and 1 nm slab with W on the central 24d site (right). PBE, preliminary.

pending) places E_F 1.24 eV above the conduction-band bottom and the flat bands that are presumably W-derived 0.07 eV above E_F , i.e. just empty. At the lower electron densities of working channels, PBE thus predicts that W releases its electrons into the host band; trapping in W 5d states cannot by itself explain why the channels switch off, and compensation by other defects or charge at the interfaces must account for the difference. A hybrid functional can lower localized d levels, so this conclusion is preliminary. The calculations do not determine the effective donor density and support the treatment of the channel donor density as a fitted, film-specific parameter. The W content of the target films is known only as “about 2 % W” without a stated basis (Chapter 3); the W fractions of the cells (3.1 and 4.2 % of the cations) are of the same order.

12.5.11 Stage 4c: the 2 nm film

In plain words. The 2 nm film is the thickness of the thinnest measured device, so it is the most important single calculation for the device model. It answers three questions: how much does the gap open at 2 nm, how much heavier do the electrons become, and is the change in the conduction band or in the valence band? The answers: the gap opens by 0.34 eV (Lin et al. found 0.33 eV), the mass rises by about 30 %, and, going from 2 to 1 nm, the whole additional opening is in the conduction band.

The 176-atom slab is the largest calculation of the study: 1328 electrons, a 45.18 Å cell, and about 24 GB of memory per k -point pool on CPUs (it was held at a 48 GB limit with four pools). Its variable-cell relaxation ran on one H100 NVL GPU with the settings of the 1 nm slab. A first run converged after 68 BFGS steps (2 October 2026) but was lost: its final SCF, with more bands than the relaxation, needed 66 GB of memory against the 32 GB requested, the job was held by the batch system, and no checkpoint of it had reached EOS (Table 12.12). The relaxation was resubmitted from the same starting geometry with 100 GB and performed the final SCF, the planar average (with the density converted for pp.x, Section 12.6.3) and the bands on the GPU itself.

Relaxation. The second run converged after 70 BFGS steps (72 SCF cycles, 41.5 h; final enthalpy -27035.66548 Ry, total residual force 0.0072 Ry/bohr). The relaxed in-plane cell is 10.318×10.309 Å

(1 nm film: 10.336×10.335 ; bulk $10.306 \text{ \AA}$). The film contracts in thickness: the H–H distance falls from 20.18 to $19.62 \text{ \AA}$ and the O–O distance from 18.34 to $18.21 \text{ \AA}$ (1 nm film: H–H 9.87 to $9.30 \text{ \AA}$). The final SCF took 56 min and the bands run 8 h 22 min on the GPU; the whole job ran for 50.8 h.

Gap and mass. The gap of the relaxed film is 1.2239 eV , at Γ like that of the 1 nm film, so

$$\Delta E_g(2.02 \text{ nm}) = 1.2239 - 0.8887 = 0.335 \text{ eV}, \quad (12.15)$$

against $+0.33 \text{ eV}$ for the 1.98 nm slab of Lin et al. [2, p. 21542] (difference 0.005 eV , inside the pre-registered 0.1 eV). The in-plane mass is $0.206 m_0$ along both $[100]$ and $[110]$ (0.2064 and 0.2059), an increment of $+0.047 m_0$ over bulk. Lin et al. give $0.23 m_0$ [2, p. 21540]: our mass is 10 % lower, against 6 % lower at 1 nm, so the increment ($+0.047$ versus $+0.06 m_0$) is 20 % lower; for the increment the rounding of their values (two digits) is of the same order as the difference. At Γ , the next empty level lies 0.72 eV above the conduction-band minimum (1.50 eV in the 1 nm film, Figure 12.12): the confinement levels move closer together as the film gets thicker.

Comparison with the device model. The thickness laws of the device model (Equation (4.3)), fitted to the published proxies, give 0.348 eV and $0.049 m_0$ at 2.02 nm ; the computed values are 4 % and 3 % lower (Figure 12.15). At 0.99 nm the laws give 0.933 eV and $0.133 m_0$, against the computed 0.900 eV and $0.122 m_0$. A refit to the computed points will therefore change the laws by a few per cent.

Where the change sits. On the vacuum scale (Figure 12.14) the relaxed 2 nm film has $\text{EA} = 4.495 \text{ eV}$ and $\text{IP} = 5.719 \text{ eV}$, with a flat vacuum plateau. Both films have the same termination and the same read-out, so their difference partitions the change of the gap between 1 and 2 nm:

$$\begin{aligned} \Delta E_c(1) - \Delta E_c(2) &= \text{EA}(2) - \text{EA}(1) = +0.567 \text{ eV}, \\ \Delta E_v(1) - \Delta E_v(2) &= \text{IP}(1) - \text{IP}(2) = -0.002 \text{ eV}, \end{aligned} \quad (12.16)$$

whose sum, $+0.565 \text{ eV}$, equals $\Delta E_g(1) - \Delta E_g(2)$. Between 1 and 2 nm the whole change of the gap is thus in the conduction band, and the ionization potential is independent of thickness within 2 meV . Two checks are open before this is used. First, a top valence level that does not move at all may belong to the surface (O–H) rather than to the film interior; a projection of that state onto the atoms will decide this. Second, the shares relative to bulk, $\Delta E_c(t)$ and $\Delta E_v(t)$ themselves, need the bulk potential at the slab’s in-plane lattice constant (two-step alignment, Section 12.3.10). The CPU cross-check of the final SCF (which also gives the projected density of states) and the 2 nm IWO slab were submitted on 4 October 2026.

12.6 How the calculations were run: the workflow

In plain words. The bulk stages fit on a laptop; the films do not. A single 2 nm calculation needs tens of gigabytes of memory and days of computing. The film and doping calculations therefore ran on CERN’s batch system, which distributes jobs over thousands of processor cores and a pool of

GPUs. A collection of scripts prepares the jobs, sends them, watches them, repairs common failures, collects the results and analyses them, so that the calculations could run day and night without manual intervention. Logging in remained a manual step done by the author: no script stores or types a password.

12.6.1 Computers and software

Stages 1–3 ran on the author’s laptop under Windows Subsystem for Linux with Quantum ESPRESSO 7.5 (conda-forge build) and 10 MPI processes. All later calculations ran on the CERN batch system, which uses HTCondor [62]: jobs are submitted from the login service lxplus, wait in a queue and are matched to free worker nodes; each job declares its CPUs, memory, disk, GPUs and a “flavour” that sets its maximum wall time (here 8 h, 1, 3 or 7 days). Files are exchanged through EOS, CERN’s disk storage. Access used a Kerberos ticket and an SSH connection that the author opened and authenticated manually; the scripts reuse this connection and never handle credentials.

The CPU jobs run Quantum ESPRESSO 7.5 from a relocatable copy of the conda environment stored on EOS and unpacked on the worker. The GPU jobs run NVIDIA’s GPU build of Quantum ESPRESSO 7.3.1 [63] from a container image on EOS, executed with Apptainer [64]. Structure building and analysis use Python with ASE, spglib, NumPy [65] and Matplotlib [66].

12.6.2 Anatomy of a job

`make_jobs.py` turns the list of calculations into a job directory (`jobs/gen_<name>/`) containing: one sub-directory per job with its input files (`*.in`, templates `*.tmpl` that receive the relaxed geometry later, and `steps.sh`, the list of steps); the pseudopotentials; `jobs.txt` (one line per job: name, CPUs, memory, disk, flavour, time limit, GPUs); `jobs_info.json` (atoms, electrons, irreducible k points); the submit description `jobs.sub`; the worker script `driver.sh`; and two helpers, `qeio.py` (geometry transfer, convergence test, Fermi level) and `dat2h5.py` (density conversion, Section 12.6.3).

On the worker, `driver.sh`:

1. unpacks Quantum ESPRESSO from EOS (and, for GPU jobs, copies the container image);
2. forces one thread per MPI process (the variables `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS` and `MKL_NUM_THREADS` are set to 1) and restricts MPI to shared-memory communication on one node;
3. runs the steps of `steps.sh`. Each `pw.x` step receives `max_seconds = time left – 30 min`, so that it stops cleanly and writes its results before the batch limit; a guard stops a run whose output exceeds 2 GB (a sign of a looping error);
4. after every step copies the outputs so far to EOS (a checkpoint), and at the end packs all outputs into one archive that HTCondor returns.

A typical slab job (`steps.sh` of `slab1r_final_cpu`) is three lines: `run_pw scf 4`, `run_ppavg slab $JOB 4.8833`, `run_pw bands 4`. The number after `run_pw` is the number of k -point pools: the MPI processes are split into groups that each handle some of the k points, which is efficient but multiplies the memory. `make_jobs.py` chooses the largest number of pools that divides the CPUs, does not exceed the irreducible k points and leaves at least four processes per pool; the 2 nm slab is

limited to two pools by memory. A relaxation step is followed by `qeio.py converged` (did it finish?) and `qeio.py newgeom`, which copies the final cell and positions into the templates of the following SCF and bands runs.

12.6.3 GPU acceleration

Before any GPU result was used, a finished CPU calculation (the unrelaxed 1 nm slab) was repeated on one H100 NVL GPU with identical input: total energies -11888.29271202 and -11888.29271155 Ry (5×10^{-7} Ry apart), the same gap and the same 46 SCF iterations, in 9 min 16 s against 1 h 16 min on 16 CPU cores, a factor of 8.2 (Figure 12.19). The difference of QE versions (7.3.1 on GPU, 7.5 on CPU) has no effect at the protocol tolerances. GPU jobs request one GPU and 8 CPU cores; one MPI process drives the GPU and the remaining cores serve as OpenMP threads; only H100 NVL and H200 nodes are accepted (A100 nodes ran about two times slower).

Two incompatibilities had to be solved. First, the PAW exact exchange of the GPU build runs on the host (Section 12.5.8). Second, the GPU build saves the density as a plain Fortran file (`charge-density.dat`), whereas the CPU build used for `pp.x` and `projwfc.x`, compiled with HDF5, reads only `charge-density.hdf5`; `pp.x` aborted on the first GPU save, and the band edges of GPU runs were lost. The converter `dat2h5.py` rewrites the density in the HDF5 layout of QE 7.5 (header attributes, Miller indices with the reciprocal vectors, the complex density as interleaved real numbers), using the `h5py` library of CERN’s software distribution on the worker. It was tested by a round trip: a QE 7.5 density converted to the plain format and back gave identical HDF5 headers and data and a byte-identical planar average from `pp.x` and `average.x`. The 2 nm final calculation is the first production use.

12.6.4 The automated workflow

All operations are commands of one documented script, `qe_workflow/dft_flow.sh` (Table 12.11), also available as tasks in Visual Studio Code. Figure 12.18 shows the flow of data.

Table 12.11. Commands of `dft_flow.sh`.

Command	What it does
<code>login, check, keepalive</code>	print the login command for the author to type; test the SSH session; keep it alive and renew the Kerberos ticket every 6 h, warning 24 h before the ticket can no longer be renewed
<code>build</code>	build and verify all structures from the relaxed bulk
<code>generate, plan, submit</code>	write job directories; show the resources each job would get; upload and submit, skipping jobs already queued or finished
<code>status, resources, hw</code>	queue and live progress (step, SCF iterations, BFGS steps, energy); free CPU slots and GPUs by model; hardware under each job
<code>fetch, analyze</code>	download finished results (and checkpoints of jobs that died); collect all numbers into <code>summary.json</code>
<code>auto</code>	every 5–10 min: status, resources, fetch, watchdog, chains, analysis
<code>dashboard</code>	live web page of the whole study
<code>local</code>	run one job on the laptop with the same driver

Allocation. `submit` decides each job’s resources from its size and the free resources at that moment: films with 90 or more atoms get one H100 NVL or H200 GPU with 8 CPU cores if one is free,

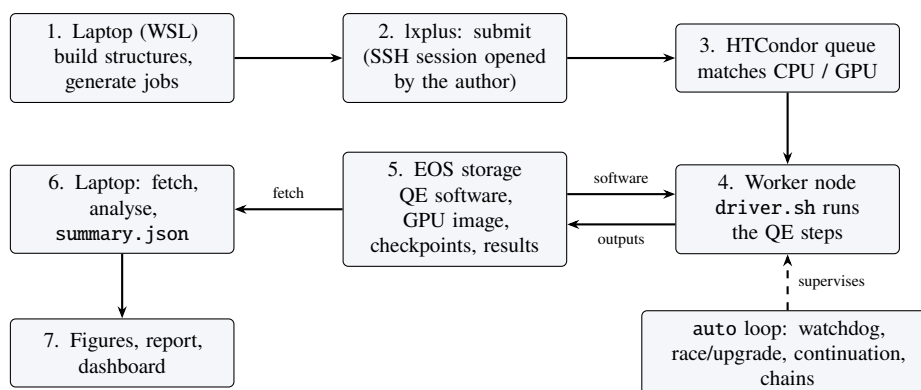


Figure 12.18. Data flow of the workflow. Jobs are generated on the laptop, submitted through the author’s Ixplus session, matched to CPU or GPU workers by HTCondor, and run by `driver.sh`, which takes the software from EOS and writes checkpoints there. Results return to the laptop for analysis. The auto loop supervises the queue.

otherwise 16 CPU cores; smaller jobs get 16 CPU cores (32 only when more than 20 such slots are free, because jobs of 32 or more cores go to a small pool of 10 slots and waited 6 h on the first day).

Supervision. The auto loop applies fixed rules:

- *race and upgrade*: copies of the same calculation (a GPU job and its CPU twin) form a group; the first copy that runs removes queued copies of equal or lower rank, and a GPU copy that starts removes a running CPU copy;
- *idle*: a GPU job queued for more than 3 h gets a CPU twin; a CPU job queued for more than 3 h is resubmitted with a smaller request;
- *stuck*: a job that runs for 60 min in an SCF step without a single SCF iteration is resubmitted (at most twice); a job without progress for 30 min is reported;
- *memory*: a job held for exceeding its memory is released with 1.5 times the memory;
- *continuation*: a relaxation that ends unfinished (time limit, eviction) is resubmitted from its last geometry as `JOB_cN` (at most five times);
- *chains*: a command attached to a relaxation runs once its finished result has been fetched; the 2 nm relaxation is chained to its final runs and to the 2 nm IWO slab.

These rules were tested with deliberate fault-injection jobs (“canaries” that simulate an unfinished relaxation, a stuck start and a race), which exposed and fixed several rule errors before the rules acted on production jobs.

12.6.5 Failures and what they taught

Table 12.12 lists the failures in the order they occurred. Each is recorded in the protocols and logs; none affected a reported number, because every result comes from a run that finished cleanly and passed its checks.

Table 12.12. Failures, causes and fixes.

Symptom	Cause	Fix
2 symmetry operations, 765 kbar, negative gap again 2 operations	inherited, never-checked bulk input spglib coordinates with QE's <code>ibrav = 3</code> basis	rebuilt from crystallographic data, automatic checks cell given explicitly (<code>ibrav = 0</code>)
32-core jobs stopped after 18 s 2 nm slab held at 95.8 GB bulk bands aborted (Davidson)	MPI launcher counted physical cores, slots expose hardware threads four k -point pools, about 24 GB each unconverged eigenvalues in a non-SCF run	<code>-map-by :OVERSUBSCRIBE</code> at most two pools CG for bulk bands; Davidson kept for slabs (CG took 13.7 h instead of 1.5 h)
jobs 17–30 times too slow	batch system set thread counts to the core count; 65 threads per process	one thread per process, all jobs resubmitted
large jobs waiting for hours	32+ cores go to a 10-slot pool	16-core CPU jobs, 8-core GPU jobs
2 nm unrelaxed SCF lost at 24 h GPU job wrote 324 GB of errors	killed at the limit in the bands step MPI tried InfiniBand on GPU nodes	checkpoint to EOS after every step transport restricted to shared memory and CUDA; 2 GB output guard
hybrid functional at 0 % GPU	PAW exact exchange on the host in the GPU build	norm-conserving pseudopotentials <code>ecutfock = 2 × ecutwfc</code>
GPU out of memory (hybrid, slab) pp.x could not read GPU saves	default exact-exchange cutoff plain vs HDF5 density format	<code>dat2h5.py</code> converter
converged 2 nm relaxation lost	final SCF needed 66 GB (32 GB requested); job held, no checkpoint on EOS (cause not determined)	memory from measured demand; no automatic release after > 1 h of running; re-run fixed before production use
workflow rule errors	logic errors found by canary jobs	

12.6.6 Computing resources

The usage to date is about 8 300 weighted CPU core-hours and 85–90 GPU-hours at CERN, of which about 3 000 core-hours and 50 GPU-hours were lost to the failures above, and about 190 core-hours on the laptop (Figure 12.19). Run on the laptop, the useful work would take about six weeks of continuous computing, and the 2 nm slab and the hybrid slab calculation (30–100 GB of memory) could not run at all. Table 12.13 gives the wall times of representative steps.

Table 12.13. Wall times of representative steps (from the output files and driver logs).

Calculation	Size	Hardware	Wall time
bulk SCF, 71 Ry, 3×3×3	40 atoms	laptop, 10 MPI	12 min
bulk vc-relax (11 BFGS steps)	40 atoms	laptop, 10 MPI	1 h 47 min
bulk bands, path (80 <i>k</i> , CG)	40 atoms	laptop, 10 MPI	11 h 16 min
1 nm slab SCF, unrelaxed	96 atoms	16 CPU cores	1 h 48 min
same, GPU benchmark	96 atoms	1 H100 NVL	9 min
1 nm vc-relax, 54 BFGS steps	96 atoms	1 A100	15 h 31 min
1 nm final SCF / bands	96 atoms	16 CPU cores	2 h 19 min / 4 h 38 min
W on 24d: relax / SCF 4 ³ / bands	80 atoms	32 CPU cores	23 h 48 min / 3 h 3 min / 18 h 36 min
HSE06 bulk / 1 nm slab	40 / 96 atoms	1 H100 NVL	12 min / 4.0 h
IWO 1 nm relax (27 BFGS steps)	96 atoms	1 H100 NVL	4 h 29 min

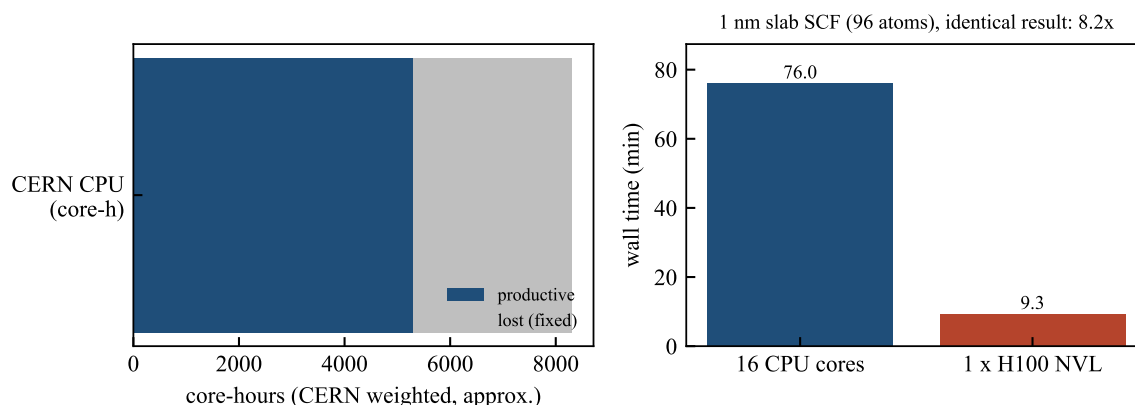


Figure 12.19. Left: CERN CPU usage (approximate weighted core-hours), productive and lost to faults that were fixed. Right: GPU benchmark, the same 96-atom SCF on 16 CPU cores and on one H100 NVL.

In plain words. Figure 12.20 is the diary of the CERN work: one row per calculation, coloured by what the calculation was doing at each moment (waiting in the queue, running on CPUs, running on a GPU). It shows the long CPU phase of the first two days, the move to GPUs once they were validated, and the many short checks that ran in parallel on 30 September.

The timeline combines three records: the exact start and end times written by `driver.sh` for every job that returned output; the queue snapshots of the monitor script every 30 min from 27 September 09:02Z; and the 5–10 min status records of the workflow from 28 September 14:20Z. Runs of the first submission that were removed without output before 09:02Z on 27 September (the thread problem, Table 12.12) are not shown, apart from the failed 18-second starts. White gaps within a row mean that no copy of that calculation was recorded as queued or running (between a removal and a resubmission, or while the monitoring was interrupted).

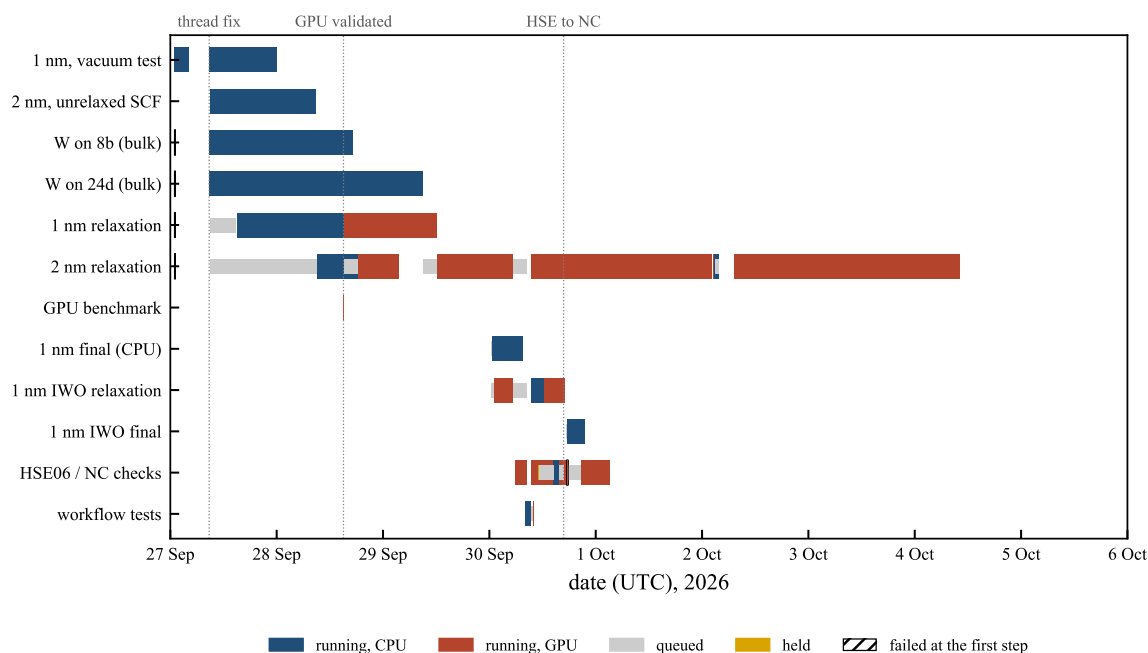


Figure 12.20. Timeline of the CERN calculations, 27 September to 1 October 2026 (UTC). Blue: running on CPUs; red: running on a GPU; grey: queued; amber: held; hatched: failed at the first step. Dotted lines: the thread fix, the GPU validation, and the move of the hybrid check to norm-conserving pseudopotentials. Several copies of a calculation (CPU twin, GPU variants, continuations) share one row.

12.7 From output files to numbers and figures

In plain words. A finished calculation leaves text files of thousands of lines. Small Python scripts read the few lines that matter (energies, eigenvalues, forces, the averaged potential), compute the derived quantities with the formulas of Section 12.3.10, and draw the figures. Nothing is copied by hand, so every number in this chapter can be regenerated from the raw outputs by running the scripts again.

12.7.1 Reading the outputs

Table 12.14 lists each quantity, the line of the output it comes from and the script that reads it. The eigenvalues of non-SCF runs are read block by block (one block per k point, Listing 12.2); k is converted from units of $2\pi/a_{\text{lat}}$ to $\text{\AA}^{-1}$ with the lattice parameter printed in the same output. This matters after a vc-relax: pw.x keeps the initial a_{lat} as its unit, so the largest k of the bulk Γ set is 0.153 and not $0.15 \text{ \AA}^{-1}$.

Listing 12.2. Reading eigenvalues and fitting the effective mass (`make_dft_figures.py`, shortened).

```

1 def eig(path, nocc, extra=8):
2     t = Path(path).read_text(errors='replace')
3     alat = float(re.search(r'lattice parameter \s*(alat)\s+=\s*([\d.]+)', t).group(1)) * BOHR
4     bl = re.findall(r'k\s*=\s*([\d. ]+?)\s*(\s*\d+ PWs)\s+\s+bands\s*(ev)\s*:\s*(.?)\s*(?=...)', t, re.S)
5     ks = np.array([...]) * 2 * np.pi / alat          # k in 1/\AA
6     ev = np.array([...])                             # eigenvalues (eV), bands 1..nocc+extra
7     return ks, ev
8
9 k = np.linalg.norm(ks[idx], axis=1); e = ev[idx, nocc] - ev[0, nocc] # CB relative to its minimum
10 sel = k <= 0.0501                                     # fit window |k| <= 0.05 1/\AA

```

Table 12.14. Where each quantity comes from.

Quantity	Source in the output	Script
total energy	line starting with “!” (last one); “Final energy/enthalpy” after a relaxation	collect_results.py, make_dft_figures.py
pressure, force	“P=” in the stress block; “Total force =”	same
VBM, CBM, gap (SCF)	“highest occupied, lowest unoccupied level (ev)”	analyze_slabs.py
bulk fundamental gap	minimum of band 177 minus maximum of band 176 over the path run	analyze_path.py, collect_results.py
m^* , α	eigenvalue blocks of the Γ -set bands run, fits Equations (12.8) and (12.9)	analyze_mass.py, collect_results.py
E_{vac} , EA, IP	<job>_avg.dat of average.x: mean of the planar average within 2 Å of the cell boundary; its spread as a flatness check	analyze_slabs.py
E_F	“the Fermi energy is”	qeio.py fermi
W–O bonds	final geometry, neighbours within 2.6 Å (ASE)	analyze_iwo.py
W 5d share	projwfc.x output and pdos/(W)_wfc##(d) files	analyze_iwo_slab.py
site energy	$[E(8b) - E(24d)] \times 13.6057 \text{ eV/Ry}$	collect_results.py
HSE ratio	gaps of the four NC runs	collect_results.py

```

11 c2 = np.polyfit(k[sel] ** 2, e[sel], 1)[0]          # slope of E against k^2
12 m_star = H2M / c2                                # H2M = hbar^2/(2 m0) = 3.80998 eV Å^2

```

All numbers are collected by `collect_results.py` into the file `summary.json` of the results directory, which feeds the live dashboard; values that are missing stay empty and are never estimated. Every value used in this chapter is also entered, with its date and context, in `RESULTS_LOG.md`.

12.7.2 Drawing the figures

All figures of this chapter are produced by one script of the report, `tools/make_dft_figures.py` (Matplotlib, Times font, PDF output), run with `python tools/make_dft_figures.py`. Each figure is one function that reads the output files directly (Table 12.15). The plotting choices follow the purpose of each figure: energies are always shown relative to a physical reference (the VBM, the CBM, E_F or E_{vac}), never as raw eigenvalues, because absolute Kohn–Sham energies have no meaning; acceptance criteria are drawn as shaded bands so that a reader sees at once whether a point passes; literature values are open symbols and values of this work filled.

12.8 How the gap opening is shared: alignment to bulk

In plain words. When the film becomes thin, the gap opens. For the transistor only the upward move of the conduction band matters, because that is where the electrons travel. To split the opening into its conduction-band and valence-band parts, the energies of the film and of the bulk crystal must be placed on one common scale. We used the average electrostatic potential deep inside the film, which there looks like bulk, as the common reference.

Method. The alignment is done in two steps (`qe_workflow/bulk_In2O3_protocol/run_bulk_potential.sh`, `bulk_alignment.py`).

Table 12.15. The figure functions of `make_dft_figures.py`: data and processing.

Figure	Data read	Processing and plotting
Figure 12.5	structure files (*.json) of the relaxed and built slabs	x - z projection; atoms drawn back to front by y , radius per element; cell outline; H-H distance
Figure 12.7	scf_ecut*_k*.out	differences to the most converged run, per atom (40 atoms) in mRy; pressure and gap differences; criteria shaded
Figure 12.8	bands_path.out, bands_path.in	distance along the path from successive $ \Delta k $, no distance across the H P jump; tick positions from the points per segment of the input; energies minus the VBM
Figure 12.9	bands_gamma.out, slab bands.out	$ k $ from Γ , $E - E_{\text{CBM}}$; least-squares slope against k^2 in the window; parabola drawn over the full range
Figure 12.10	vc-relax outputs of the CPU run and the GPU continuation	energy (log scale), gap and total force for each SCF cycle; the switch marked
Figure 12.13	slab1r_final_cpu_avg.dat, scf.out	columns z (bohr), planar and macroscopic average (Ry) converted to $\text{\AA}$ and eV; E_{vac} from the vacuum window; band edges as lines
Figure 12.15	gap of slab1r_final_cpu; literature values; law coefficients	$\Delta E_g = E_g(t) - 0.8887$ eV; HSE-corrected point $\times 1.085 \pm 0.025$ (midpoint and half-range of 1.06–1.11); laws of the device model as curves
Figure 12.16	four NC scf.out	gaps as bars, shifts and ratio in the panel
Figure 12.17	pdos_tot, W 5d PDOS files, Fermi energy	energies relative to E_F ; each curve scaled to its maximum (shapes, not absolute weights)
Figure 12.19	resource accounting of the logs; benchmark wall times	stacked bar; bar chart of the benchmark
Figure 12.11	bulk and slab outputs (as above); values of Lin et al. with page	relative difference per quantity; $\pm 5\%$ band; tolerance of the shift
Figure 12.12	eigenvalue block at Γ of bands_gamma.out and of the two slab scf.out	levels as horizontal lines relative to the highest occupied level at Γ ; occupied/empty by band index
Figure 12.14	scf.out and *_avg.dat of three slabs	E_{vac} from the vacuum window; band edges (pure films: whole k grid; doped film: Γ levels 312–314 and E_F) placed below it
Figure 12.20	monitor.log, status_history.jsonl, all *_driver.log	samples converted to bars of one sampling interval per row (copies merged); driver-log intervals drawn exactly

Step 1, bulk. The cell average of the local electrostatic potential (`pp.x`, `plot_num 11`) of the relaxed bulk crystal is $\langle V \rangle = 4.1687$ eV. Relative to it, the band edges of the bulk band structure are $E_v - \langle V \rangle = 3.9098$ eV and $E_c - \langle V \rangle = 4.7985$ eV.

Step 2, slab. The macroscopic average of the same potential is taken at the centre of each slab.

Averaging window. The planar-average period of bixbyite along [001] is $a/2$, not the $a/4$ used for the vacuum levels.

- With an $a/4$ window the slab interior oscillates by 177–205 meV. The result disagrees with the electron-affinity route by about 80 meV, so it was rejected.
- The $a/2$ window gives an interior plateau flat within 8 meV at 2 nm (1 meV with the double $a/2 * a/4$ average).

The 1 nm film. It has almost no bulk-like interior, so its values follow from the 2 nm result plus the same-termination electron-affinity differences (+0.567 and -0.002 eV). Between 1 and 2 nm the two routes agree within 15–30 meV.

Table 12.16. Band-edge shifts relative to bulk (PBE). $\Delta E_c - \Delta E_v = \Delta E_g$ in every case.

Film	Route	ΔE_c (eV)	ΔE_v (eV)	conduction-band share
2 nm	two-step, $a/2$ window	+0.281	−0.054	0.84
2 nm	double average	+0.287	−0.049	0.86
1 nm	two-step, $a/2$ window	+0.819	−0.081	0.91
1 nm	2 nm value + electron-affinity difference (adopted)	+0.848	−0.056	0.94

Caveats.

- The in-plane strain of the slabs relative to bulk (+0.29 % at 1 nm, +0.07 % at 2 nm) is not corrected; its effect is likely a few meV at 2 nm.
- The values are PBE.

The band edges are not surface states. An orbital projection of the relaxed 2 nm film (`projwfc.x`, `qe_workflow/cern_htcondor/surface_state_check.py`) checked this.

- The highest occupied state carries 0.2 % of its weight on the surface OH groups, which make up 27 % of the atoms. Its weight peaks at the centre of the film, like the lowest confined state of a box.
- The lowest empty state carries 1.4 % on the OH groups.
- A deeper valence state used as a reference carries 14 %.

Both band edges are therefore confined, bulk-like states, and the split of Table 12.16 describes them.

Consequence for the device model. Model V1 assumed the whole opening in the conduction band. The computed share is 0.84–0.94. At 2 nm the conduction band therefore moves up by 0.28 eV, not 0.35 eV. The device model takes these values over in Chapter 13.

12.9 Pending results and changes to the device model

Table 12.17. Status of the first-principles inputs of the device model.

Input	This work	Status
ΔE_g at about 1 nm	+0.90 eV (PBE); 0.95–1.00 eV (HSE-corrected)	computed; reproduces Lin et al.
Δm^* at about 1 nm	+0.122 m_0	computed
ΔE_g at about 2 nm	+0.335 eV (PBE); 0.36–0.37 eV (HSE ratio of 1 nm assumed)	computed; reproduces Lin et al.
Δm^* at about 2 nm	+0.047 m_0	computed
$\Delta E_c/\Delta E_v$ partition	relative to bulk: 2 nm $\Delta E_c = +0.281$, $\Delta E_v = -0.054$ eV (share 0.84); 1 nm +0.848, -0.056 eV (share 0.94); between 1 and 2 nm all in ΔE_c (Section 12.8)	computed; band edges bulk-like (not surface states) at 2 nm
Films of 0.75 and 0.51 nm	Lin recipe with two and one cation planes ($\text{In}_{16}\text{O}_{36}\text{H}_{24}$, $\text{In}_8\text{O}_{24}\text{H}_{24}$; H to H 0.75 and 0.51 nm before relaxation; <code>structures_v2/build_thin_slabs.py</code> , verified to reproduce the 1 nm slab exactly). The one-plane film has the composition of indium hydroxide, $\text{In}(\text{OH})_3$, and marks the limit of the recipe	relaxing on CERN (5 Oct. 2026)
W site, valence	24d, W^{6+}	computed, converged
W 5d level, moments	resonant in the conduction band, 1.2–1.8 eV above its bottom; occupied only at very high filling (Sections 12.5.9 and 12.5.10)	preliminary (PBE); corrected 5 Oct. 2026
W effect on confinement	<i>[to be completed: 2 nm IWO slab; 1 nm IWO slab mass]</i>	pending

The 1 and 2 nm points are now computed. They lie 3–8 % below the V1 laws (Section 12.5.11). With the partition relative to bulk (Section 12.8), the conduction-band shift at 2 nm is 22 % below the V1 law.

Chapter 13 refits the confinement and mass laws to the computed points (model V2) and gives them the provenance label of this work. It also calibrates the device model again and tests it on the held-out 6.3 nm film.

12.10 Limitations

Limitation. The films are idealized crystalline (001) slabs with hydrogen-passivated surfaces in vacuum; the device channels are sputtered, largely amorphous or nanocrystalline IWO on Al_2O_3 with an air-exposed top surface. The results describe trends and provide physically grounded inputs; they do not predict the measured devices directly. PBE is adequate for the confinement shift within about 10 % (Section 12.5.8) but not for absolute gaps, masses or the position of W 5d states. The W cells contain one W at 3.1–4.2 %; other positions, concentrations, complexes with oxygen defects and charge states were not computed. The hybrid check uses a reduced exact-exchange q grid, whose effect is bounded but not eliminated. The relaxations stop at forces of 10^{-3} Ry/bohr (a recorded deviation from the 10^{-4} of Lin et al.).

Summary. An independent PAW-PBE calculation reproduces the published confinement of 1 and 2 nm In_2O_3 films ($\Delta E_g = +0.90$ and $+0.335$ eV, $\Delta m^* = +0.122$ and $+0.047 m_0$, against $+0.94$ and $+0.33$ eV and $+0.13$ and $+0.06 m_0$). Between 1 and 2 nm the change of the gap lies entirely in the conduction band, and the band edges are bulk-like, not surface states. A hybrid-functional check shows that the semi-local functional gives the confinement shift to within 6–11 %, so the slab-minus-bulk construction of the device model rests on evidence. Tungsten substitutes on the 24d site as W^{6+} ; within PBE its donated electrons fill the host conduction band, and a W-5d-derived level resonant 1.2–1.8 eV above the band bottom takes up electrons only at very high filling (preliminary). Every structure, setting, run and number is traceable to scripts and output files, and the workflow that produced them is documented. The computed points lie 3–8 % below the V1 thickness laws, and 84–94 % of the gap opening is in the conduction band (V1 assumed 100 %); the device model takes these values over in Chapter 13. The W effect on confinement is pending.

Chapter 13

Model V2: first-principles inputs, off-state floors and a held-out test

Chapter 12 computed, in one consistent setup, the quantities that the device model had so far borrowed from the literature: how far the conduction band moves up when the film becomes thin, how the gap opening is shared between the conduction and the valence band, and how heavy the electrons become. This chapter puts those numbers into the device model (model V2) and asks four questions:

- V1** What produces the flat off-state currents of the measured transistors, which model V1 does not contain?
- V2** How much do the first-principles inputs change the calibrated curves?
- V3** Does the model, with nothing adjusted to the 6.3 nm film, predict that film?
- V4** What would 1, 0.75 and 0.5 nm films do?

In plain words. The device simulator needs a handful of material numbers for every film thickness. Until now the numbers describing how thinning changes the electron energies came from other people's calculations. We computed them ourselves (Chapter 12) and here swap them in. Before running anything we wrote down what each test should show and what would count as passing, and published that on the project's public repository. That way the results cannot be bent after the fact.

13.1 How the work was organised

The work followed six steps fixed with the user on 5 October 2026:

- **T0:** checks without ATLAS.
- **T1:** the off-state floor.
- **T2:** the first-principles laws in the two calibration films (2.0 and 13.2 nm).
- **T3:** the held-out 6.3 nm test.
- **T4:** the ultrathin films.
- **T5:** this chapter.

The user raised the ATLAS launch budget from 57 to 71 for these steps.

Pre-registration. Every pass/fail criterion, and every predicted value that could be computed without ATLAS, was written down before the runs. That record is `analysis_2026-10-06_V2/T0_PRECHECKS.md`, together with the SHA-256 hashes of all inputs in `T0_REGISTRATION.txt`. It was committed to the public repository as commit 7b75ee6 before the first V2 launch.

A correction to the pre-registration. The first V2 run revealed an error in one of the pre-registered calculations (Section 13.4). The correction was written, hashed and published (commit a914165) before the result of the held-out test existed. The registered files were kept unchanged, and both the registered and the corrected predictions are compared with ATLAS below.

13.2 The off-state floors

In plain words. When the measured transistors are switched off, a small current still flows. It does not depend on the gate voltage, and it is about 1400 times larger in the 13.2 nm film than in the 2 nm film. The simulated transistors switch off almost completely (about 10^{-19} A/ μm), so this current is missing from the model. We first asked which explanations are even possible, then computed whether the remaining one gives the right numbers.

What the data show. Table 13.1 lists the floors, taken as the median current between -2 and -0.5 V.

- **13.2 nm:** a constant 1.9 nA, flat within $\pm 3\%$ over 2.5 V of gate voltage.
- **6.3 nm:** flat but noisy ($\pm 30\%$, mostly random).
- **2 nm:** rises smoothly from 0.05 to 1.5 pA. This looks like a charging transient at the start of the sweep on top of a setup-level floor.

From 2 to 13.2 nm the floors grow as $t^{3.85}$.

Table 13.1. Measured off-state floors (median over $V_G = -2 \dots -0.5$ V; device width 290 μm) and their shape (`analysis_2026-10-06_V2/t0_floor_data.py`).

Film	Floor (A/ μm)	Device total	Shape over $-3 \dots -0.5$ V
2.0 nm	4.61×10^{-15}	1.3 pA	rises $0.05 \rightarrow 1.5$ pA, smooth (start-of-sweep transient on a setup-level floor)
6.3 nm	1.53×10^{-13}	44 pA	flat, $\pm 30\%$ random noise
13.2 nm	6.60×10^{-12}	1.9 nA	flat within $\pm 3\%$

What is excluded.

- **Gate leakage.** It would be largest at -3 V and would change at least threefold over the range. The 6.3 and 13.2 nm floors are flat, and the 2 nm floor is *smallest* at -3 V.
- **A current range fixed by the on-current.** The 2 and 6.3 nm devices have similar on-currents but floors 33 times apart.

- **A conducting layer at the back surface that the gate still controls.** Keeping it flat within 3 % over 2.5 V would need about $2 \times 10^{16} \text{ cm}^{-2} \text{ eV}^{-1}$ of pinning states.

What remains is a drain-to-source current outside the reach of the gate, or an instrument effect.

The ungated-strip calculation. Suppose part of the film lies outside the gate's control. Its sheet conductance follows from the model's own material values. Charge neutrality fixes the Fermi level E_F :

$$N_d t = t \left[n(E_F) + \int g_{\text{tail}}(E) f(E) dE + \int g_{\text{deep}}(E) f(E) dE \right] + N_s(E_F) - Q_s, \quad (13.1)$$

- $n = N_c F_{1/2}$ is the free-electron density;
- g_{tail} and g_{deep} are the V1 tail and deep acceptor densities of states (Sections 4.3 and 4.8);
- N_s counts electrons in surface acceptor states, and Q_s is a fixed surface charge.

The strip conductance is then $G = q n \mu_0 t (W/L)_{\text{par}}$. Its one geometric factor $(W/L)_{\text{par}}$ is fitted on the 13.2 nm floor; the 2 and 6.3 nm floors are then predicted. The band bending across the strip from a surface sheet is a few meV, much less than $k_B T$, so a uniform film is an accurate description.

Three variants were computed (Figure 13.1a):

- **N:** no surface states.
- **D:** one surface carrying the V1 interface acceptor sheet ($6.4 \times 10^{10} \text{ cm}^{-2}$ in total).
- **Q:** as D, plus the fixed charge of the channel interface ($1.73 \times 10^{12} \text{ cm}^{-2}$).

Table 13.2. Ungated-strip variants against the measured floors (one geometric factor fitted on 13.2 nm). Criterion set in the plan: both predicted floors within a factor of 3.

Variant	Surfaces	Scaling	2 nm pred/meas	6.3 nm pred/meas
N	none	$t^{2.99}$	5.07	1.21
D	one surface with the V1 interface acceptors	$t^{4.24}$	0.48	0.77
Q	D plus Q_f of the channel interface	$t^{1.25}$	134	4.76

Only variant D passes (Table 13.2). How firm is that?

- **6.3 nm:** the prediction stays within $\times 0.37$ – 0.99 when the assumed interface-state density or the donor density is halved or doubled.
- **2 nm:** the prediction swings from $\times 0.06$ to $\times 1.9$, so the 2 nm agreement is not a strong test. The 2 nm floor also sits at the setup floor, where an under-prediction would not contradict the data.

What V2 adopts. The total drain current is

$$I_D^{\text{total}}(V_G) = I_D^{\text{ATLAS}}(V_G) + G_D(t) V_D, \quad (13.2)$$

with $(W/L)_{\text{par}} = 1.03 \times 10^{-4}$ per μm of width. A gate-independent parallel conductance at fixed V_D simply adds, so Equation (13.2) is exact and costs no launch. Its status is **HYPOTHESIS**. The

location and the interface of the strip are not determined; they need the gate and source currents, a channel-length series and the layout (Section 13.8).

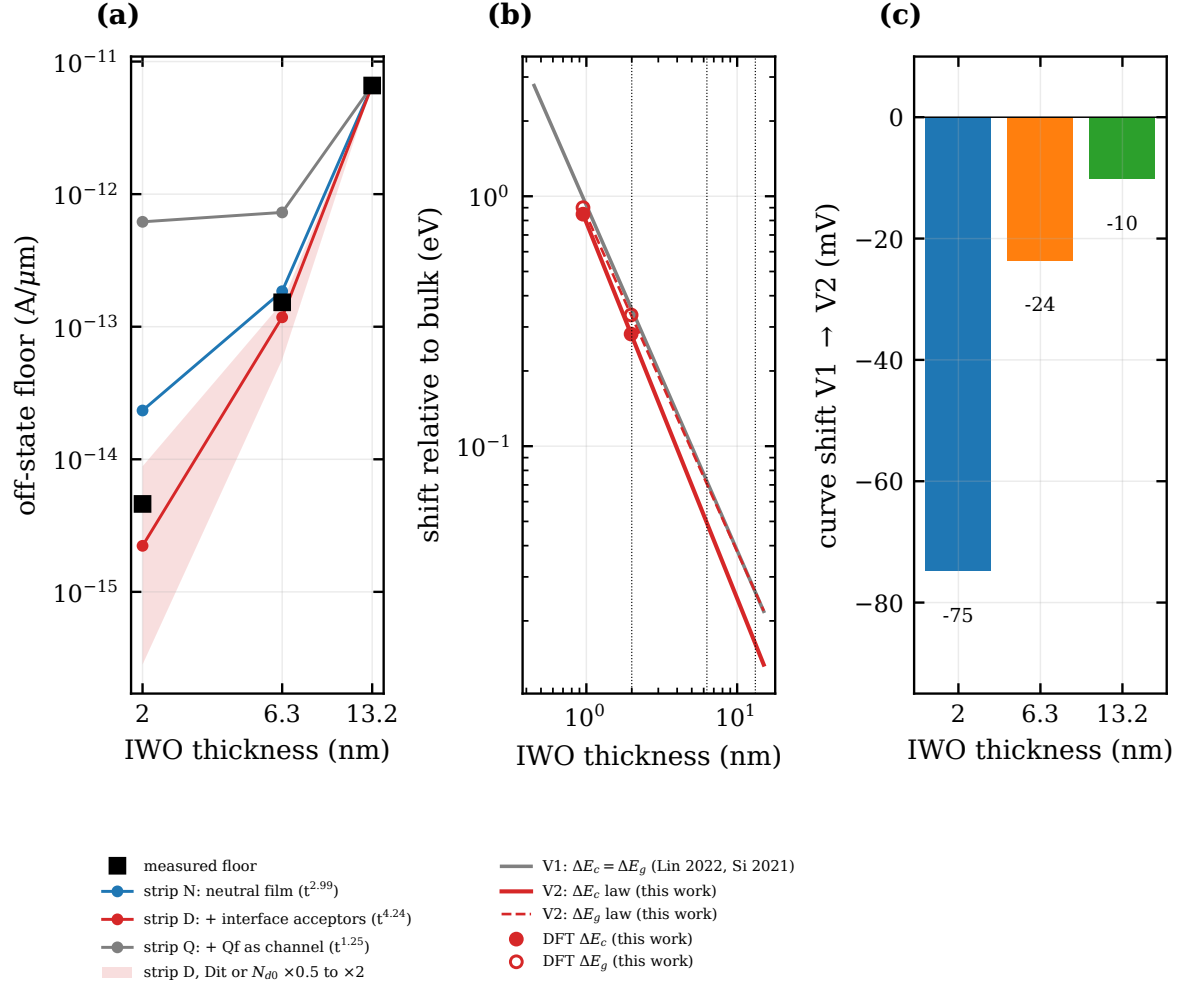


Figure 13.1. Step T0 (no ATLAS). (a) Measured off-state floors (squares) and the three ungated-strip variants, each fitted on 13.2 nm. The band shows variant D with the interface-state density or the donor density halved or doubled. (b) Conduction-band shift relative to bulk: V1 law (literature, all of the gap opening in the conduction band) and V2 laws through the DFT points of this work. Dotted lines mark the measured films. (c) Rigid shift of each calibrated transfer curve when the V1 laws are replaced by the V2 laws, before re-calibration.

13.3 First-principles inputs in the device model

In plain words. The DFT results change three numbers in the device model: how far the conduction band rises in a thin film, how much the gap opens, and how heavy the electrons are. Of these, only the conduction-band rise matters for the transfer curve, and it changes by less than one tenth of an electronvolt. That moves each calibrated curve sideways by 10 to 75 mV.

New laws. Fitted through the DFT points of Chapter 12 (PBE):

$$\Delta E_c(t) = 0.78504 t^{-1.50400} \text{ eV}, \quad \Delta E_g(t) = 0.83997 t^{-1.34569} \text{ eV}, \quad \Delta m^*(t) = 0.11416 t^{-1.29453} m_0, \quad (13.3)$$

with t in nm and $t \geq 0.95$ nm. The DFT values behind them are:

Table 13.3. DFT values behind Equation (13.3) (PBE, relative to bulk).

Film	ΔE_c (eV)	ΔE_g (eV)	Δm^* (m_0)
0.95 nm	0.848	0.900	+0.122
1.98 nm	0.281	0.335	+0.047

Two points about these values:

- **Partition.** The conduction-band share of the gap opening is 0.84 at 2 nm and 0.94 at 1 nm. V1 assumed 1.0, i.e. all of the opening in the conduction band.
- **Functional.** The HSE06/PBE ratio of the gap opening is 1.06 (range 1.06–1.11), so PBE is kept as the central value, as the DFT protocol pre-registered.

Below 0.95 nm the laws are not extrapolated; the 0.75 and 0.51 nm films enter as tabulated DFT points (Section 13.7).

Why the effect is a rigid shift. Every trap level in the decks is referenced to the conduction-band edge, and the source and drain are ideal Ohmic contacts. A change of the electron affinity therefore moves the whole transfer curve along V_G by exactly the change of ΔE_c . A change of the front fixed charge is also a rigid shift, $\Delta V = -q \Delta Q_f / C_{ox}$ (verified to 1 mV in V1). The drain current is linear in a uniform mobility.

The mass change enters only through N_c . Its effect is taken from the isolation runs 0050/0051 ($m^* \times 1.3$ shifts $V_{th,cc}$ by -0.043 V at 2 nm) and is below 2 mV here.

So the V2 curves follow from the calibrated V1 runs by exact transformations, before any launch (Table 13.4, Figure 13.1c).

Table 13.4. Conduction-band shift V1 $\rightarrow$ V2 and the resulting rigid shift of each calibrated curve.

Film	ΔE_c V1 (eV)	ΔE_c V2 (eV)	Curve shift
2.0 nm	0.3533	0.2768	−75 mV
6.3 nm	0.0724	0.0493	−24 mV
13.2 nm	0.0261	0.0162	−10 mV

V2 calibration. Re-calibration follows the V1 protocol exactly:

- **One shared front fixed charge**, chosen by least squares on the 2 and 13.2 nm curves. It moves every curve by +62 mV, giving $Q_f = 1.38345 \times 10^{12} \text{ cm}^{-2}$ (V1: 1.73×10^{12}).
- **One band mobility per calibration film**, scaled exactly to the measured on-current.

13.4 The correction to the pre-registered transformation

The exact transformation moves a calibrated ATLAS curve along V_G . The registered version clipped the curve at the end of the simulated sweep (3 V), so a curve moved to the *left* was flat over its last $|\Delta V|$.

- **2 nm** (net shift -13 mV): the first V2 run gave an on-current 0.83 % above the registered prediction, which revealed the error.
- **6.3 nm** (net shift -0.26 V): the clipping affected the registered on-state values.
- **Not affected:** curves moved to the right (13.2 nm) and the subthreshold region, including $V_{th,cc}$.

The fix extrapolates the on-state linearly from the last three simulated points. It was registered and published before the 6.3 nm result existed (`T0_CORRECTION.md`, `T0_CORRECTION_REGISTRATION.txt`).

13.5 ATLAS V2 anchors (T2)

In plain words. We ran the two calibration films again with the new numbers. ATLAS reproduced the predicted thresholds to within half a millivolt and the on-currents to within 0.1 %. One of the three agreement criteria was missed by a hair at 2 nm, for a reason we could trace.

Runs 0058 (2.0 nm) and 0059 (13.2 nm) use the V2 model file with the registered parameters. Both pass the current-balance validation (largest imbalance 1.0×10^{-16} A) without convergence flags. Table 13.5 compares them with the exact transformations.

Table 13.5. T2: ATLAS V2 anchors against the exact transformations of the V1 runs. Registered tolerance: $V_{th,cc}$ within 5 mV, I_{on} within 1 %, active RMSE within 0.005 dec. Predictions after the correction of Section 13.4; the registered I_{on} of the 2 nm film was 5.090×10^{-7} A/ μm .

Film (run)		$V_{th,cc}$ (V)	I_{on} (A/ μm)	active RMSE (dec)	verdict
2.0 nm (0058)	ATLAS	0.6677	5.132×10^{-7}	0.0661	FAIL (RMSE)
	predicted	0.6681	5.134×10^{-7}	0.0600	
13.2 nm (0059)	ATLAS	0.1033	3.073×10^{-6}	0.0968	PASS
	predicted	0.1034	3.071×10^{-6}	0.0973	

Why the 2 nm RMSE criterion was missed. The difference is not scatter. ATLAS lies smoothly above the transformation at the lowest currents of the active region: $+0.026$ dec at 10^{-13} A/ μm , falling to zero at the threshold and in the on-state.

The transformation treated the small mass change at 2 nm (-1.1 %, i.e. N_c lower by 1.7 %) as a rigid $+2$ mV shift. A change of N_c also changes the subthreshold slope slightly (isolation runs 0050/0051). About 0.6 mV/dec over four decades below the threshold current amounts to about 0.024 dec, which is the observed size.

The registered criterion is therefore missed by 0.0011 dec. The cause is an approximation in the transformation, not a defect of the V2 decks.

Final V2 calibration. As in V1, the anchor mobilities are scaled exactly to the measured on-current: $\mu_{\text{band}} = 17.544$ (2.0 nm) and 63.208 cm²/Vs (13.2 nm), factors 0.9918 and 0.9992 on runs 0058 and 0059. Against V1:

- **Active RMSE:** 0.065 dec at 2 nm (V1 0.043) and 0.097 dec at 13.2 nm (V1 0.081).
- **Constant-current threshold:** 0.668 against 0.662 V measured at 2 nm, and 0.103 against 0.083 V at 13.2 nm.

The first-principles laws thus cost a little accuracy at the anchors. The reason is that V1's literature law had put about 60 mV more of the 2–13.2 nm threshold difference into confinement than the computed band split supports.

13.6 The held-out 6.3 nm test (T3)

In plain words. This is the real test: can the model, calibrated on the 2 and 13.2 nm films, predict the 6.3 nm film it has never seen? It cannot. The predicted transistor switches on about a quarter of a volt too early, exactly as forecast before the run. Swapping in the first-principles numbers made the miss smaller by only about 0.04 V.

Run 0060 uses the V2 model with nothing adjusted to 6.3 nm. The band mobility comes from the pre-registered rule $\mu_{\text{band}}(t) = \mu_{\text{band}}(2 \text{ nm}) (t/2)^{0.675}$, i.e. $38.39 \text{ cm}^2/\text{Vs}$ (variant P). For variant E the same run is scaled exactly to the measured on-current ($\mu_{\text{band}} = 9.90$), so that only the electrostatics is tested. Table 13.6 and Figure 13.2 show the outcome.

Table 13.6. T3: held-out 6.3 nm prediction (run 0060). Pass requires all of: $|\Delta V_{th,cc}| \leq 0.10 \text{ V}$, $|\Delta V_{th,lin}| \leq 0.10 \text{ V}$, SS_{cc} within 20 %, active RMSE $\leq 0.15 \text{ dec}$, and for P also I_{on} within 30 %.

	$V_{th,cc} \text{ (V)}$	$V_{th,lin} \text{ (V)}$	$SS_{cc} \text{ (mV/dec)}$	$I_{on} \text{ (A/}\mu\text{m)}$	RMSE (dec)	verdict
measured	0.644	1.820	295.4	4.03×10^{-7}	–	–
P (predicted μ)	0.270	1.240	220.1	1.56×10^{-6}	1.039	FAIL
E (μ to I_{on})	0.417	1.240	300.2	4.03×10^{-7}	0.547	FAIL
V1, shared Q_f (for comparison)	0.359	1.180	289.7	–	0.656	–

Outcome. Both variants fail, as predicted before the run.

- **The threshold is too low:** by 0.37 V (P) and 0.23 V (E).
- **The on-state shape is wrong:** $V_{th,lin}$ is 0.58 V too low in both variants.
- **Only the subthreshold slope** of variant E meets its criterion.
- **The mobility rule over-predicts** the on-current fourfold: the 6.3 nm film has a lower apparent mobility than the 2 nm film, which no monotonic thickness law can produce.

Agreement with the pre-registered numbers. ATLAS agrees with the corrected predictions to within 0.3 mV in $V_{th,cc}$, 0.1 mV/dec in SS_{cc} and 0.001 dec in RMSE. The largest difference is 22 mV in $V_{th,lin}$, which depends on where the transconductance peaks. So the outcome of the test was fixed by the transformation before the run; the run confirms it.

Effect of the first-principles inputs. Compared like with like (variant E against V1 with the shared Q_f , corrected for their different mobilities), the first-principles laws shrink the threshold miss by about 0.04 V. They do not address its cause.

What this means. The model is a calibrated description of the 2 and 13.2 nm films. It is not yet a predictive model of the thickness dependence. Chapters 8 and 9 discuss candidate causes. The decisive data are listed in Section 13.8; the first is the device-to-device spread at each thickness.

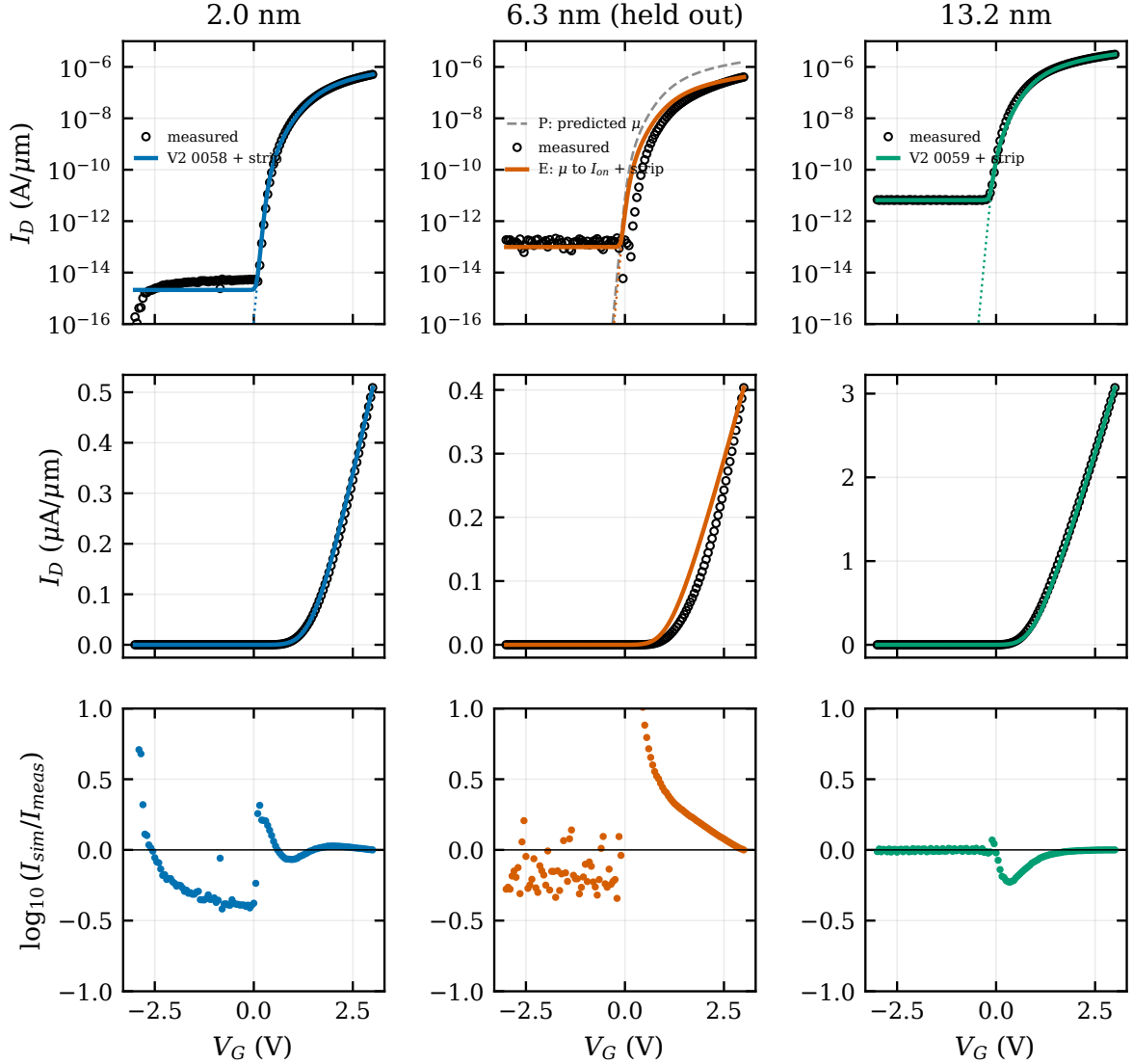


Figure 13.2. Model V2 against the measurements, including the off-state strip of Equation (13.2). Rows: logarithmic, linear, and the $\log_{10}$ ratio of simulation to measurement. Calibration films (2.0 and 13.2 nm): runs 0058 and 0059, scaled exactly to the measured on-current (factors 0.9918 and 0.9992). Held-out film (6.3 nm): run 0060, variant P dashed and variant E solid (factor 0.2579). Dotted lines: the ATLAS current without the strip. Raw data: analysis_2026-10-06_V2/data/v2_overlays_raw.csv.

The off-state strip in V2. With the final V2 mobilities, the factor fitted on 13.2 nm is $(W/L)_{\text{par}} = 1.00 \times 10^{-4}$ per μm .

- **Predicted floors:** 2.1×10^{-15} (2 nm, $\times 0.46$ of measured) and 9.8×10^{-14} A/ μm (6.3 nm, $\times 0.64$).
- **Error over all 121 points of each sweep:** it falls from about 14 dec without the strip to 0.28 (2 nm), 0.62 (6.3 nm) and 0.07 dec (13.2 nm).

The active-region numbers above are unchanged, because the strip only matters below five times the

floor.

13.7 Ultrathin films (T4)

[filled in after the 0.75 and 0.51 nm DFT films and the scenario runs]

13.8 Data that would decide the open questions

In plain words. Most of the remaining uncertainty cannot be removed by more simulation; it needs a few extra measurements on the existing wafers. Each item below names the question it would settle.

Off-state floor. (Section 13.2)

- **Gate and source currents recorded with every sweep.** If $I_S \approx -I_D$ and $I_G \approx 0$ in the off-state, the floor flows from drain to source. If $I_G \approx -I_D$, it is gate leakage.
- **A channel-length series at one thickness.** A floor that scales as $1/L$ runs through the channel region; a floor independent of L runs through a parasitic path elsewhere.
- **The instrument settings** (current range, integration time, delay), to separate a real current from the offset of a range.
- **The layout:** whether the film is patterned to the channel and whether the TiN gate covers the whole substrate.

Thickness dependence and the 6.3 nm film. (Section 13.6)

- **Several devices per thickness.** The device-to-device spread decides whether a 0.25 V difference is a model failure or ordinary variation.
- **The method and uncertainty of the thickness measurement**, for example X-ray reflectivity.
- **Transfer curves at two or more temperatures.** The 65 and 85 °C curves of the 2 nm device already exist (Januar et al. [1], SI Fig. S12).

The first-principles inputs. (Section 13.3)

- **The optical gap against thickness** (UV–Vis or ellipsometry): the measured $\Delta E_g(t)$.
- **X-ray photoelectron valence-band spectra:** the valence-band shift, i.e. the computed 0.84–0.94 share.
- **Hall measurements on unpatterned films:** carrier density and band mobility, which V2 still fits per film.

Chapter 14

Conclusions

This chapter states what the study has established about the thickness dependence of the three single IWO bottom-gate TFTs (2, 6.3 and 13.2 nm) and about the ATLAS model built to describe them. The results are ordered by strength of evidence, exactly as in `analysis_2026-09-25/SYNTHESIS.md`. Demonstrated conclusions rest on measured curves and converged simulations. Strongly supported interpretations rest on several independent lines of evidence but are not uniquely proven. Plausible hypotheses have not been verified, and rejected explanations are listed together with the quantitative inconsistency that rejects them. Every statement about data concerns one device per thickness ($N = 1$). The chapter ends with the strongest claims that the study can support in a publication, the corrections that must be applied to earlier documents of the package, and an outlook.

Overall status. The model is calibrated, not validated. The out-of-sample test of the shared parameters within the data (6.3 nm threshold, `run_0014`) failed by -0.294 V, and the registered hypothesis tests B0 and C1 for the 6.3 nm film failed or partly failed. The ID-VD and 338/358 K predictions are registered with hashes in two exact mobility variants and await independent data (Chapter 10).

Model V2 (Chapter 13) replaces the literature confinement laws by the first-principles values of this work. Its pre-registered held-out test on the 6.3 nm film fails again: the threshold is too low by 0.23 V with the mobility normalised to the on-current, and by 0.37 V with the pre-registered mobility rule.

14.1 Demonstrated conclusions

1. The measured pattern is $2 \approx 6.3 \text{ nm} \neq 13.2 \text{ nm}$ in every metric (DATA). The values are $V_{\text{th,cc}} = 0.662/0.644/0.083$ V, $\mu_{\text{FE}} = 12.15/10.95/50.17 \text{ cm}^2/\text{V s}$, $SS_{\text{cc}} = 269.9/295.4/160.0 \text{ mV/dec}$, $I_{\text{on}} = 5.09 \times 10^{-7}/4.03 \times 10^{-7}/3.07 \times 10^{-6} \text{ A}/\mu\text{m}$ and off-floor $4.6 \times 10^{-15}/1.5 \times 10^{-13}/6.6 \times 10^{-12} \text{ A}/\mu\text{m}$ (Section 3.4). Since $\mu_{\text{FE}}(6.3) < \mu_{\text{FE}}(2)$, no monotonic law can pass through all three points.
2. The work function, the electron affinity, the fixed charge and the pure confinement shift act as one number per curve (NUM). An increase of the gate WF by +0.1 eV gives a rigid shift of +0.1000 V with unchanged SS (`run_0048`, `run_0049`). Q_f is rigid within 0.6 mV over $10^{-14}..3 \times 10^{-7} \text{ A}/\mu\text{m}$ (`run_0001`→`run_0003`, `run_0002`→`run_0004`, `run_0005`→`run_0007`), and $\Delta E_c(2 \text{ nm}) = 0.353 \text{ eV}$ is equivalent to $1.97 \times 10^{12} \text{ cm}^{-2}$. The complete confinement treatment

- (ΔE_c together with $m^* \rightarrow N_c$) is only nearly rigid: the shift is 0.345 V at 10^{-14} and 0.302 V at 10^{-8} A/ μm , and SS_{cc} is 289.1 against 303.7 mV/dec (run_0012 against run_0016). Its non-rigid part is nearly reproduced by $m^* \times 0.8$ or by $N_t \times 1.09$ (log-linear scaling of run_0050 and run_0023).
3. The thresholds do not discriminate for or against confinement (NUM on DATA). A single shared offset fits with an rms of 0.1363 V with the confinement law and 0.1325 V without it; the leave-one-out values are 0.2045 and 0.1988 V, and the better variant changes with the choice of anchor film. Each reading requires one film-specific charge. With confinement, the 6.3 nm film needs $1.64 \times 10^{12} \text{ cm}^{-2}$ less positive charge (0.294 V); without confinement, the 13.2 nm film needs $1.39 \times 10^{12} \text{ cm}^{-2}$ more (0.248 V).
 4. The shared-parameter model misses the 6.3 nm threshold by -0.294 V (run_0014; OOS, failed). Because $\mu_{\text{band}} = 12.4$ is the V0 fit of the same curve, this is a test of the threshold only.
 5. No tested V1 variant reproduces the 6.3 nm on-state without degrading the subthreshold region (NUM). Eight variants of the charge and electrostatics (run_0014, run_0015, run_0030–run_0035) give a turn-on gap $V_{\text{th,lin}} - V_{\text{th,cc}}$ of 0.72–0.88 V, compared with 1.176 V measured. Variant C1 (run_0052) gives 0.939 V, and variant B0 (run_0037) gives 1.202 V but with SS_{cc} of 410.9 against 295.4 mV/dec. Smaller misses of the same sign exist at 2 nm (gap -0.145 V, $g_m -8.6\%$) and at 13.2 nm (-0.021 V, -2.6%).
 6. The constant-current threshold is confounded with the mobility (DATA/NUM). When the 13.2 nm curve is given the 2 nm mobility, $V_{\text{th,cc}}$ moves by $+0.108$ to $+0.115$ V, which is 19–20% of the measured step of 0.579 V; the confinement term of the model realises 0.292 V (50.4%).
 7. SS_{min} is not a valid metric (NUM). It moves by about 10 mV/dec between two curves with equal I_{on} , so the claim “removing confinement moves SS toward the data” is an artefact (Section 5.10).
 8. The calibrated state at DOS 384/192 is defined by run_0038, run_0039 ($\times 1.015074$) and run_0040 ($\times 1.020113$) (CAL). The measured and simulated values of $V_{\text{th,cc}}$ are 0.662/0.680, 0.644/0.653 and 0.083/0.054 V, and those of $SS_{[10^{-11}, 10^{-10}]}$ are 121.2/123.5, 124.5/124.5 and 91.8/90.9 mV/dec (13.2 nm floor-subtracted; raw value 136.1). The $10^{-10}..10^{-9}$ window is too soft by $+20/+6/+25$ mV/dec, and the fitted μ_{band} values are 17.69/11.58/61.60 $\text{cm}^2/\text{V s}$ (Section 7.3).
 9. The 300 K transfer numerics converge where this was checked (NUM). At 2 nm, refinement of the DOS changes I_{on} by $+1.98\%$ and then by $+0.48\%$ (order 2.03, Richardson residual $+0.16\%$). A mesh refinement by $\times 0.7$ passes at 6.3 nm (run_0036) and at 13.2 nm (run_0028), and I_D is linear in μ to within 6×10^{-6} (Section 6.6).
 10. The elevated-temperature runs used the silicon default $\text{tmu} = 1.5$ (NUM). For this reason the predictions are registered in two exact mobility variants (Section 5.8).
 11. run_0027 is not an overlap test. Its malformed mesh collapsed the drain, and the run is retracted (Section 7.4).
 12. The mobilities published in the paper are obtained by applying the saturation formula to these curves (DATA, provenance). The formula gives 5.09 and 27.40 $\text{cm}^2/\text{V s}$, compared with the

published 5.1 and 27.4, and it predicts $3.86 \text{ cm}^2/\text{Vs}$ at 6.3 nm (Section 5.5).

13. The computed gap opening of 1 and 2 nm In_2O_3 films lies 84–94 % in the conduction band (THEORY, PBE; Section 12.8); V1 had assumed 100 %. The band edges of the 2 nm film are bulk-like, not surface states. With these laws (model V2) the calibrated curves move rigidly by -75 , -24 and -10 mV at 2, 6.3 and 13.2 nm. Re-calibrated with one shared Q_f ($1.383 \times 10^{12} \text{ cm}^{-2}$), V2 fits the anchors with an active RMSE of 0.065 and 0.097 dec (run_0058, run_0059; CAL).
14. The pre-registered held-out test of V2 on the 6.3 nm film fails (run_0060; OOS, failed). $V_{\text{th,cc}}$ is -0.23 V off with the mobility normalised to I_{on} and -0.37 V off with the pre-registered mobility rule; $V_{\text{th,lin}}$ is -0.58 V off in both. The first-principles inputs shrink the threshold miss by about 0.04 V only. ATLAS agreed with the pre-registered (corrected) values within 0.3 mV in $V_{\text{th,cc}}$ (Section 13.6).

14.2 Strongly supported interpretations

The first interpretation (S1) is that, within the calibrated model, the rise of the apparent mobility by a factor of ≈ 4.6 between 6.3 and 13.2 nm is a change in channel transport. With respect to the contacts, the net mobility-degradation coefficient is negative in all films, g_m still rises at 3 V at 2 nm, and a thickness-independent R_{sd} would require $\rho_c \approx 0.26 \Omega \text{ cm}^2$. With respect to electrostatics, campaign A changes g_m by ≤ 2 %. The quantum capacitance changes $C_{\text{eff}}/C_{\text{ox}}$ by ≤ 1 % at 2 nm and by 3–5 % at 6.3/13.2 nm (SURRE). The roughness factor is $\times 1.05$, and the tail partition changes from 0.791 to 0.826 (SURRE). In total, 109 % of $\ln(\times 4.58)$ is located in the fitted μ_{band} . A limitation is that the bounds on R_{sd} belong to the calibration class ($\lesssim 5 \times 10^4 \Omega \mu\text{m}$ at 2 nm and $\lesssim 1\text{--}2 \times 10^4 \Omega \mu\text{m}$ at 13.2 nm). The details are given in Section 9.2.

The second interpretation (S2) is that subthreshold conduction in all films is governed by the filling of an exponential acceptor tail near E_c . One shared W_{TA} (40 meV) reproduces the slope in the $10^{-11}\text{--}10^{-10} \text{ A}/\mu\text{m}$ range within 2.3 mV/dec. ATLAS runs with ideal contacts reproduce the supralinear exponent α_H (1.59/2.14/1.29) and the negative θ through trap filling alone, and a DOS inversion validated on ATLAS curves gives measured/ATLAS ratios of 0.81–1.46 near E_c (conditional on the mobility). The specific DOS form of V1 is not supported, however: the deeper states are underestimated by $\times 2.2\text{--}2.8$, and the $10^{-10}\text{--}10^{-9}$ window is too soft. The details are given in Section 9.4.

The third interpretation (S3) is that the contacts do not create the thickness trend. The required R_{sd} would take up 83 % of V_D . At fixed ρ_c the contact fraction scales as $R_{\text{sh}}^{-1/2}$, so a thickness-independent contact would penalise the thick film, and θ_{net} has the wrong sign in all films (CORR + LIT). The details are given in Section 9.5.

The fourth interpretation (S4) is that the off-floors are caused neither by gate leakage (6.3 and 13.2 nm) nor by thermal generation. The 13.2 nm floor is flat within $\times 1.06$ while V_{gd} changes from 1.2 to 3.7 V, and the floor rises by $\times 1432$ ($\propto t^{3.85}$) on an identical gate stack. Thermal generation falls short by 13–17 decades. The origin of the floors is not determined.

The fifth interpretation (S5) is that the published mobilities are values of the saturation formula taken outside the regime in which the formula applies. The exponent of $N_t(t)$ in V1 descends from an

analysis that was built on these values.

14.3 Plausible hypotheses

The hypotheses that are plausible but not verified are collected in Table 14.1, together with the falsification criteria that were declared in advance.

14.4 Rejected explanations

The rejected explanations and the quantitative reasons for their rejection are listed in Table 14.2. The term “model-conditional” means that the rejection holds within the DOS and electrostatics of V1.

Several explanations are neither supported nor rejected and remain not determined. These are hysteresis, bias stress during the sweep, device-to-device spread, the measurement ambient, and a thickness error beyond the model-conditional bound.

14.5 Publication-ready claims

The following four statements are the strongest claims that the present study supports (SYNTHESIS.md §J). They are formulated so that they can be quoted without change.

1. (*NUM; demonstrated.*) On a single transfer curve per thickness, the confinement-induced conduction-band shift, the gate work function, the channel electron affinity and a fixed interface charge enter as one rigid threshold offset (reproduced to within 1 mV in drift-diffusion simulations). A confinement shift of about 0.3 eV at 2 nm, expected from effective-mass estimates anchored on first-principles slab calculations, is therefore an assumed input, not a result, and the measured thresholds are described equally well with or without it (single-offset rms 0.136 vs 0.133 V).
2. (*DATA + NUM; demonstrated for these devices.*) The 2 and 6.3 nm devices have similar thresholds and field-effect mobilities, whereas the 13.2 nm device shows a 0.56–0.58 V lower threshold and a 4.1–4.6-fold higher apparent mobility. With the calibrated tail-state and donor laws and a thickness-independent fixed charge, a single offset does not reproduce all three thresholds within 0.1 V, with or without the confinement term (largest residuals 0.19 and 0.18 V); at least one further thickness-dependent quantity is required, which the present data do not identify. About 0.11 V of the threshold difference follows from the constant-current threshold definition combined with the mobility difference.
3. (*Strongly supported; model-conditional.*) Within the calibrated trap-limited model, the 4.6-fold increase in apparent mobility between 6.3 and 13.2 nm cannot be attributed to source/drain resistance, interface roughness, electrostatics or tail-state filling; its physical origin, for example a thickness-dependent microstructure, is not determined by the present data.
4. (*NUM on DATA; demonstrated negative result.*) The 6.3 nm device departs from the shared calibration in two separable ways: a rigid threshold offset (0.29 V, equivalent to $1.6 \times 10^{12} \text{ cm}^{-2}$ of fixed charge) and a gate-voltage-dependent on-state shape (turn-on delay 0.31–0.36 V too

Table 14.1. Plausible but unverified hypotheses with their pre-declared falsifiers (SYNTHESIS.md §C).

#	hypothesis	for / against	falsified if
C1	Material transition between 6.3 and 13.2 nm (crystallinity, grain coalescence, percolation) raising band mobility and effective positive charge together; <i>working hypothesis</i>	co-located changes of V_{th} , μ_{FE} , SS_{cc} and floor (CORR); a 39 meV barrier difference suffices; ultrathin films lack long-range crystallinity [1, p. 2]. Against: no structural data at 6.3 or 13.2 nm; $N = 1$	GIXRD/TEM show the same phase and grain size; or Hall mobility at 6.3 nm within 30 % of 13.2 nm; or replicate mean $\mu_{FE}(6.3)$ within 30 % of $\mu_{FE}(13.2)$
C2	Donor or positive-charge step at 13.2 nm ($\approx 2.5 \times 10^{17} \rightarrow 1 \times 10^{18} \text{ cm}^{-3}$)	needed by the no-confinement reading ($+1.39 \times 10^{12} \text{ cm}^{-2}$); degenerate with confinement	C-V $V_{FB}(t)$ flat within 50 mV
C3	Confinement 0.26–0.30 V (0.14–0.38) at 2 nm	THEORY (S3 Schrodinger-Poisson + PBE); this work's DFT gives $\Delta E_c(1.98 \text{ nm}) = 0.281 \text{ eV}$ (PBE; $\times 1.06$ – 1.11 with HSE06, Section 12.8); not identifiable from the curves	optical-gap shift 2 vs 13.2 nm $< 0.1 \text{ eV}$
C4	6.3 nm rigid offset is device-specific (spread, sweep history, stress)	PBS-extrapolated sweep shift 0.003–0.24 V	replicates give $\sigma(V_{th}) < 0.1 \text{ V}$ with mean $\geq 0.2 \text{ V}$ off the shared law
C5	On-state shape needs a carrier-density-dependent (percolation-type) mobility	trade-off line of 3.2–8.5 mV gap per mV/dec of SS_{cc} ; untested	density-dependent mobility calibrated on 2 and 13.2 nm leaves the 6.3 nm gap miss $> 0.1 \text{ V}$ at $SS_{cc} \leq 330 \text{ mV/dec}$
C6	Surface + bulk trap DOS ($D_s \approx 8.5 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1} + g_b \approx 1 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$)	ATLAS-validated inversion; energy axis shifts by $k_B T \ln(\text{mobility ratio})$	multi-frequency C-V per thickness
C7	Near- E_c states as partial contributor	run_0052 recovers 33 % of the gap at $SS_{cc} + 15.6$	C1 at 768 levels loses the gain
C8	Floors are an ungated IWO path $\propto t^{3.85}$	with the model's own donor and trap values, a strip carrying the V1 interface acceptors on one surface scales as $t^{4.24}$ and, with one factor fitted on 13.2 nm, gives $\times 0.46$ and $\times 0.64$ of the 2 and 6.3 nm floors (Section 13.2); the 2 nm value is sensitive to two weakly known inputs; location not determined	$I_G \approx I_D$, or no $1/L$ scaling
C9	Q_f is an interfacial ionized oxygen-vacancy layer	electrostatically possible; Q_f absorbs $\pm 0.2 \text{ eV}$ alignment	no C-V dispersion, no NBS response
C10	MTR temperature behaviour	registered P-MTR variant; untested	$\Delta I_{on}(2 \text{ nm}, 358 \text{ K}) > +5 \%$ or $< -15 \%$

Table 14.2. Rejected explanations and the quantitative inconsistency (SYNTHESIS.md §D).

explanation	quantitative reason	class
6.3 nm thickness error (A1, run_0031)	−1 nm gives +0.036 of the required +0.294 V (12 %); closing via ΔE_c needs $t \approx 2.1$ nm	NUM, model-cond.
Fewer donors (A2, run_0032)	+0.030 V, $\times 10$ short	NUM
Front D_{it} (A4, run_0033)	$\times 5$ gives +0.051 V, gap unchanged; needs $\times 23$ –24 at 6.3 nm only	NUM
Back-surface charge alone (A3, run_0030)	matches $V_{th,cc}$ but SS_{cc} off by −64.1 mV/dec; offset at 10^{-7} worsens to +0.285 V	NUM, model-cond.
Combination A5 (run_0035)	0.215 of 0.294 V with SS_{cc} −45.9 mV/dec	NUM
S2 re-partition (B0, run_0037)	SS_{cc} 410.9 vs 295.4; $SS_{[10^{-11}, 10^{-10}]}$ +48.8 mV/dec	OOS, failed
Near- E_c band as full explanation (C1, run_0052)	gap +0.119 vs +0.28 V predicted; I_{on} +16.0 %	OOS, partly failed
Any rigid charge as whole	offset varies −0.007 to +0.141 V between 10^{-9} and 2×10^{-7} A/ μ m (run_0015)	NUM on DATA
6.3 nm miss		
Thickness-independent R_{sd} as the trend	needs $R_{sd}W \approx 1.15 \times 10^6 \Omega \mu$ m (83 % of V_D); θ_{net} wrong sign	CORR + LIT
Confinement-raised Pd/TWO barrier as the step	needs $\Delta\phi \geq 0.20$ eV; $\Delta E_c(6.3) = 0.07$ eV	CORR
Roughness factor as mobility step	$\times 1.051$ vs $\times 4.58$	CAL
Thickness-fluctuation scattering ($\propto t^6$)	implies $\mu(2) = 0.017$ cm ² /V s ($\times 700$ off)	SURR
Back-surface Coulomb scattering	$\times 2700$ short at 6.3 nm	SURR
Tail partition as mobility step	P changes +4 %; needs $N_t \times 6.7 \times 10^3$	SURR
Uniform donors as 6.3→13.2 V_{th} step	steepest slope ratio 2.35 vs 9–20 measured	SURR + DATA
Gate leakage as floor (6.3, 13.2 nm)	$\leq \times 1.06$ change while V_{gd} 1.2→3.7 V	DATA
Thermal generation as floor	13–17 decades short	calculation
Power laws in t for μ_{FE} , I_{on} , V_{th}	non-monotonic data; LOO miss $\times 2.63$ (μ_{FE}), $\times 3.76$ (I_{on})	DATA
$\Delta E_c \propto t^{-1.38}$ beyond ≈ 3 nm as physics	exceeds the infinite-well bound above 2.98 nm ($\times 1.59$ / $\times 2.51$ at 6.3 / 13.2 nm)	THEORY
“Removing confinement improves SS”	window artefact (Section 5.10); fixed-current SS within 1–2 mV/dec	NUM
“Thresholds require/refute confinement”	rms 0.136 vs 0.133 V	NUM on DATA
“Contact geometry irrelevant”	never tested (run_0027 malformed)	audit

small, transconductance 17–23 % too low). No tested change of fixed charge, donor density, interface, tail-state or near-band-edge acceptor density, film thickness or back-surface charge reproduces the shape without degrading the subthreshold slope. With one device per thickness, the offset cannot be distinguished from device-to-device variation, and the shape remains unexplained; a limitation of the constant-mobility model is the leading untested candidate.

14.6 Corrections to earlier package documents

None of the earlier documents has been edited. The corrections in Table 14.3 must be recorded separately (SYNTHESIS.md §K), and they are applied throughout this report.

14.7 Outlook

The study has turned an apparently complete thickness-resolved TCAD fit into a map of what three single transfer curves can and cannot identify. The next scientific steps follow the order of dependencies given in Section 11.3. First, the registered predictions must receive an external timestamp and must then be compared, with the unchanged extractor and without refitting, against the existing 2 nm temperature data in the supplement of the target paper. This comparison is the first genuine validation test, and it decides between the P-MTR and P-phonon transport variants. Second, replicate devices (3–5 per thickness, measured with dual sweeps) are needed before any film-specific statement about the 6.3 nm device can be made. Third, thickness metrology, the C-V flat-band voltage as a function of thickness, TLM or L-series contact measurements and a temperature series on all three films would remove the principal degeneracies, namely those between work function, affinity, fixed charge and confinement, between band mobility and tail density, and between R_{sd} and the tail exponent. Structural characterisation (GIXRD, TEM, grain size) at 6.3 and 13.2 nm is the direct test of the working hypothesis C1. A carrier-density-dependent mobility model, first-principles calculations for W-doped In_2O_3 and further thicknesses belong to a second study (Section 11.4).

Summary. The study demonstrates that the measured devices follow the pattern $2 \approx 6.3 \neq 13.2$ nm, that confinement, WF, χ and Q_f act as one rigid number per curve, and that the thresholds are neutral with respect to confinement (rms 0.136 vs 0.133 V). The 6.3 nm miss decomposes into a rigid part of 0.294 V and an on-state shape part that no tested V1 change reproduces and that remains unexplained. It is strongly supported, although conditional on the model, that the rise of the mobility by a factor of ≈ 4.6 is a change in transport within the calibrated model that is not caused by contacts, roughness, electrostatics or tail filling, and that subthreshold conduction is governed by tail filling. A material transition between 6.3 and 13.2 nm remains the working hypothesis; it is plausible but untested. The model is calibrated and non-unique but not validated, and the registered predictions (run_0041–run_0047) await independent data. Replacing the literature confinement laws by first-principles values of this work (model V2) changes the calibrated curves by at most 75 mV and does not repair the held-out 6.3 nm prediction, which fails its pre-registered test again (–0.23 V).

Table 14.3. Principal corrections to earlier package documents.

document	superseded statement	corrected wording
FINAL_STATUS	“15 real ATLAS launches”	52 recorded launches (run_0001–run_0052), of which 49 produced scored currents; run_0010 aborted, run_0011 stopped, run_0018 timed out; run_0027 malformed and retracted
FINAL_STATUS	“The final set is four runs”	final calibrated set run_0038–run_0040 (DOS 384/192; μ_{band} 17.69/11.58/61.60); run_0014 is the shared-parameter threshold test; run_0012–run_0015 superseded
FINAL_STATUS	SS_{min} column; row 0014	fixed-current SS, SS_{cc} , gap, g_{m} ; “shared-parameter out-of-sample threshold test: failed, -0.294 V”
FINAL_STATUS	“VALIDATION”	
FINAL_STATUS	“ONE fitted electrostatic value”	fitted: shared Q_{f} , 6.3 nm Q_{f} , $\mu_{\text{band}} \times 3$, N_{t} anchor and W_{TA} , N_{d} -law constants; assumed: WF, χ , D_{it} , deep Gaussian, $\varepsilon(\text{Al}_2\text{O}_3)$
FINAL_STATUS	6.3 nm “reproduced in SHAPE”	fixed-current SS reproduced (-2.2 mV/dec); on-state not (gap -0.314 V, g_{m} -16.6 %)
FINAL_STATUS	Conclusion 1: confinement “accounts for” the 0.58 V	thresholds do not discriminate (rms 0.136 vs 0.133 V); realised term 0.29 V (50 %); confinement is an assumed input
FINAL_STATUS	“publication-ready as a calibrated parameter extraction”	a calibrated, explicitly non-unique description of three single devices; not validated; submission requires Tiers 0–3 (Section 11.3)
SENSITIVITY_RESULTS	15 overlap 4um (run_0027)	RETRACTED: malformed deck
SENSITIVITY_RESULTS	“MEASURED”	
SENSITIVITY_RESULTS	WF, m^* , 6.3 nm “NOT RUN”	superseded by run_0048–run_0051 and run_0030–run_0036
NUMERICAL_CONV.	V0 checks apply to V1 decks	carried over by lineage; not re-demonstrated on the V1 2 nm deck
NUMERICAL_CONV.	DOS C “FAIL”	resolved by recalibration at 384/192; order established only at 2 nm
LIMITATIONS	#5 “ $\approx 12 \rightarrow \approx 57$ ”	$11.58 \rightarrow 61.60$ $\text{cm}^2/\text{V s}$ (DOS 384/192)
LIMITATIONS	#11 overlap insensitive	contact geometry and resistance untested
SUPERVISOR_SUMMARY	“14 launches”; “confinement story holds ... it was tested”	52 recorded launches, of which 49 produced scored currents; FINAL_STATUS Conclusion 1 wording
EVIDENCE_BRIEF	13.2 nm no-confinement	$1.434 \times 10^{12} \text{ cm}^{-2}$
	$Q_{\text{f}} \approx 1.86 \times 10^{12}$	
S2	re-partition “strongly supported”; “linear regime”	rejected (B0); regime claim withdrawn; 31.8 nm “ μ_{FE} ” is not a mobility
S3 / S1 / S6	“mildly disfavour”; A6 “favours no-QC”; P2, A6 pre-registered	“do not discriminate”; “neutral”; POST
S4	6.3 nm floor ± 3 %	± 45 %
PREDICTIONS_REGISTER	15 multipliers 1.19665/1.30426	exact 1.19669/1.30441 (tables already exact); addendum only

Appendix A

Catalogue of all simulation runs

This appendix lists every launch of the V1 package, including those that failed or were retracted, so that each simulated number of the report can be traced to its run folder and so that no run can silently disappear from the record. The catalogue (Table A.2) is generated from `results/RUN_INDEX.csv` and the `execution.json` of each run by `tools/make_appendix_assets.py`; no value in it is typed by hand. The run identifier `run_NNNN` names the folder `results/runs/run_NNNN_<film>_<label>`. The overrides column states how the run differs from the reference configuration of `config/iwo_material_model.yaml`, `config/geometry.yaml` and `config/solver.yaml` (dotted paths as accepted by `scripts/build_iwo_decks.py`) metric definitions are those of Section 5.10, and the numerical settings of Chapter 6.

A.1 Composition of the 52 launches

Table A.1. Runs by purpose (categories of `results/RUN_INDEX.csv`).

Category	Runs	Identifiers
Baseline thickness laws	2	<code>run_0001</code> (2 nm), <code>run_0002</code> (13.2 nm)
Calibration stages	7	<code>run_0003</code> , <code>run_0004</code> , <code>run_0006</code> , <code>run_0007</code> , <code>run_0008</code> , <code>run_0009</code> , <code>run_0011</code>
Held-out validation of 6.3 nm	2	<code>run_0005</code> , <code>run_0014</code>
Final calibrated, DOS 96/48	2	<code>run_0012</code> (2 nm), <code>run_0013</code> (13.2 nm)
Device-specific tuning of 6.3 nm	1	<code>run_0015</code>
Counterfactual without confinement	2	<code>run_0016</code> (2 nm), <code>run_0017</code> (13.2 nm)
One-at-a-time sensitivity (2 nm)	8	<code>run_0020</code> – <code>run_0027</code>
Numerical checks	5	<code>run_0018</code> , <code>run_0019</code> , <code>run_0028</code> , <code>run_0029</code> , <code>run_0036</code>
6.3 nm hypotheses	9	<code>run_0010</code> , <code>run_0030</code> – <code>run_0035</code> , <code>run_0037</code> , <code>run_0052</code>
DOS 384/192 recalibration	3	<code>run_0038</code> , <code>run_0039</code> , <code>run_0040</code>
Predictions, I_D – V_D	3	<code>run_0041</code> – <code>run_0043</code>
Predictions, elevated temperature	4	<code>run_0044</code> – <code>run_0047</code>
Isolation tests (gate WF, m^*)	4	<code>run_0048</code> – <code>run_0051</code>
Total	52	

The record holds 52 recorded launches, of which 49 produced scored currents: 46 carry the status `REAL_ATLAS_SCORED` and three `REAL_ATLAS_PREDICTION` (I_D – V_D families without a measurement to score against), while three produced no usable curve (`run_0010`, `run_0011`, `run_0018`); one scored run (`run_0027`) is retracted. The runs `run_0001`–`run_0009` use the reduced structure and

the pointwise sweep; every run from `run_0012` on uses the physical structure and the stepped sweep (Section 6.4). DOS 384/192 is used by the check `run_0029` and by all runs from `run_0038` to `run_0052`; all other runs, including the pre-registered B0 test `run_0037`, use 96/48.

Table A.2. Catalogue of all V1 ATLAS launches and the measured references. $SS_{[-11, -10]}$: fixed-current slope between 10^{-11} and 10^{-10} A/ μ m (mV/dec); RMSE: active-region log10 error (dec). Metrics of elevated-temperature runs are compared with the 300 K data only for reference.

run	t (nm)	label	category	$V_{th,cc}$ (V)	$V_{th,lin}$ (V)	$SS_{[-11, -10]}$	SS_{cc}	μ_{FE}	I_{on} (A/ μ m)	RMSE
<code>run_0001</code>	2	<code>v1_baseline_laws</code>	baseline laws	0.995	1.838	125.3	–	10.19	3.71e-07	3.928
<code>run_0002</code>	13.2	<code>v1_baseline_laws</code>	baseline laws	0.369	1.285	92.4	–	44.97	2.42e-06	1.030
<code>run_0003</code>	2	<code>v1_stage3_qf1p73e12</code>	calibration stage	0.686	1.561	125.0	–	10.46	4.72e-07	0.056
<code>run_0004</code>	13.2	<code>v1_stage3_qf1p73e12</code>	calibration stage	0.059	1.001	92.0	195.4	45.60	2.86e-06	0.086
<code>run_0005</code>	6.3	<code>v1_stage5_shared_qf_prediction</code>	validation (held-out 6.3 nm)	0.350	1.215	122.3	284.9	9.14	5.11e-07	0.678
<code>run_0006</code>	31.8	<code>v1_stage5_shared_qf</code>	calibration stage	-2.710	-0.168	84.9	210.2	39.90	3.38e-06	1.050
<code>run_0007</code>	6.3	<code>v1_stage5b_device_qf8p7e10</code>	calibration stage	0.644	1.474	122.8	284.7	8.95	4.28e-07	0.082
<code>run_0008</code>	2	<code>v1_stage6_mu18p13</code>	calibration stage	0.676	1.561	122.5	289.1	11.29	5.09e-07	0.037
<code>run_0009</code>	13.2	<code>v1_stage6_mu61p9</code>	calibration stage	0.053	1.001	90.8	192.5	49.00	3.07e-06	0.081
<code>run_0010</code>	31.8	<code>v1_stage6_backsheet_hyp_A</code>	6.3 nm hypothesis	–	–	–	–	–	–	–
<i>aborted in SOLVE INIT, no currents</i>										
<code>run_0011</code>	6.3	<code>v1_stage6_mu11p69</code>	calibration stage	–	–	–	–	–	–	–
<i>stopped with the chain; no currents</i>										
<code>run_0012</code>	2	<code>v2_full_stepped_final</code>	final calibrated (DOS 96/48)	0.676	1.559	122.5	289.1	11.27	5.09e-07	0.037
<code>run_0013</code>	13.2	<code>v2_full_stepped_final</code>	final calibrated (DOS 96/48)	0.053	0.999	90.8	192.5	48.95	3.07e-06	0.081
<code>run_0014</code>	6.3	<code>v2_full_stepped_validation_shared</code>	validation (held-out 6.3 nm)	0.350	1.212	122.3	284.9	9.13	5.11e-07	0.678
<code>run_0015</code>	6.3	<code>v2_full_stepped_tuned_qf8p7e10</code>	device-specific tuning	0.651	1.471	124.1	288.8	8.42	4.03e-07	0.063
<code>run_0016</code>	2	<code>sens_no_confinement</code>	counterfactual (no confinement)	0.361	1.260	132.1	303.7	11.33	6.18e-07	0.846
<code>run_0017</code>	13.2	<code>sens_no_confinement</code>	counterfactual (no confinement)	0.030	0.979	92.0	193.3	48.92	3.1e-06	0.114
<code>run_0018</code>	13.2	<code>check_A_mesh0p7</code>	numerical check	–	–	–	–	–	–	–
<i>timed out at 900 s; rerun as run_0028</i>										
<code>run_0019</code>	2	<code>check_C_dos192_96</code>	numerical check	0.677	1.541	122.7	289.3	11.35	5.19e-07	0.040
<code>run_0020</code>	2	<code>sens_chi_m0p1</code>	one-at-a-time sensitivity	0.776	1.649	122.5	289.1	11.19	4.74e-07	0.463
<code>run_0021</code>	2	<code>sens_chi_p0p1</code>	one-at-a-time sensitivity	0.576	1.467	122.5	289.1	11.34	5.45e-07	0.361
<code>run_0022</code>	2	<code>sens_nd_x2</code>	one-at-a-time sensitivity	0.667	1.550	121.5	289.8	11.28	5.13e-07	0.054
<code>run_0023</code>	2	<code>sens_nt_x1p5</code>	one-at-a-time sensitivity	0.758	1.841	144.4	357.8	10.50	3.82e-07	0.263
<code>run_0024</code>	2	<code>sens_wta_p5meV</code>	one-at-a-time sensitivity	0.696	1.557	130.0	290.0	11.30	5.11e-07	0.063
<code>run_0025</code>	2	<code>sens_dit_x3</code>	one-at-a-time sensitivity	0.701	1.582	123.9	289.7	11.25	5e-07	0.098
<code>run_0026</code>	2	<code>sens_eps_10p55</code>	one-at-a-time sensitivity	0.675	1.558	122.1	289.3	11.34	5.13e-07	0.037
<code>run_0027</code>	2	<code>sens_overlap_4um</code>	one-at-a-time sensitivity	0.675	1.564	122.2	290.0	11.07	4.98e-07	0.038
<i>MALFORMED: x.mesh ends at 24 um, drain collapsed; overlap result retracted (S5)</i>										
<code>run_0028</code>	13.2	<code>check_A_mesh0p7_t1500</code>	numerical check	0.053	0.999	90.7	192.5	48.94	3.07e-06	0.082
<code>run_0029</code>	2	<code>check_C_dos384_192</code>	numerical check	0.677	1.537	122.7	289.3	11.37	5.22e-07	0.041
<code>run_0030</code>	6.3	<code>hyp_H3_backQ_m1p64e12</code>	6.3 nm hypothesis	0.627	1.352	102.7	231.3	8.93	4.61e-07	0.237
<code>run_0031</code>	6.3	<code>hyp_H1_tphys_5p3</code>	6.3 nm hypothesis	0.386	1.265	124.5	290.5	8.96	4.87e-07	0.613

run	<i>t</i> (nm)	label	category	$V_{th,cc}$	$V_{th,lin}$	$SS_{[-11,-10]}$	SS_{cc}	μ_{FE}	I_{on}	RMSE
run_0032	6.3	hyp_H2_Nd0_1e16	6.3 nm hypothesis	0.380	1.227	122.5	280.7	9.03	5.02e-07	0.634
run_0033	6.3	hyp_H4_front_Dit_x5	6.3 nm hypothesis	0.401	1.258	127.3	286.3	9.09	4.97e-07	0.595
run_0034	6.3	hyp_H6_noQC_Qf_from_2nm	6.3 nm hypothesis	0.587	1.426	125.1	287.7	8.93	4.4e-07	0.211
run_0035	6.3	hyp_H5_combo_t5p8_Nd1e16	back-sheet hypothesis	0.565	1.336	109.6	249.5	8.81	4.59e-07	0.246
run_0036	6.3	check_L_mesh0p7_tuned	numerical check	0.651	1.471	124.2	288.9	8.42	4.03e-07	0.062
run_0037	6.3	hyp_S2_repartition_Nt2p66e19	6.3 nm hypothesis	0.654	1.856	173.3	410.9	10.40	3.73e-07	0.270
run_0038	2	recal_dos384_2p0	DOS 384/192 recalibration	0.680	1.537	123.5	290.9	11.10	5.09e-07	0.043
run_0039	6.3	recal_dos384_6p3_tuned	DOS 384/192 recalibration	0.655	1.466	124.8	290.7	8.27	3.97e-07	0.059
run_0040	13.2	recal_dos384_13p2	DOS 384/192 recalibration	0.055	0.995	91.2	193.8	47.89	3.01e-06	0.081
run_0041	2	pred_idvd_2p0	prediction ID-VD	–	–	–	–	–	–	–
<i>ID-VD prediction; output_characteristics.csv</i>										
run_0042	6.3	pred_idvd_6p3_tuned	prediction ID-VD	–	–	–	–	–	–	–
<i>ID-VD prediction; output_characteristics.csv</i>										
run_0043	13.2	pred_idvd_13p2	prediction ID-VD	–	–	–	–	–	–	–
<i>ID-VD prediction; output_characteristics.csv</i>										
run_0044	2	pred_T338K_2p0	prediction temperature	0.653	1.519	123.5	300.2	9.22	4.28e-07	0.131
<i>elevated T; ATLAS tmu=1.5 applied (P-phonon as simulated)</i>										
run_0045	2	pred_T358K_2p0	prediction temperature	0.640	1.510	125.0	304.5	8.44	3.94e-07	0.186
<i>elevated T; ATLAS tmu=1.5 applied (P-phonon as simulated)</i>										
run_0046	6.3	pred_T358K_6p3	prediction temperature	0.606	1.432	123.7	307.3	6.43	3.16e-07	0.133
<i>elevated T; ATLAS tmu=1.5 applied (P-phonon as simulated)</i>										
run_0047	13.2	pred_T358K_13p2	prediction temperature	-0.003	0.964	93.0	195.2	37.46	2.39e-06	0.182
<i>elevated T; ATLAS tmu=1.5 applied (P-phonon as simulated)</i>										
run_0048	2	sens_wf_p0p1_2p0_dos384	isolation (WF)	0.780	1.628	123.5	290.9	11.03	4.74e-07	0.470
run_0049	13.2	sens_wf_p0p1_13p2_dos384	isolation (WF)	0.155	1.087	91.2	193.8	47.69	2.86e-06	0.221
run_0050	2	sens_mstar_x1p3_2p0_dos384	isolation (m*)	0.637	1.494	113.3	273.2	11.23	5.3e-07	0.087
run_0051	13.2	sens_mstar_x1p3_13p2_dos384	isolation (m*)	0.027	0.934	86.5	180.5	49.17	3.18e-06	0.122
run_0052	6.3	hyp_S4_nearEc_band_2e12_m6.85p4	6.3 nm hypothesis	0.648	1.587	125.9	304.4	10.56	4.68e-07	0.081
run_0053	2	check_overlap_4um_meshfixed_dos384	numerical check	–	–	–	–	–	–	–
run_0054	2	check_T358K_tmun0_2p0	numerical check	0.607	1.510	117.3	285.3	11.01	5.14e-07	0.221
<i>elevated T; ATLAS tmu=1.5 applied (P-phonon as simulated)</i>										
run_0055	2	check_overlap_4um_meshfixed_dos384	numerical check	0.680	1.537	123.5	290.9	11.10	5.09e-07	0.043
run_0056	6.3	hyp_H1_tphys_5p3	6.3 nm hypothesis	–	–	–	–	–	–	–
run_0057	6.3	hyp_H2_Nd0_1e16	6.3 nm hypothesis	–	–	–	–	–	–	–
run_0058	2	v2_anchor_2p0	V2 calibration anchor	0.668	1.525	122.6	292.1	11.10	5.13e-07	0.066
run_0059	13.2	v2_anchor_13p2	V2 calibration anchor	0.103	1.042	90.5	191.7	50.07	3.07e-06	0.097
run_0060	6.3	v2_T3_prediction_6p3	validation (held-out 6.3 nm)	0.270	1.240	97.8	220.1	28.35	1.56e-06	1.039
MEASURED_6.3nm	6.3	measured	data	–	–	124.5	–	–	–	–
MEASURED_13.2nm	13.2	measured	data	–	–	136.1	–	–	–	–
MEASURED_31.8nm	31.8	measured	data	–	–	–	–	–	–	–
MEASURED_2nm	2	measured	data	–	–	121.2	–	–	–	–

A.2 Notes on non-standard runs

run_0010 (31.8 nm, back-sheet hypothesis A; status NO_EXPORT). The deck combined a back-surface donor sheet of peak $3.23 \times 10^{14} \text{ cm}^{-2} \text{ eV}^{-1}$ with $Q_f = 1.22 \times 10^{13} \text{ cm}^{-2}$ to describe the

always-on 31.8 nm device. ATLAS aborted in SOLVE INIT after 17.9 s; no current was written. The run is kept in the catalogue as a launch but contributes no number. The 31.8 nm film was subsequently excluded from the final set by the user.

run_0011 (6.3 nm, **stage 6**, $\mu_{\text{band}} = 11.69$; **status NO_EXECUTION_JSON**). The run was part of a chain that was stopped before it executed; it has no `execution.json` and no currents. The value $\mu_{\text{band}} = 11.69 \text{ cm}^2/(\text{V s})$ for 6.3 nm was later used, together with $Q_f = 8.7 \times 10^{10} \text{ cm}^{-2}$, in the physical-structure run **run_0015**.

run_0018 (13.2 nm, **mesh** $\times 0.7$; **status TIMEOUT**). The refined mesh (24 255 nodes) exceeded the default timeout of 900 s. The identical configuration was repeated with a timeout of 1500 s as **run_0028** (1068 s), which supplies check A' of Table 6.2.

run_0027 (2 nm, **contact overlap** 4 μm ; **MALFORMED, retracted**). The run is kept in the catalogue, but its malformed mesh collapsed the drain, so its scored I_{on} change is not an overlap effect and the overlap sensitivity is *not determined* (full statement in Section 7.4).

Re-scored run run_0012. The final 2 nm run at DOS 96/48 was re-scored without a new launch after the KCL acceptance rule was redefined on the median off-band current (check G' of Table 6.2); the original `execution.json` is kept next to the re-scored one.

Rescaled curves run_0039 and run_0040. The 6.3 nm and 13.2 nm recalibration runs were launched at $\mu_{\text{band}} = 11.41$ and $60.39 \text{ cm}^2/(\text{V s})$. The exact recalibrated values, 11.58 and 61.60, follow from the linearity of I_D in MUN; wherever the report shows the final curves, the run's currents are multiplied by $\times 1.015074$ and $\times 1.020113$, respectively, and the factor is stated. The catalogue lists the unscaled runs.

Temperature runs run_0044–run_0047. These runs were planned with a temperature-independent band mobility, but ATLAS applied its default constant-mobility exponent $\text{TMUN} = 1.5$ (Section 6.3). Their catalogue metrics are the as-simulated values (variant P-phonon); the P-MTR variant is obtained by multiplying the currents by $(T/300)^{1.5}$ (Chapter 10).

A.3 History: the V0 and Astra packages

V0 (IWO_ATLAS_Model_claude, 34 launches). The first package of this project fitted each thickness separately with the same generator lineage, mesh, DOS discretization, interface regularization and solver controls as V1. Its band mobilities were 16.8, 12.4, 57.6 and $84 \text{ cm}^2/(\text{V s})$ for 2.0, 6.3, 13.2 and 31.8 nm; the 31.8 nm film required an additional back-surface donor sheet and a series resistance of about $9 \times 10^4 \Omega \mu\text{m}$. V0 contributes no calibrated curve to this report, but its numerical checks (mesh $\times 0.5$ at 2 nm, mesh $\times 0.7$ at 13.2 nm, DOS 192/96, interface layer, tolerance, width convention, reproducibility; documented in its `docs/VALIDATION_CHECKS.md`) apply to the V1 decks and are included in Table 6.2, always written as “V0 run_NNNN”. V0 also established two solver facts used throughout: MOBILITY updates between SOLVE statements are ignored by ATLAS 5.28.1.R

(V0 run_0008), and the strict continuity tolerance with XANDRNORM is needed in depletion (V0 run_0006).

Astra (astra_IWO_PRIORITY_THREE_20260912, 70 solver invocations). An independent earlier package kept Conductor and Palladium volumes for the gate and the contacts and obtained the same class of fits. Its audit documents (audit/docs/SOURCES_AND_CORRECTIONS.md and the rebuild_20260912 notes) are used for three cross-checks: electrons-only versus two-carrier solutions at 2 nm (identical I_D ; check I), repeated identical runs (check H), and the thermal-generation bound on the off-current, 13 to 17 decades below the measured floors (Section 3.3). The permittivities of HfO_2 (19.57) and IWO (9.30) from the SI of Januar et al. [1] were also recorded by the Astra audit. No Astra current is used as a calibrated or predicted curve in this report.

Summary. The V1 record contains 52 recorded launches, of which 49 produced scored currents (46 scored transfer curves and 3 I_D – V_D predictions), and 3 launches without a usable curve (run_0010 aborted, run_0011 never executed, run_0018 timed out and repeated as run_0028). One scored run, run_0027, is malformed and retracted, so the contact-overlap sensitivity is not determined. Every simulated number in the report refers to one of the catalogued runs; V0 (34 launches) and Astra (70 invocations) are used only as documented cross-checks.

Appendix B

Source code

This appendix reproduces every Python script of the package that generates decks, launches the solver, extracts metrics or produces analysis numbers quoted in the report, so that each number can be re-derived from code rather than from prose. The listings are copies of the package files, made ASCII-safe for typesetting by `tools/make_appendix_assets.py`; the package files themselves (`PKG/scripts/` and `PKG/analysis_2026-09-25/scratch/`) are authoritative. The commands used to run them are collected in Chapter E.

B.1 Organisation of the code

The code falls into three groups. The first group consists of the package scripts in `scripts/` (Section B.2): the material laws, the deck generator, the runner and campaign driver that launch the real solver, the common metric extractor, and the table and plot writers. Only `run_atlas.py`, `run_campaign.py` and `sensitivity_analysis.py` launch ATLAS; all other scripts read existing files. The second group consists of the analysis scripts of 2026-09-25 in `analysis_2026-09-25/scratch/(<S1..S6, REVIEW, SYNTHESIS>)` (Section B.3), which contain the specialist analyses. None of them launches ATLAS or modifies a package file; they read run outputs and the measured data, and several import `scripts/extract_metrics.py` unchanged, so that the same extraction is used everywhere. In the listings below the file name is prefixed with the analysis folder (e.g. `S1_s1_analytic.py` is `scratch/S1/s1_analytic.py`). The third group consists of the report tools in `REPORT/tools/` (Section B.4), which build the figures, the digest and the appendix assets of this report; they are described but not listed.

Results of surrogate models (1-D Poisson, Schrödinger-Poisson, charge-sheet models) are always labelled SURR in the report. They are used for decomposition and hypothesis generation and never as a substitute for an ATLAS curve.

B.2 Package scripts

B.2.1 `material_laws.py`

The script implements `IWO_material(t, xW)`, the thickness-aware material laws of about 2 at. % W:In₂O₃ used by V1. It fits the thickness laws to the literature and DFT evidence (confinement shift,

effective mass, tail density, background donors, roughness factor) and evaluates them at a requested thickness; no simulator is run. The deck generator imports it, so that every deck is produced from the same laws (Chapters 4 and 5).

```
1  #!/usr/bin/env python3
2  """IWO_material(t, xW): thickness-aware material laws for ~2 at.% W:In2O3 (V1).
3
4  No simulator is run. This script (1) fits the thickness laws to the literature/DFT evidence
5  listed in tables/THICKNESS_LAW_EVIDENCE.csv, (2) evaluates them at the four measured
6  thicknesses, (3) writes config/iwo_material_model.yaml (consumed by build_iwo_decks.py),
7  tables/THICKNESS_LAWS.csv and the global thickness plots under plots/.
8
9  Evidence labels (docs/PARAMETER_PROVENANCE.md): MEASURED_TARGET_DEVICE, MEASURED_RELATED_MATERIAL,
10 DFT_TARGET_IWO, DFT_PURE_IN2O3_PROXY, LITERATURE_IWO, LITERATURE_IN2O3_PROXY, CALCULATED, FITTED,
11 NUMERICAL, ASSUMED.
12
13 APPROACH A (thickness-dependent effective material + classical drift-diffusion): the DFT band-edge
14 confinement enters ONLY through Eg(t), chi(t), m*(t) -> Nc(t); no BQP/Schrodinger correction is
15 applied on top (no double counting).
16 """
17 import csv, json, math
18 from pathlib import Path
19 import numpy as np
20 try:
21     import yaml
22 except ImportError: # pragma: no cover
23     yaml = None
24
25 ROOT = Path(__file__).resolve().parents[1]
26 KB = 8.617333262e-5; T = 300.0; kT = KB * T
27 H = 6.62607015e-34; ME = 9.1093837015e-31; KBJ = 1.380649e-23
28
29 def nc_3d(mstar_rel, temp=T):
30     """Effective conduction-band DOS, 3-D parabolic band, cm-3."""
31     return 2.0 * (2.0 * math.pi * mstar_rel * ME * KBJ * temp / H**2) ** 1.5 / 1e6
32
33 # ----- evidence table
34 EVIDENCE = [
35     # quantity, value, unit, thickness_nm, material, label, source, location
36     ('Eg_PBE', 0.94, 'eV', 'bulk', 'In2O3', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'Methods, p.21542'),
37     ('Eg_PBE', 1.27, 'eV', 1.98, 'In2O3 slab (H-passivated, vacuum)', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'Methods, p.21542'),
38     ('Eg_PBE', 1.88, 'eV', 0.95, 'In2O3 slab', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'Methods, p.21542'),
39     ('dEc_PBE', 0.60, 'eV', 1.5, 'In2O3 on Al2O3 slab; Ev ~unchanged', 'DFT_PURE_IN2O3_PROXY', 'Si2021_NanoLett', 'p.504, Fig.4b text'),
40     ('mstar', 0.17, 'm0', 'bulk', 'In2O3 (PBE)', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'p.21540'),
41     ('mstar', 0.19, 'm0', 3.52, 'In2O3 slab (PBE)', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'p.21540'),
42     ('mstar', 0.23, 'm0', 1.98, 'In2O3 slab (PBE)', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'p.21540'),
43     ('mstar', 0.30, 'm0', 0.95, 'In2O3 slab (PBE)', 'DFT_PURE_IN2O3_PROXY', 'Lin2022_ACSNano', 'p.21540'),
44     ('mstar_exp', 0.208, 'm0', 'bulk', 'In2O3 single crystal, n=2.8e17 (optical Hall)', 'MEASURED_RELATED_MATERIAL', 'Stokey2021_JAP', 'Abstract; Sec. IV.C'),
45     ('mstar_exp_0', 0.18, 'm0', 'bulk', 'In2O3 zero-density extrapolation (Feneberg, cited)', 'LITERATURE_IN2O3_PROXY', 'Stokey2021_JAP', 'Sec. I, Eq.(3)'),
46     ('eps_DC', 10.55, '-', 'bulk', 'In2O3 single crystal (LST/THz)', 'MEASURED_RELATED_MATERIAL', 'Stokey2021_JAP', 'Abstract; Table II'),
```

```

47 ('eps_DC', 8.9, '-', 'film', 'evaporated In203 films (Hamberg, cited)', 'LITERATURE_IN203_PROXY', '
    Si2021_NanoLett', 'p.501'),
48 ('eps_r', 9.30, '-', 'film', 'IWO of the target process (C-V, SI Fig.S1)', 'MEASURED_TARGET_DEVICE',
    'Januar2026_SI (not bundled; value recorded in Astra SOURCES_AND_CORRECTIONS)', 'SI Fig.S1b'),
49 ('eps_r', 9.03, '-', 'film', 'In203 of the target process (C-V, SI Fig.S1)', '
    MEASURED_RELATED_MATERIAL', 'Januar2026_SI (not bundled)', 'SI Fig.S1b'),
50 ('eps_r_HfO2', 19.57, '-', 'film', 'HfO2 of the target process', 'MEASURED_TARGET_DEVICE', '
    Januar2026_SI (not bundled)', 'SI Fig.S1b'),
51 ('Eg_IWO', 3.05, 'eV', 'film', 'sputtered a-IWO, different device (Fan 2021)', 'LITERATURE_IWO', '
    Fan2021_Nanomaterials (cited by Astra)', 'Sec.3'),
52 ('chi_IWO', 4.30, 'eV', 'film', 'estimate, 96/4 composition mixing (Fan 2021)', 'LITERATURE_IWO', '
    Fan2021_Nanomaterials (cited by Astra)', 'Sec.3'),
53 ('CNL_above_Ec', 0.4, 'eV', 'bulk', 'In203', 'LITERATURE_IN203_PROXY', 'Si2021_NanoLett; Lin2022;
    Wang2022', 'text'),
54 ('Dit', 6.0e11, 'cm^-2 eV^-1', 1.2, 'HfO2/In203 (from SS 63.5 mV/dec)', 'LITERATURE_IN203_PROXY', '
    Lin2022_ACSNano', 'p.21539'),
55 ('Dit', 6.3e11, 'cm^-2 eV^-1', 3.5, 'HfO2/In203 (subthreshold method)', 'LITERATURE_IN203_PROXY', '
    Wang2022_FrontMater', 'p.4'),
56 ('bulk_DOS', 3.3e20, 'cm^-3 eV^-1', 3.5, 'ALD In203 (conductance method), ND 1e20', '
    LITERATURE_IN203_PROXY', 'Wang2022_FrontMater', 'Abstract; p.4-5'),
57 ('Nt', 5.3e19, 'cm^-3', 2.0, 'IWO 2 nm (compact-model extraction, unstressed)', 'LITERATURE_IWO (
    target process, PBS device)', 'Januar2026', 'p.9'),
58 ('Nt', 3.39e18, 'cm^-3', 10.0, 'IWO 10 nm (compact-model extraction, unstressed)', 'LITERATURE_IWO (
    target process, PBS device)', 'Januar2026', 'p.9'),
59 ('Nt_ratio_13p2_to_2', 4.0, '-', '2..13.2', 'IWO thickness series', 'LITERATURE_IWO', 'Januar2026', '
    Sec.2.8, Fig.5a'),
60 ('Delta_sr', 2.87, 'A', 'series', 'IWO effective roughness (fit of mu vs t)', 'LITERATURE_IWO', '
    Januar2026', 'Sec.2.9, Fig.5f'),
61 ('mu_max', 27.4, 'cm^2/Vs', 13.2, 'IWO peak field-effect mobility', 'LITERATURE_IWO', 'Januar2026', '
    Sec.2.8'),
62 ('mu_max', 5.1, 'cm^2/Vs', 2.0, 'IWO peak field-effect mobility', 'LITERATURE_IWO', 'Januar2026', '
    Sec.2.8'),
63 ('Nbk_LFN', 2.37e18, 'cm^-3 eV^-1', 10, 'IWO (1 wt% WO3, SiO2 gate) border traps', 'LITERATURE_IWO (
    different process)', 'Kim2024_APL', 'p.6'),
64 ('Nbk_LFN', 3.12e18, 'cm^-3 eV^-1', 20, 'same', 'LITERATURE_IWO (different process)', 'Kim2024_APL',
    'p.6'),
65 ('Nbk_LFN', 14.6e18, 'cm^-3 eV^-1', 30, 'same', 'LITERATURE_IWO (different process)', 'Kim2024_APL',
    'p.6'),
66 ]
67
68 def fit_power(ts, ys):
69     """y = a * t^(-p) by log-log least squares."""
70     x = np.log(np.array(ts)); y = np.log(np.array(ys))
71     p, lna = np.polyfit(x, y, 1)
72     return float(math.exp(lna)), float(-p)
73
74 def main():
75     (ROOT / 'tables').mkdir(exist_ok=True); (ROOT / 'plots').mkdir(exist_ok=True); (ROOT / 'config').
        mkdir(exist_ok=True)
76     with (ROOT / 'tables' / 'THICKNESS_LAW_EVIDENCE.csv').open('w', newline='') as f:
77         w = csv.writer(f); w.writerow(['quantity', 'value', 'unit', 'thickness_nm', 'material', 'label',
            'source', 'location']); w.writerows(EVIDENCE)
78
79     # --- 1. Quantum-confinement band-edge shift (pure In203 DFT proxy): dEg_QC(t) = a t^-p
80     ts = [0.95, 1.5, 1.98]; dE = [1.88 - 0.94, 0.60, 1.27 - 0.94]
81     a_qc, p_qc = fit_power(ts, dE)
82     dEg = lambda t: a_qc * t ** (-p_qc)
83     # partition: Ev unchanged (Si2021) -> dEc = dEg, chi(t) = chi_bulk - dEc(t)
84     # --- 2. Effective mass: m*(t) = m*_bulk,exp + dm*(t), dm* from Lin2022 slab-minus-bulk (PBE)
85     tm = [0.95, 1.98, 3.52]; dm = [0.30 - 0.17, 0.23 - 0.17, 0.19 - 0.17]
86     b_m, q_m = fit_power(tm, dm)

```

```

87     mstar = lambda t: 0.208 + b_m * t ** (-q_m)
88     # --- 3. Tail density  $N_t(t) = N_{t2} * (2/t)^{\alpha}$ ;  $T_t$  shared
89     Nt2 = 2.0e19; alpha = 0.75; WTA = 0.040
90     Nt = lambda t: Nt2 * (2.0 / t) ** alpha
91     # --- 4. Effective background donor density (FITTED law, Kim2024 trend as qualitative support)
92     Nd0, tc, k = 2.5e17, 20.0, 4.0
93     Nd = lambda t: Nd0 * (1.0 + (t / tc) ** k)
94     # --- 5. Mobility:  $\mu_0(t) = \mu_{band}(t) * (1 - Dsr/t)^2$ ;  $\mu_{band}$  per thickness (FITTED, V0 baseline
95     values)
96     Dsr = 0.287
97     mu_band_fitted = {2.0: 16.8, 6.3: 12.4, 13.2: 57.6, 31.8: 84.0}
98     rough = lambda t: (1.0 - Dsr / t) ** 2
99
100     Eg_bulk, chi_bulk, eps_iwo = 3.05, 4.30, 9.30
101     rows = []
102     for t in [2.0, 6.3, 13.2, 31.8]:
103         m = mstar(t)
104         rows.append({'thickness_nm': t, 'dEg_QC_eV': round(dEg(t), 4), 'Eg_eV': round(Eg_bulk + dEg(t),
105         4), 'dEc_eV': round(dEc(t), 4),
106         'chi_eV': round(chi_bulk - dEg(t), 4), 'mstar_m0': round(m, 4), 'Nc300_cm3': f'{
107         nc_3d(m):.3e}',
108         'Nt_cm3': f'{Nt(t):.3e}', 'NTA_cm3_eV': f'{Nt(t) / WTA:.3e}', 'WTA_eV': WTA, 'Tt_K':
109         round(WTA / KB, 0),
110         'Nd_eff_cm3': f'{Nd(t):.3e}', 'roughness_factor': round(rough(t), 4), '
111         mu_band_cm2Vs_fitted': mu_band_fitted[t],
112         'mu0_cm2Vs': round(mu_band_fitted[t] * rough(t), 2), 'eps_r': eps_iwo})
113     with (ROOT / 'tables' / 'THICKNESS_LAWS.csv').open('w', newline='') as f:
114         w = csv.DictWriter(f, fieldnames=list(rows[0].keys())); w.writeheader(); w.writerows(rows)
115
116     model = {
117         'schema': 'IWO_material_model_v1', 'status': 'V1 physics-constrained seed; electronic values are
118         NOT ATLAS-calibrated until results/ say so',
119         'composition': {'xW_at_percent': 2.0, 'definition': 'NOT DETERMINED FROM AVAILABLE DATA: the
120         target paper states "IWO (~2% W)" without specifying atomic %, cation %, mol% WO3 or target
121         composition; treated as a material label, never converted to a donor density.'},
122         'temperature_K': {'value': 300.0, 'label': 'ASSUMED', 'note': 'measurement temperature not
123         recorded in workbook'},
124         'bulk_reference': {
125             'Eg_bulk_eV': {'value': Eg_bulk, 'label': 'LITERATURE_IWO', 'source': 'Fan2021_Nanomaterials
126             Sec.3 (different sputtered IWO device)', 'uncertainty_eV': 0.2},
127             'chi_bulk_eV': {'value': chi_bulk, 'label': 'LITERATURE_IWO', 'source': '
128             Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)', 'uncertainty_eV': 0.3,
129             'consistency_check': 'bulk In203 CNL ~0.4 eV above Ec (Si2021/Lin2022/
130             Wang2022); thin-film Ec rises by dEc(t) so TNL appears deeper below Ec -> EF farther below Ec ->
131             n_FB down, Vth up'},
132             'mstar_bulk_m0': {'value': 0.208, 'label': 'MEASURED_RELATED_MATERIAL', 'source': '
133             Stokey2021_JAP optical Hall (pure In203 crystal); W-doping increases m* (Januar2026 PBE, qualitative
134             ) -> NOT DETERMINED numerically'},
135             'eps_r': {'value': eps_iwo, 'label': 'MEASURED_TARGET_DEVICE', 'source': 'Januar2026 SI Fig.
136             S1 (C-V, IWO 9.30+/-0.04; not bundled, recorded by Astra)', 'alternatives': {'bulk_crystal_static':
137             10.55, 'evaporated_film': 8.9}, 'thickness_dependence': 'none imposed (no evidence)'},
138             'Nv300_cm3': {'value': 1.0e19, 'label': 'ASSUMED', 'note': 'holes are not solved (electrons-
139             only DD); enters only ni'},
140             'hole_mobility_cm2Vs': {'value': 0.01, 'label': 'ASSUMED'},
141             'srh_tau_s': {'value': 1.0e-6, 'label': 'ASSUMED', 'note': 'background SRH, negligible in n-
142             type wide gap'},
143         },
144         'thickness_laws': {
145             'dEg_QC_eV': {'form': 'a * t_nm^(-p)', 'a': round(a_qc, 4), 'p': round(p_qc, 4), 'label': '
146             DFT_PURE_IN203_PROXY',
147             'fit_points': [{t_nm': t, 'dE_eV': e, 'source': s} for t, e, s in zip(ts, dE,

```

```

128     ['Lin2022 slab 0.95 nm', 'Si2021 In203/Al203 1.5 nm (dEc)', 'Lin2022 slab 1.98 nm']]],
    'note': 'PBE absolute gaps are NOT used; only slab-minus-bulk shifts. Pure
In203 proxy for IWO; H-passivated/Al203-terminated slabs vs air/Al203 in device. Uncertainty ~ +/-50
%. '},
129    'band_edge_partition': {'dEc_over_dEg': 1.0, 'label': 'DFT_PURE_IN203_PROXY', 'source': '
Si2021: Ec moves up, Ev almost unchanged'},
130    'Eg_eV': 'Eg_bulk + dEg_QC(t)', 'chi_eV': 'chi_bulk - dEc(t)',
131    'mstar_m0': {'form': 'mstar_bulk_exp + b * t_nm^(-q)', 'b': round(b_m, 4), 'q': round(q_m, 4)
, 'label': 'MEASURED_RELATED_MATERIAL + DFT_PURE_IN203_PROXY increment'},
132    'Nc300_cm3': {'form': '2*(2*pi*m*kT/h^2)^1.5', 'label': 'CALCULATED', 'note': '3-D parabolic
continuum DOS; at 2 nm the film is quasi-2D, so Nc is an effective continuum parameter'},
133    'Nt_cm3': {'form': 'Nt2 * (2/t_nm)^alpha', 'Nt2': Nt2, 'alpha': alpha, 'label': 'FITTED (
anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with LITERATURE_IWO trend (Januar: ~4x from
13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention)'},
134    'WTA_eV': {'value': WTA, 'label': 'FITTED, shared across thickness', 'Tt_K': round(WTA / KB,
0), 'note': 'Januar2026 reports only dTt under stress; absolute Tt NOT DETERMINED FROM AVAILABLE
DATA'},
135    'NTA_cm3_eV': 'Nt(t)/WTA (gTA(E)=NTA*exp((E-Ec)/WTA); integral over the gap = NTA*WTA*(1-exp
(-Eg/WTA)) ~ NTA*WTA)',
136    'Nd_eff_cm3': {'form': 'Nd0 * (1 + (t_nm/tc)^k)', 'Nd0': Nd0, 'tc_nm': tc, 'k': k, 'label': '
FITTED law (effective electrically active background donor density, NOT W concentration)',
137    'support': 'Kim2024 (different IWO process): bulk/border trap density
2.4/3.1/14.6e18 at 10/20/30 nm; Januar2026: EF0 deeper for thinner films'},
138    'mu0_cm2Vs': {'form': 'mu_band(t) * (1 - Dsr/t)^2', 'Dsr_nm': Dsr, 'Dsr_label': '
LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A)',
139    'mu_band_fitted': mu_band_fitted, 'label': 'FITTED per thickness (no defensible
law; sigmoidal trend attributed to percolation/crystallinity, hypothesis)',
140    'gate_dependence': 'NOT inserted natively: ATLAS 5.28.1.R ignores MOBILITY
updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is produced self-
consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation gap (see
LIMITATIONS)'},
141    },
142    'traps': {
143        'bulk_acceptor_tail': {'atlas': 'DEFECTS NTA WTA (cm^-3/eV, eV), acceptor-like, referenced to
Ec', 'capture_cm2': 1e-15, 'capture_label': 'ASSUMED'},
144        'bulk_deep_acceptor_gaussian': {'NGA_cm3_eV': 5e16, 'EGA_from_Ec_eV': 0.6, 'WGA_eV': 0.15, '
label': 'ASSUMED (insensitive; Fan2021 range 2.5-3.7e16 for a different device)'},
145        'bulk_donor_tail': {'NTD_cm3_eV': 0, 'label': 'ASSUMED zero (valence tail has negligible
effect on n-type transfer curves; Fan2021)'},
146        'shallow_donor_gaussian': {'NGD_cm3_eV': 0, 'label': 'ASSUMED zero in V1: the effective donor
background is represented by Nd_eff(t); a donor Gaussian at fixed absolute energy (EGD referenced
to Ev) is the V2 candidate to replace Nd_eff(t) by the confinement mechanism'},
147        'interface_acceptor_sheet': {'peak_cm2_eV': 3e11, 'center_from_Ec_eV': 0.3, 'width_eV': 0.12,
'label': 'ASSUMED, shared across thickness; literature Dit(HfO2/In203) ~6e11 cm^-2/eV (
LITERATURE_IN203_PROXY; here IWO/Al203)', 'implementation': 'regularized 0.25 nm IWO layer at the
Al203 interface: volume amplitude = sheet peak / 0.25 nm (numerically verified: layer /2 changes Id
by 0.009 dec)'},
148        'fixed_interface_charge_Qf_cm2': {'value': 0.0, 'label': 'FITTED single shared value (stage
3); classified as effective electrostatic correction'},
149        'back_surface_donor_sheet': {'peak_cm2_eV': 0.0, 'center_from_Ec_eV': 0.02, 'width_eV': 0.05,
150        'label': 'HYPOTHESIS, 31.8 nm only when non-zero (device-
specific, stage 6): donor-like states at the free IWO back surface, regularized in a 0.25 nm layer (
region 5). Zero for the thin films. Not identifiable separately from Nd and Qf with one transfer
curve (V0 evidence: needed to reproduce the flat always-on baseline).'},
151    },
152    'contacts': {'gate_workfunction_eV': {'value': 4.70, 'label': 'ASSUMED (TiN 4.5-4.9 eV literature
range); degenerate with Qf and chi -> held fixed'},
153        'source_drain': {'model': 'ideal Ohmic (no workfunction)', 'label': 'ASSUMED;
underconstrained (no TLM); CNL above Ec in In203 supports low-barrier contacts (Si2021/Lin2022)'},
154        'series_resistance_ohm_um': {'2.0': 0, '6.3': 0, '13.2': 0, '31.8': 9.2e4, 'label':
'31.8 nm only, FITTED hypothesis (on-state saturation); total device 672 ohm at 3 V bounds it'}},

```

```

155     'geometry': {'L_um': 20.0, 'W_um': 290.0, 'sim_width_um': 1.0, 'hfo2_nm': 15.0, 'al2o3_nm': 2.0,
156     'contact_overlap_um': {'value': 2.0, 'label': 'ASSUMED numerical contact geometry (not reported)'},
157     'eps_hfo2': 19.57, 'eps_al2o3': {'value': 9.0, 'label': 'ASSUMED'}, 'top_of_channel'
158     : 'AIR (no cap)'},
159 }
160 if yaml:
161     (ROOT / 'config' / 'iwo_material_model.yaml').write_text(yaml.safe_dump(model, sort_keys=False,
162     width=140))
163 (ROOT / 'config' / 'iwo_material_model.json').write_text(json.dumps(model, indent=2))
164
165 # --- plots
166 try:
167     import matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
168     tt = np.linspace(0.8, 35, 400)
169     fig, ax = plt.subplots(2, 3, figsize=(13, 7.5))
170     ax[0, 0].plot(tt, Eg_bulk + dEg(tt), label='Eg(t) = 3.05 + dEg_QC (proxy)'); ax[0, 0].plot(ts, [
171     Eg_bulk + e for e in dE], 'ko', label='DFT shifts (Lin2022, Si2021)')
172     ax[0, 0].set_xscale('log'); ax[0, 0].set_ylabel('Eg (eV)'); ax[0, 0].legend(fontsize=7)
173     ax[0, 1].plot(tt, dEg(tt), label='dEc = dEg (Ev fixed)'); ax[0, 1].plot(tt, chi_bulk - dEg(tt),
174     label='chi(t) = 4.30 - dEc'); ax[0, 1].set_xscale('log'); ax[0, 1].set_ylabel('eV'); ax[0, 1].legend
175     (fontsize=7)
176     ax[0, 2].plot(tt, mstar(tt), label='m*(t) = 0.208 + dm*(t)'); ax[0, 2].plot([0.95, 1.98, 3.52],
177     [0.208 + d for d in dm], 'ko', label='PBE increments (Lin2022)'); ax[0, 2].set_xscale('log'); ax[0,
178     2].set_ylabel('m*/m0'); ax[0, 2].legend(fontsize=7)
179     ax[1, 0].semilogy(tt, [nc_3d(mstar(x)) for x in tt]); ax[1, 0].set_xscale('log'); ax[1, 0].
180     set_ylabel('Nc (cm-3)')
181     ax[1, 1].loglog(tt, Nt(tt), label='Nt(t) law'); ax[1, 1].loglog([2.0, 10.0], [5.3e19, 3.39e18], '
182     rs', label='Januar2026 (PBS devices)'); ax[1, 1].loglog([2.0, 6.3, 13.2], [2.0e19, 1.34e19, 4.6e18],
183     'ko', label='real-ATLAS fits (Claude V0)'); ax[1, 1].set_ylabel('Nt (cm-3)'); ax[1, 1].legend(
184     fontsize=7)
185     ax[1, 2].loglog(tt, Nd(tt), label='Nd_eff(t) law'); ax[1, 2].loglog([2.0, 6.3, 13.2, 31.8], [2e17
186     , 3e17, 3e17, 2.65e18], 'ko', label='real-ATLAS fits (Claude V0)'); ax[1, 2].set_ylabel('Nd_eff (cm
187     -3)'); ax[1, 2].legend(fontsize=7)
188     for a in ax.flat: a.set_xlabel('IWO thickness (nm)'); a.grid(alpha=.3)
189     fig.suptitle('IWO_material(t) V1 thickness laws: DFT proxies (pure In203), measured bulk m*,
190     fitted Nt/Nd laws')
191     fig.tight_layout(); fig.savefig(ROOT / 'plots' / 'thickness_laws_overview.png', dpi=130); plt.
192     close(fig)
193     fig, ax = plt.subplots(figsize=(6, 4)); ax.semilogx(tt, rough(tt), label='(1 - Dsr/t)2, Dsr =
194     2.87 Å')
195     for t, mu in mu_band_fitted.items(): ax.plot(t, rough(t), 'ko')
196     ax.set_xlabel('t (nm)'); ax.set_ylabel('roughness factor'); ax.legend(); ax.grid(alpha=.3); fig.
197     tight_layout(); fig.savefig(ROOT / 'plots' / 'mobility_roughness_factor.png', dpi=130); plt.close(
198     fig)
199 except Exception as e:
200     print('plots skipped:', e)
201     print(f'dEg_QC(t) = {a_qc:.3f} * t{p_qc:.3f} eV ; m*(t) = 0.208 + {b_m:.3f} * t{q_m:.3f}')
202     for r in rows: print(r)
203
204 if __name__ == '__main__':
205     main()

```

B.2.2 build_iwo_decks.py

The script generates the ATLAS decks from one material model, `IWO_material(t, xW)`, for the requested thicknesses. It accepts dotted-path overrides of the material model (`-override`), the structure form (`-structure full|reduced`) and the sweep form (`-sweep stepped|pointwise`), and it writes the provenance header of every deck (Section 6.1). All decks of Chapter C were produced

with it.

```
1 #!/usr/bin/env python3
2 """Generate the four ATLAS decks from ONE material model: IWO_material(t, xW).
3
4     python scripts/build_iwo_decks.py                # decks/iwo_{2p0,6p3,13p2,31p8}nm.in
5     python scripts/build_iwo_decks.py --thickness 2 --override traps.fixed_interface_charge_Qf_cm2.value=-3
6         e11
7
8 Reads config/iwo_material_model.yaml (thickness laws + provenance), config/geometry.yaml and
9 config/solver.yaml. Every material parameter written to a deck carries its value, law, status
10 label and source as a comment. No simulator is run by this script.
11
12 Reduced electrical deck (DC): ideal TiN gate boundary at the HfO2 bottom, ideal Pd S/D boundaries on
13 the IWO top surface. Continuous metals are equipotential in DC, so replacing their volumes by
14 Dirichlet boundaries changes nothing electrically while removing nodes (Astra kept metal volumes:
15 identical fits). --full writes decks/full_physical_structure.in with explicit Conductor/Palladium
16 volumes for TonyPlot.
17 """
18 from __future__ import annotations
19 import argparse, copy, json, math, os
20 from pathlib import Path
21 import numpy as np, yaml
22
23 ROOT = Path(__file__).resolve().parents[1]
24 EPS0 = 8.8541878128e-14; Q = 1.602176634e-19; KB = 8.617333262e-5
25 H = 6.62607015e-34; ME = 9.1093837015e-31; KBJ = 1.380649e-23
26 class _Keys(dict):
27     # 2026-10-06: thicknesses without a measured curve (V2 scenarios) get a generic key, e.g. 0.75 -> '0
28     # 75'
29     def __missing__(self, t): return f'{float(t):g}'.replace('.', 'p')
30 KEYS = _Keys({2.0: '2p0', 6.3: '6p3', 13.2: '13p2', 31.8: '31p8'})
31 # material model file (relative to ROOT); V1 by default, IWO_MODEL_FILE=config/iwo_material_model_v2.yaml
32 # selects V2
33 MODEL_FILE = os.environ.get('IWO_MODEL_FILE', 'config/iwo_material_model.yaml')
34
35 def load_configs():
36     m = yaml.safe_load((ROOT / MODEL_FILE).read_text())
37     g = yaml.safe_load((ROOT / 'config' / 'geometry.yaml').read_text())
38     s = yaml.safe_load((ROOT / 'config' / 'solver.yaml').read_text())
39     return m, g, s
40
41 def set_dotted(d, path, value):
42     """Override 'a.b.c=value'. A segment in brackets may contain a dot and matches numeric keys:
43     thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]=18.1 sets the float key 2.0 of that mapping."""
44     import re
45     keys = [a or b for a, b in re.findall(r'\[[^\]]+\]|([^\.\[\]]+)', path)]
46     def resolve(c, k):
47         if isinstance(c, list): return int(k)
48         if k in c: return k
49         try:
50             fk = float(k)
51             for kk in c:
52                 if isinstance(kk, (int, float)) and not isinstance(kk, bool) and kk == fk: return kk
53         except ValueError: pass
54     return k
55
56 cur = d
57 for k in keys[:-1]:
58     cur = cur[resolve(cur, k)]
59 last = resolve(cur, keys[-1])
60 try: value = float(value)
```

```

57     except (TypeError, ValueError): pass
58     if isinstance(cur, dict) and last in cur and isinstance(cur[last], dict) and 'value' in cur[last]:
59         cur[last]['value'] = value
60     else: cur[last] = value
61
62 def val(x):
63     return x['value'] if isinstance(x, dict) and 'value' in x else x
64
65 def nc_3d(mstar, T=300.0):
66     return 2.0 * (2.0 * math.pi * mstar * ME * KBJ * T / H ** 2) ** 1.5 / 1e6
67
68 def cox(g):
69     return EPS0 / (g['stack_bottom_to_top'][2]['thickness_nm'] * 1e-7 / g['stack_bottom_to_top'][2]['
70         eps_r'] + g['stack_bottom_to_top'][3]['thickness_nm'] * 1e-7 / g['stack_bottom_to_top'][3]['eps_r'])
71
72 def law(spec, a, p, t):
73     """a * t^p, or the tabulated value when the law lists one for this thickness (V2: DFT points below
74         the fitted range)."""
75     for k, v in (spec.get('points') or {}).items():
76         if abs(float(k) - t) < 1e-6: return float(v)
77     return spec[a] * t ** (-spec[p])
78
79 def material_at(m, t):
80     """Evaluate IWO_material(t) -> dict of ATLAS-ready values and their provenance strings."""
81     L = m['thickness_laws']; b = m['bulk_reference']
82     dEg = law(L['dEg_QC_eV'], 'a', 'p', t)
83     # V2 (2026-10-06): own conduction-band-shift law (the DFT partition dEc/dEg depends on thickness); V1
84     : dEg x partition
85     dEc = law(L['dEc_eV'], 'a', 'p', t) if 'dEc_eV' in L else dEg * L['band_edge_partition']['
86         dEc_over_dEg']
87     Eg = val(b['Eg_bulk_eV']) + dEg; chi = val(b['chi_bulk_eV']) - dEc
88     mstar = val(b['mstar_bulk_m0']) + law(L['mstar_m0'], 'b', 'q', t)
89     Nc = nc_3d(mstar)
90     Nt = L['Nt_cm3']['Nt2'] * (2.0 / t) ** L['Nt_cm3']['alpha']; WTA = val(L['WTA_eV']); NTA = Nt / WTA
91     Nd = L['Nd_eff_cm3']['Nd0'] * (1.0 + (t / L['Nd_eff_cm3']['tc_nm'])) ** L['Nd_eff_cm3']['k']
92     Dsr = L['mu0_cm2Vs']['Dsr_nm']; rough = (1.0 - Dsr / t) ** 2
93     mb = L['mu0_cm2Vs']['mu_band_fitted']; mu_band = mb.get(t, mb.get(str(t), None))
94     if mu_band is None: mu_band = float(np.interp(t, sorted(float(k) for k in mb), [mb[k] for k in sorted
95         (mb, key=float)]))
96     mu0 = mu_band * rough
97     tr = m['traps']; ct = m['contacts']
98     rs = ct['series_resistance_ohm_um']; rs_t = float(rs.get(str(t), rs.get(t, 0.0)) if isinstance(rs,
99         dict) else 0.0)
100     return {'t_nm': t, 'dEg_QC_eV': dEg, 'dEc_eV': dEc, 'Eg_eV': Eg, 'chi_eV': chi, 'mstar_m0': mstar, '
101         Nc_cm3': Nc, 'Nv_cm3': val(b['Nv300_cm3']),
102         'eps_iwo': val(b['eps_r']), 'Nt_cm3': Nt, 'NTA_cm3_eV': NTA, 'WTA_eV': WTA, 'Nd_cm3': Nd, '
103         rough': rough, 'mu_band': mu_band, 'mu0': mu0,
104         'mup': val(b['hole_mobility_cm2Vs']), 'tau_s': val(b['srh_tau_s']),
105         'NGA': tr['bulk_deep_acceptor_gaussian']['NGA_cm3_eV'], 'EGA': tr['
106         bulk_deep_acceptor_gaussian']['EGA_from_Ec_eV'], 'WGA': tr['bulk_deep_acceptor_gaussian']['WGA_eV'],
107         'NTD': tr['bulk_donor_tail']['NTD_cm3_eV'], 'NGD': tr['shallow_donor_gaussian']['NGD_cm3_eV'
108         ],
109         'EGD_from_Ec': tr['shallow_donor_gaussian'].get('EGD_from_Ec_eV', 0.1), 'WGD': tr['
110         shallow_donor_gaussian'].get('WGD_eV', 0.1),
111         'sig': tr['bulk_acceptor_tail']['capture_cm2'], 'it_peak': tr['interface_acceptor_sheet']['
112         peak_cm2_eV'], 'it_e': tr['interface_acceptor_sheet']['center_from_Ec_eV'], 'it_w': tr['
113         interface_acceptor_sheet']['width_eV'],
114         'Qf_cm2': val(tr['fixed_interface_charge_Qf_cm2']), 'gate_wf': val(ct['gate_workfunction_eV'
115         ]), 'rs_ohm_um': rs_t,
116         'bs_peak': tr.get('back_surface_donor_sheet', {}).get('peak_cm2_eV', 0.0), 'bs_e': tr.get('
117         back_surface_donor_sheet', {}).get('center_from_Ec_eV', 0.02), 'bs_w': tr.get('

```

```

back_surface_donor_sheet', {}).get('width_eV', 0.05),
102     'Qb_cm2': float(val(tr.get('back_surface_fixed_charge_cm2', 0.0)) or 0.0), # fixed charge
        at the free (air-side) IWO surface; 0 unless a hypothesis run sets it
103     'T_K': val(m['temperature_K'])}
104
105 def moment_matched_gaussian(a_b, w_b, e_b, a_i, w_i, e_i):
106     """Single Gaussian (ATLAS form  $\exp(-(E-E_0)/W)^2$ ) preserving area and first two moments of two
        overlapping Gaussians."""
107     mass = a_b * w_b + a_i * w_i
108     if mass <= 0: return e_b, w_b, 0.0
109     e = (a_b * w_b * e_b + a_i * w_i * e_i) / mass
110     var = (a_b * w_b * (w_b ** 2 / 2 + (e_b - e) ** 2) + a_i * w_i * (w_i ** 2 / 2 + (e_i - e) ** 2)) /
        mass
111     w = math.sqrt(2 * var); return e, w, mass / w
112
113 def defects_block(p, region, di_nm, interface, back=False):
114     sig = p['sig']
115     if back:
116         ngd = p['bs_peak'] / (di_nm * 1e-7); egd = p['Eg_eV'] - p['bs_e']
117         nga, ega, wga = p['NGA'], p['EGA'], p['WGA']
118     else:
119         ngd = p['NGD']; egd = p['Eg_eV'] - p['EGD_from_Ec']
120         a_i = p['it_peak'] / (di_nm * 1e-7) if interface else 0.0
121         ega, wga, nga = moment_matched_gaussian(p['NGA'], p['WGA'], p['EGA'], a_i, p['it_w'], p['it_e'])
122     return [f'# Region {region}: continuous DOS. NTA  $\text{cm}^{-3}/\text{eV}$  at  $E_c$  (tail intercept), WTA eV; NGA/NGD
        Gaussian amplitudes  $\text{cm}^{-3}/\text{eV}$ ;',
123           f'# EGA measured from  $E_c$ , EGD measured from  $E_v$  (installed manual, DEFECTS statement).' + ('
        Interface sheet regularized: NGA = sheet peak / layer thickness.' if interface else '') + (' Back-
        surface donor sheet regularized.' if back else ''),
124           f'defects region={region} continuous numa={{NUMA}} numd={{NUMD}} \\\',
125           f' nta={p["NTA_cm3_eV"]:.6g} wta={p["WTA_eV"]:.6g} ntd={p["NTD"]:.6g} wtd=0.1 \\\',
126           f' nga={nga:.6g} ega={ega:.6g} wga={wga:.6g} \\\',
127           f' ngd={ngd:.6g} egd={egd:.6g} wgd={p["WGD"]} if not back else p["bs_w"]:.6g \\\',
128           f' sigtae={sig:.3g} sigtah={sig:.3g} sigtde={sig:.3g} sigtdh={sig:.3g} \\\',
129           f' siggae={sig:.3g} siggah={sig:.3g} siggde={sig:.3g} siggdh={sig:.3g} \\\',
130           f' afile=acceptor_r{region}.dat dfile=donor_r{region}.dat']
131
132 def build_deck(t, m, g, s, meas=None, targets=None, smoke=False, full=False, vstep=None, segments=None,
        output_vgs=None, vd_segments=None):
133     """full: physical structure (Air above the channel, Pd source/drain volumes, Al2O3 and HfO2 regions,
        Conductor
134     gate volume representing TiN). segments: [(end_V, step_V), ...] -> ATLAS stepped sweep blocks
135     SOLVE VSTEP=.. VFINAL=.. NAME=gate (one block per segment, split at the METHOD switch and the Vth
        save);
136     vstep: shorthand for a single segment. Neither -> one SOLVE per point (legacy).
137     output_vgs: PREDICTION mode (2026-09-25) - ID-VD families at these gate voltages instead of the
        transfer sweep;
138     vd_segments: [(end_V, step_V), ...] drain sweep from 0 V (stepped SOLVE VSTEP/VFINAL NAME=drain
        blocks)."""
139     # Hypothesis option: the PHYSICAL film thickness (geometry AND thickness laws) may differ from the
        nominal XRR
140     # thickness t that labels the measured curve (geometry.film_thickness_override_nm). Default:
        identical.
141     t_nom = t; t = float(g.get('film_thickness_override_nm') or t_nom)
142     p = material_at(m, t); key = KEYS[t_nom]
143     T_run = 300.0
144     import re as _re
145     _mt = _re.search(r'temp=([0-9.]+)', s['models'])
146     if _mt: T_run = float(_mt.group(1))
147     if vstep and not segments: segments = [(float(g['gate_sweep_V']['stop']), float(vstep))]
148     st = g['stack_bottom_to_top']; tox_hf = st[2]['thickness_nm'] / 1000.; tox_al = st[3]['thickness_nm']

```



```

/ 1000.
149 tt = t / 1000.; di = g['mesh']['interface_layer_nm'] / 1000.; lc = val(g['contact_overlap_um']); L =
    g['channel_length_um']; xr = lc + L; xmax = xr + lc
150 Cox = cox(g); qf = p['Qf_cm2']
151 sweep = g['gate_sweep_V']; vg = np.round(np.arange(sweep['start'], sweep['stop'] + 1e-9, sweep['step']
    ), 6) if targets is None else np.asarray(targets, float)
152 if smoke: vg = np.array([-3., 0., 3.])
153 back = p['bs_peak'] > 0 and not smoke
154 ybulk = -tt + di if back else -tt
155 ysp_bulk = min(g['mesh']['y_spacing_bulk_nm'] if isinstance(g['mesh']['y_spacing_bulk_nm'], (int,
    float)) else 1.0, t / 8) / 1000.
156 ysi = g['mesh']['y_spacing_interface_nm'] / 1000.
157 ys = [(-tt, ysi), (-tt + di, ysi), (-tt + 4 * di, ysp_bulk)] if back else [(-tt, ysp_bulk)] + [(-di
    , ysi), (-di / 2, ysi), (0., ysi), (tox_al, g['mesh']['y_spacing_al2o3_nm'] / 1000.), (tox_al +
    tox_hf, g['mesh']['y_spacing_hfo2_nm'] / 1000.))
158 t_pd = st[5]['thickness_nm'] / 1000.; t_tin = st[1]['thickness_nm'] / 1000.; ytop = -tt - t_pd; ygate
    = tox_al + tox_hf + t_tin
159 if full: ys = [(ytop, 0.02)] + ys + [(ygate, 0.02)]
160 # strict-acceptance switch voltage from the measured curve
161 sa = s['strict_acceptance']; cr_strict = float(sa['cr_toler_A']); switch_v = float(sa.get('
    prediction_switch_V', 0.5)) # no measurement: prediction runs set it
162 if meas is not None:
163     mv, mi = meas; above = mv[mi >= sa['until_measured_current_A_per_um']]; switch_v = float(above
    [0]) if len(above) else float('inf')
164     off = float(np.median(mi[(mv >= -2) & (mv <= -.5)])); act = mi > s['scoring']['active_multiplier'
    ] * off if t_nom < 20 else np.ones(len(mi), bool)
165     if act.any(): cr_strict = max(cr_strict, float(sa['cr_tol_rel_active_min']) * float(mi[act].min()
    ))
166 base = f'method {s["method_base"]}' ; strict = base + f' \\n cr.toler={cr_strict:.3g} xandnorm {s["
    carriers"]}' ; relaxed = base + f' \\n cr.toler=5e-18 ^xandnorm {s["carriers"]}'
167 P = lambda name, v, unit, label, src: f'#{name} = {v} {unit} | {label} | {src}'
168 Lw = m['thickness_laws']; b = m['bulk_reference']
169 v2 = m.get('schema') != 'IWO_material_model_v1'
170 src = lambda spec, default: spec.get('source', default) if v2 else default
171 ls = ['#' + '=' * 78, f'#{ IWO (~2 at.% W:In203) FET, channel t = {t} nm --
    IWO_PHYSICS_CONSTRAINED_MODEL_V1' + (f' [{m["schema"]}]" if v2 else ''),
172     f'#{ Generated by scripts/build_iwo_decks.py from {MODEL_FILE} (ONE material model for all',
173     '# thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential
    metals, DC).',
174     '# MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current.
    Never divide by 290.',
175     f'#{ Cox = {Cox:.4e} F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = {g["drain_bias_V
    "]} V, T = {p["T_K"]} K.',
176     '# Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G
    contacts, H numerical.',
177     '#' + '=' * 78,
178     P(['A] Eg_bulk', val(b['Eg_bulk_eV']), 'eV', b['Eg_bulk_eV']['label'], b['Eg_bulk_eV']['source'
    ]),
179     P(['B] dEg_QC(t)', f'{p["dEg_QC_eV"]:.4f}', 'eV', Lw['dEg_QC_eV']['label'], f'a*t^p, a={Lw["
    dEg_QC_eV"] ["a"]}, p={Lw["dEg_QC_eV"] ["p"]} ({src(Lw["dEg_QC_eV"], "Lin2022 slabs, Si2021"))}),
180     P(['A+B] EG300', f'{p["Eg_eV"]:.4f}', 'eV', 'CALCULATED', 'Eg_bulk + dEg_QC(t)'),
181     P(['A] chi_bulk', val(b['chi_bulk_eV']), 'eV', b['chi_bulk_eV']['label'], b['chi_bulk_eV']['
    source'])]
182 if 'dEc_eV' in Lw:
183     ls.append(P(['B] dEc(t)', f'{p["dEc_eV"]:.4f}', 'eV', Lw['dEc_eV']['label'], f'a*t^p, a={Lw["
    dEc_eV"] ["a"]}, p={Lw["dEc_eV"] ["p"]} ({src(Lw["dEc_eV"], ""))}))
184 ls += [P(['A+B] AFFINITY', f'{p["chi_eV"]:.4f}', 'eV', 'CALCULATED', 'chi_bulk - dEc(t)' + (' if '
    dEc_eV' in Lw else ', dEc = dEg (Ev unchanged, Si2021)'),
185     P(['A+B] m*', f'{p["mstar_m0"]:.4f}', 'm0', Lw['mstar_m0']['label'], src(Lw['mstar_m0'], '
    Stokey2021 bulk 0.208 + Lin2022 PBE increment')),
186     P(['A+B] NC300', f'{p["Nc_cm3"]:.4e}', 'cm^-3', 'CALCULATED', '2(2 pi m* kT/h^2)^1.5; effective

```

```

continuum DOS at 2 nm'),
187     P('[A] NV300', f'{p["Nv_cm3"]:.3e}', 'cm^-3', b['Nv300_cm3']['label'], 'holes not solved'),
188     P('[A] PERMITTIVITY', p['eps_iwo'], '-', b['eps_r']['label'], b['eps_r']['source']),
189     P('[D] Nt(t)', f'{p["Nt_cm3"]:.4e}', 'cm^-3', Lw['Nt_cm3']['label'], f'Nt2*(2/t)^alpha, Nt2={Lw["Nt_cm3"]['Nt2']}, alpha={Lw["Nt_cm3"]['alpha']}'),
190     P('[D] NTA = Nt/WTa', f'{p["NTA_cm3_eV"]:.4e}', 'cm^-3/eV', 'CALCULATED', 'gTA(E)=NTA exp((E-Ec)/WTa); integral = NTA*WTa*(1-exp(-Eg/WTa))'),
191     P('[D] WTA (kTt)', p['WTA_eV'], 'eV', Lw['WTA_eV']['label'], f'Tt = {p["WTA_eV"]} / KB:.0f} K; shared across thickness'),
192     P('[D] deep acceptor Gaussian', f'{p["NGA"]:.3g} at Ec-{p["EGA"]} eV, W={p["WGA"]}', 'cm^-3/eV', m['traps']['bulk_deep_acceptor_gaussian']['label'], ''),
193     P('[D] Nd_eff(t)', f'{p["Nd_cm3"]:.4e}', 'cm^-3', Lw['Nd_eff_cm3']['label'], f'Nd0*(1+(t/tc)^k), Nd0={Lw["Nd_eff_cm3"]['Nd0']}, tc={Lw["Nd_eff_cm3"]['tc_nm']} nm, k={Lw["Nd_eff_cm3"]['k']}; NOT W concentration'),
194     P('[E] interface acceptor sheet', f'{p["it_peak"]:.3g} at Ec-{p["it_e"]} eV, W={p["it_w"]} eV, 'cm^-2/eV', m['traps']['interface_acceptor_sheet']['label'], m['traps']['interface_acceptor_sheet']['implementation']),
195     P('[E] Qf', f'{qf:.4e}', 'cm^-2', m['traps']['fixed_interface_charge_Qf_cm2']['label'], f'dVfb = -q Qf/Cox = {-Q * qf / Cox:+.4f} V'),
196     P('[F] mu_band(t)', p['mu_band'], 'cm^2/Vs', 'FITTED per thickness if not v2 else ('FITTED (anchor film) if any(abs(t - x) < 1e-6 for x in Lw['mu0_cm2Vs'].get('anchor_thicknesses_nm', [])) else 'NOT FITTED: pre-registered prediction rule or scenario value'), Lw['mu0_cm2Vs']['label']),
197     P('[F] roughness factor', f'{p["rough"]:.4f}', '-', Lw['mu0_cm2Vs']['Dsr_label'], f'(1 - Dsr/t)^2, Dsr = {Lw["mu0_cm2Vs"]['Dsr_nm']} nm'),
198     P('[F] MUN = mu_band*(1-Dsr/t)^2', f'{p["mu0"]:.4f}', 'cm^2/Vs', 'CALCULATED', Lw['mu0_cm2Vs']['gate_dependence']),
199     P('[G] gate work function', p['gate_wf'], 'eV', m['contacts']['gate_workfunction_eV']['label'], ''),
200     P('[G] S/D', m['contacts']['source_drain']['model'] + (f'; series R {p["rs_ohm_um"]:.3g} ohm.um total' if p['rs_ohm_um'] > 0 else ''), '', m['contacts']['source_drain']['label'], ''),
201     P('[H] capture cross sections', p['sig'], 'cm^2', m['traps']['bulk_acceptor_tail']['capture_label'], 'not identifiable from DC transfer curves'),
202     '# + '=' * 78, 'go atlas', f'mesh width={float(g["simulated_width_um"])}'],
203     mf = float(g['mesh'].get('spacing_factor', 1.0)) # < 1 refines every spacing (numerical convergence checks only)
204     if mf != 1.0: ls.append(f'# NUMERICAL CHECK: all mesh spacings multiplied by {mf:g}')
205     xm = g['mesh']['x_um']
206     if abs(lc - 2.0) > 1e-9: # 2026-09-27: mesh follows the contacts when the overlap differs from the 2 um default (run_0027 had a fixed 24 um mesh)
207         xm = [(0, 0.3), (lc - 0.2, 0.05), (lc, 0.01), (lc + 0.2, 0.05), (lc + 1, 0.2), (lc + L / 2, 0.7), (xr - 1, 0.2), (xr - 0.2, 0.05), (xr, 0.01), (xr + 0.2, 0.05), (xmax, 0.3)]
208     for x, sp in xm: ls.append(f'x.mesh loc={x:.10g} spac={sp * mf:.10g}')
209     for y, sp in ys: ls.append(f'y.mesh loc={y:.10g} spac={sp * mf:.10g}')
210     ls += [f'region num=1 user.material=IWO x.min=0 x.max={xmax:.9g} y.min={ybulk:.10g} y.max={-di:.10g}',
211           f'region num=2 user.material=IWO x.min=0 x.max={xmax:.9g} y.min={-di:.10g} y.max=0'],
212     if back: ls.append(f'region num=5 user.material=IWO x.min=0 x.max={xmax:.9g} y.min={-tt:.10g} y.max={ybulk:.10g}'),
213     if full:
214         ls += ['# Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70 nm S/D, air above the channel.',
215               '# TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on CONTACT (4.70 eV, ASSUMED).',
216               f'region num=3 material=Al2O3 x.min=0 x.max={xmax:.9g} y.min=0 y.max={tox_al:.10g}',
217               f'region num=4 material=HfO2 x.min=0 x.max={xmax:.9g} y.min={tox_al:.10g} y.max={tox_al + tox_hf:.10g}',
218               f'region num=6 material=Conductor x.min=0 x.max={xmax:.9g} y.min={tox_al + tox_hf:.10g} y.max={ygate:.10g}',
219               f'region num=7 material=Air x.min=0 x.max={xmax:.9g} y.min={ytop:.10g} y.max={-tt:.10g}',
220               f'electrode name=gate x.min=0 x.max={xmax:.9g} y.min={tox_al + tox_hf:.10g} y.max={ygate}']

```

```

223         :.10g}',
224         '# Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them ->
Ohmic boundary (ASSUMED, no TLM data);',
225         '# the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to
verify electrical equivalence.',
226         f'electrode name=source material=Palladium x.min=0 x.max={lc:.9g} y.min={ytop:.10g} y.max
={-tt:.10g}',
227         f'electrode name=drain material=Palladium x.min={xr:.9g} x.max={xmax:.9g} y.min={ytop:.10g
} y.max={-tt:.10g}']
228     else:
229         ls += [f'region num=3 material=oxide x.min=0 x.max={xmax:.9g} y.min=0 y.max={tox_al:.10g}',
f'region num=4 material=oxide x.min=0 x.max={xmax:.9g} y.min={tox_al:.10g} y.max={tox_al +
tox_hf:.10g}']
230         ls += [f'electrode name=source x.min=0 x.max={lc:.9g} y.min={-tt:.10g} y.max={-tt:.10g}',
f'electrode name=drain x.min={xr:.9g} x.max={xmax:.9g} y.min={-tt:.10g} y.max={-tt:.10g}',
231         f'electrode name=gate x.min=0 x.max={xmax:.9g} y.min={tox_al + tox_hf:.10g} y.max={tox_al
+ tox_hf:.10g}']
232     for r in ([1, 2, 5] if back else [1, 2]): ls.append(f'doping uniform n.type conc={p["Nd_cm3"]:.6g}
region={r}')
233 ls += ['# USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden
above.',
234         'material material=IWO user.group=semiconductor user.default=silicon \\\',
235         f' eg300={p["Eg_eV"]:.6g} affinity={p["chi_eV"]:.6g} permittivity={p["eps_iwo"]:.6g} \\\',
236         f' nc300={p["Nc_cm3"]:.6g} nv300={p["Nv_cm3"]:.6g} \\\', f' mun={p["mu0"]:.6g} mup={p["mup"]:.6
g} \\\', f' taun0={p["tau_s"]:.3g} taup0={p["tau_s"]:.3g}' + (' egalpha=0' if abs(T_run - 300) > 1e-6
else '') + (f' tmun={float(s["tmun"]):g}' if s.get('tmun') is not None else ''),
237         f'# Region 3 = Al2O3 (2 nm), eps {st[3]["eps_r"]} ({st[3]["eps_label"]}); region 4 = HfO2 (15
nm), eps {st[2]["eps_r"]} ({st[2]["eps_label"]})).',
238         '# Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT
the measured values.',
239         f'material region=3 permittivity={st[3]["eps_r"]}', f'material region=4 permittivity={st[2]["
eps_r"]}',
240         f'contact name=gate workfunction={p["gate_wf"]:.6g}']
241 if p['rs_ohm_um'] > 0: ls += ['# CONTACT RESISTANCE is in ohm.um for 2-D (installed manual); half of
the total per contact.', f'contact name=source resistance={p["rs_ohm_um"] / 2:.6g}', f'contact name=
drain resistance={p["rs_ohm_um"] / 2:.6g}']
242 ls.append(f'models {s["models"]}')
243 if not smoke:
244     ls += defects_block(p, 1, g['mesh']['interface_layer_nm'], False) + defects_block(p, 2, g['mesh'
]['interface_layer_nm'], True)
245     if back: ls += defects_block(p, 5, g['mesh']['interface_layer_nm'], False, back=True)
246 else: ls.append('# SMOKE: DOS disabled (trap-free diagnostic, not a fit).')
247 xc = (lc + xr) / 2; yc = -tt / 2
248 ls += [f'interface qf={qf:.6g} x.min=0 x.max={xmax:.9g} y.min=-0.0000001 y.max=0.0000001']
249 if p['Qb_cm2'] != 0:
250     # HYPOTHESIS (2026-09-25): fixed sheet charge at the free IWO/Air surface of the EXPOSED channel
only (between the
251     # S/D contacts; the surface under the Pd volumes is metal-covered). Same INTERFACE QF mechanism
as the front Qf.
252     ls += [f'# [E] HYPOTHESIS back-surface fixed charge Qb = {p["Qb_cm2"]:.4e} cm^-2 at the IWO/Air
surface (x = {lc:g}..{xr:g} um)',
253         f'interface qf={p["Qb_cm2"]:.6g} x.min={lc:.9g} x.max={xr:.9g} y.min={-tt - 1e-7:.10g} y.
max={-tt + 1e-7:.10g}']
254 ls += [f'# Strict acceptance (XANDRNORM, cr.toler {cr_strict:.3g}) up to Vg = {switch_v:.10g} V;
default acceptance above.', strict,
255         f'output {s["output"]}',
256         '# Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver
quantities).',
257         f'probe name=chan_mob x={xc:.6g} y={yc:.6g} n.mob dir=0', f'probe name=chan_n x={xc:.6g} y={yc
:.6g} n.conc',
258         f'probe name=avg_n region=1 average n.conc x.min={lc:.6g} x.max={xr:.6g} y.min={ybulk:.10g} y.

```

```

max={-di:.10g}',
260     'solve init', 'save outf=equilibrium.str']
261 if output_vgs:
262     # PREDICTION MODE: output characteristics at fixed gate voltages (no measured counterpart exists
yet).
263     vgs = [float(v) for v in output_vgs]; vds = [(float(e), float(st_)) for e, st_ in (vd_segments or
[(3.0, 0.1)])]
264     tags = [(f'vg' + f'{v:g}').replace('.', 'p').replace('-', 'm') for v in vgs]
265     ls += ['# PREDICTION (recorded before any ID-VD measurement): ID-VD families, default acceptance
(on-state currents).', relaxed]
266     cur = 0.0
267     for v, tag in zip(vgs, tags):
268         if abs(v - cur) > 1e-9: ls.append(f'solve vstep={0.1 if v > cur else -0.1:g} vfinal={v:.10g}
name=gate'); cur = v
269         ls += [f'log outf=idvd_{tag}.log', 'solve vdrain=0']
270         ls += [f'solve vstep={st_:g} vfinal={e:.10g} name=drain' for e, st_ in vds]
271         ls += ['log off', f'solve vstep=-0.25 vfinal=0 name=drain']
272     ls.append('save outf=final_output.str')
273     for tag in tags:
274         ls += [f'extract init infile="idvd_{tag}.log"',
275                f'extract name="IdVd_{tag}" curve(v."drain",i."drain") outfile="idvd_{tag}.dat"',
276                f'extract name="IsVd_{tag}" curve(v."drain",i."source") outfile="isvd_{tag}.dat"',
277                f'extract name="IgVd_{tag}" curve(v."drain",i."gate") outfile="igvd_{tag}.dat"']
278     ls += ['quit', '']
279     text = '\n'.join(ls).replace('{NUMA}', str(int(s['dos_levels']['numa'])))\
.replace('{NUMD}', str(
int(s['dos_levels']['numd'])))
280     meta = {'thickness_nm': t, 'key': key, 'mode': 'output_prediction', 'output_vgs': vgs, '
output_tags': tags, 'vd_segments': vds,
281            'material': {k: (float(v) if isinstance(v, (int, float, np.floating)) else v) for k, v in
p.items()}, 'Cox_F_cm2': Cox,
282            'dVfb_from_Qf_V': -Q * qf / Cox, 'full_structure': full, 'status': '
DECK_GENERATED_NOT_SIMULATED'}
283     return text, meta
284 step = s['continuation_gate_step_V']
285 if segments:
286     ls.append('# Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).')
287     if vg[0] != 0: ls.append(f'solve vstep={-step if vg[0] < 0 else step:g} vfinal={vg[0]:.10g} name=
gate')
288     ls.append(f'solve vstep=0.1 vfinal={g["drain_bias_V"]:.10g} name=drain')
289 else:
290     if vg[0] != 0:
291         n = int(math.ceil(abs(vg[0]) / step))
292         for v in np.linspace(0, vg[0], n + 1)[1:]: ls.append(f'solve vgate={v:.10g}')
293         for v in np.linspace(0, g['drain_bias_V'], int(math.ceil(g['drain_bias_V'] / .1) + 1)[1:]: ls.
append(f'solve vdrain={v:.10g}')
294 ls.append('log outf=transfer.log'); prev = float(vg[0]); relaxed_done = False; vth_saved = False
295 vth_target = None
296 if meas is not None:
297     mv, mi = meas
298     for i in range(len(mv) - 1):
299         if mi[i] > 0 and mi[i + 1] > 0 and (mi[i] - 1e-9) * (mi[i + 1] - 1e-9) <= 0: vth_target =
float(round(mv[i + 1], 6)); break
300 if segments:
301     # ATLAS stepped sweep: SOLVE [VGATE=<start>] VSTEP=<step> VFINAL=<end> NAME=gate (every step is
logged).
302     # Blocks follow the requested segments and are split at the strict->default METHOD switch and at
the
303     # measured Vth (near_vth.str save). Breakpoints inside a segment use that segment's step.
304     segs = [(float(e), float(st_)) for e, st_ in segments]
305     ls.append('# Stepped sweep (end V, step V): ' + ', '.join(f'({e:g}, {st_:g})' for e, st_ in segs)
+ ' - scored at the solved points only.')

```

```

306     extra = {float(round(switch_v, 6))} if math.isfinite(switch_v) else set()
307     if vth_target is not None: extra.add(float(vth_target))
308     bps = sorted({e for e, _ in segs} | {b for b in extra if vg[0] < b <= vg[-1]})
309     a = float(vg[0]); first = True
310     for b in bps:
311         if b < a - 1e-9: continue
312         stp = next(st_ for e, st_ in segs if b <= e + 1e-9)
313         if b > a + 1e-9 or first:
314             ls.append((f'solve vgate={a:.10g} ' if first else 'solve ') + f'vstep={stp:g} vfinal={b
: .10g} name=gate'); first = False
315             if vth_target is not None and not vth_saved and b >= vth_target - 1e-9: ls.append('save outf=
near_vth.str'); vth_saved = True
316             if not relaxed_done and b >= switch_v - 1e-9: ls += ['# On-state: default convergence
acceptance.', relaxed]; relaxed_done = True
317             a = b
318             prev = float(vg[-1])
319     else:
320         for j, v in enumerate(vg):
321             n = max(1, int(math.ceil(abs(v - prev) / step))); vv = np.linspace(prev, v, n + 1)[1:] if j else
np.array([v])
322             for z in vv:
323                 if not relaxed_done and z >= switch_v - 1e-9: ls += ['# On-state: default convergence
acceptance.', relaxed]; relaxed_done = True
324                 ls.append(f'solve vgate={z:.10g}')
325                 if vth_target is not None and not vth_saved and z >= vth_target - 1e-9: ls.append('save outf=
near_vth.str'); vth_saved = True
326                 prev = float(v)
327     ls += ['log off', 'save outf=final.str', 'output con.band val.band qfn e.mobility', 'extract init
infile="transfer.log"',
328           'extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"', 'extract name="IgVg" curve
(v."gate",i."gate") outfile="igvg.dat"',
329           'extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"',
330           'extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"',
331           'extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"',
332           'extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"']
333     for tag, fn in [(('eq', 'equilibrium.str'), ('vth', 'near_vth.str'), ('on', 'final.str'))]:
334         if tag == 'vth' and vth_target is None: continue
335         ls += [f'extract init infile="{fn}"',
336               f'extract name="cb_{tag}" curve(depth, impurity="Conduction Band Energy" material="All"
mat.occno=1 x.val={xc:.6g}) outfile="cb_{tag}.dat"',
337               f'extract name="vb_{tag}" curve(depth, impurity="Valence Band Energy" material="All" mat.
occno=1 x.val={xc:.6g}) outfile="vb_{tag}.dat"',
338               f'extract name="n_{tag}" curve(depth, impurity="Electron Conc" material="All" mat.occno=1
x.val={xc:.6g}) outfile="n_{tag}.dat"',
339               f'extract name="qfn_{tag}" curve(depth, impurity="Electron QFL" material="All" mat.occno=1
x.val={xc:.6g}) outfile="qfn_{tag}.dat"']
340     ls += ['quit', '']
341     text = '\n'.join(ls).replace('{NUMA}', str(int(s['dos_levels']['numa'])).replace('{NUMD}', str(int(s
['dos_levels']['numd'])))
342     meta = {'thickness_nm': t, 'key': key, 'material': {k: (float(v) if isinstance(v, (int, float, np.
floating)) else v) for k, v in p.items()}, 'Cox_F_cm2': Cox, 'dVfb_from_Qf_V': -Q * qf / Cox,
343           'requested_points': int(len(vg)), 'strict_cr_toler_A': cr_strict, 'strict_up_to_V': switch_v,
'smoke': smoke, 'full_structure': full, 'status': 'DECK_GENERATED_NOT_SIMULATED'}
344     return text, meta
345
346 def read_meas(t):
347     import csv
348     rows = [r for r in csv.DictReader((ROOT / 'data' / 'experimental_clean.csv').open()) if abs(float(r['
thickness_nm']) - t) < 1e-6]
349     return np.array([float(r['vg_V']) for r in rows]), np.array([float(r['id_A_per_um']) for r in rows])
350

```

```
351 def main():
352     ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.
        RawDescriptionHelpFormatter)
353     ap.add_argument('--thickness', type=float, nargs='*', default=[2.0, 6.3, 13.2, 31.8]); ap.
        add_argument('--out', type=Path, default=ROOT / 'decks')
354     ap.add_argument('--override', action='append', default=[], help='dotted path=value into
        iwo_material_model, e.g. traps.fixed_interface_charge_Qf_cm2.value=-2e11')
355     ap.add_argument('--sweep', choices=['config', 'stepped', 'pointwise'], default='config', help='
        stepped = SOLVE VSTEP/VFINAL blocks (solver.yaml sweep.segments; final form); pointwise = one SOLVE
        per 0.05 V point (runs 0001-0009)')
356     ap.add_argument('--structure', choices=['config', 'full', 'reduced'], default='config', help='full =
        physical Air/Pd/Al2O3/HfO2/Conductor volumes (final form); reduced = ideal boundaries (runs
        0001-0009)')
357     ap.add_argument('--smoke', action='store_true'); ap.add_argument('--full', action='store_true', help=
        'alias of --structure full'); a = ap.parse_args()
358     m, g, s = load_configs()
359     for o in a.override:
360         k, v = o.split('=', 1)
361         if k.startswith('solver.'): set_dotted(s, k[7:], v)
362         elif k.startswith('geometry.'): set_dotted(g, k[9:], v)
363         else: set_dotted(m, k, v)
364     full = a.full or (a.structure == 'full') or (a.structure == 'config' and str(g.get('structure', '
        reduced')).lower() == 'full')
365     stepped = (a.sweep == 'stepped') or (a.sweep == 'config' and str(s.get('sweep', {}).get('mode', '
        pointwise')) == 'stepped')
366     segs = [tuple(x) for x in s['sweep']['segments']] if stepped else None
367     a.out.mkdir(parents=True, exist_ok=True)
368     for t in a.thickness:
369         text, meta = build_deck(t, m, g, s, meas=read_meas(t), smoke=a.smoke, full=full, segments=segs)
370         name = f'iwo_{KEYS[t]}nm' + ('_smoke' if a.smoke else '') + (' if full else '_reduced') + (' if
        stepped else '_pointwise')
371         (a.out / f'{name}.in').write_text(text, encoding='ascii'); (a.out / f'{name}_metadata.json').
        write_text(json.dumps(meta, indent=2)); print(a.out / f'{name}.in')
372     if not a.smoke:
373         ttpl, _ = build_deck(2.0, m, g, s, meas=read_meas(2.0), full=full, segments=segs); (a.out / '
        master_template.in').write_text('# MASTER TEMPLATE = the 2.0 nm deck as generated (physical
        structure, stepped sweep); all decks come from the same generator/material model.\n' + ttpl,
        encoding='ascii')
374
375 if __name__ == '__main__': main()
```

B.2.3 run_atlas.py

The script launches one real ATLAS run of a generated deck through DeckBuild in batch mode. It validates the terminal currents (status, KCL, drain sign, native zeros), scores the curve against the measurement with the common extractor and appends one row to results/RUN_INDEX.csv; a Python model is never substituted for the solver. Options select the thickness, label, overrides, sweep segments, structure and timeout. The option `-rescore` repeats validation and scoring from an existing run folder without a launch, and `-output-vgs` adds the I_D - V_D families of the prediction runs (Sections 6.5 and 6.7).

```
1 #!/usr/bin/env python3
2 """Launch ONE real ATLAS run of a generated deck, validate terminal currents, score against the
3 measurement, and append to results/RUN_INDEX.csv. Never substitutes a Python model for the solver.
4
5 python scripts/run_atlas.py --thickness 2 --label v1_baseline [--override k=v ...] [--stride 2] [--vg-
    min -1] [--timeout 1500]
```

```

6   python scripts/run_atlas.py --rescore results/runs/run_0003_...
7
8   Validation (declared): |Id+Is+Ig| <= max(1e-17 A, 0.1 x min measured Id) + 1 % x sum|I|; Id above that
   floor must be
9   positive; every requested gate point exported; ATLAS completion (banner or processed EXTRACT quit); no
   parser/license error.
10  Scoring: measured region masks (all / active / subthreshold / on / low-current) frozen from the
   measurement only;
11  log10 residual = log10(max(pred, 1e-40)) - log10(meas); signed raw currents kept in terminal_currents.csv
   .
12  """
13  from __future__ import annotations
14  import argparse, csv, hashlib, json, re, shutil, subprocess, sys, threading, time
15  from datetime import datetime, timezone
16  from pathlib import Path
17  import numpy as np, yaml
18  sys.path.insert(0, str(Path(__file__).resolve().parent))
19  from build_iwo_decks import ROOT, KEYS, load_configs, set_dotted, build_deck, read_meas
20  from extract_metrics import metrics
21
22  RUNS = ROOT / 'results' / 'runs'; INDEX = ROOT / 'results' / 'RUN_INDEX.csv'
23  FIELDS = ['run_id', 'timestamp_utc', 'thickness_nm', 'label', 'deck_sha256', 'model_sha256', 'command', '
   simulator', 'exit_code', 'converged', 'elapsed_s', 'output_dir', 'notes']
24
25  def sha(s): return hashlib.sha256(s.encode()).hexdigest()
26  def rows(): return list(csv.DictReader(INDEX.open(newline=''))) if INDEX.is_file() else []
27  def next_id():
28      n = 0
29      for r in rows():
30          m = re.match(r'run_(\d+)', r.get('run_id', '')); n = max(n, int(m.group(1))) if m else n
31      return f'run_{n + 1:04d}'
32  def launches(): return len([r['run_id'] for r in rows()])
33  # parallel campaigns (2026-09-28): run IDs, the launch budget and RUN_INDEX writes are guarded by one
   lock;
34  # IDs reserved by running threads count against the budget before their first RUN_INDEX row exists
35  _LOCK = threading.RLock(); _PENDING = set()
36  def append(row):
37      with _LOCK:
38          new = not INDEX.is_file(); INDEX.parent.mkdir(parents=True, exist_ok=True)
39          with INDEX.open('a', newline='') as f:
40              w = csv.DictWriter(f, fieldnames=FIELDS); (w.writeheader() if new else None); w.writerow({k:
   row.get(k, '') for k in FIELDS})
41  def reserve(r, t, label):
42      with _LOCK:
43          used = {x['run_id'] for x in rows()} | _PENDING
44          if len(used) >= int(r['max_launches']): raise RuntimeError(f'launch budget {r["max_launches"]}
   reached; ask before increasing')
45          n = max([int(m.group(1)) for u in used for m in [re.match(r'run_(\d+)', u)] if m] + [0])
46          rid = f'run_{n + 1:04d}'; _PENDING.add(rid)
47          slug = re.sub(r'^A-Za-z0-9+', '_', label).strip('_')[:40]; dest = RUNS / f'{rid}_{KEYS[t]}_{
   slug}'; dest.mkdir(parents=True)
48          return rid, dest
49
50  def read_xy(p: Path):
51      pts = []
52      for line in p.read_text(errors='replace').splitlines():
53          w = line.replace(',', ' ').split()
54          if len(w) != 2: continue
55          try: x, y = (float(z.replace('D', 'E')) for z in w)
56          except ValueError: continue
57          pts.append((x, y))

```

```

58     if len(pts) < 3: raise RuntimeError(f'{p}: fewer than 3 numeric rows')
59     a = np.array(pts)
60     if np.any(np.diff(a[:, 0]) < -1e-9): raise RuntimeError('export is not an increasing sweep')
61     u = {float(x): float(y) for x, y in a}; xs = np.array(sorted(u)); return xs, np.array([u[x] for x in
        xs])
62
63 def scan(text):
64     lines = text.splitlines()
65     warn = [l.strip() for l in lines if re.search(r'\bwarning\b', l, re.I)][:50]
66     flags = [l.strip() for l in lines if re.search(r'fail|diverge|not converge|abort|too many trap', l, re
        .I)][:50]
67     # DeckBuild EXTRACT post-processing lines are excluded from the simulator-error scan (an EXTRACT
        syntax problem does not invalidate the ATLAS solution; missing exports are detected separately).
68     err = [l.strip() for l in lines if not l.strip().upper().startswith('EXTRACT') and any(re.search(p,
        l, re.I) for p in [r'invalid\s+parameter', r'unknown\s+(?:parameter|statement|material)', r'license
        .*(?:denied|failed|unavailable)', r'cannot\s+open', r'error\s*#', r'fatal\s+error', r'syntax error',
        r'unrecognized', r'memory allocation failure'])[:50]
69     return {'warnings': warn, 'convergence_flags': flags, 'errors': err, 'banner': bool(re.search(r'ATLAS
        version .* finished', text)), 'finished': bool(re.search(r'ATLAS version .* finished', text) or re.
        search(r'EXTRACT>\s*quit', text)), 'static_solutions': len(re.findall(r'Obtaining static solution',
        text))}
70
71 def regions(vg, meas, t, mult):
72     o = float(np.median(meas[(vg >= -2) & (vg <= -.5)])); active = meas > mult * o if t < 20 else np.ones
        (len(meas), bool)
73     on = active & (meas >= 1e-2 * meas.max()); return {'all': np.ones(len(meas), bool), 'active': active,
        'subthreshold': active & ~on, 'on': on, 'low_current': ~active}, o
74
75 def score(vg, meas, pred, t, s):
76     masks, o = regions(vg, meas, t, s['scoring']['active_multiplier']); eps = s['scoring']['
        logging_epsilon_A_per_um']
77     le = np.log10(np.maximum(pred, eps)) - np.log10(meas); lin = pred - meas
78     def reg(m):
79         if not m.any(): return {'count': 0}
80         return {'count': int(m.sum()), 'rmse_log10': float(np.sqrt(np.mean(le[m] ** 2))), 'max_abs_log10'
            : float(abs(le[m]).max()), 'mean_log10': float(le[m].mean()), 'rmse_linear_A_per_um': float(np.sqrt(
            np.mean(lin[m] ** 2))), 'median_abs_relative': float(np.median(abs(pred[m] / meas[m] - 1)))}
81     return {'regions': {k: reg(m) for k, m in masks.items()}, 'on_error_3V_percent': float(100 * (pred
        [-1] / meas[-1] - 1)), 'nonpositive_pred_count': int((pred <= 0).sum()), 'off_band_median_A_per_um':
        o,
82         'max_abs_log10_active': float(abs(le[masks['active']]).max()), 'log_treatment': f'log10(max(
        pred,{eps:g})) - log10(meas)', masks['active'], le
83
84 def plots(dest, vg, meas, pred, t, title):
85     try:
86         import matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
87     except Exception: return
88     fig, ax = plt.subplots(1, 3, figsize=(14, 4)); e = 1e-40
89     ax[0].semilogy(vg, meas, 'k.', ms=4, label='measured (workbook)'); ax[0].semilogy(vg, np.maximum(pred
        , e), 'r-', lw=1.2, label='ATLAS (this run)'); ax[0].set_ylim(1e-17, 1e-5); ax[0].set_title(f'{t} nm
        log | {title}')
90     ax[1].plot(vg, meas * 1e6, 'k.', ms=4, label='measured'); ax[1].plot(vg, pred * 1e6, 'r-', lw=1.2,
        label='ATLAS'); ax[1].set_ylabel('Id (uA/um)'); ax[1].set_title('linear')
91     ax[2].plot(vg, np.log10(np.maximum(pred, e)) - np.log10(meas), 'b.-', ms=3, lw=.8); ax[2].axhline(0,
        color='k', lw=.6); ax[2].set_ylim(-1, 1); ax[2].set_ylabel('log10(ATLAS/measured)'); ax[2].set_title
        ('residual (clipped to +/-1 dec)')
92     for a in ax: a.set_xlabel('Vg (V)'); a.grid(alpha=.3)
93     ax[0].legend(fontsize=8); ax[1].legend(fontsize=8); fig.tight_layout(); fig.savefig(dest / '
        overlay_residual.png', dpi=130); plt.close(fig)
94
95 def meas_floor(vg, meas):

```



```

96     """Measured off-band floor: median Id over Vg in [-2, -0.5] V (robust; the single-point minimum of
97     the 2 nm
98     curve, 1.1e-16 A/um at -3 V, is an outlier 40x below its own floor of 4.6e-15)."""
99     return float(np.median(meas[(vg >= -2) & (vg <= -0.5)]))
100
101 def validate(dest, xv, idr, isr, igr, floor, s):
102     r = s['runner']; abs_lim = max(float(r['kcl_absolute_limit_A']), float(r['kcl_floor_fraction']) *
103     floor); rel = float(r['kcl_relative_limit']); sign = float(r['expected_drain_sign'])
104     imb = np.abs(idr + isr + igr); sc = abs(idr) + abs(isr) + abs(igr); ok = imb <= abs_lim + rel * sc;
105     above = abs(idr) > abs_lim; signed = sign * idr; wrong = above & (signed < 0)
106     with (dest / 'terminal_currents.csv').open('w', newline='') as f:
107         w = csv.writer(f); w.writerow(['vg_V', 'id_raw_A', 'is_raw_A', 'ig_raw_A', 'id_A_per_um', '
108         kcl_abs_A', 'kcl_pass', 'above_abs_floor']); w.writerows(zip(xv, idr, isr, igr, signed, imb, ok.
109         astype(int), above.astype(int)))
110     return signed, {'kcl_abs_limit_applied_A': abs_lim, 'kcl_floor_basis': 'max(1e-17 A, 0.1 x median
111     measured Id over Vg in [-2,-0.5] V)', 'measured_floor_A_per_um': floor, 'max_kcl_abs_A': float(imb.
112     max()), 'kcl_fail_count': int((~ok).sum()), 'kcl_fail_vg': [float(v) for v in xv[~ok][:20], '
113     wrong_sign_count': int(wrong.sum()), 'nonpositive_count': int((signed <= 0).sum())}, ('SIGN_REVERSAL
114     ' if wrong.any() else 'KCL_FAILED' if not ok.all() else 'OK')
115
116 def parse_segments(text):
117     """-1:0.2,1.5:0.05,3:0.1 -> [(end, step), ...]; 'none' -> None (per-point solves)."""
118     if text is None or str(text).lower() in ('none', 'off', ''): return None
119     return [(float(a), float(b)) for a, b in (seg.split(':') for seg in str(text).split(',') )]]
120
121 def execute(t, label, overrides, stride=1, vg_min=None, timeout=None, smoke=False, vstep=None, segments='
122     config', full='config', predict=False):
123     """segments: [(end V, step V), ...] for ATLAS stepped SOLVE VSTEP/VFINAL/NAME=gate blocks ('config' =
124     solver.yaml
125     sweep.segments; None = one SOLVE per 0.05 V point). vstep: single-step shorthand. full: physical
126     structure
127     (Air / Pd volumes / Al2O3 / HfO2 / Conductor gate; 'config' = geometry.yaml structure == 'full').
128     Steps must be multiples of the 0.05 V measured grid; the run is scored at the solved points only.
129     predict (2026-10-06): no measured curve (any thickness); the Vg grid comes from geometry.yaml, KCL is
130     validated with
131     the 1e-17 A absolute limit, nothing is scored, and the curve is written to prediction.csv."""
132     m, g, s = load_configs()
133     for o in overrides:
134         k, v = o.split('=', 1) # 'solver.' / 'geometry.' prefixes address solver.yaml / geometry.yaml (
135         numerical checks); else the material model
136         if k.startswith('solver.'): set_dotted(s, k[7:], v)
137         elif k.startswith('geometry.'): set_dotted(g, k[9:], v)
138         else: set_dotted(m, k, v)
139     if full == 'config': full = str(g.get('structure', 'reduced')).lower() == 'full'
140     if segments == 'config': segments = [tuple(x) for x in s['sweep']['segments']] if str(s.get('sweep',
141     {})).get('mode', 'pointwise') == 'stepped' else None
142     if vstep: segments = [(float(g['gate_sweep_V']['stop']), float(vstep))]
143     if predict:
144         sw = g['gate_sweep_V']; vg = np.round(np.arange(sw['start'], sw['stop'] + 1e-9, sw['step']), 6);
145         meas = None
146     else:
147         vg, meas = read_meas(t)
148     base = vg[vg >= vg_min - 1e-9] if vg_min is not None else vg
149     if segments:
150         pts = [float(base[0])]; a = float(base[0])
151         for e, st_ in segments:
152             if abs(st_ / 0.05 - round(st_ / 0.05)) > 1e-6: raise ValueError('segment steps must be
153             multiples of the 0.05 V measured grid')
154             n = int(round((e - a) / st_)); pts += [round(a + i * st_, 6) for i in range(1, n + 1)]; a =
155             float(e)
156         targets = np.array(sorted(set(pts) & set(np.round(base, 6).tolist())))

```

```

139     else:
140         targets = base[:,stride] if stride > 1 else base
141         if stride > 1 and targets[-1] != vg[-1]: targets = np.append(targets, vg[-1])
142     r = s['runner']
143     rid, dest = reserve(r, t, label)
144     text, meta = build_deck(t, m, g, s, meas=None if predict else (vg, meas), targets=targets, smoke=
145         smoke, full=full, segments=segments); (dest / 'device.in').write_text(text, encoding='ascii')
146     mtxt = json.dumps(m, indent=2, default=float); (dest / 'material_model_used.json').write_text(mtxt);
147     (dest / 'metadata.json').write_text(json.dumps(meta, indent=2, default=float))
148     argv = [z.replace('{deck}', 'device.in').replace('{stdout}', 'deckbuild.out') for z in r['argv']]
149     if not Path(argv[0]).is_file() and shutil.which(argv[0]) is None: raise RuntimeError(f'executable not
150         found: {argv[0]}')
151     to = timeout or (r['timeout_seconds']['thick'] if t > 20 else r['timeout_seconds']['default'])
152     import build_iwo_decks as _b
153     row = {'run_id': rid, 'timestamp_utc': datetime.now(timezone.utc).strftime('%Y-%m-%dT%H:%M:%SZ'), '
154         thickness_nm': t, 'label': label, 'deck_sha256': sha(text), 'model_sha256': sha(mtxt), 'command': '
155         '.join(argv), 'simulator': 'DeckBuild 5.0.10.R + ATLAS 5.28.1.R', 'output_dir': str(dest.relative_to
156         (ROOT)).replace('\\', '/'), 'notes': f'STARTED points={len(targets)} timeout={to}s launch {launches
157         () + 1}/{r["max_launches"]}' + (' if _b.MODEL_FILE == 'config/iwo_material_model.yaml' else f'
158         model={_b.MODEL_FILE}') + (' PREDICTION (no measurement)' if predict else '')}
159     append(row); t0 = time.monotonic()
160     p = subprocess.Popen(argv, cwd=dest, stdout=subprocess.PIPE, stderr=subprocess.PIPE, text=True)
161     try: out, err = p.communicate(timeout=to); timed_out = False
162     except subprocess.TimeoutExpired:
163         subprocess.run(['taskkill', '/T', '/F', '/PID', str(p.pid)], capture_output=True); out, err = p.
164         communicate(); timed_out = True
165     el = time.monotonic() - t0; (dest / 'launcher_stdout.txt').write_text(out or ''); (dest / '
166         launcher_stderr.txt').write_text(err or '')
167     tr = (dest / 'deckbuild.out').read_text(errors='replace') if (dest / 'deckbuild.out').exists() else '
168         '; sc = scan((out or '') + (err or '')) + tr
169     ex = {'run_id': rid, 'thickness_nm': t, 'label': label, 'elapsed_s': el, 'returncode': p.returncode,
170         'timed_out': timed_out, 'scan': sc, 'requested_points': int(len(targets)), 'overrides': overrides}
171     def finish(status, conv, note):
172         ex['status'] = status; (dest / 'execution.json').write_text(json.dumps(ex, indent=2, default=
173             float)); append({'*row', 'timestamp_utc': datetime.now(timezone.utc).strftime('%Y-%m-%dT%H:%M:%SZ'),
174             'exit_code': 'timeout' if timed_out else p.returncode, 'converged': conv, 'elapsed_s': f'{el:.1f}',
175             'notes': note})
176     if sc['errors']: finish('SIMULATOR_ERROR', 'no', sc['errors'][0][:120]); raise RuntimeError(f'{rid}:
177         {sc["errors"][0]}')
178     if not (dest / 'idvg.dat').exists(): finish('TIMEOUT' if timed_out else 'NO_EXPORT', 'no', 'no idvg.
179         dat'); raise RuntimeError(f'{rid}: no export (timeout={timed_out}); inspect {dest}')
180     xv, idr = read_xy(dest / 'idvg.dat'); xg, igr = read_xy(dest / 'igvg.dat'); xs, isr = read_xy(dest /
181         'isvg.dat'); igr = np.interp(xv, xg, igr); isr = np.interp(xv, xs, isr)
182     if np.min(abs(xv[:, None] - targets[None, :]), axis=0).max() > 1e-6: finish('INCOMPLETE_EXPORT', '
183         partial', 'missing requested gate points'); raise RuntimeError(f'{rid}: incomplete export')
184     signed, v, verdict = validate(dest, xv, idr, isr, igr, 0.0 if predict else meas_floor(vg, meas), s);
185     ex['validation'] = v; ex['scan']['banner_present'] = sc['banner']
186     if verdict != 'OK': finish(verdict, 'yes', f'{verdict}: max KCL {v["max_kcl_abs_A"]:.2g} A'); raise
187         RuntimeError(f'{rid}: {verdict}; inspect {dest}/terminal_currents.csv')
188     if predict:
189         ex['device_metrics_sim'] = metrics(xv, signed, True, None)
190         with (dest / 'prediction.csv').open('w', newline='') as f:
191             w = csv.writer(f); w.writerow(['vg_V', 'atlas_A_per_um']); w.writerows(zip(xv, signed))
192         dm = ex['device_metrics_sim']
193         finish('REAL_ATLAS_PREDICTION', 'yes' if not sc['convergence_flags'] else 'yes-with-flags', f'
194             PREDICTION Vth_cc {dm["Vth_cc_1e-9_V"]}; Id(3V) {dm["Ion_A_per_um"]:.3e}; maxKCL {v["max_kcl_abs_A
195             "]:.1e}')
196     print(json.dumps({'run_id': rid, 'Vth_cc': dm['Vth_cc_1e-9_V'], 'Ion': dm['Ion_A_per_um'], '
197         maxKCL_A': v['max_kcl_abs_A'], 'flags': len(sc['convergence_flags']), 'elapsed_s': round(el)},
198         indent=1))
199     return dest

```

```

175     if vg_min is None:
176         pred = np.interp(vg, xv, signed); met, active, le = score(vg, meas, pred, t, s); ex['fit_metrics']
    ] = met
177         ex['device_metrics_sim'] = metrics(vg, pred, True, t); ex['device_metrics_exp'] = metrics(vg,
    meas, False, t)
178         with (dest / 'comparison.csv').open('w', newline='') as f:
179             w = csv.writer(f); w.writerow(['vg_V', 'measured_A_per_um', 'atlas_A_per_um', 'log10_error',
    'linear_error_A_per_um', 'active_fit_region']); w.writerows(zip(vg, meas, pred, le, pred - meas,
    active.astype(int)))
180         plots(dest, vg, meas, pred, t, rid)
181         finish('REAL_ATLAS_SCORED', 'yes' if not sc['convergence_flags'] else 'yes-with-flags', f'active
    RMSE {met["regions"]["active"]["rmse_log10"]:.4f} dec; Id(3V) {met["on_error_3V_percent"]:+.2f}%;
    nonpos {met["nonpositive_pred_count"]}; maxKCL {v["max_kcl_abs_A"]:.1e}')
182         print(json.dumps({'run_id': rid, 'active_rmse_log10': met['regions']['active']['rmse_log10'], '
    active_max': met['max_abs_log10_active'], 'sub_rmse': met['regions']['subthreshold'].get('rmse_log10
    '), 'on_rmse': met['regions']['on'].get('rmse_log10'), 'Id3V_err_pct': met['on_error_3V_percent'], '
    nonpositive': met['nonpositive_pred_count'], 'maxKCL_A': v['max_kcl_abs_A'], 'flags': len(sc['
    convergence_flags'])}, 'elapsed_s': round(el)), indent=1))
183     else:
184         finish('REAL_ATLAS_PARTIAL_SWEEP', 'yes' if not sc['convergence_flags'] else 'yes-with-flags', f'
    partial from {vg_min} V; maxKCL {v["max_kcl_abs_A"]:.1e}'); print(rid, 'partial sweep done', dest)
185     return dest
186
187 def execute_output(t, label, overrides, vgs, vd_segments, timeout=None):
188     """PREDICTION launch (2026-09-25): ID-VD families at fixed Vg with the same material model/structure
    as the
189     transfer runs. No measured counterpart exists, so nothing is scored; terminal-current KCL is
    validated per point
190     (limit 1e-17 A + 1 % of sum|I|) and the curves are written to output_characteristics.csv."""
191     m, g, s = load_configs()
192     for o in overrides:
193         k, v = o.split('=', 1)
194         if k.startswith('solver.'): set_dotted(s, k[7:], v)
195         elif k.startswith('geometry.'): set_dotted(g, k[9:], v)
196         else: set_dotted(m, k, v)
197     full = str(g.get('structure', 'reduced')).lower() == 'full'; r = s['runner']
198     rid, dest = reserve(r, t, label)
199     text, meta = build_deck(t, m, g, s, meas=read_meas(t), full=full, output_vgs=vgs, vd_segments=
    vd_segments); (dest / 'device.in').write_text(text, encoding='ascii')
200     mtxt = json.dumps(m, indent=2, default=float); (dest / 'material_model_used.json').write_text(mtxt);
    (dest / 'metadata.json').write_text(json.dumps(meta, indent=2, default=float))
201     argv = [z.replace('{deck}', 'device.in').replace('{stdout}', 'deckbuild.out') for z in r['argv']]
202     to = timeout or r['timeout_seconds']['thick']
203     row = {'run_id': rid, 'timestamp_utc': datetime.now(timezone.utc).strftime('%Y-%m-%dT%H:%M:%SZ'), '
    thickness_nm': t, 'label': label, 'deck_sha256': sha(text), 'model_sha256': sha(mtxt), 'command': '
    '.join(argv), 'simulator': 'DeckBuild 5.0.10.R + ATLAS 5.28.1.R', 'output_dir': str(dest.relative_to
    (ROOT)).replace('\\', '/'), 'notes': f'STARTED output prediction Vg={vgs} timeout={to}s launch {
    launches() + 1}/{r["max_launches"]}'}
204     append(row); t0 = time.monotonic()
205     p = subprocess.Popen(argv, cwd=dest, stdout=subprocess.PIPE, stderr=subprocess.PIPE, text=True)
206     try: out, err = p.communicate(timeout=to); timed_out = False
207     except subprocess.TimeoutExpired:
208         subprocess.run(['taskkill', '/T', '/F', '/PID', str(p.pid)], capture_output=True); out, err = p.
    communicate(); timed_out = True
209     el = time.monotonic() - t0; (dest / 'launcher_stdout.txt').write_text(out or ''); (dest / '
    launcher_stderr.txt').write_text(err or '')
210     tr = (dest / 'deckbuild.out').read_text(errors='replace') if (dest / 'deckbuild.out').exists() else '
    '; sc = scan((out or '') + (err or '') + tr)
211     ex = {'run_id': rid, 'thickness_nm': t, 'label': label, 'mode': 'output_prediction', 'elapsed_s': el,
    'returncode': p.returncode, 'timed_out': timed_out, 'scan': sc, 'overrides': overrides, 'output_vgs
    ': meta['output_vgs']}

```

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212     def finish(status, conv, note):
213         ex['status'] = status; (dest / 'execution.json').write_text(json.dumps(ex, indent=2, default=
float)); append(**row, 'timestamp_utc': datetime.now(timezone.utc).strftime('%Y-%m-%dT%H:%M:%SZ'),
'exit_code': 'timeout' if timed_out else p.returncode, 'converged': conv, 'elapsed_s': f'{el:.1f}',
'notes': note))
214     if sc['errors']: finish('SIMULATOR_ERROR', 'no', sc['errors'][0][:120]); raise RuntimeError(f'{rid}:
{sc["errors"][0]}')
215     rows_out, worst, fails = [], 0.0, 0
216     for v, tag in zip(meta['output_vgs'], meta['output_tags']):
217         if not (dest / f'idvd_{tag}.dat').exists(): finish('TIMEOUT' if timed_out else 'NO_EXPORT', 'no',
f'no idvd_{tag}.dat'); raise RuntimeError(f'{rid}: no export for {tag}')
218         xv, idr = read_xy(dest / f'idvd_{tag}.dat'); xs, isr = read_xy(dest / f'isvd_{tag}.dat'); xg, igr
= read_xy(dest / f'igvd_{tag}.dat')
219         isr = np.interp(xv, xs, isr); igr = np.interp(xv, xg, igr); imb = np.abs(idr + isr + igr); ok =
imb <= float(r['kcl_absolute_limit_A']) + float(r['kcl_relative_limit']) * (abs(idr) + abs(isr) +
abs(igr))
220         worst = max(worst, float(imb.max())); fails += int((~ok).sum())
221         rows_out += [(v, vd, idv, isv, igv, int(k)) for vd, idv, isv, igv, k in zip(xv, idr, isr, igr, ok
)]
222     with (dest / 'output_characteristics.csv').open('w', newline='') as f:
223         w = csv.writer(f); w.writerow(['vg_V', 'vd_V', 'id_A_per_um', 'is_A_per_um', 'ig_A_per_um', '
kcl_pass']); w.writerows(rows_out)
224     try:
225         import matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
226         fig, ax = plt.subplots(figsize=(6, 4.5)); a = np.array([r[:3] for r_ in rows_out])
227         for v in meta['output_vgs']: sel = a[:, 0] == v; ax.plot(a[sel, 1], a[sel, 2] * 1e6, '-o', ms
=2.5, label=f'Vg = {v:g} V')
228         ax.set_xlabel('Vd (V)'); ax.set_ylabel('Id (uA/um)'); ax.set_title(f'{t} nm ID-VD PREDICTION ({
rid})'); ax.grid(alpha=.3); ax.legend(fontsize=8); fig.tight_layout(); fig.savefig(dest / '
idvd_prediction.png', dpi=130); plt.close(fig)
229     except Exception: pass
230     ex['validation'] = {'max_kcl_abs_A': worst, 'kcl_fail_count': fails}
231     status = 'REAL_ATLAS_PREDICTION' if fails == 0 else 'KCL_FAILED'
232     finish(status, 'yes' if not sc['convergence_flags'] else 'yes-with-flags', f'{status}: ID-VD Vg={meta
["output_vgs"]}; points {len(rows_out)}; maxKCL {worst:.1e}')
233     print(json.dumps({'run_id': rid, 'status': status, 'points': len(rows_out), 'maxKCL_A': worst, '
elapsed_s': round(el)}, indent=1))
234     if fails: raise RuntimeError(f'{rid}: KCL failed at {fails} points; inspect {dest}/
output_characteristics.csv')
235     return dest
236
237 def rescore(run_dir, reason):
238     """Redo validation + scoring of an existing run from its exported currents (no simulator call). The
original
239     execution.json is kept as execution_before_rescore.json and a RESCORED row is appended to RUN_INDEX.
csv."""
240     dest = Path(run_dir) if Path(run_dir).is_absolute() else ROOT / run_dir
241     ex = json.loads((dest / 'execution.json').read_text()); t = float(ex['thickness_nm']); rid = ex['
run_id']
242     if not (dest / 'execution_before_rescore.json').exists(): shutil.copy(dest / 'execution.json', dest /
'execution_before_rescore.json')
243     m, g, s = load_configs(); vg, meas = read_meas(t)
244     xv, idr = read_xy(dest / 'idvg.dat'); xg, igr = read_xy(dest / 'igvg.dat'); xs, isr = read_xy(dest /
'isvg.dat'); igr = np.interp(xv, xg, igr); isr = np.interp(xv, xs, isr)
245     signed, v, verdict = validate(dest, xv, idr, isr, igr, meas_floor(vg, meas), s); ex['validation'] = v
; ex['rescore'] = {'reason': reason, 'timestamp_utc': datetime.now(timezone.utc).strftime('%Y-%m-%dT
%H:%M:%SZ'), 'previous_status': ex.get('status')}
246     row = {'run_id': rid, 'timestamp_utc': ex['rescore']['timestamp_utc'], 'thickness_nm': t, 'label': ex
['label'], 'deck_sha256': sha((dest / 'device.in').read_text()), 'model_sha256': sha((dest / '
material_model_used.json').read_text()),
247         'command': 'RESCORE (no simulator call)', 'simulator': 'DeckBuild 5.0.10.R + ATLAS 5.28.1.R',

```

```

        'exit_code': ex.get('returncode'), 'converged': 'yes', 'elapsed_s': f'{ex.get("elapsed_s", 0):.1f}',
        'output_dir': str(dest.relative_to(ROOT)).replace('\\', '/')
248     if verdict != 'OK':
249         ex['status'] = verdict; (dest / 'execution.json').write_text(json.dumps(ex, indent=2, default=
            float)); append(**row, 'notes': f'RESCORED ({reason}): still {verdict}, max KCL {v["max_kcl_abs_A
                "]:.2g} A'); raise RuntimeError(f'{rid}: {verdict} after rescore')
250     pred = np.interp(vg, xv, signed); met, active, le = score(vg, meas, pred, t, s); ex['fit_metrics'] =
        met
251     ex['device_metrics_sim'] = metrics(vg, pred, True, t); ex['device_metrics_exp'] = metrics(vg, meas,
        False, t); ex['status'] = 'REAL_ATLAS_SCORED'
252     with (dest / 'comparison.csv').open('w', newline='') as f:
253         w = csv.writer(f); w.writerow(['vg_V', 'measured_A_per_um', 'atlas_A_per_um', 'log10_error', '
            linear_error_A_per_um', 'active_fit_region']); w.writerows(zip(vg, meas, pred, le, pred - meas,
                active.astype(int)))
254     plots(dest, vg, meas, pred, t, rid); (dest / 'execution.json').write_text(json.dumps(ex, indent=2,
        default=float))
255     note = f'RESCORED ({reason}): active RMSE {met["regions"]["active"]["rmse_log10"]:.4f} dec; Id(3V) {
        met["on_error_3V_percent"]:+.2f}%; nonpos {met["nonpositive_pred_count"]}; maxKCL {v["max_kcl_abs_A
            "]:.1e}'
256     append(**row, 'notes': note); print(note); return dest
257
258 def main():
259     ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.
        RawDescriptionHelpFormatter)
260     ap.add_argument('--rescore', help='existing run directory: redo validation/scoring from its exported
        currents (no launch)'); ap.add_argument('--reason', default='rescore')
261     ap.add_argument('--thickness', type=float); ap.add_argument('--label', default='run'); ap.
        add_argument('--override', action='append', default=[])
262     ap.add_argument('--predict', action='store_true', help='PREDICTION transfer run (2026-10-06): any
        thickness, no measured curve, nothing scored')
263     ap.add_argument('--stride', type=int, default=1); ap.add_argument('--vg-min', type=float); ap.
        add_argument('--timeout', type=int); ap.add_argument('--smoke', action='store_true')
264     ap.add_argument('--vstep', type=float, help='single-step ATLAS stepped sweep, e.g. 0.1 (overrides --
        segments)')
265     ap.add_argument('--segments', default='config', help="stepped sweep 'end:step,end:step,...' e.g.
        '-1:0.2,1.5:0.05,3:0.1'; 'none' = per-point; default from solver.yaml")
266     ap.add_argument('--structure', choices=['config', 'full', 'reduced'], default='config', help='full =
        physical metal/dielectric volumes; default from geometry.yaml')
267     ap.add_argument('--output-vgs', help="PREDICTION mode: ID-VD families at these gate voltages, e.g.
        '1,2,3'")
268     ap.add_argument('--vd-segments', default='0.5:0.05,3:0.1', help="drain sweep 'end:step,...' from 0 V
        for --output-vgs"); a = ap.parse_args()
269     if a.rescore: rescore(a.rescore, a.reason); return
270     if a.thickness is None: ap.error('--thickness required')
271     if not a.predict and a.thickness not in (2.0, 6.3, 13.2, 31.8): ap.error('no measured curve for this
        thickness: use --predict')
272     if a.output_vgs:
273         execute_output(a.thickness, a.label, a.override, [float(v) for v in a.output_vgs.split(',')],
            parse_segments(a.vd_segments), a.timeout); return
274     execute(a.thickness, a.label, a.override, a.stride, a.vg_min, a.timeout, a.smoke, a.vstep,
        segments=('config' if a.segments == 'config' else parse_segments(a.segments)), full=('config'
        if a.structure == 'config' else a.structure == 'full'),
275         predict=a.predict)
276
277
278 if __name__ == '__main__':
279     try: main()
280     except RuntimeError as e: print('FAILED:', e, file=sys.stderr); sys.exit(2)

```

B.2.4 run_campaign.py

The script runs a sequential ATLAS campaign from a JSON plan (the 2026-09-25 hypothesis, convergence, sensitivity and prediction study), e.g. `python scripts/run_campaign.py config/campaign_A_6p3_hypotheses.json sequential`. Each plan entry is launched through `run_atlas.py`, one at a time, and the metrics of each run are collected in `results/campaigns/<name>.json`.

```

1  #!/usr/bin/env python3
2  """ATLAS campaign from a JSON plan (2026-09-25 hypothesis / convergence / sensitivity / prediction study)
   .
3
4  python scripts/run_campaign.py config/campaign_A_6p3_hypotheses.json sequential
5  python scripts/run_campaign.py config/campaign_A_6p3_hypotheses.json --workers 4 4 ATLAS runs in
   parallel
6  (parallel added 2026-09-28: each run has its own directory; run IDs, the launch budget and RUN_INDEX
   writes are
7  locked in run_atlas.reserve/append; entries using a {mu:key} placeholder start only after the run
   producing it)
8
9  Plan: {"campaign": name, "purpose": text, "runs": [entry, ...]}; entry keys:
10  id, thickness, label, overrides [...], mode "transfer" (default) | "output", output_vgs [...],
   vd_segments [[end, step], ...],
11  timeout (s), purpose (text), mu_used + recal_key (optional: after the run, mu_band is rescaled EXACTLY
   by
12  Ion_meas / Ion_sim - Id is linear in a constant mobility, tables/SENSITIVITY_RESULTS.md mu_band pairs -
   and stored as
13  the placeholder {mu:<recal_key>} for later entries of the same campaign).
14  Every launch goes through run_atlas.execute / execute_output (same deck generator, validation, RUN_INDEX
   row). The
15  summary results/campaigns/<plan stem>.json is rewritten after every launch; a failed launch is recorded
   and the
16  campaign continues; the campaign stops at the launch budget.
17  """
18  from __future__ import annotations
19  import json, sys, time, traceback
20  from pathlib import Path
21  sys.path.insert(0, str(Path(__file__).resolve().parent))
22  from run_atlas import execute, execute_output, ROOT
23
24  def fill(overrides, ph):
25      out = []
26      for o in overrides:
27          for k, v in ph.items(): o = o.replace('{mu:' + k + '}', f'{v:.6g}')
28          if '{mu:' in o: raise RuntimeError(f'unresolved placeholder in {o}')
29          out.append(o)
30      return out
31
32  def summarize(dest):
33      ex = json.loads((Path(dest) / 'execution.json').read_text()); s = {'status': ex.get('status'), '
   elapsed_s': round(ex.get('elapsed_s', 0), 1)}
34      if 'fit_metrics' in ex:
35          fm = ex['fit_metrics']; rg = fm['regions']
36          s.update({'active_rmse_dec': rg['active'].get('rmse_log10'), 'sub_rmse_dec': rg['subthreshold'].
   get('rmse_log10'), 'on_rmse_dec': rg['on'].get('rmse_log10'),
37                  'Id3V_err_pct': fm['on_error_3V_percent'], 'sim': ex['device_metrics_sim'], 'exp': ex['
   device_metrics_exp']})
38      if 'validation' in ex: s['max_kcl_A'] = ex['validation'].get('max_kcl_abs_A')
39      return s
40
41  def run_entry(e, res, lock):

```

```

42     rec = {'id': e['id'], 'thickness': e['thickness'], 'label': e['label'], 'purpose': e.get('purpose', '
    '), 'mode': e.get('mode', 'transfer')}
43     t0 = time.time()
44     try:
45         with lock: ov = fill(e.get('overrides', []), res['placeholders'])
46         rec['overrides'] = ov
47         if e.get('mode') == 'output':
48             dest = execute_output(float(e['thickness']), e['label'], ov, e['output_vgs'], [tuple(x) for x
    in e.get('vd_segments', [[0.5, 0.05], [3.0, 0.1]])], e.get('timeout'))
49         else:
50             dest = execute(float(e['thickness']), e['label'], ov, timeout=e.get('timeout'))
51         rec['run_dir'] = str(Path(dest).relative_to(ROOT)).replace('\\', '/'); rec['run_id'] = Path(dest)
    .name[:8]; rec.update(summarize(dest))
52         if e.get('recal_key') and 'sim' in rec:
53             mu_new = float(e['mu_used']) * rec['exp']['Ion_A_per_um'] / rec['sim']['Ion_A_per_um']
54             with lock: res['placeholders'][e['recal_key']] = mu_new
55             rec['mu_recalibrated'] = mu_new
56     except Exception as ex:
57         rec['error'] = f'{type(ex).__name__}: {ex}'; rec['traceback'] = traceback.format_exc()[-1500:]
58     rec['wall_s'] = round(time.time() - t0, 1)
59     return rec
60
61 def ready(e, res):
62     """an entry can start when every {mu:key} placeholder it uses has been produced by an earlier run"""
63     return all('{mu:' + k + '}' not in o or k in res['placeholders'] for o in e.get('overrides', []) for
    k in
64         [p.split('}')[0] for p in o.split('{mu:')[1:] or []])
65
66 def main():
67     """python run_campaign.py PLAN.json [--workers N]    (N parallel ATLAS processes, default 1 =
    sequential)"""
68     import threading
69     from concurrent.futures import ThreadPoolExecutor, FIRST_COMPLETED, wait
70     args = sys.argv[1:]; workers = 1
71     if '--workers' in args: i = args.index('--workers'); workers = int(args[i + 1]); del args[i:i + 2]
72     plan_path = Path(args[0]); plan = json.loads(plan_path.read_text())
73     out_dir = ROOT / 'results' / 'campaigns'; out_dir.mkdir(parents=True, exist_ok=True); out = out_dir /
    f'{plan_path.stem}.json'
74     res = {'campaign': plan.get('campaign', plan_path.stem), 'purpose': plan.get('purpose', ''), '
    plan_file': str(plan_path), 'placeholders': {}, 'runs': []}
75     if out.exists(): # resume: keep finished entries, skip them
76         res = json.loads(out.read_text())
77     done = {r['id'] for r in res['runs'] if r.get('run_dir')}
78     todo = [e for e in plan['runs'] if e['id'] not in done]; lock = threading.Lock(); running = {}; stop
    = False
79     with ThreadPoolExecutor(max_workers=workers) as pool:
80         while (todo and not stop) or running:
81             for e in [e for e in todo if ready(e, res)][max(0, workers - len(running))]: if not stop else
    []:
82                 todo.remove(e); running[pool.submit(run_entry, e, res, lock)] = e
83             if not running:
84                 if todo: print('STOP: remaining entries wait for placeholders that no run produced');
    break
85             continue
86     fin, _ = wait(list(running), return_when=FIRST_COMPLETED)
87     for f in fin:
88         running.pop(f); rec = f.result()
89         with lock:
90             res['runs'].append(rec); out.write_text(json.dumps(res, indent=1, default=float))
91         print(json.dumps({k: rec.get(k) for k in ('id', 'run_id', 'status', 'active_rmse_dec', '
    Id3V_err_pct', 'error', 'wall_s')}), flush=True)

```

```

92         if 'launch budget' in rec.get('error', ''): stop = True; print('STOP: budget')
93     print('CAMPAIGN DONE', out)
94
95 if __name__ == '__main__': main()

```

B.2.5 extract_metrics.py

This is the single metric extractor, applied identically to measured and simulated I_D - V_G curves; no simulator is called. Its docstring states the conventions used for $V_{th,cc}$, $V_{th,lin}$, the swings, $g_{m,max}$, μ_{FE} and I_{on} , because the target paper does not define its extraction (Section 5.10). Called with `-thickness`, it prints the metrics of a measured curve; with `-sim`, it applies the simulation conventions.

```

1  #!/usr/bin/env python3
2  """Identical metric extraction for experimental and simulated ID-VG curves (no simulator call).
3
4  Definitions (the target paper does not define its extraction; these are stated conventions):
5      Vth_cc      : constant-current threshold, Vg where Id = 1e-9 A/um (log-linear interpolation between
6                    adjacent points; no extrapolation). 1e-9 A/um x (L/W = 20/290) = 6.9e-11 A per square...
7                    reported as A/um.
8      Vth_lin     : linear extrapolation of Id at maximum transconductance (secant gm on the 0.05 V grid), Vd
9                    = 0.7 V.
10     SS           : two definitions, both applied identically to measured and simulated curves:
11                    SS_min (primary, key SS_mV_dec): minimum over 0.2 V centred windows (5 points) of dVg/
12                    dlog10(Id),
13                    restricted to Id > 5 x the MEASURED off-band median of the same thickness (applied
14                    to the
15                    simulated curve as well, so both curves are evaluated over the same current range;
16                    the solver
17                    reaches 1e-19 A/um whereas the measured floors are 5e-15..6.6e-12) and below 0.01 x
18                    Id(+3 V).
19     SS_cc        (secondary): average slope between the constant-current crossings at 1e-10 and 1e-8
20                    A/um,
21                    SS_cc = [Vg(1e-8) - Vg(1e-10)] / 2 decades, anchored to the same current band as
22                    Vth_cc.
23     mu_FE        : gm_max * L / (W Cox Vd) with W = 1 um (A/um data), L = 20 um, Cox = 8.955e-7 F/cm^2, Vd =
24                    0.7 V.
25
26     This is the linear-region estimator applied at Vd = 0.7 V; it is an apparent field-effect
27     mobility, not the band mobility.
28     Ion          : Id at Vg = +3 V.  Ioff : minimum Id over the sweep (experiment); for simulation the
29                    minimum
30                    positive Id together with the count of native zeros (never replaced by a floor).
31     Ion/Ioff     : ratio of the above (for a simulated curve with native zeros it is reported as '> Ion/
32                    Imin_positive').
33
34     Percentage errors are not reported for sign-changing or near-zero quantities (Vth): absolute differences
35     only.
36 """
37 import csv, json, math
38 from pathlib import Path
39 import numpy as np
40 ROOT = Path(__file__).resolve().parents[1]
41 COX = 8.955369814335959e-07; L_CM = 20e-4; W_CM = 1e-4; VD = 0.7; ICC = 1e-9
42
43 def crossing(vg, idv, level):
44     for i in range(len(vg) - 1):
45         a, b = idv[i], idv[i + 1]
46         if a > 0 and b > 0 and (a - level) * (b - level) <= 0 and a != b:
47             return float(vg[i] + (vg[i + 1] - vg[i]) * (math.log10(level) - math.log10(a)) / (math.log10(
48                 b) - math.log10(a)))

```



```
34     return None
35
36 _EXP_FLOOR = {}
37 def exp_off_floor(t_nm):
38     """Measured off-band median (Vg in [-2, -0.5]) of the given thickness, cached; None if unavailable."""
39     if t_nm is None: return None
40     if t_nm not in _EXP_FLOOR:
41         try:
42             vg, idv = load_curve(ROOT / 'data' / 'experimental_clean.csv', t_nm)
43             _EXP_FLOOR[t_nm] = float(np.median(idv[(vg >= -2) & (vg <= -.5)]))
44         except Exception: _EXP_FLOOR[t_nm] = None
45     return _EXP_FLOOR[t_nm]
46
47 def metrics(vg, idv, is_sim=False, t_nm=None):
48     vg = np.asarray(vg, float); idv = np.asarray(idv, float)
49     pos = idv > 0; n_zero = int((~pos).sum())
50     off = float(np.median(idv[(vg >= -2) & (vg <= -.5)])) if not is_sim else None
51     floor = off if not is_sim else exp_off_floor(t_nm) # same current range for both curves
52     v_lo, v_hi = crossing(vg, idv, ICC / 10), crossing(vg, idv, ICC * 10)
53     ss_cc = (None if (v_lo is None or v_hi is None) else (v_hi - v_lo) / 2.0 * 1000.0)
54     gm = np.gradient(idv, vg); i_gm = int(np.argmax(gm)); gm_max = float(gm[i_gm])
55     vth_lin = float(vg[i_gm] - idv[i_gm] / gm_max) if gm_max > 0 else None
56     ion = float(idv[-1])
57     thr_lo = (5 * floor) if floor is not None else None
58     lg = np.where(pos, np.log10(np.where(pos, idv, 1)), np.nan)
59     ss = None
60     for i in range(2, len(vg) - 2):
61         seg = idv[i - 2:i + 3]
62         if np.all(seg > 0) and seg[0] < 0.01 * ion and (thr_lo is None or seg[0] > thr_lo):
63             d = (lg[i + 2] - lg[i - 2]); w = vg[i + 2] - vg[i - 2]
64             if d > 0:
65                 cand = w / d * 1000.0; ss = cand if ss is None else min(ss, cand)
66     imin = float(idv[pos].min()) if pos.any() else None
67     return {'Vth_cc_1e-9_V': crossing(vg, idv, ICC), 'Vth_lin_V': vth_lin, 'SS_mV_dec': ss, '
        SS_min_window_mV_dec': ss, 'SS_cc_1e-10_1e-8_mV_dec': ss_cc, 'gm_max_A_V_um': gm_max, 'mu_FE_cm2Vs':
        gm_max * L_CM / (W_CM * COX * VD),
68         'Ion_A_per_um': ion, 'Ioff_A_per_um': imin, 'Ioff_definition': 'min Id over sweep' + (' (min
        POSITIVE; native zeros counted)' if is_sim else ''), 'native_zero_count': n_zero,
69         'Ion_over_Ioff': (ion / imin) if imin else None, 'off_band_median_A_per_um': off}
70
71 def load_curve(path_or_rows, thickness=None):
72     rows = list(csv.DictReader(Path(path_or_rows).open()))
73     if thickness is not None: rows = [r for r in rows if abs(float(r['thickness_nm']) - thickness) < 1e
        -6]
74     return np.array([float(r['vg_V']) for r in rows]), np.array([float(r[c for c in r if c.startswith('
        id') or c.startswith('atlas')][0]]) for r in rows])
75
76 if __name__ == '__main__':
77     import argparse
78     ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.
        RawDescriptionHelpFormatter)
79     ap.add_argument('--csv', default=str(ROOT / 'data' / 'experimental_clean.csv')); ap.add_argument('--
        thickness', type=float); ap.add_argument('--sim', action='store_true'); a = ap.parse_args()
80     for t in ([a.thickness] if a.thickness else [2.0, 6.3, 13.2, 31.8]):
81         vg, idv = load_curve(a.csv, t); print(t, json.dumps(metrics(vg, idv, a.sim, t), indent=1))
```

B.2.6 compare_experiment_simulation.py

The script builds `tables/EXPERIMENTAL_VS_SIMULATION.{csv,md}` and the overlay and residual plots of each thickness from the best V1 runs listed in `results/BEST_RUNS_V1.json`, with the same extraction for measured and simulated curves. No simulator is called.

```

1  #!/usr/bin/env python3
2  """Build tables/EXPERIMENTAL_VS_SIMULATION.{csv,md} and the per-thickness overlay/residual plots from the
3  best V1 runs listed in results/BEST_RUNS_V1.json (no simulator call). Same extraction algorithm
4  (scripts/extract_metrics.py) for measured and simulated curves.
5  """
6  import csv, json, sys
7  from pathlib import Path
8  import numpy as np
9  sys.path.insert(0, str(Path(__file__).resolve().parent))
10 from extract_metrics import metrics
11 ROOT = Path(__file__).resolve().parents[1]
12
13 def pct(a, b):
14     return None if (a is None or b is None or a == 0) else 100.0 * (b / a - 1.0)
15
16 def main():
17     best = json.loads((ROOT / 'results' / 'BEST_RUNS_V1.json').read_text())
18     rows = []; md = ['# Experimental vs simulation (V1 best verified runs)', '', 'Same extractor for both
19     curves (scripts/extract_metrics.py): Vth = constant current 1e-9 A/um (log interpolation); SS =
20     steepest 0.2 V window with Id above 5x the MEASURED off floor of that thickness and below 1 % of Ion
21     , applied to both curves so the same current range is evaluated (SS_cc, the average slope between
22     the 1e-10 and 1e-8 A/um crossings, is in the CSV); mu_FE = gm_max L/(W Cox Vd) at Vd = 0.7 V; Ion =
23     Id(+3 V); Ioff = min Id (simulation: min POSITIVE Id, native zeros counted). Currents in A/um. Vth
24     differences are absolute (no % for sign-changing quantities).', '',
25     '| t (nm) | run | Vth_exp (V) | Vth_sim (V) | dVth (V) | SS_exp | SS_sim (mV/dec) | mu_exp |
26     mu_sim (cm^2/Vs) | Ion_exp | Ion_sim | dIon % | Ioff_exp | Ioff_sim (min pos.) | zeros | Ion/Ioff
27     exp | Ion/Ioff sim | active log RMSE | notes |', '
28     |---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|']
29     keys = [k for k in ['2p0', '6p3', '13p2', '31p8'] if k in best] + [k for k, v in best.items() if
30     isinstance(v, dict) and 'directory' in v and k not in ('2p0', '6p3', '13p2', '31p8')]
31     for key in keys:
32         b = best[key]; d = ROOT / b['directory']; t = float(b.get('thickness_nm', key.replace('p', '.)))
33         cr = list(csv.DictReader((d / 'comparison.csv').open())); vg = np.array([float(r['vg_V']) for r
34         in cr]); me = np.array([float(r['measured_A_per_um']) for r in cr]); si = np.array([float(r['
35         atlas_A_per_um']) for r in cr])
36         ex = json.loads((d / 'execution.json').read_text()); fm = ex['fit_metrics']
37         E = metrics(vg, me, False, t); S = metrics(vg, si, True, t)
38         r = {'thickness_nm': t, 'run_id': b['run_id'], 'VDS_V': 0.7, 'Vth_exp_V': E['Vth_cc_1e-9_V'], '
39         Vth_sim_V': S['Vth_cc_1e-9_V'], 'Vth_error_V': (None if None in (E['Vth_cc_1e-9_V'], S['Vth_cc_1e-9
40         _V']) else S['Vth_cc_1e-9_V'] - E['Vth_cc_1e-9_V']),
41         'SS_exp_mV_dec': E['SS_mV_dec'], 'SS_sim_mV_dec': S['SS_mV_dec'], 'SS_error_pct': pct(E['
42         SS_mV_dec'], S['SS_mV_dec']), 'SScc_exp_mV_dec': E['SS_cc_1e-10_1e-8_mV_dec'], 'SScc_sim_mV_dec': S[
43         'SS_cc_1e-10_1e-8_mV_dec'], 'mu_exp_cm2Vs': E['mu_FE_cm2Vs'], 'mu_sim_cm2Vs': S['mu_FE_cm2Vs'], '
44         mu_error_pct': pct(E['mu_FE_cm2Vs'], S['mu_FE_cm2Vs']),
45         'Ion_exp_A_per_um': E['Ion_A_per_um'], 'Ion_sim_A_per_um': S['Ion_A_per_um'], 'Ion_error_pct
46         ': pct(E['Ion_A_per_um'], S['Ion_A_per_um']), 'Ioff_exp_A_per_um': E['Ioff_A_per_um'], '
47         Ioff_sim_minpos_A_per_um': S['Ioff_A_per_um'], 'sim_native_zero_count': S['native_zero_count'],
48         'Ioff_error_pct': (pct(E['Ioff_A_per_um'], S['Ioff_A_per_um']) if S['native_zero_count'] ==
49         0 else None), 'IonIoff_exp': E['Ion_over_Ioff'], 'IonIoff_sim': S['Ion_over_Ioff'], '
50         IonIoff_ratio_error': (None if S['native_zero_count'] else pct(E['Ion_over_Ioff'], S['Ion_over_Ioff'
51         ]))),
52         'log_RMSE_active': fm['regions']['active']['rmse_log10'], 'log_RMSE_all_positive': fm['
53         regions']['all']['rmse_log10'], 'linear_RMSE_active_A_per_um': fm['regions']['active']['
54         rmse_linear_A_per_um'], 'max_abs_log_active': fm['max_abs_log10_active'], 'notes': b.get('note', '')

```

```

    }
31     rows.append(r)
32     fmt = lambda x, f='{: .3g}': ('-' if x is None else f.format(x))
33     md.append(f'| {t} | {b[\'run_id\']} | {fmt(r[\'Vth_exp_V\'], \'{: .2f}\')} | {fmt(r[\'Vth_sim_V\'], \'{: .2f}\')} | {fmt(r[\'Vth_error_V\'], \'{:+.2f}\')} | {fmt(r[\'SS_exp_mV_dec\'], \'{: .0f}\')} | {fmt(r[\'SS_sim_mV_dec\'], \'{: .0f}\')} | {fmt(r[\'mu_exp_cm2Vs\'], \'{: .1f}\')} | {fmt(r[\'mu_sim_cm2Vs\'], \'{: .1f}\')} | {fmt(r[\'Ion_exp_A_per_um\'], \'{: .2e}\')} | {fmt(r[\'Ion_sim_A_per_um\'], \'{: .2e}\')} | {fmt(r[\'Ion_error_pct\'], \'{:+.1f}\')} | {fmt(r[\'Ioff_exp_A_per_um\'], \'{: .1e}\')} | {fmt(r[\'Ioff_sim_minpos_A_per_um\'], \'{: .1e}\')} | {S[\'native_zero_count\']} | {fmt(r[\'IonIoff_exp\'], \'{: .1e}\')} | {fmt(r[\'IonIoff_sim\'], \'{: .1e}\')} {\' (>)\'} if S[\'native_zero_count\'] else \'\' | {r[\'log_RMSE_active\']:.3f} | {r[\'notes\']} |"')
34     with (ROOT / 'tables' / 'EXPERIMENTAL_VS_SIMULATION.csv').open('w', newline='') as f:
35         w = csv.DictWriter(f, fieldnames=list(rows[0].keys())); w.writeheader(); w.writerows(rows)
36     md += [' ', 'Ioff_sim is the minimum positive simulated current; where native zeros exist the simulated Ion/Ioff is a lower bound and no percentage error is reported. The measured off-state floors are not modelled (docs/LIMITATIONS.md).']
37     (ROOT / 'tables' / 'EXPERIMENTAL_VS_SIMULATION.md').write_text('\n'.join(md) + '\n'); print('\n'.join(md))
38
39 if __name__ == '__main__': main()

```

B.2.7 make_provenance_tables.py

The script writes `tables/MATERIAL_PARAMETERS_USED.{csv,md}`, `tables/PARAMETER_PROVENANCE.csv` and `docs/PARAMETER_PROVENANCE.md` from `config/iwo_material_model.yaml`, optionally with the calibrated overrides recorded in `results/BEST_RUNS_V1.json`. These tables form the basis of Table 2.6.

```

1  #!/usr/bin/env python3
2  """Write tables/MATERIAL_PARAMETERS_USED.{csv,md}, tables/PARAMETER_PROVENANCE.csv and
3  docs/PARAMETER_PROVENANCE.md from config/iwo_material_model.yaml (+ optional calibrated overrides
4  recorded in results/BEST_RUNS_V1.json). No simulator call.
5  """
6  import csv, json, sys
7  from pathlib import Path
8  sys.path.insert(0, str(Path(__file__).resolve().parent))
9  from build_iwo_decks import ROOT, load_configs, material_at, set_dotted, cox, KEYS, Q, KB
10
11 def main():
12     m, g, s = load_configs()
13     best = json.loads((ROOT / 'results' / 'BEST_RUNS_V1.json').read_text()) if (ROOT / 'results' / 'BEST_RUNS_V1.json').exists() else {}
14     for o in best.get('shared_overrides', []): k, v = o.split('=', 1); set_dotted(m, k, v)
15     T = [2.0, 6.3, 13.2, 31.8]; P = {t: material_at(m, t) for t in T}
16     for t in T:
17         ov = best.get('per_thickness_overrides', {}).get(KEYS[t], [])
18         if ov:
19             mm = json.loads(json.dumps(m)) # one copy per thickness; ALL of its overrides applied together
20             for o in ov: k, v = o.split('=', 1); set_dotted(mm, k, v)
21             P[t] = material_at(mm, t)
22     Lw = m['thickness_laws']; b = m['bulk_reference']; C = cox(g)
23     R = []
24     def row(name, sym, kw, unit, vals, law, wdep, src_type, src, loc, label, unc, note):
25         R.append({'Parameter': name, 'Symbol': sym, 'ATLAS keyword': kw, 'Units': unit, '2.0 nm': vals[0], '6.3 nm': vals[1], '13.2 nm': vals[2], '31.8 nm': vals[3], 'Thickness law / equation': law, 'W-composition dependence': wdep, 'Temperature': '300 K', 'Source type': src_type, 'Exact source/reference': src, 'Page/Figure/Table': loc, 'Label': label, 'Uncertainty': unc, 'Notes': note})

```

```

26 f = lambda k, fmt='{:.4g}': [fmt.format(P[t][k]) for t in T]
27 row('Bandgap', 'Eg(t)', 'EG300', 'eV', f('Eg_eV'), 'Eg_bulk + dEg_QC(t), dEg_QC = %.3f t^-.3f' % (Lw
    ['dEg_QC_eV']['a'], Lw['dEg_QC_eV']['p']), 'Eg_bulk from an IWO study; W widens gap (PBE,
    qualitative)', 'literature + DFT proxy', 'Fan2021 Sec.3 (bulk); Lin2022 Methods; Si2021 Fig.4', '
    Lin2022 p.21542; Si2021 p.504', 'LITERATURE_IWO + DFT_PURE_IN203_PROXY', '+/-0.2 eV bulk; +/-50 % on
    dEg', 'PBE absolute gaps not used')
28 row('Electron affinity', 'chi(t)', 'AFFINITY', 'eV', f('chi_eV'), 'chi_bulk - dEc(t), dEc = dEg (Ev
    fixed)', 'chi_bulk composition-mixing estimate (Fan2021)', 'literature + DFT proxy', 'Fan2021 Sec.3;
    Si2021', 'Si2021 p.504', 'LITERATURE_IWO + DFT_PURE_IN203_PROXY', '+/-0.3 eV absolute', 'absolute
    vacuum alignment not available; relative dEc only')
29 row('Relative permittivity', 'eps_r', 'PERMITTIVITY', '-', f('eps_iwo'), 'shared', 'IWO 9.30 vs In203
    9.03 (same process)', 'measured (C-V) on target process', 'Januar2026 SI Fig.S1 (not bundled;
    recorded by Astra)', 'SI Fig.S1b', 'MEASURED_TARGET_DEVICE', '+/-0.04 (SI); bulk crystal 10.55 (
    Stokey2021)', 'no thickness dependence imposed')
30 row('Electron effective mass', 'm*(t)', '(none; enters Nc)', 'm0', f('mstar_m0'), '0.208 + %.3f t
    ^-.3f' % (Lw['mstar_m0']['b'], Lw['mstar_m0']['q']), 'W flattens CBM (PBE, qualitative) -> NOT
    DETERMINED', 'measured bulk + DFT proxy increment', 'Stokey2021; Lin2022', 'Stokey2021 Sec.IV.C;
    Lin2022 p.21540', 'MEASURED_RELATED_MATERIAL + DFT_PURE_IN203_PROXY', '+/-0.03 m0', 'nonparabolicity
    ignored')
31 row('Effective conduction-band DOS', 'Nc(t)', 'NC300', 'cm^-3', f('Nc_cm3', '{:.3e}'), '2(2 pi m* kT/
    h^2)^1.5', 'via m*', 'calculated', 'this work', '-', 'CALCULATED', 'x0.8-1.3', '3-D continuum value;
    quasi-2D at 2 nm')
32 row('Effective valence-band DOS', 'Nv', 'NV300', 'cm^-3', f('Nv_cm3', '{:.1e}'), 'shared', '-', '
    assumed', '-', '-', 'ASSUMED', 'x10', 'holes not solved')
33 row('Band mobility', 'mu_band(t)', '(enters MUN)', 'cm^2/Vs', f('mu_band'), 'per thickness', 'IWO
    lower than In203 (Januar2026)', 'fitted', 'this work (real ATLAS)', '-', 'FITTED', 'see
    SENSITIVITY_RESULTS', 'no defensible law; percolation/crystallinity hypothesis')
34 row('Roughness factor', '(1-Dsr/t)^2', '-', '-', f('rough'), 'Dsr = 2.87 A', 'IWO 2.87 A vs In203
    3.75 A', 'literature (target process)', 'Januar2026', 'Sec.2.9, Fig.5f', 'LITERATURE_IWO', 'unit
    ambiguity A vs nm noted', '')
35 row('Constant mobility in deck', 'mu0(t)', 'MUN', 'cm^2/Vs', f('mu0'), 'mu_band(t) (1-Dsr/t)^2', '-',
    'calculated', 'this work', '-', 'CALCULATED', '-', 'gate roll-off theta_r Vov not native (
    approximation gap)')
36 row('Effective mobility', 'mu_eff(Vg)', '(solver output: n_free/(n_free+n_trap) x mu0)', 'cm^2/Vs', [
    'solver'] * 4, 'from DOS occupancy', '-', 'simulated', '-', '-', 'CALCULATED', '-', 'reported as
    mu_FE from gm in EXPERIMENTAL_VS_SIMULATION')
37 row('Effective background donor density', 'Nd_eff(t)', 'DOPING N.TYPE CONC', 'cm^-3', f('Nd_cm3', '
    {:.3e}'), 'Nd0(1+(t/tc)^k), Nd0=2g, tc=g nm, k=g % (Lw['Nd_eff_cm3']['Nd0'], Lw['Nd_eff_cm3']['
    tc_nm'], Lw['Nd_eff_cm3']['k']), 'NOT the W concentration', 'fitted law', 'this work; Kim2024 trend'
    , 'Kim2024 p.6', 'FITTED', 'x2', 'electrically active donors only')
38 row('Flat-band carrier density', 'n_FB(t)', '(solver equilibrium n)', 'cm^-3', ['solver'] * 4, 'from
    equilibrium solution', '-', 'simulated', '-', '-', 'CALCULATED', '-', 'Ioff ~ t n_FB used only as a
    consistency check')
39 row('Bulk tail trap density', 'Nt(t)', '(NTA*WTA)', 'cm^-3', f('Nt_cm3', '{:.3e}'), 'Nt2 (2/t)^alpha,
    Nt2=1e, alpha=g % (Lw['Nt_cm3']['Nt2'], Lw['Nt_cm3']['alpha']), 'IWO lower than In203 (
    Januar2026)', 'fitted law (anchor ATLAS) + literature trend', 'this work; Januar2026', 'Januar2026
    Sec.2.8/2.10', 'FITTED', 'x2', 'Januar Nt(2 nm)=5.3e19 with theta_t convention')
40 row('Tail intercept density', 'NTA(t)', 'NTA', 'cm^-3/eV', f('NTA_cm3_eV', '{:.3e}'), 'Nt/WTA', '-',
    'calculated', '-', '-', 'CALCULATED', '-', 'gTA = NTA exp((E-Ec)/WTA)')
41 row('Tail energy width', 'WTA = kTt', 'WTA', 'eV', f('WTA_eV'), 'shared', '-', 'fitted', 'this work',
    '-', 'FITTED', '+/-0.005 eV', 'Tt = 0.0f K; absolute Tt not in the paper' % (P[2.0]['WTA_eV'] / KB)
    )
42 row('Deep acceptor density', 'NGA', 'NGA', 'cm^-3/eV', f('NGA', '{:.1e}'), 'shared', '-', 'assumed',
    'Fan2021 range', 'Fan2021 p.9', 'ASSUMED', 'x10', 'insensitive; EGA from Ec')
43 row('Interface trap density', 'Dit', 'regularized NGA in region 2 (sheet/0.25 nm)', 'cm^-2 eV^-1', f(
    'it_peak', '{:.1e}'), 'shared', '-', 'assumed / literature proxy', 'Lin2022; Wang2022 (HfO2/In203 6
    e11)', 'Lin2022 p.21539; Wang2022 p.4', 'ASSUMED', 'x3', 'Gaussian at Ec-0.3 eV, W 0.12 eV; IWO/
    Al2O3 interface here')
44 row('Fixed interface charge', 'Qf', 'INTERFACE QF', 'cm^-2', f('Qf_cm2', '{:.3e}'), 'shared value
    1.73e12; 6.3 nm DEVICE-SPECIFIC 8.7e10 (cause NOT DETERMINED)', '-', 'fitted', 'this work', '-', '
    FITTED (shared) + FITTED device-specific (6.3 nm)', '-', 'dvfb = -q Qf/Cox = %+.3f V (2 nm), %+.3f V

```

```

(6.3 nm)' % (-Q * P[2.0]['Qf_cm2'] / C, -Q * P[6.3]['Qf_cm2'] / C))
45 row('Capture cross sections', 'sigma', 'SIGTAE.. SIGGDH', 'cm^2', f('sig', '{:.0e}'), 'shared', '-',
    'assumed', '-', '-', 'ASSUMED', 'x100', 'not identifiable from DC')
46 row('Gate work function', 'Phi_TiN', 'CONTACT WORKFUNCTION', 'eV', f('gate_wf'), 'shared', '-', '
    assumed', 'TiN literature range 4.5-4.9', '-', 'ASSUMED', '+/-0.2 eV', 'degenerate with chi and Qf')
47 row('S/D contact', 'Pd/IWO', 'CONTACT (no WF) [+RESISTANCE]', 'ohm.um', [f'{P[t]["rs_ohm_um"]:.3g}'
    for t in T], 'ideal Ohmic; 31.8 nm series R', '-', 'assumed / fitted (31.8)', 'Si2022/Lin2022 (CNL)
    context', '-', 'ASSUMED (+FITTED for 31.8 nm)', 'underconstrained (no TLM)', '')
48 row('SRH lifetimes', 'tau_n, tau_p', 'TAUN0 TAUP0', 's', f('tau_s', '{:.0e}'), 'shared', '-', '
    assumed', '-', '-', 'ASSUMED', 'x100', 'negligible')
49 row('Channel length / width', 'L / W', 'geometry / MESH WIDTH', 'um', [20 / 290 (sim 1)] * 4, '-',
    '-', 'measured (figure)', 'schematic', '-', 'MEASURED_TARGET_DEVICE', '-', 'A/um normalization')
50 row('Gate-stack capacitance', 'Cox', '(from eps and thickness)', 'F/cm^2', [f'{C:.4e}'] * 4, 'eps0
    /(15 nm/19.57 + 2 nm/9.0)', '-', 'calculated', 'SI Fig.S1 (HfO2), Al2O3 eps assumed', '-', '
    CALCULATED', '+/-5 %', 'EOT 3.86 nm')
51 with (ROOT / 'tables' / 'MATERIAL_PARAMETERS_USED.csv').open('w', newline='') as fh:
52     w = csv.DictWriter(fh, fieldnames=list(R[0].keys())); w.writeheader(); w.writerows(R)
53 md = ['# Material parameters used in the simulation (V1)', '', 'Values are those written to the decks
    (see decks/*_metadata.json). Label = provenance class; no fitted value is called measured.', '',
54     '| Parameter | Symbol | ATLAS | Units | 2.0 nm | 6.3 nm | 13.2 nm | 31.8 nm | Law | Label |
    Source | Uncertainty |', '|---|---|---|---|---|---|---|---|---|---|---|']
55 for r in R: md.append(f'| {r["Parameter"]} | {r["Symbol"]} | {r["ATLAS keyword"]} | {r["Units"]} | {r
    ["2.0 nm"]} | {r["6.3 nm"]} | {r["13.2 nm"]} | {r["31.8 nm"]} | {r["Thickness law / equation"]} | {r
    ["Label"]} | {r["Exact source/reference"]} | {r["Uncertainty"]} |')
56 (ROOT / 'tables' / 'MATERIAL_PARAMETERS_USED.md').write_text('\n'.join(md) + '\n')
57 prov = [{'parameter': r['Parameter'], 'label': r['Label'], 'source': r['Exact source/reference'], '
    location': r['Page/Figure/Table'], 'notes': r['Notes']} for r in R]
58 with (ROOT / 'tables' / 'PARAMETER_PROVENANCE.csv').open('w', newline='') as fh:
59     w = csv.DictWriter(fh, fieldnames=list(prov[0].keys())); w.writeheader(); w.writerows(prov)
60 pm = ['# Parameter provenance (every model parameter carries one label)', '', 'Labels:
    MEASURED_TARGET_DEVICE, MEASURED_RELATED_MATERIAL, DFT_TARGET_IWO (none available),
    DFT_PURE_IN203_PROXY, LITERATURE_IWO, LITERATURE_IN203_PROXY, CALCULATED, FITTED, NUMERICAL, ASSUMED
    .', '', '| Parameter | Label | Source | Location | Notes |', '|---|---|---|---|---|']
61 for p in prov: pm.append(f'| {p["parameter"]} | {p["label"]} | {p["source"]} | {p["location"]} | {p[
    "notes"]} |')
62 pm += ['', '## Numerical (label NUMERICAL, from config/solver.yaml)', '', '| Item | Value | Evidence
    |', '|---|---|---|',
63     '| XANDRNORM + CR.TOLER (strict, depletion) | 1e-17 A scaled by 1e-5 x min active current |
    Claude run_0006/0009/0010/0011/0018 |', '| CR.TOLER (on-state) | 5e-18 default, ^XANDRNORM | Claude
    run_0011 |', '| IR.TOL | 1e-19 A | - |', '| MAXTRAPS / ITLIMIT / CLIMIT | 10 / 80 / 1e-6 |
    documented range |',
64     '| DOS levels NUMA/NUMD | 96/48 | x2 check: 0.006 dec (run_0016) |', '| vertical mesh | 0.25
    nm (2 nm film), 0.0625 nm at interface layer, graded to 1 nm in thick films | mesh x0.5: 0.005 dec (
    run_0015); x0.7 for 13.2 nm: 0.004 dec (run_0026) |', '| interface regularization layer | 0.25 nm |
    layer /2: 0.009 dec (run_0017) |', '| gate step | 0.05 V logged, 0.1 V continuation | - |']
65 (ROOT / 'docs' / 'PARAMETER_PROVENANCE.md').write_text('\n'.join(pm) + '\n'); print('tables written:'
    , len(R), 'parameters')
66
67 if __name__ == '__main__': main()

```

B.2.8 sensitivity_analysis.py

The script performs a one-at-a-time sensitivity analysis of the V1 model in the real solver; each case is one ATLAS launch relative to the best run of a thickness (e.g. `-thickness 2 -cases chi_m0p1 nd_x2 dit_x3`). It produced the sensitivity runs `run_0020–run_0027` (`run_0027` malformed and retracted, Section 7.4); the chain was stopped by the user when the launch budget was limited.

```
1 #!/usr/bin/env python3
```

```

2  """One-at-a-time sensitivity of the V1 model in the REAL solver (each case is one ATLAS launch).
3
4  python scripts/sensitivity_analysis.py --thickness 2 --cases chi_m0p1 nd_x2 dit_x3 --stride 2
5
6  Each case perturbs one parameter of config/iwo_material_model.yaml (via --override), runs ATLAS through
7  run_atlas.execute, and records the change of Vth_cc, SS, mu_FE, Ion, Ioff(min positive) and active log
8  RMSE
9  relative to the reference run given in results/BEST_RUNS_V1.json into tables/SENSITIVITY_RESULTS.csv.
10 Launch budget is enforced by run_atlas (solver.yaml runner.max_launches).
11 """
12 import argparse, csv, json, sys
13 from pathlib import Path
14 import numpy as np
15 sys.path.insert(0, str(Path(__file__).resolve().parent))
16 from run_atlas import execute, ROOT
17 from extract_metrics import metrics
18
19 CASES = {
20     'chi_m0p1': ['bulk_reference.chi_bulk_eV.value=4.20'], 'chi_p0p1': ['bulk_reference.chi_bulk_eV.value
21     =4.40'],
22     'wf_p0p1': ['contacts.gate_workfunction_eV.value=4.80'],
23     'nd_x2': ['thickness_laws.Nd_eff_cm3.Nd0=5e17'], 'nd_x0p5': ['thickness_laws.Nd_eff_cm3.Nd0=1.25e17'
24     ],
25     'nt_x1p5': ['thickness_laws.Nt_cm3.Nt2=3e19'], 'wta_p5meV': ['thickness_laws.WTA_eV.value=0.045'],
26     'dit_x3': ['traps.interface_acceptor_sheet.peak_cm2_eV=9e11'], 'dit_0': ['traps.
27     interface_acceptor_sheet.peak_cm2_eV=1e9'],
28     'eps_10p55': ['bulk_reference.eps_r.value=10.55'], 'mstar_bulk': ['thickness_laws.mstar_m0.b=0'],
29     'no_confinement': ['thickness_laws.dEg_QC_eV.a=0'], 'thickness_laws.mstar_m0.b=0'],
30     'overlap_4um': ['geometry.contact_overlap_um.value=4.0'],
31 }
32
33 OUT = 'SENSITIVITY_OAT.csv' # one-at-a-time results; merged into SENSITIVITY_RESULTS by
34     sensitivity_from_stage_runs.py
35
36 def main():
37     ap = argparse.ArgumentParser(description=__doc__); ap.add_argument('--thickness', type=float,
38     required=True); ap.add_argument('--cases', nargs='+', required=True); ap.add_argument('--stride',
39     type=int, default=2); ap.add_argument('--vstep', type=float, default=0.1, help='ATLAS stepped sweep
40     step (V); 0 = per-point solves with --stride'); ap.add_argument('--extra', action='append', default
41     =[]); a = ap.parse_args()
42     key = str(a.thickness).replace('.', 'p'); best = json.loads((ROOT / 'results' / 'BEST_RUNS_V1.json').
43     read_text())
44     ref = best[key]; shared = best.get('shared_overrides', []) + best.get('per_thickness_overrides', {}).
45     get(key, [])
46     rd = ROOT / ref['directory']; cr = list(csv.DictReader((rd / 'comparison.csv').open())); vg = np.
47     array([float(r['vg_V']) for r in cr]); me = np.array([float(r['measured_A_per_um']) for r in cr]);
48     r0 = np.array([float(r['atlas_A_per_um']) for r in cr])
49     M0 = metrics(vg, r0, True, a.thickness); rmse0 = json.loads((rd / 'execution.json').read_text())['
50     fit_metrics']['regions']['active']['rmse_log10']
51     out = ROOT / 'tables' / OUT; new = not out.exists()
52     for c in a.cases:
53         ov = shared + CASES[c] + a.extra
54         try: d = execute(a.thickness, f'sens_{c}', ov, stride=a.stride, vstep=(a.vstep or None))
55         except RuntimeError as e: print('case', c, 'FAILED:', e); continue
56         cr = list(csv.DictReader((d / 'comparison.csv').open())); vgs = np.array([float(r['vg_V']) for r
57         in cr]); s = np.array([float(r['atlas_A_per_um']) for r in cr]); M = metrics(vgs, s, True, a.
58         thickness)
59         if len(vgs) != len(vg): # stepped sweep: compare the reference on the same (coarser) grid
60             sel = np.isin(np.round(vg, 6), np.round(vgs, 6)); M0 = metrics(vg[sel], r0[sel], True, a.
61             thickness)
62         rm = json.loads((d / 'execution.json').read_text())['fit_metrics']['regions']['active']['
63         rmse_log10']

```

```

45     row = {'thickness_nm': a.thickness, 'case': c, 'overrides': ' '.join(CASES[c]), 'run_id': d.name.
split('_')[0] + '_' + d.name.split('_')[1], 'run_ref': ref['run_id'], 'dVth_cc_V': (M['Vth_cc_1e-9_V']
or np.nan) - (M0['Vth_cc_1e-9_V'] or np.nan), 'dSS_mV_dec': (M['SS_mV_dec'] or np.nan) - (M0['
SS_mV_dec'] or np.nan), 'dmu_FE_pct': 100 * (M['mu_FE_cm2Vs'] / M0['mu_FE_cm2Vs'] - 1), 'dIon_pct':
100 * (M['Ion_A_per_um'] / M0['Ion_A_per_um'] - 1), 'Ioff_minpos_ratio': (M['Ioff_A_per_um'] or np.
nan) / (M0['Ioff_A_per_um'] or np.nan), 'active_log_rmse': rm, 'active_log_rmse_ref': rmse0}
46     with out.open('a', newline='') as f:
47         w = csv.DictWriter(f, fieldnames=list(row.keys())); (w.writeheader() if new else None); w.
writerow(row); new = False
48         print(json.dumps(row, default=float))
49
50 if __name__ == '__main__': main()

```

B.2.9 sensitivity_from_stage_runs.py

The script derives sensitivities from the real calibration-stage runs already in `results/` (pairs of stage runs that differ in one parameter), without calling the simulator. It was written after the one-at-a-time chain had been stopped, in order to extract the remaining sensitivities from runs that already existed.

```

1  #!/usr/bin/env python3
2  """Sensitivities derived from the REAL calibration-stage runs already in results/ (no simulator call).
3
4  The one-at-a-time sensitivity chain (scripts/sensitivity_analysis.py) was stopped by the user, who
   limited the
5  work to three final launches. This script therefore records only the sensitivities that the executed
   stage runs
6  measure directly (pairs of runs that differ in exactly one parameter), plus explicit NOT RUN rows for the
   planned
7  cases so that the table is complete and honest. Output: tables/SENSITIVITY_RESULTS.csv (+ .md).
8  """
9  import csv, json
10 from pathlib import Path
11 ROOT = Path(__file__).resolve().parents[1]
12 R = ROOT / 'results' / 'runs'
13
14 import sys, numpy as np
15 sys.path.insert(0, str(Path(__file__).resolve().parent))
16 from extract_metrics import metrics
17
18 def ex(run_prefix):
19     d = next(p for p in R.iterdir() if p.name.startswith(run_prefix)); e = json.loads((d / 'execution.
   json').read_text())
20     # recompute the device metrics with the CURRENT extractor (early runs stored metrics from an earlier
   SS definition)
21     cr = list(csv.DictReader((d / 'comparison.csv').open())); vg = np.array([float(r['vg_V']) for r in cr
   ]); si = np.array([float(r['atlas_A_per_um']) for r in cr])
22     e['device_metrics_sim'] = metrics(vg, si, True, float(e['thickness_nm'])); return e, d.name
23
24 def sim(e): return e['device_metrics_sim']
25
26 PAIRS = [
27     # (thickness, parameter, from, to, run_ref, run_pert)
28     (2.0, 'Qf (fixed interface charge, cm^-2)', '0', '1.73e12', 'run_0001', 'run_0003'),
29     (13.2, 'Qf (fixed interface charge, cm^-2)', '0', '1.73e12', 'run_0002', 'run_0004'),
30     (2.0, 'mu_band (cm^2/Vs)', '16.8', '18.13', 'run_0003', 'run_0008'),
31     (13.2, 'mu_band (cm^2/Vs)', '57.6', '61.9', 'run_0004', 'run_0009'),
32     (6.3, 'Qf (fixed interface charge, cm^-2)', '1.73e12', '8.7e10', 'run_0005', 'run_0007'),
33     (2.0, 'structure + sweep form (reduced/pointwise -> physical/stepped)', 'reduced, 121 pts', 'physical
   , 76 pts', 'run_0008', 'run_0012'),

```

```

34     (13.2, 'structure + sweep form (reduced/pointwise -> physical/stepped)', 'reduced, 121 pts', '
    physical, 76 pts', 'run_0009', 'run_0013'),
35 ]
36 NOT_RUN = ['no_confinement (dEg_QC = 0, m* = bulk)', 'chi_bulk +/- 0.1 eV', 'gate work function +0.1 eV (
    same action as Qf: -q dQf/Cox)', 'Nd0 x2 / x0.5', 'Nt2 x1.5', 'WTA +5 meV', 'Dit x3 / 0', 'eps_r 9.3
    -> 10.55', 'm* = bulk only', 'contact overlap 4 um', 'any case at 6.3 nm']
37
38 def main():
39     rows = []
40     for t, par, a, b, r0, r1 in PAIRS:
41         e0, n0 = ex(r0); e1, n1 = ex(r1); s0, s1 = sim(e0), sim(e1)
42         f = lambda k: (None if s0.get(k) is None or s1.get(k) is None else s1[k] - s0[k])
43         pct = lambda k: (None if not s0.get(k) or s1.get(k) is None else 100 * (s1[k] / s0[k] - 1))
44         rows.append({'thickness_nm': t, 'parameter': par, 'from': a, 'to': b, 'run_ref': n0, 'run_pert':
            n1, 'status': 'MEASURED IN ATLAS (stage pair)',
45             'dVth_cc_V': f('Vth_cc_1e-9_V'), 'dSS_min_mV_dec': f('SS_mV_dec'), 'dmu_FE_pct': pct
            ('mu_FE_cm2Vs'), 'dIon_pct': pct('Ion_A_per_um'),
46             'active_log_rmse_ref': e0['fit_metrics']['regions']['active']['rmse_log10'], '
            active_log_rmse_pert': e1['fit_metrics']['regions']['active']['rmse_log10'],
47             'note': ''})
48     rows[0]['note'] = 'dVth = -0.309 V for dQf = 1.73e12 cm^-2: -q dQf/Cox = -0.3095 V (Cox 8.955e-7 F/cm
        ^2) -> the solver reproduces the flat-band relation exactly'
49     rows[2]['note'] = 'Ion scales ~linearly with mu_band at fixed electrostatics (+7.9 % mu -> +7.9 % Ion
        )'
50     # one-at-a-time runs executed by scripts/sensitivity_analysis.py (tables/SENSITIVITY_OAT.csv), if any
51     oat = ROOT / 'tables' / 'SENSITIVITY_OAT.csv'; done = set()
52     if oat.exists():
53         for r in csv.DictReader(oat.open()):
54             done.add(r['case'])
55             g = lambda k: (None if r.get(k) in (None, '', 'nan') else float(r[k]))
56             rows.append({'thickness_nm': float(r['thickness_nm']), 'parameter': f'OAT {r['case']}: {r[
                'overrides']}', 'from': 'final', 'to': r['case'], 'run_ref': r.get('run_ref', ''), 'run_pert': r[
                'run_id'], 'status': 'MEASURED IN ATLAS (one-at-a-time, vs the final run)',
57                 'dVth_cc_V': g('dVth_cc_V'), 'dSS_min_mV_dec': g('dSS_mV_dec'), 'dmu_FE_pct': g(
                    'dmu_FE_pct'), 'dIon_pct': g('dIon_pct'), 'active_log_rmse_ref': g('active_log_rmse_ref'), '
                    active_log_rmse_pert': g('active_log_rmse'), 'note': ''})
58     KEYMAP = {'no_confinement': 'no_confinement', 'chi_bulk': 'chi_m0p1', 'gate work': 'wf_p0p1', 'Nd0':
        'nd_x2', 'Nt2': 'nt_x1p5', 'WTA': 'wta_p5meV', 'Dit': 'dit_x3', 'eps_r': 'eps_10p55', 'm* = bulk': '
        mstar_bulk', 'overlap': 'overlap_4um', 'DOS levels': None, 'mesh x0.7': None}
59     for c in NOT_RUN:
60         key = next((v for k, v in KEYMAP.items() if k in c), None)
61         if key in done: continue
62         rows.append({'thickness_nm': '', 'parameter': c, 'from': '', 'to': '', 'run_ref': '', 'run_pert':
            '', 'status': 'NOT RUN (see docs/NUMERICAL_CONVERGENCE.md for the refinement checks; other cases
            not launched)',
63             'dVth_cc_V': None, 'dSS_min_mV_dec': None, 'dmu_FE_pct': None, 'dIon_pct': None, '
            active_log_rmse_ref': None, 'active_log_rmse_pert': None, 'note': ''})
64     out = ROOT / 'tables' / 'SENSITIVITY_RESULTS.csv'
65     with out.open('w', newline='') as fh:
66         w = csv.DictWriter(fh, fieldnames=list(rows[0].keys())); w.writeheader(); w.writerows(rows)
67         fmt = lambda x, f='{:.3g}': ('-' if x is None or x == '' else f.format(x))
68         md = ['# Sensitivity results (V1)', '', 'Rows marked MEASURED IN ATLAS come from pairs of executed
            runs that differ in exactly one input; rows marked NOT RUN are the planned one-at-a-time cases that
            were not launched (user directive). No sensitivity below is estimated analytically.', '',
69             ' | t (nm) | parameter | from | to | runs | dVth_cc (V) | dSS_min (mV/dec) | dmu_FE (%) | dIon
            (%) | active RMSE ref -> pert | status / note |', '|---|---|---|---|---|---|---|---|---|---|']
70     for r in rows:
71         md.append(f" | {r['thickness_nm']} | {r['parameter']} | {r['from']} | {r['to']} | {r['run_ref']
            '[:8] if r['run_ref'] else ''} -> {r['run_pert'][:8] if r['run_pert'] else ''} | {fmt(r['dVth_cc_V']
            ')} | {fmt(r['dSS_min_mV_dec'])} | {fmt(r['dmu_FE_pct'])} | {fmt(r['dIon_pct'])} | {fmt(r[
            'active_log_rmse_ref'], '{:.3f})} -> {fmt(r['active_log_rmse_pert'], '{:.3f})} | {r['status']}{(';

```



```
    ' + r['note']) if r['note'] else '' |")
72 (ROOT / 'tables' / 'SENSITIVITY_RESULTS.md').write_text('\n'.join(md) + '\n'); print('\n'.join(md))
73
74 if __name__ == '__main__': main()
```

B.2.10 plot_results.py

The script produces plots from the best V1 runs (results/BEST_RUNS_V1.json) and the thickness laws without calling the simulator. For each thickness it draws the overlays on logarithmic and linear axes, the residual, the band diagrams at equilibrium, threshold and on-state, and the electron density versus gate voltage. Several of these PNG files are reproduced in Chapter 4.

```
1 #!/usr/bin/env python3
2 """Plots from the best V1 runs (results/BEST_RUNS_V1.json) and the thickness laws. No simulator call.
3
4 Per thickness (plots/<key>): overlay_log.png, overlay_linear.png, residual.png, band_diagram_eq.png,
5 band_diagram_near_vth.png, electron_density_depth.png, mobility_vs_vg.png (channel-centre probe: native
6 electron mobility; with a constant band-mobility model it is flat, which is itself a finding),
7   free_vs_trapped.png
8 (if avg_n export exists). Global: metrics_vs_thickness.png (Vth, SS, mu, Ion, Ioff exp vs sim), plus the
9   law plots
10 already written by material_laws.py. Every panel states the data source in its title/legend.
11 """
12 import csv, json, sys
13 from pathlib import Path
14 import numpy as np
15 sys.path.insert(0, str(Path(__file__).resolve().parent))
16 from extract_metrics import metrics
17 ROOT = Path(__file__).resolve().parents[1]
18
19 def read2(p):
20     pts = []
21     for line in Path(p).read_text(errors='replace').splitlines():
22         w = line.replace(',', ' ').split()
23         if len(w) == 2:
24             try: pts.append((float(w[0]), float(w[1])))
25             except ValueError: pass
26     return np.array(pts) if pts else None
27
28 def main():
29     import matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
30     best = json.loads((ROOT / 'results' / 'BEST_RUNS_V1.json').read_text()); g = {}
31     for key, b in best.items():
32         if not isinstance(b, dict) or 'directory' not in b: continue
33         d = ROOT / b['directory']; t = float(b.get('thickness_nm', key.replace('p', '.'))); out = ROOT /
34         'plots' / key; out.mkdir(parents=True, exist_ok=True)
35         cr = list(csv.DictReader((d / 'comparison.csv').open())); vg = np.array([float(r['vg_V']) for r
36         in cr]); me = np.array([float(r['measured_A_per_um']) for r in cr]); si = np.array([float(r['
37         atlas_A_per_um']) for r in cr])
38         src = f'measured: workbook Sheet1 | simulated: ATLAS {b["run_id"]}'
39         # cutline depth origin: the physical-structure decks start at the top of the 70 nm Air/Pd layer;
40         shift so that the IWO top surface is depth 0
41         meta = json.loads((d / 'metadata.json').read_text()) if (d / 'metadata.json').exists() else {}
42         d_off = 70.0 if meta.get('full_structure') else 0.0
43         fig, ax = plt.subplots(figsize=(6, 4.2)); ax.semilogy(vg, me, 'k.', ms=4, label='measured'); ax.
44         semilogy(vg, np.maximum(si, 1e-40), 'r-', lw=1.3, label='ATLAS V1'); ax.set_ylim(1e-17, 1e-5); ax.
45         set_xlabel('VG (V)'); ax.set_ylabel('ID (A/um), VD = 0.7 V'); ax.set_title(f'{t} nm IWO, log scale\n
46         {src}', fontsize=8); ax.legend(); ax.grid(alpha=.3); fig.tight_layout(); fig.savefig(out / '
```

```

overlay_log.png', dpi=140); plt.close(fig)
38     fig, ax = plt.subplots(figsize=(6, 4.2)); ax.plot(vg, me * 1e6, 'k.', ms=4, label='measured'); ax
.plot(vg, si * 1e6, 'r-', lw=1.3, label='ATLAS V1'); ax.set_xlabel('VG (V)'); ax.set_ylabel('ID (uA/
um)'); ax.set_title(f'{t} nm IWO, linear scale\n{src}', fontsize=8); ax.legend(); ax.grid(alpha=.3);
fig.tight_layout(); fig.savefig(out / 'overlay_linear.png', dpi=140); plt.close(fig)
39     fig, ax = plt.subplots(1, 2, figsize=(10, 4)); le = np.log10(np.maximum(si, 1e-40)) - np.log10(me
); ax[0].plot(vg, np.clip(le, -1.5, 1.5), 'b.-', ms=3, lw=.8); ax[0].axhline(0, color='k', lw=.6);
ax[0].set_ylabel('log10(ATLAS/measured) (clipped +/-1.5)'); ax[1].plot(vg, (si - me) * 1e6, 'b.-',
ms=3, lw=.8); ax[1].axhline(0, color='k', lw=.6); ax[1].set_ylabel('ATLAS - measured (uA/um)')
40     for a in ax: a.set_xlabel('VG (V)'); a.grid(alpha=.3)
41     fig.suptitle(f'{t} nm residuals ({b["run_id"]}); native zeros where log residual is clipped',
fontsize=9); fig.tight_layout(); fig.savefig(out / 'residual.png', dpi=140); plt.close(fig)
42     for tag, title in [(('eq', 'equilibrium (VG = VD = 0)'), ('vth', 'near threshold (VG ~ Vth_cc, VD
= 0.7 V)'), ('on', 'on-state (VG = +3 V, VD = 0.7 V)')]:
43         cb, vb, n = read2(d / f'cb_{tag}.dat') if (d / f'cb_{tag}.dat').exists() else None, read2(d /
f'vb_{tag}.dat') if (d / f'vb_{tag}.dat').exists() else None, read2(d / f'n_{tag}.dat') if (d / f'
n_{tag}.dat').exists() else None
44         if cb is not None and vb is not None:
45             fig, ax = plt.subplots(figsize=(6, 4.2)); ax.plot(cb[:, 0] * 1000 - d_off, cb[:, 1], 'b-
', label='Ec'); ax.plot(vb[:, 0] * 1000 - d_off, vb[:, 1], 'r-', label='Ev')
46             qf = read2(d / f'qfn_{tag}.dat') if (d / f'qfn_{tag}.dat').exists() else None
47             if qf is not None: ax.plot(qf[:, 0] * 1000 - d_off, qf[:, 1], 'g--', label='EFn')
48             ax.axvline(t, color='gray', lw=.8, ls=':'); ax.axvline(t + 2, color='gray', lw=.8, ls=':
'); ax.set_xlim(-3, t + 17 + 3)
49             ax.set_xlabel(f'depth from film top surface at x = 12 um (nm): IWO 0..{t} nm | Al2O3 |
HfO2 to gate'); ax.set_ylabel('energy (eV)'); ax.set_title(f'{t} nm band diagram, {title}\nATLAS {b
["run_id"]} cutline export', fontsize=8); ax.legend(); ax.grid(alpha=.3); fig.tight_layout(); fig.
savefig(out / f'band_diagram_{tag}.png', dpi=140); plt.close(fig)
50             if n is not None:
51                 fig, ax = plt.subplots(figsize=(6, 4.2)); ax.semilogy(n[:, 0] * 1000 - d_off, np.maximum(
n[:, 1], 1e-5), 'k-'); ax.set_xlim(0, t); ax.set_xlabel(f'depth from film top surface (nm), IWO film
0..{t} nm; interface at {t} nm'); ax.set_ylabel('electron concentration (cm-3)'); ax.set_title(f'{
t} nm electron density vs depth, {title}\nATLAS {b["run_id"]}', fontsize=8); ax.grid(alpha=.3); fig.
tight_layout(); fig.savefig(out / f'electron_density_{tag}.png', dpi=140); plt.close(fig)
52             mob = read2(d / 'chan_mobility_vg.dat') if (d / 'chan_mobility_vg.dat').exists() else None
53             if mob is not None:
54                 gm = np.gradient(me, vg); mufe = gm * 20e-4 / (1e-4 * 8.955369814335959e-07 * 0.7)
55                 fig, ax = plt.subplots(figsize=(6, 4.2)); ax.plot(mob[:, 0], mob[:, 1], 'r-', label='ATLAS
channel-centre electron mobility (probe)'); ax.plot(vg, mufe, 'k.', ms=3, label='measured apparent
mu_FE = gm L/(W Cox Vd)'); ax.set_xlabel('VG (V)'); ax.set_ylabel('mobility (cm2/Vs)'); ax.set_ylim
(0, None); ax.set_title(f'{t} nm mobility vs VG\n{b["run_id"]}', fontsize=8); ax.legend(fontsize=7);
ax.grid(alpha=.3); fig.tight_layout(); fig.savefig(out / 'mobility_vs_vg.png', dpi=140); plt.close(
fig)
56             ne = read2(d / 'chan_electrons_vg.dat') if (d / 'chan_electrons_vg.dat').exists() else None; na =
read2(d / 'avg_electrons_vg.dat') if (d / 'avg_electrons_vg.dat').exists() else None
57             if ne is not None or na is not None:
58                 fig, ax = plt.subplots(figsize=(6, 4.2))
59                 if ne is not None: ax.semilogy(ne[:, 0], np.maximum(ne[:, 1], 1e-5), 'b-', label='free
electrons at channel centre (probe)')
60                 if na is not None: ax.semilogy(na[:, 0], np.maximum(na[:, 1], 1e-5), 'g--', label='free
electrons averaged over the film (probe)')
61                 ax.set_xlabel('VG (V)'); ax.set_ylabel('electron concentration (cm-3)'); ax.set_title(f'{t}
nm free-electron density vs VG (VD = 0.7 V)\nATLAS {b["run_id"]} probes; equilibrium value at VG = 0
is n_FB', fontsize=8); ax.legend(fontsize=7); ax.grid(alpha=.3); fig.tight_layout(); fig.savefig(
out / 'electron_density_vs_vg.png', dpi=140); plt.close(fig)
62             E = metrics(vg, me, False, t); S = metrics(vg, si, True, t)
63             if key in ('2p0', '6p3', '13p2', '31p8'): g[t] = (E, S) # global metrics plot uses the primary
entry per thickness
64         if g:
65             ts = sorted(g); fig, ax = plt.subplots(2, 3, figsize=(13, 7))
66             def pair(a, k, lab, log=False):

```

```

67         e = [g[t][0][k] if g[t][0][k] is not None else np.nan for t in ts]; s_ = [g[t][1][k] if g[t]
68         ][1][k] is not None else np.nan for t in ts]
69         a.plot(ts, e, 'ko-', label='experiment'); a.plot(ts, s_, 'rs--', label='ATLAS V1'); a.
        set_xscale('log'); a.set_xlabel('t (nm)'); a.set_ylabel(lab); a.grid(alpha=.3); a.legend(fontsize=7)
69         if log: a.set_yscale('log')
70         pair(ax[0, 0], 'Vth_cc_1e-9_V', 'Vth (const. current 1e-9 A/um) (V)'); pair(ax[0, 1], 'SS_mV_dec'
        , 'SS (mV/dec)'); pair(ax[0, 2], 'mu_FE_cm2Vs', 'mu_FE (cm^2/Vs)'); pair(ax[1, 0], 'Ion_A_per_um', '
        Ion (A/um)', True); pair(ax[1, 1], 'Ioff_A_per_um', 'Ioff (A/um; sim = min positive)', True); pair(
        ax[1, 2], 'Ion_over_Ioff', 'Ion/Ioff (sim lower bound if zeros)', True)
71         fig.suptitle('Device metrics vs thickness: experiment (workbook) vs ATLAS V1 best runs, identical
        extraction', fontsize=10); fig.tight_layout(); fig.savefig(ROOT / 'plots' / 'metrics_vs_thickness.
        png', dpi=140); plt.close(fig)
72         print('plots written for', list(g))
73
74     if __name__ == '__main__': main()

```

B.2.11 plot_sensitivity.py

The script writes `plots/sensitivity_overview.png`, which shows the no-confinement counterfactual overlays at 2 and 13.2 nm and the one-at-a-time metric changes at 2 nm from `tables/SENSITIVITY_OAT.csv`. No simulator is called.

```

1  #!/usr/bin/env python3
2  """plots/sensitivity_overview.png: (a) no-confinement counterfactual overlays at 2 and 13.2 nm, (b) one-
        at-a-time
3  metric changes at 2 nm from tables/SENSITIVITY_OAT.csv. No simulator call."""
4  import csv
5  from pathlib import Path
6  import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
7  ROOT = Path(__file__).resolve().parents[1]
8  def curve(run):
9      d = next(p for p in (ROOT / 'results' / 'runs').iterdir() if p.name.startswith(run)); r = list(csv.
        DictReader((d / 'comparison.csv').open()))
10     return np.array([float(x['vg_V']) for x in r]), np.array([float(x['measured_A_per_um']) for x in r]),
        np.array([float(x['atlas_A_per_um']) for x in r])
11  fig = plt.figure(figsize=(13, 8)); gs = fig.add_gridspec(2, 2)
12  for i, (t, ref, cf) in enumerate([(2.0, 'run_0012', 'run_0016'), (13.2, 'run_0013', 'run_0017')]):
13      ax = fig.add_subplot(gs[0, i]); vg, me, si = curve(ref); _, _, sc = curve(cf)
14      ax.semilogy(vg, me, 'k.', ms=4, label='measured'); ax.semilogy(vg, np.maximum(si, 1e-40), 'r-', label
        ='final model (confinement law on)'); ax.semilogy(vg, np.maximum(sc, 1e-40), 'b--', label='
        counterfactual: dEg = 0, m* = bulk')
15      ax.set_ylim(1e-15, 1e-5); ax.set_xlabel('VG (V)'); ax.set_ylabel('ID (A/um), VD = 0.7 V'); ax.
        set_title(f'{t} nm: confinement law switched off ({cf} vs {ref})', fontsize=9); ax.legend(fontsize
        =7); ax.grid(alpha=.3)
16  rows = [r for r in csv.DictReader((ROOT / 'tables' / 'SENSITIVITY_OAT.csv').open()) if float(r['
        thickness_nm']) == 2.0 and r['case'] != 'no_confinement']
17  lab = [r['case'] for r in rows]; x = np.arange(len(rows))
18  for j, (key, name, scale) in enumerate([(dVth_cc_V, 'dVth (mV)', 1000.0), (dSS_mV_dec, 'dSS_min (mV/
        dec)', 1.0)]):
19      ax = fig.add_subplot(gs[1, j]) if j == 0 else ax
20      if j == 0:
21          ax.bar(x - 0.2, [float(r[dVth_cc_V]) * 1000 for r in rows], 0.4, label='dVth (mV)'); ax.bar(x +
        0.2, [float(r[dSS_mV_dec]) for r in rows], 0.4, label='dSS_min (mV/dec)')
22          ax.set_xticks(x); ax.set_xticklabels(lab, rotation=30, ha='right', fontsize=8); ax.axhline(0,
        color='k', lw=.6); ax.legend(fontsize=8); ax.grid(alpha=.3, axis='y'); ax.set_title('2 nm one-at-a-
        time perturbations (runs 0020-0027 vs run_0012)', fontsize=9)
23  ax = fig.add_subplot(gs[1, 1]); ax.bar(x - 0.2, [float(r[dIon_pct]) for r in rows], 0.4, label='dIon at
        3 V (%)'); ax.bar(x + 0.2, [float(r[dmu_FE_pct]) for r in rows], 0.4, label='dmu_FE (%)')

```

```

24 ax.set_xticks(x); ax.set_xticklabels(lab, rotation=30, ha='right', fontsize=8); ax.axhline(0, color='k',
    lw=.6); ax.legend(fontsize=8); ax.grid(alpha=.3, axis='y'); ax.set_title('2 nm one-at-a-time
    perturbations: on-state', fontsize=9)
25 fig.suptitle('Counterfactual and sensitivity runs (real ATLAS, identical extraction); cases: chi -/+0.1
    eV, Nd x2, Nt x1.5, WTA +5 meV, Dit x3, eps 10.55, overlap 4 um', fontsize=9)
26 fig.tight_layout(); fig.savefig(ROOT / 'plots' / 'sensitivity_overview.png', dpi=140); print('plots/
    sensitivity_overview.png')

```

B.2.12 `qe_build_inputs.py`

The script builds an *unexecuted* Quantum ESPRESSO workflow [14, p. 1] for a matrix of W content and slab thickness, writing `pw.x` input files and a README. Quantum ESPRESSO is not installed on the project machine, so no DFT number of this report comes from this script; it documents how the DFT proxies of the thickness laws could be replaced by IWO-specific calculations.

```

1 #!/usr/bin/env python3
2 """Build the (UNEXECUTED) Quantum ESPRESSO workflow for the xW x thickness matrix.
3
4 Writes qe_workflow/<case>/pw.<step>.in files with a README. Quantum ESPRESSO is NOT installed on this
5 machine (no pw.x on Windows or in WSL; apt install needs an interactive sudo password), so nothing here
6 was run; the thickness laws in config/iwo_material_model.yaml remain labelled DFT_PURE_IN203_PROXY
7 until these inputs are executed and their results replace the proxies.
8
9 Structures: bixbyte In203 (Ia-3, No. 206), a = 10.117 Å, In1 8b (1/4,1/4,1/4), In2 24d (u,0,1/4) with
10 u = -0.0338 (0.4662), O 48e (0.3905, 0.1547, 0.3821) -- Marezio 1966 Wyckoff parameters (
    METADATA_REQUIRES_VERIFICATION
11 against the original table; verify the generated 80-atom cell stoichiometry In32048 and nearest-neighbour
    distances ~2.2 Å).
12 W substitution: one W on an In2 (24d) site in the 80-atom conventional cell gives xW = 1/32 = 3.1 cation
    %; in a 1x1x2
13 supercell (160 atoms) one W gives 1.6 %; two W give 3.1 % -- brackets the ~2 % target (definition of "2 %
    W" undetermined).
14 Slabs: (001) slabs cut from the conventional cell, O-terminated, H-passivated top/bottom (one H per
    surface O, as in
15 Lin2022/Si2021), 25 Å vacuum, dipole correction; thicknesses ~1.0, ~2.0, ~3.5 nm (1, 2, 3.5 conventional-
    cell units).
16 Settings follow the supplied papers: PBE, PAW/ONCV, ecutwfc 71 Ry / ecutrho 639 Ry (Januar2026) or 140 Ry
    ONCV (Lin2022),
17 k-grid 6x6x6 (bulk primitive) / 3x3x1 (slabs), forces < 1e-3 Ry/Bohr. Convergence tests listed in the
    README.
18 Quantities to extract: Eg (PBE, relative shifts only), Ec/Ev shifts vs bulk (vacuum-aligned via the
    planar-averaged
19 potential; pp.x + average.x), effective mass (band fit near Gamma), DOS/PDOS (W-5d near CBM), dielectric
    tensor (ph.x /
20 DFPT, bulk only), V_0 formation energy and charge-transition levels (charged supercells, Makov-Payne
    correction),
21 W substitution energy. A crystalline periodic supercell is NOT the amorphous sputtered film: results are
    trends.
22 """
23 from pathlib import Path
24 import numpy as np
25 from ase.spacegroup import crystal
26 from ase.build import surface, add_vacuum
27 from ase.io import write
28 ROOT = Path(__file__).resolve().parents[1]; OUT = ROOT / 'qe_workflow'
29
30 PSEUDO = {'In': 'In.pbe-dn-kjpaw_psl.1.0.0.UPF', 'O': 'O.pbe-n-kjpaw_psl.1.0.0.UPF', 'W': 'W.pbe-spn-
    kjpaw_psl.1.0.0.UPF', 'H': 'H.pbe-kjpaw_psl.1.0.0.UPF'}

```

```

31 BASE = {'control': {'calculation': 'vc-relax', 'prefix': 'in2o3', 'pseudo_dir': './pseudo', 'outdir': './
    tmp', 'tprnfor': True, 'tstress': True, 'forc_conv_thr': 1e-3, 'etot_conv_thr': 1e-5},
32         'system': {'ecutwfc': 71, 'ecutrho': 639, 'occupations': 'fixed', 'nbnd': None},
33         'electrons': {'conv_thr': 1e-9, 'mixing_beta': 0.4, 'ions': {}, 'cell': {'press_conv_thr': 0.5}}
34
35 def bulk():
36     return crystal(['In', 'In', 'O'], basis=[(0.25, 0.25, 0.25), (0.4662, 0.0, 0.25), (0.3905, 0.1547,
    0.3821)], spacegroup=206, cellpar=[10.117, 10.117, 10.117, 90, 90, 90])
37
38 def with_w(atoms, n_w):
39     a = atoms.copy(); idx = [i for i, s in enumerate(a.get_chemical_symbols()) if s == 'In'][:n_w]
40     for i in idx: a[i].symbol = 'W'
41     return a
42
43 def slab(atoms, layers, vac=25.0):
44     s = surface(atoms, (0, 0, 1), layers, vacuum=vac / 2); s.center(vacuum=vac / 2, axis=2)
45     # H passivation of surface O (one H per O within 1.2 Å of either surface), as in Lin2022 / Si2021
46     z = s.positions[:, 2]; zmin, zmax = z[[i for i, e in enumerate(s.get_chemical_symbols()) if e != 'H'
    ]].min(), z.max()
47     add = []
48     for i, (e, p) in enumerate(zip(s.get_chemical_symbols(), s.positions)):
49         if e == 'O' and (p[2] - zmin < 1.2 or zmax - p[2] < 1.2): add.append(p + np.array([0, 0, 0.98 if
    zmax - p[2] < 1.2 else -0.98]))
50     from ase import Atoms
51     if add: s += Atoms('H' * len(add), positions=add)
52     return s
53
54 def write_case(name, atoms, kpts, calc='vc-relax', extra=None, notes=''):
55     d = OUT / name; d.mkdir(parents=True, exist_ok=True)
56     inp = {k: dict(v) for k, v in BASE.items()}; inp['control']['calculation'] = calc; inp['control']['
    prefix'] = name
57     if extra:
58         for k, v in extra.items(): inp[k].update(v)
59     inp['system'] = {k: v for k, v in inp['system'].items() if v is not None}
60     write(d / f'pw.{calc}.in', atoms, format='espresso-in', input_data=inp, pseudopotentials=PSEUDO, kpts
    =kpts)
61     (d / 'NOTES.txt').write_text(f'{name}: {len(atoms)} atoms, formula {atoms.get_chemical_formula()}\n{
    notes}\nNOT EXECUTED (no QE on this machine).\n')
62     return len(atoms), atoms.get_chemical_formula()
63
64 def main():
65     OUT.mkdir(exist_ok=True); b = bulk(); log = []
66     log.append(('bulk_In203', *write_case('bulk_In203', b, (4, 4, 4), notes='80-atom conventional cell;
    use the 40-atom primitive (ibrav=3) with 6x6x6 for production; follow with bands (m*), DFPT (eps),
    pp.x for CBM/VBM')))
67     log.append(('bulk_IWO_1W_3p1pct', *write_case('bulk_IWO_1W_3p1pct', with_w(b, 1), (4, 4, 4), extra={
    'system': {'occupations': 'smearing', 'smearing': 'mv', 'degauss': 0.01, 'nspin': 1}}, notes='1 W on
    an In site: xW = 1/32 cation = 3.1 %; W6+ donates 3 e -> metallic occupation; compare with charged-
    cell/compensated variants')))
68     sc = b.repeat((1, 1, 2)); log.append(('bulk_IWO_1W_1p6pct_112', *write_case('bulk_IWO_1W_1p6pct_112',
    with_w(sc, 1), (4, 4, 2), extra={'system': {'occupations': 'smearing', 'smearing': 'mv', 'degauss':
    0.01}}, notes='1 W in a 160-atom 1x1x2 cell: 1.6 %; two W -> 3.1 %. W-site configuration sampling
    (8b vs 24d, W-W distance) required')))
69     for layers, tag in [(1, 'slab_1p0nm'), (2, 'slab_2p0nm'), (3, 'slab_3p0nm')]:
70         s = slab(b, layers); log.append((tag + '_In203', *write_case(tag + '_In203', s, (3, 3, 1), calc='
    relax', extra={'system': {'occupations': 'smearing', 'smearing': 'mv', 'degauss': 0.005, 'nspin':
    1}, 'control': {'tefield': True, 'dipfield': True}}, notes='(001) H-passivated slab, 25 Å vacuum,
    dipole correction; extract Ec/Ev vs vacuum (pp.x plot_num=11, average.x), Eg shift vs bulk, m* from
    bands along Gamma-X')))
71     log.append((tag + '_IWO_1W', *write_case(tag + '_IWO_1W', with_w(s, 1), (3, 3, 1), calc='relax',
    extra={'system': {'occupations': 'smearing', 'smearing': 'mv', 'degauss': 0.005}, 'control': {

```

```

    tefield': True, 'dipfield': True}}, notes='same slab with one W (xW depends on layer count; report
    cation %))'))
72  readme = ['# Quantum ESPRESSO workflow (NOT EXECUTED)', '', 'pw.x is not available on this machine.
    These inputs define the matrix xW = {0, ~1.6-3.1 %} x thickness {bulk, ~1, ~2, ~3 nm}.', '',
73      '| case | atoms | formula |', '|---|---|---|'] + [f'| {n} | {a} | {f} |' for n, a, f in log
    ] + ['',
74      '## Convergence tests required before any number replaces a proxy', '- ecutwfc 50/71/90 Ry
    (PAW) or 100/140 Ry (ONCV); k-grid 4/6/8 (bulk), 3x3x1 vs 5x5x1 (slabs); vacuum 15/25/35 Å; slab
    thickness series; W configuration (8b vs 24d, W-W separation); relaxation force < 1e-3 Ry/Bohr.',
75      '## Extraction', '- Eg: PBE gaps are underestimated (bulk ~0.9 eV vs 2.9-3.2 exp); use ONLY
    slab-minus-bulk shifts (dEg_QC). For absolute values use HSE06 on the relaxed bulk.',
76      '- dEc/dEv: vacuum-aligned CBM/VBM from the planar-averaged electrostatic potential (pp.x
    plot_num=11 -> average.x); chi = Evac - Ec directly.',
77      '- m*: parabolic fit of the lowest conduction band within 0.05 Å-1 of Gamma (bands.x);
    report nonparabolicity.',
78      '- DOS/PDOS: projwfc.x; W-5d weight at the CBM.', '- eps: ph.x DFPT (bulk only; eps_inf and
    eps_0 from the Born charges/phonons).',
79      '- V_0: neutral and +2 vacancy supercells (In2O3 and IWO), formation energy vs 0 chemical
    potential, transition levels with Makov-Payne/FNV correction; W substitution energy from the same
    references.',
80      '## Caveat', 'Crystalline periodic supercells approximate the sputtered, largely amorphous
    IWO film only in trend; results are proxies until compared with the measured devices.'])
81  (OUT / 'README.md').write_text('\n'.join(readme) + '\n'); print('\n'.join(readme[:14]))
82
83  if __name__ == '__main__': main()

```

B.2.13 export_report_pdf.py

The script renders the earlier Markdown report docs/FULL_REPORT.md to HTML with print CSS and embedded key figures, and to PDF with the installed Microsoft Edge in headless mode. It changes no report text and calls no simulator; it is not part of the present L^AT_EX report.

```

1  #!/usr/bin/env python3
2  """Render docs/FULL_REPORT.md to docs/FULL_REPORT.html (print CSS, key figures embedded) and to docs/
    FULL_REPORT.pdf
3  using the installed Microsoft Edge in headless mode (no other converter is available on this machine).
4  No simulator call; nothing in the report text is changed."""
5  import base64, html, re, subprocess, sys, tempfile
6  from pathlib import Path
7  import markdown
8  ROOT = Path(__file__).resolve().parents[1]
9  MD = ROOT / 'docs' / 'FULL_REPORT.md'; HTML = ROOT / 'docs' / 'FULL_REPORT.html'; PDF = ROOT / 'docs' / '
    FULL_REPORT.pdf'
10  EDGE = [p for p in [r'C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe', r'C:\Program Files\
    Microsoft\Edge\Application\msedge.exe'] if Path(p).is_file()]
11
12  FIGS = [('plots/2p0/overlay_log.png', 'Figure F1. 2.0 nm: measured (workbook) vs ATLAS run_0012, log
    scale.'),
13          ('plots/13p2/overlay_log.png', 'Figure F2. 13.2 nm: measured vs ATLAS run_0013, log scale.'),
14          ('plots/6p3_validation/overlay_log.png', 'Figure F3. 6.3 nm validation with the shared parameters
    : measured vs ATLAS run_0014 (threshold miss of 0.29 V).'),
15          ('plots/6p3/overlay_log.png', 'Figure F3b. 6.3 nm tuned (device-specific flat-band offset + band
    mobility): measured vs ATLAS run_0015.'),
16          ('plots/6p3/residual.png', 'Figure F3c. 6.3 nm tuned residuals (run_0015).'),
17          ('plots/2p0/residual.png', 'Figure F4. 2.0 nm residuals (log and linear).'),
18          ('plots/13p2/residual.png', 'Figure F5. 13.2 nm residuals.'),
19          ('plots/metrics_vs_thickness.png', 'Figure F6. Device metrics vs thickness, experiment vs ATLAS,
    identical extraction.'),

```

```

20     ('plots/thickness_laws_overview.png', 'Figure F7. The thickness laws of the material model with
    their evidence points.'),
21     ('plots/sensitivity_overview.png', 'Figure F7b. No-confinement counterfactual overlays (top) and
    one-at-a-time sensitivities at 2 nm (bottom), runs 0016-0027.'),
22     ('plots/2p0/band_diagram_vth.png', 'Figure F8. 2.0 nm band diagram near threshold across the
    stack (channel centre).'),
23     ('plots/13p2/band_diagram_vth.png', 'Figure F9. 13.2 nm band diagram near threshold.'),
24     ('plots/2p0/electron_density_vs_vg.png', 'Figure F10. 2.0 nm free-electron density vs VG (channel
    centre and film average; VG = 0 value is nFB).'),
25     ('plots/13p2/mobility_vs_vg.png', 'Figure F11. 13.2 nm: native probe mobility vs the measured
    apparent field-effect mobility.')]
26
27 CSS = """
28 @page { size: A4; margin: 18mm 16mm 18mm 16mm; }
29 body { font-family: Calibri, 'Segoe UI', Arial, sans-serif; font-size: 10.5pt; line-height: 1.38; color:
    #111; max-width: 100%; }
30 h1 { font-size: 20pt; margin: 0 0 4pt 0; } h2 { font-size: 14.5pt; margin-top: 18pt; border-bottom: 1px
    solid #999; padding-bottom: 2pt; page-break-after: avoid; }
31 h3 { font-size: 12pt; margin-top: 12pt; page-break-after: avoid; }
32 p { margin: 5pt 0; text-align: justify; }
33 table { border-collapse: collapse; width: 100%; margin: 6pt 0 8pt 0; font-size: 8.6pt; page-break-inside:
    auto; }
34 th, td { border: 1px solid #bbb; padding: 2.5pt 4pt; vertical-align: top; text-align: left; }
35 th { background: #eaeaea; } tr { page-break-inside: avoid; }
36 code { font-family: Consolas, 'Courier New', monospace; font-size: 8.8pt; background: #f3f3f3; padding: 0
    2pt; }
37 pre { background: #f3f3f3; border: 1px solid #ccc; padding: 6pt; font-size: 8.4pt; white-space: pre-wrap;
    word-wrap: break-word; page-break-inside: avoid; }
38 pre code { background: none; padding: 0; }
39 hr { border: 0; border-top: 1px solid #aaa; margin: 10pt 0; }
40 ol, ul { margin: 4pt 0 4pt 18pt; } li { margin: 2pt 0; }
41 figure { margin: 8pt 0; page-break-inside: avoid; text-align: center; } figure img { max-width: 92%;
    height: auto; }
42 figcaption { font-size: 9pt; color: #333; margin-top: 3pt; }
43 .cover { margin-bottom: 14pt; } .small { font-size: 9pt; color: #444; }
44 h2.fig { page-break-before: always; }
45 """
46
47 def main():
48     md_text = MD.read_text(encoding='utf-8')
49     body = markdown.markdown(md_text, extensions=['tables', 'fenced_code', 'sane_lists'])
50     figs = ['<h2 class="fig">Appendix D. Figures</h2>', '<p class="small">Rendered from plots/ of the
    package; every simulated curve is a real ATLAS run (run id in the title).</p>']
51     for rel, cap in FIGS:
52         p = ROOT / rel
53         if not p.exists(): figs.append(f'<p class="small">[{html.escape(rel)} not present]</p>');
    continue
54         b64 = base64.b64encode(p.read_bytes()).decode()
55         figs.append(f'<figure><figcaption
    >{html.escape(cap)} ({html.escape(rel)})</figcaption></figure>')
56     doc = f'<!DOCTYPE html><html><head><meta charset="utf-8"><title>IWO thickness-aware ATLAS model -
    full report</title><style>{CSS}</style></head><body>{body}{"".join(figs)}</body></html>'
57     HTML.write_text(doc, encoding='utf-8'); print('HTML', HTML, len(doc) // 1024, 'kB')
58     # Word-flavoured copy (images by absolute file path; Word's HTML import does not take data: URIs) for
    the
59     # Microsoft Word COM route used when headless Edge is unavailable (an interactive Edge session
    swallows it).
60     wfigs = ['<h2 class="fig">Appendix D. Figures</h2>']
61     for rel, cap in FIGS:
62         p = ROOT / rel
63         if p.exists(): wfigs.append(f'<p style="text-align:center"></p><p class="small" style="text-align:center">{html.escape(cap)} {html.
escape(rel)}</p>')
64 (ROOT / 'docs' / 'FULL_REPORT_word.html').write_text(f'<!DOCTYPE html><html><head><meta charset="utf
-8"><title>IWO thickness-aware ATLAS model - full report</title><style>{CSS}</style></head><body>{
body}{"".join(wfigs)}</body></html>', encoding='utf-8')
65 if '--word' in sys.argv:
66     ps = f' '$w = New-Object -ComObject Word.Application; $w.Visible = $false
67 $d = $w.Documents.Open("{(ROOT / 'docs' / 'FULL_REPORT_word.html')}"), $false, $true)
68 $d.PageSetup.TopMargin = 50; $d.PageSetup.BottomMargin = 50; $d.PageSetup.LeftMargin = 45; $d.PageSetup.
RightMargin = 45
69 $d.SaveAs2("{PDF}", 17); $d.Close($false); $w.Quit()'
70 r = subprocess.run(['powershell', '-NoProfile', '-NonInteractive', '-Command', ps],
capture_output=True, text=True, timeout=600)
71 ok = PDF.exists() and PDF.stat().st_size > 10000
72 print('PDF (Word COM)', PDF, PDF.stat().st_size // 1024 if PDF.exists() else 0, 'kB', 'OK' if ok
else 'FAILED', r.returncode, (r.stderr or '')[0:-400:])
73 return 0 if ok else 2
74 if not EDGE: print('Edge not found; PDF not produced'); return 1
75 with tempfile.TemporaryDirectory() as prof:
76     cmd = [EDGE[0], '--headless=new', '--disable-gpu', '--no-first-run', '--no-default-browser-check'
, f'--user-data-dir={prof}', '--no-pdf-header-footer', f'--print-to-pdf={PDF}', HTML.resolve().
as_uri()]
77 r = subprocess.run(cmd, capture_output=True, text=True, timeout=300)
78 ok = PDF.exists() and PDF.stat().st_size > 10000
79 print('PDF', PDF, PDF.stat().st_size // 1024 if PDF.exists() else 0, 'kB', 'OK' if ok else 'FAILED',
r.returncode)
80 return 0 if ok else 2
81
82 if __name__ == '__main__': sys.exit(main())
```

B.3 Analysis scripts of 2026-09-25

B.3.1 S1_s1_analytic.py

The script contains the order-of-magnitude analytic electrostatics of specialist S1, with constants taken from the evidence brief and the material model (THEORY). It provides the dimensional estimates against which the ATLAS-based S1 tables are checked.

```
1 """S1 analytic dimensional electrostatics (order-of-magnitude). Constants from EVIDENCE_BRIEF sec.1 and
config/iwo_material_model.yaml."""
2 import numpy as np
3 q = 1.602176634e-19; eps0 = 8.8541878128e-14; kT = 0.025852
4 COX = 8.955369814335959e-7; CQ = COX / q; es = 9.3 * eps0
5 T = [2.0, 6.3, 13.2, 31.8]
6 print(f"Cox/q = {CQ:.3e} cm^-2/V ; q/Cox = {1e12/CQ:.4f} V per 1e12 cm^-2")
7 print('\n1 Debye length L_D = sqrt(eps kT/(q^2 n)) [nm]')
8 for n in [1e16, 2.5e17, 3e17, 1e18, 3e18, 1e19]:
9     print(f" n = {n:.1e}: L_D = {np.sqrt(es*kT/(q*n))*1e7:6.2f} nm")
10 print(' trap-screening length L_t = sqrt(eps WTA/(q n_t)), WTA = 0.040 V (exponential tail, E_F below
the tail states)')
11 for nt in [1e18, 2e18, 5e18, 7e18]:
12     print(f" n_t = {nt:.1e}: L_t = {np.sqrt(es*0.040/(q*nt))*1e7:5.2f} nm")
13 print('\n2 Depletion width W = Q_s/N_d [nm] (full depletion if W >= t)')
14 for Nd in [2.5e17, 1e18, 3e18]:
15     print(f" Nd {Nd:.1e}: " + ' , '.join(f"Qs {Qs:.2g}: {Qs/Nd*1e7:7.1f}" for Qs in [1e11, 5e11, 1e12,
1.64e12, 3e12]))
16 print(' Donor sheet N_d*t (cm^-2) of the V1 law and its Vth weight q Nd t/Cox:')
17 for t in T:
```



```
18     Nd = 2.5e17 * (1 + (t/20)**4)
19     print(f"    t {t}: Nd {Nd:.2e}, Nd t = {Nd*t*1e-7:.2e} cm^-2, qNdt/Cox = {Nd*t*1e-7/CQ:.3f} V, qNd t
    ^2/(2 eps) = {q*Nd*(t*1e-7)**2/(2*es):.3f} V")
20 print('\n3) Lever arm of a sheet charge [V per 1e12 cm^-2]: front 1/Cox ; back-surface referenced to the
    back (1/Cox + t/eps_s)')
21 for t in T:
22     r = COX * t * 1e-7 / es
23     print(f"    t {t}: front {1e12/CQ:.3f} ; back {1e12/CQ*(1+r):.3f} (x{1+r:.2f}) ; IWO 'EOT' t*3.9/9.3 =
    {t*3.9/9.3:.2f} nm vs stack EOT 3.86 nm")
24 print('\n4) Gate depletion capacity between Vth and Vg = -3 V: Cox (Vth+3)/q')
25 for t, v in [(2.0, 0.662), (6.3, 0.644), (13.2, 0.083)]:
26     print(f"    t {t}: {(v+3)*CQ:.2e} cm^-2 (-> a neutral layer of sheet density above this is not
    depletable at -3 V)")
27 print('    31.8 nm always on: Id(-3 V) >= 2.6e-7 A/um needs free sheet n = I L/(q mu Vd):')
28 for mu in [50.2, 82.5]:
29     print(f"    mu {mu}: n_s = {2.6e-7*1e4*20e-4/(q*mu*0.7):.2e} cm^-2")
30 print('\n5) Measured Vth_cc steps expressed as sheet charge (q/Cox = 0.179 V per 1e12):')
31 for name, dv in [('2 -> 6.3', 0.644-0.662), ('6.3 -> 13.2', 0.083-0.644), ('2 -> 13.2', 0.083-0.662), ('
    6.3 validation miss', 0.644-0.350), ('6.3 tuned Vth_lin miss', 1.820-1.471)]:
32     print(f"    {name}: dv {dv:.3f} V = {dv*CQ:.2e} cm^-2 ; over 6.3 nm {dv*CQ/6.3e-7:.2e} cm^-3 ; over
    13.2 nm {dv*CQ/13.2e-7:.2e} cm^-3")
33 print('\n6) Confinement difference predicted between 2 and 6.3 nm: V1 law dEc(2)-dEc(6.3) = %.3f eV;
    infinite well h^2/(8 m* t^2): %.3f eV' % (
34     0.9205*2**-1.3815 - 0.9205*6.3**-1.3815, 0.3760/(0.257*4) - 0.3760/(0.218*6.3**2)))
```

B.3.2 S1_s1_atlas_tables.py

The script builds the S1 electrostatics tables only from existing ATLAS outputs and the measured data (execution.json, comparison.csv and the conduction-band, quasi-Fermi-level and electron-density profiles of the runs). No ATLAS launch is made.

```
1 """S1 (electrostatics): tables built ONLY from existing ATLAS outputs + measured data.
2 No ATLAS launch. Reads results/runs/*/execution.json, comparison.csv, cb/qfn/n profiles.
3 """
4 import os, sys, json, glob, csv
5 import numpy as np
6
7 PKG = r"C:\Users\moham\Downloads\IWO_Local_Codex_Handoff (1)\IWO_PHYSICS_CONSTRAINED_MODEL_V1"
8 OUT = os.path.join(PKG, 'analysis_2026-09-25', 'scratch', 'S1')
9 RUNS = os.path.join(PKG, 'results', 'runs')
10 sys.path.insert(0, os.path.join(PKG, 'scripts'))
11
12 def rdir(rid):
13     return glob.glob(os.path.join(RUNS, rid + '_*'))[0]
14
15 def exe(rid):
16     return json.load(open(os.path.join(rdir(rid), 'execution.json')))
17
18 def meta(rid):
19     return json.load(open(os.path.join(rdir(rid), 'metadata.json')))
20
21 def read_xy(path):
22     xs, ys = [], []
23     for line in open(path):
24         p = line.split()
25         if len(p) != 2:
26             continue
27         try:
28             xs.append(float(p[0])); ys.append(float(p[1]))
```

```

29         except ValueError:
30             pass
31         return np.array(xs), np.array(ys)
32
33 KEYS = ['Vth_cc_1e-9_V', 'Vth_lin_V', 'SS_min_window_mV_dec', 'SS_cc_1e-10_1e-8_mV_dec', 'gm_max_A_V_um',
34         'Ion_A_per_um']
35 runs = ['run_0012', 'run_0016', 'run_0014', 'run_0015', 'run_0030', 'run_0031', 'run_0032', 'run_0033',
36         'run_0034', 'run_0035', 'run_0013', 'run_0017', 'run_0020', 'run_0021', 'run_0022', 'run_0023',
37         'run_0024', 'run_0025']
38 res = {}
39 print('run      t      Qf      dEc      mu0      | Vth_cc Vth_lin gap SSmin SScc gm_max Ion      |
40       active RMSE')
41 for r in runs:
42     e = exe(r); m = meta(r)['material']
43     s = e['device_metrics_sim']; x = e['device_metrics_exp']
44     res[r] = {'sim': {k: s[k] for k in KEYS}, 'exp': {k: x[k] for k in KEYS}, 't': m['t_nm'], 'Qf': m['
45               Qf_cm2'],
46               'dEc': m['dEc_eV'], 'mu0': m['mu0'], 'Nd': m['Nd_cm3'], 'Nt': m['Nt_cm3'], 'label': e['
47               label'],
48               'rmse_active': e['fit_metrics']['regions']['active']['rmse_log10']}
49     print(f"{r} {m['t_nm']:5.1f} {m['Qf_cm2']:9.3g} {m['dEc_eV']:6.3f} {m['mu0']:6.2f} | {s['Vth_cc_1e-9
50           _V']:6.3f} {s['Vth_lin_V']:7.3f} "
51           f"{s['Vth_lin_V']-s['Vth_cc_1e-9_V']:5.3f} {s['SS_min_window_mV_dec']:6.1f} {s['SS_cc_1e-10_1e
52           -8_mV_dec']:6.1f} {s['gm_max_A_V_um']:9.3e} {s['Ion_A_per_um']:9.3e} | {res[r]['rmse_active']:.3f}
53           {e['label']}"")
54
55 print('\nMeasured (from execution.json exp blocks):')
56 for r in ['run_0012', 'run_0014', 'run_0013']:
57     x = res[r]['exp']
58     print(f"t={res[r]['t']}: Vth_cc {x['Vth_cc_1e-9_V']:.3f} Vth_lin {x['Vth_lin_V']:.3f} gap {x['
59           Vth_lin_V']-x['Vth_cc_1e-9_V']:.3f} SSmin {x['SS_min_window_mV_dec']:.1f} SScc {x['SS_cc_1e-10_1e-8
60           _mV_dec']:.1f} gm {x['gm_max_A_V_um']:.3e} Ion {x['Ion_A_per_um']:.3e}")
61
62 # Campaign A deltas vs run_0014 and vs measurement
63 print('\nCampaign A (6.3 nm) deltas vs run_0014 | vs measured')
64 base = res['run_0014']['sim']; meas = res['run_0014']['exp']
65 for r in ['run_0015', 'run_0030', 'run_0031', 'run_0032', 'run_0033', 'run_0034', 'run_0035']:
66     s = res[r]['sim']
67     d = lambda k: s[k] - base[k]
68     dm = lambda k: s[k] - meas[k]
69     print(f"{r}: dVcc {d('Vth_cc_1e-9_V'):.3f} dSSmin {d('SS_min_window_mV_dec'):.1f} dSScc {d('
70           SS_cc_1e-10_1e-8_mV_dec'):.1f} dVlin {d('Vth_lin_V'):.3f} "
71           f"dgm {100*d('gm_max_A_V_um')/base['gm_max_A_V_um']:.1f}% dIon {100*d('Ion_A_per_um')/base['
72           Ion_A_per_um']:.1f}% || vs meas: Vcc {dm('Vth_cc_1e-9_V'):.3f} "
73           f"SSmin {dm('SS_min_window_mV_dec'):.1f} SScc {dm('SS_cc_1e-10_1e-8_mV_dec'):.1f} Vlin {dm('
74           Vth_lin_V'):.3f} gm {100*dm('gm_max_A_V_um')/meas['gm_max_A_V_um']:.1f}% Ion {100*dm('Ion_A_per_um
75           ')/meas['Ion_A_per_um']:.1f}%")
76
77 # Horizontal offset dVg(I) = Vg_meas(I) - Vg_sim(I), and vertical log ratio at fixed Vg
78 def vg_at(vg, idv, level):
79     idv = np.asarray(idv); ok = idv > 0
80     for i in range(len(vg) - 1):
81         if ok[i] and ok[i+1] and idv[i] < level <= idv[i+1]:
82             a, b = np.log10(idv[i]), np.log10(idv[i+1])
83             return vg[i] + (np.log10(level) - a) / (b - a) * (vg[i+1] - vg[i])
84     return np.nan
85
86 levels = [1e-10, 1e-9, 1e-8, 3e-8, 1e-7, 2e-7, 3e-7]
87 print('\nHorizontal offset Vg_meas(I) - Vg_sim(I) [V] at I (A/um) =', levels)
88 curves = {}
89 for r in ['run_0012', 'run_0014', 'run_0015', 'run_0030', 'run_0031', 'run_0032', 'run_0033', 'run_0034',

```

```
'run_0035', 'run_0013', 'run_0016', 'run_0017']:
```

```
76 rows = list(csv.DictReader(open(os.path.join(rdir(r), 'comparison.csv'))))
77 vg = np.array([float(x['vg_V']) for x in rows]); im = np.array([float(x['measured_A_per_um']) for x
    in rows]); isim = np.array([float(x['atlas_A_per_um']) for x in rows])
78 curves[r] = (vg, im, isim)
79 off = [vg_at(vg, im, L) - vg_at(vg, isim, L) for L in levels]
80 print(f"{r} ({res[r]['t']} nm): " + ' '.join(f"{o:+.3f}" for o in off))
81
82 # measured-curve based constant-current-definition effect:  $V_g(1e-9*ratio) - V_g(1e-9)$ 
83 print('\nCC-definition effect from MEASURED curves: Vth shift if the film had the 2 nm mobility')
84 data = list(csv.DictReader(open(os.path.join(PKG, 'data', 'experimental_clean.csv'))))
85 meas = {}
86 for t in [2.0, 6.3, 13.2, 31.8]:
87     rr = [d for d in data if abs(float(d['thickness_nm']) - t) < 1e-6]
88     meas[t] = (np.array([float(d['vg_V']) for d in rr]), np.array([float(d['id_A_per_um']) for d in rr]))
89 muFE = {2.0: 12.15, 6.3: 10.95, 13.2: 50.17}
90 mu0 = {2.0: res['run_0012']['mu0'], 6.3: res['run_0015']['mu0'], 13.2: res['run_0013']['mu0']}
91 for t in [6.3, 13.2]:
92     vg, idv = meas[t]
93     for name, mu in [('mu_FE', muFE), ('mu0_band_eff', mu0)]:
94         ratio = mu[t] / mu[2.0]
95         dv = vg_at(vg, idv, 1e-9 * ratio) - vg_at(vg, idv, 1e-9)
96         print(f"t={t}: {name} ratio {ratio:.3f} -> Vth_cc(if mu were the 2 nm value) - Vth_cc = {dv:+.3f}
            V (log10 ratio {np.log10(ratio):+.3f})")
97
98 # Profiles at near_vth / on from ATLAS (channel centre x = 12 um)
99 print('\nATLAS depth profiles at x=12 um (IWO only): Ec-Efn at back/front, band bending, sheet n')
100 for r in ['run_0012', 'run_0014', 'run_0015', 'run_0030', 'run_0034', 'run_0035', 'run_0013', 'run_0017']
    ]:
101     d = rdir(r); t = res[r]['t']
102     dev = open(os.path.join(d, 'device.in')).read()
103     import re
104     vnear = re.findall(r'solve vstep=[-\\d.] + vfinal=[-\\d.]+ name=gate\\nsave outf=near_vth.str', dev)
105     for snap in ['vth', 'on']:
106         y, cb = read_xy(os.path.join(d, f'cb_{snap}.dat')); _, qf = read_xy(os.path.join(d, f'qfn_{snap}.
            dat')); _, n = read_xy(os.path.join(d, f'n_{snap}.dat'))
107         # IWO = depth 0.070 .. 0.070 + t (um); air above has n=0
108         msk = (y >= 0.07 - 1e-7) & (y <= 0.07 + t * 1e-3 + 1e-7)
109         yy, cc, qq, nn = y[msk], cb[msk], qf[msk], n[msk]
110         # remove duplicate depth points
111         _, idx = np.unique(yy, return_index=True); yy, cc, qq, nn = yy[idx], cc[idx], qq[idx], nn[idx]
112         ns = np.trapezoid(nn, yy * 1e-4)
113         cen = np.trapezoid(nn * (yy - yy[0]) * 1e3, yy) / np.trapezoid(nn, yy)
114         print(f"{r} {snap} (Vg={vnear[0] if snap=='vth' else 3.0}): Ec-Efn back {cc[0]-qq[0]:+.3f} eV,
            front {cc[-1]-qq[-1]:+.3f} eV, Ec(back)-Ec(front) {cc[0]-cc[-1]:+.3f} eV, n_s {ns:.3e} cm^-2,
            centroid from back {cen:.2f} nm of {t}")
115
116 json.dump(res, open(os.path.join(OUT, 's1_atlas_tables.json'), 'w'), indent=1)
```

B.3.3 S1_s1_poisson1d.py

The script has two parts: (A) post-processing of the ATLAS depth profiles into a trapped-charge budget at the cut $x = 12 \mu\text{m}$, and (B) a 1-D Poisson plus gradual-channel (Pao-Sah) surrogate of the V1 model, used only for decomposition and what-if questions (lever arm of a back-surface charge, neutral-layer onset, no-confinement symmetric test). The surrogate is checked against ATLAS before use; it underlies the threshold budget of Section 4.18 (SURR).

```

1  """S1 electrostatics: (A) post-processing of ATLAS depth profiles (trapped-charge budget at the cut x=12
   um),
2  (B) 1-D Poisson + gradual-channel (Pao-Sah) surrogate of the V1 model, used ONLY for decomposition / what
   -if
3  (lever arm of a back-surface charge, neutral-layer onset, no-confinement symmetric test). It is validated
   against
4  the ATLAS runs it mimics before any use; it is NOT an ATLAS result and never replaces one.
5  Physics identical to the V1 deck where possible: Fermi-Dirac electrons ( $N_c F_1/2$ ), acceptor tail NTA  $\exp((E-E_c)/WTA)$ 
6  (integral  $N_t$ ), deep acceptor Gaussian  $\exp(-(E-(E_c-EGA))/WGA)^2)$ , interface acceptor Gaussian in the
   first 0.25 nm,
7  fully ionised Nd, front fixed sheet  $Q_f$ , optional back sheet  $Q_b$  at the IWO/air surface, zero field in air,
8  series oxide  $Cox = 8.955e-7$  F/cm2, gate WF, constant mobility  $\mu_0$ ,  $V_d = 0.7$  V,  $L = 20$  um.
9  Omitted vs ATLAS: 2-D top-contact access (Pd over the film), lateral fields, contact regions.
10 """
11 import os, sys, json, glob, re
12 import numpy as np
13 from scipy.linalg import solve_banded
14
15 PKG = r"C:\Users\moham\Downloads\IWO_Local_Codex_Handoff (1)\IWO_PHYSICS_CONSTRAINED_MODEL_V1"
16 OUT = os.path.join(PKG, 'analysis_2026-09-25', 'scratch', 'S1')
17 RUNS = os.path.join(PKG, 'results', 'runs')
18 sys.path.insert(0, os.path.join(PKG, 'scripts'))
19 from extract_metrics import metrics
20
21 q = 1.602176634e-19; eps0 = 8.8541878128e-14; kT = 8.617333262e-5 * 300.0
22 COX = 8.955369814335959e-7; CQ = COX / q; L_CM = 20e-4; VD = 0.7
23
24 # ----- tables -----
25 _eta = np.linspace(-12, 70, 16401)
26 _e = np.linspace(0, 110, 22001)
27 _F = np.empty_like(_eta)
28 for i0 in range(0, len(_eta), 400):
29     et = _eta[i0:i0+400, None]
30     _F[i0:i0+400] = 2/np.sqrt(np.pi) * np.trapezoid(np.sqrt(_e)[None, :] / (1 + np.exp(np.clip(_e[None, :] - et, -700, 700)))), _e, axis=1)
31 _dF = np.gradient(_F, _eta)
32
33 def F12(x):
34     return np.where(x < -12, np.exp(x), np.interp(x, _eta, _F))
35
36 def dF12(x):
37     return np.where(x < -12, np.exp(x), np.interp(x, _eta, _dF))
38
39 _U = np.linspace(-4.0, 1.8, 5801)
40 _tab = {}
41 def occ_table(kind, a, b, Eg=3.2):
42     key = (kind, round(a, 6), round(b, 6))
43     if key in _tab:
44         return _tab[key]
45     E = np.linspace(-Eg, 0.0, 12801)
46     g = (np.exp(E / a) / a) if kind == 'tail' else np.exp(-(E + a) / b) ** 2)
47     val = np.empty_like(_U)
48     for i0 in range(0, len(_U), 200):
49         u = _U[i0:i0+200, None]
50         f = 1 / (1 + np.exp(np.clip((E[None, :] - u) / kT, -700, 700)))
51         val[i0:i0+200] = np.trapezoid(g[None, :] * f, E, axis=1)
52     _tab[key] = (val, np.gradient(val, _U))
53     return _tab[key]
54
55 def tab_eval(tb, u):

```

```

56     return np.interp(u, _U, tb[0]), np.interp(u, _U, tb[1])
57
58 # ----- parameters -----
59 def params_from_run(rid):
60     d = glob.glob(os.path.join(RUNS, rid + '_*'))[0]
61     m = json.load(open(os.path.join(d, 'metadata.json'))['material'])
62     dev = open(os.path.join(d, 'device.in')).read()
63     qb = re.findall(r'interface qf=[-+\d.eE+] x\.\min=2 x\.\max=22', dev)
64     return dict(t=m['t_nm'], chi=m['chi_eV'], wf=m['gate_wf'], eps=m['eps_iwo'] * eps0, Nc=m['Nc_cm3'],
65               Nt=m['NTA_cm3_eV'] * m['WTA_eV'],
66               WTA=m['WTA_eV'], Nd=m['Nd_cm3'], NGA=m['NGA'], EGA=m['EGA'], WGA=m['WGA'], NIT=m['it_peak']
67               / 2.5e-8, EIT=m['it_e'],
68               WIT=m['it_w'], Qf=m['Qf_cm2'], Qb=float(qb[0]) if qb else 0.0, mu=m['mu0'], Eg=m['Eg_eV']
69               ], dEc=m['dEc_eV'], mstar=m['mstar_m0'])
70
71 def law_params(t, Qf=1.73e12, mu_band=None, qc=True):
72     """V1 thickness laws (config/iwo_material_model.yaml) for thicknesses without an ATLAS run (e.g. 31.8
73     nm)."""
74     dEc = 0.9205 * t ** -1.3815 if qc else 0.0
75     ms = 0.208 + (0.1311 * t ** -1.412 if qc else 0.0)
76     Nc = 2.549759003818449e18 * (ms / 0.217746617029316) ** 1.5 # scaled from the 6.3 nm metadata value
77     rough = (1 - 0.287 / t) ** 2
78     return dict(t=t, chi=4.3 - dEc, wf=4.7, eps=9.3 * eps0, Nc=Nc, Nt=2e19 * (2 / t) ** 0.75, WTA=0.04,
79               Nd=2.5e17 * (1 + (t / 20) ** 4),
80               NGA=5e16, EGA=0.6, WGA=0.15, NIT=3e11 / 2.5e-8, EIT=0.3, WIT=0.12, Qf=Qf, Qb=0.0, mu=
81               mu_band * rough, Eg=3.05 + dEc, dEc=dEc, mstar=ms)
82
83 def no_qc(p):
84     p = dict(p); p['Nc'] = p['Nc'] * (0.208 / p['mstar']) ** 1.5; p['chi'] = 4.3; p['Eg'] = 3.05; p['dEc']
85     = 0.0; p['mstar'] = 0.208
86     return p
87
88 # ----- 1-D solver -----
89 class Film:
90     def __init__(self, p):
91         self.p = p
92         self.N = max(121, int(p['t'] / 0.04) + 1)
93         self.x = np.linspace(0, p['t'] * 1e-7, self.N)
94         self.h = self.x[1] - self.x[0]
95         self.w = np.full(self.N, self.h); self.w[0] = self.w[-1] = self.h / 2
96         self.mit = (self.x < 0.25e-7 + 1e-12).astype(float)
97         self.tt = occ_table('tail', p['WTA'], 0.0)
98         self.tg = occ_table('gauss', p['EGA'], p['WGA'])
99         self.ti = occ_table('gauss', p['EIT'], p['WIT'])
100
101     def charges(self, u):
102         p = self.p
103         n = p['Nc'] * F12(u / kT); dn = p['Nc'] * dF12(u / kT) / kT
104         t0, dt0 = tab_eval(self.tt, u); g0, dg0 = tab_eval(self.tg, u); i0, di0 = tab_eval(self.ti, u)
105         nt, ng, ni = p['Nt'] * t0, p['NGA'] * g0, p['NIT'] * i0 * self.mit
106         rho = p['Nd'] - n - nt - ng - ni
107         drho = -(dn + p['Nt'] * dt0 + p['NGA'] * dg0 + p['NIT'] * di0 * self.mit)
108         return rho, drho, n, nt, ng, ni
109
110     def solve(self, Vg, Vc, V):
111         p = self.p; a = p['eps'] / q / self.h; Vgate = Vg - p['wf']
112         for it in range(300):
113             u = V + p['chi'] - Vc
114             rho, drho = self.charges(u)[2:]
115             r = np.empty(self.N)
116             r[1:-1] = a * (V[2:] - 2 * V[1:-1] + V[:-2]) + rho[1:-1] * self.h

```

```

110         r[0] = a * (V[1] - V[0]) + CQ * (Vgate - V[0]) + p['Qf'] + rho[0] * self.h / 2
111         r[-1] = p['Qb'] - a * (V[-1] - V[-2]) + rho[-1] * self.h / 2
112         ab = np.zeros((3, self.N)); ab[0, 1:] = a; ab[2, :-1] = a
113         ab[1, 1:-1] = -2 * a + drho[1:-1] * self.h
114         ab[1, 0] = -a - CQ + drho[0] * self.h / 2; ab[1, -1] = -a + drho[-1] * self.h / 2
115         dV = solve_banded((1, 1), ab, -r)
116         mx = np.max(np.abs(dV))
117         if mx > 0.1:
118             dV *= 0.1 / mx
119         V = V + dV
120         if mx < 1e-9:
121             break
122         return V
123
124     def state(self, V, Vc):
125         u = V + self.p['chi'] - Vc
126         rho, drho, n, nt, ng, ni = self.charges(u)
127         W = self.w
128         return dict(u_front=u[0], u_back=u[-1], ns=np.sum(n * W), nt=np.sum(nt * W), ng=np.sum(ng * W),
129                    ni=np.sum(ni * W),
130                    bend=u[0] - u[-1])
131
132     def idvg(self, vgs, vcs=np.linspace(0, VD, 36)):
133         V = np.full(self.N, vgs[0] - self.p['wf'])
134         ids, st0 = [], []
135         for Vg in vgs:
136             V = self.solve(Vg, 0.0, V); V0 = V.copy()
137             ns = [self.state(V, 0.0)['ns']]; st0.append(self.state(V, 0.0))
138             Vv = V.copy()
139             for Vc in vcs[1:]:
140                 Vv = self.solve(Vg, Vc, Vv); ns.append(self.state(Vv, Vc)['ns'])
141                 ids.append(q * self.p['mu'] / L_CM * np.trapezoid(ns, vcs) * 1e-4) # A/um
142                 V = V0
143             return np.array(ids), st0
144
145     VGS = np.round(np.arange(-1.5, 3.0001, 0.05), 4)
146
147     def run1d(p, vgs=VGS):
148         f = Film(p)
149         ids, st = f.idvg(vgs)
150         m = metrics(vgs, ids, is_sim=True, t_nm=p['t'])
151         vcc = m['Vth_cc_1e-9_V']
152         # Gauss decomposition at the source-end (Vc = 0) threshold state (interpolated in Vg)
153         dec = None
154         if vcc is not None:
155             k = int(np.searchsorted(vgs, vcc)); k = min(max(k, 1), len(vgs) - 1)
156             wgt = (vcc - vgs[k-1]) / (vgs[k] - vgs[k-1])
157             s = {key: st[k-1][key] * (1 - wgt) + st[k][key] * wgt for key in st[k]}
158             tcm = p['t'] * 1e-7
159             dec = {'wf_minus_chi_bulk': p['wf'] - 4.3, 'dEc': p['dEc'], 'u_front(EFn-Ec)': s['u_front'], 'Qf':
160                   -p['Qf'] / CQ, 'Qb': -p['Qb'] / CQ,
161                   'Nd': -p['Nd'] * tcm / CQ, 'tail_trapped': s['nt'] / CQ, 'deep_gauss': s['ng'] / CQ, '
162                   interface_traps': s['ni'] / CQ,
163                   'free_electrons': s['ns'] / CQ, 'band_bending_front_minus_back_eV': s['bend'], 'nt_cm2': s
164                   ['nt'], 'ns_cm2': s['ns']}
165             dec['sum_check'] = sum(v for kk, v in dec.items() if kk in ('wf_minus_chi_bulk', 'dEc', 'Qf', 'Qb',
166                   'Nd', 'tail_trapped', 'deep_gauss', 'interface_traps', 'free_electrons')) + s['u_front'] - vcc
167         return {'metrics': {k: m[k] for k in ['Vth_cc_1e-9_V', 'Vth_lin_V', 'SS_cc_1e-10_1e-8_mV_dec', '
168                   gm_max_A_V_um', 'Ion_A_per_um']}, 'decomp': dec,
169                'Id_first': float(ids[0])}

```

```

165 def fmt(m):
166     return (f"Vcc {m['Vth_cc_1e-9_V']:.3f} Vlin {m['Vth_lin_V']:.3f} gap {m['Vth_lin_V']-m['Vth_cc_1e-9_V']:.3f} SSc {m['SS_cc_1e-10_1e-8_mV_dec']:.1f} "
167            f"gm {m['gm_max_A_V_um']:.3e} Ion {m['Ion_A_per_um']:.3e}")
168
169 if __name__ == '__main__':
170     R = {}
171     # ----- (B1) validation of the surrogate against ATLAS -----
172     print('=== (B1) 1-D surrogate vs ATLAS (same parameters) ===')
173     val_runs = ['run_0012', 'run_0016', 'run_0014', 'run_0015', 'run_0030', 'run_0031', 'run_0032', 'run_0033', 'run_0034', 'run_0035', 'run_0013', 'run_0017']
174     for rid in val_runs:
175         p = params_from_run(rid)
176         out = run1d(p)
177         ex = json.load(open(glob.glob(os.path.join(RUNS, rid + '_*'))[0] + '/execution.json'))['device_metrics_sim']
178         R[rid] = {'p': {k: v for k, v in p.items()}, '1d': out, 'atlas': {k: ex[k] for k in out['metrics']}}
179         print(f"[rid] t={p['t']} Qf={p['Qf']:.3g} Qb={p['Qb']:.3g}\n 1-D : {fmt(out['metrics'])}\n ATLAS: {fmt(ex)}")
180         d = out['decomp']
181         print(' Gauss terms at 1-D threshold (V): ' + ', '.join(f"{k} {v:+.3f}" for k, v in d.items() if k not in ('nt_cm2', 'ns_cm2')) + f", nt {d['nt_cm2']:.3g} cm-2, ns {d['ns_cm2']:.3g} cm-2")
182     json.dump(R, open(os.path.join(OUT, 's1_poisson1d_validation.json'), 'w'), indent=1, default=float)
183
184     # ----- (B2) one-at-a-time switches on the calibrated configurations -----
185     print('\n=== (B2) one-at-a-time term removal (1-D), dVth_cc = Vth(full) - Vth(term removed) ===')
186     SW = {}
187     for rid in ['run_0012', 'run_0015', 'run_0013']:
188         p0 = params_from_run(rid); base = run1d(p0)['metrics']
189         row = {'full': base['Vth_cc_1e-9_V']}
190         for name, mod in [(('Qf', {'Qf': 0.0}), ('Nd', {'Nd': 0.0}), ('tail', {'Nt': 0.0}), ('Dit', {'NIT': 0.0}), ('mu->13.30', {'mu': 13.30})]:
191             p = dict(p0); p.update(mod); v = run1d(p)['metrics']['Vth_cc_1e-9_V']
192             row[name] = base['Vth_cc_1e-9_V'] - v
193             v = run1d(no_qc(p0))['metrics']['Vth_cc_1e-9_V']; row['confinement(dEc+m*)'] = base['Vth_cc_1e-9_V'] - v
194             p = dict(p0); p['Nc'] = p0['Nc'] * (0.208 / p0['mstar']) ** 1.5; v = run1d(p)['metrics']['Vth_cc_1e-9_V']; row['m*_only(Nc)'] = base['Vth_cc_1e-9_V'] - v
195             SW[rid] = row
196             print(rid, ' '.join(f"{k} {v:+.3f}" for k, v in row.items()))
197     json.dump(SW, open(os.path.join(OUT, 's1_poisson1d_switches.json'), 'w'), indent=1)
198
199     # ----- (B3) lever arm of a back-surface sheet charge -----
200     print('\n=== (B3) back-surface charge Qb = -1e12 cm^-2: dVth_cc, dVth_lin (1-D); front-equivalent = 0.179 V ===')
201     LA = {}
202     for rid in ['run_0012', 'run_0015', 'run_0013']:
203         p0 = params_from_run(rid); base = run1d(p0)['metrics']
204         for tag, mod in [(('fullDOS', {}), ('no_tail', {'Nt': 0.0})]
205             pb = dict(p0); pb.update(mod); b0 = run1d(pb)['metrics'] if mod else base
206             pb2 = dict(pb); pb2['Qb'] = -1e12; b1 = run1d(pb2)['metrics']
207             t = p0['t']; analytic_back = (1e12 / CQ) * (1 + COX * t * 1e-7 / (9.3 * eps0))
208             LA[f'{rid}_{tag}'] = dict(dVcc=b1['Vth_cc_1e-9_V'] - b0['Vth_cc_1e-9_V'], dVlin=b1['Vth_lin_V'] - b0['Vth_lin_V'], dSScc=b1['SS_cc_1e-10_1e-8_mV_dec'] - b0['SS_cc_1e-10_1e-8_mV_dec'])
209             print(f"[rid] t={t} {tag}: dVcc {LA[f'{rid}_{tag}']['dVcc']:+.3f} dVlin {LA[f'{rid}_{tag}']['dVlin']:+.3f} dSScc {LA[f'{rid}_{tag}']['dSScc']:+.1f} | analytic front 0.179, back-referenced {analytic_back:.3f}")
210     json.dump(LA, open(os.path.join(OUT, 's1_poisson1d_leverarm.json'), 'w'), indent=1)
211
212

```

```

213 # ----- (B4) no-confinement symmetric test -----
214 print('\n=== (B4) no confinement, single Qf = 4.76e10 (A6 value), per-film mu0 ===')
215 SY = {}
216 for rid in ['run_0012', 'run_0014', 'run_0013']:
217     p = no_qc(params_from_run(rid)); p['Qf'] = 4.76e10; out = run1d(p)['metrics']; SY[rid] = out
218     print(rid, fmt(out))
219 json.dump(SY, open(os.path.join(OUT, 's1_poisson1d_symmetry.json'), 'w'), indent=1, default=float)
220
221 # ----- (B5) neutral-layer onset: Vth_cc and Id(-3 V) vs uniform Nd at 6.3 / 13.2 / 31.8 nm
222     -----
223 print('\n=== (B5) uniform Nd scan (confinement on, Qf 1.73e12), Vg from -3 V ===')
224 vgs3 = np.round(np.arange(-3.0, 3.0001, 0.05), 4)
225 NL = {}
226 for t, mu_b in [(6.3, 12.4), (13.2, 61.9), (31.8, 84.0)]:
227     for Nd in [2.5e17, 1e18, 2e18, 3e18, 5e18]:
228         p = law_params(t, mu_band=mu_b); p['Nd'] = Nd
229         o = run1d(p, vgs3)
230         NL[f"{t}_{Nd:.1e}"] = dict(Vcc=o['metrics']['Vth_cc_1e-9_V'], Id_m3V=o['Id_first'], NdT=Nd *
231             t * 1e-7)
232         vcc = o['metrics']['Vth_cc_1e-9_V']
233         print(f"t={t} Nd={Nd:.1e} (Nd*t={Nd*t*1e-7:.2e} cm-2): Vth_cc {('%3f' % vcc) if vcc is not
234             None else 'none(always on)'} Id(-3V) {o['Id_first']:.2e} A/um")
235         p = law_params(t, mu_band=mu_b); o = run1d(p, vgs3)
236         vcc = o['metrics']['Vth_cc_1e-9_V']
237         print(f"t={t} V1-law Nd={p['Nd']:.2e}: Vth_cc {('%3f' % vcc) if vcc is not None else 'none'} Id
238             (-3V) {o['Id_first']:.2e}")
239 json.dump(NL, open(os.path.join(OUT, 's1_poisson1d_neutral_layer.json'), 'w'), indent=1, default=
240     float)
241
242 # ----- (B6) pre-check of campaign-B B0 (S2 re-partition) and a pure front-Qf fit, 6.3 nm
243     -----
244 print('\n=== (B6) 6.3 nm on-state shape: B0 parameters (Nt 2.66e19, mu_band 19.6, Qf 9.8e11) in 1-D
245     ===')
246 p = params_from_run('run_0014'); p['Nt'] = 2.66e19; p['mu'] = 19.6 * 0.9109641975308642; p['Qf'] =
247     9.8e11
248 o = run1d(p)['metrics']; print('B0-1D:', fmt(o))
249 p = params_from_run('run_0015'); print('0015-1D:', fmt(run1d(p)['metrics']))
250 print('measured 6.3: Vcc 0.644 Vlin 1.820 gap 1.176 SScc 295.4 gm 3.431e-07 Ion 4.034e-07')

```

B.3.4 S1_s1_shape_test.py

The script addresses S1 task 3, namely which kind of gate-voltage-dependent charge could reproduce the 6.3 nm on-state shape ($V_{th,lin}$, $g_{m,max}$) while keeping $V_{th,cc}$, SS_{cc} and I_{on} . An extra acceptor Gaussian band (sheet density, depth below E_c , width) is added to the run_0015 configuration in the ATLAS-checked 1-D surrogate. The result is hypothesis-generating only (SURR) and motivated the pre-registered test C1 (run_0052).

```

1 """S1 task 3: what kind of gate-voltage-dependent charge reproduces the 6.3 nm on-state SHAPE (Vth_lin,
2     gm_max) while
3     keeping Vth_cc, SS_cc and Ion? 1-D surrogate (validated vs ATLAS, see s1_poisson1d.out.txt). Hypothesis-
4     generating only:
5     an extra acceptor Gaussian band (sheet density S, centre E0 below Ec, width W) is added to the run_0015
6     configuration;
7     Qf (exact rigid shift) and mu (exact scale) are then re-adjusted so that Vth_cc = 0.644 V and Ion(3 V) =
8     4.034e-7 A/um.
9     These are CALIBRATED numbers (2 free parameters matched + 1 band tried), not predictions."""
10 import numpy as np, json, os
11 import s1_poisson1d as s

```



```
8
9 class FilmBand(s.Film):
10     def __init__(self, p, band):
11         super().__init__(p)
12         self.band = band
13         self.tb = s.occ_table('gauss', band['E0'], band['W']) if band else None
14         if band:
15             self.Nb = band['S'] / (p['t'] * 1e-7 * band['W'] * np.sqrt(np.pi))
16     def charges(self, u):
17         rho, drho, n, nt, ng, ni = super().charges(u)
18         if self.band:
19             b, db = s.tab_eval(self.tb, u)
20             rho = rho - self.Nb * b; drho = drho - self.Nb * db; nt = nt + self.Nb * b
21         return rho, drho, n, nt, ng, ni
22
23 def adjust(vg, idv, vcc_t=0.644, ion_t=4.034e-7):
24     shift, k = 0.0, 1.0
25     lg = np.log10(np.clip(idv, 1e-40, None))
26     for _ in range(30):
27         cur = k * 10 ** np.interp(vg - shift, vg, lg)
28         m = s.metrics(vg, cur, is_sim=True, t_nm=6.3)
29         shift += vcc_t - m['Vth_cc_1e-9_V']
30         cur = k * 10 ** np.interp(vg - shift, vg, lg)
31         k *= ion_t / cur[-1]
32     cur = k * 10 ** np.interp(vg - shift, vg, lg)
33     return shift, k, s.metrics(vg, cur, is_sim=True, t_nm=6.3)
34
35 vg = np.round(np.arange(-1.5, 3.0001, 0.05), 4)
36 p0 = s.params_from_run('run_0015')
37 res = {}
38 print('measured 6.3: Vcc 0.644 Vlin 1.820 gap 1.176 SScc 295.4 gm 3.431e-07 Ion 4.034e-07')
39 cases = [None] + [dict(S=S, E0=E0, W=W) for S in (1.5e12, 3e12) for E0 in (0.0, 0.05, 0.10, 0.15) for W
40                  in (0.05,)]
41 for band in cases:
42     f = FilmBand(p0, band)
43     ids, _ = f.idvg(vg)
44     sh, k, m = adjust(vg, ids)
45     dQf = -sh * s.CQ # a positive Vg shift = less positive Qf
46     tag = 'no band (run_0015 config)' if band is None else f"band S={band['S']:.1e} E0=Ec-{band['E0']:.2f}
47         W={band['W']:.2f}"
48     res[tag] = dict(shift_V=sh, dQf_cm2=dQf, Qf_new=p0['Qf'] + dQf, mu_scale=k, mu_band_new=11.69 * k,
49                     **{kk: m[kk] for kk in ['Vth_cc_1e-9_V', 'Vth_lin_V', 'SS_cc_1e-10_1e-8_mV_dec', 'gm_max_A_V_um', '
50                     Ion_A_per_um']})
51     print(f"{tag}: Qf -> {p0['Qf'] + dQf:.3g} (shift {sh:+.3f} V), mu_band -> {11.69*k:.2f} | Vcc {m['
52         Vth_cc_1e-9_V']:.3f} Vlin {m['Vth_lin_V']:.3f} gap {m['Vth_lin_V']-m['Vth_cc_1e-9_V']:.3f} SScc {m['
53         SS_cc_1e-10_1e-8_mV_dec']:.1f} gm {m['gm_max_A_V_um']:.3e} Ion {m['Ion_A_per_um']:.3e}")
54 json.dump(res, open(os.path.join(s.OUT, 's1_shape_test.json'), 'w'), indent=1, default=float)
55
56 # no-QC single-(Nd, Qf) least-squares consistency check (linearised from s1_poisson1d B4/B5 sensitivities
57 )
58 r = np.array([0.665 - 0.662, 0.588 - 0.644, 0.330 - 0.083]) # B4 misses (1-D no-QC, Qf 4.76e10)
59 a = np.array([0.0036, 0.0125, 0.0283 * 1.32]) # dVth per +1e17 cm^-3 uniform Nd (B5
60             slopes; 2 nm from q t/Cox + q t^2/2eps)
61 A = np.c_[-a, np.ones(3)]
62 sol, *_ = np.linalg.lstsq(A, -r, rcond=None)
63 print('\nno-QC: best single extra uniform Nd (1e17 cm^-3) and rigid Vg offset:', sol, ' residual misses (
64     V):', r + A @ sol)
```

B.3.5 S2_s2_part1_curves.py

This is part 1 of the transport analysis of specialist S2. It extracts mobility curves from the measured and simulated transfer curves and fits the thickness laws of mobility and on-current, including the leave-one-out comparison of power law, $A + B/t$, exponential and step models (Section 5.12).

```

1  # S2 part 1: mobility decomposition, curve-shape features, degeneracy Jacobian, paper-vs-workbook
   extraction test,
2  # thickness-law form tests. Reads only local data/run files; writes s2_part1_out.txt next to this script.
3  import numpy as np, csv, json, pathlib
4  from scipy.optimize import curve_fit, least_squares
5  ROOT = pathlib.Path(__file__).resolve().parents[3]
6  OUT = open(pathlib.Path(__file__).with_name('s2_part1_out.txt'), 'w')
7  def P(*a):
8      s = ' '.join(str(x) for x in a); print(s); OUT.write(s + '\n')
9  COX = 8.955369814335959e-07; L = 20e-4; W = 1e-4; VD = 0.7
10
11  def load_exp(t):
12      vg, idv = [], []
13      for r in csv.DictReader(open(ROOT/'data'/'experimental_clean.csv')):
14          if abs(float(r['thickness_nm'])-t) < 1e-6: vg.append(float(r['vg_V'])); idv.append(float(r['
            id_A_per_um']))
15      o = np.argsort(vg); return np.array(vg)[o], np.array(idv)[o]
16  def load_sim(run):
17      d = next((ROOT/'results'/'runs').glob(run+'_*'))
18      vg, a = [], []
19      for r in csv.DictReader(open(d/'comparison.csv')):
20          vg.append(float(r['vg_V'])); a.append(max(float(r['atlas_A_per_um']), 1e-30))
21      return np.array(vg), np.array(a)
22  def feats(vg, idv):
23      gm = np.gradient(idv, vg); i = int(np.argmax(gm)); g = gm[i]
24      vlin = vg[i] - idv[i]/g
25      lg = np.log10(np.maximum(idv, 1e-30)); j = np.where(lg >= -9)[0][0]
26      vcc = vg[j-1] + (-9 - lg[j-1])*(vg[j]-vg[j-1])/(lg[j]-lg[j-1])
27      ion = idv[-1]; mu = g*L/(W*COX*VD)
28      # saturation-formula and effective-mobility variants (same curve, same Cox/W/L)
29      sq = np.gradient(np.sqrt(np.maximum(idv, 0)), vg); mu_sat = np.max(2*L/(W*COX)*sq**2)
30      ov = vg - vlin; m = ov > 0.3; mu_eff = np.max(idv[m]*L/(W*COX*ov[m]*VD)) if m.any() else np.nan
31      return dict(Vcc=vcc, Vlin=vlín, gap=vlín-vcc, gm=g, vg_gm=vg[i], mu=mu, Ion=ion, ion_over_gm=ion/g,
32                  roll=gm[-1]/g, mu_sat=mu_sat, mu_eff=mu_eff)
33
34  P('=== A. Measured curve features (Cox 8.955e-7, L 20 um, Vd 0.7 V) ===')
35  E = {}
36  for t in (2.0, 6.3, 13.2, 31.8):
37      vg, idv = load_exp(t)
38      if t == 31.8:
39          gm = np.gradient(idv, vg); g = gm.max(); P(f't=31.8 gm_max={g:.3e} A/V/um at Vg={vg[np.argmax(gm)
40              ]:.2f} -> apparent mu_FE={g*L/(W*COX*VD):.1f} cm2/Vs (always-on film, context only)'); continue
41      f = feats(vg, idv); E[t] = f
42      P(f't={t:.1f} Vcc={f['Vcc']:.3f} Vlin={f['Vlin']:.3f} gap={f['gap']:.3f} V gm_max={f['gm']:.3e} at
43          Vg={f['vg_gm']:.2f} mu_FE={f['mu']:.2f} Ion/gm={f['ion_over_gm']:.3f} V gm(3V)/gm_max={f['roll
44              ']:.3f} mu_sat-formula={f['mu_sat']:.2f} mu_eff(Id/ov)max={f['mu_eff']:.2f}')
45  P('Paper peak mu_FE 5.1 (2 nm) / 27.4 (13.2 nm); ratio paper/workbook =', round(5.1/E[2.0]['mu'],3),
46      round(27.4/E[13.2]['mu'],3))
47  for k in ('mu_sat', 'mu_eff'):
48      P(f' variant {k}: 2 nm {E[2.0][k]:.2f}, 13.2 nm {E[13.2][k]:.2f}; 13.2/2 ratio {E[13.2][k]/E[2.0][k
49          ]:.2f} (paper 13.2/2 = {27.4/5.1:.2f}; workbook linear = {E[13.2]['mu']/E[2.0]['mu']:.2f}')
50  # thickness-dependent series capacitance hypothesis (MISM model of SI Fig S1): C(t) = 1/(1/Cox + t/(eps0
51      *9.3))
52  for t in (2.0, 13.2):

```

```

47     C = 1/(1/COX + t*1e-7/(8.854e-14*9.3)); P(f' series-C(t) hypothesis t={t}: C={C:.3e}, mu would be {E
    [t]["mu"]*COX/C:.1f} (moves the wrong way)')
48
49 P('\n=== B. ATLAS runs: same features, decomposition mu_FE = mu_band x R(t) x P ===')
50 runs = {'run_0012': (2.0, 18.13), 'run_0023': (2.0, 18.13), 'run_0016': (2.0, 18.13), 'run_0014': (6.3,
    12.4), 'run_0015': (6.3, 11.69), 'run_0013': (13.2, 61.9), 'run_0017': (13.2, 61.9)}
51 S = {}
52 for r, (t, mb) in runs.items():
53     vg, a = load_sim(r); f = feats(vg, a); S[r] = f
54     R = (1-0.287/t)**2; mun = mb*R; Pp = f['mu']/mun
55     P(f' {r} t={t} Vcc={f['Vcc']:.3f} Vlin={f['Vlin']:.3f} gap={f['gap']:.3f} gm={f['gm']:.3e} mu_FE={f['
    mu']:.2f} Ion={f['Ion']:.3e} Ion/gm={f['ion_over_gm']:.3f} roll={f['roll']:.3f} | mu_band={mb} R={R
    :.3f} mun={mun:.2f} P=mu_FE/mun={Pp:.3f}')
56 P('Log-decomposition of measured mu_FE ratios (band, roughness, partition, model residual):')
57 cal = {2.0: ('run_0012', 18.13), 6.3: ('run_0015', 11.69), 13.2: ('run_0013', 61.9)}
58 def comp(t):
59     r, mb = cal[t]; R = (1-0.287/t)**2; Pp = S[r]['mu']/(mb*R); res = E[t]['mu']/S[r]['mu']; return np.
    log([mb, R, Pp, res])
60 for a, b in ((6.3, 13.2), (2.0, 13.2), (2.0, 6.3)):
61     d = comp(b) - comp(a); tot = np.log(E[b]['mu']/E[a]['mu'])
62     P(f' {a}->{b}: total ln={tot:.3f} (x{np.exp(tot):.2f}); band {d[0]:.3f} ({100*d[0]/tot:.0f}%),
    roughness {d[1]:.3f} ({100*d[1]/tot:.0f}%), partition {d[2]:.3f} ({100*d[2]/tot:.0f}%), residual {d
    [3]:.3f} ({100*d[3]/tot:.0f}%)')
63
64 P('\n=== C. Degeneracy Jacobian at 2 nm (run_0012 base; Nt x1.5 = run_0023; mu exact-linear; Qf rigid)
    ===')
65 b0, b1 = S['run_0012'], S['run_0023']; dl = np.log(1.5)
66 J = {'Vcc': (b1['Vcc']-b0['Vcc'])/dl, 'gap': (b1['gap']-b0['gap'])/dl, 'lngm': np.log(b1['gm']/b0['gm'])/
    dl, 'lnIon': np.log(b1['Ion']/b0['Ion'])/dl}
67 P(' per unit ln(Nt):', {k: round(v, 4) for k, v in J.items()})
68 P(' -> Ion is threshold-dominated: dlnIon/dlnNt =', round(J['lnIon'], 3), 'vs dln gm/dlnNt =', round(J['
    lngm'], 3))
69 P(' Nt needed to change mu_FE by x4.6 via partition alone (linearised): ln Nt change =', round(np.log
    (4.6)/abs(J['lngm']), 1), '-> factor', f'{np.exp(np.log(4.6)/abs(J['lngm'])):.1e}')
70 # 6.3 nm re-partition estimate using the 2-nm Jacobian (proxy): unknowns dlnNt, dlnmu, dQf(V)
71 t0 = S['run_0015']; ex = E[6.3]
72 res = np.array([ex['Vcc']-t0['Vcc'], ex['gap']-t0['gap'], np.log(ex['gm']/t0['gm'])])
73 A = np.array([[J['Vcc'], 0, 1], [J['gap'], 0, 0], [J['lngm'], 1, 0]])
74 x = np.linalg.solve(A, res)
75 P(f' 6.3 nm residual (exp - run_0015): dVcc={res[0]:+.3f} V, dgap={res[1]:+.3f} V, dln gm={res[2]:+.3f}'
    )
76 P(f' solution: Nt x{np.exp(x[0]):.2f} (Nt6.3 -> {8.459e18*np.exp(x[0]):.2e} cm-3), mu_band x{np.exp(x
    [1]):.2f} (-> {11.69*np.exp(x[1]):.1f}), rigid dV={x[2]:+.3f} V (Qf change {-x[2]/0.179e-12:+.2e} cm
    -2 -> Qf ~ {8.7e10 - x[2]/0.179e-12:.2e}'))
77 # sensitivity: gap Jacobian scaled by sheet-trap ratio 6.3/2 (5.3e12/4.0e12) as an alternative proxy
78 for sc in (1.0, 5.3/4.0, 2.0):
79     A2 = A.copy(); A2[1, 0] = J['gap']*sc; A2[0, 0] = J['Vcc']*sc; A2[2, 0] = J['lngm']*sc
80     x2 = np.linalg.solve(A2, res)
81     P(f' Jacobian scale {sc:.2f}: Nt x{np.exp(x2[0]):.2f}, mu_band {11.69*np.exp(x2[1]):.1f}, Qf ~
    {8.7e10 - x2[2]/0.179e-12:.2e}')
82
83 P('\n=== D. Thickness-law form tests (3 thin films) ===')
84 t3 = np.array([2.0, 6.3, 13.2])
85 series = {'mu_FE': [E[t]['mu'] for t in t3], 'Ion': [E[t]['Ion'] for t in t3], 'gm_max': [E[t]['gm'] for
    t in t3],
86     'mu_band_fit': [18.13, 11.69, 61.9], 'Vth_cc': [E[t]['Vcc'] for t in t3], 'Vth_lin': [E[t]['
    Vlin'] for t in t3]}
87 def fitpow(t, y):
88     b, la = np.polyfit(np.log(t), np.log(y), 1); return lambda tt: np.exp(la)*tt**b, f'a t^{b:.2f}'
89 def fitinv(t, y):
90     B, A = np.polyfit(1/t, y, 1); return lambda tt: A + B/tt, f'{A:.3g}+{B:.3g}/t'

```

```

91 def fitexp(t, y):
92     b, la = np.polyfit(t, np.log(y), 1); return lambda tt: np.exp(la + b*tt), f'a exp({b:.3f} t)'
93 def fitlin(t, y):
94     b, a = np.polyfit(t, y, 1); return lambda tt: a + b*tt, f'{a:.3g}+{b:.3g} t'
95 def fitstep(t, y):
96     lo = np.mean(y[:2]); return lambda tt: np.where(tt < 9.75, lo, y[2]), f'step {lo:.3g}|{y[2]:.3g} (tc
    in 6.3-13.2)'
97 for name, y in series.items():
98     y = np.array(y); pos = np.all(y > 0)
99     P(f'{name}: {np.round(y, 4).tolist()}')
100    for lab, fn in (('power', fitpow), ('A+B/t', fitinv), ('exp', fitexp), ('linear', fitlin), ('step',
        fitstep)):
101        if lab in ('power', 'exp') and not pos: continue
102        f, desc = fn(t3, y); yp = f(t3)
103        rel = (yp - y)/np.abs(y); rms = np.sqrt(np.mean(rel**2))
104        # leave-one-out: fit on 2 and 13.2, predict 6.3
105        idx = [0, 2]; f2, _ = fn(t3[idx], y[idx]) if lab != 'step' else (lambda tt: np.where(tt < 9.75, y
            [0], y[2]), '')
106        p63 = float(f2(np.array([6.3]))[0])
107        P(f'    {lab:7s} {desc:32s} rms rel resid {rms:6.3f} max {np.max(np.abs(rel)):6.3f} | L00(2,13.2)
            ->6.3: {p63:.4g} vs {y[1]:.4g} ({p63/y[1]:.2f}x)')
108    # 31.8 nm held-out check on Ion (context only)
109    vg, i318 = load_exp(31.8)
110    b = np.log(E[13.2]['Ion']/E[6.3]['Ion'])/np.log(13.2/6.3)
111    P(f'Ion power law through 6.3/13.2 (exponent {b:.2f}) predicts Ion(31.8) = {E[13.2]["Ion"]*(31.8/13.2)**b
        :.2e} vs measured {i318[-1]:.2e} A/um; pure series-R ceiling 0.7 V/9e4 ohm.um = {0.7/9e4:.1e}')
112    b2 = np.log(E[13.2]['Ion']/E[2.0]['Ion'])/np.log(13.2/2)
113    P(f'Ion power law through 2/13.2 (exponent {b2:.2f}) predicts Ion(31.8) = {E[13.2]["Ion"]*(31.8/13.2)**b2
        :.2e}')
114
115    P('\n=== E. Model laws: Nt surface+bulk decomposition, dEc vs EMA upper bound, m* ===')
116    B = (2e19 - 4.857e18)/(1/2 - 1/13.2); A = 2e19 - B/2
117    P(f'Nt = A + B/t through the V1 anchors: A={A:.2e} cm-3, B={B*1e-7:.2e} cm-2 (sheet); at 6.3: {A+B/6.3:.2
        e} vs power law 8.46e18')
118    P('Januar2026 internal Nt values: 5.3e19 (2 nm) / 3.39e18 (10 nm) -> exponent', round(np.log(5.3e19/3.39
        e18)/np.log(5), 2), '; Sec.2.8 "x4 from 13.2 to 2" -> exponent', round(np.log(4)/np.log(6.6), 2))
119    for t in (1.98, 2.0, 3.0, 6.3, 13.2):
120        law = 0.9205*t**-1.3815; ms = 0.208 + 0.1311*t**-1.412
121        ema_b = 0.376/(0.208*t*t); ema_t = 0.376/(ms*t*t)
122        P(f't={t:.2f} dEc law {law:.4f} eV | infinite-well EMA upper bound m*=0.208: {ema_b:.4f}, m*(t): {
            ema_t:.4f} | law/EMA = {law/ema_b:.2f}')
123    P('law exceeds the infinite-well parabolic bound (m*=0.208) for t >', round((0.376/(0.208*0.9205))
        *(1/0.6185), 2), 'nm')
124    OUT.close()

```

B.3.6 S2_s2_part2_mtr_temperature.py

Part 2 of S2 treats the temperature dependence of the drain current in the multiple-trapping-and-release picture, which is used to frame the two temperature variants of the predictions (Sections 5.8 and 5.9).

```

1 # S2 part 2: semi-analytic 1-D Poisson (floating thin film, Fermi-Dirac free electrons, V1 exponential
    acceptor tail,
2 # Nd_eff, Qf, TiN WF 4.70, chi(t)) + gradual-channel Id at Vd = 0.7 V. Purpose: (i) benchmark against
    ATLAS
3 # runs 0012/0015/0013 (same parameters), (ii) trap-limited (MTR) temperature expectation at 350 K with a
4 # T-independent band mobility, (iii) percolation / phonon analytic factors, (iv) Coulomb and roughness
    magnitudes.
5 import numpy as np, pathlib
6 from scipy.integrate import solve_ivp, quad

```

```

7 OUT = open(pathlib.Path(__file__).with_name('s2_part2_out.txt'), 'w')
8 def P(*a):
9     s = ' '.join(str(x) for x in a); print(s); OUT.write(s + '\n')
10 q = 1.602176634e-19; kB = 8.617333e-5; e0 = 8.8541878e-12; hbar = 1.054571817e-34; m0 = 9.1093837e-31
11 COX = 8.955369814335959e-07 * 1e4 # F/m^2
12 EPS = 9.3*e0; L = 20e-4; VD = 0.7
13
14 def F12(eta): # normalised Fermi-Dirac integral of order 1/2 (Aymerich-Humet approximation)
15     v = eta**4 + 50 + 33.6*eta*(1 - 0.68*np.exp(-0.17*(eta + 1)**2))
16     return 1/(np.exp(-eta) + 3*np.sqrt(np.pi)/4*v**(-3/8))
17
18 class Film:
19     def __init__(s, t_nm, NTA, WTA, Nc300, Nd, chi, T=300.0, Qf=1.73e12, WF=4.70):
20         s.t = t_nm*1e-9; s.NTA = NTA*1e6; s.WTA = WTA; s.T = T; s.kT = kB*T
21         s.Nc = Nc300*1e6*(T/300)**1.5; s.Nd = Nd*1e6; s.chi = chi; s.Qf = Qf*1e4; s.WF = WF
22         ug = np.linspace(-0.6, 3.0, 1441) # u = Ec - EF (eV)
23         xg = np.linspace(0, 40*WTA, 8001) # depth below Ec (eV)
24         occ = 1/(1 + np.exp(np.clip((ug[:], None] - xg[None, :])/s.kT, -700, 700)))
25         nt = np.trapezoid(np.exp(-xg/WTA)[None, :]*occ, xg, axis=1)
26         s.ug = ug; s.ntab = s.NTA*np.array(nt)
27     def n(s, u): return s.Nc*F12(-u/s.kT)
28     def nt(s, u): return np.interp(u, s.ug, s.ntab)
29     def solve(s, ubs, nstep=3000):
30         # vectorised RK4 from the back surface (x=t, du/dx=0, u=ub) to the gate interface (x=0)
31         # d2u/dx2 [V/m2] = q (Nd - n - nt)/EPS ; state y = (u, du, ns_free, ns_trap)
32         ub = np.asarray(ubs, float); h = -s.t/nstep
33         def f(y):
34             u = y[0]; n = s.n(u); nt = s.nt(u)
35             return np.array([y[1], q*(s.Nd - n - nt)/EPS, -n, -nt])
36         y = np.array([ub, np.zeros_like(ub), np.zeros_like(ub), np.zeros_like(ub)])
37         for _ in range(nstep):
38             k1 = f(y); k2 = f(y + 0.5*h*k1); k3 = f(y + 0.5*h*k2); k4 = f(y + h*k3)
39             y = y + h/6*(k1 + 2*k2 + 2*k3 + k4)
40             u0, du0, nsf, nst = y
41             Vox = (EPS*du0 - q*s.Qf)/COX
42             Vg = Vox + (s.WF - s.chi) - u0
43             return np.array([Vg, nsf, nst, u0]).T
44     def sweep(s):
45         ubs = np.concatenate([np.linspace(-0.25, 0.4, 131), np.linspace(0.4, 2.6, 111)[1:]])
46         R = s.solve(ubs)
47         for _ in range(40): # adaptive refinement: no Vg gap > 0.02 V inside [-3.6, 3.6] V
48             o = np.argsort(ubs); ubs = ubs[o]; R = R[o]
49             va, vb = R[:-1, 0], R[1:, 0]
50             m = (np.abs(va - vb) > 0.02) & (np.minimum(va, vb) < 3.6) & (np.maximum(va, vb) > -3.6) & (np
51 .diff(ubs) > 1e-12)
52             if not m.any(): break
53             new = 0.5*(ubs[:-1][m] + ubs[1:][m])
54             ubs = np.concatenate([ubs, new]); R = np.vstack([R, s.solve(new)])
55             R = R[np.isfinite(R).all(axis=1)]
56             o = np.argsort(R[:, 0]); R = R[o]
57             keep = np.concatenate([[True], np.diff(R[:, 0]) > 1e-6]); R = R[keep]
58             return R # columns Vg, ns_free (m^-2), ns_trap, u_surface
59
60 def idvg(R, mu_cm2, vg):
61     # floating thin film: n_free(Vg, V) = n_free(Vg - V, 0) -> Id/W = q mu/L * integral_{Vg-Vd}^{Vg}
62     n_free(v) dv
63     vv = np.linspace(-6, 4, 5001); nf = np.exp(np.interp(vv, R[:, 0], np.log(np.maximum(R[:, 1], 1e-30)),
64     left=-80))*1e-4 # cm^-2
65     cum = np.concatenate([[0], np.cumsum(0.5*(nf[1:] + nf[:-1])*np.diff(vv))])
66     I = q*mu_cm2/L*(np.interp(vg, vv, cum) - np.interp(vg - VD, vv, cum)) # A/cm width
67     return I*1e-4 # A/um

```

```

65 def feats(vg, idv):
66     gm = np.gradient(idv, vg); i = int(np.argmax(gm)); g = gm[i]; vlin = vg[i] - idv[i]/g
67     lg = np.log10(np.maximum(idv, 1e-40)); j = np.where(lg >= -9)[0][0]
68     vcc = vg[j-1] + (-9 - lg[j-1])*(vg[j]-vg[j-1])/(lg[j]-lg[j-1])
69     return dict(Vcc=vcc, Vlin=vlm, gm=g, mu=g*L/(1e-4*COX*1e-4*VD), Ion=idv[-1])
70
71 vg = np.round(np.arange(-3, 3.0001, 0.05), 4)
72 cases = { # t, NTA (cm-3/eV), Nc300, Nd_eff, chi, mun (cm2/Vs, = mu_band*R), Qf, ATLAS reference run
73     '2.0 (run_0012)': (2.0, 5e20, 3.27444e18, 2.5e17, 3.9467, 13.30, 1.73e12, dict(Vcc=0.676, Vlin=1.559,
74         mu=11.27, Ion=5.09e-7)),
75     '6.3 (run_0015)': (6.3, 2.11464e20, 2.54976e18, 2.525e17, 4.2276, 10.6492, 8.7e10, dict(Vcc=0.651,
76         Vlin=1.471, mu=8.42, Ion=4.03e-7)),
77     '13.2 (run_0013)': (13.2, 1.21426e20, 2.43961e18, 2.974e17, 4.2739, 59.2376, 1.73e12, dict(Vcc=0.053,
78         Vlin=0.999, mu=48.95, Ion=3.07e-6)),
79     '2.0 Nt x1.5 (run_0023)': (2.0, 7.5e20, 3.27444e18, 2.5e17, 3.9467, 13.30, 1.73e12, dict(Vcc=0.758,
80         Vlin=1.841, mu=10.50, Ion=3.815e-7)),
81 }
82 P('=== 1. Benchmark of the 1-D semi-analytic model against ATLAS (same V1 parameters; interface Gaussian,
83     deep Gaussian and 2-D contact effects omitted) ===')
84 RES = {}
85 for name, (t, nta, nc, nd, chi, mun, qf, ref) in cases.items():
86     F = Film(t, nta, 0.04, nc, nd, chi, Qf=qf); R = F.sweep(); I = idvg(R, mun, vg); f = feats(vg, I);
87     RES[name] = (F, R, I, f)
88     P(f'{name}: 1D Vcc={f['Vcc']:.3f} (ATLAS {ref['Vcc']}), Vlin={f['Vlin']:.3f} ({ref['Vlin']}), mu_FE={
89         f['mu']:.2f} ({ref['mu']}), Ion={f['Ion']:.3e} ({ref['Ion']:.3e}); partition mu_FE/mun={f['mu']/mun
90         :.3f}')
91     # free/trapped partition at Vg = Vlin(meas-ish) and 3 V
92     for v in (1.0, 2.0, 3.0):
93         nf = np.interp(v, R[:, 0], R[:, 1]); ntr = np.interp(v, R[:, 0], R[:, 2])
94         P(f'      Vg={v:.1f}: n_free={nf*1e-4:.2e} cm-2, n_trap={ntr*1e-4:.2e} cm-2, free fraction={nf/(
95             nf+ntr):.3f}')
96
97 P('\n=== 2. Pre-registered 6.3 nm re-partition candidate (from part-1 Jacobian): Nt x2.05 & x1.43,
98     mu_band 17.2, Qf 9.4e11 ===')
99 for fac in (1.43, 1.72, 2.05):
100     F = Film(6.3, 2.11464e20*fac, 0.04, 2.54976e18, 2.525e17, 4.2276, Qf=9.39e11); R = F.sweep()
101     I = idvg(R, 17.2*0.911, vg); f = feats(vg, I)
102     P(f'  Nt x{fac}: 1D Vcc={f['Vcc']:.3f} Vlin={f['Vlin']:.3f} gap={f['Vlin']-f['Vcc']:.3f} mu_FE={f['mu']
103         :.2f} Ion={f['Ion']:.3e} (meas 0.644 / 1.820 / 1.176 / 10.95 / 4.03e-7)')
104     F = Film(6.3, 2.11464e20, 0.04, 2.54976e18, 2.525e17, 4.2276, Qf=8.7e10); R = F.sweep(); I = idvg(R,
105         10.6492, vg); f = feats(vg, I)
106     P(f' reference tuned (Nt law, mu 11.69, Qf 8.7e10): Vcc={f['Vcc']:.3f} Vlin={f['Vlin']:.3f} gap={f['Vlin']
107         -f['Vcc']:.3f} mu_FE={f['mu']:.2f} Ion={f['Ion']:.3e}')
108
109 P('\n=== 3. Trap-limited (MTR) temperature expectation, T-independent mu_band, Nc ~ T^1.5, tail/WTA fixed
110     ===')
111 k1 = 1/(kB*300) - 1/(kB*350)
112 for name in ('2.0 (run_0012)', '13.2 (run_0013)'):
113     t, nta, nc, nd, chi, mun, qf, ref = cases[name]
114     F3 = Film(t, nta, 0.04, nc, nd, chi, T=350.0, Qf=qf); R3 = F3.sweep(); I3 = idvg(R3, mun, vg); f3 =
115         feats(vg, I3)
116     F0, R0, I0, f0 = RES[name]
117     P(f'{name}: 300K Vcc={f0['Vcc']:.3f} Vlin={f0['Vlin']:.3f} mu_FE={f0['mu']:.2f} Ion={f0['Ion']:.3e} |
118         350K Vcc={f3['Vcc']:.3f} Vlin={f3['Vlin']:.3f} mu_FE={f3['mu']:.2f} Ion={f3['Ion']:.3e}')
119     P(f'      dVcc={f3['Vcc']-f0['Vcc']:.3f} V, dVlin={f3['Vlin']-f0['Vlin']:.3f} V, mu_FE x{f3['mu']/f0
120         ['mu']:.3f}, Ion x{f3['Ion']/f0['Ion']:.3f}')
121     line = []
122     for v in (0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0):
123         i0 = np.interp(v, vg, I0); i3 = np.interp(v, vg, I3)
124         line.append(f'Vg={v}: Ea={np.log(i3/i0)/k1*1000:.0f}meV')
125     u0 = np.interp(v, R0[:, 0], R0[:, 3])

```

```

109         line[-1] += f'(Ec-EF_s={u0*1000:.0f}meV)'
110     P('    apparent activation of Id at fixed Vg: ' + ' ', '.join(line))
111
112 P('\n=== 4. Percolation / barrier (Kamiya-type, Gaussian barrier distribution) and phonon factors, 300 ->
    350 K ===')
113 for phi in (0.02, 0.04, 0.06):
114     for sig in (0.0, 0.01, 0.02):
115         fac = np.exp(-(phi - sig**2/(2*kB*350))/(kB*350) + (phi - sig**2/(2*kB*300))/(kB*300))
116         P(f'   phi_m={phi*1000:.0f} meV sigma={sig*1000:.0f} meV: mu(350)/mu(300) = {fac:.3f}; apparent Ea
            = {np.log(fac)/k1*1000:.1f} meV')
117     for n in (1.0, 1.5, 2.0):
118         P(f'   phonon mu ~ T^{-n}: mu(350)/mu(300) = {(350/300)**-n:.3f}; if phonons are a fraction f of 1/mu:
            factor = 1/(1-f*(350/300)**{n})')
119 P(f'   grain-boundary barrier needed for a x4.6 band-mobility step at 300 K: dPhi = kT ln 4.6 = {kB*300*np
    .log(4.6)*1000:.0f} meV;'
    f' then mu(350)/mu(300) differs between films by exp(dPhi*k1) = {np.exp(kB*300*np.log(4.6)*k1):.2f}')
120
121
122 P('\n=== 5. Coulomb scattering from a charge sheet at distance d from a 2DEG (screened, Ando-Fowler-Stern
    form; zero-thickness 2DEG = upper bound on scattering) ===')
123 def mu_coul(ns_cm2, nimp_cm2, d_nm, mstar=0.22, epsbar=9.15):
124     ns = ns_cm2*1e4; Ni = nimp_cm2*1e4; d = d_nm*1e-9; ms = mstar*m0; eb = epsbar*e0
125     kF = np.sqrt(2*np.pi*ns); qTF = q**2*ms/(2*np.pi*eb*hbar**2)
126     def integrand(th): # q = 2 kF sin(th/2)... use q = 2kF x, x in (0,1): dq/sqrt(1-x^2) -> substitute x
        = sin(a)
127         x = np.sin(th); qq = 2*kF*x
128         V = q**2/(2*eb*(qq + qTF))
129         return V**2*np.exp(-2*qq*d)*qq**2*kF
130     I = quad(integrand, 0, np.pi/2, limit=200)[0]
131     inv_tau = Ni*ms/(2*np.pi*hbar**3*kF**3)*I
132     return q/(ms*inv_tau)*1e4 # cm2/Vs
133 for ns in (3e12, 1e13):
134     P(f'   n_s={ns:.0e}: front Qf 1.73e12 at d=0: mu_C={mu_coul(ns, 1.73e12, 0.0):.0f} cm2/Vs; d=0.5 nm:
        {mu_coul(ns, 1.73e12, 0.5):.0f}')
135     for t in (2.0, 6.3, 13.2):
136         P(f'   back-surface sheet 1.64e12 at d = t - 0.5 nm (t={t}), eps_bar=(9.3+1)/2: mu_C={mu_coul(
            ns, 1.64e12, t-0.5, epsbar=5.15):.3g} cm2/Vs')
137 P(' Required to make a x4.6 step from 6.3->13.2 with Matthiessen and mu_other = 62: mu_C(6.3) ~
    1/(1/13.5 - 1/62) =', round(1/(1/13.5 - 1/62), 1))
138
139 P('\n=== 6. Roughness scattering magnitudes ===')
140 for t in (2.0, 6.3, 13.2):
141     P(f'   (1-Dsr/t)^2, Dsr=0.287 nm: t={t}: {(1-0.287/t)**2:.3f}')
142 P(f'   ratio 13.2/6.3 = {((1-0.287/13.2)/(1-0.287/6.3))**2:.3f} ; ratio 13.2/2 = {((1-0.287/13.2)
    /(1-0.287/2))**2:.3f}')
143 P(' Thickness-fluctuation (Sakaki/Uchida) mu_SR ~ t^6: if mu_SR(6.3) = 16.7 cm2/Vs (needed for x4.6 with
    mu_other 62), mu_SR(2) =', round(16.7*(2/6.3)**6, 4), 'cm2/Vs (measured ~12-18 at 2 nm) ->', round
    (12/(16.7*(2/6.3)**6)), 'x contradiction')
144 for t, ms in ((2.0, 0.257), (6.3, 0.218), (13.2, 0.211)):
145     E1 = 0.376/(ms*t*t); P(f'   t={t}: E1(inf. well)={E1*1000:.1f} meV, dE1 for a {0.287} nm thickness
        step = {2*E1*0.287/t*1000:.2f} meV')
146 OUT.close()

```

B.3.7 S2_s2_part3_repartition.py

Part 3 of S2 performs the shape-consistent re-partition of the field-effect mobility into band mobility and trap-limited factor for each film, which led to the pre-registered B0 test (run_0037; Chapter 8).

```

1 # S2 part 3: shape-consistent trap/mobility partition with the 1-D surrogate (benchmarked against ATLAS
    in part 2).

```

```

2 # For each film: scan the tail-integral factor f (Nt = f x V1 law); the gap Vth_lin - Vth_cc is
   independent of Qf
3 # (rigid shift) and of mu (exact scale) -> pick f* that reproduces the MEASURED gap; then mu_band from
   gm_max,
4 # Qf from Vth_cc. Ion and SS_min are NOT fitted and serve as internal consistency checks.
5 import numpy as np, pathlib, importlib.util, sys
6 spec = importlib.util.spec_from_file_location('p2', pathlib.Path(__file__).with_name('
   s2_part2_mtr_temperature.py'))
7 src = open(pathlib.Path(__file__).with_name('s2_part2_mtr_temperature.py')).read().split("vg = np.round(
   np.arange(-3, 3.0001, 0.05), 4)")
8 src = src.replace("OUT = open(pathlib.Path(__file__).with_name('s2_part2_out.txt'), 'w')", "OUT = None")
9 ns = {'__file__': __file__}; exec(src, ns)
10 Film, idvg = ns['Film'], ns['idvg']
11 OUT = open(pathlib.Path(__file__).with_name('s2_part3_out.txt'), 'w')
12 def P(*a):
13     s = ' '.join(str(x) for x in a); print(s, flush=True); OUT.write(s + '\n'); OUT.flush()
14 COX = 8.955369814335959e-07; L = 20e-4; VD = 0.7
15 vg = np.round(np.arange(-3, 3.0001, 0.05), 4)
16 def feats(vg, idv):
17     gm = np.gradient(idv, vg); i = int(np.argmax(gm)); g = gm[i]; vlin = vg[i] - idv[i]/g
18     lg = np.log10(np.maximum(idv, 1e-40)); j = np.where(lg >= -9)[0][0]
19     vcc = vg[j-1] + (-9 - lg[j-1])*(vg[j]-vg[j-1])/(lg[j]-lg[j-1])
20     s = np.gradient(lg, vg); m = (idv > 1e-13) & (idv < 1e-8) & (s > 0)
21     ss = 1000/np.max(s[m]) if m.any() else np.nan
22     return dict(Vcc=vcc, Vlin=vlín, gap=vlín - vcc, gm=g, mu=g*L/(1e-4*COX*VD), Ion=idv[-1], SS=ss)
23 meas = {2.0: dict(Vcc=0.662, Vlin=1.663, gap=1.001, gm=3.807e-7, Ion=5.09e-7, SS=84.5, mu=12.15),
24         6.3: dict(Vcc=0.644, Vlin=1.820, gap=1.176, gm=3.431e-7, Ion=4.03e-7, SS=114.7, mu=10.95),
25         13.2: dict(Vcc=0.083, Vlin=1.044, gap=0.962, gm=1.572e-6, Ion=3.07e-6, SS=130.8, mu=50.17)}
26 base = {2.0: (5e20, 3.27444e18, 2.5e17, 3.9467, 18.13), 6.3: (2.11464e20, 2.54976e18, 2.525e17, 4.2276,
   11.69),
27         13.2: (1.21426e20, 2.43961e18, 2.974e17, 4.2739, 61.9)}
28 scan = {2.0: [1.0, 1.25, 1.5, 2.0], 6.3: [1.0, 2.0, 2.5, 3.0, 4.0], 13.2: [0.8, 1.0, 1.25]}
29 QF0 = 1.73e12
30 P('Shape-consistent partition (1-D surrogate; WTA 0.040 eV fixed; Nd, chi, Nc from V1 laws; Qf reference
   1.73e12)')
31 summary = {}
32 for t, facs in scan.items():
33     nta, nc, nd, chi, mb = base[t]; R_t = (1 - 0.287/t)**2
34     rows = []
35     for f in facs:
36         F = Film(t, nta*f, 0.04, nc, nd, chi, Qf=QF0); R = F.sweep(); I = idvg(R, mb*R_t, vg); ft = feats
   (vg, I)
37         rows.append((f, ft)); P(f' t={t} f={f}: gap={ft["gap"]:.3f} Vcc={ft["Vcc"]:.3f} mu_FE={ft["mu
   "]:.2f} SS_min={ft["SS"]:.1f}')
38     gaps = np.array([r[1]['gap'] for r in rows]); fs = np.array(facs)
39     if meas[t]['gap'] < gaps.min() or meas[t]['gap'] > gaps.max():
40         P(f' t={t}: measured gap {meas[t]["gap"]} outside scanned range {gaps.min():.3f}-{gaps.max():.3f
   }; extrapolating log-linearly')
41     fstar = float(np.exp(np.polyval(np.polyfit(gaps, np.log(fs), 1), meas[t]['gap'])))
42     F = Film(t, nta*fstar, 0.04, nc, nd, chi, Qf=QF0); R = F.sweep(); I = idvg(R, mb*R_t, vg); ft = feats
   (vg, I)
43     k = meas[t]['gm']/ft['gm']; I2 = I*k; f2 = feats(vg, I2)
44     dV = meas[t]['Vcc'] - f2['Vcc']; Qf = QF0 - dV/0.179e-12
45     I3 = np.interp(vg - dV, vg, I2); f3 = feats(vg, I3)
46     summary[t] = (fstar, mb*k, Qf, f3)
47     P(f' ==> t={t}: f*={fstar:.2f} (Nt={fstar*nta*0.04:.2e} cm-3, sheet {fstar*nta*0.04*t*1e-7:.2e} cm
   -2), mu_band={mb*k:.2f} (V1 fit {mb}), Qf={Qf:.2e}')
48     P(f' check: Vcc {f3["Vcc"]:.3f}/{meas[t]["Vcc"]} Vlin {f3["Vlin"]:.3f}/{meas[t]["Vlin"]} mu_FE
   {f3["mu"]:.2f}/{meas[t]["mu"]} Ion {f3["Ion"]:.3e}/{meas[t]["Ion"]:.2e} ({100*(f3["Ion"]/meas[t]["
   Ion"]-1):.1f} %, not fitted) SS_min {f3["SS"]:.1f}/{meas[t]["SS"]} (not fitted)')
49 P('\nSummary: t | Nt factor | Nt (cm-3) | mu_band | Qf')

```



```
50 for t, (fs_, mb_, qf_, f3) in summary.items():
51     P(f' {t} | {fs_: .2f} | {fs_*base[t][0]*0.04:.2e} | {mb_: .2f} | {qf_: .2e}')
52 OUT.close()
```

B.3.8 S2_s2_part4_musat_location.py

Part 4 of S2 determines the gate voltage at which the saturation-formula mobility peaks. This is relevant to the interpretation of the mobility values of the paper as saturation-formula peaks near threshold (Section 5.5).

```
1 # S2 part 4: where the saturation-formula mobility (2L/(W Cox)) (d sqrt(Id)/dVg)^2 peaks on the measured
   Vd = 0.7 V curves,
2 # and the linear-formula mobility at the same Vg (paper vs workbook mobility definitions, Task 5).
3 import numpy as np, csv, pathlib
4 ROOT = pathlib.Path(__file__).resolve().parents[3]
5 COX = 8.955369814335959e-07; L = 20e-4; W = 1e-4; VD = 0.7
6 rows = list(csv.DictReader(open(ROOT/'data'/'experimental_clean.csv')))
7 out = open(pathlib.Path(__file__).with_name('s2_part4_out.txt'), 'w')
8 for t in (2.0, 6.3, 13.2):
9     d = sorted((float(r['vg_V']), float(r['id_A_per_um']))) for r in rows if abs(float(r['thickness_nm'])
   - t) < 1e-6)
10    vg = np.array([a for a, b in d]); i = np.array([b for a, b in d])
11    sq = np.gradient(np.sqrt(np.maximum(i, 0)), vg); mus = 2*L/(W*COX)*sq**2; k = int(np.argmax(mus))
12    mul = np.gradient(i, vg)*L/(W*COX*VD)
13    s = (f't={t}: mu_sat-formula peak {mus[k]:.2f} cm2/Vs at Vg={vg[k]:.2f} V (mu_lin there {mul[k]:.2f})
   ; mu_sat at 3 V {mus[-1]:.2f}; '
14    f'mu_lin at Vg=1,1.5,2,2.5,3 V: {[round(float(mul[np.argmin(abs(vg-v)])), 2) for v in (1.0, 1.5,
   2.0, 2.5, 3.0)]}')
15    print(s); out.write(s + '\n')
16 out.close()
```

B.3.9 S3_s3_e1_magnitude.py

The script addresses S3 task 1, the magnitude of electron confinement in 2, 6.3 and 13.2 nm IWO films, from pure effective-mass estimates (finite-difference eigenvalues) without fitting to device data (THEORY; Section 4.6).

```
1 """S3 task 1: magnitude of electron confinement in IWO films (2 / 6.3 / 13.2 nm).
2
3 Pure effective-mass estimates, no fitting to the device data.
4 (a) infinite well, parabolic; (b) finite asymmetric barriers (Al2O3 below, vacuum/air above),
5 BenDaniel-Duke finite differences; (c) Kane non-parabolicity  $E(1+aE)=\hbar^2 k^2/2m$  ( $a = C = 0.5 \text{ eV}^{-1}$ ,
6 Stokey2021 citing Feneberg:  $m^*(0)=0.18+/-0.02 m_0$ ,  $C=0.5+/-0.02 \text{ eV}^{-1}$ ), implemented as an energy-dependent
7 confinement mass  $m(E)=m^*(1+aK)$  iterated to self-consistency (exact for the infinite well);
8 (d) gate-field (triangular) confinement for a given electron sheet density;
9 (e) consistency of the effective-mass+NP picture with the three PBE slab points used by the ATLAS dEc law
   ;
10 (f) 'subthreshold-equivalent' rigid Ec shift = the quantity a classical DD model must put into dEc so
   that
11 the non-degenerate free sheet density equals the quantum one (includes the 2-D vs 3-D DOS factor).
12 Barrier heights are ASSUMED ranges (see report): Al2O3 CBO 2.0 / 3.4 / 4.0 eV (low guess / ATLAS loaded
   default
13 chi(Al2O3)=0.9 -> 4.3-0.9 / Si2021 '>4 eV'); vacuum barrier = chi_IWO = 4.3 eV (Fan2021 estimate, model
   yaml).
14 Barrier masses ASSUMED: Al2O3 0.4 m0, vacuum 1.0 m0.
15 """
```

```

16 import numpy as np
17 from scipy.linalg import eigh_tridiagonal
18 from scipy.optimize import least_squares
19
20 H2M = 0.0380998          #  $\hbar^2/(2 m_0)$  in eV nm2
21 KT = 0.025852           # eV at 300 K
22 Q = 1.602177e-19; EPS0 = 8.8541878e-14 # C, F/cm
23 EPS_IW0 = 9.3
24 TS = [2.0, 6.3, 13.2]
25 ATLAS_DEC = {2.0: 0.9205 * 2.0**-1.3815, 6.3: 0.9205 * 6.3**-1.3815, 13.2: 0.9205 * 13.2**-1.3815}
26 ATLAS_MSTAR = {t: 0.208 + 0.1311 * t**-1.412 for t in TS}
27
28
29 def e_inf(m, t, n=1):
30     return H2M * (n * np.pi / t) ** 2 / m
31
32
33 def kane(epar, a):
34     return epar if a == 0 else (np.sqrt(1 + 4 * a * epar) - 1) / (2 * a)
35
36
37 def fd_levels(t, m_w, a_np=0.0, vb_bot=3.4, mb_bot=0.4, vb_top=4.3, mb_top=1.0, nlev=4, dz=0.005, pad
38             =1.5):
39     """BenDaniel-Duke FD eigenvalues of a flat finite well [0,t] (energies from the well bottom).
40     Non-parabolicity: confinement mass  $m_w(1+aK)$ ,  $K = E - \langle V \rangle$ , iterated per level."""
41     z = np.arange(-pad, t + pad + dz / 2, dz)
42     V = np.where(z < 0, vb_bot, np.where(z > t, vb_top, 0.0))
43     out = []
44     for lev in range(nlev):
45         K = 0.0
46         for _ in range(60):
47             mw = m_w * (1 + a_np * K)
48             m = np.where(z < 0, mb_bot, np.where(z > t, mb_top, mw))
49             mh = 2.0 / (1.0 / m[:-1] + 1.0 / m[1:])          # harmonic mean on half nodes
50             c = H2M / (mh * dz * dz)                        # coupling
51             diag = V.copy(); diag[:-1] += c; diag[1:] += c
52             w, v = eigh_tridiagonal(diag, -c, select='i', select_range=(lev, lev))
53             psi2 = v[:, 0] ** 2; psi2 /= psi2.sum()
54             Knew = w[0] - float((psi2 * V).sum())
55             if abs(Knew - K) < 1e-7:
56                 break
57             K = Knew
58             pin = float(psi2[(z >= 0) & (z <= t)].sum())
59             out.append((w[0], K, pin))
60
61
62 def tri_E1(F_Vcm, m):
63     """Airy ground state of a triangular well, F in V/cm, m in  $m_0 \rightarrow$  eV; and  $\langle z \rangle = 2E_1/(3qF)$  in nm."""
64     F = F_Vcm * 1e-7 # V/nm
65     e1 = 2.33811 * (H2M / m) ** (1 / 3) * F ** (2 / 3)
66     return e1, 2 * e1 / (3 * F)
67
68
69 def ns_2d_rel(levels_K_E, m, t, m_cl):
70     """ratio of the non-degenerate quantum sheet density to the classical  $N_c(m_{cl})t$  one at the same ( $E - EF$ )
71     reference (bulk  $E_c$ ). levels: list of ( $E_n, m_{dos_n}$ )."""
72     # 2-D:  $\sum m_n kT / (\pi \hbar^2) \exp(-E_n/kT)$ ; 3-D:  $N_c t = 2 (m kT / (2 \pi \hbar^2))^{1.5} t$ 
73     # in units  $\hbar^2/2m_0 = H2M$ :  $m kT / (\pi \hbar^2) = m kT / (2 \pi H2M)$  [nm-2]
74     s = sum(mn * KT / (2 * np.pi * H2M) * np.exp(-En / KT) for En, mn in levels_K_E)

```

```

75     nc_t = 2 * (m_cl * KT / (4 * np.pi * H2M)) ** 1.5 * t
76     return s / nc_t
77
78
79 def main():
80     print("=== (a,c) infinite well, parabolic vs Kane NP (a=0.5 eV^-1) ; E1 in eV ===")
81     print("t_nm m* E1_inf E1_inf_NP ATLAS_dEc kT=0.0259")
82     for t in TS:
83         for m in (0.18, 0.208, 0.257, 0.35):
84             print(f"{t:5.1f} {m:5.3f} {e_inf(m, t):7.4f} {kane(e_inf(m, t), 0.5):8.4f} {ATLAS_DEC[t]:7.4f}")
85     print("\n=== (b) finite barriers (BenDaniel-Duke FD), E1 [eV], fraction of |psi|^2 inside film ===")
86     print("t m* a VbAl203 E1 K1 P_in E2")
87     rows = {}
88     for t in TS:
89         for m, a in ((0.18, 0.5), (0.18, 0.0), (0.257, 0.0), (0.35, 0.0)):
90             for vb in (2.0, 3.4, 4.0):
91                 lv = fd_levels(t, m, a, vb_bot=vb, nlev=2)
92                 rows[(t, m, a, vb)] = lv
93                 print(f"{t:4.1f} {m:5.3f} {a:3.1f} {vb:6.1f} {lv[0][0]:7.4f} {lv[0][1]:7.4f} {lv[0][2]:6.4f} {lv[1][0]:7.4f}")
94     # sensitivity: vacuum barrier 3.0 eV, barrier masses = well mass
95     for t in TS:
96         lv = fd_levels(t, 0.18, 0.5, vb_bot=3.4, vb_top=3.0, mb_bot=0.18, mb_top=0.18, nlev=1)
97         print(f"sens t={t}: Vtop=3.0 eV, barrier masses = 0.18 m0 -> E1={lv[0][0]:.4f}")
98
99     print("\n=== (e) PBE slab points vs effective-mass + NP with effective width t+delta ===")
100    pts = np.array([[0.95, 0.94], [1.5, 0.60], [1.98, 0.33]]) # model yml fit points (Lin2022, Si2021, Lin2022)
101    def model(p, t, m=0.17):
102        a, d = p
103        return np.array([kane(e_inf(m, tt + d), a) for tt in np.atleast_1d(t)])
104    fit = least_squares(lambda p: model(p, pts[:, 0]) - pts[:, 1], x0=[0.5, 0.3], bounds=([0, -0.5], [3, 2]))
105    a_fit, d_fit = fit.x
106    print(f"m=0.17 (Lin2022 PBE bulk): alpha_fit={a_fit:.3f} eV^-1, delta_fit={d_fit:.3f} nm, residuals={fit.fun}")
107    for a_fix in (0.5,):
108        f2 = least_squares(lambda p: model([a_fix, p[0]], pts[:, 0]) - pts[:, 1], x0=[0.3])
109        print(f"alpha fixed {a_fix}: delta={f2.x[0]:.3f} nm, residuals={f2.fun}")
110    for t in TS + [3.0, 4.0, 5.0]:
111        pw = 0.9205 * t ** -1.3815
112        em = model([a_fit, d_fit], t)[0]
113        em5 = model([0.5, f2.x[0]], t)[0]
114        print(f"t={t:5.2f} ATLAS power law {pw:.4f} EM+NP(fit) {em:.4f} EM+NP(a=0.5) {em5:.4f} ratio law/EM {pw/em:.2f}")
115    # local exponent of the physically shaped law
116    for t in (2.0, 6.3, 13.2):
117        e1, e2 = model([a_fit, d_fit], t * 0.99)[0], model([a_fit, d_fit], t * 1.01)[0]
118        print(f"local exponent d ln E / d ln t at {t}: {np.log(e2/e1)/np.log(1.01/0.99):.2f}")
119
120    print("\n=== (f) subthreshold-equivalent rigid shift dEc_eq = -kT ln(n2D/n3D) [eV] ===")
121    print("t case E1 dEc_eq(Nc m=0.208) dEc_eq(Nc m*(t) ATLAS) ATLAS dEc")
122    for t in TS:
123        for lab, m, a, vb in (("finite 3.4eV, m0.18, NP0.5", 0.18, 0.5, 3.4), ("finite 3.4eV, m0.257 par", 0.257, 0.0, 3.4),
124                                ("finite 3.4eV, m0.35 par", 0.35, 0.0, 3.4), ("hard wall, m0.18 NP0.5", 0.18, 0.5, None)):
125            nl = 12 if t > 6 else 6
126            if vb is None:

```

```

127         levs = [(kane(e_inf(m, t, n), a), m * (1 + 2 * a * kane(e_inf(m, t, n), a))) for n in
128         range(1, nl + 1)]
129     else:
130         lv = fd_levels(t, m, a, vb_bot=vb, nlev=nl, dz=0.01 if t > 6 else 0.005)
131         levs = [(E, m * (1 + 2 * a * K)) for E, K, _ in lv] # in-plane DOS mass at subband
132         bottom (Kane)
133         r208 = ns_2d_rel(levs, m, t, 0.208)
134         rat = ns_2d_rel(levs, m, t, ATLAS_MSTAR[t])
135         print(f"{t:4.1f} {lab:28s} {levs[0][0]:7.4f} {-KT*np.log(r208):8.4f} {-KT*np.log(
136         rat):8.4f} {ATLAS_DEC[t]:.4f}")
137
138     print("\n=== (d) gate-field (triangular) confinement: n_s -> F = q n_s/(eps_IWO eps0) at the front
139     interface ===")
140     print("n_s[cm^-2] F[V/cm] E1_tri(m0.18) <z>_tri[nm] E1_tri(m0.35) <z>")
141     for ns in (1e10, 1e11, 1e12, 3e12, 1e13, 1.3e13):
142         F = Q * ns / (EPS_IWO * EPS0)
143         ela, za = tri_E1(F, 0.18); e1b, zb = tri_E1(F, 0.35)
144         print(f"{ns:9.1e} {F:9.2e} {ela:8.4f} {za:7.2f} {e1b:8.4f} {zb:6.2f}")
145     # n_s at Vg = 3 V ~ Cox (3 - Vth_cc)/q using measured Vth_cc
146     cox = 8.955e-7
147     for t, vth in ((2.0, 0.662), (6.3, 0.644), (13.2, 0.083)):
148         print(f" t={t}: n_s(Vg=3V) ~ Cox(3-Vth_cc)/q = {cox*(3-vth)/Q:.2e} cm^-2 (upper bound; ignores
149         trapped charge)")
150
151     print("\n=== image-charge (dielectric) confinement, single air interface, eps 9.3/1 ===")
152     k = (EPS_IWO - 1) / (EPS_IWO + 1)
153     for zz in (0.5, 1.0, 2.0, 3.0):
154         vim = Q / (16 * np.pi * EPS0 * 1e2 * EPS_IWO * zz * 1e-9) * k # eps0 in F/m
155         print(f" z={zz} nm from air side: +{vim*1e3:.1f} meV")
156
157 if __name__ == "__main__":
158     main()

```

B.3.10 S3_s3_necessity_curves.py

The script addresses S3 task 3, whether confinement is required by the transfer data. It uses only existing ATLAS outputs (idvg.dat, native A/ μm) and the measured curves, and it starts with a rigid-shift test of the horizontal distance between the confinement-on and confinement-off curves versus current level (run_0012/run_0016, run_0013/run_0017).

```

1  """S3 task 3: is confinement required by the transfer data? Uses only existing real ATLAS outputs
2  (results/runs/<run>/idvg.dat, native A/um) and the measured workbook curves (data/experimental_clean.csv)
3
4  1. rigid-shift test: horizontal Vg distance between the confinement-on and confinement-off curves vs
5  current level
6  (0012 vs 0016 at 2 nm; 0013 vs 0017 at 13.2 nm). A pure chi shift + Nc(m*) change predicts a constant
7  dEc - kT ln(Nc_on/Nc_off).
8  2. local SS vs log10(Id) for both curves (5-point window, the extractor's definition) -> is the SS_min
9  change a
10 property of the curve or of where the minimum falls on the stepped grid?
11 3. resampling test: shift the confinement-on curve rigidly onto the confinement-off grid and re-extract
12 SS_min.
13 4. DOS discretisation: level spacing Eg/numa for both runs (numa = 96).
14 5. 'needed extra Vth' table: measured Vth_cc minus the no-confinement model with the shared Qf.
15 """
16
17 import sys
18 from pathlib import Path

```

```
14 import numpy as np
15 ROOT = Path(__file__).resolve().parents[3]
16 sys.path.insert(0, str(ROOT / 'scripts'))
17 from extract_metrics import metrics, load_curve # noqa: E402
18 RUNS = ROOT / 'results' / 'runs'
19 KT = 0.025852
20
21
22 def load_run(prefix):
23     d = next(p for p in RUNS.iterdir() if p.name.startswith(prefix))
24     rows = [l.split() for l in (d / 'idv.dat').read_text().splitlines()[4:] if l.strip()]
25     a = np.array([[float(x) for x in r[:2]] for r in rows])
26     # keep the drain-at-0.7 V gate sweep only: the file holds the scored transfer points (as used by the
27     # extractor)
28     return a[:, 0], a[:, 1], d.name
29
30 def logI_interp(vg, idv):
31     ok = idv > 0
32     return vg[ok], np.log10(idv[ok])
33
34
35 def vg_at_level(vg, idv, lev):
36     v, l = logI_interp(vg, idv)
37     # first upward crossing on a monotonic segment
38     for i in range(len(v) - 1):
39         if (l[i] - lev) * (l[i + 1] - lev) <= 0 and l[i + 1] > l[i]:
40             return v[i] + (v[i + 1] - v[i]) * (lev - l[i]) / (l[i + 1] - l[i])
41     return np.nan
42
43
44 def local_ss(vg, idv):
45     out = []
46     for i in range(2, len(vg) - 2):
47         seg = idv[i - 2:i + 3]
48         if np.all(seg > 0):
49             d = np.log10(seg[-1]) - np.log10(seg[0])
50             if d > 0:
51                 out.append((np.log10(idv[i]), vg[i], (vg[i + 2] - vg[i - 2]) / d * 1000))
52     return np.array(out)
53
54
55 def main():
56     pairs = [('run_0012', 'run_0016', 2.0, 0.9205 * 2 ** -1.3815, 3.27444e18, 2.38049e18),
57             ('run_0013', 'run_0017', 13.2, 0.9205 * 13.2 ** -1.3815, None, None)]
58     for on, off, t, dec, nc_on, nc_off in pairs:
59         v1, i1, n1 = load_run(on); v0, i0, n0 = load_run(off)
60         if nc_on is None:
61             m_on = 0.208 + 0.1311 * t ** -1.412; nc_on, nc_off = m_on ** 1.5, 0.208 ** 1.5
62         expect = dec - KT * np.log(nc_on / nc_off)
63         print(f"\n=== {n1} (confinement ON) vs {n0} (OFF), t = {t} nm; dEc = {dec:.4f} eV; rigid
64         expectation {expect:.4f} V ===")
65         print(" log10 Id   Vg_on   Vg_off   shift (V)")
66         for lev in np.arange(-14, -5.99, 0.5):
67             a, b = vg_at_level(v1, i1, lev), vg_at_level(v0, i0, lev)
68             print(f" {lev:6.1f} {a:7.3f} {b:7.3f} {a-b:7.3f}")
69             m1 = metrics(v1, i1, is_sim=True, t_nm=t); m0 = metrics(v0, i0, is_sim=True, t_nm=t)
70             print(f" extractor: SS_min on {m1['SS_mV_dec']:.1f} off {m0['SS_mV_dec']:.1f} ; SS_cc on {m1['
71             SS_cc_1e-10_1e-8_mV_dec']:.1f} off {m0['SS_cc_1e-10_1e-8_mV_dec']:.1f}")
72             s1, s0 = local_ss(v1, i1), local_ss(v0, i0)
73             print(" local SS (5-pt) vs log10 Id near the minimum [on | off]:")
```

```

72     for arr, lab in ((s1, 'on'), (s0, 'off')):
73         sel = arr[(arr[:, 0] > -13.5) & (arr[:, 0] < -7)]
74         print(f" {lab}: " + " "; ".join(f"{x[0]:.2f}:{x[2]:.0f}" for x in sel))
75         # resampling test: ON curve shifted rigidly by the measured Vth_cc shift, evaluated on the OFF
grid
76         sh = m1['Vth_cc_1e-9_V'] - m0['Vth_cc_1e-9_V']
77         va, la = logI_interp(v1, i1)
78         for shift in (sh, expect):
79             lg = np.interp(v0 + shift, va, la, left=np.nan, right=np.nan)
80             ii = np.isfinite(lg)
81             mm = metrics(v0[ii], 10 ** lg[ii], is_sim=True, t_nm=t)
82             print(f" ON curve shifted by {shift:.3f} V and resampled on the OFF grid: SS_min {mm['
SS_mV_dec']:~.1f}, SS_cc {mm['SS_cc_1e-10_1e-8_mV_dec']:~.1f}")
83             eg_on, eg_off = 3.05 + dec, 3.05
84             print(f" DOS discretisation (numa=96 over Eg): spacing ON {eg_on/96*1000:.1f} meV, OFF {eg_off
/96*1000:.1f} meV (WTA 40 meV, kT 25.9 meV)")
85
86         print("\n=== needed extra Vth_cc beyond the no-confinement model with the shared Qf 1.73e12 ===")
87         q_cox = 1.602177e-19 / 8.955369814335959e-07
88         meas = {2.0: 0.6618, 6.3: 0.6440, 13.2: 0.0830}
89         noqc = {2.0: 0.3608, 13.2: 0.0300}
90         # 6.3 nm no-QC at Qf 1.73e12 = A6 (run_0034, Qf 4.76e10) rigidly shifted by -q dQf/Cox (exact rigid
Qf shift, brief)
91         v34 = 0.5866667963552531
92         noqc[6.3] = v34 - q_cox * (1.73e12 - 4.76e10)
93         withqc = {2.0: 0.676, 6.3: 0.350, 13.2: 0.053}
94         for t in (2.0, 6.3, 13.2):
95             need = meas[t] - noqc[t]; qc = withqc[t] - noqc[t]
96             print(f" t={t:5.1f}: meas {meas[t]:.3f} noQC {noqc[t]:.3f} needed {need:+.3f} V (= {need/q_cox
:.2e} cm^-2 of negative charge)")
97             f" ATLAS-QC gives {qc:+.3f} residual after QC {need-qc:+.3f}")
98         print(" one-offset stories: (A) QC + Qf 1.73e12: residuals above; (B) no QC + Qf 4.76e10 (fit on 2 nm
):")
99         qf_b = 4.76e10
100         for t in (2.0, 6.3, 13.2):
101             pred = noqc[t] + q_cox * (1.73e12 - qf_b)
102             print(f" t={t:5.1f}: predicted Vth_cc {pred:.3f} vs meas {meas[t]:.3f} -> miss {pred-meas[t
]:+.3f} V")
103
104
105 if __name__ == '__main__':
106     main()

```

B.3.11 S3_s3_sp1d.py

The script addresses S3 task 2 with a minimal self-consistent 1-D Schrödinger-Poisson model, compared with a classical 3-D Fermi-Dirac model of the gate stack at the channel centre at $V_D = 0$ (equilibrium with the source/drain Fermi level). It produces the onset shifts of Section 4.6 (SURR).

```

1  """S3 task 2: minimal self-consistent 1-D Schrodinger-Poisson (SP) vs classical 3-D Fermi-Dirac model of
the
2  IWO TFT gate stack, channel centre, Vd = 0 (equilibrium with the S/D Fermi level, EF = 0).
3
4  Stack (bottom -> top): TiN gate (WF 4.70 eV, ASSUMED as in V1) | HfO2 15 nm eps 19.57 | Al2O3 2 nm eps
9.0 |
5  IWO t nm eps 9.3, chi_bulk 4.3 eV | air (no charge; Neumann dU/dy = 0 at the IWO top, as in the ATLAS air
volume).
6  Fixed charges as in V1: Qf = +1.73e12 cm^-2 at Al2O3/IWO, Nd_eff(t) uniform ionized donors.

```

```

7  Energies in eV; U(y) = local vacuum level (electron potential energy), Ec(y) = U - chi. d/dy(eps eps0 dU/
   dy) = +rho.
8
9  Electron models
10  C0 classical 3-D FD, chi = 4.3 (no confinement), Nc(m = 0.208) ~ ATLAS runs 0016 /
   0017 / 0034
11  C1 classical 3-D FD, chi = 4.3 - dEc_ATLAS(t), Nc(m*(t)) ~ ATLAS runs 0012 /
   0014 / 0013
12  Q* SP: 1-D effective-mass Schrodinger across Al2O3 | IWO | vacuum (1.5 nm), 2-D subband FD statistics,
13  in-plane DOS mass m*(1+2 a K_k), confinement mass m*(1+a K_k) (Kane, a = C), predictor-corrector
14  (Trellakis-type) Newton outer loop.
15  Q1 m*=0.18, a=0.5, finite barriers 3.4 eV (Al2O3, ASSUMED = ATLAS loaded chi 0.9) / 4.3 eV (vacuum)
   ,
16  barrier mass = well mass (central; reproduces the PBE-slab spill of ~0.15 nm per side, see s3_e1
   )
17  Q2 same, hard wall at the physical film boundaries (upper bound of confinement)
18  Q3 same, BenDaniel-Duke with heavy barrier masses 0.4 m0 (Al2O3) / 1.0 m0 (vacuum) (lower bound)
19  Q4 m*=0.257 parabolic (ATLAS m*(2 nm)), finite barriers equal mass
20  Q5 m*=0.35 parabolic, finite barriers equal mass
21  Optional traps ("T" suffix): V1 acceptor tail NTA = Nt(t)/WTA, WTA 0.04 eV and the V1 interface acceptor
   sheet
22  (3e11 cm^-2 eV^-1 at Ec-0.3 eV, W 0.12 eV) as fixed CLASSICAL DOS referenced to a local edge
23  Ec_ref = U - chi + shift_ref. shift_ref = dEc_ATLAS for C1T (as ATLAS does); for QnT two limits:
24  'conf' shift_ref = flat-band equivalent quantum shift (tail rides with the confined edge, ATLAS-like),
   and
25  'bulk' shift_ref = 0 (localized tail states unaffected by confinement). The deep Gaussian (5e16) is
   omitted
26  (< 1 mV). No image-charge, exchange-correlation, disorder, or 2-D (lateral) effects: see report.
27  ""
28  import json, sys
29  import numpy as np
30  from scipy.linalg import solve_banded, eigh_tridiagonal
31  if not hasattr(np, 'trapezoid'):
32      np.trapezoid = np.trapz
33
34  Q = 1.602177e-19; EPS0 = 8.8541878e-14; KT = 0.025852; H2M = 0.0380998
35  PHI_M, CHI = 4.70, 4.30
36  T_OX = (15.0, 2.0); EPS_OX = (19.57, 9.0); EPS_S = 9.3
37  QF = 1.73e12
38  DZ = 0.02 # nm
39  VAC_PAD = 1.5
40  LAW_DEC = lambda t: 0.9205 * t ** -1.3815
41  LAW_M = lambda t: 0.208 + 0.1311 * t ** -1.412
42  LAW_NT = lambda t: 2e19 * (2.0 / t) ** 0.75
43  LAW_ND = lambda t: 2.5e17 * (1 + (t / 20.0) ** 4)
44  WTA, DIT_PEAK, DIT_E, DIT_W = 0.04, 3e11, 0.3, 0.12
45
46
47  def F12(eta):
48      a = eta ** 4 + 33.6 * eta * (1 - 0.68 * np.exp(-0.17 * (eta + 1) ** 2)) + 50
49      return 1.0 / (np.exp(-eta) + 0.75 * np.sqrt(np.pi) * a ** (-0.375))
50
51
52  def nc3(m):
53      return 2 * (m * KT / (4 * np.pi * H2M)) ** 1.5 * 1e21 # cm^-3
54
55
56  # ---- trap occupancy tables vs x = Ec_ref - EF
   -----
57  XG = np.arange(-1.5, 3.5, 0.001)
58  _u = np.arange(0, 3.0, 0.0005)

```

```

59 TAIL_TAB = np.array([np.trapezoid(np.exp(-_u / WTA) / (1 + np.exp(np.clip((x - _u) / KT, -700, 700))), _u
    ) for x in XG])
60 _s = np.arange(-0.6, 0.6, 0.0005)
61 DIT_TAB = np.array([np.trapezoid(DIT_PEAK * np.exp(-(_s / DIT_W) ** 2) / (1 + np.exp(np.clip((x - DIT_E +
    _s) / KT, -700, 700))), _s) for x in XG])
62
63
64 def tab(tabv, x):
65     v = np.interp(x, XG, tabv)
66     d = (np.interp(x + 1e-3, XG, tabv) - np.interp(x - 1e-3, XG, tabv)) / 2e-3
67     return v, d
68
69
70 class Device:
71     def __init__(self, t, nd, nt=0.0, traps=False):
72         self.t = t
73         n_ox = int(round(sum(T_OX) / DZ)); n_s = int(round(t / DZ))
74         self.y = np.arange(n_ox + n_s + 1) * DZ
75         self.i_int = n_ox
76         yh = 0.5 * (self.y[1:] + self.y[:-1])
77         self.eps_h = np.where(yh < T_OX[0], EPS_OX[0], np.where(yh < sum(T_OX), EPS_OX[1], EPS_S))
78         self.w = np.zeros_like(self.y) # IWO volume weight (nm) per node
79         self.w[self.i_int + 1:-1] = DZ; self.w[self.i_int] = DZ / 2; self.w[-1] = DZ / 2
80         self.iwo = self.w > 0
81         self.nd, self.nt, self.traps = nd, nt, traps
82         c = EPS0 / (DZ * 1e-7) ** 2 * 1e-7 * DZ # flux coefficient per node (F/cm^2/nm-ish), see
            residual
83         self.k_h = self.eps_h * EPS0 / (DZ * 1e-7) # F/cm^2 : flux = k_h * (U_{i+1}-U_i) [C/cm^2 per V
            ]
84
85 # charge per node (C/cm^2) and derivative wrt U
86 def fixed_and_traps(self, U, shift_ref):
87     rho = np.zeros_like(U); drho = np.zeros_like(U)
88     wcm = self.w * 1e-7
89     rho += Q * self.nd * wcm
90     rho[self.i_int] += Q * QF
91     if self.traps:
92         x = U - CHI + shift_ref # Ec_ref - EF
93         v, d = tab(TAIL_TAB, x)
94         nta = self.nt / WTA
95         rho -= Q * nta * v * wcm; drho -= Q * nta * d * wcm
96         vi, di = tab(DIT_TAB, x[self.i_int])
97         rho[self.i_int] -= Q * vi; drho[self.i_int] -= Q * di
98     return rho, drho
99
100 def newton(self, U, Vg, dens, shift_ref, tol=1e-9, itmax=200):
101     """Solve A U = rho(U); dens(U) -> (n [cm^-3] per node, dn/dU)."""
102     N = len(U); U = U.copy(); U[0] = PHI_M - Vg
103     for it in range(itmax):
104         flux = self.k_h * np.diff(U)
105         R = np.zeros(N); R[:-1] += flux; R[1:] -= flux # R_i = flux_{i+1/2} - flux_{i-1/2}
106         R = -R # R_i = -(flux_right - flux_left) ; we need flux_right - flux_left - rho = 0
107         rho, drho = self.fixed_and_traps(U, shift_ref)
108         n, dn = dens(U)
109         wcm = self.w * 1e-7
110         rho = rho - Q * n * wcm; drho = drho - Q * dn * wcm
111         F = -R - rho # (flux_r - flux_l) - rho
112         F[0] = 0.0
113         # Jacobian: d(flux_r - flux_l)/dU_i = -(k_r + k_l); off-diagonals +k
114         kl = np.r_[0.0, self.k_h]; kr = np.r_[self.k_h, 0.0]
115         diag = -(kl + kr) - drho

```



```

116         ab = np.zeros((3, N))
117         ab[0, 1:] = self.k_h; ab[2, :-1] = self.k_h; ab[1] = diag
118         ab[1, 0] = 1.0; ab[0, 1] = 0.0
119         dU = solve_banded((1, 1), ab, -F)
120         dU = np.clip(dU, -0.2, 0.2)
121         U += dU
122         if np.max(np.abs(dU)) < tol:
123             return U, it
124     return U, -1
125
126
127 def classical_dens(dev, dec, m):
128     nc = nc3(m)
129     def dens(U):
130         n = np.zeros_like(U); dn = np.zeros_like(U)
131         eta = -(U[dev.iwo] - (CHI - dec)) / KT
132         f = F12(eta); fp = (F12(eta + 1e-4) - F12(eta - 1e-4)) / 2e-4
133         n[dev.iwo] = nc * f; dn[dev.iwo] = -nc * fp / KT
134     return n, dn
135 return dens
136
137
138 class SchrodingerSolver:
139     def __init__(self, dev, m, a_np, bc, v_al=3.4, mb=(None, None)):
140         self.dev, self.m, self.a, self.bc = dev, m, a_np, bc
141         self.v_al = v_al; self.mb = mb
142         i0 = dev.i_int
143         if bc == 'hard':
144             self.idx = np.arange(i0 + 1, len(dev.y) - 1) # psi = 0 at the physical boundaries
145             self.nvac = 0; self.nal = 0
146         else:
147             nal = int(round(T_OX[1] / DZ))
148             self.idx = np.arange(i0 - nal, len(dev.y)) # Al2O3 + IWO nodes (interface node = IWO side
149         )
150             self.nal = nal; self.nvac = int(round(VAC_PAD / DZ))
151
152     def potential(self, U):
153         dev = self.dev
154         V = U[self.idx] - CHI
155         region = np.zeros(len(self.idx), int) # 0 IWO, 1 Al2O3, 2 vacuum
156         if self.bc != 'hard':
157             region[:self.nal] = 1
158             V[:self.nal] = U[self.idx[:self.nal]] - (CHI - self.v_al)
159             V = np.r_[V, np.full(self.nvac, U[-1] - 0.0)] # vacuum level above the film
160             region = np.r_[region, np.full(self.nvac, 2)]
161         return V, region
162
163     def solve(self, U, emax_above_ef=0.35, kmax=40):
164         V, region = self.potential(U)
165         levels = []
166         for k in range(kmax):
167             K = 0.1
168             for _ in range(4 if self.a > 0 else 1):
169                 mw = self.m * (1 + self.a * K)
170                 m = np.where(region == 0, mw, np.where(region == 1, self.mb[0] or mw, self.mb[1] or mw))
171                 mh = 2 / (1 / m[:-1] + 1 / m[1:]); c = H2M / (mh * DZ * DZ)
172                 d = V.copy(); d[:-1] += c; d[1:] += c
173                 if self.bc == 'hard':
174                     d[0] += H2M / (mw * DZ * DZ); d[-1] += H2M / (mw * DZ * DZ)
175                 w, v = eigh_tridiagonal(d, -c, select='i', select_range=(k, k))
176                 p = v[:, 0] ** 2; p /= p.sum() * DZ

```

```

176         inside = region == 0
177         K = max(w[0] - float((p[inside] * V[inside]).sum() / p[inside].sum()), 1e-4)
178         mdos = self.m * (1 + 2 * self.a * K)
179         levels.append((w[0], p[:len(self.idx)], mdos, K))
180         if w[0] > emax_above_ef:
181             break
182     return levels
183
184
185 def quantum_dens_factory(dev, levels, U_s, idx):
186     def dens(U):
187         n = np.zeros_like(U); dn = np.zeros_like(U)
188         dU = U[idx] - U_s[idx]
189         for E, p, md, K in levels:
190             g = md * KT / (2 * np.pi * H2M) * 1e14 # cm-2
191             x = (E + dU) / KT
192             N2 = g * np.logaddexp(0, -x)
193             sig = 1 / (1 + np.exp(np.clip(x, -700, 700)))
194             n[idx] += N2 * p * 1e7 # p in nm-1 -> cm-1
195             dn[idx] += -g * sig / KT * p * 1e7
196             # only IWO nodes carry electrons (penetration into Al2O3/vacuum counted onto IWO boundary is
            # neglected;
197             # the Al2O3-side tail is dropped from Poisson, the vacuum tail is not on the grid)
198             n[~dev.iwo] = 0; dn[~dev.iwo] = 0
199         return n, dn
200     return dens
201
202
203 def sweep(dev, mode, vgs, **kw):
204     U = np.full(len(dev.y), PHI_M - vgs[0])
205     out = []
206     shift_ref = kw.get('shift_ref', 0.0)
207     for vg in vgs:
208         if mode == 'classical':
209             dens = classical_dens(dev, kw['dec'], kw['m'])
210             U, it = dev.newton(U, vg, dens, shift_ref)
211             n, _ = dens(U); levels = None
212         else:
213             ss = kw['solver']
214             for outer in range(80):
215                 levels = ss.solve(U)
216                 dens = quantum_dens_factory(dev, levels, U.copy(), ss.idx[:len(ss.idx)])
217                 Unew, it = dev.newton(U, vg, dens, shift_ref)
218                 err = np.max(np.abs(Unew - U)); U = Unew
219                 if err < 1e-6:
220                     break
221             n, _ = dens(U)
222             wcm = dev.w * 1e-7
223             ns = float((n * wcm).sum())
224             zc = float((n * dev.w * (dev.y - dev.y[dev.i_int])).sum() / max((n * dev.w).sum(), 1e-300))
225             rho, _ = dev.fixed_and_traps(U, shift_ref)
226             ntr = float(-(rho.sum() - Q * dev.nd * wcm.sum() - Q * QF) / Q) # trapped electrons, cm-2
227             # Gauss check: gate charge = -(total charge in stack)
228             qg = dev.k_h[0] * (U[1] - U[0]) # C/cm2 field at gate
229             out.append(dict(vg=float(vg), ns=ns, zc_nm=zc, ntrap=ntr, E1=(levels[0][0] if levels else None),
230                 Ec_int=float(U[dev.i_int] - CHI), gauss=float(qg + (Q * (dev.nd * wcm.sum() + QF
231                 - ns - ntr)))))
232     return out
233
234 def vg_at(res, level):

```

```

235     vg = np.array([r['vg'] for r in res]); ns = np.array([r['ns'] for r in res])
236     ok = ns > 0
237     lg = np.log10(ns[ok]); v = vg[ok]
238     if level < 10 ** lg.min() or level > 10 ** lg.max():
239         return None
240     return float(np.interp(np.log10(level), lg, v))
241
242
243 def ceff(res, v1, v2):
244     vg = np.array([r['vg'] for r in res]); ns = np.array([r['ns'] for r in res])
245     i1, i2 = np.argmin(abs(vg - v1)), np.argmin(abs(vg - v2))
246     return Q * (ns[i2] - ns[i1]) / (vg[i2] - vg[i1])
247
248
249 def main(traps=False, thick=(2.0, 6.3, 13.2)):
250     vgs = np.round(np.arange(-1.0, 3.0001, 0.04), 4)
251     summary = {}
252     for t in thick:
253         nd = LAW_ND(t); nt = LAW_NT(t)
254         dev = Device(t, nd, nt, traps=traps)
255         cases = {}
256         cases['C0'] = sweep(dev, 'classical', vgs, dec=0.0, m=0.208)
257         cases['C1'] = sweep(dev, 'classical', vgs, dec=LAW_DEC(t), m=LAW_M(t), shift_ref=LAW_DEC(t))
258         qdefs = {'Q1': (0.18, 0.5, 'finite', (None, None)), 'Q2': (0.18, 0.5, 'hard', (None, None)),
259                 'Q3': (0.18, 0.5, 'finite', (0.4, 1.0)), 'Q4': (0.257, 0.0, 'finite', (None, None)),
260                 'Q5': (0.35, 0.0, 'finite', (None, None))}
261         if traps:
262             qdefs = {'Q1': qdefs['Q1'], 'Q2': qdefs['Q2']}
263         # flat-band equivalent shift for the 'conf' tail reference: from the no-trap run at n_s=1e10
264         for name, (m, a, bc, mb) in qdefs.items():
265             ss = SchrodingerSolver(dev, m, a, bc, mb=mb)
266             if not traps:
267                 cases[name] = sweep(dev, 'quantum', vgs, solver=ss)
268             else:
269                 sref = FLAT_SHIFT.get((t, name), 0.0)
270                 cases[name + '_conf'] = sweep(dev, 'quantum', vgs, solver=ss, shift_ref=sref)
271                 cases[name + '_bulk'] = sweep(dev, 'quantum', vgs, solver=ss, shift_ref=0.0)
272         s = {}
273         for k, r in cases.items():
274             s[k] = dict(vg_1e10=vg_at(r, 1e10), vg_1e11=vg_at(r, 1e11), vg_1e12=vg_at(r, 1e12),
275                        ceff_2_3=ceff(r, 2.0, 3.0), ceff_1p5_2p5=ceff(r, 1.5, 2.5),
276                        zc_at_3V=r[-1]['zc_nm'], ns_3V=r[-1]['ns'], ntrap_3V=r[-1]['ntrap'],
277                        zc_at_1e12=float(np.interp(np.log10(1e12), np.log10([max(x['ns'], 1e-30) for x in
278                                in r])), [x['zc_nm'] for x in r])),
279                        E1_at_1e10=(float(np.interp(np.log10(1e10), np.log10([max(x['ns'], 1e-30) for x
279                                in r])), [x['E1'] - x['Ec_int'] if x['E1'] is not None else np.nan for x in r])) if r[0]['E1'] is not
280                                None else None),
281                        max_gauss_err_Ccm2=max(abs(x['gauss']) for x in r))
282         summary[str(t)] = s
283         print(f"\n===== t = {t} nm (Nd {nd:.3g}, traps={traps}, Nt {nt:.3g}) =====")
284         print("case      Vg(1e10) Vg(1e11) Vg(1e12) Ceff(2-3V)/Cox zc(1e12) zc(3V) nm ns(3V)
285                                ntrap(3V) E1-Ec_int@1e10 gaussErr")
286         cox = 1 / (T_OX[0] * 1e-7 / (EPS_OX[0] * EPS0) + T_OX[1] * 1e-7 / (EPS_OX[1] * EPS0))
287         for k, v in s.items():
288             f = lambda z: 'None' if z is None else f"{z:8.4f}"
289             print(f"{k:9s} {f(v['vg_1e10'])} {f(v['vg_1e11'])} {f(v['vg_1e12'])} {f(v['ceff_2_3']/cox
290                                :8.4f)} {f(v['zc_at_1e12']:6.3f)} {f(v['zc_at_3V']:6.3f)} {f(v['ns_3V']:9.3e)} {f(v['ntrap_3V']:9.3e)} {f(
291                                v['E1_at_1e10'])} {f(v['max_gauss_err_Ccm2']:.1e)}")
292         base = s['C0']
293         print("shifts vs C0 (V):      d@1e10      d@1e11      d@1e12")
294         for k, v in s.items():

```

```
290         if k == 'C0':
291             continue
292         d = [None if (v[x] is None or base[x] is None) else v[x] - base[x] for x in ('vg_1e10', '
vg_1e11', 'vg_1e12')]
293         print(f"  {k:9s}          " + " ".join('None' if z is None else f"{z:8.4f}" for z in d))
294         summary[str(t) + '_cox'] = cox
295         sys.stdout.flush()
296     return summary
297
298
299 FLAT_SHIFT = {}
300 if __name__ == '__main__':
301     mode = sys.argv[1] if len(sys.argv) > 1 else 'notraps'
302     if mode == 'notraps':
303         s = main(traps=False)
304         json.dump(s, open('s3_sp1d_notraps.json', 'w'), indent=1)
305     else:
306         # 'conf' tail reference = no-trap Q-vs-C0 shift at n_s = 1e10 (flat-band equivalent quantum shift)
307         prev = json.load(open('s3_sp1d_notraps.json'))
308         for t in (2.0, 6.3, 13.2):
309             for q in ('Q1', 'Q2'):
310                 FLAT_SHIFT[(t, q)] = prev[str(t)][q]['vg_1e10'] - prev[str(t)]['C0']['vg_1e10']
311         print('flat-band equivalent shifts used for tail reference:', FLAT_SHIFT)
312         s = main(traps=True)
313         json.dump(s, open('s3_sp1d_traps.json', 'w'), indent=1)
```

B.3.12 S3_s3_sp1d_series.py

The script addresses S3 task 5, the same trap-free comparison of Schrödinger-Poisson and classical models for a denser thickness series (3, 4 and 5 nm); it imports `s3_sp1d.py`.

```
1 """S3 task 5: no-trap SP vs classical for a denser thickness series (3, 4, 5 nm); same model as s3_sp1d.
   py."""
2 import json
3 import s3_sp1d as sp
4 s = sp.main(traps=False, thick=(3.0, 4.0, 5.0))
5 json.dump(s, open('s3_sp1d_series.json', 'w'), indent=1)
```

B.3.13 S4_s4_traps.py

The script contains the defect and trap analysis of specialist S4 and is read-only with respect to the runs. Its parts include the thickness law of the tail density (power law versus a surface-plus-bulk decomposition, with the numbers of Januar et al. [1, pp. 7, 9] and Kim et al. [3, pp. 173507-3, 173507-6]); the trap-spectroscopy figure of Chapter 4 (scratch/S4/s4_trap_spectroscopy.png) belongs to the same S4 analysis.

```
1 """S4 defect/trap analysis (2026-09-25). Read-only w.r.t. runs; no ATLAS launched.
2 Parts:
3 A tail-density thickness law: power law vs surface+bulk decomposition; Januar / Kim numbers
4 B subthreshold trap-DOS spectroscopy (charge-sheet inversion that removes the drain-bias and
5 free-carrier terms), VALIDATED on ATLAS runs whose DOS is known, then applied to the data
6 C 0-D film forward surrogate: SS metrics vs mobility (trap-free) and with the V1 DOS
7 D off-state floors: level, Vg dependence, thickness exponent, sheet conductance
8 E donor electrostatics: uniform-donor fits of Vth_cc(t), Kim2024 Von(t) -> Nd
9 F bias-stress magnitude scaling (Januar PBS numbers)
10 """
```

```

11 import re, os, json
12 import numpy as np
13 from scipy.interpolate import PchipInterpolator
14
15 ROOT = r"C:\Users\moham\Downloads\IWO_Local_Codex_Handoff (1)\IWO_PHYSICS_CONSTRAINED_MODEL_V1"
16 OUT = os.path.join(ROOT, r"analysis_2026-09-25\scratch\S4")
17 q = 1.602176634e-19; kT = 0.025852; LN10 = np.log(10.0)
18 Cox = 8.955e-7; CoxQ = Cox / q # cm-2 V-1
19 L = 20e-4; Vd = 0.7; eps0 = 8.8541878e-14; eps_s = 9.3 * eps0
20 res = {}
21
22 def P(*a):
23     print(*a)
24
25 # ----- A
26 P("=== A. tail-density thickness law ===")
27 law = lambda t: 2e19 * (2.0 / t) ** 0.75
28 for t in (2.0, 6.3, 13.2):
29     P(f" V1 law t={t:5.1f} nm Nt={law(t):.3e} cm-3 sheet={law(t)*t*1e-7:.3e} cm-2")
30 S2, S13 = law(2) * 2e-7, law(13.2) * 13.2e-7
31 Nb = (S13 - S2) / (11.2e-7); Ns = S2 - Nb * 2e-7
32 P(f" surface+bulk through V1 anchors: Ns={Ns:.3e} cm-2, Nb={Nb:.3e} cm-3")
33 pred63 = (Ns + Nb * 6.3e-7) / 6.3e-7
34 P(f" -> Nt(6.3) = {pred63:.3e} (law {law(6.3):.3e}; ratio {pred63/law(6.3):.3f}")
35 P(f" S2 curve-shape partition Nt(6.3)=2.66e19 (B0) is x{2.66e19/law(6.3):.2f} law, x{2.66e19/pred63:.2f} surf+bulk")
36 # Januar PBS t=0 values
37 S2j, S10j = 5.3e19 * 2e-7, 3.39e18 * 10e-7
38 Nbj = (S10j - S2j) / 8e-7
39 P(f" Januar PBS devices: sheet(2nm)={S2j:.3e}, sheet(10nm)={S10j:.3e} -> Nb={Nbj:.3e} (NEGATIVE => no non-negative surf+bulk)")
40 P(f" Januar PBS ratio Nt2/Nt10 = {5.3e19/3.39e18:.1f}; implied exponent ln(ratio)/ln(5) = {np.log(5.3e19/3.39e18)/np.log(5):.2f}")
41 P(f" Januar Fig5a 'x4 from 13.2 to 2': exponent ln4/ln6.6 = {np.log(4)/np.log(6.6):.3f}; sheet ratio 13.2/2 = {6.6/4:.3f}")
42 # Kim2024: SS-derived Nt (cm-2 eV-1) and CNF border traps (cm-3 eV-1)
43 kimT = np.array([10, 20, 30.]); kimNtSS = np.array([3.32e12, 3.96e12, 7.03e12]); kimNbt = np.array([2.37e18, 3.12e18, 14.6e18])
44 p = np.polyfit(kimT * 1e-7, kimNtSS, 1)
45 P(f" Kim2024 SS-Nt linear fit: Ds={p[1]:.3e} cm-2eV-1, gb={p[0]:.3e} cm-3eV-1; residual 20nm {kimNtSS[1]-np.polyval(p,20e-7):.2e}")
46 P(f" Kim2024 step slopes: 10->20 {(kimNtSS[1]-kimNtSS[0])/1e-6:.2e}, 20->30 {(kimNtSS[2]-kimNtSS[1])/1e-6:.2e} cm-3eV-1 (accelerating)")
47
48 # ----- helpers
49 eta_tab = np.linspace(-60, 30, 9001)
50 _e = np.linspace(0, 80, 8001)
51 F12 = np.array([2 / np.sqrt(np.pi) * np.trapezoid(np.sqrt(_e) / (1 + np.exp(np.clip(_e - et, -700, 700))), _e) for et in eta_tab[:10]])
52 eta_c = eta_tab[:10]
53 logF = PchipInterpolator(eta_c, np.log(F12))
54 def nfree(u, Nct): # sheet free density, FD, u = EF-Ec (eV)
55     return Nct * np.exp(logF(np.clip(u / kT, -60, 30)))
56 def u_of_n(n, Nct):
57     et = np.linspace(-60, 30, 20001); lf = logF(et)
58     return kT * np.interp(np.log(n / Nct), lf, et)
59
60 def model_dos(E, t_nm, NTA, WTA=0.04, NGA1=5e16, EGA1=0.6, WGA1=0.15, NGA2=1.16757e19, EGA2=0.301554, WGA2=0.123975, dit_mult=1.0):
61     """sheet DOS (cm-2 eV-1) vs E-Ec for the V1 deck: tail in both regions, deep Gaussian in region 1, regularized interface Gaussian in region 2 (0.25 nm)."""
62

```

```

63     t = t_nm * 1e-7; t2 = 0.25e-7; t1 = t - t2
64     g = t * NTA * np.exp(E / WTA) + t1 * NGA1 * np.exp(-(E + EGA1) / WGA1 ** 2) + dit_mult * t2 * NGA2
        * np.exp(-(E + EGA2) / WGA2 ** 2)
65     return np.where(E <= 0, g, 0.0)
66
67 Eg = np.linspace(-3.3, 0.0, 6601)
68 def thermal_D(u, gE): # dn_t/dEF = int g f(1-f)/kT
69     out = []
70     for uu in np.atleast_1d(u):
71         x = np.clip((Eg - uu) / kT, -700, 700); f = 1 / (1 + np.exp(x))
72         out.append(np.trapezoid(gE * f * (1 - f) / kT, Eg))
73     return np.array(out)
74 def trapped(u, gE):
75     out = []
76     for uu in np.atleast_1d(u):
77         f = 1 / (1 + np.exp(np.clip((Eg - uu) / kT, -700, 700))); out.append(np.trapezoid(gE * f, Eg))
78     return np.array(out)
79
80 # ----- data I/O
81 import csv
82 meas = {}
83 with open(os.path.join(ROOT, "data", "experimental_clean.csv")) as fh:
84     for r in csv.DictReader(fh):
85         meas.setdefault(float(r["thickness_nm"]), []).append((float(r["vg_V"]), float(r["id_A_per_um"])))
86 for k in meas:
87     a = np.array(sorted(meas[k])); meas[k] = (a[:, 0], a[:, 1])
88
89 def read_sim(run):
90     d = os.path.join(ROOT, "results", "runs", run)
91     vg, idd = [], []
92     for line in open(os.path.join(d, "idvg.dat")):
93         m = re.match(r"^\s*(-?\d.]+(?:e[-+]?[d+])?)\s+(-?\d.]+e[-+]?[d+])\s*$", line.strip(), re.I)
94         if m: vg.append(float(m.group(1))); idd.append(float(m.group(2)))
95     deck = open(os.path.join(d, "device.in")).read()
96     g = lambda key: float(re.search(key + r"=([\d.e+-]+)", deck).group(1))
97     nta = float(re.findall(r"nta=([\d.e+-]+)", deck)[0]); nga2 = float(re.findall(r"nga=([\d.e+-]+)",
        deck)[1])
98     tmin = float(re.search(r"region num=1 user.material=IW0 .*?y.min=([\d.e+-]+)", deck).group(1))
99     return np.array(vg), np.array(idd), dict(nc=g("nc300"), mun=g("mun"), nta=nta, nga2=nga2, t_nm=-tmin
        * 1e3)
100
101 VG = np.round(np.arange(-3.0, 3.0001, 0.05), 3)
102 def uniform(vg, idd, floor=1e-20):
103     ok = idd > 1e-17
104     f = PchipInterpolator(vg[ok], np.log10(idd[ok]))
105     y = np.full(VG.shape, np.log10(floor)); m = (VG >= vg[ok].min()) & (VG <= vg[ok].max()); y[m] = f(VG[
        m])
106     return 10 ** y
107
108 def locslope(y, x, w=2):
109     s = np.full_like(y, np.nan)
110     for i in range(len(x)):
111         a, b = max(0, i - w), min(len(x), i + w + 1)
112         if b - a >= 3: s[i] = np.polyfit(x[a:b], y[a:b], 1)[0]
113     return s
114
115 def spectroscopy(I, mu, Nct, floor=0.0, w=2):
116     """charge-sheet inversion: n_s0(Vg) = (L/q mu) sum_k gm(Vg - k Vd); then
117     D_trap(u) = CoxQ (dVg/du - 1) - dn_f/du (Vfb cancels)"""
118     Ich = I - floor
119     valid = Ich > max(3 * floor, 1e-16)

```

```

120     lg = np.log10(np.where(valid, Ich, 1e-30))
121     s = locslope(lg, VG, w); s[~valid] = 0.0
122     gm = LN10 * np.where(valid, Ich, 0) * s * 1e4          # A/V per cm width
123     kstep = int(round(Vd / 0.05))
124     ns = np.zeros_like(VG)
125     for j in range(len(VG)):
126         acc = 0.0; k = j
127         while k >= 0:
128             acc += gm[k]; k -= kstep
129         ns[j] = L / (q * mu) * acc
130     ok = valid & (ns > 0)
131     u = np.full_like(VG, np.nan); u[ok] = u_of_n(ns[ok], Nct)
132     dudV = locslope(np.where(ok, u, np.nan), VG, w)
133     dnf_du = np.full_like(VG, np.nan)
134     # dn_f/du from FD: numerical
135     du = 1e-4
136     dnf_du[ok] = (nfree(u[ok] + du, Nct) - nfree(u[ok] - du, Nct)) / (2 * du)
137     D = CoxQ * (1 / dudV - 1) - dnf_du
138     return dict(ns=ns, u=u, D=D, ok=ok & np.isfinite(D) & (dudV > 0), I=I)
139
140 # ----- B: validation on ATLAS
141 P("\n=== B. trap-DOS spectroscopy: validation on ATLAS runs (known DOS) ===")
142 runs = {"run_0012_2p0_v2_full_stepped_final": 2.0, "run_0015_6p3_v2_full_stepped_tuned_qf8p7e10_mu11p69":
143         6.3,
144         "run_0014_6p3_v2_full_stepped_validation_shared_params": 6.3, "
145         run_0013_13p2_v2_full_stepped_final": 13.2,
146         "run_0033_6p3_hyp_H4_front_Dit_x5": 6.3, "run_0030_6p3_hyp_H3_backQ_m1p64e12": 6.3}
147 spec = {}
148 Uq = np.array([-0.30, -0.25, -0.20, -0.15, -0.10, -0.07, -0.05])
149 def at_u(sp, uq):
150     m = sp["ok"] & (sp["I"] > 1e-13) & np.isfinite(sp["u"])
151     uu, DD = sp["u"][m], sp["D"][m]
152     o = np.argsort(uu); uu, DD = uu[o], DD[o]
153     out = []
154     for x in uq:
155         out.append(np.interp(x, uu, DD) if (uu.min() <= x <= uu.max()) else np.nan)
156     return np.array(out), (uu.min(), uu.max())
157 for run, t in runs.items():
158     vg, idd, pr = read_sim(run)
159     I = uniform(vg, idd)
160     Nct = pr["nc"] * pr["t_nm"] * 1e-7
161     sp = spectroscopy(I, pr["mun"], Nct); spec[run] = (sp, pr)
162     gE = model_dos(Eg, pr["t_nm"], pr["nta"], NGA2=pr["nga2"])
163     rec, rng = at_u(sp, Uq); tru = thermal_D(Uq, gE)
164     P(f" {run[:34]:34s} t={pr['t_nm']:.1f} mu={pr['mun']:.2f} u-range {rng[0]:.3f}..{rng[1]:.3f} eV")
165     for x, a, b in zip(Uq, rec, tru):
166         P(f"      u={x:.2f} recovered {a:10.3e} model(thermal) {b:10.3e} ratio {a/b if b>0 else np.nan
167           :6.2f}")
168
169 # ----- B: measured
170 P("\n=== B. measured films ===")
171 floors = {}
172 for t in (2.0, 6.3, 13.2):
173     vg, idd = meas[t]; m = (vg >= -2.0) & (vg <= -0.5); floors[t] = np.median(idd[m])
174 mu_sets = {"V1 mun": {2.0: 13.3, 6.3: 10.65, 13.2: 59.24}, "S2 partition": {2.0: 19.3 * 0.734, 6.3: 19.6
175     * 0.911, 13.2: 62.7 * 0.957}}
176 Nc = {2.0: 3.27444e18, 6.3: 2.54976e18, 13.2: 2.43961e18}
177 mspec = {}
178 for lab, mus in mu_sets.items():
179     P(f" mobility set: {lab} {mus}")
180     for t in (2.0, 6.3, 13.2):

```

```

177         vg, idd = meas[t]; I = np.interp(VG, vg, idd)
178         sp = spectroscopy(I, mus[t], Nc[t] * t * 1e-7, floor=floors[t]); mspec[(lab, t)] = sp
179         rec, rng = at_u(sp, Uq)
180         P(f"      t={t:5.1f}  u-range {rng[0]:.3f}...{rng[1]:.3f}")
181         for x, a in zip(Uq, rec):
182             gE = model_dos(Eg, t, 2e19 * (2 / t) ** 0.75 / 0.04)
183             P(f"      u={x:+.2f}  D_meas {a:10.3e} cm-2eV-1 per-volume {a/(t*1e-7):10.3e} cm-3eV-1
              V1-model(thermal) {thermal_D([x], gE)[0]:10.3e}")
184 # surface+bulk decomposition at fixed u (V1 mobilities)
185 P(" surface+bulk decomposition of D_meas(u) using 2 and 13.2 nm, predict 6.3 nm:")
186 for i, x in enumerate(Uq):
187     d2 = at_u(mspec[("V1 mun", 2.0)], [x])[0][0]; d13 = at_u(mspec[("V1 mun", 13.2)], [x])[0][0]; d6 =
        at_u(mspec[("V1 mun", 6.3)], [x])[0][0]
188     if np.isfinite(d2) and np.isfinite(d13):
189         gb = (d13 - d2) / 11.2e-7; Ds = d2 - gb * 2e-7
190         P(f"      u={x:+.2f}: Ds={Ds:10.3e} gb={gb:10.3e} pred6.3={Ds+gb*6.3e-7:10.3e} meas6.3={d6:10.3e}
              ")
191
192 # SS metrics of measured curves at their own points, and D at SS_min (naive formula)
193 P(" naive SS inversion (Cox/q)(SS/59.5-1):")
194 for t, ssmin, ssc in ((2.0, 84.5, 269.9), (6.3, 114.7, 295.4), (13.2, 130.8, 160.0)):
195     s0 = LN10 * kT * 1e3
196     P(f"      t={t}: SS_min {ssmin} -> D={CoxQ*(ssmin/s0-1):.3e} ({CoxQ*(ssmin/s0-1)/(t*1e-7):.2e} /cm3eV)
        ; SS_cc {sscc} -> D={CoxQ*(sscc/s0-1):.3e}")
197 for mu in (10.65, 13.3, 59.24):
198     Istar = mu * Cox * kT ** 2 / L * 1e-4
199     P(f"      free-carrier crossover current (C_free=Cox) for mu={mu}: {Istar:.2e} A/um")
200
201 # trapped-charge excess at 6.3 nm: measured vs run_0015 between Vth_cc and Vth_lin
202 P(" 6.3 nm gap: extra trapped charge needed = CoxQ*(1.820-1.471 - (0.644-0.651)) = %.3e cm-2" % (CoxQ *
        ((1.820 - 0.644) - (1.471 - 0.651))))
203
204 # ----- C: forward surrogate
205 P("\n=== C. 0-D film forward surrogate (SS metrics) ===")
206 def forward(t_nm, mu, Nc_, gE=None):
207     uu = np.linspace(-1.6, 0.35, 3901)
208     nf = nfree(uu, Nc_ * t_nm * 1e-7)
209     nt = trapped(uu, gE) if gE is not None else 0 * uu
210     x = uu + (nf + nt) / CoxQ
211     xs = np.linspace(x.min() + 0.8, x.max(), 4000)
212     nfx = np.interp(xs, x, nf)
213     cum = np.concatenate([[0], np.cumsum(0.5 * (nfx[1:] + nfx[:-1]) * np.diff(xs))])
214     vg = xs[xs >= xs[0] + Vd]
215     I = q * mu / L * (np.interp(vg, xs, cum) - np.interp(vg - Vd, xs, cum)) * 1e-4 # A/um
216     lg = np.log10(I)
217     v10, v8 = np.interp(-10, lg, vg), np.interp(-8, lg, vg)
218     ss = np.gradient(vg, lg) * 1e3
219     m = (I > 1e-13) & (I < 1e-8)
220     return (v8 - v10) / 2 * 1e3, ss[m].min()
221 for mu in (10.65, 13.3, 59.24):
222     P(f" trap-free t=2 nm mu={mu}: SS_cc={forward(2.0, mu, 3.27e18)[0]:.1f} SS_min={forward(2.0, mu,
        3.27e18)[1]:.1f}")
223 for t, mu, nta, atl in ((2.0, 13.3, 5e20, (289.1, 73.2)), (6.3, 10.65, 2.11464e20, (288.8, 110.8)),
        (13.2, 59.24, 1.21426e20, (192.5, 151.3))):
224     gE = model_dos(Eg, t, nta)
225     sc, sm = forward(t, mu, Nc[t], gE)
226     P(f" V1 DOS t={t}: surrogate SS_cc={sc:.1f} SS_min={sm:.1f} ATLAS SS_cc={atl[0]} SS_min={atl[1]}")
227     sc0, sm0 = forward(t, mu, Nc[t], None)
228     P(f"      same mu, NO traps: SS_cc={sc0:.1f} SS_min={sm0:.1f}")
229
230 # ----- D: off-state floors

```



```

231 P("\n=== D. off-state floors ===")
232 fl = {}
233 for t in (2.0, 6.3, 13.2, 31.8):
234     vg, idd = meas[t]
235     for a, b in ((-3.0, -2.0), (-2.0, -0.5), (-3.0, -0.5)):
236         m = (vg >= a) & (vg <= b)
237         sl = np.polyfit(vg[m], np.log10(np.abs(idd[m])), 1)[0]
238         P(f" t={t:5.1f} Vg[{a},{b}] median {np.median(idd[m]):.3e} min {idd[m].min():.3e} max {idd[m].max():.3e} slope {sl:+.3f} dec/V")
239     m = (vg >= -2.0) & (vg <= -0.5); fl[t] = np.median(idd[m])
240     G = fl[t] * 1e4 * L / Vd
241     P(f" sheet conductance of a parallel path G = {G:.3e} S/sq ; n*mu*t = {G/q:.3e} cm^-2 cm^2/Vs ; total device current {fl[t]*290:.2e} A")
242 for (a, b) in ((2.0, 6.3), (6.3, 13.2), (2.0, 13.2)):
243     P(f" floor exponent {a}->{b}: {np.log(fl[b]/fl[a])/np.log(b/a):.2f} (ratio {fl[b]/fl[a]:.1f})")
244 P(f" 31.8 nm Id(-3V) = {meas[31.8][1][0]:.3e} A/um -> G = {meas[31.8][1][0]*1e4*L/Vd:.3e} S/sq")
245 # gate leakage FN bound (HfO2 field), for illustration: J = A E^2 exp(-B/E)
246 Vox = np.array([1.0, 2.0, 3.0]); EOTphys = (15 + 2 * 19.57 / 9.0) * 1e-7
247 E_hf = Vox / EOTphys
248 P(f" HfO2 field at |Vg|=1,2,3 V: {E_hf/1e6} MV/cm ; Al2O3 field x{19.57/9:.2f}")
249 for phi in (1.0, 1.5, 2.0):
250     for mr in (0.2, 0.4):
251         B = 6.83e7 * np.sqrt(mr) * phi ** 1.5; A = 1.54e-6 / (mr * phi)
252         J = A * E_hf ** 2 * np.exp(-B / E_hf)
253         P(f" FN phi={phi} m*={mr}: J(1,2,3 V) = {J} A/cm^2")
254 P(" gate-leakage J needed for the floors if the leaking area were 290x100 um^2 (ASSUMED pad, not measured):")
255 for t in (2.0, 6.3, 13.2):
256     P(f" t={t}: J = {fl[t]*290/(290e-4*100e-4):.2e} A/cm^2")
257
258 # ----- E: donors
259 P("\n=== E. donor electrostatics ===")
260 def dV_donor(Nd, t_nm): # back-surface threshold shift of a fully depleted film with uniform donors
261     t = t_nm * 1e-7
262     return q * Nd * t / Cox + q * Nd * t * t / (2 * eps_s)
263 for t in (2.0, 6.3, 13.2, 31.8):
264     P(f" per 1e18 cm^-3: t={t:5.1f}: -dVth={dV_donor(1e18,t):.3f} V (sheet term {q*1e18*t*1e-7/Cox:.3f})")
265 need = 0.56 / (dV_donor(1e18, 13.2) - dV_donor(1e18, 6.3)) * 1e18
266 P(f" Nd needed for dVth_cc(6.3->13.2) = -0.56 V : {need:.2e} cm^-3 ; it predicts dVth(2->6.3) = {-(dV_donor(need,6.3)-dV_donor(need,2)):.3f} V (meas -0.018)")
267 for t in (2.0, 6.3, 13.2):
268     P(f" Qf=1.73e12 as volumetric donors in t={t}: Nd={1.73e12/(t*1e-7):.2e} cm^-3")
269 # Kim2024 Von(t) -> Nd (90 nm SiO2 per Kim; eps_s 9.3 assumed)
270 CoxK = 3.9 * eps0 / 90e-7
271 for (t1, t2, dv) in ((10, 20, 2.0), (20, 30, 2.0)):
272     a1, a2 = t1 * 1e-7, t2 * 1e-7
273     k = q * ((a2 - a1) / CoxK + (a2 ** 2 - a1 ** 2) / (2 * eps_s))
274     P(f" Kim2024 Von {t1}->{t2} nm shift -{dv} V -> Nd = {dv/k:.2e} cm^-3 (CoxK={CoxK:.2e})")
275 # 31.8 nm always-on: positive sheet that the gate at -3 V cannot remove (order of magnitude)
276 P(f" gate charge swing Cox*3V/q = {CoxQ*3:.2e} cm^-2")
277
278 # ----- F: PBS scaling
279 P("\n=== F. PBS extrapolation to a sweep ===")
280 for dv, lab in ((1.2, "2 nm IWO"), (0.72, "10 nm IWO")):
281     for beta in (0.3, 0.5, 1.0):
282         for ts in (10, 60):
283             for gamma in (1, 2):
284                 P(f" {lab}: dV(1200s,6V)={dv} beta={beta} t={ts}s gamma={gamma} (3V/6V)^g -> {dv*(ts/1200)**beta*(0.5)**gamma:.3f} V")
285

```

```

286 # ----- plot
287 import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt
288 fig, ax = plt.subplots(1, 3, figsize=(16, 5))
289 cols = {2.0: "C0", 6.3: "C1", 13.2: "C2"}
290 for t in (2.0, 6.3, 13.2):
291     sp = mspec[("V1 mun", t)]; m = sp["ok"] & (sp["I"] > 1e-13)
292     ax[0].semilogy(sp["u"][m], sp["D"][m], "o", ms=3, color=cols[t], label=f"meas {t} nm")
293     ax[1].semilogy(sp["u"][m], sp["D"][m] / (t * 1e-7), "o", ms=3, color=cols[t], label=f"meas {t} nm")
294     sp2 = mspec[("S2 partition", t)]; m2 = sp2["ok"] & (sp2["I"] > 1e-13)
295     ax[0].semilogy(sp2["u"][m2], sp2["D"][m2], "x", ms=3, color=cols[t], alpha=0.5)
296 for run, (sp, pr) in spec.items():
297     if run.startswith(("run_0012", "run_0015", "run_0013")):
298         t = pr["t_nm"]; m = sp["ok"] & (sp["I"] > 1e-13)
299         ax[0].semilogy(sp["u"][m], sp["D"][m], "-", color=cols[round(t, 1)], lw=1)
300         gE = model_dos(Eg, t, pr["nta"]); uu = np.linspace(-0.45, 0.0, 91)
301         ax[0].semilogy(uu, thermal_D(uu, gE), ":", color=cols[round(t, 1)], lw=2)
302 ax[0].set_xlabel("EF - Ec at source (eV)"); ax[0].set_ylabel("D_trap (cm^-2 eV^-1)"); ax[0].set_title("
    markers: data (o V1 mu, x S2 mu); line: inversion of ATLAS; dotted: model input")
303 ax[1].set_xlabel("EF - Ec (eV)"); ax[1].set_ylabel("D_trap / t (cm^-3 eV^-1)"); ax[1].legend()
304 for t in (2.0, 6.3, 13.2, 31.8):
305     vg, idd = meas[t]; ax[2].semilogy(vg, idd, label=f"{t} nm")
306 ax[2].set_xlabel("Vg (V)"); ax[2].set_ylabel("Id (A/um)"); ax[2].legend(); ax[2].set_title("measured
    transfer (floors)")
307 for a in ax[2:]: a.set_xlim(-0.45, 0.05); a.set_ylim(1e10, 1e15) if a is ax[0] else a.set_ylim(1e16, 1e21
    ); a.grid(alpha=.3)
308 fig.tight_layout(); fig.savefig(os.path.join(OUT, "s4_trap_spectroscopy.png"), dpi=130)
309 P("saved plot")

```

B.3.14 S4_s4_part2.py

Part 2 of S4 computes the integrated trap-charge excess at 6.3 nm by energy window (measured versus run_0015, with the same inversion), the channel current at $V_G = 0$ compared with the off-floor, and tables of the ratio of measured to ATLAS-inverted trap densities for each film.

```

1  """S4 part 2: integrated trap-charge excess at 6.3 nm by energy window (measured vs run_0015, same
    inversion),
2  channel current at Vg=0 vs floor, and ratio tables measured/ATLAS-inverted per film."""
3  import io, contextlib, runpy, os
4  import numpy as np
5  HERE = os.path.dirname(os.path.abspath(__file__))
6  buf = io.StringIO()
7  with contextlib.redirect_stdout(buf):
8      g = runpy.run_path(os.path.join(HERE, "s4_traps.py"))
9      VG, mspec, spec, meas, fl = g["VG"], g["mspec"], g["spec"], g["meas"], g["floors"]
10 pairs = {2.0: "run_0012_2p0_v2_full_stepped_final", 6.3: "
    run_0015_6p3_v2_full_stepped_tuned_qf8p7e10_mu11p69",
11         13.2: "run_0013_13p2_v2_full_stepped_final"}
12 def curve(sp):
13     m = sp["ok"] & (sp["I"] > 1e-13) & np.isfinite(sp["u"])
14     u, D = sp["u"][m], sp["D"][m]; o = np.argsort(u); return u[o], D[o]
15 print("ratio D_meas / D_ATLAS-inverted (same inversion, V1 mobilities) vs u")
16 grid = np.arange(-0.20, 0.051, 0.025)
17 for t, run in pairs.items():
18     um, Dm = curve(mspec[("V1 mun", t)]); us, Ds = curve(spec[run][0])
19     row = []
20     for x in grid:
21         ok = (um.min() <= x <= um.max()) and (us.min() <= x <= us.max())
22         row.append(f"{np.interp(x, um, Dm)/np.interp(x, us, Ds):5.2f}" if ok else " - ")
23     print(f" t={t:5.1f}: " + " ".join(row) + " (u = " + " ".join(f"{x:+.3f}" for x in grid) + ")")

```

```

24 print("\nintegrated excess trapped sheet charge at 6.3 nm, measured minus run_0015 (cm^-2):")
25 for lab in ("V1 mun", "S2 partition"):
26     um, Dm = curve(mspec[(lab, 6.3)]); us, Ds = curve(spec[pairs[6.3]][0])
27     lo, hi = max(um.min(), us.min()), min(um.max(), us.max())
28     for a, b in ((lo, -0.10), (-0.10, 0.0), (0.0, hi), (lo, hi)):
29         x = np.linspace(a, b, 200)
30         ex = np.trapezoid(np.interp(x, um, Dm) - np.interp(x, us, Ds), x)
31         print(f"  {lab:12s} u in [{a:+.3f},{b:+.3f}] : {ex:+.3e}")
32 print(" (needed from the Vth_lin-Vth_cc gap: +1.99e12)")
33 # where does Vth_lin sit in u? measured 6.3 source-end u at Vg = 1.82 V
34 for t, vl in ((2.0, 1.663), (6.3, 1.820), (13.2, 1.044)):
35     sp = mspec[("V1 mun", t)]
36     print(f"  t={t}: u(source) at Vth_lin {vl} V = {np.interp(vl, VG, sp['u']):+.3f} eV ; at Vth_cc: see
37         below")
38 for t, vc in ((2.0, 0.662), (6.3, 0.644), (13.2, 0.083)):
39     sp = mspec[("V1 mun", t)]
40     print(f"  t={t}: u(source) at Vth_cc {vc} V = {np.interp(vc, VG, sp['u']):+.3f} eV ; n_s0 = {np.
41         interp(vc, VG, sp['ns']):.2e} cm^-2")
42 print("\nfloor vs channel current at Vg = 0 V (measured):")
43 for t in (2.0, 6.3, 13.2):
44     vg, idd = meas[t]
45     i0 = idd[np.argmin(abs(vg))]
46     print(f"  t={t}: Id(0 V) = {i0:.2e} A/um, floor {fl[t]:.2e}, floor/Id(0) = {fl[t]/i0:.3f}")
47 # tail-pinned Fermi level in an isolated (ungated, neutral) film, V1 parameters, Qf ignored
48 kT = 0.025852
49 for t, Nd, Nt, mu, Nc in ((2.0, 2.5e17, 2e19, 13.3, 3.27e18), (6.3, 2.52e17, 8.46e18, 10.65, 2.55e18),
50     (13.2, 2.97e17, 4.86e18, 59.24, 2.44e18)):
51     fac = (np.pi * kT / 0.04) / np.sin(np.pi * kT / 0.04)
52     u0 = 0.04 * np.log(Nd / (Nt * fac))
53     n0 = Nc * np.exp(u0 / kT)
54     print(f"  isolated film t={t}: EF-Ec = {u0:+.3f} eV, n0 = {n0:.2e} cm^-3, q n0 mu t = {1.602e-19*n0*
55         mu*t*1e-7:.2e} S/sq")

```

B.3.15 S4_s4_part3_63shape.py

Part 3 of S4 asks which trap redistribution reproduces the 6.3 nm on-state shape. It uses a zero-dimensional charge-sheet surrogate (uniform film potential, Fermi-Dirac free electrons, thermal trap occupancy, Pao-Sah-type integral at $V_D = 0.7$ V), with each variant rigidly aligned to the measured $V_{th,cc} = 0.644$ V. It proposed the near- E_c acceptor band tested as C1 (run_0052) (SURR).

```

1 """S4 part 3: which trap redistribution reproduces the 6.3 nm on-state SHAPE?
2 0-D charge-sheet surrogate (uniform film potential, FD free electrons, thermal trap occupancy, Vd = 0.7 V
3 Pao-Sah-type integral). Each variant is rigidly aligned to the measured Vth_cc (0.644 V) - equivalent to
4 a
5 Qf change, which the brief shows is an exact rigid shift - and scored with the extractor's definitions
6 (Vth_lin from secant gm_max on the 0.05 V grid). The surrogate's absolute error is first measured on the
7 tuned ATLAS DOS (run_0015) and reported; conclusions use DIFFERENCES between variants."""
8 import io, contextlib, runpy, os
9 import numpy as np
10 HERE = os.path.dirname(os.path.abspath(__file__))
11 with contextlib.redirect_stdout(io.StringIO()):
12     g = runpy.run_path(os.path.join(HERE, "s4_traps.py"))
13     nfree, trapped, model_dos, Eg, CoxQ, q, L, Vd = (g[k] for k in ("nfree", "trapped", "model_dos", "Eg", "
14         CoxQ", "q", "L", "Vd"))
15 T = 6.3; NC = 2.54976e18; NTA0 = 2.11464e20
16 VG = np.round(np.arange(-3.0, 3.0001, 0.05), 3)
17
18 def curve(gE, mu):

```

```

17     uu = np.linspace(-1.6, 0.45, 4101)
18     nf = nfree(uu, NC * T * 1e-7); nt = trapped(uu, gE)
19     x = uu + (nf + nt) / CoxQ
20     xs = np.linspace(x.min(), x.max(), 8000); nfx = np.interp(xs, x, nf)
21     cum = np.concatenate([[0], np.cumsum(0.5 * (nfx[1:] + nfx[:-1]) * np.diff(xs))])
22     vg = xs[xs >= xs[0] + Vd]
23     I = q * mu / L * (np.interp(vg, xs, cum) - np.interp(vg - Vd, xs, cum)) * 1e-4
24     shift = 0.644 - np.interp(-9.0, np.log10(I), vg)
25     return np.interp(VG, vg + shift, I, left=1e-30, right=np.nan)
26
27 def metrics(I):
28     gm = np.gradient(I, VG); i = int(np.nanargmax(gm))
29     lg = np.log10(I)
30     v10, v8 = np.interp(-10, lg, VG), np.interp(-8, lg, VG)
31     return dict(gap=VG[i] - I[i] / gm[i] - 0.644, gm=gm[i], ion=I[-1], ssc=(v8 - v10) / 2e-3, vgm=VG[i])
32
33 def gauss(E0, sig, Ns):
34     return Ns / (sig * np.sqrt(np.pi)) * np.exp(-(Eg - E0) / sig ** 2) # sheet DOS, integral = Ns
35
36 meas = dict(gap=1.820 - 0.644, gm=3.431e-7, ion=4.034e-7, ssc=295.4)
37 atl = dict(gap=1.471 - 0.651, gm=2.639e-7, ion=4.03e-7, ssc=288.8)
38 base = model_dos(Eg, T, NTA0)
39 mb = metrics(curve(base, 10.6492))
40 print(f"surrogate on run_0015 DOS, mu 10.65: gap {mb['gap']:.3f} (ATLAS {atl['gap']:.3f}), gm {mb['gm']:.3e} (ATLAS {atl['gm']:.3e}), Ion {mb['ion']:.3e} (ATLAS {atl['ion']:.3e}), SS_cc {mb['sscc']:.0f} (ATLAS {atl['sscc']})")
41 print(f"measured 6.3 nm: gap {meas['gap']:.3f}, gm {meas['gm']:.3e}, Ion {meas['ion']:.3e}, SS_cc {meas['sscc']:.0f}")
42 print("targets for surrogate variants (surrogate baseline x measured/ATLAS ratios): ")
43     f"gap {mb['gap'] + (meas['gap']-atl['gap']):.3f}, gm {mb['gm']*meas['gm']/atl['gm']:.3e}, Ion {mb['ion']*meas['ion']/atl['ion']:.3e}"
44 tg = dict(gap=mb['gap'] + (meas['gap'] - atl['gap']), gm=mb['gm'] * meas['gm'] / atl['gm'], ion=mb['ion'] * meas['ion'] / atl['ion'])
45 rows = []
46 def score(lab, gE, mu):
47     m = metrics(curve(gE, mu))
48     err = np.sqrt(((m['gap'] - tg['gap']) / 0.05) ** 2 + (np.log(m['gm'] / tg['gm']) / 0.05) ** 2 + (np.log(m['ion'] / tg['ion']) / 0.05) ** 2)
49     rows.append((err, lab, m))
50 # (i) tail scaling (S2-type) with mobility
51 for f in (1.0, 2.0, 3.14, 4.0):
52     for mu in (10.65, 12.4, 14.0, 16.0, 17.86, 20.0):
53         score(f"tail NTA x{f:.2f}, mu {mu}", model_dos(Eg, T, NTA0 * f), mu)
54 # (ii) broader tail at fixed integral Nt (WTA 0.05/0.06)
55 for w in (0.05, 0.06):
56     for f in (1.0, 2.0):
57         for mu in (10.65, 12.4, 14.0, 17.86):
58             score(f"tail WTA {w} Nt x{f}, mu {mu}", model_dos(Eg, T, NTA0 * 0.04 / w * f, WTA=w), mu)
59 # (iii) Gaussian acceptor band (bulk or interface - identical in a 0-D film)
60 for E0 in (-0.15, -0.10, -0.07, -0.05, -0.03):
61     for sig in (0.02, 0.04):
62         for Ns in (1e12, 2e12, 3e12):
63             for mu in (10.65, 12.4, 14.0):
64                 score(f"band E0 {E0} sig {sig} Ns {Ns:.0e}, mu {mu}", base + gauss(E0, sig, Ns), mu)
65 # (iv) interface Dit x5 at Ec-0.3 (A4-like)
66 for mu in (10.65, 12.4):
67     score(f"Dit x5 (Ec-0.3), mu {mu}", model_dos(Eg, T, NTA0, dit_mult=5.0), mu)
68 rows.sort(key=lambda r: r[0])
69 print("\nbest 15 variants (score: gap/0.05 V, ln gm/0.05, ln Ion/0.05 in quadrature):")
70 for err, lab, m in rows[:15]:
71     print(f" {err:7.2f} {lab:45s} gap {m['gap']:.3f} gm {m['gm']:.3e} Ion {m['ion']:.3e} SScc {m['sscc']:.0f}")

```

```
        ']:.0f} Vg(gm_max) {m['vgm']:.2f}")
72 print("\nselected reference variants:")
73 for err, lab, m in rows:
74     if lab.startswith(("tail NTA x1.00, mu 10.65", "tail NTA x3.14, mu 17.86", "Dit x5 (Ec-0.3), mu 12.4"
75         , "tail NTA x1.00, mu 12.4")):
76         print(f" {err:7.2f} {lab:45s} gap {m['gap']:.3f} gm {m['gm']:.3e} Ion {m['ion']:.3e} SSc {m[''
77             ssc']:.0f}")
```

B.3.16 S5_s5_yfunction.py

The script performs the series-resistance, Y-function and contact analysis of the measured transfer curves and, as a method check, of ideal-contact ATLAS runs ($R_{sd} = 0$ by construction, constant μ_{band}). The inputs are the measured data and the `comparison.csv` files of the runs; no parameter is invented (Section 5.7).

```
1  """S5: series-resistance / Y-function / contact analysis on the measured IWO transfer curves
2  and on ideal-contact ATLAS runs (Rsd = 0 by construction, constant mu_band) as a method check.
3  Inputs: data/experimental_clean.csv, results/runs/<run>/comparison.csv. No parameters invented:
4  Cox, W, L, Vd from EVIDENCE_BRIEF.md section 1. Outputs to this folder."""
5  import json, pathlib
6  import numpy as np
7  from scipy.optimize import curve_fit
8  from scipy.signal import savgol_filter
9
10 ROOT = pathlib.Path(__file__).resolve().parents[3]
11 OUT = pathlib.Path(__file__).resolve().parent
12 COX = 8.955e-7          # F/cm^2 (brief)
13 L_CM = 20e-4           # cm
14 VD = 0.7               # V (schematic)
15 VTH_LIN = {2.0: 1.663, 6.3: 1.820, 13.2: 1.044} # brief table (scripts/extract_metrics.py)
16
17 def load_meas():
18     a = np.genfromtxt(ROOT / "data/experimental_clean.csv", delimiter=",", names=True)
19     d = {}
20     for t in np.unique(a["thickness_nm"]):
21         m = a["thickness_nm"] == t
22         o = np.argsort(a["vg_V"][m])
23         d[float(t)] = (a["vg_V"][m][o], a["id_A_per_um"][m][o])
24     return d
25
26 def load_sim(run):
27     a = np.genfromtxt(ROOT / f"results/runs/{run}/comparison.csv", delimiter=",", names=True)
28     return a["vg_V"], a["atlas_A_per_um"]
29
30 def gm_of(vg, idw, method):
31     if method == "grad":
32         return np.gradient(idw, vg)
33     return savgol_filter(idw, 7, 2, deriv=1, delta=vg[1] - vg[0])
34
35 def vth_lin(vg, idw):
36     gm = np.gradient(idw, vg); k = np.argmax(gm)
37     return vg[k] - idw[k] / gm[k], gm[k]
38
39 def model_theta(vg, A2, vstar, th):
40     vov = np.clip(vg - vstar, 1e-6, None)
41     return A2 * vov / (1 + th * vov)
42
43 def model_alpha_theta(vg, K, v0, alpha, th):
```

```

44     vov = np.clip(vg - v0, 1e-6, None)
45     return K * vov ** alpha / (1 + th * vov)
46
47 def analyse(vg, id_um, vstart, label, gmeth="grad"):
48     idw = id_um * 1e4          # A/cm of width
49     gm = gm_of(vg, idw, gmeth)
50     m = (vg >= vstart - 1e-9) & (vg <= 3.0 + 1e-9) & (gm > 0)
51     x, Y = vg[m], idw[m] / np.sqrt(gm[m])
52     # Y-function:  $Y = A (Vg - V^*)$ ,  $A^2 = \mu_0 \text{ Cox } Vd / L$ 
53     pY, cY = np.polyfit(x, Y, 1, cov=True)
54     A, b = pY; vstar = -b / A
55     mu0 = A ** 2 * L_CM / (COX * VD)
56     q2 = np.polyfit(x, Y, 2)
57     curv = q2[0] * (x.max() - x.min()) ** 2 / (A * (x.max() - x.min())) # quadratic term / linear span
58     # theta from  $1/\sqrt{gm} = (1 + th (Vg - V^*)) / A \rightarrow \text{slope } s = th / A$ 
59     s = np.polyfit(x, 1 / np.sqrt(gm[m]), 1)[0]
60     th_gm = s * A
61     # theta from direct Id fit (alpha fixed = 1)
62     p, c = curve_fit(model_theta, x, idw[m], p0=[A ** 2, vstar, 0.05], maxfev=20000)
63     A2f, vsf, thf = p; sd = np.sqrt(np.diag(c))
64     mu0f = A2f * L_CM / (COX * VD)
65     rsd_max = thf * L_CM / (mu0f * COX) * 1e4 if thf > 0 else 0.0 # ohm.um (theta0 = 0 bound)
66     # H-function (Cerdeira):  $H = \int Id dVg / Id$ , slope =  $1/(\alpha+1)$ , no Vt needed
67     ic = np.concatenate([[0], np.cumsum(0.5 * (idw[1:] + idw[:-1]) * np.diff(vg))])
68     H = ic[m] / idw[m]
69     sH = np.polyfit(x, H, 1)[0]
70     alphaH = 1 / sH - 1
71     # alpha + theta simultaneously (identifiability)
72     try:
73         p4, c4 = curve_fit(model_alpha_theta, x, idw[m], p0=[A2f, vsf, 1.0, max(thf, 1e-3)], maxfev=
74             =40000)
75         sd4 = np.sqrt(np.diag(c4)); corr = c4[2, 3] / (sd4[2] * sd4[3])
76     except Exception as e:
77         p4, sd4, corr = [np.nan] * 4, [np.nan] * 4, np.nan
78     ron = VD / id_um[vg.searchsorted(3.0 - 1e-9)]
79     rsh_ch = 1 / (mu0f * COX * (3.0 - vsf)) # ohm/sq at the source end (Vd/2 absorbed in V*)
80     return dict(label=label, window=[float(x.min()), float(x.max())], npts=int(m.sum()), gm_method=gmeth,
81         mu0_Y=mu0, Vstar_Y=vstar, Y_curv_rel=curv, theta_gm=th_gm,
82         mu0_fit=mu0f, Vstar_fit=vsf, theta_fit=thf, theta_fit_sd=sd[2], mu0_fit_sd=sd[0] * L_CM /
83         (COX * VD),
84         Rsd_max_ohm_um=rsd_max, Ron_ohm_um=ron, Rsd_frac_max=rsd_max / ron,
85         alpha_H=alphaH, alpha4=p4[2], theta4=p4[3], alpha4_sd=sd4[2], theta4_sd=sd4[3],
86         corr_alpha_theta=corr,
87         Rsh_ch_ohm_sq=rsh_ch)
88
89 def run_all():
90     meas = load_meas(); res = {"measured": {}, "atlas_ideal_contacts": {}}
91     # 31.8 nm: own Vth_lin
92     v31, i31 = meas[31.8]; vt31, _ = vth_lin(v31, i31 * 1e4)
93     VTH = dict(VTH_LIN); VTH[31.8] = vt31
94     for t in (2.0, 6.3, 13.2, 31.8):
95         vg, idm = meas[t]; rows = []
96         for off in (VD - 0.2, VD, VD + 0.2): # Vov > Vd window (linear regime), +/- 0.2 V
97             for gmeth in ("grad", "sg"):
98                 vs = max(VTH[t] + off, -3.0)
99                 if 3.0 - vs < 0.35: continue
100                 rows.append(analyse(vg, idm, vs, f"{t} nm Vov>{off:.1f}", gmeth))
101                 rows.append(analyse(vg, idm, VTH[t] + 0.3, f"{t} nm Vov>0.3 (includes sat. region)", "grad"))
102     res["measured"][str(t)] = rows
103     sims = {"run_0012 (2 nm cal, mu_band 18.13)": ("run_0012_2p0_v2_full_stepped_final", 2.0),
104         "run_0016 (2 nm no confinement)": ("run_0016_2p0_sens_no_confinement", 2.0),

```

```

102         "run_0015 (6.3 nm tuned, mu 11.69)": ("run_0015_6p3_v2_full_stepped_tuned_qf8p7e10_mu11p69",
103         6.3),
104         "run_0014 (6.3 nm validation, mu 12.4)": ("
105         run_0014_6p3_v2_full_stepped_validation_shared_params", 6.3),
106         "run_0013 (13.2 nm cal, mu 61.9)": ("run_0013_13p2_v2_full_stepped_final", 13.2)}
107     for k, (run, t) in sims.items():
108         vg, ids = load_sim(run); vt, _ = vth_lin(vg, ids * 1e4); rows = []
109         for off in (VD - 0.2, VD, VD + 0.2):
110             vs = vt + off
111             if 3.0 - vs < 0.35: continue
112             rows.append(analyse(vg, ids, vs, f"{k} Vov>{off:.1f} (sim Vth_lin {vt:.3f})", "grad"))
113             res["atlas_ideal_contacts"][k] = rows
114     return res, VTH
115
116 def contact_table(res):
117     """Transfer length and TLM contact resistance for a grid of specific contact resistivities
118     (rho_c NOT measured for Pd/IW0; grid spans literature-implied Ni/In2O3 to poor contacts)."""
119     out = []
120     rhoc_grid = [1e-8, 1e-7, 1e-6, 1e-5, 1e-4, 1e-3, 1e-2] # ohm.cm^2
121     for t in ("2.0", "6.3", "13.2"):
122         r0 = [r for r in res["measured"][t] if r["gm_method"] == "grad" and "Vov>0.7" in r["label"]][0]
123         rsh = r0["Rsh_ch_ohm_sq"]; ron = r0["Ron_ohm_um"]
124         for rc in rhoc_grid:
125             rc_um2 = rc * 1e8 # ohm.um^2
126             LT = np.sqrt(rc_um2 / rsh) # um
127             for Lov in (1.0, 2.0, 5.0): # overlap NOT reported -> scan
128                 Rc = np.sqrt(rc_um2 * rsh) / np.tanh(Lov / LT) # ohm.um per contact
129                 out.append(dict(t=t, rhoc_ohm_cm2=rc, Rsh_ohm_sq=rsh, LT_um=LT, Lov_um=Lov,
130                 Rc_per_contact_ohm_um=Rc, frac_of_Ron=2 * Rc / ron))
131     return out
132
133 if __name__ == "__main__":
134     res, VTH = run_all()
135     res["contact_grid"] = contact_table(res)
136     # rho_c needed to make Rsd account for the 2 nm vs 13.2 nm on-resistance gap
137     r2 = [r for r in res["measured"]["2.0"] if r["gm_method"] == "grad" and "Vov>0.7" in r["label"]][0]
138     r13 = [r for r in res["measured"]["13.2"] if r["gm_method"] == "grad" and "Vov>0.7" in r["label"]][0]
139     gap = r2["Ron_ohm_um"] - r13["Ron_ohm_um"]
140     rsh13 = r13["Rsh_ch_ohm_sq"]
141     rc_needed_um2 = (gap / 2) ** 2 / rsh13
142     res["rhoc_needed_for_gap"] = dict(Ron_gap_ohm_um=gap, per_contact=gap / 2, Rsh_used=rsh13,
143     rhoc_ohm_cm2=rc_needed_um2 * 1e-8, LT_um=np.sqrt(rc_needed_um2 /
144     rsh13),
145     IR_drop_V_at_2nm=gap * 5.08966e-7)
146     # vertical access through the film (accumulated) - specific resistance rho*t
147     q = 1.602e-19; va = {}
148     for t, mu, vov in ((2.0, r2["mu0_fit"], 3.0 - r2["Vstar_fit"]), (13.2, r13["mu0_fit"], 3.0 - r13["
149     Vstar_fit"])):
150         ns = COX * vov / q; n = ns / (t * 1e-7)
151         va[t] = dict(ns_cm2=ns, n_avg_cm3=n, rho_ohm_cm=1 / (q * n * mu), rho_t_ohm_cm2=t * 1e-7 / (q * n
152         * mu))
153     # undepleted/neutral top part of a 13.2 nm film at Nd_eff = 2.97e17 (model law, FITTED) and mu_band
154     61.9
155     va["13.2_neutral_top"] = dict(rho_t_ohm_cm2=13.2e-7 / (q * 2.97e17 * 61.9))
156     res["vertical_access"] = va
157     res["VTH_used"] = VTH
158     (OUT / "s5_results.json").write_text(json.dumps(res, indent=1, default=float))
159     # compact print
160     for grp in ("measured", "atlas_ideal_contacts"):
161         print("==", grp)
162         for k, rows in res[grp].items():

```

```

157         for r in rows:
158             print(f"{r['label']:<58s} {r['gm_method']:4s} win {r['window'][0]:.2f}-{r['window'][1]:.2f} n={r['npts']:2d} "
159                   f"mu0Y {r['mu0_Y']:6.2f} V*Y {r['Vstar_Y']:6.3f} curv {r['Y_curv_rel']:+.3f} thGm {r['theta_gm']:+.3f} | "
160                   f"mu0 {r['mu0_fit']:6.2f}+-{r['mu0_fit_sd']:.2f} th {r['theta_fit']:+.3f}+-{r['theta_fit_sd']:.3f} "
161                   f"RsdMax {r['Rsd_max_ohm_um']:.3g} Ron {r['Ron_ohm_um']:.3g} frac {r['Rsd_frac_max']:.3f} | "
162                   f"aH {r['alpha_H']:.2f} a4 {r['alpha4']:.2f}+-{r['alpha4_sd']:.2f} th4 {r['theta4']:.3f}+-{r['theta4_sd']:.3f} "
163                   f"corr {r['corr_alpha_theta']:+.2f} Rsh {r['Rsh_ch_ohm_sq']:.3g}")
164             print("== rho_c needed", res["rhoc_needed_for_gap"])
165             print("== vertical", res["vertical_access"])
166             print("== contact grid (Lov = 2 um)")
167             for c in res["contact_grid"]:
168                 if c["Lov_um"] == 2.0:
169                     print(f"t {c['t']:>4s} rho_c {c['rhoc_ohm_cm2']:.0e} LT {c['LT_um']:.3g} um Rc {c['Rc_per_contact_ohm_um']:.3g} ohm.um 2Rc/Ron {c['frac_of_Ron']:.3g}")

```

B.3.17 S5_s5_checks.py

The S5 checks comprise (1) the ideal gradual-channel baseline of the H-function exponent in the same windows (at $V_D = 0.7$ V the condition $V_D \ll V_{ov}$ does not hold), (2) the forward effect of a series resistance on the ideal curve and on a measured-like curve, and (3) the determination of which mobility definition reproduces the values 5.1 and 27.4 cm²/(V s) of the paper from the workbook curves (the provenance test of Section 3.4).

```

1  """S5 checks: (1) ideal gradual-channel baseline of the H-function exponent in the same windows (Vd = 0.7
2  V is not
3  << Vov), (2) forward effect of a series resistance on the ideal and on the measured-like curve, (3) which
4  mobility
5  definition could reproduce the paper's 5.1 / 27.4 cm2/Vs from the workbook curves (provenance test)."""
6  import numpy as np, pathlib
7  from scipy.optimize import brentq
8  ROOT = pathlib.Path(__file__).resolve().parents[3]
9  COX, L, VD, W_L = 8.955e-7, 20e-4, 0.7, 290 / 20
10
11  def gca(vov, beta, vd=VD):
12      vov = np.maximum(vov, 0)
13      return np.where(vov > vd, beta * (vov * vd - vd ** 2 / 2), beta * vov ** 2 / 2)
14
15  def alpha_H(vg, i, lo, hi):
16      ic = np.concatenate([[0], np.cumsum(0.5 * (i[1:] + i[:-1]) * np.diff(vg))])
17      m = (vg >= lo - 1e-9) & (vg <= hi + 1e-9)
18      return 1 / np.polyfit(vg[m], ic[m] / i[m], 1)[0] - 1
19
20  vg = np.round(np.arange(-3, 3.0001, 0.05), 3)
21  print("== (1) ideal GCA baseline alpha_H (Vd = 0.7 V), windows Vov in [0.7, Vov_max]")
22  for t, vt in ((2.0, 1.663), (6.3, 1.820), (13.2, 1.044)):
23      i = gca(vg - vt, 1.0)
24      print(f"t {t}: Vth {vt}: alpha_H ideal = {alpha_H(vg, i + 1e-30, vt + 0.7, 3.0):.3f}")
25
26  print("== (2) forward model: GCA + Rsd, 2 nm-like (mu 8.8, V*~1.45) and 13.2 nm-like (mu 41, Vt
27      0.85-0.35)")
28  for lab, mu, vt in (("2 nm", 8.8, 1.10), ("13.2 nm", 41.0, 0.50)):
29      beta = mu * COX / L # A/V^2 per cm width
30      for rsd_um in (0, 1e4, 3e4, 9.2e4, 3e5, 1.15e6):

```

```

28     rsd = rsd_um * 1e-4 # ohm.cm
29     I = np.array([brentq(lambda I: I - gca(v - I * rsd / 2 - vt, beta, VD - I * rsd), 0, (VD / rsd if
rsd else 10) * 0.999999) for v in vg])
30     vt1 = vt + 0.35
31     gmx = np.gradient(I, vg).max()
32     print(f"{lab} Rsd {rsd_um:8.3g} ohm.um: Ion(3V) {I[-1]*1e-4:.3e} A/um, muFE {gmx*L/(COX*VD):6.2f
}, "
33         f"alpha_H[{vt1+0.7:.2f},3] {alpha_H(vg, I + 1e-30, vt1 + 0.7, 3.0):.3f}, IR drop {I[-1]*rsd
:.3f} V")
34
35 print("== (3) mobility-definition test against paper 5.1 (2 nm) / 27.4 (13.2 nm)")
36 a = np.genfromtxt(ROOT / "data/experimental_clean.csv", delimiter=",", names=True)
37 for t, vth_lin, vth_cc in ((2.0, 1.663, 0.662), (13.2, 1.044, 0.083), (6.3, 1.820, 0.644)):
38     m = a["thickness_nm"] == t; v = a["vg_V"][m]; i = a["id_A_per_um"][m] * 1e4; o = np.argsort(v); v, i
= v[o], i[o]
39     gm = np.gradient(i, v)
40     mu_lin = gm.max() * L / (COX * VD)
41     mu_sat = (np.gradient(np.sqrt(i), v).max()) ** 2 * 2 * L / COX
42     mu_eff3_lin = i[-1] * L / (COX * VD * (3 - vth_lin))
43     mu_eff3_cc = i[-1] * L / (COX * VD * (3 - vth_cc))
44     mu_eff3_cc_half = i[-1] * L / (COX * VD * (3 - vth_cc - VD / 2))
45     print(f"t {t}: muFE_lin {mu_lin:.2f}; mu_sat-formula {mu_sat:.2f}; mu_eff(3V,Vth_lin) {mu_eff3_lin:.2
f}; "
46         f"mu_eff(3V,Vth_cc) {mu_eff3_cc:.2f}; mu_eff(3V,Vth_cc,-Vd/2) {mu_eff3_cc_half:.2f}; "
47         f"muFE_lin/paper {mu_lin/{2.0:5.1,13.2:27.4}.get(t, np.nan):.2f}")

```

B.3.18 S5_s5_plot.py

The script draws the Y-function and H-function (threshold-free exponent) figure for the measured curves and the ideal-contact ATLAS runs, reproduced as the yfunction figure.

```

1  """Figure: Y-function and H-function (threshold-free exponent) for measured curves vs ideal-contact ATLAS
runs."""
2  import pathlib, numpy as np, matplotlib
3  matplotlib.use("Agg"); import matplotlib.pyplot as plt
4  ROOT = pathlib.Path(__file__).resolve().parents[3]; OUT = pathlib.Path(__file__).resolve().parent
5  a = np.genfromtxt(ROOT / "data/experimental_clean.csv", delimiter=",", names=True)
6  runs = {2.0: "run_0012_2p0_v2_full_stepped_final", 6.3: "
run_0015_6p3_v2_full_stepped_tuned_qf8p7e10_mu11p69",
7         13.2: "run_0013_13p2_v2_full_stepped_final"}
8  fig, ax = plt.subplots(1, 3, figsize=(13, 4))
9  col = {2.0: "#1f77b4", 6.3: "#d62728", 13.2: "#2ca02c", 31.8: "#7f7f7f"}
10 for t in (2.0, 6.3, 13.2, 31.8):
11     m = a["thickness_nm"] == t; v = a["vg_V"][m]; i = a["id_A_per_um"][m] * 1e4; o = np.argsort(v); v, i
= v[o], i[o]
12     gm = np.gradient(i, v); ok = (v > 0) & (gm > 0)
13     ax[0].plot(v[ok], i[ok] / np.sqrt(gm[ok]) / np.sqrt(i[ok] / np.sqrt(gm[ok]))).max() ** 2, "o", ms=3,
color=col[t], label=f"{t} nm meas")
14     ax[1].plot(v, gm * 1e-4, "-", color=col[t], label=f"{t} nm meas")
15     ic = np.concatenate([0], np.cumsum(0.5 * (i[1:] + i[:-1]) * np.diff(v))]; H = ic / i
16     ax[2].plot(v[v > 0], H[v > 0], "-", color=col[t], label=f"{t} nm meas")
17     if t in runs:
18         s = np.genfromtxt(ROOT / f"results/runs/{runs[t]}/comparison.csv", delimiter=",", names=True)
19         vs, isim = s["vg_V"], s["atlas_A_per_um"] * 1e4; g = np.gradient(isim, vs); k = (vs > 0) & (g >
0)
20         ax[0].plot(vs[k], isim[k] / np.sqrt(g[k]) / np.sqrt(i[ok] / np.sqrt(gm[ok]))).max() ** 2, "--",
color=col[t], label=f"{t} nm ATLAS (Rsd=0)")
21         ax[1].plot(vs, g * 1e-4, "--", color=col[t])

```

```
22     ics = np.concatenate([[0], np.cumsum(0.5 * (isim[1:] + isim[:-1]) * np.diff(vs))]); ax[2].plot(vs
    [vs > 0], (ics / isim)[vs > 0], "--", color=col[t])
23 ax[0].set(xlabel="Vg (V)", ylabel="Y = Id/sqrt(gm) (normalised)", title="Y-function (linear => const. mu,
    Rsd via curvature)")
24 ax[1].set(xlabel="Vg (V)", ylabel="gm (A/V/um)", yscale="log", ylim=(1e-9, 3e-6), title="gm: no roll-off
    at 2 nm; strong roll-off at 31.8 nm")
25 ax[2].set(xlabel="Vg (V)", ylabel="H = int(Id dVg)/Id (V)", title="H-function: slope = 1/(alpha+1)")
26 ax[0].legend(fontsize=7); ax[1].legend(fontsize=7)
27 fig.tight_layout(); fig.savefig(OUT / "s5_yfunction_hfunction.png", dpi=130)
28 print("saved", OUT / "s5_yfunction_hfunction.png")
```

B.3.19 S6_s6_lib.py

The script contains the common helpers of specialist S6 (read-only): metrics from fixed-current crossings with both log-linear and PCHIP interpolation on $\log_{10} I_D$ (their difference is reported as interpolation uncertainty), fixed-current swings and on-state shape markers. The package extractor `scripts/extract_metrics.py` is imported unchanged.

```
1  """S6 common helpers (read-only use of package files; no simulator call).
2  Robust metrics: fixed-current crossings (log-linear AND PCHIP-on-log10 interpolation, difference =
    interpolation
3  uncertainty), fixed-current SS, on-state shape markers. The package extractor (scripts/extract_metrics.py
    ) is imported
4  unchanged for the legacy metrics, applied exactly as run_atlas.py does (np.interp of the native simulated
    grid onto the
5  0.05 V measurement grid).
6  """
7  import csv, json, math, sys
8  from pathlib import Path
9  import numpy as np
10 from scipy.interpolate import PchipInterpolator
11
12 PKG = Path(__file__).resolve().parents[3]
13 sys.path.insert(0, str(PKG / 'scripts'))
14 from extract_metrics import metrics as pkg_metrics, load_curve # noqa: E402
15
16 RUNS = PKG / 'results' / 'runs'
17 COX = 8.955369814335959e-07; L_CM = 20e-4; W_CM = 1e-4; VD = 0.7
18 K_B = 8.617333262e-5 # eV/K
19
20 def run_dir(rid):
21     m = list(RUNS.glob(rid + '_*'))
22     if len(m) != 1: raise FileNotFoundError(rid)
23     return m[0]
24
25 def read_idvg(rid):
26     rows = (run_dir(rid) / 'idvg.dat').read_text().split('\n')[4:]
27     xy = np.array([[float(v) for v in r.split()] for r in rows if r.strip()])
28     vg = np.round(xy[:, 0], 6); vg[np.abs(vg) < 1e-9] = 0.0
29     return vg, xy[:, 1]
30
31 def read_exec(rid):
32     return json.loads((run_dir(rid) / 'execution.json').read_text())
33
34 def meas(t):
35     return load_curve(PKG / 'data' / 'experimental_clean.csv', t)
36
37 def floor_med(t):
```

```
38     vg, i = meas(t); return float(np.median(i[(vg >= -2) & (vg <= -.5)]))
39
40 def crossing_lin(vg, idv, level):
41     for k in range(len(vg) - 1):
42         a, b = idv[k], idv[k + 1]
43         if a > 0 and b > 0 and (a - level) * (b - level) <= 0 and a != b:
44             return float(vg[k] + (vg[k + 1] - vg[k]) * (math.log10(level) - math.log10(a)) / (math.log10(
45                 b) - math.log10(a)))
46     return None
47
48 def crossing_pchip(vg, idv, level):
49     """Monotone cubic on log10(Id) over the monotone positive branch around the first crossing."""
50     k0 = None
51     for k in range(len(vg) - 1):
52         a, b = idv[k], idv[k + 1]
53         if a > 0 and b > 0 and (a - level) * (b - level) <= 0 and a != b: k0 = k; break
54     if k0 is None: return None
55     lo, hi = max(0, k0 - 3), min(len(vg), k0 + 5)
56     x, y = vg[lo:hi], idv[lo:hi]
57     if np.any(y <= 0) or np.any(np.diff(y) <= 0): return crossing_lin(vg, idv, level)
58     f = PchipInterpolator(x, np.log10(y)); xs = np.linspace(vg[k0], vg[k0 + 1], 2001); ys = f(xs)
59     j = int(np.argmin(np.abs(ys - math.log10(level)))); return float(xs[j])
60
61 def robust(vg, idv, floor_sub=None):
62     """Fixed-current markers on a curve (native grid). floor_sub: subtract a constant (measured floor)
63     first."""
64     i = np.array(idv, float) - (floor_sub or 0.0)
65     out = {}
66     for lab, lev in [('V_1e-11', 1e-11), ('V_1e-10', 1e-10), ('V_1e-9', 1e-9), ('V_1e-8', 1e-8), ('V_1e-7',
67         1e-7), ('V_3e-7', 3e-7)]:
68         out[lab] = crossing_lin(vg, i, lev); out[lab + '_pchip'] = crossing_pchip(vg, i, lev)
69     def ss(a, b, key=''):
70         va, vb = out['V_%s%s' % (a, key)], out['V_%s%s' % (b, key)]
71         if va is None or vb is None: return None
72         return (vb - va) / (math.log10(float(b)) - math.log10(float(a))) * 1000.0
73     out['SS_11_10'] = ss('1e-11', '1e-10'); out['SS_10_9'] = ss('1e-10', '1e-9'); out['SS_9_8'] = ss('1e-9',
74         '1e-8'); out['SS_10_8'] = ss('1e-10', '1e-8')
75     out['SS_11_10_pchip'] = ss('1e-11', '1e-10', '_pchip'); out['SS_10_9_pchip'] = ss('1e-10', '1e-9', '_pchip');
76     out['SS_10_8_pchip'] = ss('1e-10', '1e-8', '_pchip')
77     out['gap7'] = (out['V_1e-7'] - out['V_1e-9']) if (out['V_1e-7'] is not None and out['V_1e-9'] is not
78         None) else None
79     return out
80
81 def legacy(vg_native, id_native, t, is_sim=True):
82     """Package metrics exactly as run_atlas.py computes them for a simulation (np.interp onto the 0.05 V
83     grid)."""
84     vgm, _ = meas(t)
85     if is_sim:
86         pred = np.interp(vgm, vg_native, id_native); return pkg_metrics(vgm, pred, True, t)
87     return pkg_metrics(vg_native, id_native, False, t)
88
89 def all_metrics(vg, idv, t, is_sim=True, floor_sub=None):
90     m = legacy(vg, idv, t, is_sim); r = robust(vg, idv, floor_sub)
91     m.update(r); m['gap_lin'] = (m['Vth_lin_V'] - m['Vth_cc_1e-9_V']) if (m['Vth_lin_V'] is not None and
92         m['Vth_cc_1e-9_V'] is not None) else None
93     m['Ion_over_gm_V'] = m['Ion_A_per_um'] / m['gm_max_A_V_um']
94     return m
95
96 def fmt(x, f='{: .3f}'):
97     return '-' if x is None else f.format(x)
```

B.3.20 S6_s6_part1_numerics.py

Part 1 of S6 establishes the numerical closure from existing run outputs: linearity in μ_{band} , DOS offsets versus gate voltage and thickness, the 6.3 and 13.2 nm mesh checks, KCL, native zeros, the fragility of SS_{min} , and the uncertainty due to interpolation and to the transconductance grid. It supplies the numbers of Section 6.6.

```

1  """S6 part 1: numerical closure (linearity, DOS offsets vs Vg and thickness, 6.3/13.2 nm mesh, KCL,
    native zeros,
2  SS_min fragility, interpolation and gm-grid uncertainty). Reads only existing run outputs."""
3  import json, math
4  import numpy as np
5  from s6_lib import *
6
7  out = {}
8  P = lambda *a: print(*a)
9
10 # ----- 1. exact linearity of Id in mu_band (same DOS, same deck otherwise) -----
11 P('=== 1. Linearity of Id in mu_band ===')
12 for a, b, mua, mub, lab in [('run_0038', 'run_0029', 17.69, 18.13, '2 nm DOS 384/192 (B1 vs check C)'),
13                             ('run_0003', 'run_0008', 16.8, 18.13, '2 nm DOS 96/48 reduced deck (stage
    pair)')]:
14     va, ia = read_idvg(a); vb, ib = read_idvg(b)
15     common = np.intersect1d(va, vb); ia2 = np.array([ia[va == v][0] for v in common]); ib2 = np.array([ib
    [vb == v][0] for v in common])
16     msk = ia2 > 1e-14; r = ib2[msk] / ia2[msk]
17     exp_r = mub / mua
18     P(f'{lab}: {b}/{a} expected {exp_r:.6f}; observed mean {r.mean():.6f}, min {r.min():.6f}, max {r.max
    ():.6f} over {msk.sum()} pts with Id>1e-14 (Vg {common[msk][0]}..{common[msk][-1]})')
19     out[f'linearity_{a}_{b}'] = dict(expected=exp_r, mean=float(r.mean()), min=float(r.min()), max=float(
    r.max()), n=int(msk.sum()))
20
21 # ----- 2. DOS discretization offsets vs Vg -----
22 P('\n=== 2. DOS 96/48 -> 384/192 offset (same mu after exact rescale) ===')
23 cases = [('2.0', 'run_0012', 'run_0029', 1.0, '96/48 -> 384/192 (both mu 18.13)'),
24          ('2.0', 'run_0012', 'run_0019', 1.0, '96/48 -> 192/96 (both mu 18.13)'),
25          ('6.3', 'run_0015', 'run_0039', 11.69 / 11.41, 'run_0015 (mu 11.69, 96/48) vs B2 (mu 11.41,
    384/192) x 11.69/11.41'),
26          ('13.2', 'run_0013', 'run_0040', 61.9 / 60.39, 'run_0013 (mu 61.9, 96/48) vs B3 (mu 60.39,
    384/192) x 61.9/60.39')]
27 dos = {}
28 for t, a, b, sc, lab in cases:
29     va, ia = read_idvg(a); vb, ib = read_idvg(b); ib = ib * sc
30     rows = []
31     for v in [0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0]:
32         x, y = ia[va == v][0], ib[vb == v][0]; rows.append((v, y / x - 1))
33     ta = float(t); ra = robust(va, ia); rb = robust(vb, ib)
34     la = legacy(va, ia, ta); lb = legacy(vb, ib, ta)
35     d = dict(dIon_pct=100 * (ib[-1] / ia[-1] - 1), dVcc_mV=1000 * (rb['V_1e-9'] - ra['V_1e-9']), dSS_10_9
    =rb['SS_10_9'] - ra['SS_10_9'], dSS_11_10=rb['SS_11_10'] - ra['SS_11_10'],
36             dSS_10_8=rb['SS_10_8'] - ra['SS_10_8'], dVlin_mV=1000 * (lb['Vth_lin_V'] - la['Vth_lin_V']),
    dgm_pct=100 * (lb['gm_max_A-V_um'] / la['gm_max_A-V_um'] - 1), dgap7_mV=1000 * (rb['gap7'] - ra['
    gap7']))
37     dos[f'{t}_{a}_{b}'] = d
38     P(f'{t} nm {lab}:')
39     P('  dId/Id (%) at Vg: ' + ', '.join(f'{v:g}: {100*r:+.2f}' for v, r in rows))
40     P('  ' + ', '.join(f'{k} {v:+.2f}' for k, v in d.items()))
41 out['dos'] = dos
42 # Richardson-type estimate of the residual beyond 384/192 at 2 nm (96->192->384)
43 va, i96 = read_idvg('run_0012'); _, i192 = read_idvg('run_0019'); _, i384 = read_idvg('run_0029')

```

```

44 e1 = i192[-1] / i96[-1] - 1; e2 = i384[-1] / i192[-1] - 1; p = math.log(e1 / e2, 2); resid = e2 / (2 ** p
    - 1)
45 P(f'\n2 nm Ion: 96->192 {100*e1:+.3f} %, 192->384 {100*e2:+.3f} %, observed order p = {p:.2f}; Richardson
    residual beyond 384/192 = {100*resid:+.3f} %')
46 out['richardson_2nm'] = dict(e96_192=e1, e192_384=e2, order=p, residual=resid)
47 for v in [0.7, 1.0, 1.5, 2.0]:
48     a, b, c = i96[va == v][0], i192[va == v][0], i384[va == v][0]
49     e1v, e2v = b / a - 1, c / b - 1
50     P(f'    Vg {v}: 96->192 {100*e1v:+.3f} %, 192->384 {100*e2v:+.3f} %')
51
52 # ----- 3. mesh checks -----
53 P(f'\n=== 3. Mesh x0.7 ===')
54 for t, a, b in [(6.3, 'run_0015', 'run_0036'), (13.2, 'run_0013', 'run_0028')]:
55     va, ia = read_idvg(a); vb, ib = read_idvg(b); ta = float(t)
56     fl = floor_med(ta); msk = (ia > 5 * fl) & (ib > 5 * fl)
57     dl = np.abs(np.log10(ib[msk]) - np.log10(ia[msk]))
58     ra, rb = robust(va, ia), robust(vb, ib); la, lb = legacy(va, ia, ta), legacy(vb, ib, ta)
59     d = dict(max_dlog=float(dl.max()), rms_dlog=float(np.sqrt((dl ** 2).mean()))), dVcc_mV=1000 * (rb['
        V_1e-9'] - ra['V_1e-9']), dSS_11_10=rb['SS_11_10'] - ra['SS_11_10'], dSS_10_9=rb['SS_10_9'] - ra['
        SS_10_9'],
60     dSS_10_8=rb['SS_10_8'] - ra['SS_10_8'], dVlin_mV=1000 * (lb['Vth_lin_V'] - la['Vth_lin_V']),
        dgm_pct=100 * (lb['gm_max_A-V_um'] / la['gm_max_A-V_um'] - 1), dIon_pct=100 * (ib[-1] / ia[-1] - 1)
        , dSSmin=lb['SS_mV_dec'] - la['SS_mV_dec'])
61     out[f'mesh_{t}'] = d
62     P(f'{t} nm {b} vs {a}: ' + ', '.join(f'{k} {v:+.4f}' for k, v in d.items()))
63
64 # ----- 4. KCL and native zeros -----
65 P(f'\n=== 4. KCL and native zeros (runs 0012-0052 with idvg.dat) ===')
66 kcl = []
67 for n in range(12, 53):
68     rid = f'run_{n:04d}'
69     try: d = run_dir(rid)
70     except FileNotFoundError: continue
71     if not (d / 'execution.json').exists(): continue
72     ex = read_exec(rid); v = ex.get('validation', {})
73     rec = dict(run=rid, max_kcl=v.get('max_kcl_abs_A'), limit=v.get('kcl_abs_limit_applied_A'), fails=v.
        get('kcl_fail_count'))
74     if (d / 'idvg.dat').exists():
75         vg, i = read_idvg(rid); npz = i <= 0
76         rec.update(nonpos=int(npz.sum()), vg_max_nonpos=float(vg[npz].max()) if npz.any() else None,
77             max_abs_below=float(np.abs(i[vg <= (vg[npz].max() if npz.any() else -9)]).max()) if
78             npz.any() else None,
79             first_vg_above_1e15=float(vg[np.argmax(i > 1e-15)]))
80     kcl.append(rec)
81 for r in kcl: P(r)
82 out['kcl'] = kcl
83 tr = [r for r in kcl if r.get('nonpos') is not None]
84 P('max KCL over transfer runs: %.3g A; max |Id| at or below the highest-Vg native zero: %.3g A/um;
    highest Vg of a native zero: %.2f V' % (
85     max(r['max_kcl'] for r in tr), max(r['max_abs_below'] for r in tr if r['max_abs_below'] is not None),
86     max(r['vg_max_nonpos'] for r in tr if r['vg_max_nonpos'] is not None)))
87
88 # ----- 5. SS_min fragility -----
89 P(f'\n=== 5. SS_min window phase: run_0012 vs B1 (run_0038), run_0003 vs run_0008 ===')
90 for t, runs in [(2.0, ['run_0012', 'run_0038', 'run_0029', 'run_0019', 'run_0016', 'run_0003', 'run_0008'
    ]), (13.2, ['run_0013', 'run_0040', 'run_0017', 'run_0028'])]:
91     fl = floor_med(t); thr = 5 * fl
92     for rid in runs:
93         vg, i = read_idvg(rid); m = legacy(vg, i, t); r = robust(vg, i)
94         first = vg[np.argmax(i > thr)]
95         P(f'{t} nm {rid}: 5x floor {thr:.3g}; first native point above = {first:.2f} V (Id {i[vg==first'

```

```

] [0]:.3g}); previous point {i[vg==round(first-0.05,2)][0]:.3g}); SS_min {m["SS_mV_dec"]:.1f};
SS_11_10 {fmt(r["SS_11_10"], "{:.1f}")}; SS_10_9 {r["SS_10_9"]:.1f}; SS_cc {r["SS_10_8"]:.1f}')
94
95 # ----- 6. interpolation uncertainty (log-linear vs PCHIP) -----
96 P('\n=== 6. Crossing interpolation: log-linear minus PCHIP (mV) ===')
97 interp = {}
98 for lab, t, src in [('meas', 2.0, None), ('meas', 6.3, None), ('meas', 13.2, None), ('B1', 2.0, 'run_0038
    '), ('B2', 6.3, 'run_0039'), ('B3', 13.2, 'run_0040'), ('C1', 6.3, 'run_0052')]:
99     vg, i = meas(t) if src is None else read_idvg(src); r = robust(vg, i)
100     d = {k: 1000 * (r[k] - r[k + '_pchip']) for k in ['V_1e-11', 'V_1e-10', 'V_1e-9', 'V_1e-8', 'V_1e-7']}
        if r[k] is not None and r[k + '_pchip'] is not None}
101     dss = {k: r[k] - r[k + '_pchip'] for k in ['SS_11_10', 'SS_10_9', 'SS_10_8'] if r[k] is not None and
        r[k + '_pchip'] is not None}
102     interp[f'{lab}_{t}'] = dict(dV_mV=d, dSS=dss)
103     P(f'{lab} {t}: ' + ', '.join(f'{k} {v:+.2f}' for k, v in d.items()) + ' | ' + ', '.join(f'{k} {v:+.2f}
        ') for k, v in dss.items()))
104 out['interp'] = interp
105
106 # ----- 7. gm grid effect: measured curve re-sampled as the simulation grid, then np.interp back
        -----
107 P('\n=== 7. gm_max / Vth_lin grid effect (measured curves passed through the simulation grid) ===')
108 vg_sim, _ = read_idvg('run_0038')
109 gridfx = {}
110 for t in [2.0, 6.3, 13.2]:
111     vg, i = meas(t); base = pkg_metrics(vg, i, False, t)
112     sub = np.array([i[np.argmin(np.abs(vg - v))] for v in vg_sim]); back = np.interp(vg, vg_sim, sub)
113     m2 = pkg_metrics(vg, back, False, t)
114     gridfx[t] = dict(dgm_pct=100 * (m2['gm_max_A_V_um'] / base['gm_max_A_V_um'] - 1), dVlin_mV=1000 * (m2
        ['Vth_lin_V'] - base['Vth_lin_V']), dVcc_mV=1000 * (m2['Vth_cc_1e-9_V'] - base['Vth_cc_1e-9_V']))
115     P(f'{t} nm: gm_max {100*(m2["gm_max_A_V_um"]/base["gm_max_A_V_um"]-1):+.2f} %, Vth_lin {1000*(m2["
        Vth_lin_V"]-base["Vth_lin_V"]):+.1f} mV, Vth_cc {1000*(m2["Vth_cc_1e-9_V"]-base["Vth_cc_1e-9_V"]):
        :+.2f} mV')
116 out['gm_grid'] = gridfx
117
118 # ----- 8. ID-VD vs transfer consistency at Vd = 0.7 V -----
119 P('\n=== 8. Path independence: ID-VD (Vd = 0.7 V) vs transfer curve at the same Vg, same mu ===')
120 import csv as _csv
121 path = {}
122 for t, rt, ro, sc in [(2.0, 'run_0038', 'run_0041', 17.6897 / 17.69), (6.3, 'run_0039', 'run_0042',
    11.582 / 11.41), (13.2, 'run_0040', 'run_0043', 61.6046 / 60.39)]:
123     vg, i = read_idvg(rt); i = i * sc
124     rows = list(_csv.DictReader((run_dir(ro) / 'output_characteristics.csv').open()))
125     diffs = []
126     for vgv in [1.0, 1.5, 2.0, 2.5, 3.0]:
127         idd = [float(r['id_A_per_um']) for r in rows if abs(float(r['vg_V']) - vgv) < 1e-9 and abs(float(
            r['vd_V']) - 0.7) < 1e-9][0]
128         itr = i[np.argmin(np.abs(vg - vgv))]; diffs.append(100 * (idd / itr - 1))
129     path[t] = diffs
130     P(f'{t} nm {ro} vs {rt} x{sc:.6f}: dId % at Vg 1/1.5/2/2.5/3 = ' + ', '.join(f'{d:+.3f}' for d in
        diffs))
131 out['path'] = path
132
133 json.dump(out, open(Path(__file__).with_name('s6_part1_out.json'), 'w'), indent=1, default=float)

```

B.3.21 S6_s6_part2_robust.py

Part 2 of S6 produces the tables of fixed-current metrics for the final converged comparison, the isolation tests B4–B7 (run_0048–run_0051) with a rigid-shift test, and the closure of the 6.3 nm

hypotheses A1–A7, B0 and C1. Curves are rescaled by the exact linearity in μ_{band} (verified in part 1 to 10^{-5}).

```

1  """S6 part 2: robust-metric tables (final converged comparison, isolation tests B4-B7 with rigid-shift
    test, 6.3 nm
2  hypothesis closure A1-A7, B0, C1). Rescaling by exact linearity in mu_band (verified in part 1 to 1e-5)."""
3  import json
4  import numpy as np
5  from s6_lib import *
6
7  MU = {2.0: 17.6896871948524, 6.3: 11.581986680320135, 13.2: 61.60456093550155}
8  KEYS = ['V_1e-9', 'SS_11_10', 'SS_10_9', 'SS_10_8', 'SS_mV_dec', 'Vth_lin_V', 'gap_lin', 'V_1e-7', 'gap7',
    , 'gm_max_A-V_um', 'mu_FE_cm2Vs', 'Ion_A_per_um', 'Ion_over_gm_V']
9  res = {}
10
11  def row(lab, m):
12      return (f"| {lab} | {fmt(m['V_1e-9'])} | {fmt(m['SS_11_10'], '{:.1f}')} | {fmt(m['SS_10_9'], '{:.1f}')} |
    | {fmt(m['SS_10_8'], '{:.1f}')} | {fmt(m['SS_mV_dec'], '{:.1f}')} | "
13      f"{fmt(m['Vth_lin_V'])} | {fmt(m['gap_lin'])} | {fmt(m['V_1e-7'])} | {fmt(m['gap7'])} | {m['
    gm_max_A-V_um']:.4g} | {m['mu_FE_cm2Vs']:.2f} | {m['Ion_A_per_um']:.4g} | {m['Ion_over_gm_V']:.3f} |
    ")
14  HDR = ('| curve | Vth_cc 1e-9 (V) | SS 1e-11..1e-10 | SS 1e-10..1e-9 | SS 1e-10..1e-8 (SS_cc) | SS_min (
    fragile) | Vth_lin (V) | gap Vlin-Vcc (V) | V(1e-7) (V) | V(1e-7)-Vcc (V) | gm_max (A/V/um) | mu_FE
    | Ion (A/um) | Ion/gm (V) |\n'
15      '|---|---|---|---|---|---|---|---|---|---|---|---|')
16
17  print('=== A. Final converged comparison (DOS 384/192, recalibrated mu; exact linear rescale) ===')
18  print(HDR)
19  final = {}
20  for t, rid, mu_run in [(2.0, 'run_0038', 17.69), (6.3, 'run_0039', 11.41), (13.2, 'run_0040', 60.39)]:
21      vg, i = meas(t); me = all_metrics(vg, i, t, False); final[f'meas_{t}'] = me; print(row(f'measured {t}
    nm', me))
22      if t == 13.2:
23          fl = floor_med(t); ms = all_metrics(vg, i, t, False, floor_sub=fl); final[f'meas_{t}_floorsub'] =
    ms
24          print(row(f'measured {t} nm, floor {fl:.2g} subtracted (fixed-current markers only)', ms))
25          vg, i = read_idvg(rid); sc = MU[t] / mu_run; m = all_metrics(vg, i * sc, t); final[f'final_{t}'] = m;
    final[f'final_{t}_scale'] = sc
26          print(row(f'{rid} x{sc:.5f} (mu {MU[t]:.3f})', m))
27          old = {2.0: 'run_0012', 6.3: 'run_0015', 13.2: 'run_0013'}[t]; vg, i = read_idvg(old); mo =
    all_metrics(vg, i, t); final[f'old_{t}'] = mo
28          print(row(f'old (DOS 96/48, reference)', mo))
29  res['final'] = final
30  print('\nSim - measured (final converged):')
31  for t in [2.0, 6.3, 13.2]:
32      s, e = final[f'final_{t}'], final[f'meas_{t}']
33      d = {k: (s[k] - e[k]) if (s[k] is not None and e[k] is not None) else None for k in ['V_1e-9', '
    SS_11_10', 'SS_10_9', 'SS_10_8', 'Vth_lin_V', 'gap_lin', 'V_1e-7', 'gap7']}
34      d['gm_pct'] = 100 * (s['gm_max_A-V_um'] / e['gm_max_A-V_um'] - 1); d['Ion_pct'] = 100 * (s['
    Ion_A_per_um'] / e['Ion_A_per_um'] - 1)
35      print(t, {k: (None if v is None else round(v, 4)) for k, v in d.items()})
36      res[f'diff_{t}'] = d
37  for k in ['SS_11_10', 'SS_10_9', 'SS_10_8']:
38      s, e = final[f'final_13.2'][k], final[f'meas_13.2_floorsub'][k]
39      print(f'13.2 nm {k}: sim {s:.1f} vs floor-subtracted measured {fmt(e, "{:.1f}")}')
40
41  print('\n=== B. Isolation tests (DOS 384/192, same mu as baseline) ===')
42  print(HDR)
43  iso = {}
44  for lab, t, base, pert in [('WF +0.1 eV', 2.0, 'run_0038', 'run_0048'), ('WF +0.1 eV', 13.2, 'run_0040',

```

```

    'run_0049'), ('m* x1.3', 2.0, 'run_0038', 'run_0050'), ('m* x1.3', 13.2, 'run_0040', 'run_0051'),
45     ('chi -0.1 eV (DOS 96/48)', 2.0, 'run_0012', 'run_0020'), ('chi +0.1 eV (DOS
    96/48)', 2.0, 'run_0012', 'run_0021'))]:
46 vb, ib = read_idvg(base); vp, ip = read_idvg(pert); mb = all_metrics(vb, ib, t); mp = all_metrics(vp,
    ip, t)
47 print(row(f'{base} base {t}', mb)); print(row(f'{pert} {lab}', mp))
48 d = {k: (mp[k] - mb[k]) for k in ['V_1e-9', 'SS_11_10', 'SS_10_9', 'SS_10_8', 'Vth_lin_V', 'gap_lin',
    'gap7']}
49 d['Ion_pct'] = 100 * (mp['Ion_A_per_um'] / mb['Ion_A_per_um'] - 1); d['gm_pct'] = 100 * (mp['
    gm_max_A_V_um'] / mb['gm_max_A_V_um'] - 1)
50 # rigid-shift test: best shift s (multiple of 0.05 V) such that  $I_p(V_g) = I_b(V_g - s)$  on common grid
    points
51 best = None
52 for s in np.round(np.arange(-0.3, 0.301, 0.05), 3):
53     pts = [(v, ip[k]) for k, v in enumerate(vp) if np.any(np.abs(vb - (v - s)) < 1e-9) and ip[k] > 1e
    -13]
54     if len(pts) < 10: continue
55     dl = np.array([abs(np.log10(y) - np.log10(ib[np.abs(vb - (v - s)) < 1e-9][0])) for v, y in pts])
56     if best is None or dl.max() < best[1]: best = (float(s), float(dl.max()), float(np.median(dl)),
    len(pts))
57 d['rigid_best_shift_V'], d['rigid_max_dlog'], d['rigid_median_dlog'], d['rigid_npts'] = best
58 iso[f'{lab}_{t}'] = d
59 print(f'    delta: ' + ', '.join(f'{k} {v:+.4g}' for k, v in d.items() if v is not None))
60 res['iso'] = iso
61 # thickness dependence of the 2 -> 13.2 differences
62 for lab, b2, p2, b13, p13 in [('WF', 'run_0038', 'run_0048', 'run_0040', 'run_0049'), ('mstar', 'run_0038
    ', 'run_0050', 'run_0040', 'run_0051')]:
63     M = {r: all_metrics(*read_idvg(r), t) for r, t in [(b2, 2.0), (p2, 2.0), (b13, 13.2), (p13, 13.2)]}
64     dv_b = M[b13]['V_1e-9'] - M[b2]['V_1e-9']; dv_p = M[p13]['V_1e-9'] - M[p2]['V_1e-9']
65     ir_b = M[b13]['Ion_A_per_um'] / M[b2]['Ion_A_per_um']; ir_p = M[p13]['Ion_A_per_um'] / M[p2]['
    Ion_A_per_um']
66     ss_b = M[b13]['SS_10_8'] - M[b2]['SS_10_8']; ss_p = M[p13]['SS_10_8'] - M[p2]['SS_10_8']
67     mr_b = M[b13]['mu_FE_cm2Vs'] / M[b2]['mu_FE_cm2Vs']; mr_p = M[p13]['mu_FE_cm2Vs'] / M[p2]['
    mu_FE_cm2Vs']
68     print(f'{lab}: dVth_cc(2->13.2) {dv_b:+.4f} -> {dv_p:+.4f} (change {1000*(dv_p-dv_b):+.1f} mV); Ion
    ratio 13.2/2 {ir_b:.4f} -> {ir_p:.4f} ({100*(ir_p/ir_b-1):+.2f} %); '
69     f'SS_cc difference 13.2-2 {ss_b:+.1f} -> {ss_p:+.1f}; mu_FE ratio {mr_b:.4f} -> {mr_p:.4f}')
70     res[f'tdep_{lab}'] = dict(dv_base=dv_b, dv_pert=dv_p, Iratio_base=ir_b, Iratio_pert=ir_p, dSS_base=
    ss_b, dSS_pert=ss_p, muratio_base=mr_b, muratio_pert=mr_p)
71 # rigid-shift test for Qf pairs
72 for lab, base, pert, t in [('Qf 0 -> 1.73e12 (2 nm)', 'run_0001', 'run_0003', 2.0), ('Qf 0 -> 1.73e12
    (13.2 nm)', 'run_0002', 'run_0004', 13.2), ('Qf 1.73e12 -> 8.7e10 (6.3 nm)', 'run_0005', 'run_0007',
    6.3)]:
73     vb, ib = read_idvg(base); vp, ip = read_idvg(pert)
74     ma, mb2 = robust(vb, ib), robust(vp, ip)
75     print(f'{lab}: dV at 1e-11/1e-10/1e-9/1e-8/1e-7 = ' + ', '.join(f'{1000*(mb2[k]-ma[k]):+.1f}' for k
    in ['V_1e-11', 'V_1e-10', 'V_1e-9', 'V_1e-8', 'V_1e-7'] if ma[k] is not None and mb2[k] is not None)
    + ' mV')
76
77 print('\n=== C. 6.3 nm hypothesis closure (robust metrics) ===')
78 print(HDR)
79 hyp = {}
80 vg, i = meas(6.3); hyp['meas'] = all_metrics(vg, i, 6.3, False); print(row('measured 6.3 nm', hyp['meas'
    ]))
81 for lab, rid, sc in [('run_0014 validation (Qf 1.73e12, mu 12.4, 96/48)', 'run_0014', 1.0), ('run_0015
    tuned (Qf 8.7e10, mu 11.69, 96/48)', 'run_0015', 1.0),
82     ('B2 run_0039 x1.01507 (tuned, 384/192)', 'run_0039', MU[6.3] / 11.41), ('A7
    run_0036 mesh x0.7 of run_0015', 'run_0036', 1.0),
83     ('A3 run_0030 back -1.64e12', 'run_0030', 1.0), ('A1 run_0031 t=5.3 nm', 'run_0031',
    1.0), ('A2 run_0032 Nd0 1e16', 'run_0032', 1.0),
84     ('A4 run_0033 Dit x5', 'run_0033', 1.0), ('A6 run_0034 no QC, Qf from 2 nm (held-out

```

```

    ), 'run_0034', 1.0), ('A5 run_0035 combination', 'run_0035', 1.0),
85     ('B0 run_0037 S2 re-partition (96/48)', 'run_0037', 1.0), ('C1 run_0052 S4 near-Ec
    band + mu 15.4 (384/192)', 'run_0052', 1.0)]:
86     vg, i = read_idvg(rid); m = all_metrics(vg, i * sc, 6.3); hyp[rid] = m; print(row(lab, m))
87     res['hyp'] = hyp
88     json.dump(res, open(Path(__file__).with_name('s6_part2_out.json'), 'w'), indent=1, default=float)

```

B.3.22 S6_s6_part3_predictions.py

Part 3 of S6 performs the computations of the prediction register from real ATLAS outputs: I_D – V_D (B8–B10, run_0041–run_0043) and elevated temperature (B11–B14, run_0044–run_0047). The 300 K references B1–B3 are rescaled exactly to the mobility used in the prediction runs, in the two temperature variants P-phonon (as simulated, $TMUN = 1.5$) and P-MTR (temperature-independent band mobility).

```

1  """S6 part 3: prediction register computations from real ATLAS outputs (B8-B10 ID-VD; B11-B14 elevated T;
    300 K
2  references B1/B2/B3 rescaled exactly to the mu used in the prediction runs). Two temperature variants:
3  P-phonon = as simulated (ATLAS silicon-default TMUN 1.5: mu(T) = mu300 (T/300)^-1.5, verified in
    run_0045 deckbuild.out)
4  P-MTR    = simulated currents x (T/300)^+1.5 (T-independent band mobility; exact because Id is linear
    in a spatially
5  uniform constant mobility, verified in part 1 to 1e-5).
6  Writes s6_part3_out.json and the canonical rows (s6_canonical_rows.json) consumed by s6_write_register.py
    ."""
7  import csv, json, math
8  import numpy as np
9  from s6_lib import *
10
11  MU_PRED = {2.0: 17.6897, 6.3: 11.582, 13.2: 61.6046}          # mu_band in B8-B14 decks (campaign_B
    overrides)
12  REF = {2.0: ('run_0038', 17.69), 6.3: ('run_0039', 11.41), 13.2: ('run_0040', 60.39)}
13  IDVD = {2.0: 'run_0041', 6.3: 'run_0042', 13.2: 'run_0043'}
14  TRUNS = {(2.0, 338.15): 'run_0044', (2.0, 358.15): 'run_0045', (6.3, 358.15): 'run_0046', (13.2, 358.15):
    'run_0047'}
15  FILMS = [2.0, 6.3, 13.2]; VGS_OUT = [1.0, 1.5, 2.0, 2.5, 3.0]; VDS = [0.05, 0.1, 0.5, 1.0, 2.0, 3.0]
16  VG_EA = [0.0, 0.5, 1.0, 1.5, 2.0, 3.0]
17  tag = lambda t: {2.0: '2p0', 6.3: '6p3', 13.2: '13p2'}[t]
18  canon = []; out = {}
19  def C(film, T, var, vg, vd, q, val, unit, rid):
20      canon.append(dict(film=f'{film:g} nm', T_K=T, variant=var, Vg=vg, Vd=vd, quantity=q, value=val, unit=
    unit, run_id=rid))
21
22  # ----- ID-VD
23  print('=== ID-VD predictions (300 K, ideal Ohmic Pd, constant mu, Vd sweep 0-3 V) ===')
24  idvd = {}
25  for t in FILMS:
26      rid = IDVD[t]; rows = list(csv.DictReader((run_dir(rid) / 'output_characteristics.csv').open()))
27      for vg in VGS_OUT:
28          pts = sorted([(float(r['vd_V']), float(r['id_A_per_um'])) for r in rows if abs(float(r['vg_V']) -
    vg) < 1e-9])
29          vd = np.array([p[0] for p in pts]); i = np.array([p[1] for p in pts])
30          at = lambda x: float(i[np.argmin(np.abs(vd - x))])
31          h = 0.05; I1, I2 = at(0.05), at(0.1); b = (I2 - 2 * I1) / (2 * h * h); a = (4 * I1 - I2) / (2 * h
    )
32          g = np.gradient(i, vd); g0 = a
33          vsat = None
34          for k in range(1, len(vd)):

```

```

35         if g[k] <= 0.1 * g0 and vd[k] > 0:
36             vsat = float(vd[k - 1] + (vd[k] - vd[k - 1]) * (g[k - 1] - 0.1 * g0) / (g[k - 1] - g[k]))
           ; break
37         gd3 = (at(3.0) - at(2.9)) / 0.1
38         d = dict(Id={x: at(x) for x in VDS}, r_lin=I2 / I1, curv_b=b, gd0=g0, vdsat10=vsat, gd3=gd3,
           gd3_over_gd0=gd3 / g0, Id07=at(0.7))
39         idvd[(t, vg)] = d
40         for x in vd[vd > 0]: C(t, 300.0, 'as_simulated_300K', vg, float(x), 'Id', float(i[vd == x][0]), '
           A/um', rid)
41         C(t, 300.0, 'as_simulated_300K', vg, '0.1/0.05', 'Id_ratio_Vd0p1_over_Vd0p05', I2 / I1, '-', rid)
42         C(t, 300.0, 'as_simulated_300K', vg, '0-0.1', 'gd0_quadratic_fit', g0, 'S/um', rid)
43         C(t, 300.0, 'as_simulated_300K', vg, '0-0.1', 'd2Id_dVd2_at_0', 2 * b, 'A/um/V^2', rid)
44         C(t, 300.0, 'as_simulated_300K', vg, '-', 'Vd_sat_gd10pct_of_gd0', vsat if vsat is not None else
           '>3', 'V', rid)
45         C(t, 300.0, 'as_simulated_300K', vg, 3.0, 'gd_output_conductance', gd3, 'S/um', rid)
46         C(t, 300.0, 'as_simulated_300K', vg, 3.0, 'gd3_over_gd0', gd3 / g0, '-', rid)
47         print(f'{t:>4} nm Vg {vg}: ' + ', '.join(f'Id({x})={d["Id"][x]:.4g}' for x in VDS) + f'; r
           (0.1/0.05)={I2/I1:.4f}; d2Id/dVd2(0)={2*b:.3g}; gd0={g0:.4g}; Vdsat10={fmt(vsat,"{:.2f}")}; gd(3V)={
           gd3:.3g} ({100*gd3/g0:.2f} % of gd0)')
48 out['idvd'] = {f'{k[0]}_{k[1]}': v for k, v in idvd.items()}
49 print('\nFilm ratios Id(t)/Id(2 nm):')
50 for t in [6.3, 13.2]:
51     for vg in VGS_OUT:
52         rr = {x: idvd[(t, vg)]['Id'][x] / idvd[(2.0, vg)]['Id'][x] for x in VDS}
53         for x in VDS: C(t, 300.0, 'as_simulated_300K', vg, x, f'Id_ratio_{tag(t)}_over_2p0', rr[x], '-',
           f'{IDVD[t]}/{IDVD[2.0]}')
54         print(f'{t} / 2.0 nm, Vg {vg}: ' + ', '.join(f'{x}: {rr[x]:.3f}' for x in VDS))
55         out[f'ratio_{t}_{vg}'] = rr
56
57 # ----- temperature
58 print('\n=== Temperature predictions (Vd 0.7 V) ===')
59 curves = {}
60 for t in FILMS:
61     rid, mu_run = REF[t]; vg, i = read_idvg(rid); sc = MU_PRED[t] / mu_run
62     curves[(t, 300.0, 'ref')] = (vg, i * sc, f'{rid} x{sc:.6f}')
63 for (t, T), rid in TRUNS.items():
64     vg, i = read_idvg(rid); f = (T / 300.0) ** 1.5
65     curves[(t, T, 'P-phonon')] = (vg, i, rid); curves[(t, T, 'P-MTR')] = (vg, i * f, f'{rid} x(T/300)
           ^1.5={f:.5f}')
66 M = {k: all_metrics(v[0], v[1], k[0]) for k, v in curves.items()}
67 temp = {}
68 for (t, T, var), m in M.items():
69     if var == 'ref': continue
70     r = M[(t, 300.0, 'ref')]
71     d = dict(Vth_cc=m['V_1e-9'], dVth_cc_mV=1000 * (m['V_1e-9'] - r['V_1e-9']), dV1e7_mV=1000 * (m['V_1e
           -7'] - r['V_1e-7']), dV1e11_mV=1000 * (m['V_1e-11'] - r['V_1e-11']),
72             SS_11_10=m['SS_11_10'], dSS_11_10=m['SS_11_10'] - r['SS_11_10'], SS_10_9=m['SS_10_9'],
           dSS_10_9=m['SS_10_9'] - r['SS_10_9'], SS_10_8=m['SS_10_8'], dSS_10_8=m['SS_10_8'] - r['SS_10_8'],
73             Vth_lin=m['Vth_lin_V'], dVth_lin_mV=1000 * (m['Vth_lin_V'] - r['Vth_lin_V']), mu_FE=m['
           mu_FE_cm2Vs'], mu_FE_ratio=m['mu_FE_cm2Vs'] / r['mu_FE_cm2Vs'],
74             Ion=m['Ion_A_per_um'], dIon_pct=100 * (m['Ion_A_per_um'] / r['Ion_A_per_um'] - 1), gap=m['
           gap_lin'], dgap_mV=1000 * (m['gap_lin'] - r['gap_lin']))
75     temp[(t, T, var)] = d
76     rid = curves[(t, T, var)][2]
77     for q, u in [(('Vth_cc', 'V'), ('dVth_cc_mV', 'mV'), ('SS_11_10', 'mV/dec'), ('dSS_11_10', 'mV/dec'),
           ('SS_10_9', 'mV/dec'), ('dSS_10_9', 'mV/dec'), ('SS_10_8', 'mV/dec'), ('dSS_10_8', 'mV/dec'),
78             ('Vth_lin', 'V'), ('dVth_lin_mV', 'mV'), ('mu_FE', 'cm2/Vs'), ('mu_FE_ratio', '-'), ('
           Ion', 'A/um'), ('dIon_pct', '%'))]:
79         C(t, T, var, '-' if q not in ('Ion', 'dIon_pct') else 3.0, 0.7, q, d[q], u, rid)
80     print(f'{t:>4} nm {T} K {var:8s}: Vcc {d["Vth_cc"]:.4f} (d {d["dVth_cc_mV"]:.1f} mV); dV(1e-11) {d["
           dV1e11_mV"]:.1f}; dV(1e-7) {d["dV1e7_mV"]:.1f}; SS 1e-11..1e-10 {d["SS_11_10"]:.1f} ({d["dSS_11_10

```

```

    "]:+.1f}); '
81     f'1e-10..1e-9 {d["SS_10_9"]:.1f} ({d["dSS_10_9"]:.1f}); SS_cc {d["SS_10_8"]:.1f} ({d["dSS_10_8"]:.1f}); Vlin {d["Vth_lin"]:.3f} ({d["dVth_lin_mV"]:.1f}); mu_FE {d["mu_FE"]:.3f} (x{d["mu_FE_ratio"]:.4f}); Ion {d["Ion"]:.4g} ({d["dIon_pct"]:.2f} %)')
82 for (t, T, var), (vg, i, rid) in curves.items():
83     for k, v in enumerate(vg):
84         if v >= -1.0 and i[k] > 1e-16:
85             C(t, T, 'reference_300K' if var == 'ref' else var, float(v), 0.7, 'Id', float(i[k]), 'A/um',
86               rid)
87         if var == 'ref':
88             m = M[(t, T, var)]
89             for q, key, u in [('Vth_cc', 'V_1e-9', 'V'), ('SS_11_10', 'SS_11_10', 'mV/dec'), ('SS_10_9', 'SS_10_9', 'mV/dec'), ('SS_10_8', 'SS_10_8', 'mV/dec'), ('Vth_lin', 'Vth_lin_V', 'V'), ('mu_FE', 'mu_FE_cm2Vs', 'cm2/Vs'), ('Ion', 'Ion_A_per_um', 'A/um')]:
90                 C(t, 300.0, 'reference_300K', 3.0 if q == 'Ion' else '-', 0.7, q, m[key], u, rid)
91 out['temp'] = {f'{k[0]}_{k[1]}_{k[2]}': v for k, v in temp.items()}
92 out['ref'] = {f'{t}': {k: M[(t, 300.0, 'ref')][k] for k in ['V_1e-9', 'SS_11_10', 'SS_10_9', 'SS_10_8', 'Vth_lin_V', 'mu_FE_cm2Vs', 'Ion_A_per_um', 'V_1e-7', 'V_1e-11']} for t in FILMS}
93
94 # mu_FE and Ion thickness ratios at 358 K
95 for var in ['P-phonon', 'P-MTR']:
96     r300 = M[(13.2, 300.0, 'ref')]['mu_FE_cm2Vs'] / M[(2.0, 300.0, 'ref')]['mu_FE_cm2Vs']; r358 = M[(13.2, 358.15, var)]['mu_FE_cm2Vs'] / M[(2.0, 358.15, var)]['mu_FE_cm2Vs']
97     i300 = M[(13.2, 300.0, 'ref')]['Ion_A_per_um'] / M[(2.0, 300.0, 'ref')]['Ion_A_per_um']; i358 = M[(13.2, 358.15, var)]['Ion_A_per_um'] / M[(2.0, 358.15, var)]['Ion_A_per_um']
98     print(f'{var}: mu_FE(13.2)/mu_FE(2) 300 K {r300:.4f} -> 358 K {r358:.4f} ({100*(r358/r300-1):.2f} %) ; Ion ratio {i300:.4f} -> {i358:.4f}')
99     out[f'ratio_T_{var}'] = dict(mu300=r300, mu358=r358, ion300=i300, ion358=i358)
100     C(13.2, 358.15, var, '-', 0.7, 'mu_FE_ratio_13p2_over_2p0', r358, '-', 'run_0047/run_0045'); C(13.2, 300.0, 'reference_300K', '-', 0.7, 'mu_FE_ratio_13p2_over_2p0', r300, '-', 'run_0040/run_0038')
101
102 # ----- activation energies
103 print('\n=== Apparent activation energy Ea(Vg) = -k dln(Id)/d(1/T) (meV), Vd 0.7 V ===')
104 ea = {}
105 def idat(key, v):
106     vg, i, _ = curves[key]; return float(i[np.argmin(np.abs(vg - v))])
107 for t in FILMS:
108     Ts = [300.0, 338.15, 358.15] if t == 2.0 else [300.0, 358.15]
109     for var in ['P-phonon', 'P-MTR']:
110         for v in VG_EA:
111             I = [idat((t, 300.0, 'ref'), v)] + [idat((t, T, var), v) for T in Ts[1:]]
112             if min(I) <= 1e-16: ea[(t, var, v)] = None; continue
113             x = np.array([1 / (K_B * T) for T in Ts]); y = np.log(np.array(I))
114             e_all = -np.polyfit(x, y, 1)[0] * 1000
115             e_pair = [-(y[k + 1] - y[k]) / (x[k + 1] - x[k]) * 1000 for k in range(len(Ts) - 1)]
116             ea[(t, var, v)] = dict(Ea_meV=float(e_all), pairs=[float(e) for e in e_pair], I=I, measurable=bool(I[0] > 5 * floor_med(t)))
117             rid = '+'.join([curves[(t, 300.0, 'ref')][2]] + [curves[(t, T, var)][2] for T in Ts[1:]])
118             C(t, '/'.join(f'{T:g}' for T in Ts), var, v, 0.7, 'Ea_apparent', float(e_all), 'meV', rid)
119             print(f'{t}>4 nm {var:8s}: ' + '; '.join(f'Vg {v}: {ea[(t,var,v)]["Ea_meV"]:.1f}' + (f' (pairs {", ".join(f'{e:.1f}' for e in ea[(t,var,v)]["pairs"])}' if t == 2.0 else '') + (' if ea[(t,var,v)]["measurable"] else ' [below 5x measured floor]') if ea[(t, var, v)] else f'Vg {v}: n/a' for v in VG_EA))
120
121 out['ea'] = {f'{k[0]}_{k[1]}_{k[2]}': v for k, v in ea.items()}
122
123 # ----- comparison with S2 analytic (350 K)
124 print('\n=== ATLAS P-MTR interpolated to 350 K vs S2 analytic MTR expectation ===')
125 s2 = {2.0: dict(dVcc=-64, mu=0.993, ion=1.004, dVlin=-15, ea={0.0: 504, 0.5: 120, 1.0: 56, 1.5: 23, 2.0: 7, 3.0: 1}),
126       13.2: dict(dVcc=-66, mu=0.999, ion=1.011, dVlin=-24, ea={0.0: 156, 0.5: 61, 1.0: 20, 1.5: 8, 2.0: 4, 3.0: 2})}

```

```

125 cmp = {}
126 for t in [2.0, 13.2]:
127     if t == 2.0: Ta, Tb = 338.15, 358.15; A = temp[(t, Ta, 'P-MTR')]; B = temp[(t, Tb, 'P-MTR')]
128     else: Ta, Tb = 300.0, 358.15; A = dict(dVth_cc_mV=0.0, mu_FE_ratio=1.0, dIon_pct=0.0, dVth_lin_mV
129         =0.0); B = temp[(t, Tb, 'P-MTR')]
129     w = (350.0 - Ta) / (Tb - Ta); li = lambda k: A[k] + w * (B[k] - A[k])
130     c = dict(dVcc=li('dVth_cc_mV'), mu=li('mu_FE_ratio'), ion=1 + li('dIon_pct') / 100, dVlin=li('
131         dVth_lin_mV'))
131     cmp[t] = c
132     print(f'{t} nm @350 K (linear-in-T interpolation {Ta}-{Tb}): dVth_cc {c["dVcc"]:.1f} mV (S2 {s2[t]["
133         dVcc"]}); mu_FE x{c["mu"]:.4f} (S2 x{s2[t]["mu"]}); Ion x{c["ion"]:.4f} (S2 x{s2[t]["ion"]});
134         dVth_lin {c["dVlin"]:.1f} mV (S2 {s2[t]["dVlin"]})')
135     print(' Ea (ATLAS P-MTR, 300-358 K fit) vs S2 (300-350 K): ' + ' '.join(f'Vg {v}: {ea[(t, "P-MTR", v
136         )]["Ea_meV"]:.0f} vs {s2[t]["ea"][v]}' for v in VG_EA if ea[(t, 'P-MTR', v)]))
137 out['s2_compare'] = cmp
138 json.dump(out, open(Path(__file__).with_name('s6_part3_out.json'), 'w'), indent=1, default=float)
139 json.dump(canon, open(Path(__file__).with_name('s6_canonical_rows.json'), 'w'), indent=0, default=float)
140 print('canonical rows:', len(canon))

```

B.3.23 S6_s6_write_register.py

The script writes PREDICTIONS_REGISTER.md and predictions_canonical.csv from the outputs of part 3, together with hashes; it makes no simulator call and modifies no package file (Chapter D).

```

1 """Writes analysis_2026-09-25/PREDICTIONS_REGISTER.md and predictions_canonical.csv from the part-3
2     outputs
3 (s6_part3_out.json, s6_canonical_rows.json). No simulator call; no package file is modified."""
4 import csv, hashlib, json, math
5 from datetime import datetime, timezone
6 from pathlib import Path
7 import numpy as np
8 from s6_lib import PKG, run_dir, meas, fmt
9
10 HERE = Path(__file__).resolve().parent; AN = HERE.parents[1] if HERE.name == 'S6' else None
11 AN = PKG / 'analysis_2026-09-25'
12 o = json.loads((HERE / 's6_part3_out.json').read_text()); canon = json.loads((HERE / 's6_canonical_rows.
13     json').read_text())
14 FILMS = [2.0, 6.3, 13.2]; VGS = [1.0, 1.5, 2.0, 2.5, 3.0]; VDS = ['0.05', '0.1', '0.5', '1.0', '2.0', '
15     3.0']
16 IDVD = {2.0: 'run_0041', 6.3: 'run_0042', 13.2: 'run_0043'}
17 VLIN300 = {t: o['ref'][f'{t}']['Vth_lin_V'] for t in FILMS}
18
19 # normalised output curves Id(Vd)/Id(Vd=0.7) (removes the 300 K transfer-calibration residual) ->
20     canonical rows
21 for t in FILMS:
22     rows = list(csv.DictReader((run_dir(IDVD[t]) / 'output_characteristics.csv').open()))
23     for vg in VGS:
24         pts = {round(float(r['vd_V']), 3): float(r['id_A_per_um']) for r in rows if abs(float(r['vg_V'])
25             - vg) < 1e-9}
26         for vd, i in sorted(pts.items()):
27             if vd > 0: canon.append(dict(film=f'{t:g} nm', T_K=300.0, variant='as_simulated_300K', Vg=vg,
28                 Vd=vd, quantity='Id_over_Id_at_Vd0p7', value=i / pts[0.7], unit='-', run_id=IDVD[t]))
29 # calibration residual at Vd = 0.7 V (model / measured transfer curve at the same Vg) - context, not a
30     prediction
31 calres = {}
32 for t in FILMS:
33     vg, i = meas(t)
34     for v in VGS:
35         im = float(i[np.argmax(np.abs(vg - v))]); calres[(t, v)] = o['idvd'][f'{t}_{v}']['Id07'] / im

```

```

29
30 csv_path = AN / 'predictions_canonical.csv'
31 with csv_path.open('w', newline='') as f:
32     w = csv.DictWriter(f, fieldnames=['film', 'T_K', 'variant', 'Vg', 'Vd', 'quantity', 'value', 'unit',
33         'run_id']); w.writeheader()
34     for r in canon:
35         r = dict(r); v = r['value']
36         if isinstance(v, float): r['value'] = f'{v:.6g}'
37         w.writerow(r)
38
39 def sha(p):
40     return hashlib.sha256(Path(p).read_bytes()).hexdigest()
41
42 files = ['data/experimental_clean.csv', 'scripts/extract_metrics.py', 'config/iwo_material_model.yaml', '
43     config/campaign_B_recal_predictions.json', 'results/campaigns/campaign_B_recal_predictions.json']
44 for rid in ['run_0038', 'run_0039', 'run_0040']:
45     d = run_dir(rid).relative_to(PKG).as_posix(); files += [f'{d}/idvg.dat', f'{d}/execution.json', f'{d}
46         /device.in']
47 for rid in ['run_0041', 'run_0042', 'run_0043']:
48     d = run_dir(rid).relative_to(PKG).as_posix(); files += [f'{d}/output_characteristics.csv', f'{d}/
49         execution.json', f'{d}/device.in']
50 for rid in ['run_0044', 'run_0045', 'run_0046', 'run_0047']:
51     d = run_dir(rid).relative_to(PKG).as_posix(); files += [f'{d}/idvg.dat', f'{d}/execution.json', f'{d}
52         /device.in', f'{d}/deckbuild.out']
53 files += ['analysis_2026-09-25/scratch/S6/s6_lib.py', 'analysis_2026-09-25/scratch/S6/
54     s6_part3_predictions.py', 'analysis_2026-09-25/scratch/S6/s6_write_register.py',
55     'analysis_2026-09-25/scratch/S6/s6_part3_out.txt', 'analysis_2026-09-25/predictions_canonical.
56     csv']
57 hashes = [(p, sha(PKG / p)) for p in files]
58 now = datetime.now(timezone.utc).strftime('%Y-%m-%dT%H:%M:%SZ')
59
60 L = []; A = L.append
61 A('# Prediction register: IWO V1 ATLAS model, ID-VD and elevated-temperature transfer predictions')
62 A('')
63 A(f'***Registered (UTC): {now}.** Registrant: specialist S6, analysis_2026-09-25 (no ATLAS launch by S6;
64     every number below is read from the real ATLAS runs listed, or derived from them by the exact
65     operations stated). ')
66
67 'None of the predicted quantities was used in calibration, and no measurement of them was in the
68 project\'s possession at registration time. Januar et al. 2026 SI Fig. S12 (2 nm transfer curves at
69 65 and 85 C) exists but is NOT in the package; it is the first intended test.')
70
71 A('')
72 A('## 1. Model state that is being registered')
73 A('- Model V1 (config/iwo_material_model.yaml), physical structure, classical electrons-only drift-
74     diffusion with Fermi statistics, acceptor tail NTA/NTA (NTA 0.040 eV), deep and interface Gaussians,
75     fixed front charge Qf, uniform Nd_eff(t), confinement law dEc = 0.9205 t^-1.3815 eV, Nc(m*(t)),
76     constant spatially uniform band mobility, ideal Ohmic Pd contacts, DOS levels 384/192.')
77 A('- Calibrated (300 K, Vd 0.7 V, one ID-VG per film): mu_band = 17.6897 (2 nm), 11.582 (6.3 nm), 61.6046
78     cm^2/Vs (13.2 nm); Qf = 1.73e12 cm^-2 (2 and 13.2 nm, shared) and 8.7e10 cm^-2 (6.3 nm, device-
79     specific "tuned" configuration).')
80 A('- 300 K references: run_0038 (B1) x 17.6897/17.69 = x0.999983, run_0039 (B2) x 11.582/11.41 = x1
81     .015074, run_0040 (B3) x 61.6046/60.39 = x1.020113 (exact: Id is linear in the uniform constant
82     mobility; verified run_0029/run_0038 ratio 1.024879 vs 1.024873 expected over 43 points; ID-VD at Vd
83     0.7 V equals the rescaled transfer curves to <0.001 % at all 15 (film, Vg) points).')
84 A('- Temperature assumptions (campaign B plan): Nc, Nv proportional to T^1.5 (ATLAS); Eg T-independent (
85     egalpha = 0); NTA/NTA, Dit, deep states, Qf, Nd_eff T-independent; tail occupancy at the lattice
86     temperature (thermal equilibrium, isothermal, no self-heating).')
87 A('- Known calibration residuals the predictions inherit (final converged comparison, S6 report section
88     1): 2 nm gm_max -8.6 %, Vth_lin -0.127 V; 6.3 nm gm_max -23 %, Vth_lin -0.354 V (on-state shape NOT
89     reproduced; the 6.3 nm predictions are lower-confidence); 13.2 nm gm_max -2.6 %.)')
90
91 A('')
92 A('## 2. Temperature caveat (tmu) and the two registered variants')
93 A('ATLAS applied its silicon-default constant-mobility temperature exponent (printed "tmu = 1.5" in `

```

```

run_0045/deckbuild.out`, REGIONAL MOBILITY MODEL SUMMARY, lines ~384-423: "mu = 12.977 @ 358 K, tmu
= 1.5"), i.e. mu_band(T) = mu_band(300 K)(T/300)^-1.5, although the plan declared a T-independent
band mobility. Because the mobility is uniform and constant, the current is exactly proportional to
that factor, so two variants are exact:')
67 A('- **P-phonon** = as simulated (band mobility falls as T^-1.5).')
68 A('- **P-MTR** = simulated currents x (T/300)^+1.5 = x1.19665 (338.15 K) and x1.30426 (358.15 K): T-
independent band mobility, multiple-trapping-and-release through the equilibrium tail only.')
69 A('The as-simulated runs must never be quoted as "T-independent mobility". A measured result between the
variants is read through an effective exponent: mu_band proportional to T^-g with g = -ln(R_meas/
R_MTR)/ln(T/300), where R is the measured and P-MTR-predicted ratio of the same current quantity (
Ion or mu_FE) at T vs 300 K.')
```

70 A('')

71 A('## 3. Extraction definitions (apply identically to future measured curves)')

72 A('`scripts/extract\_metrics.py` (unchanged) for Vth_cc (Id = 1e-9 A/um, log-linear), Vth_lin (secant gm
on the 0.05 V grid, linear extrapolation at gm_max), mu_FE = gm_max L/(W Cox Vd) (L 20 um, Cox 8.955
e-7 F/cm^2, Vd 0.7 V), Ion = Id(3 V). Simulated curves are first interpolated (np.interp) from the
native grid (0.05 V for -1..1.5 V, 0.1 V above) onto the 0.05 V measurement grid, as in `scripts/
run_atlas.py`. Added fixed-current metrics (S6): V(I) = Vg at Id = I (log-linear), SS(I1..I2) = [V(
I2)-V(I1)]/log10(I2/I1) for 1e-11..1e-10, 1e-10..1e-9 and 1e-10..1e-8 A/um. SS_min is NOT registered
(window-phase artefact, S6 report 1.4). Ea(Vg) = -k_B dln(Id)/d(1/T) at fixed Vg and Vd 0.7 V (
least squares over the available temperatures). **The primary registered quantities are the changes
(dVth, dSS, ratios, Ea), to be compared with changes measured on the same device relative to its own
300 K curve**; absolute values carry the 300 K calibration residual.')

73 A('')

74 A('## 4. ID-VD predictions (300 K; runs B8 run_0041, B9 run_0042, B10 run_0043; Vd 0-0.5 V in 0.05 V
steps, 0.5-3 V in 0.1 V steps)')

75 A('')

76 A('### 4.1 Drain current (A/um)')

77 A('| film | Vg (V) | Id @0.05 | Id @0.1 | Id @0.5 | Id @1 | Id @2 | Id @3 V | Id @0.7 (= calibrated
transfer) | model/measured at Vd 0.7 (calibration residual, context) |')

78 A('|---|---|---|---|---|---|---|---|---|---|')

79 for t in FILMS:

80 for vg in VGS:

81 d = o['idvd'][f'{t}_{vg}']; I = d['Id']

82 A(f'| {t:g} nm | {vg:g} | ' + ' | '.join(f'{I[k]:.4g}' for k in VDS) + f' | {d["Id07"]:.4g} | {
calres[t, vg]:.3f} |')

83 A('')

84 A('### 4.2 Output-curve shape (the genuine prediction content: normalised to the calibrated Vd = 0.7 V
point)')

85 A('| film | Vg | Vg - Vth_lin(300 K model) (V) | Id(0.1)/Id(0.05) | d2Id/dVd2 at Vd->0 (A/um/V^2) | gd0 (
S/um, quadratic fit 0-0.1 V) | Vd_sat: gd = 10 % of gd0 (V) | gd at Vd 3 V (S/um) | gd(3 V)/gd0 | Id
(3)/Id(0.7) | Id(0.1)/Id(0.7) |')

86 A('|---|---|---|---|---|---|---|---|---|---|')

87 for t in FILMS:

88 for vg in VGS:

89 d = o['idvd'][f'{t}_{vg}']; I = d['Id']

90 A(f'| {t:g} nm | {vg:g} | {vg - VLIN300[t]:.2f} | {d["r_lin"]:.4f} | {2*d["curv_b"]:.3g} | {d["
gd0"]:.4g} | {fmt(d["vdsat10"], "{:.2f}")} | {d["gd3"]:.3g} | {100*d["gd3_over_gd0"]:.2f} % | {I
["3.0"]/d["Id07"]:.4f} | {I["0.1"]/d["Id07"]:.4f} |')

91 A('')

92 A('Gradual-channel reference: Id(0.1)/Id(0.05) = 2 x [1 - 0.05/Vov]/[1 - 0.025/Vov] < 2 for an Ohmic
contact (Vov = effective overdrive); every registered value is 1.80-1.98 with negative low-Vd
curvature (no S-shape). Output conductance at 3 V is 0.01-0.68 % of gd0 (L = 20 um, no channel-
length modulation, no parallel path in the model).')

93 A('')

94 A('### 4.3 Film ratios Id(t)/Id(2 nm) at equal Vg and Vd')

95 A('| ratio | Vg | Vd 0.05 | 0.1 | 0.5 | 1 | 2 | 3 V |')

96 A('|---|---|---|---|---|---|---|---|')

97 for t in [6.3, 13.2]:

98 for vg in VGS:

99 rr = o[f'ratio_{t}_{vg}']

```

100     A(f'| {t:g}/2 nm | {vg:g} | ' + ' | '.join(f'{rr[k]:.3f}' for k in VDS) + ' |')
101 A('')
102 A('## 5. Elevated-temperature transfer predictions (Vd 0.7 V; B11 run_0044 338.15 K 2 nm; B12 run_0045,
    B13 run_0046, B14 run_0047 at 358.15 K)')
103 A('')
104 A('### 5.1 300 K references (rescaled B1/B2/B3)')
105 A('| film | Vth_cc (V) | V(1e-11) (V) | V(1e-7) (V) | SS 1e-11..1e-10 | SS 1e-10..1e-9 | SS 1e-10..1e-8 (
    mV/dec) | Vth_lin (V) | mu_FE (cm^2/Vs) | Ion (A/um) |')
106 A('|---|---|---|---|---|---|---|---|---|')
107 for t in FILMS:
108     r = o['ref'][f'{t}']
109     A(f'| {t:g} nm | {r["V_1e-9"]:.4f} | {r["V_1e-11"]:.4f} | {r["V_1e-7"]:.4f} | {r["SS_11_10"]:.1f} | {
        r["SS_10_9"]:.1f} | {r["SS_10_8"]:.1f} | {r["Vth_lin_V"]:.4f} | {r["mu_FE_cm2Vs"]:.3f} | {r["
        Ion_A_per_um"]:.4g} |')
110 A('')
111 A('### 5.2 Changes relative to 300 K')
112 A('| film | T (K) | variant | Vth_cc (V) | dVth_cc (mV) | dV(1e-11) (mV) | dV(1e-7) (mV) | SS 1e-11..1e
    -10 (d) | SS 1e-10..1e-9 (d) | SS 1e-10..1e-8 (d) | dVth_lin (mV) | mu_FE (ratio) | Ion (A/um) |
    dIon (%) |')
113 A('|---|---|---|---|---|---|---|---|---|---|---|---|')
114 for key in ['2.0_338.15', '2.0_358.15', '6.3_358.15', '13.2_358.15']:
115     for var in ['P-phonon', 'P-MTR']:
116         d = o['temp'][f'{key}_{var}']; t, T = key.split('_')
117         A(f'| {float(t):g} nm | {T} | {var} | {d["Vth_cc"]:.4f} | {d["dVth_cc_mV"]:.1f} | {d["dV1e11_mV"]
            ":.1f} | {d["dV1e7_mV"]:.1f} | {d["SS_11_10"]:.1f} ({d["dSS_11_10"]:.1f}) | {d["SS_10_9"]:.1f}
            ({d["dSS_10_9"]:.1f}) | {d["SS_10_8"]:.1f} ({d["dSS_10_8"]:.1f}) | {d["dVth_lin_mV"]:.1f} | {d["
            mu_FE"]:.3f} (x{d["mu_FE_ratio"]:.4f}) | {d["Ion"]:.4g} | {d["dIon_pct"]:.2f} |')
118 A('')
119 rp, rm = o['ratio_T_P-phonon'], o['ratio_T_P-MTR']
120 A(f'Thickness ratio mu_FE(13.2)/mu_FE(2): {rp["mu300"]:.3f} (300 K) -> {rm["mu358"]:.3f} (358.15 K) in
    both variants (+{100*(rm["mu358"]/rm["mu300"]-1):.1f} %); Ion(13.2)/Ion(2): {rp["ion300"]:.3f} -> {
    rp["ion358"]:.3f}. The tmu factor is common to all films, so the ratios are variant-independent.')
121 A('')
122 A('### 5.3 Apparent activation energy of Id at fixed Vg (meV; Vd 0.7 V; 2 nm fit over 300/338.15/358.15 K
    with the two pair values in brackets, 6.3 and 13.2 nm over 300/358.15 K)')
123 A('| film | variant | Vg 0 | 0.5 | 1.0 | 1.5 | 2.0 | 3.0 V |')
124 A('|---|---|---|---|---|---|---|---|')
125 for t in FILMS:
126     for var in ['P-phonon', 'P-MTR']:
127         cells = []
128         for v in [0.0, 0.5, 1.0, 1.5, 2.0, 3.0]:
129             e = o['ea'][f'{t}_{var}_{v}']
130             if not e: cells.append('n/a (Id < 1e-16)'); continue
131             s = f'{e["Ea_meV"]:.1f} + (f' ({e["pairs"][0]:.1f}, {e["pairs"][1]:.1f})' if t == 2.0
                else '') + (' if e["measurable"] else ' *)'
132             cells.append(s)
133             A(f'| {t:g} nm | {var} | ' + ' | '.join(cells) + ' |')
134 A('')
135 A('\* below 5x the measured 300 K off-floor of that film: not measurable on the existing devices. The P-
    MTR minus P-phonon difference is the tmu term, +1.5 k_B T_eff = +42.3 meV for 300-358.15 K, at every
    Vg.')
136 A('')
137 A('### 5.4 Cross-check against the pre-registered analytic MTR expectation (S2 section 4, 1-D surrogate,
    300 -> 350 K, T-independent mu)')
138 s2 = {2.0: dict(dVcc=-64, mu=0.993, ion=1.004, dVlin=-15), 13.2: dict(dVcc=-66, mu=0.999, ion=1.011,
    dVlin=-24)}
139 A('| film | quantity at 350 K | S2 surrogate (registered 2026-09-24T18:57Z) | ATLAS P-MTR (interpolated
    linearly in T) | difference |')
140 A('|---|---|---|---|---|')
141 for t in [2.0, 13.2]:
142     c = o['s2_compare'][f'{t}']

```

```

143     A(f' | {t:g} nm | dVth_cc (mV) | {s2[t]["dVcc"]} | {c["dVcc"]:+.1f} | {c["dVcc"]-s2[t]["dVcc"]:+.1f} |
      ')
144     A(f' | {t:g} nm | mu_FE ratio | {s2[t]["mu"]} | {c["mu"]:.4f} | {c["mu"]-s2[t]["mu"]:+.4f} |')
145     A(f' | {t:g} nm | Ion ratio | {s2[t]["ion"]} | {c["ion"]:.4f} | {c["ion"]-s2[t]["ion"]:+.4f} |')
146     A(f' | {t:g} nm | dVth_lin (mV) | {s2[t]["dVlin"]} | {c["dVlin"]:+.1f} | {c["dVlin"]-s2[t]["dVlin"]
      "]:+.1f} |')
147     A('Ea (P-MTR, 300-358 K fit) vs S2 (300-350 K): 2 nm 124/57/23/8/2 vs 120/56/23/7/1 meV at Vg
      0.5/1/1.5/2/3 V; 13.2 nm 163/63/21/8/5/2 vs 156/61/20/8/4/2 meV at Vg 0/0.5/1/1.5/2/3 V. This is a
      model-vs-surrogate consistency test (numerical), not an experimental validation.')
148     A('')
149     A('## 6. What each outcome would falsify')
150     A('| registered prediction | model value(s) | measured outcome that falsifies | assumption falsified |')
151     A('|---|---|---|---|')
152     A('| Low-Vd linearity, all films and Vg | Id(0.1)/Id(0.05) = 1.80-1.98; d2Id/dVd2 < 0 at Vd -> 0 | ratio
      > 2.0 or upward (S-shaped) curvature below ~0.2 V, in particular at 2 nm only | ideal Ohmic Pd/IWO
      injection (S5 M2: confinement-raised barrier); also the mu_band = intrinsic lumping |')
153     n01 = ' / '.join(f'{o["idvd"]}[f"{t}_3.0"] Id["0.1"] / o["idvd"]}[f"{t}_3.0"] Id["07"]:.3f}' for t in
      FILMS)
154     A(f'| Normalised output Id(0.1)/Id(0.7) at Vg 3 V | {n01} (2 / 6.3 / 13.2 nm) | measured value lower by
      more than the repeatability, growing with Id | negligible Rsd (series resistance compresses the low-
      Vd end); thickness ordering of the deficit identifies which film carries Rc |')
155     A('| Saturation | Vd_sat(10 %) 0.43-1.80 V (2 nm), 0.44-1.90 (6.3), 0.68-2.30 (13.2) for Vg 1-3 V | quasi
      -saturation well below these (> ~0.3 V lower) | constant mobility and zero Rsd (field-dependent
      mobility or Rsd) |')
156     A('| Output conductance at Vd 3 V | 0.01-0.19 % of gd0 (2, 6.3 nm), 0.03-0.68 % (13.2 nm) | several % of
      gd0, or rising with t | no parallel/ungated path, long-channel electrostatics (S4 floor path; back-
      channel conduction) |')
157     A('| Film ratios at low Vd | Id(13.2)/Id(2) at Vd 0.1 V = 26.5 / 10.8 / 7.44 / 6.29 / 5.75 (Vg 1-3 V) |
      low-Vd ratio differing from the Vd 0.7 V ratio beyond the predicted Vd dependence | thickness-
      independent (zero) contact resistance |')
158     A('| On-current vs T (2 nm, 358 K) | P-MTR +0.99 %, P-phonon -22.6 % (mu_FE x0.992 / x0.760) | dIon > ~+5
      % (g < -0.2) | T-independent or phonon-like band mobility: activated band mobility / percolation (
      S2 4b predicts mu x1.03-1.39) |')
159     A('| On-current vs T | as above | dIon near -22 % (g ~ +1.5) | T-independent band mobility (band-like
      phonon-limited transport instead) |')
160     A('| Thickness ratio mu_FE(13.2)/mu_FE(2) | 4.40 -> 4.44 at 358 K (both variants) | shrinks toward ~3.3 (
      or 2.7-3.3) | common band-transport mechanism in all films (S2: barrier-limited thin films, phonon-
      limited 13.2 nm) |')
161     A('| dVth_cc at 358 K | P-MTR -73 / -79 / -76 mV; P-phonon -41 / -47 / -57 mV (2 / 6.3 / 13.2 nm);
      dVth_lin -27 / -35 / -30 mV in both | at 358 K, outside the -35 ... -85 mV envelope of the two
      variants beyond the (NOT DETERMINED, to be measured) repeatability, or a thickness spread >> 20 mV |
      equilibrium tail filling as the only T-dependent charge (e.g. thermally ionized V_0 donors, T-
      dependent Qf, hysteresis) |')
162     A('| dVth_cc at 338 K (2 nm) | P-MTR -49 mV; P-phonon -28 mV | as above | as above |')
163     A('| Subthreshold Ea at fixed Vg | P-MTR 124 / 128 / 63 meV at 0.5 V; 57 / 53 / 21 meV at 1 V (2 / 6.3 /
      13.2 nm) | ordering Ea(2) ~ Ea(6.3) > Ea(13.2) violated beyond ~10 meV | tail/EF alignment (Vth)
      ordering of the model; a different EF-Ec at the 6.3 nm film |')
164     A('| On-state Ea (Vg 3 V) | P-MTR +1.5 / +3.5 / +2.4 meV; P-phonon -40.6 / -38.8 / -39.9 meV | thin-film
      Ea(3 V) exceeding the 13.2 nm value by > ~10 meV | thickness-independent transport mechanism (
      structural/percolation step, S2) |')
165     A('| SS at fixed current, 358 K | P-MTR: SS 1e-10..1e-9 -12.1 / -12.5 / -10.6 mV/dec; P-phonon +2.0 /
      +2.4 / -3.3 | increase > ~20 mV/dec | T-independent tail (WTA) and Dit; thermal-equilibrium trapping
      |')
166     A('')
167     A('## 7. Numerical uncertainty of the registered numbers')
168     A('- Id, Ion, mu_FE ratios: DOS residual beyond 384/192 +0.16 % at 2 nm (Richardson, observed order 2.0
      from runs 0012/0019/0029), smaller for 6.3/13.2 nm (offsets scale ~Nt); mesh x0.7 <= 0.013 % (runs
      0036, 0028); linear rescale exact to 1e-5. DOS convergence was verified at 300 K only (not at
      338/358 K).')
169     A('- Vth_cc, V(I): +/-1 mV (DOS residual < 0.5 mV, mesh 0.3-0.4 mV, log-linear vs PCHIP interpolation <=
      1.5 mV). Fixed-current SS: +/-1 mV/dec. Vth_lin: grid-limited, -5 to -12 mV bias of the simulation

```



```
    grid relative to a 0.05 V measurement (0.1 V solver steps above 1.5 V); gm_max: -0.4 to -0.7 %.')
170 A('- Ea: < 0.5 meV (numerical); ID-VD: Vd_sat +/-0.05 V (0.1 V grid); gd(3 V) at Vg 1 V is set by current
    differences of ~5e-13 A/um per 0.1 V and carries up to ~5 % numerical uncertainty (KCL residual up
    to 1e-14 A in runs 0041/0042).')
171 A('- Physical/model-form uncertainty is NOT included in these numbers (S6 report sections 3-4).')
172 A('')
173 A('## 8. Run IDs')
174 A('300 K references: run_0038 (B1, 2 nm), run_0039 (B2, 6.3 nm tuned), run_0040 (B3, 13.2 nm). ID-VD:
    run_0041 (B8), run_0042 (B9), run_0043 (B10). Elevated T: run_0044 (B11, 2 nm, 338.15 K), run_0045 (
    B12, 2 nm, 358.15 K), run_0046 (B13, 6.3 nm, 358.15 K), run_0047 (B14, 13.2 nm, 358.15 K). Linearity
    check: run_0029 vs run_0038. All launches are logged in `results/RUN_INDEX.csv` (finish rows
    2026-09-24T19:28:46Z to 21:54:12Z).')
175 A('')
176 A('## 9. Canonical machine-readable file')
177 A(f'`analysis_2026-09-25/predictions_canonical.csv` ({len(canon)} rows; columns film, T_K, variant, Vg,
    Vd, quantity, value, unit, run_id). Variants: as_simulated_300K (ID-VD), reference_300K, P-phonon, P
    -MTR. It holds every ID-VD point (Vd > 0), the normalised output curves, the derived output metrics,
    the full predicted transfer curves (Vg >= -1 V, Id > 1e-16 A/um) for every film, temperature and
    variant, the metric changes and the Ea values.')
178 A('')
179 A('## 10. Source files and SHA-256 (computed with Python hashlib at registration)')
180 A('| file (relative to the package root) | SHA-256 |')
181 A('|---|---|')
182 for p, h in hashes: A(f'| {p} | {h} |')
183 A('')
184 A('Any later change to these files, to the extraction definitions or to the model state invalidates the
    prospective status of this register; a revised register must be a new, separately timestamped file.'
    )
185 (AN / 'PREDICTIONS_REGISTER.md').write_text('\n'.join(L) + '\n', encoding='utf-8')
186 print('written', now, len(canon), 'rows'); print('\n'.join(f'{p} {h}' for p, h in hashes))
```

B.3.24 REVIEW_review_verify.py

The script contains the referee's re-computation of the load-bearing specialist claims of REVIEW.md. It reads only raw files (measured data, run outputs and deckbuild.out) and imports scripts/extract_metrics.py unchanged; no ATLAS launch is made.

```
1 #!/usr/bin/env python3
2 """Referee re-computation of load-bearing specialist claims (REVIEW.md).
3 Reads only raw files (data/experimental_clean.csv, results/runs/*/comparison.csv,idvg.dat,execution.json
  ,metadata.json},
4 deckbuild.out). Uses scripts/extract_metrics.py unchanged (imported). No ATLAS launch, no package file
  modified."""
5 import sys, json, math, csv, glob
6 from pathlib import Path
7 import numpy as np
8
9 ROOT = Path(__file__).resolve().parents[3]
10 sys.path.insert(0, str(ROOT / 'scripts'))
11 import extract_metrics as em # unchanged package extractor
12
13 RUNS = ROOT / 'results' / 'runs'
14 Q = 1.602176634e-19
15 COX = em.COX
16 qC = Q / COX * 1e12 # V per 1e12 cm^-2
17 out = {}
18
19 def rdir(rid):
20     return Path(glob.glob(str(RUNS / f'{rid}_*'))[0])
```

```

21
22 def exe(rid):
23     return json.loads((rdir(rid) / 'execution.json').read_text())
24
25 def meta(rid):
26     return json.loads((rdir(rid) / 'metadata.json').read_text())
27
28 def comp(rid):
29     rows = list(csv.DictReader((rdir(rid) / 'comparison.csv').open()))
30     return (np.array([float(r['vg_V']) for r in rows]), np.array([float(r['measured_A_per_um']) for r in
31         rows])),
32         np.array([float(r['atlas_A_per_um']) for r in rows]))
33
34 def native(rid, name='idvg.dat'):
35     lines = (rdir(rid) / name).read_text().split('\n')[4:]
36     xy = np.array([float(v) for v in l.split()] for l in lines if l.strip())
37     return xy[:, 0], xy[:, 1]
38
39 def meas(t):
40     return em.load_curve(ROOT / 'data' / 'experimental_clean.csv', t)
41
42 def V_at(vg, idv, I):
43     return em.crossing(vg, idv, I)
44
45 def p(*a):
46     print(*a)
47
48 # ----- 0. measured metrics
49 p('=== 0. measured metrics (extract_metrics.py unchanged)')
50 M = {}
51 for t in (2.0, 6.3, 13.2, 31.8):
52     vg, idv = meas(t); M[t] = em.metrics(vg, idv, False, t)
53     m = M[t]
54     p(t, {k: (float('%5g' % v) if isinstance(v, (float, np.floating)) else v) for k, v in m.items() if k
55         in ('Vth_cc_1e-9_V', 'Vth_lin_V', 'SS_mV_dec', 'SS_cc_1e-10_1e-8_mV_dec', 'gm_max_A_V_um', '
56         mu_FE_cm2Vs', 'Ion_A_per_um', 'off_band_median_A_per_um')})
57     gm = np.gradient(idv, vg); p('    gm argmax at Vg =', vg[int(np.argmax(gm))], ' gm(3V)/gm_max =',
58         round(gm[-1] / gm.max(), 4))
59
60 # ----- 1. run metrics recomputed from
61     comparison.csv
62 p('\n=== 1. simulated metrics recomputed from comparison.csv (atlas column, already np.interp onto the
63     0.05 V grid)')
64 RUNLIST = ['run_0001', 'run_0002', 'run_0003', 'run_0004', 'run_0005', 'run_0007', 'run_0012', 'run_0013',
65     'run_0014', 'run_0015', 'run_0016', 'run_0017',
66     'run_0019', 'run_0029', 'run_0030', 'run_0031', 'run_0032', 'run_0033', 'run_0034', 'run_0035',
67     'run_0036', 'run_0037', 'run_0038',
68     'run_0039', 'run_0040', 'run_0048', 'run_0049', 'run_0050', 'run_0051', 'run_0052']
69 S = {}
70 for rid in RUNLIST:
71     try:
72         ex = exe(rid); t = ex['thickness_nm']; vg, mm, pr = comp(rid)
73         m = em.metrics(vg, pr, True, t); S[rid] = dict(m, t=t, rmse=ex['fit_metrics']['regions']['active']
74             ['rmse_log10'], ion_err=ex['fit_metrics']['on_error_3V_percent'])
75         dj = ex['device_metrics_sim']
76         keys = [k for k in ('Vth_cc_1e-9_V', 'SS_cc_1e-10_1e-8_mV_dec', 'gm_max_A_V_um', 'Ion_A_per_um')
77             if k in dj]
78         same = all(abs((m[k] or 0) - (dj[k] or 0)) < 1e-9 * max(1, abs(dj[k] or 0)) for k in keys) if
79             keys else 'n/a (older execution.json schema)'
80         p(rid, t, 'Vcc %.4f Vlin %.4f SSmin %s SScc %.1f gm %.4e muFE %.3f Ion %.4e RMSE %.4f IonErr %+.2
81             f%% | matches execution.json: %s' % (

```

```

70         m['Vth_cc_1e-9_V'], m['Vth_lin_V'], (%.1f' % m['SS_mV_dec']) if m['SS_mV_dec'] else None, m[
    'SS_cc_1e-10_1e-8_mV_dec'] or float('nan'), m['gm_max_A_V_um'], m['mu_FE_cm2Vs'], m['Ion_A_per_um'],
    S[rid]['rmse'], S[rid]['ion_err'], same))
71     except Exception as e:
72         p(rid, 'ERROR', e)
73
74     def miss(rid):
75         return S[rid]['Vth_cc_1e-9_V'] - M[S[rid]['t']]['Vth_cc_1e-9_V']
76
77     p('\n--- V1: A6 (run_0034) vs run_0014 Vth_cc miss')
78     p('run_0014 miss %+.4f V ; run_0034 (A6) miss %+.4f V ; run_0015 miss %+.4f V' % (miss('run_0014'), miss(
    'run_0034'), miss('run_0015')))
79     p('gap (Vth_lin - Vth_cc): meas 6.3 %.4f ; run_0014 %.4f ; run_0015 %.4f ; run_0034 %.4f ; B0 %.4f ; C1
    %.4f' % (
80         M[6.3]['Vth_lin_V'] - M[6.3]['Vth_cc_1e-9_V'], *S[r]['Vth_lin_V'] - S[r]['Vth_cc_1e-9_V'] for r in (
    'run_0014', 'run_0015', 'run_0034', 'run_0037', 'run_0052'))))
81
82     p('\n--- V2: B0 run_0037')
83     b0 = S['run_0037']
84     p('B0 Vcc %.4f Vlin %.4f SS_cc %.2f SS_min %.2f muFE %.3f Ion err %+.2f%%; meas SS_cc %.2f SS_min %.2f' %
    (
85         b0['Vth_cc_1e-9_V'], b0['Vth_lin_V'], b0['SS_cc_1e-10_1e-8_mV_dec'], b0['SS_mV_dec'], b0['mu_FE_cm2Vs
    '], b0['ion_err'], M[6.3]['SS_cc_1e-10_1e-8_mV_dec'], M[6.3]['SS_mV_dec']))
86     p('B0 SS_cc - run_0015 SS_cc = %+.1f ; B0 SS_min - run_0015 SS_min = %+.1f' % (b0['SS_cc_1e-10_1e-8
    _mV_dec'] - S['run_0015']['SS_cc_1e-10_1e-8_mV_dec'], b0['SS_mV_dec'] - S['run_0015']['SS_mV_dec']))
87
88     # ----- 3. saturation-formula reproduction
89     p('\n=== V3: saturation-formula mobility mu_sat = 2L/(W Cox) (d sqrt(Id)/dVg)^2 with several derivative
    schemes')
90     K = 2 * em.L_CM / (em.W_CM * COX)
91     try:
92         from scipy.signal import savgol_filter
93     except Exception:
94         savgol_filter = None
95     musat = {}
96     for t in (2.0, 6.3, 13.2):
97         vg, idv = meas(t); s = np.sqrt(idv); h = vg[1] - vg[0]
98         sch = {}
99         sch['central np.gradient(sqrt Id)'] = np.gradient(s, vg)
100        sch['forward diff'] = np.append(np.diff(s) / h, np.nan)
101        sch['backward diff'] = np.insert(np.diff(s) / h, 0, np.nan)
102        sch['gm/(2 sqrt Id), central gm'] = np.gradient(idv, vg) / (2 * s)
103        if savgol_filter is not None:
104            for w in (5, 7, 9):
105                sch[f'Savitzky-Golay w{w} o2'] = savgol_filter(s, w, 2, deriv=1, delta=h)
106        res = {}
107        for k, d in sch.items():
108            v = K * d ** 2; v = np.where(np.isfinite(v), v, -1); i = int(np.argmax(v)); res[k] = (float(v[i])
    , float(vg[i]))
109        musat[t] = res
110        p(t, ' '.join(f'[{k}: {v[0]:.2f} @ {v[1]:.2f} V]' for k, v in res.items()))
111        vcc, vlin = M[t]['Vth_cc_1e-9_V'], M[t]['Vth_lin_V']; vp = res['central np.gradient(sqrt Id)'][1]
112        p('central-scheme peak at %.2f V: Vov vs Vth_cc = %.2f V, vs Vth_lin = %.2f V (Vd 0.7 V); mu_sat/
    mu_FE(lin) = %.3f' % (vp, vp - vcc, vp - vlin, res['central np.gradient(sqrt Id)'][0] / M[t]['
    mu_FE_cm2Vs']))
113    p('ratio 13.2/2 (central) = %.3f ; paper 27.4/5.1 = %.3f' % (musat[13.2]['central np.gradient(sqrt Id)
    '][0] / musat[2.0]['central np.gradient(sqrt Id)'][0], 27.4 / 5.1))
114
115    # ----- 4. floors
116    p('\n=== V4: off-state floors')
117    fl = {}

```

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118 for t in (2.0, 6.3, 13.2):
119     vg, idv = meas(t)
120     for lo, hi in ((-2, -0.5), (-3, -0.5), (-3, -2)):
121         sel = (vg >= lo - 1e-9) & (vg <= hi + 1e-9); x = vg[sel]; y = idv[sel]
122         med = float(np.median(y)); sl = np.polyfit(x, np.log10(y), 1)[0]
123         dev = y / med
124         p(f'{t} [{lo},{hi}] n={sel.sum()} median {med:.3e} min {y.min():.3e} max {y.max():.3e} max/min {y
            .max()/y.min():.2f} '
125             f'p10/p90 of I/median {np.percentile(dev,10):.3f}/{np.percentile(dev,90):.3f} slope {sl:+.4f}
            dec/V; total current (x290 um) {med*290:.2e} A')
126         if (lo, hi) == (-2, -0.5): fl[t] = med
127         # half-window medians to test flatness vs Vg
128         a = np.median(idv[(vg >= -3) & (vg <= -1.75)]); b = np.median(idv[(vg >= -1.75) & (vg <= -0.5)])
129         p(f' median(-3..-1.75)/median(-1.75..-0.5) = {a/b:.3f}')
130 p('floor ratio 13.2/2 = %.1f (exponent %.3f); 6.3/2 = %.1f (exp %.3f); 13.2/6.3 = %.1f (exp %.3f)' % (
131     fl[13.2] / fl[2.0], math.log(fl[13.2] / fl[2.0]) / math.log(6.6), fl[6.3] / fl[2.0], math.log(fl[6.3]
        / fl[2.0]) / math.log(3.15), fl[13.2] / fl[6.3], math.log(fl[13.2] / fl[6.3]) / math.log(13.2 /
        6.3)))
132 for t in (2.0, 6.3, 13.2):
133     p(f' floor/Ion {t}: {fl[t]/M[t]["Ion_A_per_um"]:.2e}')
134
135 # ----- 5. q/Cox and rigid-shift equivalence
136 p('\n=== V5: q/Cox rigid shifts')
137 p('q/Cox = %.5f V per 1e12 cm^-2 ; dqf 1.73e12-8.7e10 -> %.4f V ; 1.73e12 -> %.4f V ; 1.73e12-4.76e10 ->
    %.4f V' % (qC, 1.643 * qC, 1.73 * qC, (1.73 - 0.0476) * qC))
138 dEc = lambda t: 0.9205 * t ** -1.3815
139 p('dEc law: 2 nm %.4f, 6.3 nm %.4f, 13.2 nm %.4f eV; dEc(2) as charge %.3e cm^-2' % (dEc(2), dEc(6.3),
    dEc(13.2), dEc(2) / qC * 1e12))
140 def hshift(ra, rb, currents):
141     va, ia = native(ra); vb, ib = native(rb); r = []
142     for I in currents:
143         a = V_at(va, ia, I); b = V_at(vb, ib, I); r.append(None if (a is None or b is None) else a - b)
144     return r
145 cur = [1e-14, 1e-13, 1e-12, 1e-11, 1e-10, 1e-9, 1e-8, 1e-7, 3e-7]
146 for ra, rb, lab in (('run_0012', 'run_0016', '2 nm QC on - off'), ('run_0013', 'run_0017', '13.2 nm QC on
    - off'), ('run_0003', 'run_0001', '2 nm Qf 1.73e12 - 0'),
147     ('run_0004', 'run_0002', '13.2 nm Qf 1.73e12 - 0'), ('run_0007', 'run_0005', '6.3 nm
    Qf 8.7e10 - 1.73e12'), ('run_0034', 'run_0014', '6.3 nm A6 - run_0014')):
148     try:
149         p(lab, ' '.join(f'{I:.0e}:{(d if d is None else round(d,4))}' for I, d in zip(cur, hshift(ra, rb,
            cur))))
150     except Exception as e:
151         p(lab, 'ERROR', e)
152 p('Vth_cc realised confinement: 2 nm (0012-0016) %.4f ; 13.2 nm (0013-0017) %.4f ; 6.3 nm (0014 - (0034 -
    Qf shift)) %.4f' % (
153     S['run_0012']['Vth_cc_1e-9_V'] - S['run_0016']['Vth_cc_1e-9_V'], S['run_0013']['Vth_cc_1e-9_V'] - S[
        'run_0017']['Vth_cc_1e-9_V'],
154     S['run_0014']['Vth_cc_1e-9_V'] - (S['run_0034']['Vth_cc_1e-9_V'] - (1.73 - 0.0476) * qC)))
155 p('SS_cc 0012 / 0016 / meas2: %.1f / %.1f / %.1f' % (S['run_0012']['SS_cc_1e-10_1e-8_mV_dec'], S[
    'run_0016']['SS_cc_1e-10_1e-8_mV_dec'], M[2.0]['SS_cc_1e-10_1e-8_mV_dec']))
156
157 # ----- 6. DOS offset
158 p('\n=== V6: DOS 96/48 -> 384/192 Ion offset at equal mu')
159 mu = lambda rid: meta(rid)['material']['mu_band']
160 for a, b in (('run_0012', 'run_0029'), ('run_0012', 'run_0038'), ('run_0015', 'run_0039'), ('run_0013', '
    run_0040')):
161     ia, ib = S[a]['Ion_A_per_um'], S[b]['Ion_A_per_um']; ma, mb = mu(a), mu(b)
162     p(f'{a} (mu {ma}) Ion {ia:.5e} ; {b} (mu {mb}) Ion {ib:.5e} ; offset at equal mu = {100*(ib*ma/mb/ia
        -1):+.3f} %')
163 i96, i192, i384 = S['run_0012']['Ion_A_per_um'], S['run_0019']['Ion_A_per_um'], S['run_0029']['
    Ion_A_per_um']

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164 p('mu 0012/0019/0029: %s %s %s' % (mu('run_0012'), mu('run_0019'), mu('run_0029')))
165 d1 = i192 / i96 - 1; d2 = i384 / i192 - 1; pr = math.log(d1 / d2) / math.log(2)
166 p(f'refinement 96->192 {100*d1:+.3f} %, 192->384 {100*d2:+.3f} %, observed order p = {pr:.2f}, Richardson
    residual beyond 384 = {100*d2/(2**pr-1):+.3f} %')
167 p('mu recal: 2 nm %.4f 6.3 nm %.4f 13.2 nm %.4f' % (18.13 * M[2.0]['Ion_A_per_um'] / (S['run_0029']['
    Ion_A_per_um']), 11.41 * M[6.3]['Ion_A_per_um'] / S['run_0039']['Ion_A_per_um'], 60.39 * M[13.2]['
    Ion_A_per_um'] / S['run_0040']['Ion_A_per_um']))
168 va, ia = native('run_0029'); vb, ib = native('run_0038'); sel = (ia > 1e-14) & (ib > 1e-14)
169 r = ia[sel] / ib[sel]; p(f'linearity run_0029/run_0038 over {sel.sum()} pts: mean {r.mean():.6f} min {r.
    min():.6f} max {r.max():.6f} ; mu ratio {mu("run_0029")/mu("run_0038"):.6f}')
170
171 # ----- 7. tmu
172 p('\n=== V7: tmu factor')
173 for T in (338.15, 358.15):
174     p(f'T {T}: (T/300)^-1.5 = {(T/300)**-1.5:.5f} ; (T/300)^+1.5 = {(T/300)**1.5:.5f}')
175     rough2 = (1 - 0.287 / 2) ** 2
176 p('2 nm mun at 300 K = mu_band 17.6897 x (1-0.287/2)^2 = %.4f (deckbuild prints "mu = 12.977 @ 358 K, tmu
    1.5")' % (17.6897 * rough2))
177 for rid, ref, T, t in (('run_0045', 'run_0038', 358.15, 2.0), ('run_0044', 'run_0038', 338.15, 2.0), ('
    run_0046', 'run_0039', 358.15, 6.3), ('run_0047', 'run_0040', 358.15, 13.2)):
178     vg, idv = native(rid); ion = idv[-1]; vr, ir = native(ref); ionr = ir[-1]
179     ex = exe(rid); mm = ex.get('device_metrics_sim', {})
180     p(f'{rid} ({T} K) Ion_native {ion:.4e} / {ref} {ionr:.4e} = {ion/ionr:.4f} ; x(T/300)^1.5 = {ion/ionr
        *(T/300)**1.5:.4f} ; mu_band run/ref {mu(rid)}/{mu(ref)}')
181
182 # ----- 8. S1 decomposition
183 p('\n=== V8: S1 "86 % of the 2->6.3 drop is confinement"')
184 dm = S['run_0012']['Vth_cc_1e-9_V'] - S['run_0014']['Vth_cc_1e-9_V']
185 p('ATLAS 2->6.3 model drop (0012-0014) %.4f V ; bare dEc difference %.4f eV -> %.1f %% ; realised (
    counterfactual) confinement difference %.4f V -> %.1f %%' % (
186     dm, dEc(2) - dEc(6.3), 100 * (dEc(2) - dEc(6.3)) / dm,
187     (S['run_0012']['Vth_cc_1e-9_V'] - S['run_0016']['Vth_cc_1e-9_V']) - (S['run_0014']['Vth_cc_1e-9_V'] -
        (S['run_0034']['Vth_cc_1e-9_V'] - (1.73 - 0.0476) * qc)),
188     100 * ((S['run_0012']['Vth_cc_1e-9_V'] - S['run_0016']['Vth_cc_1e-9_V']) - (S['run_0014']['Vth_cc_1e
        -9_V'] - (S['run_0034']['Vth_cc_1e-9_V'] - (1.73 - 0.0476) * qc))) / dm))
189 d213m = M[2.0]['Vth_cc_1e-9_V'] - M[13.2]['Vth_cc_1e-9_V']; qc213 = (S['run_0012']['Vth_cc_1e-9_V'] - S['
    run_0016']['Vth_cc_1e-9_V']) - (S['run_0013']['Vth_cc_1e-9_V'] - S['run_0017']['Vth_cc_1e-9_V'])
190 p('measured 2->13.2 Vth_cc step %.4f V ; model step (0012-0013) %.4f ; realised confinement share %.4f V
    = %.1f %% of the measured step' % (
191     d213m, S['run_0012']['Vth_cc_1e-9_V'] - S['run_0013']['Vth_cc_1e-9_V'], qc213, 100 * qc213 / d213m))
192 p('runs 0001/0002 offsets vs measured: 2 nm %+.4f, 13.2 nm %+.4f (diff %.4f V)' % (miss('run_0001'), miss
    ('run_0002'), miss('run_0001') - miss('run_0002'))))
193 p('runs 0003/0004 offsets vs measured: 2 nm %+.4f, 13.2 nm %+.4f' % (miss('run_0003'), miss('run_0004')))
194 for rid in ('run_0001', 'run_0002', 'run_0003', 'run_0004'):
195     mt = meta(rid)['material']; p(f' {rid}: Qf {mt.get("Qf_cm2")} mu_band {mt.get("mu_band")}
        full_structure {meta(rid).get("full_structure")}')
196
197 # ----- 9. SS_min
198 p('\n=== V9: SS_min window phase (2 nm)')
199 thr = 5 * fl[2.0]
200 for rid in ('run_0012', 'run_0038', 'run_0029'):
201     vg, mm, pr = comp(rid); i = int(np.argmax(abs(vg - 0.15)))
202     p(f'{rid}: Id(0.15 V) {pr[i]:.4e} vs 5xfloor {thr:.4e} -> window starts {"0.15" if pr[i] > thr else
        ">=0.20"} ; SS_min {S[rid]["SS_mV_dec"]:.2f}')
203 p('run_0013 / run_0017 SS_min %.1f / %.1f' % (S['run_0013']['SS_mV_dec'], S['run_0017']['SS_mV_dec']))
204
205 # ----- 10. fixed-current SS with / without
    floor subtraction
206 p('\n=== V10: fixed-current SS (measured), with and without subtracting the off-band median')
207 def ssfix(vg, idv, i1, i2):
208     a = V_at(vg, idv, i1); b = V_at(vg, idv, i2)

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209     return None if (a is None or b is None) else (b - a) / math.log10(i2 / i1) * 1000
210 for t in (2.0, 6.3, 13.2):
211     vg, idv = meas(t); sub = idv - fl[t]; sub = np.where(sub > 0, sub, np.nan)
212     # crossing() needs positive values; replace nan by tiny
213     sub2 = np.where(np.isfinite(sub), sub, 1e-30)
214     p(t, 'SS 1e-11..1e-10: raw %s sub %s ; 1e-10..1e-9: raw %s sub %s' % tuple(
215         (None if v is None else round(v, 1)) for v in (ssfix(vg, idv, 1e-11, 1e-10), ssfix(vg, sub2, 1e
216             -11, 1e-10), ssfix(vg, idv, 1e-10, 1e-9), ssfix(vg, sub2, 1e-10, 1e-9))))
217 for rid in ('run_0038', 'run_0039', 'run_0040'):
218     vg, ii = native(rid); p(rid, 'sim SS 1e-11..1e-10 %.1f ; 1e-10..1e-9 %.1f' % (ssfix(vg, ii, 1e-11, 1e
219         -10), ssfix(vg, ii, 1e-10, 1e-9)))
220 # ----- 11. symmetric leave-one-out, QC vs no
221     QC
222 p('\n=== V11: one-offset Vth_cc stories, QC vs no-QC (rigid Qf shift exact)')
223 Vm = {t: M[t]['Vth_cc_1e-9_V'] for t in (2.0, 6.3, 13.2)}
224 QC = {2.0: S['run_0012']['Vth_cc_1e-9_V'], 6.3: S['run_0014']['Vth_cc_1e-9_V'], 13.2: S['run_0013']['
225     Vth_cc_1e-9_V']} # Qf 1.73e12
226 NQ = {2.0: S['run_0016']['Vth_cc_1e-9_V'], 6.3: S['run_0034']['Vth_cc_1e-9_V'] - (1.73 - 0.0476) * qC,
227     13.2: S['run_0017']['Vth_cc_1e-9_V']} # all at Qf 1.73e12
228 for name, mod in (('QC', QC), ('noQC', NQ)):
229     res = {t: mod[t] - Vm[t] for t in Vm}
230     p(name, 'raw residuals at Qf 1.73e12:', {t: round(v, 4) for t, v in res.items()})
231     for anchor in (2.0, 6.3, 13.2):
232         off = -res[anchor]; r = {t: res[t] + off for t in Vm if t != anchor}
233         rms = math.sqrt(np.mean([v ** 2 for v in r.values()]))
234         p(f'   Qf fitted on {anchor} nm only -> held-out misses {dict((k, round(v,4)) for k,v in r.items
235             ()): rms {rms:.4f} V ; Qf = {(1.73 - off/qC):.3f}e12 cm^-2')
236     off = -np.mean(list(res.values())); r = {t: res[t] + off for t in Vm}; p('   LSQ offset over all 3 ->
237     residuals', {k: round(v, 4) for k, v in r.items()}, 'rms %.4f V' % math.sqrt(np.mean([v ** 2 for v
238         in r.values()])))
239 # leave-one-out with LSQ offset on the other two
240 loo = {}
241 for h in Vm:
242     others = [t for t in Vm if t != h]; off = -np.mean([res[t] for t in others]); loo[h] = res[h] +
243     off
244 p('   L00 (offset from the other two) held-out misses', {k: round(v, 4) for k, v in loo.items()}, '
245     rms %.4f' % math.sqrt(np.mean([v ** 2 for v in loo.values()])))
246 # ----- 12. S2 L00 power law vs step
247 p('\n=== V12: L00 on mu_FE (fit 2 & 13.2, predict 6.3)')
248 mf = {t: M[t]['mu_FE_cm2Vs'] for t in (2.0, 6.3, 13.2)}
249 b = math.log(mf[13.2] / mf[2.0]) / math.log(6.6); pred = mf[2.0] * (6.3 / 2) ** b
250 p(f'power-law exponent {b:.4f}; predicted mu(6.3) {pred:.2f} vs {mf[6.3]:.2f} -> x{pred/mf[6.3]:.2f}')
251 p(f'step with t_c in (6.3,13.2): predicted mu(6.3) = mu(2) = {mf[2.0]:.2f} -> x{mf[2.0]/mf[6.3]:.3f};
252     step with t_c in (2,6.3): predicted {mf[13.2]:.2f} -> x{mf[13.2]/mf[6.3]:.2f}')
253 p('Nt exponent ln4/ln6.6 = %.4f ; Nt law check 2e19 (2/6.3)^0.75 = %.3e, (2/13.2)^0.75 = %.3e' % (math.
254     log(4) / math.log(6.6), 2e19 * (2 / 6.3) ** 0.75, 2e19 * (2 / 13.2) ** 0.75))
255 # ----- 13. CC-definition mobility effect (S1)
256 p('\n=== V13: CC-definition effect on measured 13.2 nm (S1: +0.108 / +0.115 V)')
257 vg, idv = meas(13.2)
258 for lab, ratio in (('mu_FE', mf[13.2] / mf[2.0]), ('mu0 4.45', 4.45)):
259     p(lab, 'ratio %.3f Vg(1e-9 x ratio) - Vg(1e-9) = %.4f V' % (ratio, V_at(vg, idv, 1e-9 * ratio) -
260         V_at(vg, idv, 1e-9)))
261 # ----- 14. 31.8 nm apparent mu_FE
262 p('\n=== V14: 31.8 nm')
263 vg, idv = meas(31.8); gm = np.gradient(idv, vg); i = int(np.argmax(gm))
264 p('31.8 nm gm_max %.4e at Vg %.2f V -> mu_FE %.2f ; Id(-3 V) %.3e ; local gm maxima at Vg: %s' % (gm[i],
265     vg[i], M[31.8]['mu_FE_cm2Vs'], idv[0],

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256     [float(vg[k]) for k in range(1, len(gm) - 1) if gm[k] > gm[k - 1] and gm[k] >= gm[k + 1] and gm[k] >
        0.5 * gm.max())])
257
258 # ----- 15. isolation Ion (brief 5b vs S6)
259 p('\n=== V15: isolation runs Ion vs same-mu baselines')
260 for rid, ref in (('run_0048', 'run_0038'), ('run_0049', 'run_0040'), ('run_0050', 'run_0038'), ('run_0051',
    'run_0040')):
261     p(f'{rid} (mu {mu(rid)}) vs {ref} (mu {mu(ref)}): dIon {100*(S[rid]["Ion_A_per_um"]/S[ref]["Ion_A_per_um"]-1):.2f} % ; vs measured {S[rid]["ion_err"]:.2f} % ; dVth_cc {S[rid]["Vth_cc_1e-9_V"]-S[ref]["Vth_cc_1e-9_V"]:.4f} V ; dSS_cc {S[rid]["SS_cc_1e-10_1e-8_mV_dec"]-S[ref]["SS_cc_1e-10_1e-8_mV_dec"]:.1f}')
262
263 # ----- 16. A7 mesh, active-region spans
264 p('\n=== V16: A7 mesh check and active-region spans')
265 p('run_0036 RMSE %.5f Ion err %+.3f%% ; run_0015 RMSE %.5f ; dVth_cc %.5f V' % (S['run_0036']['rmse'], S['run_0036']['ion_err'], S['run_0015']['rmse'], S['run_0036']['Vth_cc_1e-9_V'] - S['run_0015']['Vth_cc_1e-9_V']))
266 for rid in ('run_0012', 'run_0015', 'run_0013', 'run_0038', 'run_0039', 'run_0040'):
267     vg, mm, pr = comp(rid); o = float(np.median(mm[(vg >= -2) & (vg <= -.5)])); act = mm > 5 * o #
        config/solver.yaml scoring.active_multiplier = 5.0
268     cnt = exe(rid)['fit_metrics']['regions']['active']['count']
269     p(f'{rid}: active region Vg {vg[act].min():.2f}..{vg[act].max():.2f} V, n={act.sum()} (execution.json count {cnt}), Id {mm[act].min():.2e}..{mm[act].max():.2e} ({math.log10(mm[act].max()/mm[act].min()):.2f} dec)')
270
271 # ----- 17. paper SS claim 60-70 mV/dec
272 p('\n=== V17: smallest point-to-point SS on the measured curves (paper claims 60-70 mV/dec)')
273 for t in (2.0, 6.3, 13.2):
274     vg, idv = meas(t); lg = np.log10(idv); d = np.diff(lg); ss = np.where(d > 0, 0.05 / np.where(d > 0, d, 1) * 1000, np.inf)
275     j = int(np.argmin(ss)); p(f'{t}: min 2-point SS {ss[j]:.1f} mV/dec between Vg {vg[j]:.2f} and {vg[j+1]:.2f} (Id {idv[j]:.2e} -> {idv[j+1]:.2e}); SS_min(5-pt, floor-gated) {M[t]["SS_mV_dec"]:.1f}')
276     # 5-point without floor gate
277     best = min(((vg[i+2]-vg[i-2])/(lg[i+2]-lg[i-2])*1000, vg[i]) for i in range(2, len(vg)-2) if lg[i+2] > lg[i-2])
278     p(f'    5-pt SS without floor gate: {best[0]:.1f} at Vg {best[1]:.2f}')
279
280 # ----- 18. tuned run: overdrive vs mobility
        share of the +26.8 % Ion
281 p('\n=== V18: share of run_0014 Ion error due to mobility choice')
282 i14 = S['run_0014']['Ion_A_per_um']; i15 = S['run_0015']['Ion_A_per_um']; m14 = mu('run_0014'); m15 = mu('run_0015')
283 p(f'run_0014 Ion {i14:.4e} (mu {m14}); run_0015 {i15:.4e} (mu {m15}); run_0015 rescaled to mu {m14}: {i15*m14/m15:.4e} -> {100*(i15*m14/m15/M[6.3]["Ion_A_per_um"]-1):.2f} % ; log share of mobility in the +26.8 %: {math.log(m14/m15)/math.log(i14/M[6.3]["Ion_A_per_um"])*100:.1f} %')
284 p('run_0015 gap %.4f vs meas %.4f ; gm %.4e vs %.4e (%+.1f%%) ; Vth_lin miss %+.4f' % (S['run_0015']['Vth_lin_V'] - S['run_0015']['Vth_cc_1e-9_V'], M[6.3]['Vth_lin_V'] - M[6.3]['Vth_cc_1e-9_V'], S['run_0015']['gm_max_A_V_um'], M[6.3]['gm_max_A_V_um'], 100 * (S['run_0015']['gm_max_A_V_um'] / M[6.3]['gm_max_A_V_um'] - 1), S['run_0015']['Vth_lin_V'] - M[6.3]['Vth_lin_V']))
285
286 p('\n=== V19: small analytic checks quoted in REVIEW.md')
287 E1 = lambda t, m: 0.376 / (m * t * t) # infinite-well parabolic bound, eV (t in nm), as used by S2
288 tx = (0.376 / (0.208 * 0.9205)) ** (1 / (2 - 1.3815))
289 p(f'dEc law vs infinite-well bound (m* 0.208): 6.3 nm {dEc(6.3):.4f} vs {E1(6.3,0.208):.4f} eV (x{dEc(6.3)/E1(6.3,0.208):.2f}); 13.2 nm {dEc(13.2):.4f} vs {E1(13.2,0.208):.4f} (x{dEc(13.2)/E1(13.2,0.208):.2f}); crossover t = {tx:.3f} nm')
290 p('WF +/-0.2 eV as Qf-equivalent: %.3e cm^-2' % (0.2 / qC * 1e12))
291 for t in (2.0, 13.2):
292     vp = musat[t]['central np.gradient(sqrt Id)'][1]; vov = vp - M[t]['Vth_cc_1e-9_V']
293     p(f'{t} nm: S2 heuristic Vd/(2Vov-Vd) with Vov vs Vth_cc = {0.7/(2*vov-0.7):.3f} ; actual mu_sat/mu_FE = {musat[t]["central np.gradient(sqrt Id)"][0]/M[t]["mu_FE_cm2Vs"]:.3f}')
294 p('no-QC Qf difference 13.2 vs 2 nm: %.3e cm^-2 (%.4f V)' % ((NQ[2.0] - Vm[2.0] - (NQ[13.2] - Vm[13.2])))

```

```

    / qC * 1e12, (NQ[2.0] - Vm[2.0]) - (NQ[13.2] - Vm[13.2])))
295 p('B1 / B3 (as run) Vth_cc misses: %+.4f / %+.4f V' % (miss('run_0038'), miss('run_0040')))
296 p('B1 on-state: gap %.4f vs meas %.4f (%+.4f V) ; gm %+.2f %%' % (S['run_0038']['Vth_lin_V'] - S['
    run_0038']['Vth_cc_1e-9_V'], M[2.0]['Vth_lin_V'] - M[2.0]['Vth_cc_1e-9_V'],
297     (S['run_0038']['Vth_lin_V'] - S['run_0038']['Vth_cc_1e-9_V']) - (M[2.0]['Vth_lin_V'] - M[2.0]['
    Vth_cc_1e-9_V']), 100 * (S['run_0038']['gm_max_A_V_um'] / M[2.0]['gm_max_A_V_um'] - 1)))
298 p('C1 share of run_0015 gap miss: %.1f %%' % (100 * ((S['run_0052']['Vth_lin_V'] - S['run_0052']['
    Vth_cc_1e-9_V']) - (S['run_0015']['Vth_lin_V'] - S['run_0015']['Vth_cc_1e-9_V'])) / ((M[6.3]['
    Vth_lin_V'] - M[6.3]['Vth_cc_1e-9_V']) - (S['run_0015']['Vth_lin_V'] - S['run_0015']['Vth_cc_1e-9_V'
    ]))))))
299 p('mu_band values in metadata: ' + ', '.join(f'{r}:{mu(r)}' for r in ('run_0012', 'run_0013', 'run_0014',
    'run_0015', 'run_0034', 'run_0037', 'run_0052'))))

```

B.3.25 SYNTHESIS_syn_checks.py

The script contains the read-only checks of the synthesis stage. It is run from the package root with the Python environment of the project, and its outputs are written to `scratch/SYNTHESIS/syn_checks_out.txt`.

```

1  """Synthesis-stage read-only checks (no ATLAS, no package file modified).
2  Run from the package root with ..\IW0_ATLAS_Model_claude\\.venv\Scripts\python.exe
3  Outputs: analysis_2026-09-25/scratch/SYNTHESIS/syn_checks_out.txt
4  (a) horizontal offsets Vg_meas(I) - Vg_sim(I) at fixed currents (rigid vs shape part of misses)
5  (b) one-offset Vth_cc stories with/without confinement (re-derivation of REVIEW V11 from execution.json)
6  (c) SHA-256 of every file hashed in PREDICTIONS_REGISTER.md section 10, compared with the registered
    value
7  """
8  import csv, hashlib, json, math, re, sys
9  from pathlib import Path
10 import numpy as np
11
12 ROOT = Path(__file__).resolve().parents[3]
13 sys.path.insert(0, str(ROOT / 'scripts'))
14 OUT = Path(__file__).with_name('syn_checks_out.txt')
15 lines = []
16 def p(*a):
17     s = ' '.join(str(x) for x in a); lines.append(s); print(s)
18
19 RUNS = {d.name[:8]: d for d in (ROOT / 'results' / 'runs').iterdir() if d.is_dir()}
20
21 def load_meas(t):
22     vg, idv = [], []
23     with open(ROOT / 'data' / 'experimental_clean.csv') as f:
24         for r in csv.DictReader(f):
25             if abs(float(r['thickness_nm']) - t) < 1e-6:
26                 vg.append(float(r['vg_V'])); idv.append(float(r['id_A_per_um']))
27     return np.array(vg), np.array(idv)
28
29 def load_sim(run, scale=1.0):
30     vg, idv = [], []
31     with open(RUNS[run] / 'comparison.csv') as f:
32         for r in csv.DictReader(f):
33             vg.append(float(r['vg_V'])); idv.append(float(r['atlas_A_per_um']) * scale)
34     return np.array(vg), np.array(idv)
35
36 def v_at(vg, idv, level):
37     # first upward crossing, log-linear (same rule as extract_metrics.crossing)
38     for i in range(len(vg) - 1):
39         a, b = idv[i], idv[i + 1]
40         if a > 0 and b > 0 and (a - level) * (b - level) <= 0 and a != b:

```



```

41         return vg[i] + (vg[i + 1] - vg[i]) * (math.log10(level) - math.log10(a)) / (math.log10(b) -
    math.log10(a))
42     return None
43
44 LEVELS = [1e-11, 1e-10, 1e-9, 1e-8, 3e-8, 1e-7, 2e-7, 3e-7]
45 CASES = [
46     (2.0, 'run_0012', 1.0, '2 nm calibrated (96/48)'),
47     (2.0, 'run_0038', 17.6897 / 17.69, '2 nm B1 recal (384/192)'),
48     (2.0, 'run_0016', 1.0, '2 nm no confinement, Qf 1.73e12'),
49     (6.3, 'run_0014', 1.0, '6.3 nm shared params (mu 12.4)'),
50     (6.3, 'run_0015', 1.0, '6.3 nm tuned (Qf 8.7e10, mu 11.69)'),
51     (6.3, 'run_0039', 11.582 / 11.41, '6.3 nm B2 recal'),
52     (6.3, 'run_0034', 1.0, '6.3 nm A6 no-QC, Qf 4.76e10'),
53     (6.3, 'run_0030', 1.0, '6.3 nm A3 back charge'),
54     (6.3, 'run_0037', 1.0, '6.3 nm B0 re-partition'),
55     (6.3, 'run_0052', 1.0, '6.3 nm C1 near-Ec band'),
56     (13.2, 'run_0013', 1.0, '13.2 nm calibrated'),
57     (13.2, 'run_0040', 61.6046 / 60.39, '13.2 nm B3 recal'),
58     (13.2, 'run_0017', 1.0, '13.2 nm no confinement'),
59 ]
60 p('=== (a) horizontal offset Vg_meas(I) - Vg_sim(I) in V (positive: measured curve reaches I later)')
61 p('levels (A/um):', ' '.join(f'{x:.0e}' for x in LEVELS))
62 for t, run, sc, lab in CASES:
63     vm, im = load_meas(t); vs, isim = load_sim(run, sc)
64     row = []
65     for lv in LEVELS:
66         a, b = v_at(vm, im, lv), v_at(vs, isim, lv)
67         row.append('  n/a' if (a is None or b is None) else f'{a - b:+.3f}')
68     p(f'{run} {lab:38s}', ' '.join(row))
69
70 p('\n=== (a2) 6.3 nm run_0014 miss split: rigid (offset at 1e-9) and shape (offset(I) - offset(1e-9))')
71 vm, im = load_meas(6.3)
72 for run in ['run_0014', 'run_0015', 'run_0034']:
73     vs, isim = load_sim(run)
74     o9 = v_at(vm, im, 1e-9) - v_at(vs, isim, 1e-9)
75     shape = {lv: (v_at(vm, im, lv) - v_at(vs, isim, lv)) - o9 for lv in [1e-8, 3e-8, 1e-7, 2e-7, 3e-7]}
76     p(run, f'rigid(1e-9) {o9:+.3f} V =', f'{o9 / 0.17891:+.2f}e12 cm^-2 ;', 'shape part:', ' ', '.join(f'{k
    :.0e}: {v:+.3f}' for k, v in shape.items()))
77
78 p('\n=== (b) one-offset Vth_cc stories from execution.json (rigid Qf shift exact, 0.17891 V per 1e12)')
79 def ex(run):
80     return json.load(open(RUNS[run] / 'execution.json'))
81 def vs(run): return ex(run)['device_metrics_sim']['Vth_cc_1e-9_V']
82 def vm(run): return ex(run)['device_metrics_exp']['Vth_cc_1e-9_V']
83 QOC = 0.17891 # V per 1e12 cm^-2
84 # residual (model - measured) at the shared Qf 1.73e12
85 qc = {2.0: vs('run_0012') - vm('run_0012'), 6.3: vs('run_0014') - vm('run_0014'), 13.2: vs('run_0013') -
    vm('run_0013')}
86 # no-QC at Qf 1.73e12: 2 nm run_0016, 13.2 nm run_0017, 6.3 nm run_0034 shifted back from Qf 4.76e10 to
    1.73e12
87 noqc = {2.0: vs('run_0016') - vm('run_0016'), 6.3: vs('run_0034') - QOC * (1.73 - 0.0476) - vm('run_0034'
    ), 13.2: vs('run_0017') - vm('run_0017')}
88 for name, res in (('with confinement', qc), ('without confinement', noqc)):
89     r = np.array([res[t] for t in (2.0, 6.3, 13.2)])
90     lsq = r - r.mean()
91     loo = [r[i] - np.mean(np.delete(r, i)) for i in range(3)]
92     anch = {t: [r[j] - r[i] for j in range(3) if j != i] for i, t in enumerate((2.0, 6.3, 13.2))}
93     p(f'{name}: raw residuals at Qf 1.73e12 {dict((t, round(float(x), 4)) for t, x in zip((2.0, 6.3,
    13.2), r))}')
94     p(f'  LSQ one-offset rms {np.sqrt(np.mean(lsq ** 2)):.4f} V ; L00 rms {np.sqrt(np.mean(np.square(loo
    )):.4f} V')

```

```

95     for t, m in anch.items():
96         p(f'    anchor {t} nm -> held-out misses {[round(float(x), 4) for x in m]} rms {np.sqrt(np.mean(np
          .square(m)):.4f} V')
97
98 p('\n=== (c) SHA-256 of PREDICTIONS_REGISTER.md section-10 files vs registered values')
99 reg = (ROOT / 'analysis_2026-09-25' / 'PREDICTIONS_REGISTER.md').read_text(encoding='utf-8')
100 n_ok = n_bad = 0
101 for rel, h in re.findall(r'\\s*`([^\`]+)`\\s*`\\s*`([0-9a-f]{64})`\\s*`|', reg):
102     fp = ROOT / rel.replace('/', '\\') if sys.platform.startswith('win') else ROOT / rel
103     if not fp.exists():
104         p('MISSING', rel); n_bad += 1; continue
105     now = hashlib.sha256(fp.read_bytes()).hexdigest()
106     ok = now == h
107     n_ok += ok; n_bad += (not ok)
108     if not ok: p('CHANGED', rel, 'registered', h[:12], 'now', now[:12])
109 p(f'files matching registered hash: {n_ok}; changed/missing: {n_bad}')
110 p('register file itself SHA-256:', hashlib.sha256((ROOT / 'analysis_2026-09-25' / 'PREDICTIONS_REGISTER.
          md').read_bytes()).hexdigest())
111
112 OUT.write_text('\n'.join(lines) + '\n', encoding='utf-8')

```

B.3.26 SYNTHESIS_syn_leverage.py

The script makes a read-only comparison of the leverage of the confinement package with that of the fitted and assumed parameters at 2 nm on the same metrics: fixed-current swings (10^{-11} – 10^{-10} , 10^{-10} – 10^{-9}), SS_{cc} (10^{-10} – 10^{-8}) and the non-rigidity of the horizontal shift (shift at 10^{-8} minus shift at 10^{-11} A/ μ m), for the one-at-a-time pairs relative to run_0012 or run_0038.

```

1  """Read-only: same-metric leverage of the confinement package vs fitted/assumed parameters at 2 nm.
2  Compares fixed-current SS (1e-11..1e-10, 1e-10..1e-9), SS_cc (1e-10..1e-8) and the non-rigidity of the
3  horizontal shift (shift at 1e-8 minus shift at 1e-11) for one-at-a-time pairs vs run_0012 (or run_0038).
4  Output: syn_leverage_out.txt next to this script."""
5  import csv, math
6  from pathlib import Path
7  import numpy as np
8  ROOT = Path(__file__).resolve().parents[3]
9  RUNS = {d.name[:8]: d for d in (ROOT / 'results' / 'runs').iterdir() if d.is_dir()}
10 out = []
11 def p(s): out.append(s); print(s)
12 def load(run):
13     vg, i = [], []
14     with open(RUNS[run] / 'comparison.csv') as f:
15         for r in csv.DictReader(f):
16             vg.append(float(r['vg_V'])); i.append(float(r['atlas_A_per_um']))
17     return np.array(vg), np.array(i)
18 def v_at(vg, idv, lv):
19     for k in range(len(vg) - 1):
20         a, b = idv[k], idv[k + 1]
21         if a > 0 and b > 0 and (a - lv) * (b - lv) <= 0 and a != b:
22             return vg[k] + (vg[k + 1] - vg[k]) * (math.log10(lv) - math.log10(a)) / (math.log10(b) - math
          .log10(a))
23     return None
24 def mets(run):
25     vg, i = load(run)
26     v = {lv: v_at(vg, i, lv) for lv in (1e-11, 1e-10, 1e-9, 1e-8)}
27     return {'V1e-9': v[1e-9], 'SS11_10': (v[1e-10] - v[1e-11]) * 1e3, 'SS10_9': (v[1e-9] - v[1e-10]) * 1
          e3,
28         'SScc': (v[1e-8] - v[1e-10]) / 2 * 1e3, 'v': v}

```

```

29 PAIRS = [('run_0012', 'run_0016', 'confinement package removed (dEc + m)'),
30          ('run_0012', 'run_0024', 'WTA +5 meV'),
31          ('run_0012', 'run_0025', 'Dit x3'),
32          ('run_0012', 'run_0023', 'Nt x1.5'),
33          ('run_0012', 'run_0022', 'Nd x2'),
34          ('run_0038', 'run_0050', 'm* x1.3 (DOS 384/192)'),
35          ('run_0038', 'run_0048', 'WF +0.1 eV (DOS 384/192)')]
36 p('2 nm one-at-a-time pairs: deltas (pert - ref) of V(1e-9) [mV], SS 1e-11..1e-10, SS 1e-10..1e-9, SS_cc
    [mV/dec], non-rigidity = shift(1e-8) - shift(1e-11) [mV]')
37 for ref, pert, lab in PAIRS:
38     a, b = mets(ref), mets(pert)
39     nonrig = ((b['v'][1e-8] - a['v'][1e-8]) - (b['v'][1e-11] - a['v'][1e-11])) * 1e3
40     p(f'{pert} vs {ref} {lab:40s} dV(1e-9) {1e3 * (b["V1e-9"] - a["V1e-9"]):+7.1f} dSS11_10 {b["SS11_10"]
        - a["SS11_10"]:+6.1f} '
41     f'dSS10_9 {b["SS10_9"] - a["SS10_9"]:+6.1f} dSScc {b["SScc"] - a["SScc"]:+6.1f} non-rigidity {
        nonrig:+6.1f}')
42 (Path(__file__).with_name('syn_leverage_out.txt')).write_text('\n'.join(out) + '\n', encoding='utf-8')

```

B.4 Collection script of this report

The report itself is assembled from package files by scripts in `REPORT/tools/`; they read package files and write only inside the report folder. In stage 0a, `tools/build_digest.py` builds `REPORT/digest/`, which contains the paper texts split by printed page (Section 2.1), the list of literature items used, the citations found in the analysis reports, the ATLAS manual pages of the statements used by the decks, the run table `digest/runs.csv` and a list of key numbers, so that writers never re-read raw sources. In stage 0b, `tools/make_figures.py` builds every registry figure from real run and data files and writes `figures/<stem>.pdf`, a PNG copy and `figures/MANIFEST.md` with the sources and key numbers of each figure; the call `python tools/make_figures.py [stem ...]` rebuilds selected figures. In stage 0c, `tools/make_appendix_assets.py` copies code, decks, configuration and data into the report (ASCII-sanitised) and generates `appendices/A_runs_table.tex` (Table A.2) and `appendices/D_hashes_table.tex` (re-verified hashes). Finally, `tools/lint_latex.py` is a static checker of the $\text{\LaTeX}$ sources, since no TeX installation is available on the project machine; it prints a report and exits with code 1 if any error is found.

Summary. All deck generation, solver launches, metric extraction and analysis are performed by the scripts listed here. Only three package scripts launch ATLAS, and each launch is logged. All analysis scripts are read-only with respect to the runs and share one metric extractor, so that measured and simulated metrics are computed identically. Surrogate models are confined to decomposition and hypothesis generation (SURR).

Appendix C

ATLAS input decks

The decks are the exact inputs given to the solver; every simulated curve of the report is the solution of one of them (or of a deck generated in the same way and archived in its run folder). This appendix reproduces eleven representative decks, chosen so that every kind of change made during the study appears at least once: the final calibrated decks of the three films, the 6.3 nm shared-parameter test and tuned decks, three 6.3 nm hypothesis decks, the B0 and C1 pre-registered tests, and one deck of each prediction type. The configuration files from which all decks are generated are listed at the end. The decks are copies of `results/runs/<run>/device.in`; the statement-by-statement explanation with manual pages is in Section 6.1.

C.1 The deck set

Table C.1. The eleven decks reproduced in this appendix (files in `code/decks/`; run metrics in Table A.2).

File	Film	Role
<code>run_0038_final_2p0nm_dos384.in</code>	2.0 nm	final calibrated deck (B1), DOS 384/192
<code>run_0039_final_6p3nm_tuned_dos384.in</code>	6.3 nm	final device-specific tuned deck (B2), DOS 384/192
<code>run_0040_final_13p2nm_dos384.in</code>	13.2 nm	final calibrated deck (B3), DOS 384/192
<code>run_0014_validation_6p3nm_shared.in</code>	6.3 nm	held-out threshold test with the shared parameters (failed), DOS 96/48
<code>run_0015_tuned_6p3nm_dos96.in</code>	6.3 nm	device-specific tuning, DOS 96/48
<code>run_0030_hyp_backsurface_charge.in</code>	6.3 nm	hypothesis A3: fixed charge on the exposed back surface
<code>run_0034_hyp_no_confinement.in</code>	6.3 nm	hypothesis A6: no confinement, Q_f fitted on 2 nm only
<code>run_0037_B0_repartition.in</code>	6.3 nm	B0: pre-registered trap/mobility re-partition of S2
<code>run_0052_C1_nearEc_band.in</code>	6.3 nm	C1: pre-registered near- E_c acceptor band of S4
<code>run_0041_pred_idvd_2p0nm.in</code>	2.0 nm	prediction: I_D – V_D families (B8)
<code>run_0045_pred_T358K_2p0nm.in</code>	2.0 nm	prediction: transfer curve at 358.15 K (B12)

C.2 Anatomy of a V1 deck

All eleven decks share one skeleton (147 lines; 149 for `run_0030` and 168 for `run_0041`), generated by `scripts/build_iwo_decks.py`: provenance header; mesh; IWO regions 1 (bulk) and 2 (0.25 nm interface layer), Al_2O_3 (region 3), HfO_2 (region 4), Conductor gate (region 6) and Air (region 7); Pd source/drain electrode volumes; uniform background donors; the user-defined IWO material with explicit oxide permittivities and the gate work function 4.7 eV; `models fermi srh`; continuous defect DOS in regions 1 and 2; the fixed front charge; the strict METHOD; probes; the bias ramp; the logged stepped sweep with the switch to default acceptance; and the extraction of curves and band profiles. Thickness enters only through the values written by the generator: band gap, affinity and N_c (confinement laws), N_{TA} (tail law), $N_{\text{d,eff}}$ (donor law), MUN (band mobility times the roughness factor $(1 - 0.287/t)^2 = 0.7336, 0.9111$ and 0.9565 at 2.0, 6.3 and 13.2 nm), the interface-layer DOS, the vertical mesh and the position of the switch between strict and default acceptance, which follows the measured current of the film (Section 6.3).

C.3 Parameter lines that differ between the decks

Table C.2 lists, for each deck, the overrides relative to the reference configuration as recorded in the run catalogue. “Reference” means the value of `config/iwo_material_model.yaml` at the film thickness; for the 6.3 nm film the reference band mobility is $12.4 \text{ cm}^2/(\text{V s})$.

Table C.2. Overrides of the reproduced decks (Q_f in cm^{-2} , μ_{band} in $\text{cm}^2/(\text{V s})$).

Run	Film	Q_f	μ_{band}	DOS	Other overrides
<code>run_0038</code>	2.0	1.73×10^{12}	17.69	384/192	–
<code>run_0039</code>	6.3	8.7×10^{10}	11.41	384/192	curve rescaled $\times 1.015074$ to $\mu_{\text{band}} = 11.58$ in the report
<code>run_0040</code>	13.2	1.73×10^{12}	60.39	384/192	curve rescaled $\times 1.020113$ to $\mu_{\text{band}} = 61.60$ in the report
<code>run_0014</code>	6.3	1.73×10^{12}	reference	96/48	– (all parameters shared with 2 and 13.2 nm)
<code>run_0015</code>	6.3	8.7×10^{10}	11.69	96/48	–
<code>run_0030</code>	6.3	1.73×10^{12}	reference	96/48	back-surface fixed charge $-1.64 \times 10^{12} \text{ cm}^{-2}$ on the IWO/air surface (additional INTERFACE statement)
<code>run_0034</code>	6.3	4.76×10^{10}	reference	96/48	confinement removed: ΔE_g^{QC} amplitude $a = 0$ and m^* increment $b = 0$
<code>run_0037</code>	6.3	9.8×10^{11}	19.6	96/48	tail density thickness-independent: $N_t = 2.66 \times 10^{19} \text{ cm}^{-3}$ (exponent $\alpha = 0$)
<code>run_0052</code>	6.3	8.7×10^{10}	15.4	384/192	bulk Gaussian acceptor band set to $N_{\text{GA}} = 9.0 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$ at $E_{\text{GA}} = 0.03 \text{ eV}$, $W_{\text{GA}} = 0.02 \text{ eV}$ (replacing the deep level at $E_c - 0.6 \text{ eV}$); registered as a sheet of $2 \times 10^{12} \text{ cm}^{-2}$
<code>run_0041</code>	2.0	1.73×10^{12}	17.6897	384/192	after the transfer sweep: I_D – V_D families at $V_G = 1, 1.5, 2, 2.5, 3 \text{ V}$, V_D from 0 to 3 V

Run	Film	Q_f	μ_{band}	DOS	Other overrides
run_0045	2.0	1.73×10^{12}	17.6897	384/192	models fermi srh temp=358.15 print; TMUN not set, so the default 1.5 applies (Section 6.3)

The consequences of these differences are analysed where the runs are used: the final decks in Chapter 7, the 6.3 nm decks in Chapter 8, and the prediction decks in Chapter 10.

C.4 Deck listings

C.4.1 run_0038: final 2 nm deck

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 2.0 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
   divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
   numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.3533 eV | DFT_PURE_IN2O3_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.4033 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 3.9467 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2573 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
   0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 3.2744e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.0000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
   bundled, recorded by Astra)
18 # [D] Nt(t) = 2.0000e+19 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
   LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
   *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 5.0000e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
   *(1-exp(-Eg/WTA))
20 # [D] WTA (kTt) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
   2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5002e+17 cm^-3 | FITTED law (effective electrically active background donor density,
   NOT W concentration) | Nd*(1+(t/tc)^k), Nd=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
   thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
   regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
   numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
   correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 17.69 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
   sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.7336 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
   0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 12.9772 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
   ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is

```

```

    produced self-consistently by the DOS occupancy; the Ando roll-off  $\theta_{r*Vov}$  is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2O21/Lin2O22) |
30 # [H] capture cross sections =  $1e-15 \text{ cm}^2$  | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.072 spac=0.02
46 y.mesh loc=-0.002 spac=0.00025
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.002 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.072 y.max=-0.002
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.072 y.max=-0.002
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.072 y.max=-0.002
66 doping uniform n.type conc=2.50025e+17 region=1
67 doping uniform n.type conc=2.50025e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.40331 affinity=3.94669 permittivity=9.3 \
71 nc300=3.27444e+18 nv300=1e+19 \
72 mun=12.9772 mup=0.01 \
73 taun0=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7

```

```

79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=384 numd=192 \
83   nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
84   nga=5e+16 ega=0.6 wga=0.15 \
85   ngd=0 egd=3.30331 wgd=0.1 \
86   sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87   siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88   afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm-3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=384 numd=192 \
92   nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
93   nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94   ngd=0 egd=3.30331 wgd=0.1 \
95   sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96   siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97   afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.00000001 y.max=0.00000001
99 # Strict acceptance (XANDRNORM, cr.toler 1e-17) up to Vg = 0.7 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
101   cr.toler=1e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.001 n.mob dir=0
105 probe name=chan_n x=12 y=-0.001 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.002 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.7 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
   outfile="cb_eq.dat"
134 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
   outfile="vb_eq.dat"

```



```

135 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
136 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"
137 extract init infile="near_vth.str"
138 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"
139 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
140 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
141 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_vth.dat"
142 extract init infile="final.str"
143 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
144 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
145 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
146 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
147 quit

```

C.4.2 run_0039: final tuned 6.3 nm deck

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
    divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
    numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0724 eV | DFT_PURE_IN2O3_PROXY | a*t^p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.1224 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2276 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2177 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.5498e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 8.4586e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 2.1146e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |

```

```

        regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
        numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 8.7000e+10 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
        correction | dVfb = -q Qf/Cox = -0.0156 V
25 # [F] mu_band(t) = 11.41 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
        sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
        0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 10.3941 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
        ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
        produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
        gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
        -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
        supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
        nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
        CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
        ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
        electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.1224 affinity=4.2276 permittivity=9.3 \

```

```

71  nc300=2.54976e+18 nv300=1e+19 \
72  mun=10.3941 mup=0.01 \
73  taun0=1e-06 taup0=1e-06
74  # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75  # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76  material region=3 permittivity=9.0
77  material region=4 permittivity=19.57
78  contact name=gate workfunction=4.7
79  models fermi srh temp=300 print
80  # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81  # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82  defects region=1 continuous numa=384 numd=192 \
83  nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
84  nga=5e+16 ega=0.6 wga=0.15 \
85  ngd=0 egd=3.0224 wgd=0.1 \
86  sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87  siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88  afile=acceptor_r1.dat dfile=donor_r1.dat
89  # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90  # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91  defects region=2 continuous numa=384 numd=192 \
92  nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
93  nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94  ngd=0 egd=3.0224 wgd=0.1 \
95  sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96  siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97  afile=acceptor_r2.dat dfile=donor_r2.dat
98  interface qf=8.7e+10 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99  # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
101  cr.toler=2.28e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
105 probe name=chan_n x=12 y=-0.00315 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.65 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
119  cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"

```

```

127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_eq.dat"
134 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_eq.dat"
135 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
136 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"
137 extract init infile="near_vth.str"
138 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"
139 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
140 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
141 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile=
    "qfn_vth.dat"
142 extract init infile="final.str"
143 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
144 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
145 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
146 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
147 quit

```

C.4.3 run_0040: final 13.2 nm deck

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 13.2 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
    divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
    numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0261 eV | DFT_PURE_IN2O3_PROXY | a*t^p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.0761 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2739 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2114 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.4396e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 4.8570e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2

```

```

      *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/NTA = 1.2143e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/NTA); integral = NTA*NTA
      *(1-exp(-Eg/NTA))
20 # [D] WTA (kTt) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
      2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.9744e+17 cm^-3 | FITTED law (effective electrically active background donor density,
      NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
      thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
      regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
      numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
      correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 60.39 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
      sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9570 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
      0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 57.7925 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
      ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
      produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
      gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
      -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
      supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0832 spac=0.02
46 y.mesh loc=-0.0132 spac=0.001
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0132 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
      nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
      CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0832 y.max=-0.0132
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067

```

```

62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0832 y.max=-0.0132
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0832 y.max=-0.0132
66 doping uniform n.type conc=2.97437e+17 region=1
67 doping uniform n.type conc=2.97437e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.07606 affinity=4.27394 permittivity=9.3 \
71 nc300=2.43961e+18 nv300=1e+19 \
72 mun=57.7925 mup=0.01 \
73 taun0=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=384 numd=192 \
83 nta=1.21426e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=5e+16 ega=0.6 wga=0.15 \
85 ngd=0 egd=2.97606 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88 afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=384 numd=192 \
92 nta=1.21426e+20 wta=0.04 ntd=0 wtd=0.1 \
93 nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94 ngd=0 egd=2.97606 wgd=0.1 \
95 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97 afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99 # Strict acceptance (XANDRNORM, cr.toler 7.88e-16) up to Vg = 0.1 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
101 cr.toler=7.88e-16 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.0066 n.mob dir=0
105 probe name=chan_n x=12 y=-0.0066 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0132 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.1 name=gate

```

```
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
134   outfile="cb_eq.dat"
135 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
136   outfile="vb_eq.dat"
137 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
138   n_eq.dat"
139 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
140   qfn_eq.dat"
141 extract init infile="near_vth.str"
142 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
143   outfile="cb_vth.dat"
144 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
145   outfile="vb_vth.dat"
146 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
147   n_vth.dat"
148 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
149   qfn_vth.dat"
150 extract init infile="final.str"
151 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
152   outfile="cb_on.dat"
153 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
154   outfile="vb_on.dat"
155 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
156   n_on.dat"
157 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
158   qfn_on.dat"
159 quit
```

C.4.4 run_0014: 6.3 nm held-out test with shared parameters

```
1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
6   divide by 290.
7 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
8 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
9   numerical.
10 #=====
11 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
12 # [B] dEg_QC(t) = 0.0724 eV | DFT_PURE_IN2O3_PROXY | a*t^p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
```

```

11 # [A+B] EG300 = 3.1224 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2276 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2177 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN203_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.5498e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 8.4586e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 2.1146e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN203_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 12.4 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 11.2960 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVEs (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025

```



```

54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.1224 affinity=4.2276 permittivity=9.3 \
71 nc300=2.54976e+18 nv300=1e+19 \
72 mun=11.296 mup=0.01 \
73 taun0=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=96 numd=48 \
83 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=5e+16 ega=0.6 wga=0.15 \
85 ngd=0 egd=3.0224 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88 afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=96 numd=48 \
92 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
93 nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94 ngd=0 egd=3.0224 wgd=0.1 \
95 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97 afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.00000001 y.max=0.00000001
99 # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
101 cr.toler=2.28e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
105 probe name=chan_n x=12 y=-0.00315 n.conc

```

```
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.65 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
134   outfile="cb_eq.dat"
135 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
136   outfile="vb_eq.dat"
137 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
138   n_eq.dat"
139 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
140   qfn_eq.dat"
141 extract init infile="near_vth.str"
142 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
143   outfile="cb_vth.dat"
144 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
145   outfile="vb_vth.dat"
146 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
147   n_vth.dat"
148 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
149   qfn_vth.dat"
150 extract init infile="final.str"
151 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
152   outfile="cb_on.dat"
153 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
154   outfile="vb_on.dat"
155 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
156   n_on.dat"
157 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
158   qfn_on.dat"
159 quit
```

C.4.5 run_0015: 6.3 nm device-specific tuning

```
1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
```

```

3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
  divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
  numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0724 eV | DFT_PURE_IN2O3_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.1224 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2276 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2177 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
  0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.5498e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
  bundled, recorded by Astra)
18 # [D] Nt(t) = 8.4586e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
  LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
  *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 2.1146e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
  *(1-exp(-Eg/WTA))
20 # [D] WTA (kTt) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
  2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm^-3 | FITTED law (effective electrically active background donor density,
  NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
  thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
  regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
  numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 8.7000e+10 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
  correction | dVfb = -q Qf/Cox = -0.0156 V
25 # [F] mu_band(t) = 11.69 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
  sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
  0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 10.6492 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
  ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
  produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
  gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
  -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
  supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05

```

```

44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.1224 affinity=4.2276 permittivity=9.3 \
71 nc300=2.54976e+18 nv300=1e+19 \
72 mun=10.6492 mup=0.01 \
73 taun0=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    -3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=96 numd=48 \
83 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=5e+16 ega=0.6 wga=0.15 \
85 ngd=0 egd=3.0224 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88 afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    -3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=96 numd=48 \
92 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
93 nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94 ngd=0 egd=3.0224 wgd=0.1 \
95 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \

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```

96  siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97  afile=acceptor_r2.dat dfile=donor_r2.dat
98  interface qf=8.7e+10 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99  # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
101   cr.toler=2.28e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
105 probe name=chan_n x=12 y=-0.00315 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.65 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
134   outfile="cb_eq.dat"
134 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
135   outfile="vb_eq.dat"
135 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
136   n_eq.dat"
136 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
137   qfn_eq.dat"
137 extract init infile="near_vth.str"
138 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
139   outfile="cb_vth.dat"
139 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
140   outfile="vb_vth.dat"
140 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
141   n_vth.dat"
141 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
142   qfn_vth.dat"
142 extract init infile="final.str"
143 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
144   outfile="cb_on.dat"
144 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
145   outfile="vb_on.dat"
145 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
146   n_on.dat"

```

```

146 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
147 quit

```

C.4.6 run_0030: hypothesis A3, back-surface charge

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
    divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
    numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0724 eV | DFT_PURE_IN2O3_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.1224 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2276 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2177 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.5498e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.0000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 8.4586e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 2.1146e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 12.4 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 11.2960 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVEs (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0

```

```

34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.1224 affinity=4.2276 permittivity=9.3 \
71 nc300=2.54976e+18 nv300=1e+19 \
72 mun=11.296 mup=0.01 \
73 taun0=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm^-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=96 numd=48 \
83 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=5e+16 ega=0.6 wga=0.15 \
85 ngd=0 egd=3.0224 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \

```

```

88  afile=acceptor_r1.dat dfile=donor_r1.dat
89  # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90  # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91  defects region=2 continuous numa=96 numd=48 \
92  nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
93  nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94  ngd=0 egd=3.0224 wgd=0.1 \
95  sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96  siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97  afile=acceptor_r2.dat dfile=donor_r2.dat
98  interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99  # [E] HYPOTHESIS back-surface fixed charge Qb = -1.6400e+12 cm-2 at the IWO/Air surface (x = 2.22 um)
100 interface qf=-1.64e+12 x.min=2 x.max=22 y.min=-0.0063001 y.max=-0.0062999
101 # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
102 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
103 cr.toler=2.28e-17 xandrnorm carriers=1 electrons
104 output con.band val.band e.mobility
105 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
106 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
107 probe name=chan_n x=12 y=-0.00315 n.conc
108 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
109 solve init
110 save outf=equilibrium.str
111 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
112 solve vstep=-0.1 vfinal=-3 name=gate
113 solve vstep=0.1 vfinal=0.7 name=drain
114 log outf=transfer.log
115 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
116 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
117 solve vstep=0.05 vfinal=0.65 name=gate
118 save outf=near_vth.str
119 # On-state: default convergence acceptance.
120 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
121 cr.toler=5e-18 ^xandrnorm carriers=1 electrons
122 solve vstep=0.05 vfinal=1.5 name=gate
123 solve vstep=0.1 vfinal=3 name=gate
124 log off
125 save outf=final.str
126 output con.band val.band qfn e.mobility
127 extract init infile="transfer.log"
128 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
129 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
130 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
131 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
132 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
133 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
134 extract init infile="equilibrium.str"
135 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_eq.dat"
136 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_eq.dat"
137 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
138 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"
139 extract init infile="near_vth.str"
140 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"

```

```

141 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
142 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
143 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_vth.dat"
144 extract init infile="final.str"
145 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
146 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
147 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
148 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
149 quit

```

C.4.7 run_0034: hypothesis A6, no confinement

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
    divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
    numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0000 eV | DFT_PURE_IN2O3_PROXY | a*t^p, a=0.0, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.0500 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.3000 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2080 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.3805e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.0000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 8.4586e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 2.1146e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kTt) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 4.7600e+10 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.0085 V
25 # [F] mu_band(t) = 12.4 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)

```

```

26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 Å) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 11.2960 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.05 affinity=4.3 permittivity=9.3 \
71 nc300=2.38049e+18 nv300=1e+19 \
72 mun=11.296 mup=0.01 \
73 taun0=1e-06 taup0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).

```

```

75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=96 numd=48 \
83   nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
84   nga=5e+16 ega=0.6 wga=0.15 \
85   ngd=0 egd=2.95 wgd=0.1 \
86   sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87   siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88   afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=96 numd=48 \
92   nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
93   nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94   ngd=0 egd=2.95 wgd=0.1 \
95   sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96   siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97   afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=4.76e+10 x.min=0 x.max=24 y.min=-0.00000001 y.max=0.00000001
99 # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
101   cr.toler=2.28e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
105 probe name=chan_n x=12 y=-0.00315 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.65 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"

```

```

132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_eq.dat"
134 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_eq.dat"
135 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
136 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"
137 extract init infile="near_vth.str"
138 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"
139 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
140 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
141 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile=
    "qfn_vth.dat"
142 extract init infile="final.str"
143 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
144 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
145 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
146 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
147 quit

```

C.4.8 run_0037: B0, trap/mobility re-partition

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
    divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
    numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0724 eV | DFT_PURE_IN2O3_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.1224 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2276 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2177 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 2.5498e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.0000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 2.6600e+19 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2.66e+19, alpha=0.0
19 # [D] NTA = Nt/WTA = 6.6500e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kTt) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness

```

```

21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 9.8000e+11 cm-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.1753 V
25 # [F] mu_band(t) = 19.6 cm2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 17.8549 cm2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063

```

```

65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IW0 user.group=semiconductor user.default=silicon \
70 eg300=3.1224 affinity=4.2276 permittivity=9.3 \
71 nc300=2.54976e+18 nv300=1e+19 \
72 mun=17.8549 mup=0.01 \
73 taun0=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=96 numd=48 \
83 nta=6.65e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=5e+16 ega=0.6 wga=0.15 \
85 ngd=0 egd=3.0224 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88 afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=96 numd=48 \
92 nta=6.65e+20 wta=0.04 ntd=0 wtd=0.1 \
93 nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94 ngd=0 egd=3.0224 wgd=0.1 \
95 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97 afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=9.8e+11 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99 # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
101 cr.toler=2.28e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
105 probe name=chan_n x=12 y=-0.00315 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.65 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
119 cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate

```

```

121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_eq.dat"
134 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_eq.dat"
135 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
136 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"
137 extract init infile="near_vth.str"
138 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"
139 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
140 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
141 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_vth.dat"
142 extract init infile="final.str"
143 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
144 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
145 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
146 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
147 quit

```

C.4.9 run_0052: C1, near- E_c acceptor band

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 6.3 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
    divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
    numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.0724 eV | DFT_PURE_IN2O3_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.1224 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 4.2276 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2177 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment

```

```

15 # [A+B] NC300 = 2.5498e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 8.4586e+18 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 2.1146e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 9e+19 at Ec-0.03 eV, W=0.02 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5246e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 8.7000e+10 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.0156 V
25 # [F] mu_band(t) = 15.4 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.9110 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 14.0288 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.0763 spac=0.02
46 y.mesh loc=-0.0063 spac=0.0007875
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.0063 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).

```



```

57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.0763 y.max=-0.0063
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.0763 y.max=-0.0063
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.0763 y.max=-0.0063
66 doping uniform n.type conc=2.52461e+17 region=1
67 doping uniform n.type conc=2.52461e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.1224 affinity=4.2276 permittivity=9.3 \
71 nc300=2.54976e+18 nv300=1e+19 \
72 mun=14.0288 mup=0.01 \
73 taun=1e-06 tau0=1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    -3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=384 numd=192 \
83 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=9e+19 ega=0.03 wga=0.02 \
85 ngd=0 egd=3.0224 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88 afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    -3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=384 numd=192 \
92 nta=2.11464e+20 wta=0.04 ntd=0 wtd=0.1 \
93 nga=1.56938e+19 ega=0.15 wga=0.206452 \
94 ngd=0 egd=3.0224 wgd=0.1 \
95 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97 afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=8.7e+10 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99 # Strict acceptance (XANDRNORM, cr.toler 2.28e-17) up to Vg = 0.65 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
101 cr.toler=2.28e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.00315 n.mob dir=0
105 probe name=chan_n x=12 y=-0.00315 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.0063 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate

```

```
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.65 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
134   outfile="cb_eq.dat"
135 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
136   outfile="vb_eq.dat"
137 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
138   n_eq.dat"
139 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
140   qfn_eq.dat"
141 extract init infile="near_vth.str"
142 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
143   outfile="cb_vth.dat"
144 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
145   outfile="vb_vth.dat"
146 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
147   n_vth.dat"
148 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
149   qfn_vth.dat"
150 extract init infile="final.str"
151 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
152   outfile="cb_on.dat"
153 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
154   outfile="vb_on.dat"
155 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
156   n_on.dat"
157 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
158   qfn_on.dat"
159 quit
```

C.4.10 run_0041: prediction of the 2 nm output characteristics

```
1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 2.0 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
6   divide by 290.
7 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 300.0 K.
```

```

7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
  numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.3533 eV | DFT_PURE_IN203_PROXY | a*t^p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.4033 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 3.9467 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2573 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN203_PROXY increment | Stokey2021 bulk
    0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 3.2744e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
    bundled, recorded by Astra)
18 # [D] Nt(t) = 2.0000e+19 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
    LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
    *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 5.0000e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
    *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
    2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5002e+17 cm^-3 | FITTED law (effective electrically active background donor density,
    NOT W concentration) | Nd0*(1+(t/tc)^k), Nd0=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
    thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN203_PROXY; here IWO/Al2O3) |
    regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
    numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
    correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 17.6897 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
    sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.7336 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
    0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 12.9770 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
    ignores MOBILITY updates between SOLVES (Claude run_0008); the free/trapped partition of Eq.(5) is
    produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
    gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
    -> held fixed |
29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.072 spac=0.02
46 y.mesh loc=-0.002 spac=0.00025
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05

```

```

49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.002 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.072 y.max=-0.002
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.072 y.max=-0.002
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.072 y.max=-0.002
66 doping uniform n.type conc=2.50025e+17 region=1
67 doping uniform n.type conc=2.50025e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70 eg300=3.40331 affinity=3.94669 permittivity=9.3 \
71 nc300=3.27444e+18 nv300=1e+19 \
72 mun=12.977 mup=0.01 \
73 taun=1e-06 tau=0.1e-06
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=300 print
80 # Region 1: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    -3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).
82 defects region=1 continuous numa=384 numd=192 \
83 nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
84 nga=5e+16 ega=0.6 wga=0.15 \
85 ngd=0 egd=3.30331 wgd=0.1 \
86 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88 afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    -3/eV;
90 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=384 numd=192 \
92 nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
93 nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94 ngd=0 egd=3.30331 wgd=0.1 \
95 sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96 siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97 afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001
99 # Strict acceptance (XANDRNORM, cr.toler 1e-17) up to Vg = 0.7 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \

```

```
101  cr.toler=1e-17 xandrnorm carriers=1 electrons
102  output con.band val.band e.mobility
103  # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104  probe name=chan_mob x=12 y=-0.001 n.mob dir=0
105  probe name=chan_n x=12 y=-0.001 n.conc
106  probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.002 y.max=-0.00025
107  solve init
108  save outf=equilibrium.str
109  # PREDICTION (recorded before any ID-VD measurement): ID-VD families, default acceptance (on-state
      currents).
110  method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 \
111  cr.toler=5e-18 ^xandrnorm carriers=1 electrons
112  solve vstep=0.1 vfinal=1 name=gate
113  log outf=idvd_vg1.log
114  solve vdrain=0
115  solve vstep=0.05 vfinal=0.5 name=drain
116  solve vstep=0.1 vfinal=3 name=drain
117  log off
118  solve vstep=-0.25 vfinal=0 name=drain
119  solve vstep=0.1 vfinal=1.5 name=gate
120  log outf=idvd_vg1p5.log
121  solve vdrain=0
122  solve vstep=0.05 vfinal=0.5 name=drain
123  solve vstep=0.1 vfinal=3 name=drain
124  log off
125  solve vstep=-0.25 vfinal=0 name=drain
126  solve vstep=0.1 vfinal=2 name=gate
127  log outf=idvd_vg2.log
128  solve vdrain=0
129  solve vstep=0.05 vfinal=0.5 name=drain
130  solve vstep=0.1 vfinal=3 name=drain
131  log off
132  solve vstep=-0.25 vfinal=0 name=drain
133  solve vstep=0.1 vfinal=2.5 name=gate
134  log outf=idvd_vg2p5.log
135  solve vdrain=0
136  solve vstep=0.05 vfinal=0.5 name=drain
137  solve vstep=0.1 vfinal=3 name=drain
138  log off
139  solve vstep=-0.25 vfinal=0 name=drain
140  solve vstep=0.1 vfinal=3 name=gate
141  log outf=idvd_vg3.log
142  solve vdrain=0
143  solve vstep=0.05 vfinal=0.5 name=drain
144  solve vstep=0.1 vfinal=3 name=drain
145  log off
146  solve vstep=-0.25 vfinal=0 name=drain
147  save outf=final_output.str
148  extract init infile="idvd_vg1.log"
149  extract name="IdVd_vg1" curve(v."drain",i."drain") outfile="idvd_vg1.dat"
150  extract name="IsVd_vg1" curve(v."drain",i."source") outfile="isvd_vg1.dat"
151  extract name="IgVd_vg1" curve(v."drain",i."gate") outfile="igvd_vg1.dat"
152  extract init infile="idvd_vg1p5.log"
153  extract name="IdVd_vg1p5" curve(v."drain",i."drain") outfile="idvd_vg1p5.dat"
154  extract name="IsVd_vg1p5" curve(v."drain",i."source") outfile="isvd_vg1p5.dat"
155  extract name="IgVd_vg1p5" curve(v."drain",i."gate") outfile="igvd_vg1p5.dat"
156  extract init infile="idvd_vg2.log"
157  extract name="IdVd_vg2" curve(v."drain",i."drain") outfile="idvd_vg2.dat"
158  extract name="IsVd_vg2" curve(v."drain",i."source") outfile="isvd_vg2.dat"
159  extract name="IgVd_vg2" curve(v."drain",i."gate") outfile="igvd_vg2.dat"
160  extract init infile="idvd_vg2p5.log"
```

```

161 extract name="IdVd_vg2p5" curve(v."drain",i."drain") outfile="idvd_vg2p5.dat"
162 extract name="IsVd_vg2p5" curve(v."drain",i."source") outfile="isvd_vg2p5.dat"
163 extract name="IgVd_vg2p5" curve(v."drain",i."gate") outfile="igvd_vg2p5.dat"
164 extract init infile="idvd_vg3.log"
165 extract name="IdVd_vg3" curve(v."drain",i."drain") outfile="idvd_vg3.dat"
166 extract name="IsVd_vg3" curve(v."drain",i."source") outfile="isvd_vg3.dat"
167 extract name="IgVd_vg3" curve(v."drain",i."gate") outfile="igvd_vg3.dat"
168 quit

```

C.4.11 run_0045: prediction of the 2 nm transfer curve at 358.15 K

```

1 #=====
2 # IWO (~2 at.% W:In2O3) FET, channel t = 2.0 nm -- IWO_PHYSICS_CONSTRAINED_MODEL_V1
3 # Generated by scripts/build_iwo_decks.py from config/iwo_material_model.yaml (ONE material model for all
4 # thicknesses). Reduced electrical deck: ideal TiN gate / Pd S/D boundaries (equipotential metals, DC).
5 # MESH WIDTH = 1 um -> currents are A/um. Total 290-um device current = 290 x logged current. Never
   divide by 290.
6 # Cox = 8.9554e-07 F/cm^2 (HfO2 15 nm eps 19.57 + Al2O3 2 nm eps 9.0). Vd = 0.7 V, T = 358.15 K.
7 # Category labels: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H
   numerical.
8 #=====
9 # [A] Eg_bulk = 3.05 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
10 # [B] dEg_QC(t) = 0.3533 eV | DFT_PURE_IN2O3_PROXY | a*t^p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
11 # [A+B] EG300 = 3.4033 eV | CALCULATED | Eg_bulk + dEg_QC(t)
12 # [A] chi_bulk = 4.3 eV | LITERATURE_IWO | Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
13 # [A+B] AFFINITY = 3.9467 eV | CALCULATED | chi_bulk - dEc(t), dEc = dEg (Ev unchanged, Si2021)
14 # [A+B] m* = 0.2573 m0 | MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment | Stokey2021 bulk
   0.208 + Lin2022 PBE increment
15 # [A+B] NC300 = 3.2744e+18 cm^-3 | CALCULATED | 2(2 pi m* kT/h^2)^1.5; effective continuum DOS at 2 nm
16 # [A] NV300 = 1.0000e+19 cm^-3 | ASSUMED | holes not solved
17 # [A] PERMITTIVITY = 9.3 - | MEASURED_TARGET_DEVICE | Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not
   bundled, recorded by Astra)
18 # [D] Nt(t) = 2.0000e+19 cm^-3 | FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with
   LITERATURE_IWO trend (Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2 nm with theta_t convention) | Nt2
   *(2/t)^alpha, Nt2=2e+19, alpha=0.75
19 # [D] NTA = Nt/WTA = 5.0000e+20 cm^-3/eV | CALCULATED | gTA(E)=NTA exp((E-Ec)/WTA); integral = NTA*WTA
   *(1-exp(-Eg/WTA))
20 # [D] WTA (kT) = 0.04 eV | FITTED, shared across thickness | Tt = 464 K; shared across thickness
21 # [D] deep acceptor Gaussian = 5e+16 at Ec-0.6 eV, W=0.15 cm^-3/eV | ASSUMED (insensitive; Fan2021 range
   2.5-3.7e16 for a different device) |
22 # [D] Nd_eff(t) = 2.5002e+17 cm^-3 | FITTED law (effective electrically active background donor density,
   NOT W concentration) | Nd*(1+(t/tc)^k), Nd=2.5e+17, tc=20.0 nm, k=4.0; NOT W concentration
23 # [E] interface acceptor sheet = 3e+11 at Ec-0.3 eV, W=0.12 eV cm^-2/eV | ASSUMED, shared across
   thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (LITERATURE_IN2O3_PROXY; here IWO/Al2O3) |
   regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak / 0.25 nm (
   numerically verified: layer /2 changes Id by 0.009 dec)
24 # [E] Qf = 1.7300e+12 cm^-2 | FITTED single shared value (stage 3); classified as effective electrostatic
   correction | dVfb = -q Qf/Cox = -0.3095 V
25 # [F] mu_band(t) = 17.6897 cm^2/Vs | FITTED per thickness | FITTED per thickness (no defensible law;
   sigmoidal trend attributed to percolation/crystallinity, hypothesis)
26 # [F] roughness factor = 0.7336 - | LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A) | (1 - Dsr/t)^2, Dsr =
   0.287 nm
27 # [F] MUN = mu_band*(1-Dsr/t)^2 = 12.9770 cm^2/Vs | CALCULATED | NOT inserted natively: ATLAS 5.28.1.R
   ignores MOBILITY updates between SOLVEs (Claude run_0008); the free/trapped partition of Eq.(5) is
   produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an approximation
   gap (see LIMITATIONS)
28 # [G] gate work function = 4.7 eV | ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi
   -> held fixed |

```

```

29 # [G] S/D = ideal Ohmic (no workfunction) | ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3
    supports low-barrier contacts (Si2021/Lin2022) |
30 # [H] capture cross sections = 1e-15 cm^2 | ASSUMED | not identifiable from DC transfer curves
31 #=====
32 go atlas
33 mesh width=1.0
34 x.mesh loc=0 spac=0.3
35 x.mesh loc=1.8 spac=0.05
36 x.mesh loc=2 spac=0.01
37 x.mesh loc=2.2 spac=0.05
38 x.mesh loc=3 spac=0.2
39 x.mesh loc=12 spac=0.7
40 x.mesh loc=21 spac=0.2
41 x.mesh loc=21.8 spac=0.05
42 x.mesh loc=22 spac=0.01
43 x.mesh loc=22.2 spac=0.05
44 x.mesh loc=24 spac=0.3
45 y.mesh loc=-0.072 spac=0.02
46 y.mesh loc=-0.002 spac=0.00025
47 y.mesh loc=-0.00025 spac=6.25e-05
48 y.mesh loc=-0.000125 spac=6.25e-05
49 y.mesh loc=0 spac=6.25e-05
50 y.mesh loc=0.002 spac=0.00025
51 y.mesh loc=0.017 spac=0.001
52 y.mesh loc=0.067 spac=0.02
53 region num=1 user.material=IWO x.min=0 x.max=24 y.min=-0.002 y.max=-0.00025
54 region num=2 user.material=IWO x.min=0 x.max=24 y.min=-0.00025 y.max=0
55 # Physical stack of the measured device (bottom gate): TiN 50 nm / HfO2 15 nm / Al2O3 2 nm / IWO / Pd 70
    nm S/D, air above the channel.
56 # TiN is not in the ATLAS metal table: the gate is a Conductor volume whose work function is set on
    CONTACT (4.70 eV, ASSUMED).
57 region num=3 material=Al2O3 x.min=0 x.max=24 y.min=0 y.max=0.002
58 region num=4 material=HfO2 x.min=0 x.max=24 y.min=0.002 y.max=0.017
59 region num=6 material=Conductor x.min=0 x.max=24 y.min=0.017 y.max=0.067
60 region num=7 material=Air x.min=0 x.max=24 y.min=-0.072 y.max=-0.002
61 electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067
62 # Pd S/D volumes on the IWO top surface. No CONTACT work function is set for them -> Ohmic boundary (
    ASSUMED, no TLM data);
63 # the 2 nm run of this structure is compared with the reduced-boundary deck (run_0008) to verify
    electrical equivalence.
64 electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.072 y.max=-0.002
65 electrode name=drain material=Palladium x.min=22 x.max=24 y.min=-0.072 y.max=-0.002
66 doping uniform n.type conc=2.50025e+17 region=1
67 doping uniform n.type conc=2.50025e+17 region=2
68 # USER.DEFAULT=silicon is only the parser template; every relevant parameter is overridden above.
69 material material=IWO user.group=semiconductor user.default=silicon \
70   eg300=3.40331 affinity=3.94669 permittivity=9.3 \
71   nc300=3.27444e+18 nv300=1e+19 \
72   mun=12.977 mup=0.01 \
73   taun0=1e-06 tau0=1e-06 egalp=0
74 # Region 3 = Al2O3 (2 nm), eps 9.0 (ASSUMED (not reported)); region 4 = HfO2 (15 nm), eps 19.57 (
    MEASURED_TARGET_DEVICE (SI Fig.S1)).
75 # Set explicitly for the named materials too: ATLAS built-in permittivity defaults are NOT the measured
    values.
76 material region=3 permittivity=9.0
77 material region=4 permittivity=19.57
78 contact name=gate workfunction=4.7
79 models fermi srh temp=358.15 print
80 # Region 1: continuous DOS. NTA cm^-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
81 # EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement).

```

```

82 defects region=1 continuous numa=384 numd=192 \
83   nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
84   nga=5e+16 ega=0.6 wga=0.15 \
85   ngd=0 egd=3.30331 wgd=0.1 \
86   sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
87   siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
88   afile=acceptor_r1.dat dfile=donor_r1.dat
89 # Region 2: continuous DOS. NTA cm-3/eV at Ec (tail intercept), WTA eV; NGA/NGD Gaussian amplitudes cm
    ^-3/eV;
90 #   EGA measured from Ec, EGD measured from Ev (installed manual, DEFECTS statement). Interface sheet
    regularized: NGA = sheet peak / layer thickness.
91 defects region=2 continuous numa=384 numd=192 \
92   nta=5e+20 wta=0.04 ntd=0 wtd=0.1 \
93   nga=1.16757e+19 ega=0.301554 wga=0.123975 \
94   ngd=0 egd=3.30331 wgd=0.1 \
95   sigtae=1e-15 sigtah=1e-15 sigtde=1e-15 sigtdh=1e-15 \
96   siggae=1e-15 siggah=1e-15 siggde=1e-15 siggdh=1e-15 \
97   afile=acceptor_r2.dat dfile=donor_r2.dat
98 interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.00000001 y.max=0.00000001
99 # Strict acceptance (XANDRNORM, cr.toler 1e-17) up to Vg = 0.7 V; default acceptance above.
100 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
101   cr.toler=1e-17 xandrnorm carriers=1 electrons
102 output con.band val.band e.mobility
103 # Interior probes (channel centre) for mobility / free-electron exports vs Vg (native solver quantities).
104 probe name=chan_mob x=12 y=-0.001 n.mob dir=0
105 probe name=chan_n x=12 y=-0.001 n.conc
106 probe name=avg_n region=1 average n.conc x.min=2 x.max=22 y.min=-0.002 y.max=-0.00025
107 solve init
108 save outf=equilibrium.str
109 # Bias ramp to the sweep start with stepped SOLVE blocks (no per-point statements).
110 solve vstep=-0.1 vfinal=-3 name=gate
111 solve vstep=0.1 vfinal=0.7 name=drain
112 log outf=transfer.log
113 # Stepped sweep (end V, step V): (-1, 0.2), (1.5, 0.05), (3, 0.1) - scored at the solved points only.
114 solve vgate=-3 vstep=0.2 vfinal=-1 name=gate
115 solve vstep=0.05 vfinal=0.7 name=gate
116 save outf=near_vth.str
117 # On-state: default convergence acceptance.
118 method newton trap maxtraps=10 itlimit=80 climt=1e-6 ir.tol=1e-19 \
119   cr.toler=5e-18 ^xandrnorm carriers=1 electrons
120 solve vstep=0.05 vfinal=1.5 name=gate
121 solve vstep=0.1 vfinal=3 name=gate
122 log off
123 save outf=final.str
124 output con.band val.band qfn e.mobility
125 extract init infile="transfer.log"
126 extract name="IdVg" curve(v."gate",i."drain") outfile="idvg.dat"
127 extract name="IgVg" curve(v."gate",i."gate") outfile="igvg.dat"
128 extract name="IsVg" curve(v."gate",i."source") outfile="isvg.dat"
129 extract name="mob_vg" curve(v."gate",probe."chan_mob") outfile="chan_mobility_vg.dat"
130 extract name="n_vg" curve(v."gate",probe."chan_n") outfile="chan_electrons_vg.dat"
131 extract name="navg_vg" curve(v."gate",probe."avg_n") outfile="avg_electrons_vg.dat"
132 extract init infile="equilibrium.str"
133 extract name="cb_eq" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_eq.dat"
134 extract name="vb_eq" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_eq.dat"
135 extract name="n_eq" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_eq.dat"
136 extract name="qfn_eq" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_eq.dat"

```



```
137 extract init infile="near_vth.str"
138 extract name="cb_vth" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_vth.dat"
139 extract name="vb_vth" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_vth.dat"
140 extract name="n_vth" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_vth.dat"
141 extract name="qfn_vth" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    "qfn_vth.dat"
142 extract init infile="final.str"
143 extract name="cb_on" curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="cb_on.dat"
144 extract name="vb_on" curve(depth, impurity="Valence Band Energy" material="All" mat.occno=1 x.val=12)
    outfile="vb_on.dat"
145 extract name="n_on" curve(depth, impurity="Electron Conc" material="All" mat.occno=1 x.val=12) outfile="
    n_on.dat"
146 extract name="qfn_on" curve(depth, impurity="Electron QFL" material="All" mat.occno=1 x.val=12) outfile="
    qfn_on.dat"
147 quit
```

C.5 Configuration files

The three YAML files define the reference configuration; the three JSON files are the campaign plans of 2026-09-25 (Section 6.7). Every run's overrides in Table A.2 are dotted paths into these files.

C.5.1 config/iwo_material_model.yaml

The single material model: bulk reference values, thickness laws, trap and charge parameters, contacts and temperature, each with its provenance label (Chapters 4 and 5).

```
1 schema: IWO_material_model_v1
2 status: V1 physics-constrained seed; electronic values are NOT ATLAS-calibrated until results/ say so
3 composition:
4   xW_at_percent: 2.0
5   definition: 'NOT DETERMINED FROM AVAILABLE DATA: the target paper states "IWO (~2% W)" without
6     specifying atomic %, cation %, mol% WO3 or target
7     composition; treated as a material label, never converted to a donor density.'
7 temperature_K:
8   value: 300.0
9   label: ASSUMED
10  note: measurement temperature not recorded in workbook
11 bulk_reference:
12   Eg_bulk_eV:
13     value: 3.05
14     label: LITERATURE_IWO
15     source: Fan2021_Nanomaterials Sec.3 (different sputtered IWO device)
16     uncertainty_eV: 0.2
17   chi_bulk_eV:
18     value: 4.3
19     label: LITERATURE_IWO
20     source: Fan2021_Nanomaterials Sec.3 (composition-mixing estimate)
21     uncertainty_eV: 0.3
22     consistency_check: bulk In203 CNL ~0.4 eV above Ec (Si2021/Lin2022/Wang2022); thin-film Ec rises by
23       dEc(t) so TNL appears deeper below Ec
24       -> EF farther below Ec -> n_FB down, Vth up
24 mstar_bulk_m0:
25   value: 0.208
```

```

26     label: MEASURED_RELATED_MATERIAL
27     source: Stokey2021_JAP optical Hall (pure In2O3 crystal); W-doping increases m* (Januar2026 PBE,
           qualitative) -> NOT DETERMINED numerically
28     eps_r:
29         value: 9.3
30         label: MEASURED_TARGET_DEVICE
31         source: Januar2026 SI Fig.S1 (C-V, IWO 9.30+/-0.04; not bundled, recorded by Astra)
32         alternatives:
33             bulk_crystal_static: 10.55
34             evaporated_film: 8.9
35             thickness_dependence: none imposed (no evidence)
36     Nv300_cm3:
37         value: 1.0e+19
38         label: ASSUMED
39         note: holes are not solved (electrons-only DD); enters only ni
40     hole_mobility_cm2Vs:
41         value: 0.01
42         label: ASSUMED
43     srh_tau_s:
44         value: 1.0e-06
45         label: ASSUMED
46         note: background SRH, negligible in n-type wide gap
47     thickness_laws:
48         dEg_QC_eV:
49             form: a * t_nm^(-p)
50             a: 0.9205
51             p: 1.3815
52             label: DFT_PURE_IN2O3_PROXY
53             fit_points:
54                 - t_nm: 0.95
55                   dE_eV: 0.94
56                   source: Lin2022 slab 0.95 nm
57                 - t_nm: 1.5
58                   dE_eV: 0.6
59                   source: Si2021 In2O3/Al2O3 1.5 nm (dEc)
60                 - t_nm: 1.98
61                   dE_eV: 0.33000000000000007
62                   source: Lin2022 slab 1.98 nm
63             note: PBE absolute gaps are NOT used; only slab-minus-bulk shifts. Pure In2O3 proxy for IWO; H-
64                 passivated/Al2O3-terminated slabs vs air/Al2O3
65                 in device. Uncertainty ~ +/-50 %.
66     band_edge_partition:
67         dEc_over_dEg: 1.0
68         label: DFT_PURE_IN2O3_PROXY
69         source: 'Si2021: Ec moves up, Ev almost unchanged'
70     Eg_eV: Eg_bulk + dEg_QC(t)
71     chi_eV: chi_bulk - dEc(t)
72     mstar_m0:
73         form: mstar_bulk_exp + b * t_nm^(-q)
74         b: 0.1311
75         q: 1.4121
76         label: MEASURED_RELATED_MATERIAL + DFT_PURE_IN2O3_PROXY increment
77     Nc300_cm3:
78         form: 2*(2*pi*m*kT/h^2)^1.5
79         label: CALCULATED
80         note: 3-D parabolic continuum DOS; at 2 nm the film is quasi-2D, so Nc is an effective continuum
81             parameter
82     Nt_cm3:
83         form: Nt2 * (2/t_nm)^alpha
84         Nt2: 2.0e+19
85         alpha: 0.75

```

```

84     label: 'FITTED (anchor from real-ATLAS 2 nm fit: NTA*WTA) consistent with LITERATURE_IWO trend (
      Januar: ~4x from 13.2 to 2 nm; 5.3e19 at 2
85       nm with theta_t convention)'
86   WTA_eV:
87     value: 0.04
88     label: FITTED, shared across thickness
89     Tt_K: 464.0
90     note: Januar2026 reports only dTt under stress; absolute Tt NOT DETERMINED FROM AVAILABLE DATA
91   NTA_cm3_eV: Nt(t)/WTA (gTA(E)=NTA*exp((E-Ec)/WTA); integral over the gap = NTA*WTA*(1-exp(-Eg/WTA)) ~
      NTA*WTA)
92   Nd_eff_cm3:
93     form: Nd0 * (1 + (t_nm/tc)^k)
94     Nd0: 2.5e+17
95     tc_nm: 20.0
96     k: 4.0
97     label: FITTED law (effective electrically active background donor density, NOT W concentration)
98     support: 'Kim2024 (different IWO process): bulk/border trap density 2.4/3.1/14.6e18 at 10/20/30 nm;
      Januar2026: EF0 deeper for thinner films'
99   mu0_cm2Vs:
100     form: mu_band(t) * (1 - Dsr/t)^2
101     Dsr_nm: 0.287
102     Dsr_label: LITERATURE_IWO (Januar2026 Fig.5f, 2.87 A)
103     mu_band_fitted:
104       2.0: 16.8
105       6.3: 12.4
106       13.2: 57.6
107       31.8: 84.0
108     label: FITTED per thickness (no defensible law; sigmoidal trend attributed to percolation/
      crystallinity, hypothesis)
109     gate_dependence: 'NOT inserted natively: ATLAS 5.28.1.R ignores MOBILITY updates between SOLVEs (
      Claude run_0008); the free/trapped partition
110       of Eq.(5) is produced self-consistently by the DOS occupancy; the Ando roll-off theta_r*Vov is an
      approximation gap (see LIMITATIONS)'
111   traps:
112     bulk_acceptor_tail:
113       atlas: DEFECTS NTA WTA (cm^-3/eV, eV), acceptor-like, referenced to Ec
114       capture_cm2: 1.0e-15
115       capture_label: ASSUMED
116     bulk_deep_acceptor_gaussian:
117       NGA_cm3_eV: 5.0e+16
118       EGA_from_Ec_eV: 0.6
119       WGA_eV: 0.15
120       label: ASSUMED (insensitive; Fan2021 range 2.5-3.7e16 for a different device)
121     bulk_donor_tail:
122       NTD_cm3_eV: 0
123       label: ASSUMED zero (valence tail has negligible effect on n-type transfer curves; Fan2021)
124     shallow_donor_gaussian:
125       NGD_cm3_eV: 0
126       label: 'ASSUMED zero in V1: the effective donor background is represented by Nd_eff(t); a donor
      Gaussian at fixed absolute energy (EGD referenced
127       to Ev) is the V2 candidate to replace Nd_eff(t) by the confinement mechanism'
128     interface_acceptor_sheet:
129       peak_cm2_eV: 300000000000.0
130       center_from_Ec_eV: 0.3
131       width_eV: 0.12
132       label: ASSUMED, shared across thickness; literature Dit(HfO2/In2O3) ~6e11 cm^-2/eV (
      LITERATURE_IN2O3_PROXY; here IWO/Al2O3)
133     implementation: 'regularized 0.25 nm IWO layer at the Al2O3 interface: volume amplitude = sheet peak
      / 0.25 nm (numerically verified: layer
134       /2 changes Id by 0.009 dec)'
135     fixed_interface_charge_Qf_cm2:

```

```

136     value: 0.0
137     label: FITTED single shared value (stage 3); classified as effective electrostatic correction
138 back_surface_donor_sheet:
139     peak_cm2_eV: 0.0
140     center_from_Ec_eV: 0.02
141     width_eV: 0.05
142     label: 'HYPOTHESIS, 31.8 nm only when non-zero (device-specific, stage 6): donor-like states at the
143           free IWO back surface, regularized in
144           a 0.25 nm layer (region 5). Zero for the thin films. Not identifiable separately from Nd and Qf
145           with one transfer curve (V0 evidence: needed
146           to reproduce the flat always-on baseline).'
147 contacts:
148     gate_workfunction_eV:
149         value: 4.7
150         label: ASSUMED (TiN 4.5-4.9 eV literature range); degenerate with Qf and chi -> held fixed
151     source_drain:
152         model: ideal Ohmic (no workfunction)
153         label: ASSUMED; underconstrained (no TLM); CNL above Ec in In2O3 supports low-barrier contacts (
154             Si2O21/Lin2O22)
155     series_resistance_ohm_um:
156         '2.0': 0
157         '6.3': 0
158         '13.2': 0
159         '31.8': 92000.0
160         label: 31.8 nm only, FITTED hypothesis (on-state saturation); total device 672 ohm at 3 V bounds it
161 geometry:
162     L_um: 20.0
163     W_um: 290.0
164     sim_width_um: 1.0
165     hfo2_nm: 15.0
166     al2o3_nm: 2.0
167     contact_overlap_um:
168         value: 2.0
169         label: ASSUMED numerical contact geometry (not reported)
170     eps_hfo2: 19.57
171     eps_al2o3:
172         value: 9.0
173         label: ASSUMED
174     top_of_channel: AIR (no cap)

```

C.5.2 config/geometry.yaml

Stack, structure form, channel dimensions, width convention, overlap, drain bias, sweep range and mesh (Sections 3.2 and 6.2).

```

1 # Device geometry (STAGE 1: locked). Sources: Januar2026 Sec.2.1; user schematic (inputs/
2   device_and_transfer_curves.png, PPTX slide).
3 stack_bottom_to_top:
4   - {layer: n+ Si support, role: screened by the continuous TiN gate; omitted from the electrical domain,
5     label: MEASURED_TARGET_DEVICE}
6   - {layer: TiN gate, thickness_nm: 50, role: ideal equipotential boundary (reduced deck) / Conductor
7     volume (full deck), label: MEASURED_TARGET_DEVICE}
8   - {layer: HfO2, thickness_nm: 15.0, eps_r: 19.57, eps_label: MEASURED_TARGET_DEVICE (SI Fig.S1), label:
9     MEASURED_TARGET_DEVICE}
10  - {layer: Al2O3, thickness_nm: 2.0, eps_r: 9.0, eps_label: ASSUMED (not reported), label:
11    MEASURED_TARGET_DEVICE}
12  - {layer: IWO channel, thickness_nm: [2.0, 6.3, 13.2, 31.8], label: MEASURED_TARGET_DEVICE}
13  - {layer: Pd source/drain, thickness_nm: 70, role: ideal contacts on the IWO top surface (reduced) /
14    Palladium volumes (full), label: MEASURED_TARGET_DEVICE}

```

```

9   - {layer: top of channel, material: AIR, note: no cap layer is specified by the source; none is added}
10  structure: full          # 'full' = physical volumes: Air above the channel, Pd S/D volumes, Al2O3
    + HfO2 regions, Conductor gate volume (TiN); 'reduced' = ideal boundaries (runs 0001-0009,
    electrically equivalent in DC)
11  channel_length_um: 20.0
12  physical_width_um: 290.0
13  simulated_width_um: 1.0      # MESH WIDTH=1 -> logged currents are numerically A/um; I_total = 290 x I(
    A/um). Never divide by 290.
14  contact_overlap_um: {value: 2.0, label: ASSUMED numerical contact geometry (overlap not reported);
    sensitivity test listed in LIMITATIONS}
15  drain_bias_V: 0.7           # from the schematic (workbook has no units)
16  gate_sweep_V: {start: -3.0, stop: 3.0, step: 0.05}
17  mesh:
18    x_um: [[0, 0.3], [1.8, 0.05], [2.0, 0.01], [2.2, 0.05], [3.0, 0.2], [12.0, 0.7], [21.0, 0.2], [21.8,
    0.05], [22.0, 0.01], [22.2, 0.05], [24.0, 0.3]]
19    interface_layer_nm: 0.25   # regularization layer for the interface sheet DOS (numerically verified
    : halving changes Id by 0.009 dec)
20    y_spacing_interface_nm: 0.0625
21    y_spacing_bulk_nm: "min(1.0, t/8)" # 2 nm: 0.25 nm uniform; thick films graded (32-bit ATLAS memory
    limit ~30k nodes)
22    y_spacing_al2o3_nm: 0.25
23    y_spacing_hfo2_nm: 1.0

```

C.5.3 config/solver.yaml

Models, carriers, METHOD settings, strict acceptance, DOS levels, sweep segments, runner recipe, timeouts, launch budget, KCL limits and scoring rules (Sections 6.3 to 6.5).

```

1  # Solver / models (NUMERICAL category). Every enabled model has a stated reason (docs/PHYSICS_MODEL.md).
2  models: "fermi srh temp=300 print"          # Fermi statistics (degenerate accumulation, low Nc); SRH
    background only
3  carriers: "carriers=1 electrons"            # holes numerically zero in the 3.05 eV n-type oxide (Jp 1e
    -38..1e-55 A in every Claude/Astra run); no hole conduction modelled
4  method_base: "newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19"
5  strict_acceptance:
6    cr_toler_A: 1.0e-17                    # XANDRNORM + tightened continuity tolerance in depletion (
    prevents 1e-15 A glitches: Claude run_0006)
7    cr_tol_rel_active_min: 1.0e-5          # strict CR.TOLER = max(cr_toler_A, 1e-5 x smallest measured
    active current) (thick films: run_0018)
8    until_measured_current_A_per_um: 1.0e-9 # above this measured current the default acceptance (^
    xandrnorm, cr.toler 5e-18) is used (on-state RHS floor scales with current: run_0011)
9    dos_levels: {numa: 96, numd: 48}        # x2 check changes Id by 0.006 dec (Claude run_0016)
10   continuation_gate_step_V: 0.1          # ramp step (stepped SOLVE VSTEP block to the sweep start)
11   sweep:
12     mode: stepped                        # ATLAS SOLVE VSTEP=.. VFINAL=.. NAME=gate blocks (user
    directive 2026-09-21); 'pointwise' = one SOLVE per 0.05 V point (runs 0001-0009)
13     segments: [[-1.0, 0.2], [1.5, 0.05], [3.0, 0.1]] # [end V, step V] from the -3 V start: coarse in
    deep depletion, 0.05 V around turn-on (Vth/SS extraction), 0.1 V in the on-state
14   output: "con.band val.band e.mobility"
15   runner:
16     argv: ["C:/sedatools/exe/deckbuild.exe", "-run", "{deck}", "-outfile", "{stdout}"] # verified batch
    recipe (cwd = run dir, relative names)
17     timeout_seconds: {default: 900, thick: 1500}
18     max_launches: 71                    # +14 on 2026-10-05 for model V2 (user-approved "approved, go
    ahead with all steps": off-current path, DFT inputs, predictive 6.3 nm test, 1/0.7/0.5 nm); +2 on
    2026-09-28 for the parallel pilot (user-approved); raised to 54 on 2026-09-27 (overlap redo +
    explicit tmun check, user-approved); earlier raised from 30 on 2026-09-25 at the user's explicit
    request (6.3 nm hypothesis study, DOS 384/192 recalibration, mesh, sensitivity, predictions: ~20
    launches)

```

```

19 kcl_absolute_limit_A: 1.0e-17
20 kcl_floor_fraction: 0.1          # applied absolute limit = max(1e-17 A, 0.1 x smallest
    measured current of the device)
21 kcl_relative_limit: 0.01
22 expected_drain_sign: 1
23 scoring:
24   active_multiplier: 5.0          # active = measured Id > 5 x median(measured Id, -2..-0.5 V);
    all points for t >= 20 nm
25   logging_epsilon_A_per_um: 1.0e-40 # native zeros are reported, never replaced by a floor

```

C.5.4 config/campaign_A_6p3_hypotheses.json

```

1 {
2   "campaign": "A_6p3_hypotheses",
3   "purpose": "Controlled one-mechanism-at-a-time study of the 6.3 nm validation miss (run_0014: shared Qf
    1.73e12, mu_band 12.4, DOS 96/48, physical structure, stepped sweep; Vth_cc 0.35 vs measured 0.64 V)
    . Every entry changes only the named mechanism relative to run_0014 (A7 is relative to the tuned
    run_0015). Parameter values are chosen from physical reasoning BEFORE the run, not fitted.
    2026-09-25, user request.",
4   "runs": [
5     {"id": "A3_back_charge", "thickness": 6.3, "label": "hyp_H3_backQ_mlp64e12",
6      "purpose": "Back-surface (IWO/Air, exposed channel only) fixed NEGATIVE sheet charge equal to the
        front-charge deficit of the tuned run (1.73e12 - 8.7e10 = 1.64e12 cm^-2), e.g. chemisorbed O2-/OH-.
        Tests whether the same charge placed at the air-side surface reproduces the curve (location
        identifiability).",
7      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=1.73e12", "traps.
        back_surface_fixed_charge_cm2=-1.64e12"]},
8     {"id": "A1_thickness", "thickness": 6.3, "label": "hyp_H1_tphys_5p3",
9      "purpose": "Thickness error: physical film 5.3 nm (-1.0 nm, -16 %, larger than a typical XRR/
        ellipsometry error) with ALL thickness laws (dEc, m*, Nt, Nd, roughness) evaluated at 5.3 nm;
        mu_band held at the 6.3 nm validation value 12.4.",
10      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=1.73e12", "geometry.
        film_thickness_override_nm=5.3", "thickness_laws.mu0_cm2Vs.mu_band_fitted[5.3]=12.4"]},
11     {"id": "A2_donor", "thickness": 6.3, "label": "hyp_H2_Nd0_1e16",
12      "purpose": "Reduced donor density: effective background donors removed (Nd0 2.5e17 -> 1e16). Analytic
        bound: q*Nd*t/Cox = 0.028 V at 2.5e17, so this mechanism alone cannot exceed ~+0.03 V.",
13      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=1.73e12", "thickness_laws.Nd_eff_cm3.Nd0=1e16
        "]}},
14     {"id": "A4_front_dit", "thickness": 6.3, "label": "hyp_H4_front_Dit_x5",
15      "purpose": "Energy-distributed front-interface acceptor traps instead of a fixed charge: Dit peak 3e11
        -> 1.5e12 cm^-2 eV^-1 (x5). Discriminating prediction: traps shift Vth AND degrade SS; a fixed
        charge shifts Vth only. Measured SS 115 mV/dec bounds it.",
16      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=1.73e12", "traps.interface_acceptor_sheet.
        peak_cm2_eV=1.5e12"]},
17     {"id": "A6_no_confinement_storyB", "thickness": 6.3, "label": "hyp_H6_noQC_Qf_from_2nm",
18      "purpose": "Confinement-necessity test (held-out prediction, symmetric to run_0014): NO confinement (
        dEg a=0, m* b=0) and the single Qf that makes the no-confinement 2 nm run (run_0016) match the
        measured 2 nm Vth_cc exactly (Qf shift is an exact rigid shift: 1.73e12 - (0.6618-0.3608) V x Cox/q
        = 4.76e10). Tests whether classical physics calibrated on 2 nm alone predicts 6.3 nm.",
19      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=4.76e10", "thickness_laws.dEg_QC_eV.a=0", "
        thickness_laws.mstar_m0.b=0"]},
20     {"id": "A5_combination", "thickness": 6.3, "label": "hyp_H5_combo_t5p8_Nd1e16_backQ_m1e12",
21      "purpose": "Combination at individually plausible values: film 5.8 nm (-0.5 nm), donors removed (Nd0 1
        e16), back-surface charge -1.0e12 cm^-2 (typical adsorbate scale). Tests whether plausible
        magnitudes add up to the +0.29 V miss without any single implausible value.",
22      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=1.73e12", "geometry.
        film_thickness_override_nm=5.8", "thickness_laws.mu0_cm2Vs.mu_band_fitted[5.8]=12.4", "
        thickness_laws.Nd_eff_cm3.Nd0=1e16", "traps.back_surface_fixed_charge_cm2=-1.0e12"]},
23     {"id": "A7_mesh", "thickness": 6.3, "label": "check_L_mesh0p7_tuned", "timeout": 1500,

```

```

24     "purpose": "Numerical check L (open): 6.3 nm tuned configuration (run_0015) with all mesh spacings x0
      .7 (x0.5 would exceed the 32-bit node limit, as at 13.2 nm).",
25     "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=8.7e10", "thickness_laws.mu0_cm2Vs.
      mu_band_fitted[6.3]=11.69", "geometry.mesh.spacing_factor=0.7"]]
26   ]
27 }

```

C.5.5 config/campaign_B_recal_predictions.json

```

1  {
2    "campaign": "B_recal_predictions",
3    "purpose": "(1) DOS 96/48 -> 384/192 recalibration of the three final configurations (one converged run
      per film; mu_band pre-scaled by 1/1.025 from the 2 nm run_0029 offset, then rescaled EXACTLY by
      Ion_meas/Ion_sim - Id is linear in a constant mobility); (2) PROSPECTIVE predictions recorded before
      any measurement: ID-VD families (Vg = 1, 1.5, 2, 2.5, 3 V; Vd 0-3 V) and transfer curves at 65 C
      (338.15 K, 2 nm) and 85 C (358.15 K, all films) - the temperatures of the Januar SI Fig. S12
      elevated-temperature transfer curves of the 2 nm device, which are NOT in our possession (prediction
      recorded before we see them) with the recalibrated parameters; (3) parameter-isolation tests: gate
      work function +0.1 eV alone and m* x1.3 alone at 2 and 13.2 nm, relative to the B1/B3 baselines (
      same mu). Temperature runs keep mu_band, NTA/NTA, Qf and Eg temperature-independent (egalpha=0); Nc
      scales as T^1.5 in ATLAS; these are the stated assumptions of the prediction. 2026-09-25, user
      request. B0 (added after specialist S2): pre-registered test of the S2 trap/mobility re-partition at
      6.3 nm (Nt(6.3) = Nt(2 nm)-like 2.66e19, mu_band 19.6, Qf 9.8e11; S2 prediction before the run:
      Vth_cc 0.64+/-0.03 V, Vth_lin 1.82-1.90 V, mu_FE 10.5-11.0, Ion -5.-10 %), DOS 96/48 for
      comparability with campaign A.",
4    "runs": [
5      {
6        "id": "B0_repartition_6p3",
7        "thickness": 6.3,
8        "label": "hyp_S2_repartition_Nt2p66e19_mu19p6",
9        "purpose": "S2 pre-registered prediction test: is the 6.3 nm on-state shape miss (Vth_lin -0.35 V,
      gm -23 %) a trap/mobility mis-partition of the Nt(t) law? Nt fixed at 2.66e19 (alpha 0), mu_band
      19.6, Qf 9.8e11, DOS 96/48.",
10       "overrides": [
11         "traps.fixed_interface_charge_Qf_cm2.value=9.8e11",
12         "thickness_laws.Nt_cm3.alpha=0",
13         "thickness_laws.Nt_cm3.Nt2=2.66e19",
14         "thickness_laws.mu0_cm2Vs.mu_band_fitted[6.3]=19.6"
15       ]
16     },
17     {
18       "id": "B1_dos384_2p0",
19       "thickness": 2.0,
20       "label": "recal_dos384_2p0",
21       "mu_used": 17.69,
22       "recal_key": "2p0",
23       "purpose": "Converged-DOS (384/192) recalibration of the 2 nm film: mu_band 18.13/1.0249 = 17.69 (
      run_0029 offset +2.49 %).",
24       "overrides": [
25         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
26         "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]=17.69",
27         "solver.dos_levels.numa=384",
28         "solver.dos_levels.numd=192"
29       ]
30     },
31     {
32       "id": "B2_dos384_6p3",
33       "thickness": 6.3,
34       "label": "recal_dos384_6p3_tuned",

```

```

35     "mu_used": 11.41,
36     "recal_key": "6p3",
37     "purpose": "Converged-DOS recalibration of the 6.3 nm tuned configuration (device-specific Qf 8.7
e10): mu_band 11.69/1.025 = 11.41.",
38     "overrides": [
39         "traps.fixed_interface_charge_Qf_cm2.value=8.7e10",
40         "thickness_laws.mu0_cm2Vs.mu_band_fitted[6.3]=11.41",
41         "solver.dos_levels.numa=384",
42         "solver.dos_levels.numd=192"
43     ]
44 },
45 {
46     "id": "B3_dos384_13p2",
47     "thickness": 13.2,
48     "label": "recal_dos384_13p2",
49     "mu_used": 60.39,
50     "recal_key": "13p2",
51     "timeout": 1500,
52     "purpose": "Converged-DOS recalibration of the 13.2 nm film: mu_band 61.9/1.025 = 60.39.",
53     "overrides": [
54         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
55         "thickness_laws.mu0_cm2Vs.mu_band_fitted[13.2]=60.39",
56         "solver.dos_levels.numa=384",
57         "solver.dos_levels.numd=192"
58     ]
59 },
60 {
61     "id": "B8_idvd_2p0",
62     "thickness": 2.0,
63     "label": "pred_idvd_2p0",
64     "mode": "output",
65     "output_vgs": [
66         1.0,
67         1.5,
68         2.0,
69         2.5,
70         3.0
71     ],
72     "vd_segments": [
73         [
74             0.5,
75             0.05
76         ],
77         [
78             3.0,
79             0.1
80         ]
81     ],
82     "timeout": 2000,
83     "purpose": "PREDICTION: ID-VD at Vg = 1, 1.5, 2, 2.5, 3 V, 2 nm, recalibrated parameters (ideal
Ohmic contacts, constant mobility).",
84     "overrides": [
85         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
86         "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]={mu:2p0}",
87         "solver.dos_levels.numa=384",
88         "solver.dos_levels.numd=192"
89     ]
90 },
91 {
92     "id": "B9_idvd_6p3",
93     "thickness": 6.3,

```



```
94     "label": "pred_idvd_6p3_tuned",
95     "mode": "output",
96     "output_vgs": [
97         1.0,
98         1.5,
99         2.0,
100        2.5,
101        3.0
102    ],
103    "vd_segments": [
104        [
105            0.5,
106            0.05
107        ],
108        [
109            3.0,
110            0.1
111        ]
112    ],
113    "timeout": 2000,
114    "purpose": "PREDICTION: ID-VD at Vg = 1, 1.5, 2, 2.5, 3 V, 6.3 nm tuned configuration, recalibrated",
115    "overrides": [
116        "traps.fixed_interface_charge_Qf_cm2.value=8.7e10",
117        "thickness_laws.mu0_cm2Vs.mu_band_fitted[6.3]={mu:6p3}",
118        "solver.dos_levels.numa=384",
119        "solver.dos_levels.numd=192"
120    ]
121 },
122 {
123     "id": "B10_idvd_13p2",
124     "thickness": 13.2,
125     "label": "pred_idvd_13p2",
126     "mode": "output",
127     "output_vgs": [
128         1.0,
129         1.5,
130         2.0,
131         2.5,
132         3.0
133     ],
134     "vd_segments": [
135         [
136            0.5,
137            0.05
138        ],
139        [
140            3.0,
141            0.1
142        ]
143    ],
144    "timeout": 3000,
145    "purpose": "PREDICTION: ID-VD at Vg = 1, 1.5, 2, 2.5, 3 V, 13.2 nm, recalibrated.",
146    "overrides": [
147        "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
148        "thickness_laws.mu0_cm2Vs.mu_band_fitted[13.2]={mu:13p2}",
149        "solver.dos_levels.numa=384",
150        "solver.dos_levels.numd=192"
151    ]
152 },
153 {
```

```

154     "id": "B11_T338_2p0",
155     "thickness": 2.0,
156     "label": "pred_T338K_2p0",
157     "timeout": 1500,
158     "purpose": "PREDICTION: transfer curve (Vd 0.7 V) at 65 C (338.15 K), 2p0, recalibrated DOS
384/192; mu_band, NTA/WTa, Qf, Dit T-independent; Eg fixed (egalpha=0); Nc ~ T^1.5 (ATLAS).",
159     "overrides": [
160         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
161         "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]={mu:2p0}",
162         "solver.dos_levels.numa=384",
163         "solver.dos_levels.numd=192",
164         "solver.models=fermi srh temp=338.15 print",
165         "temperature_K.value=338.15"
166     ]
167 },
168 {
169     "id": "B12_T358_2p0",
170     "thickness": 2.0,
171     "label": "pred_T358K_2p0",
172     "timeout": 1500,
173     "purpose": "PREDICTION: transfer curve (Vd 0.7 V) at 85 C (358.15 K), 2p0, recalibrated DOS
384/192; mu_band, NTA/WTa, Qf, Dit T-independent; Eg fixed (egalpha=0); Nc ~ T^1.5 (ATLAS).",
174     "overrides": [
175         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
176         "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]={mu:2p0}",
177         "solver.dos_levels.numa=384",
178         "solver.dos_levels.numd=192",
179         "solver.models=fermi srh temp=358.15 print",
180         "temperature_K.value=358.15"
181     ]
182 },
183 {
184     "id": "B13_T358_6p3",
185     "thickness": 6.3,
186     "label": "pred_T358K_6p3",
187     "timeout": 1500,
188     "purpose": "PREDICTION: transfer curve (Vd 0.7 V) at 85 C (358.15 K), 6p3, recalibrated DOS
384/192; mu_band, NTA/WTa, Qf, Dit T-independent; Eg fixed (egalpha=0); Nc ~ T^1.5 (ATLAS).",
189     "overrides": [
190         "traps.fixed_interface_charge_Qf_cm2.value=8.7e10",
191         "thickness_laws.mu0_cm2Vs.mu_band_fitted[6.3]={mu:6p3}",
192         "solver.dos_levels.numa=384",
193         "solver.dos_levels.numd=192",
194         "solver.models=fermi srh temp=358.15 print",
195         "temperature_K.value=358.15"
196     ]
197 },
198 {
199     "id": "B14_T358_13p2",
200     "thickness": 13.2,
201     "label": "pred_T358K_13p2",
202     "timeout": 1500,
203     "purpose": "PREDICTION: transfer curve (Vd 0.7 V) at 85 C (358.15 K), 13p2, recalibrated DOS
384/192; mu_band, NTA/WTa, Qf, Dit T-independent; Eg fixed (egalpha=0); Nc ~ T^1.5 (ATLAS).",
204     "overrides": [
205         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
206         "thickness_laws.mu0_cm2Vs.mu_band_fitted[13.2]={mu:13p2}",
207         "solver.dos_levels.numa=384",
208         "solver.dos_levels.numd=192",
209         "solver.models=fermi srh temp=358.15 print",
210         "temperature_K.value=358.15"

```

```

211     ]
212 },
213 {
214     "id": "B4_wf_2p0",
215     "thickness": 2.0,
216     "label": "sens_wf_p0p1_2p0_dos384",
217     "purpose": "Isolation: gate work function 4.70 -> 4.80 eV alone at 2 nm (baseline B1, same mu
17.69).",
218     "overrides": [
219         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
220         "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]=17.69",
221         "solver.dos_levels.numa=384",
222         "solver.dos_levels.numd=192",
223         "contacts.gate_workfunction_eV.value=4.8"
224     ]
225 },
226 {
227     "id": "B5_wf_13p2",
228     "thickness": 13.2,
229     "label": "sens_wf_p0p1_13p2_dos384",
230     "timeout": 1500,
231     "purpose": "Isolation: gate work function +0.1 eV alone at 13.2 nm (baseline B3, same mu 60.39).",
232     "overrides": [
233         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
234         "thickness_laws.mu0_cm2Vs.mu_band_fitted[13.2]=60.39",
235         "solver.dos_levels.numa=384",
236         "solver.dos_levels.numd=192",
237         "contacts.gate_workfunction_eV.value=4.8"
238     ]
239 },
240 {
241     "id": "B6_mstar_2p0",
242     "thickness": 2.0,
243     "label": "sens_mstar_x1p3_2p0_dos384",
244     "purpose": "Isolation: m* x1.3 alone (bulk 0.208 -> 0.2704, increment b 0.1311 -> 0.1704; enters
ATLAS only through Nc) at 2 nm; dEc law unchanged.",
245     "overrides": [
246         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
247         "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]=17.69",
248         "solver.dos_levels.numa=384",
249         "solver.dos_levels.numd=192",
250         "bulk_reference.mstar_bulk_m0.value=0.2704",
251         "thickness_laws.mstar_m0.b=0.17043"
252     ]
253 },
254 {
255     "id": "B7_mstar_13p2",
256     "thickness": 13.2,
257     "label": "sens_mstar_x1p3_13p2_dos384",
258     "timeout": 1500,
259     "purpose": "Isolation: m* x1.3 alone at 13.2 nm (baseline B3).",
260     "overrides": [
261         "traps.fixed_interface_charge_Qf_cm2.value=1.73e12",
262         "thickness_laws.mu0_cm2Vs.mu_band_fitted[13.2]=60.39",
263         "solver.dos_levels.numa=384",
264         "solver.dos_levels.numd=192",
265         "bulk_reference.mstar_bulk_m0.value=0.2704",
266         "thickness_laws.mstar_m0.b=0.17043"
267     ]
268 }
269 ]

```

270 }

C.5.6 config/campaign_C_6p3_nearEc_band.json

```

1 {
2   "campaign": "C_6p3_nearEc_band",
3   "purpose": "Pre-registered test of specialist S4's hypothesis for the 6.3 nm ON-STATE SHAPE miss (tuned
      run_0015 / B2: Vth_lin-Vth_cc gap too small by ~0.35 V, gm_max -23 %): an extra ~2e12 cm^-2 of
      acceptor-like states within ~1 kT of Ec plus a higher band mobility. Deck: tuned 6.3 nm
      configuration at DOS 384/192 (baseline B2_dos384_6p3) with the bulk acceptor Gaussian replaced by
      NGA = 2e12/(6.3e-7 x 0.02 x sqrt(pi)) = 9.0e19 cm^-3 eV^-1, EGA = 0.03 eV below Ec, WGA = 0.02 eV,
      and mu_band 15.4 (S4 surrogate). Qf is NOT refitted in ATLAS: a Qf change is an exact rigid shift (
      runs 0001->0003, 0005->0007), so the shape metrics (gap, gm_max, SS_cc) are read directly and Vth_cc
      is corrected analytically. S4 PREDICTION recorded before the run (0-D surrogate, relative to
      run_0015): gap +0.28 V, gm_max +24 %, SS_cc +23 mV/dec. S1's competing expectation for gap-state
      bands (1-D): the gap does not open without SS_cc rising well above the measured 295 mV/dec.
      Approximation: in the 0.25 nm interface layer the band is moment-matched with the interface sheet (
      generator), and the assumed deep Gaussian (5e16 at Ec-0.6 eV, insensitive) is replaced.
      2026-09-25.",
4   "runs": [
5     {"id": "C1_nearEc_band_6p3", "thickness": 6.3, "label": "hyp_S4_nearEc_band_2e12_mu15p4",
6      "purpose": "S4 near-Ec acceptor band (2e12 cm^-2 at Ec-0.03 eV, W 0.02 eV) + mu_band 15.4 on the tuned
        6.3 nm configuration, DOS 384/192.",
7      "overrides": ["traps.fixed_interface_charge_Qf_cm2.value=8.7e10", "thickness_laws.mu0_cm2Vs.
        mu_band_fitted[6.3]=15.4", "solver.dos_levels.numa=384", "solver.dos_levels.numd=192",
8        "traps.bulk_deep_acceptor_gaussian.NGA_cm3_eV=9.0e19", "traps.
        bulk_deep_acceptor_gaussian.EGA_from_Ec_eV=0.03", "traps.bulk_deep_acceptor_gaussian.WGA_eV=0.02"]}
9   ]
10  }
```

Summary. All decks share one generated skeleton; a run differs from the reference only through the overrides recorded in the run catalogue, and thickness enters only through the values written by the generator from the material laws. The eleven listed decks cover every kind of change made in the study: the final DOS 384/192 calibration, the 6.3 nm shared-parameter test, tuning and hypothesis tests, the pre-registered B0 and C1 tests, and the I_D - V_D and temperature predictions.

Appendix D

Prediction register: files and hashes

This appendix documents the integrity record of the prediction register discussed in Chapter 10. The register `analysis_2026-09-25/PREDICTIONS_REGISTER.md` was written at 2026-09-24T23:24:53Z (UTC) by specialist S6. Every number in it was read from the real ATLAS runs `run_0038–run_0047` or derived from them by exact operations (linear mobility rescaling, the $(T/300)^{1.5}$ factor, metric extraction with the unchanged `scripts/extract_metrics.py`). None of the predicted quantities was used in calibration, and no measurement of them was in the project’s possession at registration time.

Registration statement. At registration, the SHA-256 hashes of 44 files were computed with Python `hashlib`: the measured data, the extractor, the material model and campaign configuration, the input deck, execution record and output of every reference and prediction run (`run_0038–run_0047`), the S6 analysis scripts and outputs, and the canonical machine-readable prediction file `analysis_2026-09-25/predictions_canonical.csv` (1917 rows; columns `film`, `T_K`, `variant`, `Vg`, `Vd`, `quantity`, `value`, `unit`, `run_id`). On 2026-09-25 the synthesis re-computed all 44 hashes (`scratch/SYNTHESIS/syn_checks.py`, check [SYN-c]); all 44 are identical to the registered values (Table D.1). The SHA-256 of the register file itself is `836ead1f731745f95e339dedff351c2cc4dae5c6cdd3979d5c`.

Known textual error. The register prose misprints the exact P-MTR multipliers 1.19669 and 1.30441 as 1.19665 and 1.30426 while the code, tables and canonical CSV are exact; following Section 10.3 the register is not edited, and the details are in Section 5.8.

Limitations. The timestamp is internal. No third-party timestamp (repository commit, OSF/Zenodo deposit, or e-mail to the supervisor) exists yet; checklist item 0.2 (Section 11.3) requires one before the comparison data are requested or opened. Any later change to the registered files, the extraction definitions or the model state invalidates the prospective status; a revised register must be a new, separately timestamped file.

Table D.1. Files of the prediction register with their SHA-256 as registered on 2026-09-24T23:24:53Z (UTC) and re-verified at 2026-10-05T18:54:30Z.

file (relative to the package root)	SHA-256 (first 32 hex)	re-check
<code>data/experimental_clean.csv</code>	<code>e27a53e8c72169d5e19cdf666c9203f8</code>	identical

file	SHA-256	re-check
scripts/extract_metrics.py	a7d0b236ecfa4f67433fc842a2b9cebe	identical
config/iwo_material_model.yaml	ab063e8d20b7e57bff498096294183b1	identical
config/campaign_B_recal_predictions.json	3247b99c427801d417c3e3388e80289d	identical
results/campaigns/campaign_B_recal_predictions.json	8b9be148cd9b6fb5dd327d35a5c36a6b	identical
results/runs/run_0038_2p0_recal_dos384_2p0/idvg.dat	cefecae05b2e76ee97c36763f099230c	identical
results/runs/run_0038_2p0_recal_dos384_2p0/execution.json	d6e68576e7471d3e8451a6b0c9eb9bb9	identical
results/runs/run_0038_2p0_recal_dos384_2p0/device.in	165bbbb59bd379b6c310f5a3469b4b8f	identical
results/runs/run_0039_6p3_recal_dos384_6p3_tuned/idvg.dat	bb7d26ec07baf7068ef3a9d92d7eb08c	identical
results/runs/run_0039_6p3_recal_dos384_6p3_tuned/execution.json	d3695bc38c7a0443821838c9bca1d6b	identical
results/runs/run_0039_6p3_recal_dos384_6p3_tuned/device.in	184189690ce24b31a1a0270d14e0f3ad	identical
results/runs/run_0040_13p2_recal_dos384_13p2/idvg.dat	3c0f81cf430abf0bb923b0b665272786	identical
results/runs/run_0040_13p2_recal_dos384_13p2/execution.json	edd53139cc9351b16b98b8bcf597599c	identical
results/runs/run_0040_13p2_recal_dos384_13p2/device.in	11020ebbbea8ae1a494e753c28e473b	identical
results/runs/run_0041_2p0_pred_idvd_2p0/output_characteristics.csv	44447d5f2a6d530f23eaf1f5138f1b979	identical
results/runs/run_0041_2p0_pred_idvd_2p0/execution.json	8dd4220c82167f041af8f03fc04d7fc3	identical
results/runs/run_0041_2p0_pred_idvd_2p0/device.in	80b4962a3e46230bd736e8b4b3d1fa67	identical
results/runs/run_0042_6p3_pred_idvd_6p3_tuned/output_characteristics.csv	746f90c63af72995fbc7a5d35bbca1	identical
results/runs/run_0042_6p3_pred_idvd_6p3_tuned/execution.json	d85f19f36ed6ac60d5e3151d973f9ca2	identical
results/runs/run_0042_6p3_pred_idvd_6p3_tuned/device.in	1132e600cd9374ee037f81281e48dc3e	identical
results/runs/run_0043_13p2_pred_idvd_13p2/output_characteristics.csv	1c7708f4e7f0106217e735c4fccd45c	identical
results/runs/run_0043_13p2_pred_idvd_13p2/execution.json	6159ada850361d8e1b84d9cd3bf214fd	identical
results/runs/run_0043_13p2_pred_idvd_13p2/device.in	b0eac4011200c8bf282a879b40a3ced8	identical
results/runs/run_0044_2p0_pred_T338K_2p0/idvg.dat	0b4cef05172bfd34337b50fcc2f74021	identical
results/runs/run_0044_2p0_pred_T338K_2p0/execution.json	7124ae3006202185ce312c8d0a3c5364	identical
results/runs/run_0044_2p0_pred_T338K_2p0/device.in	26b7083e160859384e5127f3e5becd63	identical
results/runs/run_0044_2p0_pred_T338K_2p0/deckbuild.out	a5f900792998d2b19f94a4ff55b863bf	identical
results/runs/run_0045_2p0_pred_T358K_2p0/idvg.dat	47fffa91c6dac0b8a0237c015b226a3a	identical
results/runs/run_0045_2p0_pred_T358K_2p0/execution.json	484be4c2b88daea4e08e910142a12b87	identical
results/runs/run_0045_2p0_pred_T358K_2p0/device.in	5f6e39b9c4f1ea9a424b7abcac65b133	identical
results/runs/run_0045_2p0_pred_T358K_2p0/deckbuild.out	1706847d78970fe11e9ddd7855dbc7cd	identical
results/runs/run_0046_6p3_pred_T358K_6p3/idvg.dat	ecfe1f99b0e0156c9d7c026d929e5309	identical
results/runs/run_0046_6p3_pred_T358K_6p3/execution.json	ac6566a376ea032689654967609565df	identical
results/runs/run_0046_6p3_pred_T358K_6p3/device.in	2eb29a95c9319b70d82835d274d8f4ce	identical
results/runs/run_0046_6p3_pred_T358K_6p3/deckbuild.out	72bd89d391696d3f1839d01ac4bd443d	identical
results/runs/run_0047_13p2_pred_T358K_13p2/idvg.dat	102890b1657e435fd534bea02475fed0	identical
results/runs/run_0047_13p2_pred_T358K_13p2/execution.json	4cddb3db0f5169e9967f5334d54b739	identical
results/runs/run_0047_13p2_pred_T358K_13p2/device.in	df4fedf5850342c77541df502667d29a	identical
results/runs/run_0047_13p2_pred_T358K_13p2/deckbuild.out	f1f5c586bd3f9d48224500c4fafbe58f	identical
analysis_2026-09-25/scratch/S6/s6_lib.py	19af8ef0d57ee2e8d7680e5d50351c61	identical
analysis_2026-09-25/scratch/S6/s6_part3_predictions.py	c8b13ef7258ff85801ac7de3e56bcf59	identical
analysis_2026-09-25/scratch/S6/s6_write_register.py	b35fb2a261c423deac9873c1ce85fb6b	identical
analysis_2026-09-25/scratch/S6/s6_part3_out.txt	0ecbc5ca2425424f239f4c254c32b96e	identical
analysis_2026-09-25/predictions_canonical.csv	5c9950be9a3909210364a8d0592725d6	identical

Appendix E

Reproducibility

A simulation study is reproducible only if a reader can regenerate every deck, rerun every launch, recompute every metric and rebuild every figure from archived inputs, and can check that the archived inputs are the ones that were used. This appendix states the software environment, the commands for each stage, the launch budget that constrained the study, the layout of the archive, the hashes of the data files and the integrity rules that governed the work. Paths prefixed `PKG/` are relative to the package root `IWO_PHYSICS_CONSTRAINED_MODEL_V1`; paths prefixed `REPORT/` are relative to `PKG/report_latex_full`.

E.1 Software environment

The device simulator is Silvaco ATLAS 5.28.1.R, driven in batch mode by DeckBuild (`C:/sedatools/exe/deckbuild`); the manual is `C:/sedatools/lib/atlas/5.28.1.R/docs/atlas_users1.pdf` [22]. One 32-bit licence is available, so launches are strictly sequential, and the memory of the executable limits the mesh to about 30 000 nodes (Section 6.2). The batch recipe is `deckbuild.exe -run <deck> -outfile <stdout>`, executed with the run directory as working directory and with relative file names (`runner.argv` in `config/solver.yaml`). Python is used from the virtual environment `IWO_ATLAS_Model_claude/.venv` (with `numpy`, `scipy`, `matplotlib`, `pyyaml` and `pypdf`), in which every script of Chapter B runs. Two tools are not available: Quantum ESPRESSO is not installed (the workflow of `scripts/qe_build_inputs.py` is unexecuted), and no TeX installation exists on the project machine, so the report is checked statically by `tools/lint_latex.py` and compiled elsewhere with `pdflatex` and `BibTeX`. The operating system is Windows, and all paths in the configuration use forward slashes.

E.2 Regenerating the simulation campaigns

All commands are run from `PKG/` with the Python of the environment above. The three stages are independent: decks can be generated and inspected without launching, and a launched run can be re-scored without relaunching.

1. Decks.

```
python scripts/build_iwo_decks.py                                # decks/iwo_{2p0,6p3,13p2,31p8}nm.in
python scripts/build_iwo_decks.py --thickness 6.3 \
```

```
--override traps.fixed_interface_charge_Qf_cm2.value=8.7e10 \
--structure full --sweep stepped
```

2. Single launches. One launch per call; the run is validated, scored and appended to `results/RUN_INDEX.csv`. The overrides of every run are those of Table A.2; for example, `run_0038` corresponds to

```
python scripts/run_atlas.py --thickness 2.0 --label recal_dos384_2p0 \
--override traps.fixed_interface_charge_Qf_cm2.value=1.73e12 \
--override "thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]=17.69" \
--override solver.dos_levels.numa=384 --override solver.dos_levels.numd=192
python scripts/run_atlas.py --rescore results/runs/<run folder> --reason "<why>" # no launch
```

Prediction runs of output characteristics add `-output-vgs "1,1.5,2,2.5,3"` (drain segments `-vd-segments 0.5:0.05,3:0.1` by default); thick films need `-timeout 1500`.

3. Campaigns of 2026-09-25.

```
python scripts/run_campaign.py config/campaign_A_6p3_hypotheses.json # run_0030-0036
python scripts/run_campaign.py config/campaign_B_recad_predictions.json # run_0037-0051
python scripts/run_campaign.py config/campaign_C_6p3_nearEc_band.json # run_0052
```

A campaign launches its entries one after another and writes `results/campaigns/<name>.json`. Rerunning a campaign creates *new* run identifiers; the archived runs are never overwritten.

4. Tables and plots of the package.

```
python scripts/extract_metrics.py --thickness 2.0 # measured metrics of one film
python scripts/compare_experiment_simulation.py # tables/EXPERIMENTAL_VS_SIMULATION.*
python scripts/make_provenance_tables.py # tables/PARAMETER_PROVENANCE.csv etc.
python scripts/plot_results.py
python scripts/plot_sensitivity.py
```

E.3 Regenerating the analysis, the figures and the report

Specialist analyses. The scripts of Section B.3 live in `PKG/analysis_2026-09-25/scratch/<S1..S6, REVIEW, SYNTHESIS>/` and are run from `PKG/` with the same Python; they read run outputs and the measured data and write their outputs next to themselves (e.g. `scratch/SYNTHESIS/syn_checks_out.txt`). None of them launches ATLAS.

Report assets. From `REPORT/`:

```
python tools/build_digest.py # digest/: papers by printed page, manual pages, runs.csv, numbers
python tools/make_figures.py # figures/<stem>.pdf, figures/png/<stem>.png, figures/MANIFEST.md
python tools/make_figures.py measured_transfer numerics_convergence # selected stems only
python tools/make_appendix_assets.py # code/, data/ copies; appendices/A_runs_table.tex, D_hashes_table.tex
python tools/lint_latex.py # static check; exit code 1 on any error
```

Report build. `main.tex` with `preamble.tex` and `references.bib` compiles with the standard sequence (Overleaf default):

```
pdflatex main
bibtex main
pdflatex main
pdflatex main
```

E.4 Launch budget

The number of ATLAS launches was capped by the user (`runner.max_launches` in `config/solver.yaml`). The cap was 30 and was raised to 52 on 2026-09-25 at the user's explicit request, for the 6.3 nm hypothesis study, the DOS 384/192 recalibration, the mesh and sensitivity checks and the predictions. All 52 launches were used (52/52), and 49 of the 52 recorded launches produced scored currents (Section A.1). The one remaining numerical item that would have required a launch, the 31.8 nm mesh refinement (check K), was closed because the film was excluded. Because of the cap, several questions are answered with exact rescaling (linearity of I_D in MUN, Section 6.6) or with surrogates (SURR) instead of new runs, and only one sensitivity per parameter exists. A launch takes between about 2 and 32 minutes (`run_0012`: 125 s; `run_0043`: 1926 s); the default timeout is 900 s, 1500 s for thick films.

E.5 File map of the archive

Table E.1. Where to find each kind of evidence.

Path	Content
PKG/config/	material model, geometry, solver (YAML); campaign plans A, B, C (JSON)
PKG/scripts/	generator, runner, campaign driver, extractor, table and plot scripts (Section B.2)
PKG/data/experimental_clean.csv	the measured transfer curves (<code>thickness_nm</code> , <code>source_row</code> , <code>vg_V</code> , <code>id_A_per_um</code>); <code>data_summary.json</code>
PKG/results/RUN_INDEX.csv	launch record of all 52 recorded launches (49 with scored currents)
PKG/results/runs/run_NNNN_*/	per run: <code>device.in</code> , <code>execution.json</code> (simulated and measured metrics, fit metrics, validation), <code>comparison.csv</code> , <code>idvg.dat</code> , <code>terminal_currents.csv</code> , <code>output_characteristics.csv</code> (I_D - V_D runs), <code>deckbuild.out</code> , band-profile and probe exports
PKG/results/campaigns/	per-campaign metric summaries
PKG/results/BEST_RUNS_V1.json	runs designated as best per film
PKG/tables/	provenance, material parameters, thickness-law evidence, sensitivity and experiment-vs-simulation tables
PKG/docs/	physics model, parameter extraction and provenance, device structure, numerical convergence, limitations, earlier reports
PKG/plots/	package plots per film (overlays, band diagrams, electron density)

Path	Content
PKG/inputs/papers_text/	literature texts with page markers
PKG/inputs/astra_IWO_PRIORITY_THREE/	the 2026-09-25 package and its audit (cross-checks only)
PKG/analysis_2026-09-25/	evidence brief, specialist reports S1–S6, review, synthesis, prediction register and scratch/ scripts
REPORT/chapters/, REPORT/appendices/ REPORT/figures/ REPORT/digest/	the L ^A T _E X sources of this report registry figures (PDF, PNG) and MANIFEST.md writer digest: papers by printed page, manual pages, run table, numbers
REPORT/code/, REPORT/data/	ASCII-sanitised copies of scripts, analysis scripts, decks, configuration and data used in the appendices
REPORT/tools/	report tools (digest, figures, appendix assets, lint)
IWO_ATLAS_Model_claude/ (sibling folder)	the V0 package (34 launches) and the Python environment

E.6 SHA-256 of the data files

Table E.2 gives the leading 16 hexadecimal digits of the SHA-256 digest of the data and configuration files as copied into the report (computed on 2026-09-25 with PowerShell `Get-FileHash`). Because the copies were made ASCII-safe for typesetting, a copy’s digest can differ from that of the package original; the digests of the package files registered with the predictions are given in Table D.1 (Chapter D), and those are the reference for integrity checks.

Table E.2. SHA-256 (first 16 hex digits) and size of the report copies of the data and configuration files.

File (in REPORT/)	SHA-256 (first 16)	Bytes
data/experimental_clean.csv	e27a53e8c72169d5	12362
data/data_summary.json	027f14cfebb77642	1349
data/RUN_INDEX.csv	d281afb916883703	41866
data/runs_catalogue.csv	ba4d4a27dc43c27c	19251
data/BEST_RUNS_V1.json	72b931f71676be01	2201
data/campaign_A_6p3_hypotheses.json	725e49b715886544	15010
data/campaign_B_recal_predictions.json	8b9be148cd9b6fb5	28133
data/campaign_C_6p3_nearEc_band.json	2103b7ca98e69fda	3479
data/predictions_canonical.csv	5c9950be9a390921	142801
data/PREDICTIONS_REGISTER.md	836ead1f731745f9	24985
code/config/iwo_material_model.yaml	8cbd89968f5b8695	6892
code/config/geometry.yaml	9ac32b09a1570e02	2231
code/config/solver.yaml	e315bad35ff021c1	2436
code/config/campaign_A_6p3_hypotheses.json	32b9b4dbbe4793cc	3948
code/config/campaign_B_recal_predictions.json	6c426538bd1a2382	10310
code/config/campaign_C_6p3_nearEc_band.json	a42c0a6b4b74186a	1816

The files `data/campaign_*.json` are the campaign result summaries (`results/campaigns/`), whereas `code/config/campaign_*.json` are the campaign plans; the two sets have the same names but different content.

E.7 Integrity rules

The following rules, set by the user and the supervisor, governed the study and this report.

1. *Real solver only.* Every simulated number comes from a real ATLAS run listed in `results/RUN_INDEX.csv`. No Python model is substituted for the solver; surrogate results are labelled SURR and never shown as ATLAS curves.
2. *Nothing invented.* No parameter, source or run is invented. Where evidence is missing, the report states “not determined from the available data”.
3. *Labels.* Every parameter carries a provenance label, “calibrated” and “validated” are kept distinct, and every claim carries an evidence class (??).
4. *No manufactured agreement.* Native numerical zeros are never replaced by a floor; the measured off-state plateaus remain a limitation of the model.
5. *Complete record.* Failed, timed-out and malformed runs remain in the catalogue with their status (Section A.2); a re-scored run keeps its original `execution.json`, and retracted results (e.g. `run_0027`) are stated as retracted, together with the correction.
6. *Exact rescaling only.* A curve is rescaled without a new launch only where the operation is exact (linearity of I_D in the constant uniform mobility; the $(T/300)^{1.5}$ factor of the temperature variants), and the factor is stated wherever the rescaled curve is used.
7. *Sequential launches by the main session.* With a single 32-bit licence, only the main session launched ATLAS, one run at a time; the specialist agents worked read-only on existing outputs.
8. *Registration before testing.* Predictions were registered with a timestamp (2026-09-24T23:24:53Z) and file hashes before any comparison with measurement; none of the predicted quantities was used in calibration (Chapter D).
9. *Page-exact literature.* Every statement taken from a paper or the manual carries the printed page.

Summary. Every stage can be rerun from the archive: decks from three YAML files and the recorded overrides, launches with one command per run (ATLAS 5.28.1.R, DeckBuild batch, one 32-bit licence), campaigns from three JSON plans, analysis and figures from read-only scripts, and the report from its sources. The launch budget (52/52) is exhausted, which is why exact rescaling and surrogates were used where a launch was not available. Hashes of the archived data and the integrity rules allow a reader to verify that the archived inputs are those used.

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